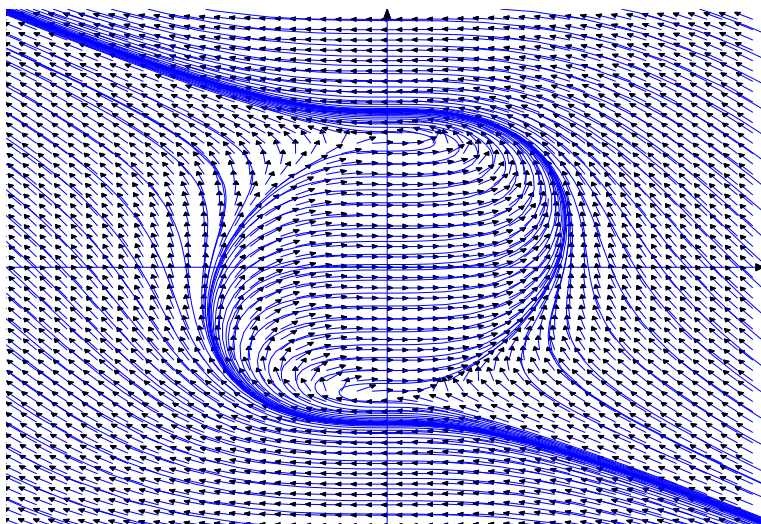


STUDENT SOLUTIONS MANUAL FOR ELEMENTARY DIFFERENTIAL EQUATIONS



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Chapter 1

Introduction

1.2 Basic Concepts

1.2.2. (a) If $y = ce^{2x}$, then $y' = 2ce^{2x} = 2y$.

(b) If $y = \frac{x^2}{3} + \frac{c}{x}$, then $y' = \frac{2x}{3} - \frac{c}{x^2}$, so $xy' + y = \frac{2x^2}{3} - \frac{c}{x} + \frac{x^2}{3} + \frac{c}{x} = x^2$.

(c) If

$$y = \frac{1}{2} + ce^{-x^2}, \quad \text{then} \quad y' = -2xce^{-x^2}$$

and

$$y' + 2xy = -2xce^{-x^2} + 2x\left(\frac{1}{2} + ce^{-x^2}\right) = -2xce^{-x^2} + x + 2xce^{-x^2} = x.$$

(d) If

$$y = \frac{1 + ce^{-x^2/2}}{1 - ce^{-x^2/2}}$$

then

$$\begin{aligned} y' &= \frac{(1 - ce^{-x^2/2})(-cxe^{-x^2/2}) - (1 + ce^{-x^2/2})cxe^{-x^2/2}}{(1 - ce^{-x^2/2})^2} \\ &= \frac{-2cxe^{-x^2/2}}{(1 - ce^{-x^2/2})^2} \end{aligned}$$

and

$$\begin{aligned} y^2 - 1 &= \left(\frac{1 + ce^{-x^2/2}}{1 - ce^{-x^2/2}}\right)^2 - 1 \\ &= \frac{(1 + ce^{-x^2/2})^2 - (1 - ce^{-x^2/2})^2}{(1 - ce^{-x^2/2})^2} \end{aligned}$$

$$= \frac{4ce^{-x^2/2}}{(1 - ce^{-x^2/2})^2},$$

so

$$2y' + x(y^2 - 1) = \frac{-4cx + 4cx}{(1 - ce^{-x^2/2})^2} = 0.$$

(e) If $y = \tan\left(\frac{x^3}{3} + c\right)$, then

$$y' = x^2 \sec^2\left(\frac{x^3}{3} + c\right) = x^2 \left(1 + \tan^2\left(\frac{x^3}{3} + c\right)\right) = x^2(1 + y^2).$$

(f) If

$$\begin{aligned} y &= (c_1 + c_2x)e^x + \sin x + x^2, \quad \text{then} \\ y' &= (c_1 + 2c_2x)e^x + \cos x + 2x, \\ y' &= (c_1 + 3c_2x)e^x - \sin x + 2, \end{aligned}$$

and

$$\begin{aligned} y'' - 2y' + y &= c_1e^x(1 - 2 + 1) + c_2xe^x(3 - 4 + 1) \\ &\quad - \sin x - 2\cos x + \sin x + 2 - 4x + x^2 \\ &= -2\cos x + x^2 - 4x + 2. \end{aligned}$$

(g) If $y = c_1e^x + c_2x + \frac{2}{x}$, then $y' = c_1e^x + c_2 - \frac{2}{x^2}$ and $y'' = c_1e^x + \frac{4}{x^3}$, so

$$(1 - x)y'' + xy' - y = c_1(1 - x + x - 1) + c_2(x - x) + \frac{4(1 - x)}{x^3} - \frac{2}{x} - \frac{2}{x} = \frac{4(1 - x - x^2)}{x^3}$$

(h) If $y = \frac{c_1 \sin x + c_2 \cos x}{x^{1/2}} + 4x + 8$ then $y' = \frac{c_1 \cos x - c_2 \sin x}{x^{1/2}} - \frac{c_1 \sin x + c_2 \cos x}{2x^{3/2}} + 4$

$$\text{and } y'' = -\frac{c_1 \sin x + c_2 \cos x}{x^{1/2}} - \frac{c_1 \sin x - c_2 \cos x}{x^{3/2}} + \frac{3}{4} \frac{c_1 \sin x + c_2 \cos x}{x^{5/2}}, \text{ so}$$

$$\begin{aligned} x^2 y'' + xy' + \left(x^2 - \frac{1}{4}\right)y &= c_1 \left(-x^{-3/2} \sin x - x^{1/2} \cos x + \frac{3}{4}x^{-1/2} \sin x + x^{1/2} \cos x - \right. \\ &\quad \left. \frac{1}{2}x^{-1/2} \sin x + x^{3/2} \sin x - \frac{1}{4}x^{-1/2} \sin x\right) + c_2 \left(-x^{-3/2} \cos x + x^{1/2} \sin x + \frac{3}{4}x^{-1/2} \cos x \right. \\ &\quad \left. - x^{1/2} \sin x - \frac{1}{2}x^{-1/2} \cos x + x^{3/2} \cos x - \frac{1}{4}x^{-1/2} \cos x\right) + 4x + \left(x^2 - \frac{1}{4}\right)(4x + 8) = \\ &= 4x^3 + 8x^2 + 3x - 2. \end{aligned}$$

1.2.4. (a) If $y' = -xe^x$, then $y = -xe^x + \int e^x dx + c = (1 - x)e^x + c$, and $y(0) = 1 \Rightarrow 1 = 1 + c$, so $c = 0$ and $y = (1 - x)e^x$.

(b) If $y' = x \sin x^2$, then $y = -\frac{1}{2} \cos x^2 + c$; $y\left(\sqrt{\frac{\pi}{2}}\right) = 1 \Rightarrow 1 = 0 + c$, so $c = 1$ and $y = 1 - \frac{1}{2} \cos x^2$.

(c) Write $y' = \tan x = \frac{\sin x}{\cos x} = -\frac{1}{\cos x} \frac{d}{dx}(\cos x)$. Integrating this yields

$$\begin{aligned} y &= -\ln|\cos x| + c; \quad y(\pi/4) = 3 \Rightarrow 3 = -\ln(\cos(\pi/4)) + c, \text{ or } 3 = \ln\sqrt{2} + c, \text{ so} \\ c &= 3 - \ln\sqrt{2}, \text{ so } y = -\ln(|\cos x|) + 3 - \ln\sqrt{2} = 3 - \ln(\sqrt{2}|\cos x|). \end{aligned}$$

- (d) If $y'' = x^4$, then $y' = \frac{x^5}{5} + c_1$; $y'(2) = -1 \Rightarrow \frac{32}{5} + c_1 = -1 \Rightarrow c_1 = -\frac{37}{5}$, so
 $y' = \frac{x^5}{5} - \frac{37}{5}$. Therefore, $y = \frac{x^6}{30} - \frac{37}{15}(x-2) + c_2$;
 $y(2) = -1 \Rightarrow \frac{64}{30} + c_2 = -1 \Rightarrow c_2 = -\frac{47}{15}$, so $y = -\frac{47}{15} - \frac{37}{5}(x-2) + \frac{x^6}{30}$.
- (e) (A) $\int x e^{2x} dx = \frac{x e^{2x}}{2} - \frac{1}{2} \int e^{2x} dx = \frac{x e^{2x}}{2} - \frac{e^{2x}}{4}$. Therefore, $y' = \frac{x e^{2x}}{2} - \frac{e^{2x}}{4} + c_1$;
 $y'(0) = 1 \Rightarrow -\frac{1}{4} + c_1 = \frac{5}{4} \Rightarrow c_1 = \frac{5}{4}$, so $y' = \frac{x e^{2x}}{2} - \frac{e^{2x}}{4} + \frac{5}{4}$; Using (A) again,
 $y = \frac{x e^{2x}}{4} - \frac{e^{2x}}{8} - \frac{e^{2x}}{8} + \frac{5}{4}x + c_2 = \frac{x e^{2x}}{4} - \frac{e^{2x}}{4} + \frac{5}{4}x + c_2$;
 $y(0) = 7 \Rightarrow -\frac{1}{4} + c_2 = 7 \Rightarrow c_2 = \frac{29}{4}$, so $y = \frac{x e^{2x}}{4} - \frac{e^{2x}}{4} + \frac{5}{4}x + \frac{29}{4}$.
- (f) (A) $\int x \sin x dx = -x \cos x + \int \cos x dx = -x \cos x + \sin x$ and (B)
 $\int x \cos x dx = x \sin x - \int \sin x dx = x \sin x + \cos x$. If $y'' = -x \sin x$, then (A) implies
that $y' = x \cos x - \sin x + c_1$; $y'(0) = -3 \Rightarrow c_1 = -3$, so $y' = x \cos x - \sin x - 3$. Now
(B) implies that $y = x \sin x + \cos x + \cos x - 3x + c_2 = x \sin x + 2 \cos x - 3x + c_2$;
 $y(0) = 1 \Rightarrow 2 + c_2 = 1 \Rightarrow c_2 = -1$, so $y = x \sin x + 2 \cos x - 3x - 1$.
- (g) If $y''' = x^2 e^x$, then $y'' = \int x^2 e^x dx = x^2 e^x - 2 \int x e^x dx = x^2 e^x - 2x e^x + 2e^x + c_1$;
 $y''(0) = 3 \Rightarrow 2 + c_1 = 3 \Rightarrow c_1 = 1$, so (A) $y'' = (x^2 - 2x + 2)e^x + 1$. Since
 $\int (x^2 - 2x + 2)e^x dx = (x^2 - 2x + 2)e^x - \int (2x - 2)e^x dx =$
 $(x^2 - 2x + 2)e^x - (2x - 2)e^x + 2e^x = (x^2 - 4x + 6)e^x$, (A) implies that
 $y' = (x^2 - 4x + 6)e^x + x + c_2$; $y'(0) = -2 \Rightarrow 6 + c_2 = -2 \Rightarrow c_2 = -8$, so (B)
 $y' = (x^2 - 4x + 6)e^x + x - 8$; Since $\int (x^2 - 4x + 6)e^x dx =$
 $(x^2 - 4x + 6)e^x - \int (2x - 4)e^x dx = (x^2 - 4x + 6)e^x - (2x - 4)e^x + 2e^x = (x^2 - 6x + 12)e^x$,
(B) implies that $y = (x^2 - 6x + 12)e^x + \frac{x^2}{2} - 8x + c_3$;
 $y(0) = 1 \Rightarrow 12 + c_3 = 1 \Rightarrow c_3 = -11$, so $y = (x^2 - 6x + 12)e^x + \frac{x^2}{2} - 8x - 11$.
- (h) If $y''' = 2 + \sin 2x$, then $y'' = 2x - \frac{\cos 2x}{2} + c_1$; $y''(0) = 3 \Rightarrow -\frac{1}{2} + c_1 = 3 \Rightarrow c_1 = \frac{7}{2}$, so
 $y'' = 2x - \frac{\cos 2x}{2} + \frac{7}{2}$. Then $y' = x^2 - \frac{\sin 2x}{4} + \frac{7}{2}x + c_2$; $y'(0) = -6 \Rightarrow c_2 = -6$, so
 $y' = x^2 - \frac{\sin 2x}{4} + \frac{7}{2}x - 6$. Then $y = \frac{x^3}{3} + \frac{\cos 2x}{8} + \frac{7}{4}x^2 - 6x + c_3$;
 $y(0) = 1 \Rightarrow \frac{1}{8} + c_3 = 1 \Rightarrow c_3 = \frac{7}{8}$, so $y = \frac{x^3}{3} + \frac{\cos 2x}{8} + \frac{7}{4}x^2 - 6x + \frac{7}{8}$.
- (i) If $y''' = 2x + 1$, then $y'' = x^2 + x + c_1$; $y''(2) = 7 \Rightarrow 6 + c_1 = 7 \Rightarrow c_1 = 1$; so
 $y'' = x^2 + x + 1$. Then $y' = \frac{x^3}{3} + \frac{x^2}{2} + (x-2) + c_2$;
 $y'(2) = -4 \Rightarrow \frac{14}{3} + c_2 = -4 \Rightarrow c_2 = -\frac{26}{3}$, so $y' = \frac{x^3}{3} + \frac{x^2}{2} + (x-2) - \frac{26}{3}$. Then
 $y = \frac{x^4}{12} + \frac{x^3}{6} + \frac{1}{2}(x-2)^2 - \frac{26}{3}(x-2) + c_3$; $y(2) = 1 \Rightarrow \frac{8}{3} + c_3 = 1 \Rightarrow c_3 = -\frac{5}{3}$, so
 $y = \frac{x^4}{12} + \frac{x^3}{6} + \frac{1}{2}(x-2)^2 - \frac{26}{3}(x-2) - \frac{5}{3}$.

- 1.2.6. (a) If $y = x^2(1 + \ln x)$, then $y(e) = e^2(1 + \ln e) = 2e^2$; $y' = 2x(1 + \ln x) + x = 3x + 2x \ln x$, so $y'(e) = 3e + 2e \ln e = 5e$; (A) $y'' = 3 + 2 + 2 \ln x = 5 + 2 \ln x$. Now, $3xy' - 4y = 3x(3x + 2x \ln x) - 4x^2(1 + \ln x) = 5x^2 + 2x^2 \ln x = x^2 y''$, from (A).
- (b) If $y = \frac{x^2}{3} + x - 1$, then $y(1) = \frac{1}{3} + 1 - 1 = \frac{1}{3}$; $y' = \frac{2}{3}x + 1$, so $y'(1) = \frac{2}{3} + 1 = \frac{5}{3}$; (A) $y'' = \frac{2}{3}$. Now $x^2 - xy' + y + 1 = x^2 - x\left(\frac{2}{3}x + 1\right) + \frac{x^2}{3} + x - 1 + 1 = \frac{2}{3}x^2 = x^2 y''$, from (A).
- (c) If $y = (1 + x^2)^{-1/2}$, then $y(0) = (1 + 0^2)^{-1/2} = 1$; $y' = -x(1 + x^2)^{-3/2}$, so $y'(0) = 0$; (A) $y'' = (2x^2 - 1)(1 + x^2)^{-5/2}$. Now, $(x^2 - 1)y - x(x^2 + 1)y' = (x^2 - 1)(1 + x^2)^{-1/2} - x(x^2 + 1)(-x)(1 + x^2)^{-3/2} = (2x^2 - 1)(1 + x^2)^{-1/2} = y''(1 + x^2)^2$ from (A), so $y'' = \frac{(x^2 - 1)y - x(x^2 + 1)y'}{(1 + x^2)^2}$.
- (d) If $y = \frac{x^2}{1 - x}$, then $y(1/2) = \frac{1/4}{1 - 1/2} = \frac{1}{2}$; $y' = -\frac{x(x - 2)}{(1 - x)^2}$, so $y'(1/2) = \frac{(-1/2)(-3/2)}{(1 - 1/2)^2} = 3$; (A) $y'' = \frac{2}{(1 - x)^3}$. Now, (B) $x + y = x + \frac{x^2}{1 - x} = \frac{x}{1 - x}$ and (C) $xy' - y = -\frac{x^2(x - 2)}{(1 - x)^2} - \frac{x^2}{1 - x} = \frac{x^2}{(1 - x)^2}$. From (B) and (C), $(x + y)(xy' - y) = \frac{x^3}{(1 - x)^3} = \frac{x^3}{2} y''$, so $y'' = \frac{2(x + y)(xy' - y)}{x^3}$.
- 1.2.8. (a) $y = (x - c)^a$ is defined and $x - c = y^{1/a}$ on (c, ∞) ; moreover, $y' = a(x - c)^{a-1} = a(y^{1/a})^{a-1} = ay^{(a-1)/a}$.
- (b) if $a > 1$ or $a < 0$, then $y \equiv 0$ is a solution of (B) on $(-\infty, \infty)$.
- 1.2.10. (a) Since $y' = c$ we must show that the right side of (B) reduces to c for all values of x in some interval. If $y = c^2 + cx + 2c + 1$,

$$\begin{aligned} x^2 + 4x + 4y &= x^2 + 4x + 4c^2 + 4cx + 8c + 4 \\ &= x^2 + 4(1 + c)x + 4(c^2 + 2c + 1) \\ &= x^2 + 4(1 + c) + 2(c + 1)^2 = (x + 2c + 2)^2. \end{aligned}$$

Therefore, $\sqrt{x^2 + 4x + 4y} = x + 2c + 2$ and the right side of (B) reduces to c if $x > -2c - 2$.

- (b) If $y_1 = -\frac{x(x + 4)}{4}$, then $y_1' = -\frac{x + 2}{2}$ and $x^2 + 4x + 4y = 0$ for all x . Therefore, y_1 satisfies (A) on $(-\infty, \infty)$.

Chapter 2

First Order Equations

2.1 Linear First Order Equations

$$2.1.2. \quad \frac{y'}{y} = -3x^2; \quad |\ln|y|| = -x^3 + k; \quad y = ce^{-x^3}. \quad y = ce^{-(\ln x)^2/2}.$$

$$2.1.4. \quad \frac{y'}{y} = -\frac{3}{x}; \quad \ln|y| = -3\ln|x| + k = -\ln|x|^3 + k; \quad y = \frac{c}{x^3}.$$

$$2.1.6. \quad \frac{y'}{y} = -\frac{1+x}{x} = -\frac{1}{x} - 1; \quad |\ln|y|| = -\ln|x| - x + k; \quad y = \frac{ce^{-x}}{x}; \quad y(1) = 1 \Rightarrow c = e;$$
$$y = \frac{e^{-(x-1)}}{x}.$$

$$2.1.8. \quad \frac{y'}{y} = -\frac{1}{x} - \cot x; \quad |\ln|y|| = -\ln|x| - \ln|\sin x| + k = -\ln|x \sin x| + k; \quad y = \frac{c}{x \sin x};$$
$$y(\pi/2) = 2 \Rightarrow c = \pi; \quad y = \frac{\pi}{x \sin x}.$$

$$2.1.10. \quad \frac{y'}{y} = -\frac{k}{x}; \quad |\ln|y|| = -k\ln|x| + k_1 = \ln|x^{-k}| + k_1; \quad y = c|x|^{-k}; \quad y(1) = 3 \Rightarrow c = 3;$$
$$y = 3x^{-k}.$$

$$2.1.12. \quad \frac{y_1'}{y_1} = -3; \quad \ln|y_1| = -3x; \quad y_1 = e^{-3x}; \quad y = ue^{-3x}; \quad u'e^{-3x} = 1; \quad u' = e^{3x}; \quad u = \frac{e^{3x}}{3} + c;$$
$$y = \frac{1}{3} + ce^{-3x}.$$

$$2.1.14. \quad \frac{y_1'}{y_1} = -2x; \quad \ln|y_1| = -x^2; \quad y_1 = e^{-x^2}; \quad y = ue^{-x^2}; \quad u'e^{-x^2} = xe^{-x^2}; \quad u' = x;$$
$$u = \frac{x^2}{2} + c; \quad y = e^{-x^2} \left(\frac{x^2}{2} + c \right).$$

$$2.1.16. \quad \frac{y_1'}{y_1} = -\frac{1}{x}; \quad \ln|y_1| = -\ln|x|; \quad y_1 = \frac{1}{x}; \quad y = \frac{u}{x}; \quad \frac{u'}{x} = \frac{7}{x^2} + 3; \quad u' = \frac{7}{x} + 3x;$$

$$u = 7 \ln|x| + \frac{3x^2}{2} + c; \quad y = \frac{7 \ln|x|}{x} + \frac{3x}{2} + \frac{c}{x}.$$

$$2.1.18. \quad \frac{y_1'}{y_1} = -\frac{1}{x} - 2x; \quad \ln|y_1| = -\ln|x| - x^2; \quad y_1 = \frac{e^{-x^2}}{x}; \quad y = \frac{ue^{-x^2}}{x}; \quad \frac{u'e^{-x^2}}{x} = x^2e^{-x^2};$$

$$u' = x^3; \quad u = \frac{x^4}{4} + c; \quad y = e^{-x^2} \left(\frac{x^3}{4} + \frac{c}{x} \right).$$

$$2.1.20. \quad \frac{y_1'}{y_1} = -\tan x; \quad \ln|y_1| = \ln|\cos x|; \quad y_1 = \cos x; \quad y = u \cos x; \quad u' \cos x = \cos x; \quad u' = 1;$$

$$u = x + c; \quad y = (x + c) \cos x.$$

$$2.1.22. \quad \frac{y_1'}{y_1} = \frac{4x-3}{(x-2)(x-1)} = \frac{5}{x-2} - \frac{1}{x-1}; \quad \ln|y_1| = 5 \ln|x-2| - \ln|x-1| = \ln \left| \frac{(x-2)^5}{x-1} \right|;$$

$$y_1 = \frac{(x-2)^5}{x-1}; \quad y = \frac{u(x-2)^5}{x-1}; \quad \frac{u'(x-2)^5}{x-1} = \frac{(x-2)^2}{x-1}; \quad u' = \frac{1}{(x-2)^3};$$

$$u = -\frac{1}{2} \frac{1}{(x-2)^2} + c; \quad y = -\frac{1}{2} \frac{(x-2)^3}{(x-1)} + c \frac{(x-2)^5}{(x-1)}.$$

$$2.1.24. \quad \frac{y_1'}{y_1} = -\frac{3}{x}; \quad \ln|y_1| = -3 \ln|x| = \ln|x|^{-3}; \quad y_1 = \frac{1}{x^3}; \quad y = \frac{u}{x^3}; \quad \frac{u'}{x^3} = \frac{e^x}{x^2}; \quad u' = xe^x;$$

$$u = xe^x - e^x + c; \quad y = \frac{e^x}{x^2} - \frac{e^x}{x^3} + \frac{c}{x^3}.$$

$$2.1.26. \quad \frac{y_1'}{y_1} = -\frac{4x}{1+x^2}; \quad \ln|y_1| = -2 \ln(1+x^2) = \ln(1+x^2)^{-2}; \quad y_1 = \frac{1}{(1+x^2)^2}; \quad y = \frac{u}{(1+x^2)^2};$$

$$\frac{u'}{(1+x^2)^2} = \frac{2}{(1+x^2)^2}; \quad u' = 2; \quad u = 2x + c; \quad y = \frac{2x+c}{(1+x^2)^2}; \quad y(0) = 1 \Rightarrow c = 1;$$

$$y = \frac{2x+1}{(1+x^2)^2}.$$

$$2.1.28. \quad \frac{y_1'}{y_1} = -\cot x; \quad \ln|y_1| = -\ln|\sin x|; \quad y_1 = \frac{1}{\sin x}; \quad y = \frac{u}{\sin x}; \quad \frac{u'}{\sin x} = \cos x;$$

$$u' = \sin x \cos x; \quad u = \frac{\sin^2 x}{2} + c; \quad y = \frac{\sin x}{2} + c \csc x; \quad y(\pi/2) = 1 \Rightarrow c = \frac{1}{2};$$

$$y = \frac{1}{2}(\sin x + \csc x).$$

$$2.1.30. \quad \frac{y_1'}{y_1} = -\frac{3}{x-1}; \quad \ln|y_1| = -3 \ln|x-1| = \ln|x-1|^{-3}; \quad y_1 = \frac{1}{(x-1)^3}; \quad y = \frac{u}{(x-1)^3};$$

$$\frac{u'}{(x-1)^3} = \frac{1}{(x-1)^4} + \frac{\sin x}{(x-1)^3}; \quad u' = \frac{1}{x-1} + \sin x; \quad u = \ln|x-1| - \cos x + c;$$

$$y = \frac{\ln|x-1| - \cos x + c}{(x-1)^3}; \quad y(0) = 1 \Rightarrow c = 0; \quad y = \frac{\ln|x-1| - \cos x}{(x-1)^3}.$$

$$2.1.32. \quad \frac{y_1'}{y_1} = -\frac{2}{x}; \quad \ln|y_1| = 2 \ln|x| = \ln(x^2); \quad y_1 = x^2; \quad y = ux^2; \quad u'x^2 = -x; \quad u' = -\frac{1}{x};$$

$$u = -\ln|x| + c; \quad y = x^2(c - \ln|x|); \quad y(1) = 1 \Rightarrow c = 1; \quad y = x^2(1 - \ln x).$$

$$\begin{aligned}
2.1.34. \quad \frac{y_1'}{y_1} &= -\frac{3}{x-1}; \quad \ln|y_1| = -3\ln|x-1| = \ln|x-1|^{-3}; \quad y_1 = \frac{1}{(x-1)^3}; \quad y = \frac{u}{(x-1)^3}; \\
\frac{u'}{(x-1)^3} &= \frac{1+(x-1)\sec^2 x}{(x-1)^4}; \quad u' = \frac{1}{x-1} + \sec^2 x; \quad u = \ln|x-1| + \tan x + c; \\
y &= \frac{\ln|x-1| + \tan x + c}{(x-1)^3}; \quad y(0) = -1 \Rightarrow c = 1; \quad y = \frac{\ln|x-1| + \tan x + 1}{(x-1)^3}.
\end{aligned}$$

$$\begin{aligned}
2.1.36. \quad \frac{y_1'}{y_1} &= \frac{2x}{x^2-1}; \quad \ln|y_1| = \ln|x^2-1|; \quad y_1 = x^2-1; \quad y = u(x^2-1); \quad u'(x^2-1) = x; \\
u' &= \frac{x}{x^2-1}; \quad u = \frac{1}{2}\ln|x^2-1| + c; \quad y = (x^2-1)\left(\frac{1}{2}\ln|x^2-1| + c\right); \quad y(0) = 4 \Rightarrow c = -4; \\
y &= (x^2-1)\left(\frac{1}{2}\ln|x^2-1| - 4\right).
\end{aligned}$$

$$\begin{aligned}
2.1.38. \quad \frac{y_1'}{y_1} &= -2x; \quad \ln|y_1| = -x^2; \quad y_1 = e^{-x^2}; \quad y = ue^{-x^2}; \quad u'e^{-x^2} = x^2; \quad u' = x^2e^{x^2}; \\
u &= c + \int_0^x t^2 e^{t^2} dt; \quad y = e^{-x^2}\left(c + \int_0^x t^2 e^{t^2} dt\right); \quad y(0) = 3 \Rightarrow c = 3; \\
y &= e^{-x^2}\left(3 + \int_0^x t^2 e^{t^2} dt\right).
\end{aligned}$$

$$\begin{aligned}
2.1.40. \quad \frac{y_1'}{y_1} &= -1; \quad \ln|y_1| = -x; \quad y_1 = e^{-x}; \quad y = ue^{-x}; \quad u'e^{-x} = \frac{e^{-x}\tan x}{x}; \quad u' = \frac{\tan x}{x}; \\
u &= c + \int_1^x \frac{\tan t}{t} dt; \quad y = e^{-x}\left(c + \int_1^x \frac{\tan t}{t} dt\right); \quad y(1) = 0 \Rightarrow c = 0; \quad y = e^{-x} \int_1^x \frac{\tan t}{t} dt.
\end{aligned}$$

$$\begin{aligned}
2.1.42. \quad \frac{y_1'}{y_1} &= -1 - \frac{1}{x}; \quad \ln|y_1| = -x - \ln|x|; \quad y_1 = \frac{e^{-x}}{x}; \quad y = \frac{ue^{-x}}{x}; \quad \frac{u'e^{-x}}{x} = \frac{e^{x^2}}{x}; \quad u' = e^x e^{x^2}; \\
u &= c + \int_1^x e^t e^{t^2} dt; \quad y = \frac{e^{-x}}{x}\left(c + \int_1^x e^t e^{t^2} dt\right); \quad y(1) = 2 \Rightarrow c = 2e; \\
y &= \frac{1}{x}\left(2e^{-(x-1)} + e^{-x} \int_1^x e^t e^{t^2} dt\right).
\end{aligned}$$

2.1.44. (b) Eqn. (A) is equivalent to

$$y' - \frac{2}{x} = -\frac{1}{x} \quad (\text{B})$$

on $(-\infty, 0)$ and $(0, \infty)$. Here $\frac{y_1'}{y_1} = \frac{2}{x}$; $\ln|y_1| = 2\ln|x|$; $y_1 = x^2$; $y = ux^2$;

$u'x^2 = -\frac{1}{x}$; $u' = -\frac{1}{x^3}$; $u = \frac{1}{2x^2} + c$, so $y = \frac{1}{2} + cx^2$ is the general solution of (A) on $(-\infty, 0)$ and $(0, \infty)$.

(c) From the proof of (b), any solution of (A) must be of the form

$$y = \begin{cases} \frac{1}{2} + c_1 x^2, & x \geq 0, \\ \frac{1}{2} + c_2 x^2, & x < 0, \end{cases} \quad (\text{C})$$

for $x \neq 0$, and any function of the form (C) satisfies (A) for $x \neq 0$. To complete the proof we must show that any function of the form (C) is differentiable and satisfies (A) at $x = 0$. By definition,

$$y'(0) = \lim_{x \rightarrow 0} \frac{y(x) - y(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{y(x) - 1/2}{x}$$

if the limit exists. But

$$\frac{y(x) - 1/2}{x} = \begin{cases} c_1 x, & x > 0 \\ c_2 x, & x < 0, \end{cases}$$

so $y'(0) = 0$. Since $0y'(0) - 2y(0) = 0 \cdot 0 - 2(1/2) = -1$, any function of the form (C) satisfies (A) at $x = 0$.

- (d) From (b) any solution y of (A) on $(-\infty, \infty)$ is of the form (C), so $y(0) = 1/2$.
- (e) If $x_0 > 0$, then every function of the form (C) with $c_1 = \frac{y_0 - 1/2}{x_0^2}$ and c_2 arbitrary is a solution of the initial value problem on $(-\infty, \infty)$. Since these functions are all identical on $(0, \infty)$, this does not contradict Theorem 2.1.1, which implies that (B) (so (A)) has exactly one solution on $(0, \infty)$ such that $y(x_0) = y_0$. A similar argument applies if $x_0 < 0$.

2.1.46. (a) Let $y = c_1 y_1 + c_2 y_2$. Then

$$\begin{aligned} y' + p(x)y &= (c_1 y_1 + c_2 y_2)' + p(x)(c_1 y_1 + c_2 y_2) \\ &= c_1 y_1' + c_2 y_2' + c_1 p(x)y_1 + c_2 p(x)y_2 \\ &= c_1 (y_1' + p(x)y_1) + c_2 (y_2' + p(x)y_2) = c_1 f_1(x) + c_2 f_2(x). \end{aligned}$$

(b) Let $f_1 = f_2 = f$ and $c_1 = -c_2 = 1$.

(c) Let $f_1 = f$, $f_2 = 0$, and $c_1 = c_2 = 1$.

2.1.48. (a) If $z = \tan y$, then $z' = (\sec^2 y)y'$, so $z' - 3z = -1$; $z_1 = e^{3x}$; $z = ue^{3x}$; $u'e^{3x} = -1$;

$$u' = -e^{-3x}; \quad u = \frac{e^{-3x}}{3} + c; \quad z = \frac{1}{3} + ce^{3x} = \tan y; \quad y = \tan^{-1} \left(\frac{1}{3} + ce^{3x} \right).$$

(b) If $z = e^{y^2}$, then $z' = 2yy'e^{y^2}$, so $z' + \frac{2}{x}z = \frac{1}{x^2}$; $z_1 = \frac{1}{x^2}$; $z = \frac{u}{x^2}$; $\frac{u'}{x^2} = \frac{1}{x^2}$; $u' = 1$;

$$u = x + c; \quad z = \frac{1}{x} + \frac{c}{x^2} = e^{y^2}; \quad y = \pm \left[\ln \left(\frac{1}{x} + \frac{c}{x^2} \right) \right]^{1/2}.$$

(c) Rewrite the equation as $\frac{y'}{y} + \frac{2}{x} \ln y = 4x$. If $z = \ln y$, then $z' = \frac{y'}{y}$, so $z' + \frac{2}{x}z = 4x$;

$$z_1 = \frac{1}{x^2}; \quad z = \frac{u}{x^2}; \quad \frac{u'}{x^2} = 4x; \quad u' = 4x^3; \quad u = x^4 + c; \quad z = x^2 + \frac{c}{x^2} = \ln y;$$

$$y = \exp \left(x^2 + \frac{c}{x^2} \right).$$

(d) If $z = -\frac{1}{1+y}$, then $z' = \frac{y'}{(1+y)^2}$, so $z' + \frac{1}{x}z = -\frac{3}{x^2}$; $z_1 = \frac{1}{x}$; $z = \frac{u}{x}$; $\frac{u'}{x} = -\frac{3}{x^2}$;

$$u' = -\frac{3}{x}; \quad u = -3 \ln |x| - c; \quad z = -\frac{3 \ln |x| + c}{x} = -\frac{1}{1+y}; \quad y = -1 + \frac{x}{3 \ln |x| + c}.$$

2.2 Separable Equations

2.2.2. By inspection, $y \equiv k\pi$ (k = integer) is a constant solution. Separate variables to find others:

$$\left(\frac{\cos y}{\sin y}\right)y' = -\sin x; \ln(|\sin y|) = \cos x + c.$$

2.2.4. $y \equiv 0$ is a constant solution. Separate variables to find others: $\left(\frac{\ln y}{y}\right)y' = -x^2$;

$$\frac{(\ln y)^2}{2} = -\frac{x^3}{3} + c.$$

2.2.6. $y \equiv 1$ and $y \equiv -1$ are constant solutions. For others, separate variables:

$$(y^2 - 1)^{-3/2}yy' = \frac{1}{x^2}; \quad -(y^2 - 1)^{-1/2} = -\frac{1}{x} - c = -\left(\frac{1 + cx}{x}\right); \quad (y^2 - 1)^{1/2} = \left(\frac{x}{1 + cx}\right);$$

$$(y^2 - 1) = \left(\frac{x}{1 + cx}\right)^2; \quad y^2 = 1 + \left(\frac{x}{1 + cx}\right)^2; \quad y = \pm \left(1 + \left(\frac{x}{1 + cx}\right)^2\right)^{1/2}.$$

2.2.8. By inspection, $y \equiv 0$ is a constant solution. Separate variables to find others:

$$\frac{y'}{y} = -\frac{x}{1 + x^2}; \quad \ln|y| = -\frac{1}{2}\ln(1 + x^2) + k; \quad y = \frac{c}{\sqrt{1 + x^2}}, \text{ which includes the constant solution } y \equiv 0.$$

$$2.2.10. (y - 1)^2y' = 2x + 3; \quad \frac{(y - 1)^3}{3} = x^2 + 3x + c; \quad (y - 1)^3 = 3x^2 + 9x + c;$$

$$y = 1 + (3x^2 + 9x + c)^{1/3}.$$

$$2.2.12. \frac{y'}{y(y + 1)} = -x; \quad \left[\frac{1}{y} - \frac{1}{y + 1}\right]y' = -x; \quad \ln\left|\frac{y}{y + 1}\right| = -\frac{x^2}{2} + k; \quad \frac{y}{y + 1} = ce^{-x^2/2};$$

$$y(2) = 1 \Rightarrow c = \frac{e^2}{2}; \quad y = (y + 1)ce^{-x^2/2}; \quad y(1 - ce^{-x^2/2}) = ce^{-x^2/2}; \quad y = \frac{ce^{-x^2/2}}{1 - ce^{-x^2/2}}; \text{ setting}$$

$$c = \frac{e^2}{2} \text{ yields } y = \frac{e^{-(x^2-4)/2}}{2 - e^{-(x^2-4)/2}}.$$

$$2.2.14. \frac{y'}{(y + 1)(y - 1)(y - 2)} = -\frac{1}{x + 1}; \quad \left[\frac{1}{6}\frac{1}{y + 1} - \frac{1}{2}\frac{1}{y - 1} + \frac{1}{3}\frac{1}{y - 2}\right]y' = -\frac{1}{x + 1};$$

$$\left[\frac{1}{y + 1} - \frac{3}{y - 1} + \frac{2}{y - 2}\right]y' = -\frac{6}{x + 1}; \quad \ln|y + 1| - 3\ln|y - 1| + 2\ln|y - 2| = -6\ln|x + 1| + k;$$

$$\frac{(y + 1)(y - 2)^2}{(y - 1)^3} = \frac{c}{(x + 1)^6}; \quad y(1) = 0 \Rightarrow c = -256; \quad \frac{(y + 1)(y - 2)^2}{(y - 1)^3} = -\frac{256}{(x + 1)^6}.$$

$$2.2.16. \frac{y'}{y(1 + y^2)} = 2x; \quad \left[\frac{1}{y} - \frac{y}{y^2 + 1}\right]y' = 2x; \quad \ln\left(\frac{|y|}{\sqrt{y^2 + 1}}\right) = x^2 + k; \quad \frac{y}{\sqrt{y^2 + 1}} = ce^{x^2};$$

$$y(0) = 1 \Rightarrow c = \frac{1}{\sqrt{2}}; \quad \frac{y}{\sqrt{y^2 + 1}} = \frac{e^{x^2}}{\sqrt{2}}; \quad 2y^2 = (y^2 + 1)e^{x^2}; \quad y^2(2 - e^{x^2}) = e^{2x^2};$$

$$y = \frac{1}{\sqrt{2e^{-2x^2} - 1}}.$$

2.2.18. $\frac{y'}{(y-1)(y-2)} = -2x$; $\left[\frac{1}{y-2} - \frac{1}{y-1}\right]y' = -2x$; $\ln\left|\frac{y-2}{y-1}\right| = -x^2 + k$; $\frac{y-2}{y-1} = ce^{-x^2}$;
 $y(0) = 3 \Rightarrow c = \frac{1}{2}$; $\frac{y-2}{y-1} = \frac{e^{-x^2}}{2}$; $y-2 = \frac{e^{-x^2}}{2}(y-1)$; $y\left(1 - \frac{e^{-x^2}}{2}\right) = 2 - \frac{e^{-x^2}}{2}$;
 $y = \frac{4 - e^{-x^2}}{2 - e^{-x^2}}$. The interval of validity is $(-\infty, \infty)$.

2.2.20. $\frac{y'}{y(y-2)} = -1$; $\frac{1}{2}\left[\frac{1}{y-2} - \frac{1}{y}\right]y' = -1$; $\left[\frac{1}{y-2} - \frac{1}{y}\right]y' = -2$; $\ln\left|\frac{y-2}{y}\right| = -2x + k$;
 $\frac{y-2}{y} = ce^{-2x}$; $y(0) = 1 \Rightarrow c = -1$; $\frac{y-2}{y} = -e^{-2x}$; $y-2 = -ye^{-2x}$; $y(1 + e^{-2x}) = 2$;
 $y = \frac{2}{1 + e^{-2x}}$. The interval of validity is $(-\infty, \infty)$.

2.2.22. $y \equiv 2$ is a constant solution of the differential equation, and it satisfies the initial condition. Therefore, $y \equiv 2$ is a solution of the initial value problem. The interval of validity is $(-\infty, \infty)$.

2.2.24. $\frac{y'}{1+y^2} = \frac{1}{1+x^2}$; $\tan^{-1}y = \tan^{-1}x + k$; $y = \tan(\tan^{-1}x + k)$. Now use the identity
 $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$ with $A = \tan^{-1}x$ and $B = \tan^{-1}c$ to rewrite y as $y = \frac{x+c}{1-cx}$,
 where $c = \tan k$.

2.2.26. $(\sin y)y' = \cos x$; $-\cos y = \sin x + c$; $y(\pi) = \frac{\pi}{2} \Rightarrow c = 0$, so (A) $\cos y = -\sin x$. To obtain y explicitly we note that $-\sin x = \cos(x + \pi/2)$, so (A) can be rewritten as $\cos y = \cos(x + \pi/2)$. This equation holds if and only if one of the following conditions holds for some integer k :

$$(B) \quad y = x + \frac{\pi}{2} + 2k\pi; \quad (C) \quad y = -x - \frac{\pi}{2} + 2k\pi.$$

Among these choices the only way to satisfy the initial condition is to let $k = 1$ in (C), so
 $y = -x + \frac{3\pi}{2}$.

2.2.28. Rewrite the equation as $P' = -\alpha P(P - 1/\alpha)$. By inspection, $P \equiv 0$ and $P \equiv 1/\alpha$ are constant solutions. Separate variables to find others: $\frac{P'}{P(P - 1/\alpha)} = -\alpha\alpha$;

$$\left[\frac{1}{P - 1/\alpha} - \frac{1}{P}\right]P' = -\alpha; \ln\left|\frac{P - 1/\alpha}{P}\right| = -\alpha t + k; (A) \frac{P - 1/\alpha}{P} = ce^{-\alpha t};$$

$$P(1 - ce^{-\alpha t}) = 1/\alpha; (B) P = \frac{1}{\alpha(1 - ce^{-\alpha t})}. \text{ From (A), } P(0) = P_0 \Rightarrow c = \frac{P_0 - 1/\alpha}{P_0}.$$

Substituting this into (B) yields $P = \frac{P_0}{\alpha P_0 + (1 - \alpha P_0)e^{-\alpha t}}$. From this $\lim_{t \rightarrow \infty} P(t) = 1/\alpha$.

2.2.30. If $q = rS$ the equation for I reduces to $I' = -rI^2$, so $\frac{I'}{I^2} = -r$; $-\frac{1}{I} = -rt - \frac{1}{I_0}$; so

$$I = \frac{I_0}{1 + rI_0 t} \text{ and } \lim_{t \rightarrow \infty} I(t) = 0. \text{ If } q \neq rS, \text{ then rewrite the equation for } I \text{ as}$$

$I' = -rI(I - \alpha)$ with $\alpha = S - \frac{q}{r}$. Separating variables yields $\frac{I'}{I(I - \alpha)} = -r$;
 $\left[\frac{1}{I - \alpha} - \frac{1}{I} \right] I' = -r\alpha$; $\ln \left| \frac{I - \alpha}{I} \right| = -r\alpha t + k$; (A) $\frac{I - \alpha}{I} = ce^{-r\alpha t}$; $I(1 - ce^{-r\alpha t}) = \alpha$; (B)
 $I = \frac{\alpha}{1 - ce^{-r\alpha t}}$. From (A), $I(0) = I_0 \Rightarrow c = \frac{I_0 - \alpha}{I_0}$. Substituting this into (B) yields
 $I = \frac{\alpha I_0}{I_0 + (\alpha - I_0)e^{-r\alpha t}}$. If $q < rS$, then $\alpha > 0$ and $\lim_{t \rightarrow \infty} I(t) = \alpha = S - \frac{q}{r}$. If $q > rS$, then
 $\alpha < 0$ and $\lim_{t \rightarrow \infty} I(t) = 0$.

2.2.34. The given equation is separable if $f = ap$, where a is a constant. In this case the equation is

$$y' + p(x)y = ap(x). \quad (\text{A})$$

Let P be an antiderivative of p ; that is, $P' = p$.

SOLUTION BY SEPARATION OF VARIABLES. $y' = -p(x)(y - a)$; $\frac{y'}{y - a} = -p(x)$;

$\ln |y - a| = -P(x) + k$; $y - a = ce^{-P(x)}$; $y = a + ce^{-P(x)}$.

SOLUTION BY VARIATION OF PARAMETERS. $y_1 = e^{-P(x)}$ is a solution of the complementary equation, so solutions of (A) are of the form $y = ue^{-P(x)}$ where $u'e^{-P(x)} = ap(x)$. Hence, $u' = ap(x)e^{P(x)}$; $u = ae^{P(x)} + c$; $y = a + ce^{-P(x)}$.

2.2.36. Rewrite the given equation as (A) $y' - \frac{2}{x}y = \frac{x^5}{y + x^2}$. $y_1 = x^2$ is a solution of $y' - \frac{2}{x}y = 0$.

Look for solutions of (A) of the form $y = ux^2$. Then $u'x^2 = \frac{x^5}{(u+1)x^2} = \frac{x^3}{u+1}$; $u' = \frac{x}{u+1}$;
 $(u+1)u' = x$; $\frac{(1+u)^2}{2} = \frac{x^2}{2} + \frac{c}{2}$; $u = -1 \pm \sqrt{x^2 + c}$; $y = x^2(-1 \pm \sqrt{x^2 + c})$.

2.2.38. $y_1 = e^{2x}$ is a solution of $y' - 2y = 0$. Look for solutions of the nonlinear equation of the form $y = ue^{2x}$. Then $u'e^{2x} = \frac{xe^{2x}}{1-u}$; $u' = \frac{x}{1-u}$; $(1-u)u' = x$; $-\frac{(1-u)^2}{2} = \frac{1}{2}(x^2 - c)$;
 $u = 1 \pm \sqrt{c - x^2}$; $y = e^{2x}(1 \pm \sqrt{c - x^2})$.

2.3 Existence and Uniqueness of Solutions of Nonlinear Equations

2.3.2. $f(x, y) = \frac{e^x + y}{x^2 + y^2}$ and $f_y(x, y) = \frac{1}{x^2 + y^2} - \frac{2y(e^x + y)}{(x^2 + y^2)^2}$ are both continuous at all $(x, y) \neq (0, 0)$. Hence, Theorem 2.3.1 implies that if $(x_0, y_0) \neq (0, 0)$, then the initial value problem has a unique solution on some open interval containing x_0 . Theorem 2.3.1 does not apply if $(x_0, y_0) = (0, 0)$.

2.3.4. $f(x, y) = \frac{x^2 + y^2}{\ln xy}$ and $f_y(x, y) = \frac{2y}{\ln xy} - \frac{x^2 + y^2}{x(\ln xy)^2}$ are both continuous at all (x, y) such that $xy > 0$ and $xy \neq 1$. Hence, Theorem 2.3.1 implies that if $x_0 y_0 > 0$ and $x_0 y_0 \neq 1$, then

the initial value problem has unique solution on an open interval containing x_0 .

Theorem 2.3.1 does not apply if $x_0 y_0 \leq 0$ or $x_0 y_0 = 1$.

- 2.3.6. $f(x, y) = 2xy$ and $f_y(x, y) = 2x$ are both continuous at all (x, y) . Hence, Theorem 2.3.1 implies that if (x_0, y_0) is arbitrary, then the initial value problem has a unique solution on some open interval containing x_0 .
- 2.3.8. $f(x, y) = \frac{2x+3y}{x-4y}$ and $f_y(x, y) = \frac{3}{x-4y} + 4\frac{2x+3y}{(x-4y)^2}$ are both continuous at all (x, y) such that $x \neq 4y$. Hence, Theorem 2.3.1 implies that if $x_0 \neq 4y_0$, then the initial value problem has a unique solution on some open interval containing x_0 . Theorem 2.3.1 does not apply if $x_0 = 4y_0$.
- 2.3.10. $f(x, y) = x(y^2 - 1)^{2/3}$ is continuous at all (x, y) , but $f_y(x, y) = \frac{4}{3}xy(y^2 - 1)^{1/3}$ is continuous at (x, y) if and only if $y \neq \pm 1$. Hence, Theorem 2.3.1 implies that if $y_0 \neq \pm 1$, then the initial value problem has a unique solution on some open interval containing x_0 , while if $y_0 = \pm 1$, then the initial value problem has at least one solution (possibly not unique on any open interval containing x_0).
- 2.3.12. $f(x, y) = (x+y)^{1/2}$ and $f_y(x, y) = \frac{1}{2(x+y)^{1/2}}$ are both continuous at all (x, y) such that $x+y > 0$. Hence, Theorem 2.3.1 implies that if $x_0 + y_0 > 0$, then the initial value problem has a unique solution on some open interval containing x_0 . Theorem 2.3.1 does not apply if $x_0 + y_0 \leq 0$.
- 2.3.14. To apply Theorem 2.3.1, rewrite the given initial value problem as (A)
 $y' = f(x, y)$, $y(x_0) = y_0$, where $f(x, y) = -p(x)y + q(x)$ and $f_y(x, y) = -p(x)$. If p and f are continuous on some open interval (a, b) containing x_0 , then f and f_y are continuous on some open rectangle containing (x_0, y_0) , so Theorem 2.3.1 implies that (A) has a unique solution on *some* open interval containing x_0 . The conclusion of Theorem 2.1.2 is more specific: the solution of (A) exists and is unique on (a, b) . For example, in the extreme case where $(a, b) = (-\infty, \infty)$, Theorem 2.3.1 still implies only existence and uniqueness on *some* open interval containing x_0 , while Theorem 2.1.2 implies that the solution exists and is unique on $(-\infty, \infty)$.
- 2.3.16. First find solutions of (A) $y' = y^{2/5}$. Obviously $y \equiv 0$ is a solution. If $y \neq 0$, then we can separate variables on any open interval where y has no zeros: $y^{-2/5}y' = 1$; $\frac{5}{3}y^{3/5} = x + c$; $y = \left(\frac{3}{5}(x + c)^{5/3}\right)$. (Note that this solution is also defined at $x = -c$, even though $y(-c) = 0$. To satisfy the initial condition, let $c = 1$. Thus, $y = \left(\frac{3}{5}(x + 1)^{5/3}\right)$ is a solution of the initial value problem on $(-\infty, \infty)$; moreover, since $f(x, y) = y^{2/5}$ and $f_y(x, y) = \frac{2}{5}y^{-3/5}$ are both continuous at all (x, y) such that $y \neq 0$, this is the only solution on $(-5/3, \infty)$, by an argument similar to that given in Example 2.3.7, the function

$$y = \begin{cases} 0, & -\infty < x \leq -\frac{5}{3} \\ \left(\frac{3}{5}x + 1\right)^{5/3}, & -\frac{5}{3} < x < \infty \end{cases}$$

(To see that y satisfies $y' = y^{2/5}$ at $x = -\frac{5}{3}$ use an argument similar to that of

Discussion 2.3.15-2) For every $a \geq \frac{5}{3}$, the following function is also a solution:

$$y = \begin{cases} \left(\frac{3}{5}(x+a)\right)^{5/3}, & -\infty < x < -a, \\ 0, & -a \leq x \leq -\frac{5}{3} \\ \left(\frac{3}{5}x+1\right)^{5/3}, & -\frac{5}{3} < x < \infty. \end{cases}$$

2.3.18. Obviously, $y_1 \equiv 1$ is a solution. From Discussion 2.3.18 (taking $c = 0$ in the two families of solutions) yields $y_2 = 1 + |x|^3$ and $y_3 = 1 - |x|^3$. Other solutions are $y_4 = 1 + x^3$, $y_5 = 1 - x^3$,

$$y_6 = \begin{cases} 1 + x^3, & x \geq 0, \\ 1, & x < 0, \end{cases}; \quad y_7 = \begin{cases} 1 - x^3, & x \geq 0, \\ 1, & x < 0, \end{cases};$$

$$y_8 = \begin{cases} 1, & x \geq 0, \\ 1 + x^3, & x < 0, \end{cases}; \quad y_9 = \begin{cases} 1, & x \geq 0, \\ 1 - x^3, & x < 0 \end{cases}$$

It is straightforward to verify that all these functions satisfy $y' = 3x(y-1)^{1/3}$ for all $x \neq 0$.

Moreover, $y'_i(0) = \lim_{x \rightarrow 0} \frac{y_i(x) - 1}{x} = 0$ for $1 \leq i \leq 9$, which implies that they also satisfy the equation at $x = 0$.

2.3.20. Let y be any solution of (A) $y' = 3x(y-1)^{1/3}$, $y(3) = -7$. By continuity, there is some open interval I containing $x_0 = 3$ on which $y(x) < 1$. From Discussion 2.3.18, $y = 1 + (x^2 + c)^{3/2}$ on I ; $y(3) = -7 \Rightarrow c = -5$; (B) $y = 1 - (x^2 - 5)^{3/2}$. It now follows that every solution of (A) satisfies $y(x) < 1$ and is given by (B) on $(\sqrt{5}, \infty)$; that is, (B) is the unique solution of (A) on $(\sqrt{5}, \infty)$. This solution can be extended uniquely to $(0, \infty)$ as

$$y = \begin{cases} 1, & 0 < x \leq \sqrt{5}, \\ 1 - (x^2 - 5)^{3/2}, & \sqrt{5} < x < \infty \end{cases}$$

It can be extended to $(-\infty, \infty)$ in infinitely many ways. Thus,

$$y = \begin{cases} 1, & -\infty < x \leq \sqrt{5}, \\ 1 - (x^2 - 5)^{3/2}, & \sqrt{5} < x < \infty \end{cases}$$

is a solution of the initial value problem on $(-\infty, \infty)$. Moreover, if $\alpha \geq 0$, then

$$y = \begin{cases} 1 + (x^2 - \alpha^2)^{3/2}, & -\infty < x < -\alpha, \\ 1, & -\alpha \leq x \leq \sqrt{5}, \\ 1 - (x^2 - 5)^{3/2}, & \sqrt{5} < x < \infty, \end{cases}$$

and

$$y = \begin{cases} 1 - (x^2 - \alpha^2)^{3/2}, & -\infty < x < -\alpha, \\ 1, & -\alpha \leq x \leq \sqrt{5}, \\ 1 - (x^2 - 5)^{3/2}, & \sqrt{5} < x < \infty, \end{cases}$$

are also solutions of the initial value problem on $(-\infty, \infty)$.

2.4 Transformation of Nonlinear Equations into Separable Equations

2.4.2. Rewrite as $y' - \frac{2}{7x}y = -\frac{x}{7y^6}$. Then $\frac{y'_1}{y_1} = \frac{2}{7x}$; $\ln|y_1| = \frac{2}{7}\ln|x| = \ln|x|^{2/7}$; $y_1 = x^{2/7}$;
 $y = ux^{2/7}$; $u'x^{2/7} = -\frac{1}{7u^6x^{5/7}}$; $u^6u' = -\frac{1}{7x}$; $\frac{u^7}{7} = -\frac{1}{7}\ln|x| + \frac{c}{7}$; $u = (c - \ln|x|)^{1/7}$;
 $y = x^{2/7}(c - \ln|x|)^{1/7}$.

2.4.4. Rewrite as $y' + \frac{2x}{1+x^2}y = \frac{1}{(1+x^2)^2y}$. Then $\frac{y'_1}{y_1} = -\frac{2x}{1+x^2}$; $\ln|y_1| = -\ln(1+x^2)$;
 $y_1 = \frac{1}{1+x^2}$; $y = \frac{u}{1+x^2}$; $\frac{u'}{1+x^2} = \frac{1}{u(1+x^2)}$; $u'u = 1$; $\frac{u^2}{2} = x + \frac{c}{2}$; $u = \pm\sqrt{2x+c}$;
 $y = \pm\frac{\sqrt{2x+c}}{1+x^2}$.

2.4.6. $\frac{y'_1}{y_1} = \frac{1}{3}\left(\frac{1}{x} + 1\right)$; $\ln|y_1| = \frac{1}{3}(\ln|x| + x)$; $y_1 = x^{1/3}e^{x/3}$; $y = ux^{1/3}e^{x/3}$;
 $u'x^{1/3}e^{x/3} = x^{4/3}e^{4x/3}u^4$; $\frac{u'}{u^4} = xe^x$; $-\frac{1}{3u^3} = (x-1)e^x - \frac{c}{3}$; $u = \frac{1}{[3(1-x)e^x + c]^{1/3}}$;
 $y = \left[\frac{x}{3(1-x) + ce^{-x}}\right]^{1/3}$.

2.4.8. $\frac{y'_1}{y_1} = x$; $\ln|y_1| = \frac{x^2}{2}$; $y_1 = e^{x^2/2}$; $y = ue^{x^2/2}$; $u'e^{x^2/2} = xu^{3/2}e^{3x^2/4}$; $\frac{u'}{u^{3/2}} = xe^{x^2/4}$; (A)
 $-\frac{2}{u^{1/2}} = 2e^{x^2/4} + 2c$; $u^{1/2} = -\frac{1}{c + e^{x^2/4}}$; $u = \frac{1}{(c + e^{x^2/4})^2}$; $y = \frac{1}{(1 + ce^{-x^2/4})^2}$. Because of
 (A) we must choose c so that $y(1) = 4$ and $1 + ce^{-1/4} < 0$. This implies that $c = -3e^{1/4}$;
 $y = \left[1 - \frac{3}{2}e^{-(x^2-1)/4}\right]^{-2}$.

2.4.10. $\frac{y'_1}{y_1} = 2$; $\ln|y_1| = 2x$; $y_1 = e^{2x}$; $y = ue^{2x}$; $u'e^{2x} = 2u^{1/2}e^x$; $u^{-1/2}u' = 2e^{-x}$;
 $2u^{1/2} = -2e^{-x} + 2c$; $u^{1/2} = c - e^{-x} > 0$; $y(0) = 1 \Rightarrow u(0) = 1 \Rightarrow c = 2$; $u = (2 - e^{-x})^2$;
 $y = (2e^x - 1)^2$.

2.4.12. Rewrite as $y' + \frac{2}{x}y = \frac{y^3}{x^2}$. Then $\frac{y'_1}{y_1} = -\frac{2}{x}$; $\ln|y_1| = -2\ln|x| = \ln x^{-2}$; $y_1 = \frac{1}{x^2}$; $y = \frac{u}{x^2}$;
 $\frac{u'}{x^2} = \frac{u^3}{x^8}$; $\frac{u'}{u^3} = \frac{1}{x^6}$; $-\frac{1}{2u^2} = -\frac{1}{5x^5} + c$; $y(1) = \frac{1}{\sqrt{2}} \Rightarrow u(1) = \frac{1}{\sqrt{2}} \Rightarrow c = -\frac{4}{5}$;
 $u = \left[\frac{5x^5}{2(1+4x^5)}\right]^{1/2}$; $y = \left[\frac{5x}{2(1+4x^5)}\right]^{1/2}$.

2.4.14. $P = ue^{at}$; $u'e^{at} = -a\alpha u^2e^{2at}$; $\frac{u'}{u^2} = -a\alpha e^{at}$; $-\frac{1}{u} = -a \int_0^t \alpha(\tau)e^{a\tau} d\tau - \frac{1}{P_0}$;

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$$P = \frac{P_0 e^{at}}{1 + aP_0 \int_0^t \alpha(\tau) e^{a\tau} d\tau}, \text{ which can also be written as } P = \frac{P_0}{e^{-at} + aP_0 e^{-at} \int_0^t \alpha(\tau) e^{a\tau} d\tau}.$$

$$\text{Therefore, } \lim_{t \rightarrow \infty} P(t) = \begin{cases} \infty & \text{if } L = 0, \\ 0 & \text{if } L = \infty, \\ 1/aL & \text{if } 0 < L < \infty. \end{cases}$$

2.4.16. $y = ux$; $u'x + u = u^2 + 2u$; (A) $u'x = u(u+1)$. Since $u \equiv 0$ and $u \equiv -1$ are constant solutions of (A), $y \equiv 0$ and $y = -x$ are solutions of the given equation. The nonconstant solutions of (A) satisfy $\frac{u'}{u(u+1)} = \frac{1}{x}$; $\left[\frac{1}{u} - \frac{1}{u+1}\right] u' = \frac{1}{x}$; $\ln \left| \frac{u}{u+1} \right| = \ln |x| + k$;

$$\frac{u}{u+1} = cx; u = (u+1)cx; u(1-cx) = cx; u = \frac{cx}{1-cx}; y = \frac{cx^2}{1-cx}.$$

2.4.18. $y = ux$; $u'x + u = u + \sec u$; $u'x = \sec u$; $(\cos u)u' = \frac{1}{x}$; $\sin u = \ln |x| + c$;
 $u = \sin^{-1}(\ln |x| + c)$; $y = x \sin^{-1}(\ln |x| + c)$.

2.4.20. Rewrite the given equation as $y' = \frac{x^2 + 2y^2}{xy}$; $y = ux$; $u'x + u = \frac{1}{u} + 2u$; $u'x = \frac{1+u^2}{u}$;
 $\frac{uu'}{1+u^2} = \frac{1}{x}$; $\frac{1}{2} \ln(1+u^2) = \ln |x| + k$; $\ln \left(1 + \frac{y^2}{x^2}\right) = \ln x^2 + 2k$; $1 + \frac{y^2}{x^2} = cx^2$;
 $x^2 + y^2 = cx^4$; $y = \pm x \sqrt{cx^2 - 1}$.

2.4.22. $y = ux$; $u'x + u = u + u^2$; $u'x = u^2$; $\frac{u'}{u^2} = \frac{1}{x}$; $-\frac{1}{u} = \ln |x| + c$;
 $y(-1) = 2 \Rightarrow u(-1) = -2 \Rightarrow c = \frac{1}{2}$; $u = -\frac{1}{2 \ln |x| + 1}$; $y = -\frac{2x}{2 \ln |x| + 1}$.

2.4.24. Rewrite the given equation as $y' = -\frac{x^2 + y^2}{xy}$; $y = ux$; $u'x + u = -\frac{1}{u} - u$;
 $u'x = -\frac{1+2u^2}{u}$; $-\frac{uu'}{1+2u^2} = \frac{1}{x}$; $-\frac{1}{4} \ln(1+2u^2) = \ln |x| + k$; $x^4(1+2u^2) = c$;
 $y(1) = 2 \Rightarrow u(1) = 2 \Rightarrow c = 9$; $x^4(1+2u^2) = 9$; $u^2 = \frac{9-x^4}{2x^4}$; $u = \frac{1}{x^2} \left(\frac{9-x^4}{2} \right)^{1/2}$;
 $y = \frac{1}{x} \left(\frac{9-x^4}{2} \right)^{1/2}$.

2.4.26. Rewrite the given equation as $y' = 2 + \frac{y^2}{x^2} + 4\frac{y}{x}$; $y = ux$; $u'x + u = 2 + u^2 + 4u$;
 $u'x = u^2 + 3u + 2 = (u+1)(u+2)$; $\frac{u'}{(u+1)(u+2)} = \frac{1}{x}$; $\left[\frac{1}{u+1} - \frac{1}{u+2}\right] u' = \frac{1}{x}$;
 $\ln \left| \frac{u+1}{u+2} \right| = \ln |x| + k$; $\frac{u+1}{u+2} = cx$; $y(1) = 1 \Rightarrow u(1) = 1 \Rightarrow c = \frac{2}{3}$; $\frac{u+1}{u+2} = \frac{2}{3}x$;
 $u+1 = \frac{2}{3}x(u+2)$; $u\left(1 - \frac{2}{3}x\right) = -1 + \frac{4}{3}x$; $u = -\frac{4x-3}{2x-3}$; $y = -\frac{x(4x-3)}{2x-3}$.

$$\begin{aligned}
2.4.28. \quad & y = ux; \quad u'x + u = \frac{1+u}{1-u}; \quad u'x = \frac{1+u^2}{1-u}; \quad \frac{(1-u)u'}{1+u^2} = \frac{1}{x}; \\
& \tan^{-1} u - \frac{1}{2} \ln(1+u^2) = \ln|x| + c; \quad \tan^{-1} \frac{y}{x} - \frac{1}{2} \ln \left(1 + \frac{y^2}{x^2} \right) = \ln|x| + c; \\
& \tan^{-1} \frac{y}{x} - \frac{1}{2} \ln(x^2 + y^2) = c.
\end{aligned}$$

$$\begin{aligned}
2.4.30. \quad & y = ux; \quad u'x + u = \frac{u^3 + 2u^2 + u + 1}{(u+1)^2} = \frac{u(u+1)^2 + 1}{(u+1)^2} = u + \frac{1}{(u+1)^2}; \quad u'x = \frac{1}{(u+1)^2}; \\
& (u+1)^2 u' = \frac{1}{x}; \quad \frac{(u+1)^3}{3} = \ln|x| + c; \quad (u+1)^3 = 3(\ln|x| + c); \quad \left(\frac{y}{x} + 1 \right)^3 = 3(\ln|x| + c); \\
& (y+x)^3 = 3x^3(\ln|x| + c).
\end{aligned}$$

$$\begin{aligned}
2.4.32. \quad & y = ux; \quad u'x + u = \frac{u}{u-2}; \quad (\text{A}) \quad u'x = \frac{u(u-3)}{2-u}; \quad \text{Since } u \equiv 0 \text{ and } u \equiv 3 \text{ are constant} \\
& \text{solutions of (A), } y \equiv 0 \text{ and } y = 3x \text{ are solutions of the given equation. The nonconstant} \\
& \text{solutions of (A) satisfy } \frac{(2-u)u'}{u(u-3)} = \frac{1}{x}; \quad \left[\frac{1}{u-3} + \frac{2}{u} \right] u' = -\frac{3}{x}; \\
& \ln|u-3| + 2 \ln|u| = -3 \ln|x| + k; \quad u^2(u-3) = \frac{c}{x^3}; \quad y^2(y-3x) = c.
\end{aligned}$$

$$\begin{aligned}
2.4.34. \quad & y = ux; \quad u'x + u = \frac{1+u+3u^3}{1+3u^2} = u + \frac{1}{1+3u^2}; \quad (1+3u^2)u' = \frac{1}{x}; \quad u + u^3 = \ln|x| + c; \\
& \frac{y}{x} + \frac{y^3}{x^3} = \ln|x| + c.
\end{aligned}$$

$$\begin{aligned}
2.4.36. \quad & \text{Rewrite the given equation as } y' = \frac{x^2 - xy + y^2}{xy}; \quad y = ux; \quad u'x + u = \frac{1}{u} - 1 + u; \\
& u'x = \frac{1-u}{u}; \quad \frac{uu'}{u-1} = -\frac{1}{x}; \quad \left[1 + \frac{1}{u-1} \right] u' = -\frac{1}{x}; \quad u + \ln|u-1| = -\ln|x| + k; \\
& e^u(u-1) = \frac{c}{x}; \quad e^{y/x}(y-x) = c.
\end{aligned}$$

$$\begin{aligned}
2.4.38. \quad & y = ux; \quad u'x + u = 1 + \frac{1}{u} + u; \quad (\text{A}) \quad u'x = \frac{u+1}{u}. \quad \text{Since (A) has the constant solution} \\
& u = -1; \quad y = -x \text{ is a solution of the given equation. The nonconstant solutions of (A) satisfy} \\
& \frac{uu'}{u+1} = \frac{1}{x}; \quad \left[1 - \frac{1}{u+1} \right] u' = \frac{1}{x}; \quad u - \ln|u+1| = \ln|x| + c; \quad \frac{y}{x} - \ln \left| \frac{y}{x} + 1 \right| = \ln|x| + c; \\
& y - x \ln|y-x| = cy.
\end{aligned}$$

$$2.4.40. \quad \text{If } x = X - X_0 \text{ and } y = Y - Y_0, \text{ then } \frac{dy}{dx} = \frac{dY}{dX} = \frac{dY}{dX} \frac{dX}{dx} = \frac{dY}{dX}, \text{ so } y = y(x) \text{ satisfies the} \\
\text{given equation if and only if } Y = Y(X) \text{ satisfies}$$

$$\frac{dY}{dX} = \frac{a(X - X_0) + b(Y - Y_0) + \alpha}{c(X - X_0) + d(Y - Y_0) + \beta},$$

which reduces to the nonlinear homogeneous equation

$$\frac{dY}{dX} = \frac{aX + bY}{cX + dY}$$

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if and only if

$$\begin{aligned} aX_0 + bY_0 &= \alpha \\ cX_0 + dY_0 &= \beta. \end{aligned} \tag{B}$$

We will now show that if $ad - bc \neq 0$, then it is possible (for any choice of α and β) to solve (B). Multiplying the first equation in (B) by d and the second by b yields

$$\begin{aligned} daX_0 + dbY_0 &= d\alpha \\ bcX_0 + bdY_0 &= b\beta. \end{aligned}$$

Subtracting the second of these equations from the first yields $(ad - bc)X_0 = \alpha d - \beta b$. Since $ad - bc \neq 0$, this implies that $X_0 = \frac{\alpha d - \beta b}{ad - bc}$. Multiplying the first equation in (B) by c and the second by a yields

$$\begin{aligned} caX_0 + cbY_0 &= c\alpha \\ acX_0 + adY_0 &= a\beta. \end{aligned}$$

Subtracting the first of these equations from the second yields $(ad - bc)Y_0 = \alpha c - \beta a$. Since $ad - bc \neq 0$ this implies that $Y_0 = \frac{\alpha c - \beta a}{ad - bc}$.

2.4.42. For the given equation, (B) of Exercise 2.4.40 is

$$\begin{aligned} 2X_0 + Y_0 &= 1 \\ X_0 + 2Y_0 &= -4. \end{aligned}$$

Solving this pair of equations yields $X_0 = 2$ and $Y_0 = -3$. The transformed differential equation is

$$\frac{dY}{dX} = \frac{2X + Y}{X + 2Y}. \tag{A}$$

Let $Y = uX$; $u'X + U = \frac{2+u}{1+2u}$; (B) $u'X = -\frac{2(u-1)(u+1)}{2u+1}$. Since $u \equiv 1$ and $u \equiv -1$ satisfy (B), $Y = X$ and $Y = -X$ are solutions of (A). Since $X = x + 2$ and $Y = y - 3$, it follows that $y = x + 5$ and $y = -x + 1$ are solutions of the given equation. The nonconstant solutions of (B) satisfy $\frac{(2u+1)u'}{(u-1)(u+1)} = -\frac{2}{X}$; $\left[\frac{1}{u+1} + \frac{3}{u-1}\right]u' = -\frac{4}{X}$; $\ln|u+1| + 3\ln|u-1| = -4\ln|X| + k$; $(u+1)(u-1)^3 = \frac{c}{X^4}$; $(Y+X)(Y-X)^3 = c$; Setting $X = x + 2$ and $Y = y - 3$ yields $(y+x-1)(y-x-5)^3 = c$.

2.4.44. Rewrite the given equation as $y' = \frac{y^3 + x}{3xy^2}$; $y = ux^{1/3}$; $u'x^{1/3} + \frac{1}{3x^{2/3}}u = \frac{u^3 + 1}{3u^2x^{2/3}}$; $u'x^{1/3} = \frac{1}{3x^{2/3}u^2}$; $u^2u' = \frac{1}{3x}$; $\frac{u^3}{3} = \frac{1}{3}(\ln|x| + c)$; $u = (\ln|x| + c)^{1/3}$; $y = x^{1/3}(\ln|x| + c)^{1/3}$.

2.4.46. Rewrite the given equation as $y' = \frac{2(y^2 + x^2y - x^4)}{x^3}$; $y = ux^2$; $u'x^2 + 2xu = 2x(u^2 + u - 1)$; (A) $u'x^2 = 2x(u^2 - 1)$. Since $u \equiv 1$ and $u \equiv -1$ are constant

solutions of (A), $y = x^2$ and $y = -x^2$ are solutions of the given equation. The nonconstant solutions of (A) satisfy $\frac{u'}{u^2-1} = \frac{2}{x}$; $\left[\frac{1}{u-1} - \frac{1}{u+1}\right] u' = \frac{4}{x}$; $\ln\left|\frac{u-1}{u+1}\right| = 4\ln|x| + k$;
 $\frac{u-1}{u+1} = cx^4$; $(u-1) = (u+1)cx^4$; $u(1-cx^4) = 1+cx^4$; $u = \frac{1+cx^4}{1-cx^4}$; $y = \frac{x^2(1+cx^4)}{1-cx^4}$.

2.4.48. $y = u \tan x$; $u' \tan x + u \sec^2 x = (u^2 + u + 1) \sec^2 x$; $u' \tan x = (u^2 + 1) \sec^2 x$;
 $\frac{u'}{u^2+1} = \sec^2 x \cot x = \cot x + \tan x$; $\tan^{-1} u = \ln|\sin x| - \ln|\cos x| + c = \ln|\tan x| + c$;
 $u = \tan(\ln|\tan x| + c)$; $y = \tan x \tan(\ln|\tan x| + c)$.

2.4.50. Rewrite the given equation as $y' = \frac{(y + \sqrt{x})^2}{2x(y + 2\sqrt{x})}$; $y = ux^{1/2}$;
 $u'x^{1/2} + \frac{1}{2\sqrt{x}}u = \frac{(u+1)^2}{2\sqrt{x}(u+2)}$; $u'x^{1/2} = \frac{1}{2\sqrt{x}(u+2)}$; $(u+2)u' = \frac{1}{2x}$;
 $\frac{(u+2)^2}{2} = \frac{1}{2}(\ln|x| + c)$; $(u+2)^2 = \ln|x| + c$; $u = -2 \pm \sqrt{\ln|x| + c}$;
 $y = x^{1/2}(-2 \pm \sqrt{\ln|x| + c})$.

2.4.52. $y_1 = \frac{1}{x^2}$ is a solution of $y' + \frac{2}{x}y = 0$. Let $y = \frac{u}{x^2}$; then

$$\frac{u'}{x^2} = \frac{3x^2(u^2/x^4) + 6x(u/x^2) + 2}{x^2(2x(u/x^2) + 3)} = \frac{3(u/x)^2 + 6(u/x) + 2}{x^2(2(u/x) + 3)},$$

so (A) $u' = \frac{3(u/x)^2 + 6(u/x) + 2}{2(u/x) + 3}$. Since (A) is a homogeneous nonlinear equation, we now

substitute $u = vx$ into (A). This yields $v'x + v = \frac{3v^2 + 6v + 2}{2v + 3}$; $v'x = \frac{(v+1)(v+2)}{2v+3}$;

$$\frac{(2v+3)v'}{(v+1)(v+2)} = \frac{1}{x}; \left[\frac{1}{v+1} + \frac{1}{v+2}\right] v' = \frac{1}{x}; \ln|(v+1)(v+2)| = \ln|x| + k; \text{ (B)}$$

$(v+1)(v+2) = cx$. Since $y(2) = 2 \Rightarrow u(2) = 8 \Rightarrow v(2) = 4$, (B) implies that $c = 15$.

$(v+1)(v+2) = 15x$; $v^2 + 3v + 2 - 15x = 0$. From the quadratic formula,

$$v = \frac{-3 + \sqrt{1+60x}}{2}; u = vx = \frac{x(-3 + \sqrt{1+60x})}{2}; y = \frac{u}{x^2} = \frac{-3 + \sqrt{1+60x}}{2x}.$$

2.4.54. Differentiating (A) $y_1(x) = \frac{y(ax)}{a}$ yields (B) $y_1'(x) = \frac{1}{a}y'(ax) \cdot a = y'(ax)$. Since $y'(x) = q(y(x)/x)$ on some interval I , (C) $y'(ax) = q(y(ax)/ax)$ on some interval J . Substituting (A) and (B) into (C) yields $y_1'(x) = q(y_1(x)/x)$ on J .

2.4.56. If $y = z + 1$, then $z' + z = xz^2$; $z = ue^{-x}$; $u'e^{-x} = xu^2e^{-2x}$; $\frac{u'}{u^2} = xe^{-x}$;
 $-\frac{1}{u} = -e^{-x}(x+1) - c$; $u = \frac{1}{e^{-x}(x+1) + c}$; $z = \frac{1}{x+1+ce^x}$; $y = 1 + \frac{1}{x+1+ce^x}$.

2.4.58. If $y = z + 1$, then $z' + \frac{2}{x}z = z^2$; $z_1 = \frac{1}{x^2}$; $z = \frac{u}{x^2}$; $\frac{u'}{x^2} = \frac{u^2}{x^4}$; $\frac{u'}{u^2} = \frac{1}{x^2}$;
 $-\frac{1}{u} = -\frac{1}{x} + c = -\frac{1-cx}{x}$; $u = -\frac{x}{1-cx}$; $z = -\frac{1}{x(1-cx)}$; $y = 1 - \frac{1}{x(1-cx)}$.

2.5 Exact Equations

2.5.2. $M(x, y) = 3y \cos x + 4xe^x + 2x^2e^x$; $N(x, y) = 3 \sin x + 3$; $M_y(x, y) = 3 \cos x = N_x(x, y)$, so the equation is exact. We must find F such that (A) $F_x(x, y) = 3y \cos x + 4xe^x + 2x^2e^x$ and (B) $F_y(x, y) = 3 \sin x + 3$. Integrating (B) with respect to y yields (C) $F(x, y) = 3y \sin x + 3y + \psi(x)$. Differentiating (C) with respect to x yields (D) $F_x(x, y) = 3y \cos x + \psi'(x)$. Comparing (D) with (A) shows that (E) $\psi'(x) = 4xe^x + 2x^2e^x$. Integration by parts yields $\int xe^x dx = xe^x - e^x$ and $\int x^2e^x dx = x^2e^x - 2xe^x + 2e^x$.

Substituting from the last two equations into (E) yields $\psi(x) = 2x^2e^x$. Substituting this into (C) yields $F(x, y) = 3y \sin x + 3y + 2x^2e^x$. Therefore, $3y \sin x + 3y + 2x^2e^x = c$.

2.5.4. $M(x, y) = 2x - 2y^2$; $N(x, y) = 12y^2 - 4xy$; $M_y(x, y) = -4y = N_x(x, y)$, so the equation is exact. We must find F such that (A) $F_x(x, y) = 2x - 2y^2$ and (B) $F_y(x, y) = 12y^2 - 4xy$. Integrating (A) with respect to x yields (C) $F(x, y) = x^2 - 2xy^2 + \phi(y)$. Differentiating (C) with respect to y yields (D) $F_y(x, y) = -4xy + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = 12y^2$, so we take $\phi(y) = 4y^3$. Substituting this into (C) yields $F(x, y) = x^2 - 2xy^2 + 4y^3$. Therefore, $x^2 - 2xy^2 + 4y^3 = c$.

2.5.6. $M(x, y) = 4x + 7y$; $N(x, y) = 3x + 4y$; $M_y(x, y) = 7 \neq 3 = N_x(x, y)$, so the equation is not exact.

2.5.8. $M(x, y) = 2x + y$; $N(x, y) = 2y + 2x$; $M_y(x, y) = 1 \neq 2 = N_x(x, y)$, so the equation is not exact.

2.5.10. $M(x, y) = 2x^2 + 8xy + y^2$; $N(x, y) = 2x^2 + \frac{xy^3}{3}$;

$M_y(x, y) = 8x + 2y \neq 4x + \frac{y^3}{3} = N_x(x, y)$, so the equation is not exact.

2.5.12. $M(x, y) = y \sin xy + xy^2 \cos xy$; $N(x, y) = x \sin xy + xy^2 \cos xy$;
 $M_y(x, y) = 3xy \cos xy + (1 - x^2y^2) \sin xy \neq (xy + y^2) \cos xy + (1 - xy^3) \sin xy = N_x(x, y)$, so the equation is not exact.

2.5.14. $M(x, y) = e^x(x^2y^2 + 2xy^2) + 6x$; $N(x, y) = 2x^2ye^x + 2$;
 $M_y(x, y) = 2xye^x(x + 2) = N_x(x, y)$, so the equation is exact. We must find F such that (A) $F_x(x, y) = e^x(x^2y^2 + 2xy^2) + 6x$ and (B) $F_y(x, y) = 2x^2ye^x + 2$. Integrating (B) with respect to y yields (C) $F(x, y) = x^2y^2e^x + 2y + \psi(x)$. Differentiating (C) with respect to x yields (D) $F_x(x, y) = e^x(x^2y^2 + 2xy^2) + \psi'(x)$. Comparing (D) with (A) shows that $\psi'(x) = 6x$, so we take $\psi(x) = 3x^2$. Substituting this into (C) yields $F(x, y) = x^2y^2e^x + 2y + 3x^2$. Therefore, $x^2y^2e^x + 2y + 3x^2 = c$.

2.5.16. $M(x, y) = e^{xy}(x^4y + 4x^3) + 3y$; $N(x, y) = x^5e^{xy} + 3x$;
 $M_y(x, y) = x^4e^{xy}(xy + 5) + 3 = N_x(x, y)$, so the equation is exact. We must find F such that (A) $F_x(x, y) = e^{xy}(x^4y + 4x^3) + 3y$ and (B) $F_y(x, y) = x^5e^{xy} + 3x$. Integrating (B) with respect to y yields (C) $F(x, y) = x^4e^{xy} + 3xy + \psi(x)$. Differentiating (C) with respect to x yields (D) $F_x(x, y) = e^{xy}(x^4y + 4x^3) + 3y + \psi'(x)$. Comparing (D) with (A) shows that $\psi'(x) = 0$, so we take $\psi(x) = 0$. Substituting this into (C) yields $F(x, y) = x^4e^{xy} + 3xy$. Therefore, $x^4e^{xy} + 3xy = c$.

2.5.18. $M(x, y) = 4x^3y^2 - 6x^2y - 2x - 3$; $N(x, y) = 2x^4y - 2x^3$;
 $M_y(x, y) = 8x^3y - 6x^2 = N_x(x, y)$, so the equation is exact. We must find F such that (A) $F_x(x, y) = 4x^3y^2 - 6x^2y - 2x - 3$ and (B) $F_y(x, y) = 2x^4y - 2x^3$. Integrating (A) with respect to x yields (C) $F(x, y) = x^4y^2 - 2x^3y - x^2 - 3x + \phi(y)$. Differentiating (C) with respect to y yields (D) $F_y(x, y) = 2x^4y - 2x^3 + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = 0$, so we take $\phi(y) = 0$. Substituting this into (C) yields $F(x, y) = x^4y^2 - 2x^3y - x^2 - 3x$. Therefore, $x^4y^2 - 2x^3y - x^2 - 3x = c$. Since $y(1) = 3 \Rightarrow c = -1$, $x^4y^2 - 2x^3y - x^2 - 3x + 1 = 0$ is an implicit solution of the initial value problem. Solving this for y by means of the quadratic formula yields $y = \frac{x + \sqrt{2x^2 + 3x - 1}}{x^2}$.

2.5.20. $M(x, y) = (y^3 - 1)e^x$; $N(x, y) = 3y^2(e^x + 1)$; $M_y(x, y) = 3y^2e^x = N_x(x, y)$, so the equation is exact. We must find F such that (A) $F_x(x, y) = (y^3 - 1)e^x$ and (B) $F_y(x, y) = 3y^2(e^x + 1)$. Integrating (A) with respect to x yields (C) $F(x, y) = (y^3 - 1)e^x + \phi(y)$. Differentiating (C) with respect to y yields (D) $F_y(x, y) = 3y^2e^x + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = 3y^2$, so we take $\phi(y) = y^3$. Substituting this into (C) yields $F(x, y) = (y^3 - 1)e^x + y^3$. Therefore, $(y^3 - 1)e^x + y^3 = c$. Since $y(0) = 0 \Rightarrow c = -1$, $(y^3 - 1)e^x + y^3 = -1$ is an implicit solution of the initial value problem. Therefore, $y^3(e^x + 1) = e^x - 1$, so $y = \left(\frac{e^x - 1}{e^x + 1}\right)^{1/3}$.

2.5.22. $M(x, y) = (2x - 1)(y - 1)$; $N(x, y) = (x + 2)(x - 3)$; $M_y(x, y) = 2x - 1 = N_x(x, y)$, so the equation is exact. We must find F such that (A) $F_x(x, y) = (2x - 1)(y - 1)$ and (B) $F_y(x, y) = (x + 2)(x - 3)$. Integrating (A) with respect to x yields (C) $F(x, y) = (x^2 - x)(y - 1) + \phi(y)$. Differentiating (C) with respect to y yields (D) $F_y(x, y) = x^2 - x + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = -6$, so we take $\phi(y) = -6y$. Substituting this into (C) yields $F(x, y) = (x^2 - x)(y - 1) - 6y$. Therefore, $(x^2 - x)(y - 1) - 6y = c$. Since $y(1) = -1 \Rightarrow c = 6$, $(x^2 - x)(y - 1) - 6y = 6$ is an implicit solution of the initial value problem. Therefore, $(x^2 - x - 6)y = x^2 - x + 6$, so $y = \frac{x^2 - x + 6}{(x - 3)(x + 2)}$.

2.5.24. $M(x, y) = e^x(x^4y^2 + 4x^3y^2 + 1)$; $N(x, y) = 2x^4ye^x + 2y$;
 $M_y(x, y) = 2x^3ye^x(x + 4) = N_x(x, y)$, so the equation is exact. We must find F such that (A) $F_x(x, y) = e^x(x^4y^2 + 4x^3y^2 + 1)$ and (B) $F_y(x, y) = 2x^4ye^x + 2y$. Integrating (B) with respect to y yields (C) $F(x, y) = x^4y^2e^x + y^2 + \psi(x)$. Differentiating (C) with respect to x yields (D) $F_x(x, y) = e^xy^2(x^4 + 4x^3) + \psi'(x)$. Comparing (D) with (A) shows that $\psi'(x) = e^x$, so we take $\psi(x) = e^x$. Substituting this into (C) yields $F(x, y) = (x^4y^2 + 1)e^x + y^2$. Therefore, $(x^4y^2 + 1)e^x + y^2 = c$.

2.5.28. $M(x, y) = x^2 + y^2$; $N(x, y) = 2xy$; $M_y(x, y) = 2y = N_x(x, y)$, so the equation is exact. We must find F such that (A) $F_x(x, y) = x^2 + y^2$ and (B) $F_y(x, y) = 2xy$. Integrating (A) with respect to x yields (C) $F(x, y) = \frac{x^3}{3} + xy^2 + \phi(y)$. Differentiating (C) with respect to y yields (D) $F_y(x, y) = 2xy + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = 0$, so we take $\phi(y) = 0$. Substituting this into (C) yields $F(x, y) = \frac{x^3}{3} + xy^2$. Therefore, $\frac{x^3}{3} + xy^2 = c$.

2.5.30. (a) Exactness requires that

$$N_x(x, y) = M_y(x, y) = \frac{\partial}{\partial y}(x^3y^2 + 2xy + 3y^2) = 2x^3y + 2x + 6y. \text{ Hence,}$$

$$N(x, y) = \frac{x^4y}{4} + x^2 + 6xy + g(x), \text{ where } g \text{ is differentiable.}$$

(b) Exactness requires that $N_x(x, y) = M_y(x, y) = \frac{\partial}{\partial y}(\ln xy + 2y \sin x) = \frac{1}{y} + 2 \sin x$.

$$\text{Hence, } N(x, y) = \frac{x}{y} - 2 \cos x + g(x), \text{ where } g \text{ is differentiable.}$$

(c) Exactness requires that $N_x(x, y) = M_y(x, y) = \frac{\partial}{\partial y}(x \sin x + y \sin y) = y \cos y + \sin y$.

$$\text{Hence, } N(x, y) = x(y \cos y + \sin y) + g(x), \text{ where } g \text{ is differentiable.}$$

2.5.32. The assumptions imply that $\frac{\partial M_1}{\partial y} = \frac{\partial N_1}{\partial x}$ and $\frac{\partial M_2}{\partial y} = \frac{\partial N_2}{\partial x}$. Therefore,

$$\frac{\partial}{\partial y}(M_1 + M_2) = \frac{\partial M_1}{\partial y} + \frac{\partial M_2}{\partial y} = \frac{\partial N_1}{\partial x} + \frac{\partial N_2}{\partial x} = \frac{\partial}{\partial x}(N_1 + N_2), \text{ which implies that}$$

$$(M_1 + M_2) dx + (N_1 + N_2) dy = 0 \text{ is exact on } R.$$

2.5.34. Here $M(x, y) = Ax^2 + Bxy + Cy^2$ and $N(x, y) = Dx^2 + Exy + Fy^2$. Since $M_y = Bx + 2Cy$ and $N_x = 2Dx + Ey$, the equation is exact if and only if $B = 2D$ and $E = 2C$.

2.5.36. Differentiating (A) $F(x, y) = \int_{y_0}^y N(x_0, s) ds + \int_{x_0}^x M(t, y) dt$ with respect to x yields

$F_x(x, y) = M(x, y)$, since the first integral in (A) is independent of x and $M(t, y)$ is a continuous function of t for each fixed y . Differentiating (A) with respect to y and using the

assumption that $M_y = N_x$ yields $F_y(x, y) = N(x_0, y) + \int_{x_0}^x \frac{\partial M}{\partial y}(t, y) dt =$

$$N(x_0, y) + \int_{x_0}^x \frac{\partial N}{\partial x}(t, y) dt = N(x_0, y) + N(x, y) - N(x_0, y) = N(x, y).$$

2.5.38. $y_1 = \frac{1}{x^2}$ is a solution of $y' + \frac{2}{x}y = 0$. Let $y = \frac{u}{x^2}$; then

$$\frac{u'}{x^2} = -\frac{2x(u/x^2)}{(x^2 + 2x^2(u/x^2) + 1)} = -\frac{2xu}{x^2(x^2 + 2u + 1)},$$

so $u' = -\frac{2xu}{x^2 + 2u + 1}$, which can be rewritten as (A) $2xu dx + (x^2 + 2u + 1) du = 0$. Since

$\frac{\partial}{\partial u}(2xu) = \frac{\partial}{\partial x}(x^2 + 2u + 1) = 2x$, (A) is exact. To solve (A) we must find F such that (A) $F_x(x, u) = 2xu$ and (B) $F_u(x, u) = x^2 + 2u + 1$. Integrating (A) with respect to x yields (C) $F(x, u) = x^2u + \phi(u)$. Differentiating (C) with respect to u yields (D) $F_u(x, u) = x^2 + \phi'(u)$. Comparing (D) with (B) shows that $\phi'(u) = 2u + 1$, so we take $\phi(u) = u^2 + u$. Substituting this into (C) yields $F(x, u) = x^2u + u^2 + u = u(x^2 + u + 1)$. Therefore, $u(x^2 + u + 1) = c$.

Since $y(1) = -2 \Rightarrow u(1) = -2, c = 0$. Therefore, $u(x^2 + u + 1) = 0$. Since $u \equiv 0$ does not satisfy $u(1) = -2$, it follows that $u = -x^2 - 1$ and $y = -1 - \frac{1}{x^2}$.

2.5.40. $y_1 = e^{-x^2}$ is a solution of $y' + 2xy = 0$. Let $y = ue^{-x^2}$; then $u'e^{-x^2} = -e^{-x^2} \left(\frac{3x+2u}{2x+3u} \right)$, so

$$u' = -\frac{3x+2u}{2x+3u}, \text{ which can be rewritten as (A) } (3x+2u)dx + (2x+3u)du = 0. \text{ Since}$$

$$\frac{\partial}{\partial u}(3x+2u) = \frac{\partial}{\partial x}(2x+3u) = 2, \text{ (A) is exact. To solve (A) we must find } F \text{ such that (A)}$$

$$F_x(x, u) = 3x+2u \text{ and (B) } F_u(x, u) = 2x+3u. \text{ Integrating (A) with respect to } x \text{ yields (C)}$$

$$F(x, u) = \frac{3x^2}{2} + 2xu + \phi(u). \text{ Differentiating (C) with respect to } u \text{ yields (D)}$$

$$F_u(x, u) = 2x + \phi'(u). \text{ Comparing (D) with (B) shows that } \phi'(u) = 3u, \text{ so we take}$$

$$\phi(u) = \frac{3u^2}{2}. \text{ Substituting this into (C) yields } F(x, u) = \frac{3x^2}{2} + 2xu + \frac{3u^2}{2}. \text{ Therefore,}$$

$$\frac{3x^2}{2} + 2xu + \frac{3u^2}{2} = c. \text{ Since } y(0) = -1 \Rightarrow u(0) = -1, c = \frac{3}{2}. \text{ Therefore, } 3x^2 + 4xu + 3u^2 = 3$$

is an implicit solution of the initial value problem. Rewriting this as

$$3u^2 + 4xu + (3x^2 - 3) = 0 \text{ and solving for } u \text{ by means of the quadratic formula yields}$$

$$u = -\left(\frac{2x + \sqrt{9 - 5x^2}}{3} \right), \text{ so } y = -e^{-x^2} \left(\frac{2x + \sqrt{9 - 5x^2}}{3} \right).$$

2.5.42. Since $Mdx + Ndy = 0$ is exact, (A) $M_y = N_x$. Since $-Ndx + Mdy = 0$ is exact, (B) $M_x = -N_y$. Differentiating (A) with respect to y and (B) with respect to x yields (C) $M_{yy} = N_{xy}$ and (D) $M_{xx} = -N_{yx}$. Since $N_{xy} = N_{yx}$, adding (C) and (D) yields $M_{xx} + M_{yy} = 0$. Differentiating (A) with respect to x and (B) with respect to y yields (E) $M_{yx} = N_{xx}$ and (F) $M_{xy} = -N_{yy}$. Since $M_{xy} = M_{yx}$, subtracting (F) from (E) yields $N_{xx} + N_{yy} = 0$.

- 2.5.44. (a) If $F(x, y) = x^2 - y^2$, then $F_x(x, y) = 2x$, $F_y(x, y) = -2y$, $F_{xx}(x, y) = 2$, and $F_{yy}(x, y) = -2$. Therefore, $F_{xx} + F_{yy} = 0$, and G must satisfy (A) $G_x(x, y) = 2y$ and (B) $G_y(x, y) = 2x$. Integrating (A) with respect to x yields (C) $G(x, y) = 2xy + \phi(y)$. Differentiating (C) with respect to y yields (D) $G_y(x, y) = 2x + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = 0$, so we take $\phi(y) = c$. Substituting this into (C) yields $G(x, y) = 2xy + c$.
- (b) If $F(x, y) = e^x \cos y$, then $F_x(x, y) = e^x \cos y$, $F_y(x, y) = -e^x \sin y$, $F_{xx}(x, y) = e^x \cos y$, and $F_{yy}(x, y) = -e^x \cos y$. Therefore, $F_{xx} + F_{yy} = 0$, and G must satisfy (A) $G_x(x, y) = e^x \sin y$ and (B) $G_y(x, y) = e^x \cos y$. Integrating (A) with respect to x yields (C) $G(x, y) = e^x \sin y + \phi(y)$. Differentiating (C) with respect to y yields (D) $G_y(x, y) = e^x \cos y + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = 0$, so we take $\phi(y) = c$. Substituting this into (C) yields $G(x, y) = e^x \sin y + c$.
- (c) If $F(x, y) = x^3 - 3xy^2$, then $F_x(x, y) = 3x^2 - 3y^2$, $F_y(x, y) = -6xy$, $F_{xx}(x, y) = 6x$, and $F_{yy}(x, y) = -6x$. Therefore, $F_{xx} + F_{yy} = 0$, and G must satisfy (A) $G_x(x, y) = 6xy$ and (B) $G_y(x, y) = 3x^2 - 3y^2$. Integrating (A) with respect to x yields (C) $G(x, y) = 3x^2y + \phi(y)$. Differentiating (C) with respect to y yields (D) $G_y(x, y) = 3x^2 + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = -3y^2$, so we take $\phi(y) = -y^3 + c$. Substituting this into (C) yields $G(x, y) = 3x^2y - y^3 + c$.
- (d) If $F(x, y) = \cos x \cosh y$, then $F_x(x, y) = -\sin x \cosh y$, $F_y(x, y) = \cos x \sinh y$, $F_{xx}(x, y) = -\cos x \cosh y$, and $F_{yy}(x, y) = \cos x \cosh y$. Therefore, $F_{xx} + F_{yy} = 0$, and

G must satisfy (A) $G_x(x, y) = -\cos x \sinh y$ and (B) $G_y(x, y) = -\sin x \cosh y$. Integrating (A) with respect to x yields (C) $G(x, y) = -\sin x \sinh y + \phi(y)$. Differentiating (C) with respect to y yields (D) $G_y(x, y) = -\sin x \cosh y + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = 0$, so we take $\phi(y) = c$. Substituting this into (C) yields $G(x, y) = -\sin x \sinh y + c$.

- (e) If $F(x, y) = \sin x \cosh y$, then $F_x(x, y) = \cos x \cosh y$, $F_y(x, y) = \sin x \sinh y$, $F_{xx}(x, y) = -\sin x \cosh y$, and $F_{yy}(x, y) = \sin x \cosh y$. Therefore, $F_{xx} + F_{yy} = 0$, and G must satisfy (A) $G_x(x, y) = -\sin x \sinh y$ and (B) $G_y(x, y) = \cos x \cosh y$. Integrating (A) with respect to x yields (C) $G(x, y) = \cos x \sinh y + \phi(y)$. Differentiating (C) with respect to y yields (D) $G_y(x, y) = \cos x \cosh y + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = 0$, so we take $\phi(y) = c$. Substituting this into (C) yields $G(x, y) = \cos x \sinh y + c$.

2.6 Exact Equations

2.6.2. (a and b) To show that $\mu(x, y) = \frac{1}{(x-y)^2}$ is an integrating factor for (A) and that (B) is

exact, it suffices to observe that $\frac{\partial}{\partial x} \left(\frac{xy}{x-y} \right) = -\frac{y^2}{(x-y)^2}$ and

$\frac{\partial}{\partial y} \left(\frac{xy}{x-y} \right) = \frac{x^2}{(x-y)^2}$. By Theorem 2.5.1 this also shows that (C) is an implicit solution of (B). Since $\mu(x, y)$ is never zero, any solution of (B) is a solution of (A).

- (c) If we interpret (A) as $-y^2 + x^2 y' = 0$, then substituting $y = x$ yields $-x^2 + x^2 \cdot 1 = 0$.

(NOTE: In Exercises 2.6.3–2.6.23, the given equation is multiplied by an integrating factor to produce an exact equation, and an implicit solution is found for the latter. For a complete analysis of the relationship between the sets of solutions of the two equations it is necessary to check for additional solutions of the given equation “along which” the integrating factor is undefined, or for solutions of the exact equation “along which” the integrating factor vanishes. In the interests of brevity we omit these tedious details except in cases where there actually is a difference between the sets of solutions of the two equations.)

2.6.4. $M(x, y) = 3x^2 y$; $N(x, y) = 2x^3$; $M_y(x, y) - N_x(x, y) = 3x^2 - 6x^2 = -3x^2$;

$$p(x) = \frac{M_y(x, y) - N_x(x, y)}{N(x, y)} = -\frac{3x^2}{2x^3} = -\frac{3}{2x}; \quad \int p(x) dx = -\frac{3}{2} \ln|x|; \quad \mu(x) = P(x) = x^{-3/2};$$

therefore $3x^{1/2} y dx + 2x^{3/2} dy = 0$ is exact. We must find F such that (A) $F_x(x, y) = 3x^{1/2} y$ and (B) $F_y(x, y) = 2x^{3/2}$. Integrating (A) with respect to x yields (C)

$F(x, y) = 2x^{3/2} y + \phi(y)$. Differentiating (C) with respect to y yields (D)

$F_y(x, y) = 2x^{3/2} + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = 0$, so we take

$\phi(y) = 0$. Substituting this into (C) yields $F(x, y) = 2x^{3/2} y$, so $x^{3/2} y = c$.

2.6.6. $M(x, y) = 5xy + 2y + 5$; $N(x, y) = 2x$; $M_y(x, y) - N_x(x, y) = (5x + 2) - 2 = 5x$;

$$p(x) = \frac{M_y(x, y) - N_x(x, y)}{N(x, y)} = \frac{5x}{2x} = \frac{5}{2}; \quad \int p(x) dx = \frac{5x}{2}; \quad \mu(x) = P(x) = e^{5x/2}; \text{ therefore}$$

$e^{5x/2}(5xy + 2y + 5) dx + 2xe^{5x/2} dy = 0$ is exact. We must find F such that (A)

$F_x(x, y) = e^{5x/2}(5xy + 2y + 5)$ and (B) $F_y(x, y) = 2xe^{5x/2}$. Integrating (B) with respect to y yields (C) $F(x, y) = 2xye^{5x/2} + \psi(x)$. Differentiating (C) with respect to x yields (D) $F_x(x, y) = 5xye^{5x/2} + 2ye^{5x/2} + \psi'(x)$. Comparing (D) with (A) shows that $\psi'(x) = 5e^{5x/2}$, so we take $\psi(x) = 2e^{5x/2}$. Substituting this into (C) yields $F(x, y) = 2e^{5x/2}(xy + 1)$, so $e^{5x/2}(xy + 1) = c$.

2.6.8. $M(x, y) = 27xy^2 + 8y^3$; $N(x, y) = 18x^2y + 12xy^2$;
 $M_y(x, y) - N_x(x, y) = (54xy + 24y^2) - (36xy + 12y^2) = 18xy + 12y^2$;
 $p(x) = \frac{M_y(x, y) - N_x(x, y)}{N(x, y)} = \frac{18xy + 12y^2}{18x^2y + 12y^2x} = \frac{1}{x}$; $\int p(x) dx = \ln|x|$; $\mu(x) = P(x) = x$;
 therefore $(27x^2y^2 + 8xy^3) dx + (18x^3y + 12x^2y^2) dy = 0$ is exact. We must find F such that (A) $F_x(x, y) = 27x^2y^2 + 8xy^3$ and (B) $F_y(x, y) = 18x^3y + 12x^2y^2$. Integrating (A) with respect to x yields (C) $F(x, y) = 9x^3y^2 + 4x^2y^3 + \phi(y)$. Differentiating (C) with respect to y yields (D) $F_y(x, y) = 18x^3y + 12x^2y^2 + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = 0$, so we take $\phi(y) = 0$. Substituting this into (C) yields $F(x, y) = 9x^3y^2 + 4x^2y^3$, so $x^2y^2(9x + 4y) = c$.

2.6.10. $M(x, y) = y^2$; $N(x, y) = \left(xy^2 + 3xy + \frac{1}{y}\right)$;
 $M_y(x, y) - N_x(x, y) = 2y - (y^2 + 3y) = -y(y + 1)$;
 $q(y) = \frac{N_x(x, y) - M_y(x, y)}{M(x, y)} = \frac{y(y + 1)}{y^2} = 1 + \frac{1}{y}$; $\int q(y) dy = y \ln|y|$; $\mu(y) = Q(y) = ye^y$;
 therefore $y^3e^y dx + e^y(xy^3 + 3xy^2 + 1) dy = 0$ is exact. We must find F such that (A) $F_x(x, y) = y^3e^y$ and (B) $F_y(x, y) = e^y(xy^3 + 3xy^2 + 1)$. Integrating (A) with respect to x yields (C) $F(x, y) = xy^3e^y + \phi(y)$. Differentiating (C) with respect to y yields (D) $F_y(x, y) = xy^3e^y + 3xy^2e^y + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = e^y$, so we take $\phi(y) = e^y$. Substituting this into (C) yields $F(x, y) = xy^3e^y + e^y$, so $e^y(xy^3 + 1) = c$.

2.6.12. $M(x, y) = x^2y + 4xy + 2y$; $N(x, y) = x^2 + x$;
 $M_y(x, y) - N_x(x, y) = (x^2 + 4x + 2) - (2x + 1) = x^2 + 2x + 1 = (x + 1)^2$;
 $p(x) = \frac{M_y(x, y) - N_x(x, y)}{N(x, y)} = \frac{(x + 1)^2}{x(x + 1)} = 1 + \frac{1}{x}$; $\int p(x) dx = x + \ln|x|$;
 $\mu(x) = P(x) = xe^x$; therefore $e^x(x^3y + 4x^2y + 2xy) dx + e^x(x^3 + x^2) dy = 0$ is exact. We must find F such that (A) $F_x(x, y) = e^x(x^3y + 4x^2y + 2xy)$ and (B) $F_y(x, y) = e^x(x^3 + x^2)$. Integrating (B) with respect to y yields (C) $F(x, y) = y(x^3 + x^2)e^x + \psi(x)$. Differentiating (C) with respect to x yields (D) $F_x(x, y) = e^x(x^3y + 4x^2y + 2xy) + \psi'(x)$. Comparing (D) with (A) shows that $\psi'(x) = 0$, so we take $\psi(x) = 0$. Substituting this into (C) yields $F(x, y) = y(x^3 + x^2)e^x = x^2y(x + 1)e^x$, so $x^2y(x + 1)e^x = c$.

2.6.14. $M(x, y) = \cos x \cos y$; $N(x, y) = \sin x \cos y - \sin x \sin y + y$;
 $M_y(x, y) - N_x(x, y) = -\cos x \sin y - (\cos x \cos y - \cos x \sin y) = -\cos x \cos y$;
 $q(y) = \frac{N_x(x, y) - M_y(x, y)}{M(x, y)} = \frac{\cos x \cos y}{\cos x \cos y} = 1$; $\int q(y) dy = 1$; $\mu(y) = Q(y) = e^y$; therefore $e^y \cos x \cos y dx + e^y(\sin x \cos y - \sin x \sin y + y) dy = 0$ is exact. We must find F such that (A) $F_x(x, y) = e^y \cos x \cos y$ and (B) $F_y(x, y) = e^y(\sin x \cos y - \sin x \sin y + y)$. Integrating (A) with respect to x yields (C) $F(x, y) = e^y \sin x \cos y + \phi(y)$. Differentiating (C) with

respect to y yields (D) $F_y(x, y) = e^y(\sin x \cos y - \sin x \sin y) + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = ye^y$, so we take $\phi(y) = e^y(y - 1)$. Substituting this into (C) yields $F(x, y) = e^y(\sin x \cos y + y - 1)$, so $e^y(\sin x \cos y + y - 1) = c$.

2.6.16. $M(x, y) = y \sin y$; $N(x, y) = x(\sin y - y \cos y)$;
 $M_y(x, y) - N_x(x, y) = (y \cos y + \sin y) - (\sin y - y \cos y) = 2y \cos y$;
 $q(y) = \frac{N_x(x, y) - M_y(x, y)}{N(x, y)} = -\frac{2 \cos y}{\sin y}$; $\int q(y) dy = -2 \ln |\sin y|$; $\mu(y) = Q(y) = \frac{1}{\sin^2 y}$;
 therefore $\left(\frac{y}{\sin y}\right) dx + x \left(\frac{1}{\sin y} - \frac{y \cos y}{\sin^2 y}\right) dy = 0$ is exact. We must find F such that (A)
 $F_x(x, y) = \frac{y}{\sin y}$ and (B) $F_y(x, y) = x \left(\frac{1}{\sin y} - \frac{y \cos y}{\sin^2 y}\right)$. Integrating (A) with respect to x
 yields (C) $F(x, y) = \frac{xy}{\sin y} + \phi(y)$. Differentiating (C) with respect to y yields (D)
 $F_y(x, y) = x \left(\frac{1}{\sin y} - \frac{y \cos y}{\sin^2 y}\right) + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = 0$, so we
 take $\phi(y) = 0$. Substituting this into (C) yields $F(x, y) = \frac{xy}{\sin y}$, so $\frac{xy}{\sin y} = c$. In addition,
 the given equation has the constant solutions $y = k\pi$, where k is an integer.

2.6.18. $M(x, y) = \alpha y + \gamma xy$; $N(x, y) = \beta x + \delta xy$; $M_y(x, y) - N_x(x, y) = (\alpha + \gamma x) - (\beta + \delta y)$; and
 $p(x)N(x, y) - q(y)M(x, y) = p(x)x(\beta + \delta y) - q(y)y(\alpha + \gamma x)$. so exactness requires that
 $(\alpha + \gamma x) - (\beta + \delta y) = p(x)x(\beta + \delta y) - q(y)y(\alpha + \gamma x)$, which holds if $p(x)x = -1$ and
 $q(y)y = -1$. Thus $p(x) = -\frac{1}{x}$; $q(y) = -\frac{1}{y}$; $\int p(x) dx = -\ln |x|$; $\int q(y) dy = -\ln |y|$;
 $P(x) = \frac{1}{x}$; $Q(y) = \frac{1}{y}$; $\mu(x, y) = \frac{1}{xy}$. Therefore, $\left(\frac{\alpha}{x} + \gamma\right) dx + \left(\frac{\beta}{y} + \delta\right) dy = 0$ is exact. We
 must find F such that (A) $F_x(x, y) = \frac{\alpha}{x} + \gamma$ and (B) $F_y(x, y) = \frac{\beta}{y} + \delta$. Integrating (A) with
 respect to x yields (C) $F(x, y) = \alpha \ln |x| + \gamma x + \phi(y)$. Differentiating (C) with respect to y
 yields (D) $F_y(x, y) = \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = \frac{\beta}{y} + \delta$, so we take
 $\phi(y) = \beta \ln |y| + \delta y$. Substituting this into (C) yields $F(x, y) = \alpha \ln |x| + \gamma x + \beta \ln |y| + \delta y$, so
 $|x|^\alpha |y|^\beta e^{\gamma x} e^{\delta y} = c$. The given equation also has the solutions $x \equiv 0$ and $y \equiv 0$.

2.6.20. $M(x, y) = 2y$; $N(x, y) = 3(x^2 + x^2 y^3)$; $M_y(x, y) - N_x(x, y) = 2 - (6x + 6xy^3)$; and
 $p(x)N(x, y) - q(y)M(x, y) = 3p(x)(x^2 + x^2 y^3) - 2q(y)y$. so exactness requires that (A)
 $2 - 6x - 6xy^3 = 3p(x)x(x + xy^3) - 2q(y)y$. To obtain similar terms on the two sides of (A)
 we let $p(x)x = a$ and $q(y)y = b$ where a and b are constants such that
 $2 - 6x - 6xy^3 = 3a(x + xy^3) - 2b$, which holds if $a = -2$ and $b = -1$. Thus, $p(x) = -\frac{2}{x}$;
 $q(y) = -\frac{1}{y}$; $\int p(x) dx = -2 \ln |x|$; $\int q(y) dy = -\ln |y|$; $P(x) = \frac{1}{x^2}$; $Q(y) = \frac{1}{y}$;
 $\mu(x, y) = \frac{1}{x^2 y}$. Therefore, $\frac{2}{x^2} dx + 3\left(\frac{1}{y} + y^2\right) dy = 0$ is exact. We must find F such that
 (B) $F_x(x, y) = \frac{2}{x^2}$ and (C) $F_y(x, y) = 3\left(\frac{1}{y} + y^2\right)$. Integrating (B) with respect to x yields
 (D) $F(x, y) = -\frac{2}{x} + \phi(y)$. Differentiating (D) with respect to y yields (E) $F_y(x, y) = \phi'(y)$.

Comparing (E) with (C) shows that $\phi'(y) = 3\left(\frac{1}{y} + y^2\right)$, so we take $\phi(y) = y^3 + 3 \ln |y|$.

Substituting this into (D) yields $F(x, y) = -\frac{2}{x} + y^3 + 3 \ln |y|$, so $-\frac{2}{x} + y^3 + 3 \ln |y| = c$. The given equation also has the solutions $x \equiv 0$ and $y \equiv 0$.

- 2.6.22. $M(x, y) = x^4 y^4$; $N(x, y) = x^5 y^3$; $M_y(x, y) - N_x(x, y) = 4x^4 y^3 - 5x^4 y^3 = -x^4 y^3$; and $p(x)N(x, y) - q(y)M(x, y) = p(x)x^5 y^3 - q(y)x^4 y^4$. so exactness requires that $-x^4 y^3 = p(x)x^5 y^3 - q(y)x^4 y^4$, which is equivalent to $p(x)x - q(y)y = -1$. This holds if $p(x)x = a$ and $q(y)y = a + 1$ where a is an arbitrary real number. Thus, $p(x) = \frac{a}{x}$; $q(y) = \frac{a+1}{y}$; $\int p(x) dx = a \ln |x|$; $\int q(y) dy = (a+1) \ln |y|$; $P(x) = |x|^a$; $Q(y) = |y|^{a+1}$; $\mu(x, y) = |x|^a |y|^{a+1}$. Therefore, $|x|^a |y|^{a+1} (x^4 y^4 dx + x^5 y^3 dy) = 0$ is exact for any choice of a . For simplicity we let $a = -4$, so (A) is equivalent to $y dx + x dy = 0$. We must find F such that (B) $F_x(x, y) = y$ and (C) $F_y(x, y) = x$. Integrating (B) with respect to x yields (D) $F(x, y) = xy + \phi(y)$. Differentiating (D) with respect to y yields (E) $F_y(x, y) = x + \phi'(y)$. Comparing (E) with (C) shows that $\phi'(y) = 0$, so we take $\phi(y) = 0$. Substituting this into (D) yields $F(x, y) = xy$, so $xy = c$.

- 2.6.24. $M(x, y) = x^4 y^3 + y$; $N(x, y) = x^5 y^2 - x$;
 $M_y(x, y) - N_x(x, y) = (3x^4 y^2 + 1) - (5x^4 y^2 - 1) = -2x^4 y^2 + 2$;
 $p(x) = \frac{M_y(x, y) - N_x(x, y)}{N(x, y)} = -\frac{2x^4 y^2 - 2}{x^5 y^2 - x} = -\frac{2}{x}$; $\int p(x) dx = -2 \ln |x|$; $\mu(x) = P(x) = \frac{1}{x^2}$;
therefore $\left(x^2 y^3 + \frac{y}{x^2}\right) dx + \left(x^3 y^2 - \frac{1}{x}\right) dy = 0$ is exact. We must find F such that (A) $F_x(x, y) = \left(x^2 y^3 + \frac{y}{x^2}\right)$ and (B) $F_y(x, y) = \left(x^3 y^2 - \frac{1}{x}\right)$. Integrating (A) with respect to x yields (C) $F(x, y) = \frac{x^3 y^3}{3} - \frac{y}{x} + \phi(y)$. Differentiating (C) with respect to y yields (D) $F_y(x, y) = x^3 y^2 - \frac{1}{x} + \phi'(y)$. Comparing (D) with (B) shows that $\phi'(y) = 0$, so we take $\phi(y) = 0$. Substituting this into (C) yields $F(x, y) = \frac{x^3 y^3}{3} - \frac{y}{x}$, so $\frac{x^3 y^3}{3} - \frac{y}{x} = c$.

- 2.6.26. $M(x, y) = 12xy + 6y^3$; $N(x, y) = 9x^2 + 10xy^2$;
 $M_y(x, y) - N_x(x, y) = (12x + 18y^2) - (18x + 10y^2) = -6x + 8y^2$; and
 $p(x)N(x, y) - q(y)M(x, y) = p(x)x(9x + 10y^2) - q(y)y(12x + 6y^2)$, so exactness requires that (A) $-6x + 8y^2 = p(x)x(9x + 10y^2) - q(y)y(12x + 6y^2)$. To obtain similar terms on the two sides of (A) we let $p(x)x = a$ and $q(y)y = b$ where a and b are constants such that $-6x + 8y^2 = a(9x + 10y^2) - b(12x + 6y^2)$, which holds if $9a - 12b = -6$, $10a - 6b = 8$; that is, $a = b = 2$. Thus $p(x) = \frac{2}{x}$; $q(y) = \frac{2}{y}$; $\int p(x) dx = 2 \ln |x|$; $\int q(y) dy = 2 \ln |y|$; $P(x) = x^2$; $Q(y) = y^2$; $\mu(x, y) = x^2 y^2$. Therefore, $(12x^3 y^3 + 6x^2 y^5) dx + (9x^4 y^2 + 10x^3 y^4) dy = 0$ is exact. We must find F such that (B) $F_x(x, y) = 12x^3 y^3 + 6x^2 y^5$ and (C) $F_y(x, y) = 9x^4 y^2 + 10x^3 y^4$. Integrating (B) with respect to x yields (D) $F(x, y) = 3x^4 y^3 + 2x^3 y^5 + \phi(y)$. Differentiating (D) with respect to y yields (E) $F_y(x, y) = 9x^4 y^2 + 10x^3 y^4 + \phi'(y)$. Comparing (E) with (C) shows that $\phi'(y) = 0$, so we take $\phi(y) = 0$. Substituting this into (D) yields $F(x, y) = 3x^4 y^3 + 2x^3 y^5$, so $x^3 y^3 (3x + 2y^2) = c$.

2.6.28. $M(x, y) = ax^m y + by^{n+1}$; $N(x, y) = cx^{m+1} + dxy^n$;
 $M_y(x, y) - N_x(x, y) = [ax^{m+1} + (n+1)by^n] - [(m+1)cx^m + dy^n]$;
 $p(x)N(x, y) - q(y)M(x, y) = xp(x)(cx^m + dy^n) - yp(y)(ax^m + by^n)$. Let (A) $xp(x) = \alpha$ and
 (B) $yp(y) = \beta$, where α and β are to be chosen so that
 $[ax^{m+1} + (n+1)by^n] - [(m+1)cx^m + dy^n] = \alpha(cx^m + dy^n) - \beta(ax^m + by^n)$, which will
 hold if

$$\begin{aligned} c\alpha - a\beta &= a - (m+1)c =_{\text{df}} A \\ d\alpha - b\beta &= -d + (n+1)b =_{\text{df}} B. \end{aligned} \tag{C}$$

Since $ad - bc \neq 0$ it can be verified that $\alpha = \frac{aB - bA}{ad - bc}$ and $\beta = \frac{cB - dA}{ad - bc}$ satisfy (C). From
 (A) and (B), $p(x) = \frac{\alpha}{x}$ and $q(y) = \frac{\beta}{y}$, so $\mu(x, y) = x^\alpha y^\beta$ is an integrating factor for the given
 equation.

2.6.30. (a) Since $M(x, y) = p(x)y - f(x)$ and $N(x, y) = 1$, $\frac{M_y(x, y) - N_x(x, y)}{N(x, y)} = p(x)$ and

Theorem 2.6.1 implies that $\mu(x) \pm e^{\int p(x) dx}$ is an integrating factor for (C).

(b) Multiplying (A) through $\mu = \pm e^{\int p(x) dx}$ yields (D) $\mu(x)y' + \mu'(x)y = \mu(x)f(x)$, which
 is equivalent to $(\mu(x)y)' = \mu(x)f(x)$. Integrating this yields

$\mu(x)y = c + \int \mu(x)f(x) dx$, so $y = \frac{1}{\mu(x)} \left(c + \int \mu(x)f(x) dx \right)$, which is equivalent to

(B) since $y_1 = \frac{1}{\mu}$ is a nontrivial solution of $y' + p(x)y = 0$.

Chapter 3

Numerical Methods

3.1 Euler's Method

3.1.2. $y_1 = 1.200000000$, $y_2 = 1.440415946$, $y_3 = 1.729880994$

3.1.4. $y_1 = 2.962500000$, $y_2 = 2.922635828$, $y_3 = 2.880205639$

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	Exact
3.1.6.	0.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
	0.1	2.100 000 000	2.169 990 965	2.202 114 518	2.232 642 918
	0.2	2.514 277 288	2.649 377 900	2.713 011 720	2.774 352 565
	0.3	3.317 872 752	3.527 672 599	3.628 465 025	3.726 686 582
	0.4	4.646 592 772	4.955 798 226	5.106 379 369	5.254 226 636
	0.5	6.719 737 638	7.171 467 977	7.393 322 991	7.612 186 259
	0.6	9.876 155 616	10.538 384 528	10.865 186 799	11.188 475 269
	0.7	14.629 532 397	15.605 686 107	16.088 630 652	16.567 103 199
	0.8	21.751 925 418	23.197 328 550	23.913 328 531	24.623 248 150
	0.9	32.399 118 931	34.545 932 627	35.610 005 377	36.665 439 956
	1.0	48.298 147 362	51.492 825 643	53.076 673 685	54.647 937 102

	x	$h = 0.05$	$h = 0.025$	$h = 0.0125$	Exact
3.1.8.	1.00	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
	1.05	2.250 000 000	2.259 280 190	2.264 490 570	2.270 158 103
	1.10	2.536 734 694	2.559 724 746	2.572 794 280	2.587 150 838
	1.15	2.867 950 854	2.910 936 426	2.935 723 355	2.963 263 785
	1.20	3.253 613 825	3.325 627 715	3.367 843 117	3.415 384 615
	1.25	3.706 750 613	3.820 981 064	3.889 251 900	3.967 391 304
	1.30	4.244 700 641	4.420 781 829	4.528 471 927	4.654 198 473
	1.35	4.891 020 001	5.158 883 503	5.327 348 558	5.528 980 892
	1.40	5.678 467 290	6.085 075 790	6.349 785 943	6.676 923 077
	1.45	6.653 845 988	7.275 522 641	7.698 316 221	8.243 593 315
	1.50	7.886 170 437	8.852 463 793	9.548 039 907	10.500 000 000

	x	Approximate Solutions			Residuals		
		$h = 0.1$	$h = 0.05$	$h = 0.025$	$h = 0.1$	$h = 0.05$	$h = 0.025$
3.1.10.	1.0	1.000 000 000	1.000 000 000	1.000 000 000	0.0000	0.0000	0.0000
	1.1	0.920 000 000	0.921 898 275	0.922 822 717	-0.0384	-0.0189	-0.0094
	1.2	0.847 469 326	0.851 018 464	0.852 746 371	-0.0745	-0.0368	-0.0183
	1.3	0.781 779 403	0.786 770 087	0.789 197 876	-0.1092	-0.0540	-0.0268
	1.4	0.722 453 556	0.728 682 209	0.731 709 712	-0.1428	-0.0707	-0.0351
	1.5	0.669 037 867	0.676 299 618	0.679 827 306	-0.1752	-0.0868	-0.0432
	1.6	0.621 054 176	0.629 148 585	0.633 080 163	-0.2062	-0.1023	-0.0509
	1.7	0.578 000 416	0.586 740 390	0.590 986 601	-0.2356	-0.1170	-0.0583
	1.8	0.539 370 187	0.548 588 902	0.553 070 392	-0.2631	-0.1310	-0.0653
	1.9	0.504 674 296	0.514 228 603	0.518 877 246	-0.2889	-0.1441	-0.0719
	2.0	0.473 456 737	0.483 227 470	0.487 986 391	-0.3129	-0.1563	-0.0781

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	"Exact"
3.1.12.	1.0	0.000 000 000	0.000 000 000	0.000 000 000	0.000 000 000
	1.1	-0.100 000 000	-0.099 875 000	-0.099 780 455	-0.099 664 000
	1.2	-0.199 000 000	-0.198 243 434	-0.197 800 853	-0.197 315 517
	1.3	-0.294 996 246	-0.293 129 862	-0.292 110 713	-0.291 036 003
	1.4	-0.386 095 345	-0.382 748 403	-0.380 986 158	-0.379 168 221
	1.5	-0.470 695 388	-0.465 664 569	-0.463 078 857	-0.460 450 590
	1.6	-0.547 627 491	-0.540 901 018	-0.537 503 081	-0.534 085 626
	1.7	-0.616 227 665	-0.607 969 574	-0.603 849 795	-0.599 737 720
	1.8	-0.676 329 533	-0.666 833 345	-0.662 136 956	-0.657 473 792
	1.9	-0.728 190 908	-0.717 819 639	-0.712 718 751	-0.707 670 533
	2.0	-0.772 381 768	-0.761 510 960	-0.756 179 726	-0.750 912 371

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	"Exact"
3.1.14. Euler's method:	2.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
	2.1	2.420 000 000	2.440 610 764	2.451 962 006	2.464 119 569
	2.2	2.922 484 288	2.972 198 224	2.999 753 046	3.029 403 212
	2.3	3.524 104 434	3.614 025 082	3.664 184 099	3.718 409 925
	2.4	4.244 823 572	4.389 380 160	4.470 531 822	4.558 673 929
	2.5	5.108 581 185	5.326 426 396	5.449 503 467	5.583 808 754
	2.6	6.144 090 526	6.459 226 591	6.638 409 411	6.834 855 438
	2.7	7.385 795 229	7.828 984 275	8.082 588 076	8.361 928 926
	2.8	8.875 017 001	9.485 544 888	9.837 137 672	10.226 228 709
	2.9	10.661 332 618	11.489 211 987	11.969 020 902	12.502 494 409
	3.0	12.804 226 135	13.912 944 662	14.559 623 055	15.282 004 826

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
Euler semilinear method:	2.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
	2.1	2.467 233 571	2.465 641 081	2.464 871 435	2.464 119 569
	2.2	3.036 062 650	3.032 657 307	3.031 011 316	3.029 403 212
	2.3	3.729 169 725	3.723 668 026	3.721 008 466	3.718 409 925
	2.4	4.574 236 356	4.566 279 470	4.562 432 696	4.558 673 929
	2.5	5.605 052 990	5.594 191 643	5.588 940 276	5.583 808 754
	2.6	6.862 874 116	6.848 549 921	6.841 623 814	6.834 855 438
	2.7	8.398 073 101	8.379 595 572	8.370 660 695	8.361 928 926
	2.8	10.272 163 096	10.248 681 420	10.237 326 199	10.226 228 709
	2.9	12.560 265 110	12.530 733 531	12.516 452 106	12.502 494 409
	3.0	15.354 122 287	15.317 257 705	15.299 429 421	15.282 004 826

	x	$h = 0.2$	$h = 0.1$	$h = 0.05$	“Exact”
3.1.16. Euler’s method:	1.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
	1.2	1.768 294 197	1.786 514 499	1.794 412 375	1.801 636 774
	1.4	1.603 028 371	1.628 427 487	1.639 678 822	1.650 102 616
	1.6	1.474 580 412	1.502 563 111	1.515 157 063	1.526 935 885
	1.8	1.368 349 549	1.396 853 671	1.409 839 229	1.422 074 283
	2.0	1.276 424 761	1.304 504 818	1.317 421 794	1.329 664 953
	2.2	1.194 247 156	1.221 490 111	1.234 122 458	1.246 155 344
	2.4	1.119 088 175	1.145 348 276	1.157 607 418	1.169 334 346
	2.6	1.049 284 410	1.074 553 688	1.086 419 453	1.097 812 069
	2.8	0.983 821 745	1.008 162 993	1.019 652 023	1.030 719 114
	3.0	0.922 094 379	0.945 604 800	0.956 752 868	0.967 523 153

	x	$h = 0.2$	$h = 0.1$	$h = 0.05$	“Exact”
Euler semilinear method:	1.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
	1.2	1.806 911 831	1.804 304 958	1.802 978 526	1.801 636 774
	1.4	1.659 738 603	1.654 968 381	1.652 547 436	1.650 102 616
	1.6	1.540 257 861	1.533 652 916	1.530 308 405	1.526 935 885
	1.8	1.438 532 932	1.430 361 800	1.426 232 584	1.422 074 283
	2.0	1.348 782 285	1.339 279 577	1.334 486 249	1.329 664 953
	2.2	1.267 497 415	1.256 876 924	1.251 528 766	1.246 155 344
	2.4	1.192 497 494	1.180 958 765	1.175 157 264	1.169 334 346
	2.6	1.122 416 379	1.110 147 777	1.103 988 310	1.097 812 069
	2.8	1.056 405 906	1.043 585 743	1.037 158 237	1.030 719 114
	3.0	0.993 954 754	0.980 751 307	0.974 140 320	0.967 523 153

3.1.18. Euler's method:

x	$h = 0.2$	$h = 0.1$	$h = 0.05$	"Exact"
0.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
0.2	1.200 000 000	1.186 557 290	1.179 206 574	1.171 515 153
0.4	1.333 543 409	1.298 441 890	1.280 865 289	1.263 370 891
0.6	1.371 340 142	1.319 698 328	1.295 082 088	1.271 251 278
0.8	1.326 367 357	1.270 160 237	1.243 958 980	1.218 901 287
1.0	1.233 056 306	1.181 845 667	1.158 064 902	1.135 362 070
1.2	1.122 359 136	1.080 477 477	1.060 871 608	1.042 062 625
1.4	1.013 100 262	0.981 124 989	0.965 917 496	0.951 192 532
1.6	0.914 000 211	0.890 759 107	0.879 460 404	0.868 381 328
1.8	0.827 848 558	0.811 673 612	0.803 582 000	0.795 518 627
2.0	0.754 572 560	0.743 869 878	0.738 303 914	0.732 638 628

Euler semilinear method:

x	$h = 0.2$	$h = 0.1$	$h = 0.05$	"Exact"
0.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
0.2	1.153 846 154	1.162 906 599	1.167 266 650	1.171 515 153
0.4	1.236 969 953	1.250 608 357	1.257 097 924	1.263 370 891
0.6	1.244 188 456	1.258 241 892	1.264 875 987	1.271 251 278
0.8	1.195 155 456	1.207 524 076	1.213 335 781	1.218 901 287
1.0	1.115 731 189	1.125 966 437	1.130 768 614	1.135 362 070
1.2	1.025 938 754	1.034 336 918	1.038 283 392	1.042 062 625
1.4	0.937 645 707	0.944 681 597	0.948 002 346	0.951 192 532
1.6	0.856 581 823	0.862 684 171	0.865 583 126	0.868 381 328
1.8	0.784 832 910	0.790 331 183	0.792 963 532	0.795 518 627
2.0	0.722 610 454	0.727 742 966	0.730 220 211	0.732 638 628

3.1.20. Euler's method:

x	$h = 0.1$	$h = 0.05$	$h = 0.025$	"Exact"
0.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
0.1	0.700 000 000	0.725 841 563	0.736 671 690	0.746 418 339
0.2	0.498 330 000	0.532 982 493	0.547 988 831	0.561 742 917
0.3	0.356 272 689	0.392 592 562	0.408 724 303	0.423 724 207
0.4	0.254 555 443	0.289 040 639	0.304 708 942	0.319 467 408
0.5	0.181 440 541	0.212 387 189	0.226 758 594	0.240 464 879
0.6	0.128 953 069	0.155 687 255	0.168 375 130	0.180 626 161
0.7	0.091 393 543	0.113 851 516	0.124 744 976	0.135 394 692
0.8	0.064 613 612	0.083 076 641	0.092 230 966	0.101 293 057
0.9	0.045 585 102	0.060 505 907	0.068 068 776	0.075 650 324
1.0	0.032 105 117	0.043 997 045	0.050 159 310	0.056 415 515

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
Euler semilinear method:	0.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
	0.1	0.740 818 221	0.743 784 320	0.745 143 557	0.746 418 339
	0.2	0.555 889 275	0.558 989 106	0.560 410 719	0.561 742 917
	0.3	0.418 936 461	0.421 482 025	0.422 642 541	0.423 724 207
	0.4	0.315 890 439	0.317 804 400	0.318 668 549	0.319 467 408
	0.5	0.237 908 421	0.239 287 095	0.239 902 094	0.240 464 879
	0.6	0.178 842 206	0.179 812 811	0.180 239 888	0.180 626 161
	0.7	0.134 165 506	0.134 840 668	0.135 133 367	0.135 394 692
	0.8	0.100 450 939	0.100 918 118	0.101 117 514	0.101 293 057
	0.9	0.075 073 968	0.075 396 974	0.075 532 643	0.075 650 324
	1.0	0.056 020 154	0.056 243 980	0.056 336 491	0.056 415 515

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
3.1.22. Euler’s method:	2.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
	2.1	1.000 000 000	1.005 062 500	1.007 100 815	1.008 899 988
	2.2	1.020 500 000	1.026 752 091	1.029 367 367	1.031 723 469
	2.3	1.053 489 840	1.059 067 423	1.061 510 137	1.063 764 243
	2.4	1.093 521 685	1.097 780 573	1.099 748 225	1.101 614 730
	2.5	1.137 137 554	1.140 059 654	1.141 496 651	1.142 903 776
	2.6	1.182 269 005	1.184 090 031	1.185 056 276	1.186 038 851
	2.7	1.227 745 005	1.228 755 801	1.229 350 441	1.229 985 178
	2.8	1.272 940 309	1.273 399 187	1.273 721 920	1.274 092 525
	2.9	1.317 545 833	1.317 651 554	1.317 786 528	1.317 967 533
	3.0	1.361 427 907	1.361 320 824	1.361 332 589	1.361 383 810

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
Euler semilinear method:	2.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
	2.1	0.982 476 904	0.996 114 142	1.002 608 435	1.008 899 988
	2.2	0.988 105 346	1.010 577 663	1.021 306 044	1.031 723 469
	2.3	1.009 495 813	1.037 358 814	1.050 731 634	1.063 764 243
	2.4	1.041 012 955	1.071 994 816	1.086 964 414	1.101 614 730
	2.5	1.078 631 301	1.111 346 365	1.127 262 285	1.142 903 776
	2.6	1.119 632 590	1.153 300 133	1.169 781 376	1.186 038 851
	2.7	1.162 270 287	1.196 488 725	1.213 325 613	1.229 985 178
	2.8	1.205 472 927	1.240 060 456	1.257 146 091	1.274 092 525
	2.9	1.248 613 584	1.283 506 001	1.300 791 772	1.317 967 533
	3.0	1.291 345 518	1.326 535 737	1.344 004 102	1.361 383 810

3.2 The Improved Euler Method and Related Methods

3.2.2. $y_1 = 1.220207973$, $y_2 = 1.489578775$ $y_3 = 1.819337186$

3.2.4. $y_1 = 2.961317914$; $y_2 = 2.920132727$; $y_3 = 2.876213748$.

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	Exact
3.2.6.	0.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
	0.1	2.257 138 644	2.238 455 342	2.234 055 168	2.232 642 918
	0.2	2.826 004 666	2.786 634 110	2.777 340 360	2.774 352 565
	0.3	3.812 671 926	3.747 167 263	3.731 674 025	3.726 686 582
	0.4	5.387 430 580	5.285 996 803	5.261 969 043	5.254 226 636
	0.5	7.813 298 361	7.660 199 197	7.623 893 064	7.612 186 259
	0.6	11.489 337 756	11.260 349 005	11.206 005 869	11.188 475 269
	0.7	17.015 861 211	16.674 352 914	16.593 267 820	16.567 103 199
	0.8	25.292 140 630	24.783 149 862	24.662 262 731	24.623 248 150
	0.9	37.662 496 723	36.903 828 191	36.723 608 928	36.665 439 956
	1.0	56.134 480 009	55.003 390 448	54.734 674 836	54.647 937 102

	x	$h = 0.05$	$h = 0.025$	$h = 0.0125$	Exact
3.2.8.	1.00	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
	1.05	2.268 367 347	2.269 670 336	2.270 030 868	2.270 158 103
	1.10	2.582 607 299	2.585 911 295	2.586 827 341	2.587 150 838
	1.15	2.954 510 022	2.960 870 733	2.962 638 822	2.963 263 785
	1.20	3.400 161 788	3.411 212 150	3.414 293 964	3.415 384 615
	1.25	3.942 097 142	3.960 434 900	3.965 570 792	3.967 391 304
	1.30	4.612 879 780	4.642 784 826	4.651 206 769	4.654 198 473
	1.35	5.461 348 619	5.510 188 575	5.524 044 591	5.528 980 892
	1.40	6.564 150 753	6.645 334 756	6.668 600 859	6.676 923 077
	1.45	8.048 579 617	8.188 335 998	8.228 972 215	8.243 593 315
	1.50	10.141 969 585	10.396 770 409	10.472 502 111	10.500 000 000

	x	Approximate Solutions			Residuals		
		$h = 0.1$	$h = 0.05$	$h = 0.025$	$h = 0.1$	$h = 0.05$	$h = 0.025$
3.2.10.	1.0	1.000 000 000	1.000 000 000	1.000 000 000	0.000 00	0.000 000	0.000 000
	1.1	0.923 734 663	0.923 730 743	0.923 730 591	0.000 04	0.000 001	-0.000 001
	1.2	0.854 475 600	0.854 449 616	0.854 444 697	0.000 35	0.000 068	0.000 015
	1.3	0.791 650 344	0.791 596 016	0.791 584 634	0.000 78	0.000 167	0.000 039
	1.4	0.734 785 779	0.734 703 826	0.734 686 010	0.001 25	0.000 277	0.000 065
	1.5	0.683 424 095	0.683 318 666	0.683 295 308	0.001 71	0.000 384	0.000 091
	1.6	0.637 097 057	0.636 973 423	0.636 945 710	0.002 13	0.000 483	0.000 115
	1.7	0.595 330 359	0.595 193 634	0.595 162 740	0.002 50	0.000 572	0.000 137
	1.8	0.557 658 422	0.557 513 000	0.557 479 947	0.002 83	0.000 650	0.000 156
	1.9	0.523 638 939	0.523 488 343	0.523 453 958	0.003 11	0.000 718	0.000 173
	2.0	0.492 862 999	0.492 709 931	0.492 674 855	0.003 35	0.000 777	0.000 187

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
3.2.12.	1.0	0.000 000 000	0.000 000 000	0.000 000 000	0.000 000 000
	1.1	−0.099 500 000	−0.099 623 114	−0.099 653 809	−0.099 664 000
	1.2	−0.196 990 313	−0.197 235 180	−0.197 295 585	−0.197 315 517
	1.3	−0.290 552 949	−0.290 917 718	−0.291 006 784	−0.291 036 003
	1.4	−0.378 532 718	−0.379 013 852	−0.379 130 237	−0.379 168 221
	1.5	−0.459 672 297	−0.460 262 848	−0.460 404 546	−0.460 450 590
	1.6	−0.533 180 153	−0.533 868 468	−0.534 032 512	−0.534 085 626
	1.7	−0.598 726 853	−0.599 496 413	−0.599 678 824	−0.599 737 720
	1.8	−0.656 384 109	−0.657 214 624	−0.657 410 640	−0.657 473 792
	1.9	−0.706 530 934	−0.707 400 266	−0.707 604 759	−0.707 670 533
	2.0	−0.749 751 364	−0.750 637 632	−0.750 845 571	−0.750 912 371

3.2.14. Improved Euler method:

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
	2.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
	2.1	2.461 242 144	2.463 344 439	2.463 918 368	2.464 119 569
	2.2	3.022 367 633	3.027 507 237	3.028 911 026	3.029 403 212
	2.3	3.705 511 610	3.714 932 709	3.717 507 170	3.718 409 925
	2.4	4.537 659 565	4.553 006 531	4.557 202 414	4.558 673 929
	2.5	5.551 716 960	5.575 150 456	5.581 560 437	5.583 808 754
	2.6	6.787 813 853	6.822 158 665	6.831 558 101	6.834 855 438
	2.7	8.294 896 222	8.343 829 180	8.357 227 947	8.361 928 926
	2.8	10.132 667 135	10.200 955 596	10.219 663 917	10.226 228 709
	2.9	12.373 954 732	12.467 758 807	12.493 470 722	12.502 494 409
	3.0	15.107 600 968	15.234 856 000	15.269 755 072	15.282 004 826

Improved Euler semilinear method:

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
	2.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
	2.1	2.464 261 688	2.464 155 139	2.464 128 464	2.464 119 569
	2.2	3.029 706 047	3.029 479 005	3.029 422 165	3.029 403 212
	2.3	3.718 897 663	3.718 531 995	3.718 440 451	3.718 409 925
	2.4	4.559 377 397	4.558 849 990	4.558 717 956	4.558 673 929
	2.5	5.584 766 724	5.584 048 510	5.583 868 709	5.583 808 754
	2.6	6.836 116 246	6.835 170 986	6.834 934 347	6.834 855 438
	2.7	8.363 552 464	8.362 335 253	8.362 030 535	8.361 928 926
	2.8	10.228 288 880	10.226 744 312	10.226 357 645	10.226 228 709
	2.9	12.505 082 132	12.503 142 042	12.502 656 361	12.502 494 409
	3.0	15.285 231 726	15.282 812 424	15.282 206 780	15.282 004 826

3.2.16. Improved Euler method:

x	$h = 0.2$	$h = 0.1$	$h = 0.05$	"Exact"
1.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
1.2	1.801 514 185	1.801 606 135	1.801 629 115	1.801 636 774
1.4	1.649 911 580	1.650 054 870	1.650 090 680	1.650 102 616
1.6	1.526 711 768	1.526 879 870	1.526 921 882	1.526 935 885
1.8	1.421 841 570	1.422 016 119	1.422 059 743	1.422 074 283
2.0	1.329 441 172	1.329 609 020	1.329 650 971	1.329 664 953
2.2	1.245 953 205	1.246 104 819	1.246 142 713	1.246 155 344
2.4	1.169 162 994	1.169 291 515	1.169 323 639	1.169 334 346
2.6	1.097 677 870	1.097 778 523	1.097 803 683	1.097 812 069
2.8	1.030 626 179	1.030 695 880	1.030 713 305	1.030 719 114
3.0	0.967 473 721	0.967 510 790	0.967 520 062	0.967 523 153

Improved Euler semilinear method:

x	$h = 0.2$	$h = 0.1$	$h = 0.05$	"Exact"
1.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
1.2	1.801 514 185	1.801 606 135	1.801 629 115	1.801 636 774
1.4	1.649 911 580	1.650 054 870	1.650 090 680	1.650 102 616
1.6	1.526 711 768	1.526 879 870	1.526 921 882	1.526 935 885
1.8	1.421 841 570	1.422 016 119	1.422 059 743	1.422 074 283
2.0	1.329 441 172	1.329 609 020	1.329 650 971	1.329 664 953
2.2	1.245 953 205	1.246 104 819	1.246 142 713	1.246 155 344
2.4	1.169 162 994	1.169 291 515	1.169 323 639	1.169 334 346
2.6	1.097 677 870	1.097 778 523	1.097 803 683	1.097 812 069
2.8	1.030 626 179	1.030 695 880	1.030 713 305	1.030 719 114
3.0	0.967 473 721	0.967 510 790	0.967 520 062	0.967 523 153

3.2.18. Improved Euler method:

x	$h = 0.2$	$h = 0.1$	$h = 0.05$	"Exact"
0.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
0.2	1.166 771 705	1.170 394 902	1.171 244 037	1.171 515 153
0.4	1.255 835 116	1.261 642 355	1.262 958 788	1.263 370 891
0.6	1.263 517 157	1.269 528 214	1.270 846 761	1.271 251 278
0.8	1.212 551 997	1.217 531 648	1.218 585 457	1.218 901 287
1.0	1.130 812 573	1.134 420 589	1.135 150 284	1.135 362 070
1.2	1.039 104 333	1.041 487 727	1.041 938 536	1.042 062 625
1.4	0.949 440 052	0.950 888 923	0.951 132 561	0.951 192 532
1.6	0.867 475 787	0.868 263 999	0.868 364 849	0.868 381 328
1.8	0.795 183 973	0.795 523 696	0.795 530 315	0.795 518 627
2.0	0.732 679 223	0.732 721 613	0.732 667 905	0.732 638 628

Improved Euler semilinear method:

x	$h = 0.2$	$h = 0.1$	$h = 0.05$	“Exact”
0.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
0.2	1.170 617 859	1.171 292 452	1.171 459 576	1.171 515 153
0.4	1.261 629 934	1.262 938 347	1.263 262 919	1.263 370 891
0.6	1.269 173 253	1.270 734 290	1.271 122 186	1.271 251 278
0.8	1.216 926 014	1.218 409 355	1.218 778 420	1.218 901 287
1.0	1.133 688 235	1.134 944 960	1.135 257 876	1.135 362 070
1.2	1.040 721 691	1.041 728 386	1.041 979 126	1.042 062 625
1.4	0.950 145 706	0.950 931 597	0.951 127 345	0.951 192 532
1.6	0.867 573 431	0.868 179 975	0.868 331 028	0.868 381 328
1.8	0.794 899 034	0.795 364 245	0.795 480 063	0.795 518 627
2.0	0.732 166 678	0.732 521 078	0.732 609 267	0.732 638 628

3.2.20. Improved Euler method:

x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
0.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
0.1	0.749 165 000	0.747 022 742	0.746 561 141	0.746 418 339
0.2	0.565 942 699	0.562 667 885	0.561 961 242	0.561 742 917
0.3	0.428 618 351	0.424 803 657	0.423 978 964	0.423 724 207
0.4	0.324 556 426	0.320 590 918	0.319 732 571	0.319 467 408
0.5	0.245 417 735	0.241 558 658	0.240 723 019	0.240 464 879
0.6	0.185 235 654	0.181 643 813	0.180 866 303	0.180 626 161
0.7	0.139 546 094	0.136 310 496	0.135 610 749	0.135 394 692
0.8	0.104 938 506	0.102 096 319	0.101 482 503	0.101 293 057
0.9	0.078 787 731	0.076 340 645	0.075 813 072	0.075 650 324
1.0	0.059 071 894	0.056 999 028	0.056 553 023	0.056 415 515

Improved Euler semilinear method:

x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
0.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
0.1	0.745 595 127	0.746 215 164	0.746 368 056	0.746 418 339
0.2	0.560 827 568	0.561 515 647	0.561 686 492	0.561 742 917
0.3	0.422 922 083	0.423 524 585	0.423 674 586	0.423 724 207
0.4	0.318 820 339	0.319 306 259	0.319 427 337	0.319 467 408
0.5	0.239 962 317	0.240 339 716	0.240 433 757	0.240 464 879
0.6	0.180 243 441	0.180 530 866	0.180 602 470	0.180 626 161
0.7	0.135 106 416	0.135 322 934	0.135 376 855	0.135 394 692
0.8	0.101 077 312	0.101 239 368	0.101 279 714	0.101 293 057
0.9	0.075 489 492	0.075 610 310	0.075 640 381	0.075 650 324
1.0	0.056 295 914	0.056 385 765	0.056 408 124	0.056 415 515

3.2.22. Improved Euler method:

x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
2.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
2.1	1.010 250 000	1.009 185 754	1.008 965 733	1.008 899 988
2.2	1.033 547 273	1.032 105 322	1.031 811 002	1.031 723 469
2.3	1.065 562 151	1.064 135 919	1.063 849 094	1.063 764 243
2.4	1.103 145 347	1.101 926 450	1.101 685 553	1.101 614 730
2.5	1.144 085 693	1.143 140 125	1.142 957 158	1.142 903 776
2.6	1.186 878 796	1.186 202 854	1.186 075 600	1.186 038 851
2.7	1.230 530 804	1.230 088 035	1.230 007 943	1.229 985 178
2.8	1.274 404 357	1.274 147 657	1.274 104 430	1.274 092 525
2.9	1.318 104 153	1.317 987 551	1.317 971 490	1.317 967 533
3.0	1.361 395 309	1.361 379 259	1.361 382 239	1.361 383 810

Improved Euler semilinear method:

x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
2.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
2.1	1.012 802 674	1.009 822 081	1.009 124 116	1.008 899 988
2.2	1.038 431 870	1.033 307 426	1.032 108 359	1.031 723 469
2.3	1.072 484 834	1.065 821 457	1.064 263 950	1.063 764 243
2.4	1.111 794 329	1.104 013 534	1.102 197 168	1.101 614 730
2.5	1.154 168 041	1.145 554 968	1.143 547 198	1.142 903 776
2.6	1.198 140 189	1.188 883 373	1.186 728 849	1.186 038 851
2.7	1.242 762 459	1.232 984 559	1.230 712 361	1.229 985 178
2.8	1.287 441 845	1.277 221 941	1.274 850 828	1.274 092 525
2.9	1.331 821 976	1.321 210 992	1.318 753 047	1.317 967 533
3.0	1.375 699 933	1.364 730 937	1.362 193 997	1.361 383 810

3.2.24.

x	$h = 0.1$	$h = 0.05$	$h = 0.025$	Exact
1.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
1.1	1.151 019 287	1.153 270 661	1.153 777 957	1.153 937 085
1.2	1.238 798 618	1.241 884 421	1.242 580 821	1.242 799 540
1.3	1.289 296 258	1.292 573 128	1.293 313 355	1.293 546 032
1.4	1.317 686 801	1.320 866 599	1.321 585 242	1.321 811 247
1.5	1.333 073 855	1.336 036 248	1.336 705 820	1.336 916 440
1.6	1.341 027 170	1.343 732 006	1.344 343 232	1.344 535 503
1.7	1.345 001 345	1.347 446 389	1.347 998 652	1.348 172 348
1.8	1.347 155 352	1.349 355 473	1.349 852 082	1.350 008 229
1.9	1.348 839 325	1.350 816 158	1.351 261 995	1.351 402 121
2.0	1.350 890 736	1.352 667 599	1.353 067 951	1.353 193 719

	x	$h = 0.05$	$h = 0.025$	$h = 0.0125$	Exact
3.2.26.	1.00	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
	1.05	2.268 496 358	2.269 703 943	2.270 043 628	2.270 158 103
	1.10	2.582 897 367	2.585 985 695	2.586 855 275	2.587 150 838
	1.15	2.954 995 034	2.960 992 388	2.962 683 751	2.963 263 785
	1.20	3.400 872 342	3.411 384 294	3.414 355 862	3.415 384 615
	1.25	3.943 047 906	3.960 651 794	3.965 644 965	3.967 391 304
	1.30	4.614 039 436	4.643 018 510	4.651 277 424	4.654 198 473
	1.35	5.462 568 051	5.510 357 362	5.524 069 547	5.528 980 892
	1.40	6.564 985 580	6.645 224 236	6.668 472 955	6.676 923 077
	1.45	8.047 824 947	8.187 384 679	8.228 413 044	8.243 593 315
	1.50	10.136 329 642	10.393 419 681	10.470 731 411	10.500 000 000

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
3.2.28.	0.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
	0.1	0.984 142 840	0.984 133 302	0.984 130 961	0.984 130 189
	0.2	0.965 066 124	0.965 044 455	0.965 039 117	0.965 037 353
	0.3	0.942 648 578	0.942 611 457	0.942 602 279	0.942 599 241
	0.4	0.916 705 578	0.916 648 569	0.916 634 423	0.916 629 732
	0.5	0.886 970 525	0.886 887 464	0.886 866 778	0.886 859 904
	0.6	0.853 066 054	0.852 948 011	0.852 918 497	0.852 908 668
	0.7	0.814 458 249	0.814 291 679	0.814 249 848	0.814 235 883
	0.8	0.770 380 571	0.770 143 777	0.770 083 998	0.770 063 987
	0.9	0.719 699 643	0.719 355 385	0.719 267 905	0.719 238 519
	1.0	0.660 658 411	0.660 136 630	0.660 002 840	0.659 957 689

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
3.2.30.	1.0	0.000 000 000	0.000 000 000	0.000 000 000	0.000 000 000
	1.1	-0.099 666 667	-0.099 665 005	-0.099 664 307	-0.099 674 132
	1.2	-0.197 322 275	-0.197 317 894	-0.197 316 222	-0.197 355 914
	1.3	-0.291 033 227	-0.291 036 361	-0.291 036 258	-0.291 123 993
	1.4	-0.379 131 069	-0.379 160 444	-0.379 166 504	-0.379 315 647
	1.5	-0.460 350 276	-0.460 427 667	-0.460 445 166	-0.460 662 347
	1.6	-0.533 897 316	-0.534 041 581	-0.534 075 026	-0.534 359 685
	1.7	-0.599 446 325	-0.599 668 984	-0.599 721 072	-0.600 066 382
	1.8	-0.657 076 288	-0.657 379 719	-0.657 450 947	-0.657 845 646
	1.9	-0.707 175 010	-0.707 553 135	-0.707 641 993	-0.708 072 516
	2.0	-0.750 335 016	-0.750 775 571	-0.750 879 100	-0.751 331 499

- 3.2.32. (a) Let $x_i = a + ih$, $i = 0, 1, \dots, n$. If y is the solution of the initial value problem $y' = f(x)$, $y(a) = 0$, then $y(b) = \int_a^b f(x) dx$. The improved Euler method yields $y_{i+1} = y_i + .5h(f(a + ih) + f(a + (i + 1)h))$, $i = 0, 1, \dots, n - 1$, where $y_0 = a$ and y_n is

an approximation to $\int_a^b f(x) dx$. But

$$y_n = \sum_{i=0}^{n-1} (y_{i+1} - y_i) = .5h(f(a) + f(b)) + h \sum_{i=1}^{n-1} f(a + ih).$$

- (c) The local truncation error is a multiple of $y'''(\tilde{x}_i) = f''(\tilde{x}_i)$, where $x_i < \tilde{x}_i < x_{i+1}$. Therefore, the quadrature formula is exact if f is a polynomial of degree < 2 .
- (d) Let $E(f) = \int_a^b f(x) dx - y_n$. Note that E is linear. If f is a polynomial of degree 2, then $f(x) = f_0(x) + K(x-a)^2$ where $\deg(f_0) \leq 1$. Since $E(f_0) = 0$ from (c) and

$$\begin{aligned} E((x-a)^2) &= \frac{(b-a)^3}{3} - \frac{(b-a)^2 h}{2} - h^3 \sum_{i=1}^{n-1} i^2 \\ &= h^3 \left[\frac{n^3}{3} - \frac{n^2}{2} - \frac{n(n-1)(2n-1)}{6} \right] = -\frac{nh^3}{6} = -\frac{(b-a)h^2}{6}, \end{aligned}$$

$$E(f) = -\frac{K(b-a)h^2}{6}; \text{ therefore the error is proportional to } h^2.$$

3.3 The Runge-Kutta Method

3.3.2. $y_1 = 1.221551366$; $y_2 = 1.492920208$

3.3.4. $y_1 = 2.961316248$; $y_2 = 2.920128958$.

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	Exact
	0.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
	0.1	2.232 752 507	2.232 649 573	2.232 643 327	2.232 642 918
	0.2	2.774 582 759	2.774 366 625	2.774 353 431	2.774 352 565
	0.3	3.727 068 686	3.726 710 028	3.726 688 030	3.726 686 582
3.3.6.	0.4	5.254 817 388	5.254 263 005	5.254 228 886	5.254 226 636
	0.5	7.613 077 020	7.612 241 222	7.612 189 662	7.612 186 259
	0.6	11.189 806 778	11.188 557 546	11.188 480 365	11.188 475 269
	0.7	16.569 088 310	16.567 225 975	16.567 110 808	16.567 103 199
	0.8	24.626 206 255	24.623 431 201	24.623 259 496	24.623 248 150
	0.9	36.669 848 687	36.665 712 858	36.665 456 874	36.665 439 956
	1.0	54.654 509 699	54.648 344 019	54.647 962 328	54.647 937 102

x	$h = 0.05$	$h = 0.025$	$h = 0.0125$	Exact
1.00	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
1.05	2.270 153 785	2.270 157 806	2.270 158 083	2.270 158 103
1.10	2.587 139 846	2.587 150 083	2.587 150 789	2.587 150 838
1.15	2.963 242 415	2.963 262 317	2.963 263 689	2.963 263 785
3.3.8. 1.20	3.415 346 864	3.415 382 020	3.415 384 445	3.415 384 615
1.25	3.967 327 077	3.967 386 886	3.967 391 015	3.967 391 304
1.30	4.654 089 950	4.654 191 000	4.654 197 983	4.654 198 473
1.35	5.528 794 615	5.528 968 045	5.528 980 049	5.528 980 892
1.40	6.676 590 929	6.676 900 116	6.676 921 569	6.676 923 077
1.45	8.242 960 669	8.243 549 415	8.243 590 428	8.243 593 315
1.50	10.498 658 198	10.499 906 266	10.499 993 820	10.500 000 000

3.3.10.

x	Approximate Solutions			Residuals		
	$h = 0.1$	$h = 0.05$	$h = 0.025$	$h = 0.1$	$h = 0.05$	$h = 0.025$
1.0	1	1	1	0	0	0
1.1	0.923 730 622	0.923 730 677	0.923 730 681	-6.08×10^{-7}	-3.89×10^{-8}	-2.45×10^{-9}
1.2	0.854 443 253	0.854 443 324	0.854 443 328	-8.19×10^{-7}	-5.29×10^{-8}	-3.35×10^{-9}
1.3	0.791 581 155	0.791 581 218	0.791 581 222	-7.53×10^{-7}	-4.95×10^{-8}	-3.16×10^{-9}
1.4	0.734 680 497	0.734 680 538	0.734 680 541	-5.23×10^{-7}	-3.59×10^{-8}	-2.33×10^{-9}
1.5	0.683 288 034	0.683 288 051	0.683 288 052	-2.24×10^{-7}	-1.78×10^{-8}	-1.22×10^{-9}
1.6	0.636 937 046	0.636 937 040	0.636 937 040	7.9×10^{-8}	6×10^{-10}	-9×10^{-11}
1.7	0.595 153 053	0.595 153 029	0.595 153 028	3.51×10^{-7}	1.71×10^{-8}	9.3×10^{-10}
1.8	0.557 469 558	0.557 469 522	0.557 469 520	5.78×10^{-7}	3.09×10^{-8}	1.79×10^{-9}
1.9	0.523 443 129	0.523 443 084	0.523 443 081	7.60×10^{-7}	4.21×10^{-8}	2.48×10^{-9}
2.0	0.492 663 789	0.492 663 738	0.492 663 736	9.02×10^{-7}	5.08×10^{-8}	3.02×10^{-9}

3.3.12.

x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
1.0	0.000 000 000	0.000 000 000	0.000 000 000	0.000 000 000
1.1	-0.099 663 901	-0.099 663 994	-0.099 664 000	-0.099 664 000
1.2	-0.197 315 322	-0.197 315 504	-0.197 315 516	-0.197 315 517
1.3	-0.291 035 700	-0.291 035 983	-0.291 036 001	-0.291 036 003
1.4	-0.379 167 790	-0.379 168 194	-0.379 168 220	-0.379 168 221
1.5	-0.460 450 005	-0.460 450 552	-0.460 450 587	-0.460 450 590
1.6	-0.534 084 875	-0.534 085 579	-0.534 085 623	-0.534 085 626
1.7	-0.599 736 802	-0.599 737 663	-0.599 737 717	-0.599 737 720
1.8	-0.657 472 724	-0.657 473 726	-0.657 473 788	-0.657 473 792
1.9	-0.707 669 346	-0.707 670 460	-0.707 670 529	-0.707 670 533
2.0	-0.750 911 103	-0.750 912 294	-0.750 912 367	-0.750 912 371

		x	$h = 0.1$	$h = 0.05$	$h = 0.025$	"Exact"
3.3.14. Runge-Kutta method:		2.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
		2.1	2.464 113 907	2.464 119 185	2.464 119 544	2.464 119 569
		2.2	3.029 389 360	3.029 402 271	3.029 403 150	3.029 403 212
		2.3	3.718 384 519	3.718 408 199	3.718 409 812	3.718 409 925
		2.4	4.558 632 516	4.558 671 116	4.558 673 746	4.558 673 929
		2.5	5.583 745 479	5.583 804 456	5.583 808 474	5.583 808 754
		2.6	6.834 762 639	6.834 849 135	6.834 855 028	6.834 855 438
		2.7	8.361 796 619	8.361 919 939	8.361 928 340	8.361 928 926
		2.8	10.226 043 942	10.226 216 159	10.226 227 891	10.226 228 709
		2.9	12.502 240 429	12.502 477 158	12.502 493 285	12.502 494 409
		3.0	15.281 660 036	15.281 981 407	15.282 003 300	15.282 004 826

Runge-Kutta semilinear method:

x	$h = 0.1$	$h = 0.05$	$h = 0.025$	"Exact"
2.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
2.1	2.464 119 623	2.464 119 573	2.464 119 570	2.464 119 569
2.2	3.029 403 325	3.029 403 219	3.029 403 212	3.029 403 212
2.3	3.718 410 105	3.718 409 936	3.718 409 925	3.718 409 925
2.4	4.558 674 188	4.558 673 945	4.558 673 930	4.558 673 929
2.5	5.583 809 105	5.583 808 776	5.583 808 755	5.583 808 754
2.6	6.834 855 899	6.834 855 467	6.834 855 440	6.834 855 438
2.7	8.361 929 516	8.361 928 963	8.361 928 928	8.361 928 926
2.8	10.226 229 456	10.226 228 756	10.226 228 712	10.226 228 709
2.9	12.502 495 345	12.502 494 468	12.502 494 413	12.502 494 409
3.0	15.282 005 990	15.282 004 899	15.282 004 831	15.282 004 826

		x	$h = 0.2$	$h = 0.1$	$h = 0.05$	"Exact"
3.3.16. Runge-Kutta method:		1.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
		1.2	1.801 636 785	1.801 636 775	1.801 636 774	1.801 636 774
		1.4	1.650 102 633	1.650 102 617	1.650 102 616	1.650 102 616
		1.6	1.526 935 904	1.526 935 886	1.526 935 885	1.526 935 885
		1.8	1.422 074 302	1.422 074 284	1.422 074 283	1.422 074 283
		2.0	1.329 664 970	1.329 664 954	1.329 664 953	1.329 664 953
		2.2	1.246 155 357	1.246 155 345	1.246 155 344	1.246 155 344
		2.4	1.169 334 355	1.169 334 347	1.169 334 346	1.169 334 346
		2.6	1.097 812 074	1.097 812 070	1.097 812 069	1.097 812 069
		2.8	1.030 719 113	1.030 719 114	1.030 719 114	1.030 719 114
		3.0	0.967 523 147	0.967 523 152	0.967 523 153	0.967 523 153

Runge-Kutta semilinear method:

x	$h = 0.2$	$h = 0.1$	$h = 0.05$	“Exact”
1.0	2.000 000 000	2.000 000 000	2.000 000 000	2.000 000 000
1.2	1.801 636 785	1.801 636 775	1.801 636 774	1.801 636 774
1.4	1.650 102 633	1.650 102 617	1.650 102 616	1.650 102 616
1.6	1.526 935 904	1.526 935 886	1.526 935 885	1.526 935 885
1.8	1.422 074 302	1.422 074 284	1.422 074 283	1.422 074 283
2.0	1.329 664 970	1.329 664 954	1.329 664 953	1.329 664 953
2.2	1.246 155 357	1.246 155 345	1.246 155 344	1.246 155 344
2.4	1.169 334 355	1.169 334 347	1.169 334 346	1.169 334 346
2.6	1.097 812 074	1.097 812 070	1.097 812 069	1.097 812 069
2.8	1.030 719 113	1.030 719 114	1.030 719 114	1.030 719 114
3.0	0.967 523 147	0.967 523 152	0.967 523 153	0.967 523 153

3.3.18. Runge-Kutta method:

x	$h = 0.2$	$h = 0.1$	$h = 0.05$	“Exact”
0.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
0.2	1.171 515 610	1.171 515 156	1.171 515 152	1.171 515 153
0.4	1.263 365 845	1.263 370 556	1.263 370 869	1.263 370 891
0.6	1.271 238 957	1.271 250 529	1.271 251 232	1.271 251 278
0.8	1.218 885 528	1.218 900 353	1.218 901 230	1.218 901 287
1.0	1.135 346 772	1.135 361 174	1.135 362 016	1.135 362 070
1.2	1.042 049 558	1.042 061 864	1.042 062 579	1.042 062 625
1.4	0.951 181 964	0.951 191 920	0.951 192 495	0.951 192 532
1.6	0.868 372 923	0.868 380 842	0.868 381 298	0.868 381 328
1.8	0.795 511 927	0.795 518 241	0.795 518 603	0.795 518 627
2.0	0.732 633 229	0.732 638 318	0.732 638 609	0.732 638 628

Runge-Kutta semilinear method:

x	$h = 0.2$	$h = 0.1$	$h = 0.05$	“Exact”
0.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
0.2	1.171 517 316	1.171 515 284	1.171 515 161	1.171 515 153
0.4	1.263 374 485	1.263 371 110	1.263 370 904	1.263 370 891
0.6	1.271 254 636	1.271 251 485	1.271 251 291	1.271 251 278
0.8	1.218 903 802	1.218 901 442	1.218 901 297	1.218 901 287
1.0	1.135 363 869	1.135 362 181	1.135 362 077	1.135 362 070
1.2	1.042 063 952	1.042 062 706	1.042 062 630	1.042 062 625
1.4	0.951 193 560	0.951 192 595	0.951 192 536	0.951 192 532
1.6	0.868 382 157	0.868 381 378	0.868 381 331	0.868 381 328
1.8	0.795 519 315	0.795 518 669	0.795 518 629	0.795 518 627
2.0	0.732 639 212	0.732 638 663	0.732 638 630	0.732 638 628

		x	$h = 0.1$	$h = 0.05$	$h = 0.025$	"Exact"
3.3.20. Runge-Kutta method:		0.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
		0.1	0.746 430 962	0.746 418 992	0.746 418 376	0.746 418 339
		0.2	0.561 761 987	0.561 743 921	0.561 742 975	0.561 742 917
		0.3	0.423 746 057	0.423 725 371	0.423 724 274	0.423 724 207
		0.4	0.319 489 811	0.319 468 612	0.319 467 478	0.319 467 408
		0.5	0.240 486 460	0.240 466 046	0.240 464 947	0.240 464 879
		0.6	0.180 646 105	0.180 627 244	0.180 626 225	0.180 626 161
		0.7	0.135 412 569	0.135 395 665	0.135 394 749	0.135 394 692
		0.8	0.101 308 709	0.101 293 911	0.101 293 107	0.101 293 057
		0.9	0.075 663 769	0.075 651 059	0.075 650 367	0.075 650 324
		1.0	0.056 426 886	0.056 416 137	0.056 415 552	0.056 415 515

Runge-Kutta semilinear method:

x	$h = 0.1$	$h = 0.05$	$h = 0.025$	"Exact"
0.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
0.1	0.746 416 306	0.746 418 217	0.746 418 332	0.746 418 339
0.2	0.561 740 647	0.561 742 780	0.561 742 908	0.561 742 917
0.3	0.423 722 193	0.423 724 084	0.423 724 199	0.423 724 207
0.4	0.319 465 760	0.319 467 308	0.319 467 402	0.319 467 408
0.5	0.240 463 579	0.240 464 800	0.240 464 874	0.240 464 879
0.6	0.180 625 156	0.180 626 100	0.180 626 158	0.180 626 161
0.7	0.135 393 924	0.135 394 645	0.135 394 689	0.135 394 692
0.8	0.101 292 474	0.101 293 021	0.101 293 055	0.101 293 057
0.9	0.075 649 884	0.075 650 297	0.075 650 322	0.075 650 324
1.0	0.056 415 185	0.056 415 495	0.056 415 514	0.056 415 515

		x	$h = 0.1$	$h = 0.05$	$h = 0.025$	"Exact"
3.3.22. Runge-Kutta method:		2.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
		2.1	1.008 912 398	1.008 900 636	1.008 900 025	1.008 899 988
		2.2	1.031 740 789	1.031 724 368	1.031 723 520	1.031 723 469
		2.3	1.063 781 819	1.063 765 150	1.063 764 295	1.063 764 243
		2.4	1.101 630 085	1.101 615 517	1.101 614 774	1.101 614 730
		2.5	1.142 915 917	1.142 904 393	1.142 903 811	1.142 903 776
		2.6	1.186 047 678	1.186 039 295	1.186 038 876	1.186 038 851
		2.7	1.229 991 054	1.229 985 469	1.229 985 194	1.229 985 178
		2.8	1.274 095 992	1.274 092 692	1.274 092 535	1.274 092 525
		2.9	1.317 969 153	1.317 967 605	1.317 967 537	1.317 967 533
		3.0	1.361 384 082	1.361 383 812	1.361 383 809	1.361 383 810

Runge-Kutta semilinear method:

x	$h = 0.1$	$h = 0.05$	$h = 0.025$	“Exact”
2.0	1.000 000 000	1.000 000 000	1.000 000 000	1.000 000 000
2.1	1.008 913 934	1.008 900 843	1.008 900 041	1.008 899 988
2.2	1.031 748 526	1.031 725 001	1.031 723 564	1.031 723 469
2.3	1.063 798 300	1.063 766 321	1.063 764 371	1.063 764 243
2.4	1.101 656 264	1.101 617 259	1.101 614 886	1.101 614 730
2.5	1.142 951 721	1.142 906 691	1.142 903 955	1.142 903 776
2.6	1.186 092 475	1.186 042 105	1.186 039 051	1.186 038 851
2.7	1.230 043 983	1.229 988 742	1.229 985 397	1.229 985 178
2.8	1.274 156 172	1.274 096 377	1.274 092 762	1.274 092 525
2.9	1.318 035 787	1.317 971 658	1.317 967 787	1.317 967 533
3.0	1.361 456 502	1.361 388 196	1.361 384 079	1.361 383 810

x	$h = 0.1$	$h = 0.05$	$h = 0.025$	Exact
1.00	0.142 854 841	0.142 857 001	0.142 857 134	0.142 857 143
1.10	0.053 340 745	0.053 341 989	0.053 342 066	0.053 342 071
1.20	-0.046 154 629	-0.046 153 895	-0.046 153 849	-0.046 153 846
1.30	-0.153 363 206	-0.153 362 764	-0.153 362 736	-0.153 362 734
1.40	-0.266 397 049	-0.266 396 779	-0.266 396 762	-0.266 396 761
1.50	-0.383 721 107	-0.383 720 941	-0.383 720 931	-0.383 720 930
1.60	-0.504 109 696	-0.504 109 596	-0.504 109 589	-0.504 109 589
1.70	-0.626 598 326	-0.626 598 268	-0.626 598 264	-0.626 598 264
1.80	-0.750 437 351	-0.750 437 320	-0.750 437 318	-0.750 437 318
1.90	-0.875 050 587	-0.875 050 574	-0.875 050 573	-0.875 050 573
2.00	-1.000 000 000	-1.000 000 000	-1.000 000 000	-1.000 000 000

	x	$h = 0.1$	$h = 0.05$	$h = 0.025$	Exact
3.3.26.	0.50	-8.954 103 230	-8.954 063 245	-8.954 060 698	-8.954 060 528
	0.60	-5.059 648 314	-5.059 633 293	-5.059 632 341	-5.059 632 277
	0.70	-2.516 755 942	-2.516 749 850	-2.516 749 465	-2.516 749 439
	0.80	-0.752 508 672	-0.752 506 238	-0.752 506 084	-0.752 506 074
	0.90	0.530 528 482	0.530 529 270	0.530 529 319	0.530 529 323
	1.00	1.500 000 000	1.500 000 000	1.500 000 000	1.500 000 000
	1.10	2.256 519 743	2.256 519 352	2.256 519 328	2.256 519 326
	1.20	2.863 543 039	2.863 542 454	2.863 542 417	2.863 542 415
	1.30	3.362 731 379	3.362 730 700	3.362 730 658	3.362 730 655
	1.40	3.782 361 948	3.782 361 231	3.782 361 186	3.782 361 183
1.50	4.142 171 279	4.142 170 553	4.142 170 508	4.142 170 505	

3.3.28. (a) Let $x_i = a + ih$, $i = 0, 1, \dots, n$. If y is the solution of the initial value problem

$y' = f(x)$, $y(a) = 0$, then $y(b) = \int_a^b f(x) dx$. The Runge-Kutta method yields

$$y_{i+1} = y_i + \frac{h}{6} (f(a + ih) + 4f(a + (2i + 1)h/2) + f(a + (i + 1)h)), \quad i = 0, 1, \dots, n - 1,$$

where $y_0 = a$ and y_n is an approximation to $\int_a^b f(x) dx$. But

$$y_n = \sum_{i=0}^{n-1} (y_{i+1} - y_i) = \frac{h}{6}(f(a) + f(b)) + \frac{h}{3} \sum_{i=1}^{n-1} f(a + ih) + \frac{2h}{3} \sum_{i=1}^n f(a + (2i-1)h/2).$$

- (c) The local truncation error is a multiple of $y^{(5)}(\tilde{x}_i) = f^{(4)}(\tilde{x}_i)$, where $x_i < \tilde{x}_i < x_{i+1}$. Therefore, the quadrature formula is exact if f is a polynomial of degree < 4 .
- (d) Let $E(f) = \int_a^b f(x) dx - y_n$. Note that E is linear. If f is a polynomial of degree 4, then $f(x) = f_0(x) + K(x-a)^4$ where $\deg(f_0) \leq 3$ and K is constant. Since $E(f_0) = 0$ from (c) and

$$\begin{aligned} E((x-a)^4) &= \frac{(b-a)^5}{5} - \frac{(b-a)^4 h}{6} - \frac{h^5}{3} \sum_{i=1}^{n-1} i^4 - \frac{2h^5}{3} \sum_{i=1}^n (i-1/2)^4 \\ &= h^5 \left[\frac{n^5}{5} - \frac{n^4}{6} - \left(\frac{n^5}{15} - \frac{n^4}{6} + \frac{n^3}{9} - \frac{n}{90} \right) - \left(\frac{2n^5}{15} - \frac{n^3}{9} + \frac{7n}{360} \right) \right] \\ &= -\frac{nh^5}{120} = -\frac{(b-a)h^4}{120}, \end{aligned}$$

$$E(f) = -\frac{(b-a)h^4}{120}; \text{ thus, the error is proportional to } h^4.$$

Chapter 4

Applications of First Order Equations

4.1 Growth and Decay

4.1.2. $k\tau = \ln 2$ and $\tau = 2 \Rightarrow k = \frac{\ln 2}{2}$; $Q(t) = Q_0 e^{-t \ln 2 / 2}$; if $Q(T) = \frac{Q_0}{10}$, then $\frac{Q_0}{10} = Q_0 e^{-T \ln 2 / 2}$;
 $\ln 10 = \frac{T \ln 2}{2}$; $T = \frac{2 \ln 10}{\ln 2}$ days.

4.1.4. Let t_1 be the elapsed time since the tree died. Since $p(t) = e^{-(t \ln 2) \tau}$, it follows that
 $p_1 = p_0 e^{-(t_1 \ln 2) / \tau}$, so $\ln \left(\frac{p_1}{p_0} \right) = -\frac{t_1}{\tau} \ln 2$ and $t_1 = \tau \frac{\ln(p_0/p_1)}{\ln 2}$.

4.1.6. $Q = Q_0 e^{-kt}$; $Q_1 = Q_0 e^{-kt_1}$; $Q_2 = Q_0 e^{-kt_2}$; $\frac{Q_2}{Q_1} = e^{-k(t_2 - t_1)}$; $\ln \left(\frac{Q_1}{Q_2} \right) = k(t_2 - t_1)$;
 $k = \frac{1}{t_2 - t_1} \ln \left(\frac{Q_1}{Q_2} \right)$.

4.1.8. $Q' = .06Q$, $Q(0) = Q_0$; $Q = Q_0 e^{.06t}$. We must find τ such that $Q(\tau) = 2Q_0$; that is,
 $Q_0 e^{.06\tau} = 2Q_0$, so $.06\tau = \ln 2$ and $\tau = \frac{\ln 2}{.06} = \frac{50 \ln 2}{3}$ yr.

4.1.10. (a) If T is the time to triple the value, then $Q(T) = Q_0 e^{.05T} = 3Q_0$, so $e^{.05T} = 3$.
Therefore, $.05T = \ln 3$ and $T = 20 \ln 3$.

(b) If $Q(10) = 100000$, then $Q_0 e^{.5} = 100000$, so $Q_0 = 100000 e^{-.5}$

4.1.12. $Q' = -\frac{Q^2}{2}$, $Q(0) = 50$; $\frac{Q'}{Q^2} = -\frac{1}{2}$; $-\frac{1}{Q} = -\frac{t}{2} + c$; $Q(0) = 50 \Rightarrow c = -\frac{1}{50}$;
 $\frac{1}{Q} = \frac{t}{2} + \frac{1}{50} = \frac{1 + 25t}{50}$; $Q = \frac{50}{1 + 25t}$. Now
 $Q(T) = 25 \Rightarrow 1 + 25T = 2 \Rightarrow 25T = 1 \Rightarrow T = \frac{1}{25}$ years.

4.1.14. Since $\tau = 1500$, $k = \frac{\ln 2}{1500}$; hence $Q = Q_0 e^{-(t \ln 2)/1500}$. If $Q(t_1) = \frac{3Q_0}{4}$, then

$$e^{-(t_1 \ln 2)/1500} = \frac{3}{4}; \quad -t_1 \frac{\ln 2}{1500} = \ln \left(\frac{3}{4} \right) = -\ln \left(\frac{4}{3} \right); \quad t_1 = 1500 \frac{\ln \left(\frac{4}{3} \right)}{\ln 2}. \text{ Finally,}$$

$$Q(2000) = Q_0 e^{-\frac{4}{3} \ln 2} = 2^{-4/3} Q_0.$$

4.1.16. (A) $S' = 1 - \frac{S}{10}$, $S(0) = 20$. Rewrite the differential equation in (A) as (B) $S' + \frac{S}{10} = 1$. Since $S_1 = e^{-t/10}$ is a solution of the complementary equation, the solutions of (B) are given by $S = u e^{-t/10}$, where $u' e^{-t/10} = 1$. Therefore, $u' = e^{t/10}$; $u = 10 e^{t/10} + c$; $S = 10 + c e^{-t/10}$. Now $S(0) = 20 \Rightarrow c = 10$, so $S = 10 + 10 e^{-t/10}$ and $\lim_{t \rightarrow \infty} S(t) = 10$ g.

4.1.18. (A) $V' = -750 + \frac{V}{20}$, $V(0) = 25000$. Rewrite the differential equation in (A) as (B) $V' - \frac{V}{20} = -750$. Since $V_1 = e^{t/20}$ is a solution of the complementary equation, the solutions of (B) are given by $V = u e^{t/20}$, where $u' e^{t/20} = -750$. Therefore, $u' = -750 e^{-t/20}$; $u = 15000 e^{-t/20} + c$; $V = 15000 + c e^{t/20}$; $V(0) = 25000 \Rightarrow c = 10000$. Therefore, $V = 15000 + 10000 e^{t/20}$.

4.1.20. $p' = \frac{p}{2} - \frac{p^2}{8} = -\frac{1}{8} p(p-4)$; $\frac{p'}{p(p-4)} = -\frac{1}{8}$; $\frac{1}{4} \left[\frac{1}{p-4} - \frac{1}{p} \right] p' = -\frac{1}{8}$;
 $\left[\frac{1}{p-4} - \frac{1}{p} \right] p' = -\frac{1}{2}$; $\left| \frac{p-4}{p} \right| = -\frac{t}{2} + k$; $\frac{p-4}{p} = c e^{-t/2}$; $p(0) = 100 \Rightarrow c = \frac{24}{25}$;
 $\frac{p-4}{p} = \frac{24}{25} e^{-t/2}$; $p-4 = \frac{24}{25} p e^{-t/2}$; $p \left(1 - \frac{24}{25} e^{-t/2} \right) = 4$; $p = \frac{4}{1 - \frac{24}{25} e^{-t/2}} = \frac{100}{24 - 24 e^{-t/2}}$.

4.1.22. (a) $P' = rP - 12M$.

(b) $P = u e^{rt}$; $u' e^{rt} = -12M$; $u' = -12M e^{-rt}$; $u = \frac{12M}{r} e^{-rt} + c$; $P = \frac{12M}{r} + c e^{rt}$;
 $P(0) = P_0 \Rightarrow c = P_0 - \frac{12M}{r}$; $P = \frac{12M}{r} (1 - e^{rt}) + P_0 e^{rt}$.

(c) Since $P(N) = 0$, the answer to (b) implies that $M = \frac{r P_0}{12(1 - e^{-rN})}$.

4.1.24. The researcher's salary is the solution of the initial value problem $S' = aS$, $S(0) = S_0$. Therefore, $S = S_0 e^{at}$. If $P = P(t)$ is the value of the trust fund, then $P' = -S_0 e^{at} + rP$, or $P' - rP = -S_0 e^{at}$. Therefore, (A) $P = u e^{rt}$, where $u' e^{rt} = -S_0 e^{at}$, so (B) $u' = -S_0 e^{(a-r)t}$. If $a \neq r$, then (B) implies that $u = \frac{S_0}{r-a} e^{(a-r)t} + c$, so (A) implies that $P = \frac{S_0}{r-a} e^{at} + c e^{rt}$. Now $P(0) = P_0 \Rightarrow c = P_0 - \frac{S_0}{r-a}$; therefore $P = \frac{S_0}{r-a} e^{at} + \left(P_0 - \frac{S_0}{r-a} \right) e^{rt}$. We must choose P_0 so that $P(T) = 0$; that is, $P = \frac{S_0}{r-a} e^{aT} + \left(P_0 - \frac{S_0}{r-a} \right) e^{rT} = 0$. Solving this for P_0 yields $P_0 = \frac{S_0(1 - e^{(a-r)T})}{r-a}$. If $a = r$, then (B) becomes $u' = -S_0$, so $u = -S_0 t + c$ and (A) implies that $P = (-S_0 t + c) e^{rt}$. Now $P(0) = P_0 \Rightarrow c = P_0$; therefore $P = (-S_0 t + P_0) e^{rt}$. To make $P(T) = 0$ we must take $P_0 = S_0 T$.

4.1.26. $Q' = \frac{at}{1+btQ^2} - kQ$; $\lim_{t \rightarrow \infty} Q(t) = (a/bk)^{1/3}$.

4.2 Cooling and Mixing

4.2.2. Since $T_0 = 100$ and $T_M = -10$, $T = -10 + 110e^{-kt}$. Now $T(1) = 80 \Rightarrow 80 = -10 + 110e^{-k}$, so $e^{-k} = \frac{9}{11}$ and $k = \ln \frac{11}{9}$. Therefore, $T = -10 + 110e^{-t \ln \frac{11}{9}}$.

4.2.4. Let T be the thermometer reading. Since $T_0 = 212$ and $T_M = 70$, $T = 70 + 142e^{-kt}$. Now $T(2) = 125 \Rightarrow 125 = 70 + 142e^{-2k}$, so $e^{-2k} = \frac{55}{142}$ and $k = \frac{1}{2} \ln \frac{142}{55}$. Therefore, (A)
 $T = 70 + 142e^{-\frac{t}{2} \ln \frac{142}{55}}$.

(a) $T(2) = 70 + 142e^{-2 \ln \frac{142}{55}} = 70 + 142 \left(\frac{55}{142} \right)^2 \approx 91.30^\circ\text{F}$.

(b) Let τ be the time when $T(\tau) = 72$, so $72 = 70 + 142e^{-\frac{\tau}{2} \ln \frac{142}{55}}$, or $e^{-\frac{\tau}{2} \ln \frac{142}{55}} = \frac{1}{71}$.
 Therefore, $\tau = 2 \frac{\ln 71}{\ln \frac{142}{55}} \approx 8.99$ min.

(c) Since (A) implies that $T > 70$ for all $t > 0$, the thermometer will never read 69°F .

4.2.6. Since $T_M = 20$, $T = 20 + (T_0 - 20)e^{-kt}$. Now $T_0 - 5 = 20 + (T_0 - 20)e^{-4k}$ and $T_0 - 7 = 20 + (T_0 - 20)e^{-8k}$. Therefore, $\frac{T_0 - 25}{T_0 - 20} = e^{-4k}$ and $\left(\frac{T_0 - 27}{T_0 - 20} \right) = e^{-8k}$, so $\frac{T_0 - 27}{T_0 - 20} = \left(\frac{T_0 - 25}{T_0 - 20} \right)^2$, which implies that $(T_0 - 20)(T_0 - 27) = (T_0 - 25)^2$, or $T_0^2 - 47T_0 + 540 = T_0^2 - 50T_0 + 625$; hence $3T_0 = 85$ and $T_0 = (85/3)^\circ\text{C}$.

4.2.8. $Q' = 3 - \frac{3}{40}Q$, $Q(0) = 0$. Rewrite the differential equation as (A) $Q' + \frac{3}{40}Q = 3$. Since $Q_1 = e^{-3t/40}$ is a solution of the complementary equation, the solutions of (A) are given by $Q = ue^{-3t/40}$ where $u'e^{-3t/40} = 3$. Therefore, $u' = 3e^{3t/40}$, $u = 40e^{3t/40} + c$, and $Q = 40 + ce^{-3t/40}$. Now $Q(0) = 0 \Rightarrow c = -40$, so $Q = 40(1 - e^{-3t/40})$.

4.2.10. $Q' = \frac{3}{2} - \frac{Q}{20}$, $Q(0) = 10$. Rewrite the differential equation as (A) $Q' + \frac{Q}{20} = \frac{3}{2}$. Since $Q_1 = e^{-t/20}$ is a solution of the complementary equation, the solutions of (A) are given by $Q = ue^{-t/20}$ where $u'e^{-t/20} = \frac{3}{2}$. Therefore, $u' = \frac{3}{2}e^{t/20}$, $u = 30e^{t/20} + c$, and $Q = 30 + ce^{-t/20}$. Now $Q(0) = 10 \Rightarrow c = -20$, so $Q = 30 - 20e^{-t/20}$ and $K = \frac{Q}{100} = .3 - .2e^{-t/20}$.

4.2.12. $Q' = 10 - \frac{Q}{5}$, or (A) $Q' + \frac{Q}{5} = 10$. Since $Q_1 = e^{-t/5}$ is a solution of the complementary equation, the solutions of (A) are given by $Q = ue^{-t/5}$ where $u'e^{-t/5} = 10$. Therefore, $u' = 10e^{t/5}$, $u = 50e^{t/5} + c$, and $Q = 50 + ce^{-t/5}$. Since $\lim_{t \rightarrow \infty} Q(t) = 50$, the minimum capacity is 50 gallons.

4.2.14. Since there are $2t + 600$ gallons of mixture in the tank at time t and mixture is being drained at 4 gallons/min, $Q' = 3 - \frac{2}{t+300}Q$, $Q(0) = 40$. Rewrite the differential equation as (A) $Q' + \frac{2}{t+300}Q = 3$. Since $Q_1 = \frac{1}{(t+300)^2}$ is a solution of the complementary equation, the solutions of (A) are given by $Q = \frac{u}{(t+300)^2}$ where $\frac{u'}{(t+300)^2} = 3$. Therefore, $u' = 3(t+300)^2$, $u = (t+300)^3 + c$, and $Q = t + 300 + \frac{c}{(t+300)^2}$. Now $Q(0) = 40 \Rightarrow c = -234 \times 10^5$, so $Q = t + 300 - \frac{234 \times 10^5}{(t+300)^2}$, $0 \leq t \leq 300$.

4.2.16. (a) $S' = -k_m(S - T_m)$, $S(0) = 0$, so (A) $S = T_m + (S_0 - T_m)e^{-k_mt}$.
 $T' = -k(T - S) = -k(T - T_m - (S_0 - T_m)e^{-k_mt})$, from (A).
 Therefore, $T' + kT = kT_m + k(S_0 - T_m)e^{-k_mt}$; $T = ue^{-kt}$; (B)
 $u' = kT_me^{kt} + k(S_0 - T_m)e^{(k-k_m)t}$; $u = T_me^{kt} + \frac{k}{k-k_m}(S_0 - T_m)e^{(k-k_m)t} + c$;
 $T(0) = T_0 \Rightarrow c = T_0 - T_m - \frac{k}{k-k_m}(S_0 - T_m)$;
 $u = T_me^{kt} + \frac{k}{k-k_m}(S_0 - T_m)e^{(k-k_m)t} + T_0 - T_m - \frac{k}{k-k_m}(S_0 - T_m)$;
 $T = T_m + (T_0 - T_m)e^{-kt} + \frac{k(S_0 - T_m)}{(k-k_m)}(e^{-k_mt} - e^{-kt})$.
 (b) If $k = k_m$ (B) becomes (B) $u' = kT_me^{kt} + k(S_0 - T_m)$; $u = T_me^{kt} + k(S_0 - T_m)t + c$;
 $T(0) = T_0 \Rightarrow c = T_0 - T_m$; $u = T_me^{kt} + k(S_0 - T_m)t + (T_0 - T_m)$;
 $T = T_m + k(S_0 - T_m)te^{-kt} + (T_0 - T_m)e^{-kt}$.
 (c) $\lim_{t \rightarrow \infty} T(t) = \lim_{t \rightarrow \infty} S(t) = T_m$ in either case.

4.2.18. $V' = aV - bV^2 = -bV(V - a/b)$; $\frac{V'}{V(V - a/b)} = -b$; $\left[\frac{1}{V - a/b} - \frac{1}{V} \right] V' = -a$;
 $\ln \left| \frac{V - a/b}{V} \right| = -at + k$; (A) $\frac{V - a/b}{V} = ce^{-at}$; (B) $V = \frac{a}{b} \frac{1}{1 - ce^{-at}}$. Since $V(0) = V_0$, (A)
 $\Rightarrow c = \frac{V_0 - a/b}{V_0}$. Substituting this into (B) yields $V = \frac{a}{b} \frac{V_0}{V_0 - (V_0 - a/b)e^{-at}}$ so
 $\lim_{t \rightarrow \infty} V(t) = a/b$

4.2.20. If $Q_n(t)$ is the number of pounds of salt in T_n at time t , then
 $Q'_{n+1} + \frac{r}{W}Q_{n+1} = rc_n(t)$, $n = 0, 1, \dots$, where $c_0(t) \equiv c$. Therefore, $Q_{n+1} = u_{n+1}e^{-rt/W}$; (A)
 $u'_{n+1} = re^{rt/W}c_n(t)$. In particular, with $n = 0$, $u_1 = cW(e^{rt/W} - 1)$, so $Q_1 = cW(1 - e^{-rt/W})$
 and $c_1 = c(1 - e^{-rt/W})$. We will shown by induction that $c_n = c \left(1 - e^{-rt/W} \sum_{j=0}^{n-1} \frac{1}{j!} \left(\frac{rt}{W} \right)^j \right)$.
 This is true for $n = 1$; if it is true for a given n , then, from (A),

$$u'_{n+1} = cre^{rt/W} \left(1 - e^{-rt/W} \sum_{j=0}^{n-1} \frac{1}{j!} \left(\frac{rt}{W} \right)^j \right) = cre^{rt/W} - cr \sum_{j=0}^{n-1} \frac{1}{j!} \left(\frac{rt}{W} \right)^j,$$

so (since $Q_{n+1}(0) = 0$),

$$u_{n+1} = cW(e^{rt/W} - 1) - c \sum_{j=0}^{n-1} \frac{1}{(j+1)!} \frac{r^{j+1}}{W^j} t^{j+1}.$$

Therefore,

$$c_{n+1} = \frac{1}{W} u_{n+1} e^{-rt/W} = c \left(1 - e^{-rt/W} \sum_{j=0}^n \frac{1}{j!} \left(\frac{rt}{W} \right)^j \right),$$

which completes the induction. From this, $\lim_{t \rightarrow \infty} c_n(t) = c$.

4.2.22. Since the incoming solution contains 1/2 lb of salt per gallon and there are always 600 gallons in the tank, we conclude intuitively that $\lim_{t \rightarrow \infty} Q(t) = 300$. To verify this

rigorously, note that $Q_1(t) = \exp \left(-\frac{1}{150} \int_0^t a(\tau) d\tau \right)$ is a solution of the complementary equation, (A) $Q_1(0) = 1$, and (B) $\lim_{t \rightarrow \infty} Q_1(t) = 0$ (since $\lim_{t \rightarrow \infty} a(t) = 1$).

Therefore, $Q = Q_1 u$; $Q_1 u' = 2$; $u' = \frac{2}{Q_1}$; $u = Q_0 + 2 \int_0^t \frac{d\tau}{Q_1(\tau)}$ (see (A)), and

$Q(t) = Q_0 Q_1(t) + 2 Q_1(t) \int_0^t \frac{d\tau}{Q_1(\tau)}$. From (B), $\lim_{t \rightarrow \infty} Q(t) = 2 \lim_{t \rightarrow \infty} Q_1(t) \int_0^t \frac{d\tau}{Q_1(\tau)}$, a $0 \cdot \infty$ indeterminate form. By L'Hôpital's rule,

$$\lim_{t \rightarrow \infty} Q(t) = 2 \lim_{t \rightarrow \infty} \frac{1}{Q_1(t)} \bigg/ \left(\frac{-Q_1'(t)}{Q_1^2(t)} \right) = -2 \lim_{t \rightarrow \infty} \frac{Q_1(t)}{Q_1'(t)} = 300.$$

4.3 Elementary Mechanics

4.3.2. The firefighter's mass is $m = \frac{192}{32} = 6$ sl, so $6v' = -192 - kv$, or (A) $v' + \frac{k}{6}v = -32$. Since $v_1 = e^{-kt/6}$ is a solution of the complementary equation, the solutions of (A) are $v = ue^{-kt/6}$ where $u'e^{-kt/6} = -32$. Therefore, $u' = -32e^{kt/6}$; $u = -\frac{192}{k}e^{kt/6} + c$; $v = -\frac{192}{k} + ce^{-kt/6}$.

Now $v(0) = 0 \Rightarrow c = \frac{192}{k}$. Therefore, $v = -\frac{192}{k}(1 - e^{-kt/6})$ and $\lim_{t \rightarrow \infty} v(t) = -\frac{192}{k} = -16$ ft/s, so $k = 12$ lb-s/ft and $v = -16(1 - e^{-2t})$.

4.3.3. $m = \frac{64000}{32} = 2000$, so $2000v' = 50000 - 2000v$, or (A) $v' + v = 25$. Since $v_1 = e^{-t}$ is a solution of the complementary equation, the solutions of (A) are $v = ue^{-t}$ where $u'e^{-t} = 25$. Therefore, $u' = 25e^t$; $u = 25e^t + c$; $v = 25 + ce^{-t}$. Now $v(0) = 0 \Rightarrow c = -25$. Therefore, $v = 25(1 - e^{-t})$ and $\lim_{t \rightarrow \infty} v(t) = 25$ ft/s.

4.3.4. $20v' = 10 - \frac{1}{2}v$, or (A) $v' + \frac{1}{20}v = \frac{1}{2}$. Since $v_1 = e^{-t/40}$ is a solution of the complementary equation, the solutions of (A) are $v = ue^{-t/40}$ where $u'e^{-t/40} = \frac{1}{2}$. Therefore, $u' = \frac{e^{t/40}}{2}$; $u = 20e^{t/40} + c$; $v = 20 + ce^{-t/40}$. Now $v(0) = -7 \Rightarrow c = -27$. Therefore, $v = 20 - 27e^{-t/40}$.

4.3.6. $m = \frac{3200}{32} = 100$ sl. The component of the gravitational force in the direction of motion is $-3200 \cos(\pi/3) = -1600$ lb. Therefore, $100v' = -1600 + v^2$. Separating variables yields $\frac{v'}{(v-40)(v+40)} = \frac{1}{100}$, or $\left[\frac{1}{v-40} - \frac{1}{v+40} \right] = \frac{4}{5}$. Therefore, $\ln \left| \frac{v-40}{v+40} \right| = \frac{4t}{5} + k$ and $\frac{v-40}{v+40} = ce^{4t/5}$. Now $v(0) = -64 \Rightarrow c = \frac{13}{3}$; therefore $\frac{v-40}{v+40} = \frac{13e^{4t/5}}{3}$, so $v = \frac{40(3 + 13e^{4t/5})}{3 - 13e^{4t/5}}$, or $v = -\frac{40(13 + 3e^{-4t/5})}{13 - 3e^{-4t/5}}$.

4.3.8. From Example 4.3.1, (A) $v = -\frac{mg}{k} + \left(v_0 + \frac{mg}{k}\right)e^{-kt/m}$. Integrating this yields (B) $y = -\frac{mgt}{k} - \frac{m}{k} \left(v_0 + \frac{mg}{k}\right)e^{-kt/m} + c$. Now $y(0) = y_0 \Rightarrow c = y_0 + \frac{m}{k} \left(v_0 + \frac{mg}{k}\right)$. Substituting this into (B) yields

$$\begin{aligned} y &= -\frac{mgt}{k} - \frac{m}{k} \left(v_0 + \frac{mg}{k}\right)e^{-kt/m} + y_0 + \frac{m}{k} \left(v_0 + \frac{mg}{k}\right) \\ &= y_0 + \frac{m}{k} \left(v_0 - gt + \frac{mg}{k} - \left(v_0 + \frac{mg}{k}\right)e^{-kt/m}\right) \\ &= y_0 + \frac{m}{k}(v_0 - v - gt) \end{aligned}$$

where the last equality follows from (A).

4.3.10. $m = \frac{256}{32} = 8$ sl. Since the resisting force is 1 lb when $|v| = 4$ ft/s, $k = \frac{1}{16}$. Therefore, $8v' = -256 + \frac{1}{16}v^2 = \frac{1}{16}(v^2 - (64)^2)$. Separating variables yields $\frac{v'}{(v-64)(v+64)} = \frac{1}{128}$, or $\left[\frac{1}{v-64} - \frac{1}{v+64} \right]v' = 1$. Therefore, $\ln \left| \frac{v-64}{v+64} \right| = t + k$ and $\frac{v-64}{v+64} = ce^t$. Now $v(0) = 0 \Rightarrow c = -1$; therefore $\frac{v-64}{v+64} = -e^t$, so $v = \frac{64(1-e^t)}{1+e^t}$, or $v = -\frac{64(1-e^{-t})}{1+e^{-t}}$. Therefore, $\lim_{t \rightarrow \infty} v(t) = -64$.

4.3.12. (a) $mv' = -mg - kv^2 = -mg(1 + \gamma^2 v^2)$, where $\gamma = \sqrt{\frac{k}{mg}}$. Therefore, (A) $\frac{v'}{1 + \gamma^2 v^2} = -g$.

With the substitution $u = \gamma v$, $\int \frac{dv}{1 + \gamma^2 v^2} = \frac{1}{\gamma} \int \frac{du}{1 + u^2} = \frac{1}{\gamma} \tan^{-1} u = \frac{1}{\gamma} \tan^{-1}(\gamma v)$.

Therefore, $\frac{1}{\gamma} \tan^{-1}(\gamma v) = -gt + c$. Now $v(0) = v_0 \Rightarrow c = \frac{1}{\gamma} \tan^{-1}(\gamma v_0)$, so

$\frac{1}{\gamma} \tan^{-1}(\gamma v) = -gt + \frac{1}{\gamma} \tan^{-1}(\gamma v_0)$. Since $v(T) = 0$, it follows that

$$T = \frac{1}{\gamma g} \tan^{-1} \gamma v_0 = \sqrt{\frac{m}{kg}} \tan^{-1} \left(v_0 \sqrt{\frac{k}{mg}} \right).$$

(b) Replacing t by $t - T$ and setting $v_0 = 0$ in the answer to the previous exercise yields

$$v = -\sqrt{\frac{mg}{k}} \frac{1 - e^{-2\sqrt{\frac{gk}{m}}(t-T)}}{1 + e^{-2\sqrt{\frac{gk}{m}}(t-T)}}.$$

- 4.3.14. (a) $mv' = -mg + f(|v|)$; since $s = |v| = -v$, (A) $ms' = mg - f(s)$.
- (b) Since f is increasing and $\lim_{t \rightarrow \infty} f(s) \leq mg$, $mg - f(s) > 0$ for all s . This and (A) imply that s is an increasing function of t , so either (B) $\lim_{t \rightarrow \infty} s(t) = \infty$ or (C) $\lim_{t \rightarrow \infty} s(t) = \bar{s} < \infty$. However, (A) and (C) imply that $s'(t) > K = g - f(\bar{s})/m$ for all $t > 0$. Consequently, $s(t) > s_0 + Kt$ for all $t > 0$, which contradicts (C) because $K > 0$.
- (c) There is a unique positive number s_T such that $f(s_T) = mg$, and $s \equiv s_T$ is a constant solution of (A). Now suppose that $s(0) < s_T$. Then Theorem 2.3.1 implies that (D) $s(t) < s_T$ for all $t > 0$, so (A) implies that s is strictly increasing. This and (D) imply that $\lim_{t \rightarrow \infty} s(t) = \bar{s} \leq s_T$. If $\bar{s} < s_T$ then (A) implies that $s'(t) > K = g - f(\bar{s})/m$. Consequently, $s(t) > s(0) + Kt$, which contradicts (D) because $K > 0$. Therefore, $s(0) < s_T \Rightarrow \lim_{t \rightarrow \infty} s(t) = s_T$. A similar proof with inequalities reversed shows that $s(0) > s_T \Rightarrow \lim_{t \rightarrow \infty} s(t) = s_T$.
- 4.3.16. (a) (A) $mv' = -mg + k\sqrt{|v|}$; since the magnitude of the resistance is 64 lb when $v = 16$ ft/s, $4k = 64$, so $k = 16 \text{ lb s}^{0.5}/\text{ft}^{0.5}$. Since $m = 2$ and $g = 32$, (A) becomes $2v' = -64 + 16\sqrt{|v|}$, or $v' = -32 + 8\sqrt{|v|}$.
- (b) From Exercise 4.3.14(c), v_T is the negative number such that $-32 + 8\sqrt{|v_T|} = 0$; thus, $v_T = -16$ ft/s.
- 4.3.18. With $h = 0$, $v_e = \sqrt{2gR}$, where R is the radius of the moon and g is the acceleration due to gravity at the moon's surface. With length in miles, $g = \frac{5.31}{5280} \text{ mi/s}^2$, so
- $$v_e = \sqrt{\frac{2 \cdot 5.31 \cdot 1080}{5280}} \approx 1.47 \text{ mi/s}.$$
- 4.3.20. Suppose that there is a number y_m such that $y(t) \leq y_m$ for all $t \geq 0$ and let
- $$\alpha = \frac{gR^2}{(y_m + R)^2}.$$
- Then $\frac{d^2y}{dt^2} \leq -\alpha$ for all $t \geq 0$. Integrating this inequality from $t = 0$ to $t = T > 0$ yields $v(T) - v_0 \leq -\alpha T$, or $v(T) \leq v_0 - \alpha T$, so $v(T) < 0$ for $T > \frac{v_0}{\alpha}$. This implies that the vehicle must eventually fall back to Earth, which contradicts the assumption that it continues to climb forever.

4.4 Autonomous Second Order Equations

- 4.4.1. $\bar{y} = 0$ is a stable equilibrium. The phase plane equivalent is $v \frac{dv}{dy} + y^3 = 0$, so the trajectories are $v^2 + \frac{y^4}{4} = c$.
- 4.4.2. $\bar{y} = 0$ is an unstable equilibrium. The phase plane equivalent is $v \frac{dv}{dy} + y^2 = 0$, so the trajectories are $v^2 + \frac{2y^3}{3} = c$.
- 4.4.4. $\bar{y} = 0$ is a stable equilibrium. The phase plane equivalent is $v \frac{dv}{dy} + ye^{-y} = 0$, so the trajectories are $v^2 - e^{-y}(y + 1) = c$.

4.4.6. $p(y) = y^3 - 4y = (y+2)y(y-2)$, so the equilibria are $-2, 0, 2$. Since

$$\begin{aligned} y(y-2)(y+2) &< 0 && \text{if } y < -2 \text{ or } 0 < y < 2, \\ &> 0 && \text{if } -2 < y < 0 \text{ or } y > 2, \end{aligned}$$

0 is unstable and $-2, 2$ are stable. The phase plane equivalent is $v \frac{dv}{dy} + y^3 - 4y = 0$, so the trajectories are $2v^2 + y^4 - 8y^2 = c$. Setting $(y, v) = (0, 0)$ yields $c = 0$, so the equation of the separatrix is $2v^2 - y^4 + 8y^2 = 0$.

4.4.8. $p(y) = y(y-2)(y-1)(y+2)$, so the equilibria are $-2, 0, 1, 2$. Since

$$\begin{aligned} y(y-2)(y-1)(y+2) &> 0 && \text{if } y < -2 \text{ or } 0 < y < 1 \text{ or } y > 2, \\ &< 0 && \text{if } -2 < y < 0 \text{ or } 1 < y < 2, \end{aligned}$$

0, 2 are stable and $-2, 1$ are unstable. The phase plane equivalent is

$v \frac{dv}{dy} + y(y-2)(y-1)(y+2) = 0$, so the trajectories are

$30v^2 + y^2(12y^3 - 15y^2 - 80y + 120) = c$. Setting $(y, v) = (-2, 0)$ and $(y, v) = (1, 0)$ yields $c = 496$ and $c = 37$ respectively, so the equations of the separatrices are $30v^2 + y^2(12y^3 - 15y^2 - 80y + 120) = 496$ and $30v^2 + y^2(12y^3 - 15y^2 - 80y + 120) = 37$.

4.4.10. $p(y) = y^3 - ay$. If $a \leq 0$, then $p(0) = 0$, $p(y) > 0$ if $y > 0$, and $p(y) < 0$ if $y < 0$, so 0 is stable. If $a > 0$, then

$$\begin{aligned} y^3 - ay = y(y - \sqrt{a})(y + \sqrt{a}) &> 0 && \text{if } -\sqrt{a} < y < 0 \text{ or } y > \sqrt{a}, \\ &< 0 && \text{if } y < -\sqrt{a} \text{ or } 0 < y < \sqrt{a}, \end{aligned}$$

so $-\sqrt{a}$ and $\sqrt{a}$ are stable and 0 is unstable. We say that $a = 0$ is a critical value because it separates the two cases.

4.4.12. $p(y) = y - ay^3$. If $a \leq 0$, then $p(0) = 0$, $p(y) > 0$ if $y > 0$, and $p(y) < 0$ if $y < 0$, so 0 is stable. If $a > 0$, then

$$\begin{aligned} y - ay^3 = -ay(y - 1/\sqrt{a})(y + 1/\sqrt{a}) &> 0 && \text{on } (-\infty, -1/\sqrt{a}) \cup (0, 1/\sqrt{a}) \\ &< 0 && \text{on } (-1/\sqrt{a}, 0) \cup (1/\sqrt{a}, \infty) \end{aligned}$$

so $-\sqrt{a}$ and $\sqrt{a}$ are unstable and 0 is stable. We say that $a = 0$ is a critical value because it separates the two cases.

4.4.24. (a) Since $v' = -p(y) \geq k$ and $v(0) = 0$, $v \geq kt$ and therefore $y \geq y_0 + kt^2/2$ for $0 \leq t < T$.

(b) Let $0 < \epsilon < \rho$. Suppose that y is the solution of the initial value problem (A) $y'' + p(y) = 0$, $y(0) = y_0$, $y'(0) = 0$, where $\bar{y} < y_0 < \bar{y} + \epsilon$. Now let $Y = y - \bar{y}$ and $P(Y) = p(Y + \bar{y})$. Then $P(0) = 0$ and $P(Y) < 0$ if $0 < Y \leq \rho$. Moreover, Y is the solution of $Y'' + P(Y) = 0$, $Y(0) = Y_0$, $Y'(0) = 0$, where $Y_0 = y_0 - \bar{y}$, so $0 < Y_0 < \epsilon$. From (a), $Y(t) \geq \epsilon$ for some $t > 0$. Therefore, $y(t) > \bar{y} + \epsilon$ for some $t > 0$, so $\bar{y}$ is an unstable equilibrium of $y'' + p(y) = 0$.

4.5 Applications to Curves

4.5.2. Differentiating (A) $e^{xy} = cy$ yields (B) $(xy' + y)e^{xy} = cy'$. From (A), $c = \frac{e^{xy}}{y}$. Substituting

$$\text{this into (B) and cancelling } e^{xy} \text{ yields } xy' + y = \frac{y'}{y}, \text{ so } y' = -\frac{y^2}{(xy - 1)}.$$

4.5.4. Solving $y = x^{1/2} + cx$ for c yields $c = \frac{y}{x} - x^{-1/2}$, and differentiating yields

$$0 = \frac{y'}{x} - \frac{y}{x^2} + \frac{x^{-3/2}}{2}, \text{ or } xy' - y = -\frac{x^{1/2}}{2}.$$

4.5.6. Rewriting $y = x^3 + \frac{c}{x}$ as $xy = x^4 + c$ and differentiating yields $xy' + y = 4x^3$.

4.5.8. Rewriting $y = e^x + c(1 + x^2)$ as $\frac{y}{1 + x^2} = \frac{e^x}{1 + x^2} + c$ and differentiating yields

$$\frac{y'}{1 + x^2} - \frac{2xy}{(1 + x^2)^2} = \frac{e^x}{1 + x^2} - \frac{2xe^x}{(1 + x^2)^2}, \text{ so } (1 + x^2)y' - 2xy = (1 - x)^2e^x.$$

4.5.10. If (A) $y = f + cg$, then (B) $y' = f' + cg'$. Multiplying (A) by g' and (B) by g yields (C) $yy' = fg' + cgg'$ and (D) $y'g = f'g + cg'g$, and subtracting (C) from (D) yields $y'g - yg' = f'g - fg'$.

4.5.12. Let (x_0, y_0) be the center and r be the radius of a circle in the family. Since $(-1, 0)$ and $(1, 0)$ are on the circle, $(x_0 + 1)^2 + y_0^2 = (x_0 - 1)^2 + y_0^2$, which implies that $x_0 = 0$. Therefore, the equation of the circle is (A) $x^2 + (y - y_0)^2 = r^2$. Since $(1, 0)$ is on the circle, $r^2 = 1 + y_0^2$. Substituting this into (A) shows that the equation of the circle is

$$x^2 + y^2 - 2yy_0 = 1, \text{ so } 2y_0 = \frac{x^2 + y^2 - 1}{y}. \text{ Differentiating } y(2x + 2yy') - y'(x^2 + y^2 - 1) = 0, \text{ so } y'(y^2 - x^2 + 1) + 2xy = 0.$$

4.5.14. From Example 4.5.6 the equation of the line tangent to the parabola at (x_0, x_0^2) is (A) $y = -x_0^2 + 2x_0x$.

- From (A), $(x, y) = (5, 9)$ is on the tangent line through (x_0, x_0^2) if and only if $9 = -x_0^2 + 10x_0$, or $x_0^2 - 10x_0 + 9 = (x_0 - 1)(x_0 - 9) = 0$. Letting $x_0 = 1$ in (A) yields the line $y = -1 + 2x$, tangent to the parabola at $(x_0, x_0^2) = (1, 1)$. Letting $x_0 = 9$ in (A) yields the line $y = -81 + 18x$, tangent to the parabola at $(x_0, x_0^2) = (9, 81)$.
- From (A), $(x, y) = (6, 11)$ is on the tangent line through (x_0, x_0^2) if and only if $11 = -x_0^2 + 12x_0$, or $x_0^2 - 12x_0 + 11 = (x_0 - 1)(x_0 - 11) = 0$. Letting $x_0 = 1$ in (A) yields the line $y = -1 + 2x$, tangent to the parabola at $(x_0, x_0^2) = (1, 1)$. Letting $x_0 = 11$ in (A) yields the line $y = -121 + 22x$, tangent to the parabola at $(x_0, x_0^2) = (11, 121)$.
- From (A), $(x, y) = (-6, 20)$ is on the tangent line through (x_0, x_0^2) if and only if $20 = -x_0^2 - 12x_0$, or $x_0^2 + 12x_0 + 20 = (x_0 + 2)(x_0 + 10) = 0$. Letting $x_0 = -2$ in (A) yields the line $y = -4 - 4x$, tangent to the parabola at $(x_0, x_0^2) = (-2, 4)$. Letting $x_0 = -10$ in (A) yields the line $y = -100 - 20x$, tangent to the parabola at $(x_0, x_0^2) = (-10, 100)$.

- (d) From (A), $(x, y) = (-3, 5)$ is on the tangent line through (x_0, x_0^2) if and only if $5 = -x_0^2 - 6x_0$, or $x_0^2 + 6x_0 + 5 = (x_0 + 1)(x_0 + 5) = 0$. Letting $x_0 = -1$ in (A) yields the line $y = -1 - 2x$, tangent to the parabola at $(x_0, x_0^2) = (-1, 1)$. Letting $x_0 = -5$ in (A) yields the line $y = -25 - 10x$, tangent to the parabola at $(x_0, x_0^2) = (-5, 25)$.
- 4.5.15. (a) If (x_0, y_0) is any point on the circle such that $x_0 \neq \pm 1$ (and therefore $y_0 \neq 0$), then differentiating (A) yields $2x_0 + 2y_0y'_0 = 0$, so $y'_0 = -\frac{x_0}{y_0}$. Therefore, the equation of the tangent line is $y = y_0 - \frac{x_0}{y_0}(x - x_0)$. Since $x_0^2 + y_0^2 = 1$, this is equivalent to (B).
- (b) Since $y' = -\frac{x_0}{y_0}$ on the tangent line, we can rewrite (B) as $y - xy' = \frac{1}{y_0}$. Hence (F) $\frac{1}{(y - xy')^2} = y_0^2$ and (G) $x_0^2 = 1 - y_0^2 = \frac{(y - xy')^2 - 1}{(y - xy')^2}$. Since $(y')^2 = \frac{x_0^2}{y_0^2}$, (F) and (G) imply that $(y')^2 = (y - xy')^2 - 1$, which implies (C).
- (c) Using the quadratic formula to solve (C) for y' yields

$$y' = \frac{xy \pm \sqrt{x^2 + y^2 - 1}}{x^2 - 1} \quad (\text{H})$$

if (x, y) is on a tangent line with slope y' . If $y = \frac{1 - x_0x}{y_0}$, then

$$x^2 + y^2 - 1 = x^2 + \left(\frac{1 - x_0x}{y_0}\right)^2 - 1 = \left(\frac{x - x_0}{y_0}\right)^2 \quad (\text{since } x_0^2 + y_0^2 = 1).$$

Since $y' = -\frac{x_0}{y_0}$, this implies that (H) is equivalent to $-\frac{x_0}{y_0} = \frac{1}{x^2 - 1} \left[\frac{x(1 - x_0x)}{y_0} \pm \left| \frac{x - x_0}{y_0} \right| \right]$, which

holds if and only if we choose the “ $\pm$ ” so that $\pm \left| \frac{x - x_0}{y_0} \right| = -\left(\frac{x - x_0}{y_0} \right)$. Therefore, we must choose $\pm = -$ if $\frac{x - x_0}{y_0} > 0$, so (H) reduces to (D), or $\pm = +$ if $\frac{x - x_0}{y_0} < 0$, so (H) reduces to (E).

- (d) Differentiating (A) yields $2x + 2yy' = 0$, so $y' = -\frac{x}{y}$ on either semicircle. Since (D) and (E) both reduce to $y' = \frac{xy}{1 - x^2} = -\frac{x}{y}$ (since $x^2 + y^2 = 1$) on both semicircles, the conclusion follows.
- (e) From (D) and (E) the slopes of tangent lines from $(5, 5)$ tangent to the circle are $y' = \frac{25 \pm \sqrt{49}}{24} = \frac{3}{4}, \frac{4}{3}$. Therefore, tangent lines are $y = 5 + \frac{3}{4}(x - 5) = \frac{1 + 3x/5}{4/5}$ and $y = 5 + \frac{4}{3}(x - 5) = \frac{1 - 4x/5}{-3/5}$, which intersect the circle at $(-3/5, 4/5)$ $(4/5, -3/5)$, respectively. (See (B)).

- 4.5.16. (a) If (x_0, y_0) is any point on the parabola such that $x_0 > 0$ (and therefore $y_0 \neq 0$), then differentiating (A) yields $1 = 2y_0y'_0$, so $y'_0 = \frac{1}{2y_0}$. Therefore, the equation of the tangent

line is $y = y_0 + \frac{1}{2y_0}(x - x_0)$. Since $x_0 = y_0^2$, this is equivalent to (B).

- (b) Since $y' = \frac{1}{2y_0}$ on the tangent line, we can rewrite (B) as $\frac{y_0}{2} = y - xy'$. Substituting this into (B) yields $y = (y - xy') + \frac{x}{4(y - xy')}$, which implies (C).
- (c) Using the quadratic formula to solve (C) for y' yields

$$y' = \frac{y \pm \sqrt{y^2 - x}}{2x} \quad (\text{F})$$

if (x, y) is on a tangent line with slope y' . If $y = \frac{y_0}{2} + \frac{x}{2y_0}$, then $y^2 - x = \frac{1}{4} \left(y_0 - \frac{x}{y_0} \right)^2$

so (F) is equivalent to $\frac{1}{2y_0} = \frac{y_0 + \frac{x}{y_0} \pm \left| y_0 - \frac{x}{y_0} \right|}{4x}$ which holds if and only if we choose

the “ $\pm$ ” so that $\pm \left| y_0 - \frac{x}{y_0} \right| = - \left(y_0 - \frac{x}{y_0} \right)$. Therefore, we must choose $\pm = +$ if $x > y_0^2 = x_0$, so (F) reduces to (D), or $\pm = -$ if $x < y_0^2 = x_0$, so (F) reduces to (E).

- (d) Differentiating (A) yields $1 = 2yy'$, so $y' = \frac{1}{2y}$ on either half of the parabola. Since (D) and (E) both reduce to this if $x = y^2$, the conclusion follows.

4.5.18. The equation of the line tangent to the curve at $(x_0, y(x_0))$ is $y = y(x_0) + y'(x_0)(x - x_0)$.

Since $y(x_0/2) = 0$, $y(x_0) - \frac{y'(x_0)x_0}{2} = 0$. Since x_0 is arbitrary, it follows that $y' = \frac{2y}{x}$, so

$\frac{y'}{y} = \frac{2}{x}$, $\ln |y| = 2 \ln |x| + k$, and $y = cx^2$. Since $(1, 2)$ is on the curve, $c = 2$.

Therefore, $y = 2x^2$.

4.5.20. The equation of the line tangent to the curve at $(x_0, y(x_0))$ is $y = y(x_0) + y'(x_0)(x - x_0)$.

Since (x_1, y_1) is on the line, $y(x_0) + y'(x_0)(x_1 - x_0) = y_1$. Since x_0 is arbitrary, it follows that

$y + y'(x_1 - x) = y_1$, so $\frac{y'}{y - y_1} = \frac{1}{x - x_1}$, $\ln |y - y_1| = \ln |x - x_1| + k$, and $y - y_1 = c(x - x_1)$.

4.5.22. The equation of the line tangent to the curve at $(x_0, y(x_0))$ is $y = y(x_0) + y'(x_0)(x - x_0)$.

Since $y(0) = x_0$, $x_0 = y(x_0) - y'(x_0)x_0$. Since x_0 is arbitrary, it follows that $x = y - xy'$, so

(A) $y' - \frac{y}{x} = -1$. The solutions of (A) are of the form $y = ux$, where $u'x = -1$, so $u' = -\frac{1}{x}$.

Therefore, $u = -\ln |x| + c$ and $y = -x \ln |x| + cx$.

4.5.24. The equation of the line normal to the curve at (x_0, y_0) is $y = y(x_0) - \frac{x - x_0}{y'(x_0)}$. Since

$y(0) = 2y(x_0)$, $y(x_0) + \frac{x_0}{y'(x_0)} = 2y(x_0)$. Since x_0 is arbitrary, it follows that $y'y = x$, so (A)

$\frac{y^2}{2} = \frac{x^2}{2} + \frac{c}{2}$ and $y^2 = x^2 + c$. Now $y(2) = 1 \leftrightarrow c = -3$. Therefore, $y = \sqrt{x^2 - 3}$.

4.5.26. Differentiating the given equation yields $2x + 4y + 4xy' + 2yy' = 0$, so $y' = -\frac{x+2y}{2x+y}$ is a differential equation for the given family, and (A) $y' = \frac{2x+y}{x+2y}$ is a differential equation for the orthogonal trajectories. Substituting $y = ux$ in (A) yields $u'x + u = \frac{2+u}{1+2u}$, so $u'x = -\frac{2(u^2-1)}{1+2u}$ and $\frac{1+2u}{(u-1)(u+1)}u' = -\frac{2}{x}$, or $\left[\frac{3}{u-1} + \frac{1}{u+1}\right]u' = -\frac{4}{x}$. Therefore, $3\ln|u-1| + \ln|u+1| = -4\ln|x| + K$, so $(u-1)^3(u+1) = \frac{k}{x^4}$. Substituting $u = \frac{y}{x}$ yields the orthogonal trajectories $(y-x)^3(y+x) = k$.

4.5.28. Differentiating yields $ye^{x^2}(1+2x^2) + xe^{x^2}y' = 0$, so $y' = -\frac{y(1+2x^2)}{x}$ is a differential equation for the given family. Therefore, (A) $y' = -\frac{x}{y(1+2x^2)}$ is a differential equation for the orthogonal trajectories. From (A), $yy' = -\frac{x}{1+2x^2}$, so $\frac{y^2}{2} = -\frac{1}{4}\ln(1+2x^2) + \frac{k}{2}$, and the orthogonal trajectories are given by $y^2 = -\frac{1}{2}\ln(1+2x^2) + k$.

4.5.30. Differentiating (A) $y = 1 + cx^2$ yields (B) $y' = 2cx$. From (C), $c = \frac{y-1}{x^2}$. Substituting this into (B) yields the differential equation $y' = \frac{2(y-1)}{x}$ for the given family of parabolas. Therefore, $y' = -\frac{x}{2(y-1)}$ is a differential equation for the orthogonal trajectories. Separating variables yields $2(y-1)y' = -x$, so $(y-1)^2 = -\frac{x^2}{2} + k$. Now $y(-1) = 3 \leftrightarrow k = \frac{9}{2}$, so $(y-1)^2 = -\frac{x^2}{2} + \frac{9}{2}$. Therefore, (D) $y = 1 + \sqrt{\frac{9-x^2}{2}}$. This curve intersects the parabola (A) if and only if the equation (C) $cx^2 = \sqrt{\frac{9-x^2}{2}}$ has a solution x^2 in $(0, 9)$. Therefore, $c > 0$ is a necessary condition for intersection. We will show that it is also sufficient. Squaring both sides of (C) and simplifying yields $2c^2x^4 + x^2 - 9 = 0$. Using the quadratic formula to solve this for x^2 yields $x^2 = \frac{-1 + \sqrt{1+72c^2}}{4c^2}$. The condition $x^2 < 9$ holds if and only if $-1 + \sqrt{1+72c^2} < 36c^4$, which is equivalent to $1 + 72c^2 < (1 + 36c^2)^2 = 1 + 72c^2 + 1296c^4$, which holds for all $c > 0$.

4.5.32. The angles θ and θ_1 from the x -axis to the tangents to C and C_1 satisfy $\tan \theta = f(x_0, y_0)$ and $\tan \theta_1 = \frac{f(x_0, y_0) + \tan \alpha}{1 - f(x_0, y_0) \tan \alpha} = \frac{\tan \theta + \tan \alpha}{1 - \tan \theta \tan \alpha} = \tan(\theta + \alpha)$. Therefore, assuming θ and θ_1 are both in $[0, 2\pi)$, $\theta_1 = \theta + \alpha$.

4.5.34. Circles centered at the origin are given by $x^2 + y^2 = r^2$. Differentiating yields $2x + 2yy' = 0$, so $y' = -\frac{x}{y}$ is a differential equation for the given family, and $y' = \frac{-(x/y) + \tan \alpha}{1 + (x/y) \tan \alpha}$ is a differential equation for the desired family. Substituting $y = ux$

yields $u'x + u = \frac{-1/u + \tan \alpha}{1 + (1/u) \tan \alpha} = \frac{-1 + u \tan \alpha}{u + \tan \alpha}$. Therefore, $u'x = -\frac{1 + u^2}{u + \tan \alpha}$,
 $\frac{u + \tan \alpha}{1 + u^2} u' = -\frac{1}{x}$ and $\frac{1}{2} \ln(1 + u^2) + \tan \alpha \tan^{-1} u = -\ln |x| + k$. Substituting $u = \frac{y}{x}$ yields
 $\frac{1}{2} \ln(x^2 + y^2) + (\tan \alpha) \tan^{-1} \frac{y}{x} = k$.

Chapter 5

Linear Second Order Equations

5.1 Homogeneous Linear Equations

- 5.1.2. (a) If $y_1 = e^x \cos x$, then $y_1' = e^x(\cos x - \sin x)$ and $y_1'' = e^x(\cos x - \sin x - \sin x - \cos x) = -2e^x \sin x$, so $y_1'' - 2y_1' + 2y_1 = e^x(-2 \sin x - 2 \cos x + 2 \sin x + 2 \cos x) = 0$. If $y_2 = e^x \sin x$, then $y_2' = e^x(\sin x + \cos x)$ and $y_2'' = e^x(\sin x + \cos x + \cos x - \sin x) = 2e^x \cos x$, so $y_2'' - 2y_2' + 2y_2 = e^x(2 \cos x - 2 \sin x - 2 \cos x + 2 \sin x) = 0$.

- (b) If (B) $y = e^x(c_1 \cos x + c_2 \sin x)$, then

$$y' = e^x(c_1(\cos x - \sin x) + c_2(\sin x + \cos x)) \quad (C)$$

and

$$\begin{aligned} y'' &= c_1 e^x(\cos x - \sin x - \sin x - \cos x) \\ &\quad + c_2 e^x(\sin x + \cos x + \cos x - \sin x) \\ &= 2e^x(-c_1 \sin x + c_2 \cos x), \end{aligned}$$

so

$$\begin{aligned} y'' - 2y' + 2y &= c_1 e^x(-2 \sin x - 2 \cos x + 2 \sin x + 2 \cos x) \\ &\quad + c_2 e^x(2 \cos x - 2 \sin x - 2 \cos x + 2 \sin x) = 0. \end{aligned}$$

- (c) We must choose c_1 and c_2 in (B) so that $y(0) = 3$ and $y'(0) = -2$. Setting $x = 0$ in (B) and (C) shows that $c_1 = 3$ and $c_1 + c_2 = -2$, so $c_2 = -5$. Therefore, $y = e^x(3 \cos x - 5 \sin x)$.
- (d) We must choose c_1 and c_2 in (B) so that $y(0) = k_0$ and $y'(0) = k_1$. Setting $x = 0$ in (B) and (C) shows that $c_1 = k_0$ and $c_1 + c_2 = k_1$, so $c_2 = k_1 - k_0$. Therefore, $y = e^x(k_0 \cos x + (k_1 - k_0) \sin x)$.

- 5.1.4. (a) If $y_1 = \frac{1}{x-1}$, then $y_1' = -\frac{1}{(x-1)^2}$ and $y_1'' = \frac{2}{(x-1)^3}$, so

$$(x^2 - 1)y_1'' + 4xy_1' + 2y_1 = \frac{2(x^2 - 1)}{(x-1)^3} - \frac{4x}{(x-1)^2} + \frac{2}{x-1}$$

$$= \frac{2(x+1) - 4x + 2(x-1)}{(x-1)^2} = 0.$$

Similar manipulations show that $(x^2 - 1)y_2'' + 4xy_2' + 2y_2 = 0$. The general solution on each of the intervals $(-\infty, -1)$, $(-1, 1)$, and $(1, \infty)$ is (B) $y = \frac{c_1}{x-1} + \frac{c_2}{x+1}$.

- (b) Differentiating (B) yields (C) $y' = -\frac{c_1}{(x-1)^2} - \frac{c_2}{(x+1)^2}$. We must choose c_1 and c_2 in (B) so that $y(0) = -5$ and $y'(0) = 1$. Setting $x = 0$ in (B) and (C) shows that $-c_1 + c_2 = -5$, $-c_1 - c_2 = 1$. Therefore, $c_1 = 2$ and $c_2 = -3$, so $y = \frac{2}{x-1} - \frac{3}{x+1}$ on $(-1, 1)$.

- (d) The Wronskian of $\{y_1, y_2\}$ is

$$W(x) = \begin{vmatrix} \frac{1}{x-1} & \frac{1}{x+1} \\ -\frac{1}{(x-1)^2} & -\frac{1}{(x+1)^2} \end{vmatrix} = \frac{2}{(x^2-1)^2}, \quad (\text{D})$$

so $W(0) = 2$. Since $p(x) = \frac{4x}{x^2-1}$, so $\int_0^x p(t) dt = \int_0^x \frac{4t}{t^2-1} dt = \ln(x^2-1)^2$, Abel's formula implies that $W(x) = W(0)e^{-\ln(x^2-1)^2} = \frac{2}{(x^2-1)^2}$, consistent with (D).

5.1.6. From Abel's formula, $W(x) = W(\pi)e^{-3\int_\pi^x (t^2+1) dt} = 0 \cdot e^{-3\int_\pi^x (t^2+1) dt} = 0$.

5.1.8. $p(x) = \frac{1}{x}$; therefore $\int_1^x p(t) dt = \int_1^x \frac{dt}{t} = \ln x$, so Abel's formula yields $W(x) = W(1)e^{-\ln x} = \frac{1}{x}$.

5.1.10. $p(x) = -2$; $P(x) = -2x$; $y_2 = uy_1 = ue^{3x}$; $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{Ke^{2x}}{e^{6x}} = Ke^{-4x}$;
 $u = -\frac{K}{4}e^{-4x}$. Choose $K = -4$; then $y_2 = e^{-4x}e^{3x} = e^{-x}$.

5.1.12. $p(x) = -2a$; $P(x) = -2ax$; $y_2 = uy_1 = ue^{ax}$; $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{Ke^{2ax}}{e^{2ax}} = K$; $u = Kx$.
 Choose $K = 1$; then $y_2 = xe^{ax}$.

5.1.14. $p(x) = -\frac{1}{x}$; $P(x) = -\ln x$; $y_2 = uy_1 = ux$; $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{Kx}{x^2} = \frac{K}{x}$; $u = K \ln x$.
 Choose $K = 1$; then $y_2 = x \ln x$.

5.1.16. $p(x) = -\frac{1}{x}$; $P(x) = -\ln|x|$; $y_2 = uy_1 = ux^{1/2}e^{2x}$; $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{Kx}{xe^{4x}} = e^{-4x}$;
 $u = -\frac{Ke^{-4x}}{4}$. Choose $K = -4$; then $y_2 = e^{-4x}(x^{1/2}e^{2x}) = x^{1/2}e^{-2x}$.

5.1.18. $p(x) = -\frac{2}{x}$; $P(x) = -2 \ln |x|$; $y_2 = uy_1 = ux \cos x$; $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{Kx^2}{x^2 \cos^2 x} = K \sec^2 x$;
 $u = K \tan x$. Choose $K = 1$; then $y_2 = \tan x(x \cos x) = x \sin x$.

5.1.20. $p(x) = -\frac{3x+2}{3x-1} = -1 - \frac{3}{3x-1}$; $P(x) = -x - \ln |3x-1|$; $y_2 = uy_1 = ue^{2x}$;
 $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{K(3x-1)e^x}{e^{4x}} = K(3x-1)e^{-3x}$; $u = -Kxe^{-3x}$. Choose $K = -1$; then
 $y_2 = xe^{-3x}e^{2x} = xe^{-x}$.

5.1.22. $p(x) = -\frac{2(2x^2-1)}{x(2x+1)} = -2 - \frac{2}{2x+1} + \frac{2}{x}$; $P(x) = -2x - \ln |2x+1| + 2 \ln |x|$;
 $y_2 = uy_1 = \frac{u}{x}$; $u' = \frac{Ke^{-P(x)}}{y_1^2(x)} = \frac{K(2x+1)e^{2x}}{x^2}x^2 = K(2x+1)e^{2x}$; $u = Kxe^{2x}$. Choose
 $K = 1$; then $y_2 = \frac{xe^{2x}}{x} = e^{2x}$.

5.1.24. Suppose that $y \equiv 0$ on (a, b) . Then $y' \equiv 0$ and $y'' \equiv 0$ on (a, b) , so y is a solution of (A)
 $y'' + p(x)y' + q(x)y = 0$, $y(x_0) = 0$, $y'(x_0) = 0$ on (a, b) . Since Theorem 5.1.1 implies that
 (A) has only one solution on (a, b) , the conclusion follows.

5.1.26. If $\{z_1, z_2\}$ is a fundamental set of solutions of (A) on (a, b) , then every solution y of (A) on
 (a, b) is a linear combination of $\{z_1, z_2\}$; that is,
 $y = c_1 z_1 + c_2 z_2 = c_1(\alpha y_1 + \beta y_2) + c_2(\gamma y_1 + \delta y_2) = (c_1 \alpha + c_2 \gamma)y_1 + (c_1 \beta + c_2 \delta)y_2$, which
 shows that every solution of (A) on (a, b) can be written as a linear combination of $\{y_1, y_2\}$.
 Therefore, $\{y_1, y_2\}$ is a fundamental set of solutions of (A) on (a, b) .

5.1.28. The Wronskian of $\{y_1, y_2\}$ is

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} y_1 & ky_1 \\ y_1' & ky_1' \end{vmatrix} = k(y_1 y_1' - y_1' y_1) = 0.$$

nor y_2 can be a solution of $y'' + p(x)y' + q(x)y = 0$ on (a, b) .

5.1.30. $W(x_0) = (y_1(x_0)y_2'(x_0) - y_1'(x_0)y_2(x_0)) = 0$ if either $y_1(x_0) = y_2(x_0) = 0$ or
 $y_1'(x_0) = y_2'(x_0) = 0$, and Theorem 5.1.6 implies that $\{y_1, y_2\}$ is linearly dependent on (a, b) .

5.1.32. Let x_0 be an arbitrary point in (a, b) . By the motivating argument preceding
 Theorem 5.1.4, (B) $W(x_0) = y_1(x_0)y_2'(x_0) - y_1'(x_0)y_2(x_0) \neq 0$. Now let y be the solution of
 $y'' + p(x)y' + q(x)y = 0$, $y(x_0) = y_1(x_0)$, $y'(x_0) = y_1'(x_0)$. By assumption, y is a linear
 combination of $\{y_1, y_2\}$ on (a, b) ; that is, $y = c_1 y_1 + c_2 y_2$, where

$$\begin{aligned} c_1 y_1(x_0) + c_2 y_2(x_0) &= y_1(x_0) \\ c_1 y_1'(x_0) + c_2 y_2'(x_0) &= y_1'(x_0). \end{aligned}$$

Solving this system by Cramer's rule yields

$$c_1 = \frac{1}{W(x_0)} \begin{vmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{vmatrix} = 1 \text{ and } c_2 = \frac{1}{W(x_0)} \begin{vmatrix} y_1(x_0) & y_1(x_0) \\ y_1'(x_0) & y_1'(x_0) \end{vmatrix} = 0.$$

Therefore, $y = y_1$, which shows that y_1 is a solution of (A). A similar argument shows that
 y_2 is a solution of (A).

5.1.34. Expanding the determinant by cofactors of its first column shows that the first equation in the exercise can be written as

$$\frac{y}{W} \begin{vmatrix} y_1' & y_2' \\ y_1'' & y_2'' \end{vmatrix} - \frac{y'}{W} \begin{vmatrix} y_1 & y_2 \\ y_1'' & y_2'' \end{vmatrix} + \frac{y''}{W} \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = 0,$$

which is of the form (A) with

$$p = -\frac{1}{W} \begin{vmatrix} y_1 & y_2 \\ y_1'' & y_2'' \end{vmatrix} \quad \text{and} \quad q = \frac{1}{W} \begin{vmatrix} y_1' & y_2' \\ y_1'' & y_2'' \end{vmatrix}.$$

5.1.36. Theorem 5.1.6 implies that there are constants c_1 and c_2 such that (B)

$y = c_1 y_1 + c_2 y_2$ on (a, b) . To see that c_1 and c_2 are unique, assume that (B) holds, and let x_0 be a point in (a, b) . Then (C) $y' = c_1 y_1' + c_2 y_2'$. Setting $x = x_0$ in (B) and (C) yields

$$\begin{aligned} c_1 y_1(x_0) + c_2 y_2(x_0) &= y(x_0) \\ c_1 y_1'(x_0) + c_2 y_2'(x_0) &= y'(x_0). \end{aligned}$$

Since Theorem 5.1.6 implies that $y_1(x_0)y_2'(x_0) - y_1'(x_0)y_2(x_0) \neq 0$, the argument preceding Theorem 5.1.4 implies that c_1 and c_2 are given uniquely by

$$c_1 = \frac{y_2'(x_0)y(x_0) - y_2(x_0)y'(x_0)}{y_1(x_0)y_2'(x_0) - y_1'(x_0)y_2(x_0)} \quad c_2 = \frac{y_1(x_0)y'(x_0) - y_1'(x_0)y(x_0)}{y_1(x_0)y_2'(x_0) - y_1'(x_0)y_2(x_0)}.$$

5.1.38. The general solution of $y'' = 0$ is $y = c_1 + c_2 x$, so $y' = c_2$. Imposing the stated initial conditions on $y_1 = c_1 + c_2 x$ yields $c_1 + c_2 x_0 = 1$ and $c_2 = 0$; therefore $c_1 = 1$, so $y_1 = 1$. Imposing the stated initial conditions on $y_2 = c_1 + c_2 x$ yields $c_1 + c_2 x_0 = 0$ and $c_2 = 1$; therefore $c_1 = -x_0$, so $y_2 = x - x_0$. The solution of the general initial value problem is $y = k_0 + k_1(x - x_0)$.

5.1.40. Let $y_1 = a_1 \cos \omega x + a_2 \sin \omega x$ and $y_2 = b_1 \cos \omega x + b_2 \sin \omega x$. Then

$$\begin{aligned} a_1 \cos \omega x_0 + a_2 \sin \omega x_0 &= 1 \\ \omega(-a_1 \sin \omega x_0 + a_2 \cos \omega x_0) &= 0 \end{aligned}$$

and

$$\begin{aligned} b_1 \cos \omega x_0 + b_2 \sin \omega x_0 &= 0 \\ \omega(-b_1 \sin \omega x_0 + b_2 \cos \omega x_0) &= 1. \end{aligned}$$

Solving these systems yields $a_1 = \cos \omega x_0$, $a_2 = \sin \omega x_0$, $b_1 = -\frac{\sin \omega x_0}{\omega}$, and $b_2 = \frac{\cos \omega x_0}{\omega}$.

Therefore, $y_1 = \cos \omega x_0 \cos \omega x + \sin \omega x_0 \sin \omega x = \cos \omega(x - x_0)$ and

$y_2 = \frac{1}{\omega}(-\sin \omega x_0 \cos \omega x + \cos \omega x_0 \sin \omega x) = \frac{1}{\omega} \sin \omega(x - x_0)$. The solution of the general

initial value problem is $y = k_0 \cos \omega(x - x_0) + \frac{k_1}{\omega} \sin \omega(x - x_0)$.

5.1.42. (a) If $y_1 = x^2$, then $y_1' = 2x$ and $y_1'' = 2$, so $x^2 y_1'' - 4x y_1' + 6y_1 = x^2(2) - 4x(2x) + 6x^2 = 0$ for x in $(-\infty, \infty)$. If $y_2 = x^3$, then $y_2' = 3x^2$ and $y_2'' = 6x$, so $x^2 y_2'' - 4x y_2' + 6y_2 = x^2(6x) - 4x(3x^2) + 6x^3 = 0$ for x in $(-\infty, \infty)$. If $x \neq 0$, then $y_2(x)/y_1(x) = x$, which is nonconstant on $(-\infty, 0)$ and $(0, \infty)$, so Theorem 5.1.6 implies that $\{y_1, y_2\}$ is a fundamental set of solutions of (A) on each of these intervals.

(b) Theorem 5.1.6 and (a) imply that y satisfies (A) on $(-\infty, 0)$ and on $(0, \infty)$ if and only if $y = \begin{cases} a_1 x^2 + a_2 x^3, & x > 0, \\ b_1 x^2 + b_2 x^3, & x < 0. \end{cases}$ Since $y(0) = 0$ we can complete the proof that y is a solution of (A) on $(-\infty, \infty)$ by showing that $y'(0)$ and $y''(0)$ both exist if and only if $a_1 = b_1$. Since

$$\frac{y(x) - y(0)}{x - 0} = \begin{cases} a_1 x + a_2 x^2, & \text{if } x > 0, \\ b_1 x + b_2 x^2, & \text{if } x < 0, \end{cases}$$

it follows that $y'(0) = \lim_{x \rightarrow 0} \frac{y(x) - y(0)}{x - 0} = 0$. Therefore, $y' = \begin{cases} 2a_1 x + 3a_2 x^2, & x \geq 0, \\ 2b_1 x + 3b_2 x^2, & x < 0. \end{cases}$

Since $\frac{y'(x) - y'(0)}{x - 0} = \begin{cases} 2a_1 + 3a_2 x, & \text{if } x > 0, \\ 2b_1 + 3b_2 x, & \text{if } x < 0, \end{cases}$ it follows that

$y''(0) = \lim_{x \rightarrow 0} \frac{y'(x) - y'(0)}{x - 0}$ exists if and only if $a_1 = b_1$. By renaming $a_1 = b_1 = c_1$, $a_2 = c_2$, and $b_2 = c_3$ we see that y is a solution of (A) on $(-\infty, \infty)$ if and only if $y = \begin{cases} c_1 x^2 + c_2 x^3, & x \geq 0, \\ c_1 x^2 + c_3 x^3, & x < 0. \end{cases}$

(c) We have shown that $y(0) = y'(0) = 0$ for any choice of c_1 and c_2 in (C). Therefore, the given initial value problem has a solution if and only if $k_0 = k_1 = 0$, in which case every function of the form (C) is a solution.

(d) If $x_0 > 0$, then c_1 and c_2 in (C) are uniquely determined by k_0 and k_1 , but c_3 can be chosen arbitrarily. Therefore, (B) has a unique solution on $(0, \infty)$, but infinitely many solutions on $(-\infty, \infty)$. If $x_0 < 0$, then c_1 and c_3 in (C) are uniquely determined by k_0 and k_1 , but c_2 can be chosen arbitrarily. Therefore, (B) has a unique solution on $(-\infty, 0)$, but infinitely many solutions on $(-\infty, \infty)$.

5.1.44. (a) If $y_1 = x^3$, then $y_1' = 3x^2$ and $y_1'' = 6x$, so $x^2 y_1'' - 6x y_1' + 12y_1 = x^2(6x) - 6x(3x^2) + 12x^3 = 0$ for x in $(-\infty, \infty)$. If $y_2 = x^4$, then $y_2' = 4x^3$ and $y_2'' = 12x^2$, so $x^2 y_2'' - 6x y_2' + 12y_2 = x^2(12x^2) - 6x(4x^3) + 12x^4 = 0$ for x in $(-\infty, \infty)$. If $x \neq 0$, then $y_2(x)/y_1(x) = x$, which is nonconstant on $(-\infty, 0)$ and $(0, \infty)$, so Theorem 5.1.6 implies that $\{y_1, y_2\}$ is a fundamental set of solutions of (A) on each of these intervals.

(b) Theorem 5.1.2 and (a) imply that y satisfies (A) on $(-\infty, 0)$ and on $(0, \infty)$ if and only if (C) $y = \begin{cases} a_1 x^3 + a_2 x^4, & x > 0, \\ b_1 x^3 + b_2 x^4, & x < 0. \end{cases}$ Since $y(0) = 0$ we can complete the proof that y is a solution of (A) on $(-\infty, \infty)$ by showing that $y'(0)$ and $y''(0)$ both exist for any choice of a_1, a_2, b_1 , and b_2 . Since $\frac{y(x) - y(0)}{x - 0} = \begin{cases} a_1 x^2 + a_2 x^3, & \text{if } x > 0, \\ b_1 x^2 + b_2 x^3, & \text{if } x < 0, \end{cases}$ it follows that

$$y'(0) = \lim_{x \rightarrow 0} \frac{y(x) - y(0)}{x - 0} = 0. \text{ Therefore, } y' = \begin{cases} 3a_1x^2 + 4a_2x^3, & x \geq 0, \\ 3b_1x^2 + 4b_2x^3, & x < 0. \end{cases} \text{ Since}$$

$$\frac{y'(x) - y'(0)}{x - 0} = \begin{cases} 3a_1x + 4a_2x^2, & \text{if } x > 0, \\ 3b_1x + 4b_2x^2, & \text{if } x < 0, \end{cases} \text{ it follows that}$$

$$y''(0) = \lim_{x \rightarrow 0} \frac{y'(x) - y'(0)}{x - 0} = 0. \text{ Therefore, (B) is a solution of (A) on } (-\infty, \infty).$$

- (c) We have shown that $y(0) = y'(0) = 0$ for any choice of a_1 , a_2 , b_1 , and b_2 in (B). Therefore, the given initial value problem has a solution if and only if $k_0 = k_1 = 0$, in which case every function of the form (B) is a solution.
- (d) If $x_0 > 0$, then a_1 and a_2 in (B) are uniquely determined by k_0 and k_1 , but b_1 and b_2 can be chosen arbitrarily. Therefore, (C) has a unique solution on $(0, \infty)$, but infinitely many solutions on $(-\infty, \infty)$. If $x_0 < 0$, then b_1 and b_2 in (B) are uniquely determined by k_0 and k_1 , but a_1 and a_2 can be chosen arbitrarily. Therefore, (C) has a unique solution on $(-\infty, 0)$, but infinitely many solutions on $(-\infty, \infty)$.

5.2 Constant Coefficient Homogeneous Equations

$$5.2.2. \quad p(r) = r^2 - 4r + 5 = (r - 2)^2 + 1; \quad y = e^{2x}(c_1 \cos x + c_2 \sin x).$$

$$5.2.4. \quad p(r) = r^2 - 4r + 4 = (r - 2)^2; \quad y = e^{2x}(c_1 + c_2x).$$

$$5.2.6. \quad p(r) = r^2 + 6r + 10 = (r + 3)^2 + 1; \quad y = e^{-3x}(c_1 \cos x + c_2 \sin x).$$

$$5.2.8. \quad p(r) = r^2 + r = r(r + 1); \quad y = c_1 + c_2e^{-x}.$$

$$5.2.10. \quad p(r) = r^2 + 6r + 13y = (r + 3)^2 + 4; \quad y = e^{-3x}(c_1 \cos 2x + c_2 \sin 2x).$$

$$5.2.12. \quad p(r) = 10r^2 - 3r - 1 = (2r - 1)(5r + 1) = 10(r - 1/2)(r + 1/5); \quad y = c_1e^{-x/5} + c_2e^{x/2}.$$

$$5.2.14. \quad p(r) = 6r^2 - r - 1 = (2r - 1)(3r + 1) = 6(r - 1/2)(r + 1/3); \quad y = c_1e^{-x/3} + c_2e^{x/2};$$

$$y' = -\frac{c_1}{3}e^{-x/3} + \frac{c_2}{2}e^{x/2}; \quad y(0) = 10 \Rightarrow c_1 + c_2 = 10, \quad y'(0) = 0 \Rightarrow -\frac{c_1}{3} + \frac{c_2}{2} = 0;$$

$$c_1 = 6, c_2 = 4; \quad y = 4e^{x/2} + 6e^{-x/3}.$$

$$5.2.16. \quad p(r) = 4r^2 - 4r - 3 = (2r - 3)(2r + 1) = 4(r - 3/2)(r + 1/2); \quad y = c_1e^{-x/2} + c_2e^{3x/2};$$

$$y' = -\frac{c_1}{2}e^{-x/2} + \frac{3c_2}{2}e^{3x/2}; \quad y(0) = \frac{13}{12} \Rightarrow c_1 + c_2 = \frac{13}{12}, \quad y'(0) = \frac{23}{24} \Rightarrow -\frac{c_1}{2} + \frac{3c_2}{2} = \frac{23}{24};$$

$$c_1 = \frac{1}{3}, c_2 = \frac{3}{4}; \quad y = \frac{e^{-x/2}}{3} + \frac{3e^{3x/2}}{4}.$$

$$5.2.18. \quad p(r) = r^2 + 7r + 12 = (r + 3)(r + 4); \quad y = c_1e^{-4x} + c_2e^{-3x}; \quad y' = -4c_1e^{-4x} - 3c_2e^{-3x};$$

$$y(0) = -1 \Rightarrow c_1 + c_2 = -1, \quad y'(0) = 0 \Rightarrow -4c_1 - 3c_2 = 0; \quad c_1 = 3, c_2 = -4;$$

$$y = 3e^{-4x} - 4e^{-3x}.$$

$$5.2.20. \quad p(r) = 36r^2 - 12r + 1 = (6r - 1)^2 = 36(r - 1/6)^2; \quad y = e^{x/6}(c_1 + c_2x);$$

$$y' = \frac{e^{x/6}}{6}(c_1 + c_2x) + c_2e^{x/6}; \quad y(0) = 3 \Rightarrow c_1 = 3, \quad y'(0) = \frac{5}{2} \Rightarrow \frac{c_1}{6} + c_2 = \frac{5}{2} \Rightarrow c_2 = 2;$$

$$y = e^{x/6}(3 + 2x).$$

5.2.22. (a) From (A), $ay''(x) + by'(x) + cy(x) = 0$ for all x . Replacing x by $x - x_0$ yields (C) $ay''(x - x_0) + by'(x - x_0) + cy(x - x_0) = 0$. If $z(x) = y(x - x_0)$, then the chain rule implies that $z'(x) = y'(x - x_0)$ and $z''(x) = y''(x - x_0)$, so (C) is equivalent to $az'' + bz' + cz = 0$.

(b) If $\{y_1, y_2\}$ is a fundamental set of solutions of (A) then Theorem 5.1.6 implies that y_2/y_1 is nonconstant. Therefore, $\frac{z_2(x)}{z_1(x)} = \frac{y_2(x - x_0)}{y_1(x - x_0)}$ is also nonconstant, so Theorem 5.1.6 implies that $\{z_1, z_2\}$ is a fundamental set of solutions of (A).

(c) Let $p(r) = ar^2 + br + c$ be the characteristic polynomial of (A). Then:

- If $p(r) = 0$ has distinct real roots r_1 and r_2 , then the general solution of (A) is

$$y = c_1 e^{r_1(x-x_0)} + c_2 e^{r_2(x-x_0)}.$$

- If $p(r) = 0$ has a repeated root r_1 , then the general solution of (A) is

$$y = e^{r_1(x-x_0)}(c_1 + c_2(x - x_0)).$$

- If $p(r) = 0$ has complex conjugate roots $r_1 = \lambda + i\omega$ and $r_2 = \lambda - i\omega$ (where $\omega > 0$), then the general solution of (A) is

$$y = e^{\lambda(x-x_0)}(c_1 \cos \omega(x - x_0) + c_2 \sin \omega(x - x_0)).$$

5.2.24. $p(r) = r^2 - 6r - 7 = (r - 7)(r + 1)$;

$$\begin{aligned} y &= c_1 e^{-(x-2)} + c_2 e^{7(x-2)}; \\ y' &= -c_1 e^{-(x-2)} + 7c_2 e^{7(x-2)}; \end{aligned}$$

$$\begin{aligned} y(2) &= -\frac{1}{3} \Rightarrow c_1 + c_2 = -\frac{1}{3}, \quad y'(2) = -5 \Rightarrow -c_1 + 7c_2 = -5; \quad c_1 = \frac{1}{3}, c_2 = -\frac{2}{3}; \\ y &= \frac{1}{3} e^{-(x-2)} - \frac{2}{3} e^{7(x-2)}. \end{aligned}$$

5.2.26. $p(r) = 9r^2 + 6r + 1 = (3r + 1)^2 = 9(r + 1/3)^2$;

$$\begin{aligned} y &= e^{-(x-2)/3} (c_1 + c_2(x - 2)); \\ y' &= -\frac{1}{3} e^{-(x-2)/3} (c_1 + c_2(x - 2)) + c_2 e^{-(x-2)/3}; \end{aligned}$$

$$\begin{aligned} y(2) &= 2 \Rightarrow c_1 = 2, \quad y'(2) = -\frac{14}{3} \Rightarrow -\frac{c_1}{3} + c_2 = -\frac{14}{3} \Rightarrow c_2 = -4; \\ y &= e^{-(x-2)/3} (2 - 4(x - 2)). \end{aligned}$$

5.2.28. $p(r) = r^2 + 3$;

$$\begin{aligned} y &= c_1 \cos \sqrt{3} \left(x - \frac{\pi}{3}\right) + c_2 \sin \sqrt{3} \left(x - \frac{\pi}{3}\right); \\ y' &= -\sqrt{3} c_1 \sin \sqrt{3} \left(x - \frac{\pi}{3}\right) + \sqrt{3} c_2 \cos \sqrt{3} \left(x - \frac{\pi}{3}\right); \end{aligned}$$

$$y(\pi/3) = 2 \Rightarrow c_1 = 2, \quad y'(\pi/3) = -1 \Rightarrow c_2 = -\frac{1}{\sqrt{3}};$$

$$y = 2 \cos \sqrt{3} \left(x - \frac{\pi}{3} \right) - \frac{1}{\sqrt{3}} \sin \sqrt{3} \left(x - \frac{\pi}{3} \right).$$

5.2.30. y is a solution of $ay'' + by' + cy = 0$ if and only if

$$\begin{aligned} y &= c_1 e^{r_1(x-x_0)} + c_2 e^{r_2(x-x_0)} \\ y' &= r_1 c_1 e^{r_1(x-x_0)} + r_2 c_2 e^{r_2(x-x_0)}. \end{aligned}$$

Now $y_1(x_0) = k_0$ and $y'_1(x_0) = k_1 \Rightarrow c_1 + c_2 = k_0, r_1 c_1 + r_2 c_2 = k_1$. Therefore, $c_1 = \frac{r_2 k_0 - k_1}{r_2 - r_1}$ and $c_2 = \frac{k_1 - r_1 k_0}{r_2 - r_1}$. Substituting c_1 and c_2 into the above equations for y and y' yields

$$\begin{aligned} y &= \frac{r_2 k_0 - k_1}{r_2 - r_1} e^{r_1(x-x_0)} + \frac{k_1 - r_1 k_0}{r_2 - r_1} e^{r_2(x-x_0)} \\ &= \frac{k_0}{r_2 - r_1} (r_2 e^{r_1(x-x_0)} - r_1 e^{r_2(x-x_0)}) + \frac{k_1}{r_2 - r_1} (e^{r_2(x-x_0)} - e^{r_1(x-x_0)}). \end{aligned}$$

5.2.32. y is a solution of $ay'' + by' + cy = 0$ if and only if

$$y = e^{\lambda(x-x_0)} (c_1 \cos \omega(x-x_0) + c_2 \sin \omega(x-x_0)) \quad (\text{A})$$

and

$$\begin{aligned} y' &= \lambda e^{\lambda(x-x_0)} (c_1 \cos \omega(x-x_0) + c_2 \sin \omega(x-x_0)) \\ &\quad + \omega e^{\lambda(x-x_0)} (-c_1 \sin \omega(x-x_0) + c_2 \cos \omega(x-x_0)). \end{aligned}$$

Now $y_1(x_0) = k_0 \Rightarrow c_1 = k_0$ and $y'_1(x_0) = k_1 \Rightarrow \lambda c_1 + \omega c_2 = k_1$, so $c_2 = \frac{k_1 - \lambda k_0}{\omega}$. Substituting c_1 and c_2 into (A) yields

$$y = e^{\lambda(x-x_0)} \left[k_0 \cos \omega(x-x_0) + \left(\frac{k_1 - \lambda k_0}{\omega} \right) \sin \omega(x-x_0) \right].$$

5.2.34. (b)

$$\begin{aligned} e^{i\theta_1} e^{i\theta_2} &= (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= (\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2) \\ &= \cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2) = e^{i(\theta_1 + \theta_2)}. \end{aligned}$$

(c)

$$e^{z_1 + z_2} = e^{(\alpha_1 + i\beta_1) + (\alpha_2 + i\beta_2)} = e^{(\alpha_1 + \alpha_2) + i(\beta_1 + \beta_2)}$$

$$\begin{aligned}
&= e^{(\alpha_1+\alpha_2)} e^{i(\beta_1+\beta_2)} \text{ (from (F) with } \alpha = \alpha_1 + \alpha_2 \text{ and } \beta = \beta_1 + \beta_2) \\
&= e^{\alpha_1} e^{\alpha_2} e^{i(\beta_1+\beta_2)} \text{ (property of the real-valued exponential function)} \\
&= e^{\alpha_1} e^{\alpha_2} e^{i\beta_1} e^{i\beta_2} \text{ (from (b))} \\
&= e^{\alpha_1} e^{i\beta_1} e^{\alpha_2} e^{i\beta_2} = e^{\alpha_1+i\beta_1} e^{\alpha_2+i\beta_2} = e^{z_1} e^{z_2}.
\end{aligned}$$

- (d) The real and imaginary parts of $z_1 = e^{(\lambda+i\omega)x}$ are $u_1 = e^{\lambda x} \cos \omega x$ and $v_1 = e^{\lambda x} \sin \omega x$, which are both solutions of $ay'' + by' + cy = 0$, by Theorem 5.2.1(c). Similarly, the real and imaginary parts of $z_2 = e^{(\lambda-i\omega)x}$ are $u_2 = e^{\lambda x} \cos(-\omega x) = e^{\lambda x} \cos \omega x$ and $v_2 = e^{\lambda x} \sin(-\omega x) = -e^{\lambda x} \sin \omega x$, which are both solutions of $ay'' + by' + cy = 0$, by Theorem 5.2.1(c).

5.3 Nonhomogeneous Linear Equations

5.3.2. The characteristic polynomial of the complementary equation is

$p(r) = r^2 - 4r + 5 = (r-2)^2 + 1$, so $\{e^{2x} \cos x, e^{2x} \sin x\}$ is a fundamental set of solutions for the complementary equation. Let $y_p = A + Bx$; then
 $y_p'' - 4y_p' + 5y_p = -4B + 5(A + Bx) = 1 + 5x$. Therefore, $5B = 5$, $-4B + 5A = 1$, so $B = 1$, $A = 1$. Therefore, $y_p = 1 + x$ is a particular solution and $y = 1 + x + e^{2x}(c_1 \cos x + c_2 \sin x)$ is the general solution.

5.3.4. The characteristic polynomial of the complementary equation is

$p(r) = r^2 - 4r + 4 = (r-2)^2$, so $\{e^{2x}, xe^{2x}\}$ is a fundamental set of solutions for the complementary equation. Let $y_p = A + Bx + Cx^2$; then
 $y_p'' - 4y_p' + 4y_p = 2C - 4(B + 2Cx) + 4(A + Bx + Cx^2) =$
 $(2C - 4B + 4A) + (-8C + 4B)x + 4Cx^2 = 2 + 8x - 4x^2$.
Therefore, $4C = -4$, $-8C + 4B = 8$, $2C - 4B + 4A = 2$, so $C = -1$, $B = 0$, and $A = 1$.
Therefore, $y_p = 1 - x^2$ is a particular solution and $y = 1 - x^2 + e^{2x}(c_1 + c_2 x)$ is the general solution.

5.3.6. The characteristic polynomial of the complementary equation is

$p(r) = r^2 + 6r + 10 = (r+3)^2 + 1$, so $\{e^{-3x} \cos x, e^{-3x} \sin x\}$ is a fundamental set of solutions for the complementary equation. Let $y_p = A + Bx$; then
 $y_p'' + 6y_p' + 10y_p = 6B + 10(A + Bx) = 22 + 20x$. Therefore, $10B = 20$, $6B + 10A = 22$, so $B = 2$, $A = 1$. Therefore, $y_p = 1 + 2x$ is a particular solution and (A)
 $y = 1 + 2x + e^{-3x}(c_1 \cos x + c_2 \sin x)$ is the general solution. Now
 $y(0) = 2 \Rightarrow 2 = 1 + c_1 \Rightarrow c_1 = 1$. Differentiating (A) yields
 $y' = 2 - 3e^{-3x}(c_1 \cos x + c_2 \sin x) + e^{-3x}(-c_1 \sin x + c_2 \cos x)$, so
 $y'(0) = -2 \Rightarrow -2 = 2 - 3c_1 + c_2 \Rightarrow c_2 = -1$. $y = 1 + 2x + e^{-3x}(\cos x - \sin x)$ is the solution of the initial value problem.

5.3.8. If $y_p = \frac{A}{x}$, then $x^2 y_p'' + 7x y_p' + 8y_p = A \left(x^2 \left(\frac{2}{x^3} \right) + 7x \left(\frac{-1}{x^2} \right) + \left(\frac{8}{x} \right) \right) = \frac{3A}{x} = \frac{6}{x}$ if $A = 2$.

Therefore, $y_p = \frac{2}{x}$ is a particular solution.

5.3.10. If $y_p = Ax^3$, then $x^2y_p'' - xy_p' + y_p = A(x^2(6x) - x(3x^2) + x^3) = 4Ax^3 = 2x^3$ if $A = \frac{1}{2}$.

Therefore, $y_p = \frac{x^3}{2}$ is a particular solution.

5.3.12. If $y_p = Ax^{1/3}$, then

$$x^2y_p'' + xy_p' + y_p = A\left(x^2\left(\frac{-2x^{-5/3}}{9}\right) + x\left(\frac{x^{-2/3}}{3}\right) + x^{1/3}\right) = \frac{10A}{9}x^{1/3} = 10x^{1/3} \text{ if } A = 9.$$

Therefore, $y_p = 9x^{1/3}$ is a particular solution.

5.3.14. If $y_p = \frac{A}{x^3}$, then $x^2y_p'' + 3xy_p' - 3y_p = A\left(x^2\left(\frac{12}{x^5}\right) + 3x\left(\frac{-3}{x^4}\right) + \frac{3}{x^3}\right) = 0$. Therefore, y_p is not a solution of the given equation for any choice of A .

5.3.16. The characteristic polynomial of the complementary equation is

$p(r) = r^2 + 5r - 6 = (r + 6)(r - 1)$, so $\{e^{-6x}, e^x\}$ is a fundamental set of solutions for the complementary equation. Let $y_p = Ae^{3x}$; then $y_p'' + 5y_p' - 6y_p = p(3)Ae^{3x} = 18Ae^{3x} = 6e^{3x}$

if $A = \frac{1}{3}$. Therefore, $y_p = \frac{e^{3x}}{3}$ is a particular solution and $y = \frac{e^{3x}}{3} + c_1e^{-6x} + c_2e^x$ is the general solution.

5.3.18. The characteristic polynomial of the complementary equation is

$p(r) = r^2 + 8r + 7 = (r + 1)(r + 7)$, so $\{e^{-7x}, e^{-x}\}$ is a fundamental set of solutions for the complementary equation. Let $y_p = Ae^{-2x}$; then

$y_p'' + 8y_p' + 7y_p = p(-2)Ae^{-2x} = -5Ae^{-2x} = 10e^{-2x}$ if $A = -2$. Therefore, $y_p = -2e^{-2x}$ is a particular solution and (A) $y = -2e^{-2x} + c_1e^{-7x} + c_2e^{-x}$ is the general solution.

Differentiating (A) yields $y' = 4e^{-2x} - 7c_1e^{-7x} - c_2e^{-x}$. Now

$y(0) = -2 \Rightarrow -2 = -2 + c_1 + c_2$ and $y'(0) = 10 \Rightarrow 10 = 4 - 7c_1 - c_2$. Therefore, $c_1 = -1$ and $c_2 = 1$, so $y = -2e^{-2x} - e^{-7x} + e^{-x}$ is the solution of the initial value problem.

5.3.20. The characteristic polynomial of the complementary equation is

$p(r) = r^2 + 2r + 10 = (r + 1)^2 + 9$, so $\{e^{-x} \cos 3x, e^{-x} \sin 3x\}$ is a fundamental set of solutions for the complementary equation. If $y_p = Ae^{x/2}$, then

$$y_p'' + 2y_p' + 10y_p = p(1/2)Ae^{x/2} = \frac{45}{4}Ae^{x/2} = e^{x/2} \text{ if } A = \frac{4}{45}.$$

Therefore, $y_p = \frac{4}{45}e^{x/2}$ is a particular solution and $y = \frac{4}{45}e^{x/2} + e^{-x}(c_1 \cos 3x + c_2 \sin 3x)$ is the general solution.

5.3.22. The characteristic polynomial of the complementary equation is

$p(r) = r^2 - 7r + 12 = (r - 4)(r - 3)$. If $y_p = Ae^{4x}$, then

$y_p'' - 7y_p' + 12y_p = p(4)Ae^{4x} = 0 \cdot e^{4x} = 0$, so $y_p'' - 7y_p' + 12y_p \neq 5e^{4x}$ for any choice of A .

5.3.24. The characteristic polynomial of the complementary equation is

$p(r) = r^2 - 8r + 16 = (r - 4)^2$, so $\{e^{4x}, xe^{4x}\}$ is a fundamental set of solutions for the complementary equation. If $y_p = A \cos x + B \sin x$, then

$y_p'' - 8y_p' + 16y_p = -(A \cos x + B \sin x) - 8(-A \sin x + B \cos x) + 16(A \cos x + B \sin x) = (15A - 8B) \cos x + (8A + 15B) \sin x$, so $15A - 8B = 23$, $8A + 15B = -7$, which implies that $A = 1$ and $B = -1$. Hence $y_p = \cos x - \sin x$ and $y = \cos x - \sin x + e^{4x}(c_1 + c_2x)$ is the general solution.

5.3.26. The characteristic polynomial of the complementary equation is

$p(r) = r^2 - 2r + 3 = (r - 1)^2 + 2$, so $\{e^x \cos \sqrt{2}x, e^x \sin \sqrt{2}x\}$ is a fundamental set of solutions for the complementary equation. If $y_p = A \cos 3x + B \sin 3x$, then
 $y_p'' - 2y_p' + 3y_p = -9(A \cos 3x + B \sin 3x) - 6(-A \sin 3x + B \cos 3x) + 3(A \cos 3x + B \sin 3x) =$
 $-(6A + 6B) \cos 3x + (6A - 6B) \sin 3x$, so $-6A - 6B = -6$, $6A - 6B = 6$, which implies that
 $A = 1$ and $B = 0$. Hence $y_p = \cos 3x$ is a particular solution and
 $y = \cos 3x + e^x(c_1 \cos \sqrt{2}x + c_2 \sin \sqrt{2}x)$ is the general solution.

5.3.28. The characteristic polynomial of the complementary equation is

$p(r) = r^2 + 7r + 12 = (r + 3)(r + 4)$, so $\{e^{-4x}, e^{-3x}\}$ is a fundamental set of solutions for the complementary equation. If $y_p = A \cos 2x + B \sin 2x$, then
 $y_p'' + 7y_p' + 12y_p = -4(A \cos 2x + B \sin 2x) + 14(-A \sin 2x + B \cos 2x) + 12(A \cos 2x + B \sin 2x) =$
 $(8A + 14B) \cos 2x + (8B - 14A) \sin 2x$, so $8A + 14B = -2$, $-14A + 8B = 36$, which implies
that $A = -2$ and $B = 1$. Hence $y_p = -2 \cos 2x + \sin 2x$ is a particular solution and (A)
 $y = -2 \cos 2x + \sin 2x + c_1 e^{-4x} + c_2 e^{-3x}$ is the general solution. Differentiating (A) yields
 $y' = 2 \sin 2x + 2 \cos 2x - 4c_1 e^{-4x} - 3c_2 e^{-3x}$. Now $y(0) = -3 \Rightarrow -3 = -2 + c_1 + c_2$ and
 $y'(0) = 3 \Rightarrow 3 = 2 - 4c_1 - 3c_2$. Therefore, $c_1 = 2$ and $c_2 = -3$, so
 $y = -2 \cos 2x + \sin 2x + 2e^{-4x} - 3e^{-3x}$ is the solution of the initial value problem.

5.3.30. $\{\cos \omega_0 x, \sin \omega_0 x\}$ is a fundamental set of solutions of the complementary equation. If

$y_p = A \cos \omega x + B \sin \omega x$, then
 $y_p'' + \omega_0^2 y_p = -\omega^2(A \cos \omega x + B \sin \omega x) + \omega_0^2(A \cos \omega x + B \sin \omega x) =$
 $(\omega_0^2 - \omega^2)(A \cos \omega x + B \sin \omega x) = M \cos \omega x + N \sin \omega x$ if $A = \frac{M}{\omega_0^2 - \omega^2}$ and $B = \frac{N}{\omega_0^2 - \omega^2}$.

Therefore,

$$y_p = \frac{1}{\omega_0^2 - \omega^2} (M \cos \omega x + N \sin \omega x)$$

is a particular solution of the given equation and

$$y = \frac{1}{\omega_0^2 - \omega^2} (M \cos \omega x + N \sin \omega x) + c_1 \cos \omega_0 x + c_2 \sin \omega_0 x$$

is the general solution.

5.3.32. If $y_p = A \cos \omega x + B \sin \omega x$, then

$ay_p'' + by_p' + cy_p = -a\omega^2(A \cos \omega x + B \sin \omega x) + b\omega(-A \sin \omega x + B \cos \omega x) + c(A \cos \omega x +$
 $B \sin \omega x) = [(c - a\omega^2)A + b\omega B] \cos \omega x + [-b\omega A + (c - a\omega^2)B] \sin \omega x$. Therefore, y_p is a
solution of (A) if and only if the set of equations (B)
 $(c - a\omega^2)A + b\omega B = M$, $-b\omega A + (c - a\omega^2)B = N$ has a solution. If $(c - a\omega^2)^2 + (b\omega)^2 \neq 0$,
then (B) has the solution $A = \frac{(c - a\omega^2)M - b\omega N}{(c - a\omega^2)^2 + (b\omega)^2}$, $B = \frac{(c - a\omega^2)N + b\omega M}{(c - a\omega^2)^2 + (b\omega)^2}$, and
 $y_p = A \cos \omega x + B \sin \omega x$ is a solution of (A). If $(c - a\omega^2)^2 + (b\omega)^2 = 0$ (which is true if and
only if the left side of (A) is of the form $a(y'' + \omega^2 y)$), then the coefficients of A and B in (B)
are all zero, so (B) does not have a solution, so (A) does not have a solution of the form
 $y_p = A \cos \omega x + B \sin \omega x$.

5.3.34. From Exercises 5.3.2 and 5.3.17, $y_{p_1} = 1 + x$ and $y_{p_2} = e^{2x}$ are particular solutions of
 $y'' - 4y' + 5y = 1 + 5x$ and $y'' - 4y' + 5y = e^{2x}$ respectively, and $\{e^{2x} \cos x, e^{2x} \sin x\}$ is a

fundamental set of solutions of the complementary equation.

Therefore, $y_p = y_{p_1} + y_{p_2} = 1 + x + e^{2x}$ is a particular solution of the given equation, and $y = 1 + x + e^{2x}(1 + c_1 \cos x + c_2 \sin x)$ is the general solution.

5.3.36. From Exercises 5.3.4 and 5.3.19, $y_{p_1} = 1 - x^2$ and $y_{p_2} = e^x$ are particular solutions of $y'' - 4y' + 4y = 2 + 8x - 4x^2$ and $y'' - 4y' + 4y = e^x$ respectively, and $\{e^{2x}, xe^{2x}\}$ is a fundamental set of solutions of the complementary equation.

Therefore, $y_p = y_{p_1} + y_{p_2} = 1 - x^2 + e^x$ is a particular solution of the given equation, and $y = 1 - x^2 + e^x + e^{2x}(c_1 + c_2 x)$ is the general solution.

5.3.38. From Exercises 5.3.6 and 5.3.21, $y_{p_1} = 1 + 2x$ and $y_{p_2} = e^{-3x}$ are particular solutions of $y'' + 6y' + 10y = 22 + 20x$ and $y'' + 6y' + 10y = e^{-3x}$ respectively, and $\{e^{-3x} \cos x, e^{-3x} \sin x\}$ is a fundamental set of solutions of the complementary equation.

Therefore, $y_p = y_{p_1} + y_{p_2} = 1 + 2x + e^{-3x}$ is a particular solution of the given equation, and $y = 1 + 2x + e^{-3x}(1 + c_1 \cos x + c_2 \sin x)$ is the general solution.

5.3.40. Letting $c_1 = c_2 = 0$ shows that (A) $y_p'' + p(x)y_p' + q(x)y_p = f$. Letting $c_1 = 1$ and $c_2 = 0$ shows that (B) $(y_1 + y_p)'' + p(x)(y_1 + y_p)' + q(x)(y_1 + y_p) = f$. Now subtract (A) from (B) to see that $y_1'' + p(x)y_1' + q(x)y_1 = 0$. Letting $c_1 = 0$ and $c_2 = 1$ shows that (C) $(y_2 + y_p)'' + p(x)(y_2 + y_p)' + q(x)(y_2 + y_p) = f$. Now subtract (A) from (C) to see that $y_2'' + p(x)y_2' + q(x)y_2 = 0$.

5.4 The Method of Undetermined Coefficients I

5.4.2. If $y = ue^{-3x}$, then $y'' - 6y' + 5y = e^{-3x}[(u'' - 6u' + 9u) - 6(u' - 3u) + 5u] = e^{-3x}(35 - 8x)$, so $u'' - 12u' + 32u = 35 - 8x$ and $u_p = A + Bx$, where $-12B + 32(A + Bx) = 35 - 8x$.

Therefore, $32B = -8$, $32A - 12B = 35$, so $B = -\frac{1}{4}$, $A = 1$, and $u_p = 1 - \frac{x}{4}$. Therefore,

$$y_p = e^{-3x} \left(1 - \frac{x}{4} \right).$$

5.4.4. If $y = ue^{2x}$, then

$y'' + 2y' + y = e^{2x}[(u'' + 4u' + 4u) + 2(u' + 2u) + u] = e^{2x}(-7 - 15x + 9x^2)$ so $u'' + 6u' + 9u = -7 - 15x + 9x^2$ and $u_p = A + Bx + Cx^2$, where

$2C + 6(B + 2Cx) + 9(A + Bx + Cx^2) = -7 - 15x + 9x^2$. Therefore, $9C = 9$,

$9B + 12C = -15$, $9A + 6B + 2C = -7$, so $C = 1$, $B = -3$, $A = 1$, and $u_p = 1 - 3x + x^2$.

Therefore, $y_p = e^{2x}(1 - 3x + x^2)$.

5.4.6. If $y = ue^x$, then $y'' - y' - 2y = e^x[(u'' + 2u' + u) - (u' + u) - 2u] = e^x(9 + 2x - 4x^2)$ so $u'' + u' - 2u = 9 + 2x - 4x^2$, and $u_p = A + Bx + Cx^2$, where

$2C + (B + 2Cx) - 2(A + Bx + Cx^2) = 9 + 2x - 4x^2$. Therefore, $-2C = -4$, $-2B + 2C = 2$, $-2A + B + 2C = 9$, so $C = 2$, $B = 1$, $A = -2$, and $u_p = -2 + x + 2x^2$. Therefore,

$$y_p = e^x(-2 + x + 2x^2).$$

5.4.8. If $y = ue^x$, then $y'' - 3y' + 2y = e^x[(u'' + 2u' + u) - 3(u' + u) + 2u] = e^x(3 - 4x)$, so $u'' - u' = 3 - 4x$ and $u_p = Ax + Bx^2$, where $2B - (A + 2Bx) = 3 - 4x$. Therefore, $-2B = -4$, $-A + 2B = 3$, so $B = 2$, $A = 1$, and $u_p = x(1 + 2x)$. Therefore, $y_p = xe^x(1 + 2x)$.

5.4.10. If $y = ue^{2x}$, then

$$\begin{aligned} 2y'' - 3y' - 2y &= e^{2x} [2(u'' + 4u' + 4u) - 3(u' + 2u) - 2u] = e^{2x}(-6 + 10x), \text{ so} \\ 2u'' + 5u' &= -6 + 10x \text{ and } u_p = Ax + Bx^2, \text{ where } 2(2B) + 5(A + 2Bx) = -6 + 10x. \\ \text{Therefore, } 10B &= 10, 5A + 4B = -6, \text{ so } B = 1, A = -2, \text{ and } u_p = x(-2 + x). \\ \text{Therefore, } y_p &= xe^{2x}(-2 + x). \end{aligned}$$

5.4.12. If $y = ue^x$, then $y'' - 2y' + y = e^x [(u'' + 2u' + u) - 2(u' + u) + u] = e^x(1 - 6x)$, so $u'' = 1 - 6x$. Integrating twice and taking the constants of integration to be zero yields $u_p = x^2 \left(\frac{1}{2} - x \right)$. Therefore, $y_p = x^2 e^x \left(\frac{1}{2} - x \right)$.

5.4.14. If $y = ue^{-x/3}$, then

$$\begin{aligned} 9y'' + 6y' + y &= e^{-x/3} \left[9 \left(u'' - \frac{2u'}{3} + \frac{u}{9} \right) + 6 \left(u' - \frac{u}{3} \right) + u \right] = e^{-x/3}(2 - 4x + 4x^2), \text{ so} \\ 9u'' &= 2 - 4x + 4x^2, \text{ or } u'' = \frac{1}{9}(2 - 4x + 4x^2). \text{ Integrating twice and taking the constants of} \\ \text{integration to be zero yields } u_p &= \frac{x^2}{27}(3 - 2x + x^2). \text{ Therefore, } y_p = \frac{x^2 e^{-x/3}}{27}(3 - 2x + x^2). \end{aligned}$$

5.4.16. If $y = ue^x$, then $y'' - 6y' + 8y = e^x [(u'' + 2u' + u) - 6(u' + u) + 8u] = e^x(11 - 6x)$, so $u'' - 4u' + 3u = 11 - 6x$ and $u_p = A + Bx$, where $-4B + 3(A + Bx) = 11 - 6x$. Therefore, $3B = -6$, $3A - 4B = 11$, so $B = -2$, $A = 1$ and $u_p = 1 - 2x$. Therefore, $y_p = e^x(1 - 2x)$. The characteristic polynomial of the complementary equation is $p(r) = r^2 - 6r + 8 = (r - 2)(r - 4)$, so $\{e^{2x}, e^{4x}\}$ is a fundamental set of solutions of the complementary equation. Therefore, $y = e^x(1 - 2x) + c_1 e^{2x} + c_2 e^{4x}$ is the general solution of the nonhomogeneous equation.

5.4.18. If $y = ue^x$, then $y'' + 2y' - 3y = e^x [(u'' + 2u' + u) + 2(u' + u) - 3u] = -16xe^x$, so $u'' + 4u' = -16x$ and $u_p = Ax + Bx^2$, where $2B + 4(A + 2Bx) = -16x$. Therefore, $8B = -16$, $4A + 2B = 0$, so $B = -2$, $A = 1$, and $u_p = x(1 - 2x)$. Therefore, $y_p = xe^x(1 - 2x)$. The characteristic polynomial of the complementary equation is $p(r) = r^2 + 2r - 3 = (r + 3)(r - 1)$, so $\{e^x, e^{-3x}\}$ is a fundamental set of solutions of the complementary equation. Therefore, $y = xe^x(1 - 2x) + c_1 e^x + c_2 e^{-3x}$ is the general solution of the nonhomogeneous equation.

5.4.20. If $y = ue^{2x}$, then $y'' - 4y' - 5y = e^{2x} [(u'' + 4u' + 4u) - 4(u' + 2u) - 5u] = 9e^{2x}(1 + x)$, so $u'' - 9u = 9 + 9x$ and $u_p = A + Bx$, where $-9(A + Bx) = 9 + 9x$. Therefore, $-9B = -9$, $-9A = 9$, so $B = -1$, $A = -1$, and $u_p = -1 - x$. Therefore, $y_p = -e^{2x}(1 + x)$. The characteristic polynomial of the complementary equation is $p(r) = r^2 - 4r - 5 = (r - 5)(r + 1)$, so $\{e^{-x}, e^{5x}\}$ is a fundamental set of solutions of the complementary equation. Therefore, (A) $y = -e^{2x}(1 + x) + c_1 e^{-x} + c_2 e^{5x}$ is the general solution of the nonhomogeneous equation. Differentiating (A) yields $y' = -2e^{2x}(1 + x) - e^{2x} - c_1 e^{-x} + 5c_2 e^{5x}$. Now $y(0) = 0$, $y'(0) = -10 \Rightarrow 0 = -1 + c_1 + c_2$, $-10 = -3 - c_1 + 5c_2$, so $c_1 = 2$, $c_2 = -1$. Therefore, $y = -e^{2x}(1 + x) + 2e^{-x} - e^{5x}$ is the solution of the initial value problem.

5.4.22. If $y = ue^{-x}$, then $y'' + 4y' + 3y = e^{-x} [(u'' - 2u' + u) + 4(u' - u) + 3u] = -e^{-x}(2 + 8x)$, so $u'' + 2u' = -2 - 8x$ and $u_p = Ax + Bx^2$, where $2B + 2(A + 2Bx) = -2 - 8x$.

Therefore, $4B = -8$, $2A + 2B = -2$, so $B = -2$, $A = 1$, and $u_p = x(1 - 2x)$.

Therefore, $y_p = xe^{-x}(1 - 2x)$. The characteristic polynomial of the complementary equation is $p(r) = r^2 + 4r + 3 = (r + 3)(r + 1)$, so $\{e^{-x}, e^{-3x}\}$ is a fundamental set of solutions of the complementary equation. Therefore, (A) $y = xe^{-x}(1 - 2x) + c_1e^{-x} + c_2e^{-3x}$ is the general solution of the nonhomogeneous equation. Differentiating (A) yields

$$y' = -xe^{-x}(1 - 2x) + e^{-x}(1 - 4x) - c_1e^{-x} - 3c_2e^{-3x}. \text{ Now}$$

$$y(0) = 1, y'(0) = 2 \Rightarrow 1 = c_1 + c_2, 2 = 1 - c_1 - 3c_2, \text{ so } c_1 = 2, c_2 = -1.$$

Therefore, $y = e^{-x}(2 + x - 2x^2) - e^{-3x}$ is the solution of the initial value problem.

5.4.24. We must find particular solutions y_{p_1} and y_{p_2} of (A) $y'' + y' + y = xe^x$ and (B)

$y'' + y' + y = e^{-x}(1 + 2x)$, respectively. To find a particular solution of (A) we write

$$y = ue^x. \text{ Then } y'' + y' + y = e^x[(u'' + 2u' + u) + (u' + u) + u] = xe^x \text{ so } u'' + 3u' + 3u = x$$

and $u_p = A + Bx$, where $3B + 3(A + Bx) = x$. Therefore, $3B = 1$, $3A + 3B = 0$, so $B = \frac{1}{3}$,

$A = -\frac{1}{3}$, and $u_p = -\frac{1}{3}(1 - x)$, so $y_{p_1} = -\frac{e^x}{3}(1 - x)$. To find a particular solution of (B) we

write $y = ue^{-x}$. Then $y'' + y' + y = e^{-x}[(u'' - 2u' + u) + (u' - u) + u] = e^{-x}(1 + 2x)$, so $u'' - u' + u = 1 + 2x$ and $u_p = A + Bx$, where $-B + (A + Bx) = 1 + 2x$. Therefore, $B = 2$, $A - B = 1$, so $A = 3$, and $u_p = 2 + 3x$, so $y_{p_2} = e^{-x}(3 + 2x)$. Now

$$y_p = y_{p_1} + y_{p_2} = -\frac{e^x}{3}(1 - x) + e^{-x}(3 + 2x).$$

5.4.26. We must find particular solutions y_{p_1} and y_{p_2} of (A) $y'' - 8y' + 16y = 6xe^{4x}$ and (B)

$y'' - 8y' + 16y = 2 + 16x + 16x^2$, respectively. To find a particular solution of (A) we write

$$y = ue^{4x}. \text{ Then } y'' - 8y' + 16y = e^x[(u'' + 8u' + 16u) - 8(u' + 4u) + 16u] = 6xe^{4x}, \text{ so}$$

$u'' = 6x$, $u_p = x^3$, and $y_{p_1} = x^3e^{4x}$. To find a particular solution of (B) we write

$$y_p = A + Bx + Cx^2. \text{ Then } y_p'' - 8y_p' + 16y_p = 2C - 8(B + 2Cx) + 16(A + Bx + Cx^2) = (16A - 8B + 2C) + (16B - 16C)x + 16Cx^2 = 2 + 16x + 16x^2 \text{ if } 16C = 16, 16B - 16C = 16, 16A - 8B + 2C = 2. \text{ Therefore, } C = 1, B = 2, A = 1, \text{ and } y_{p_2} = 1 + 2x + x^2. \text{ Now}$$

$$y_p = y_{p_1} + y_{p_2} = x^3e^{4x} + 1 + 2x + x^2.$$

5.4.28. We must find particular solutions y_{p_1} and y_{p_2} of (A) $y'' - 2y' + 2y = e^x(1 + x)$ and (B)

$y'' - 2y' + 2y = e^{-x}(2 - 8x + 5x^2)$, respectively. To find a particular solution of (A) we write

$$y = ue^x. \text{ Then } y'' - 2y' + 2y = e^x[(u'' + 2u' + u) - 2(u' + u) + 2u] = e^x(1 + x), \text{ so}$$

$u'' + u = 1 + x$ and $u_p = 1 + x$, so $y_{p_1} = e^x(1 + x)$. To find a particular solution of (B) we

write $y = ue^{-x}$. Then

$$y'' - 2y' + 2y = e^{-x}[(u'' - 2u' + u) - 2(u' - u) + 2u] = e^{-x}(2 - 8x + 5x^2), \text{ so}$$

$$u'' - 4u' + 5u = 2 - 8x + 5x^2 \text{ and } u_p = A + Bx + Cx^2, \text{ where}$$

$$2C - 4(B + 2Cx) + 5(A + Bx + Cx^2) = 2 - 8x + 5x^2. \text{ Therefore, } 5C = 5, 5B - 8C = -8,$$

$$5A - 4B + 2C = 2, \text{ so } C = 1, B = 0, A = 0, \text{ and } u_p = x^2. \text{ Therefore, } y_{p_2} = x^2e^{-x}. \text{ Now}$$

$$y_p = y_{p_1} + y_{p_2} = e^x(1 + x) + x^2e^{-x}.$$

5.4.30. (a) If $y = ue^{\alpha x}$, then $ay'' + by' + cy = e^{\alpha x}[a(u'' + 2\alpha u' + \alpha^2 u) + b(u' + \alpha u) + cu] = e^{\alpha x}[au'' + (2a\alpha + b)u' + (a\alpha^2 + b\alpha + c)u] = e^{\alpha x}(au'' + p'(\alpha)u' + p(\alpha)u)$.

Therefore, $ay'' + by' + cy = e^{\alpha x}G(x)$ if and only if $au'' + p'(\alpha)u' + p(\alpha)u = G(x)$.

(b) Substituting $u_p = A + Bx + Cx^2 + Dx^3$ into (B) yields

$$a(2C + 6Dx) + p'(\alpha)(B + 2Cx + 3Dx^2) + p(\alpha)(A + Bx + Cx^2 + Dx^3)$$

$$= [p(\alpha)A + p'(\alpha)B + 2aC] + [p(\alpha)B + 2p'(\alpha)C + 6aD]x \\ + [p(\alpha)C + 3p'(\alpha)D]x^2 + p(\alpha)Dx^3 = g_0 + g_1x + g_2x^2 + g_3x^3$$

if

$$\begin{aligned} p(\alpha)D &= g_3 \\ p(\alpha)C + 3p'(\alpha)D &= g_2 \\ p(\alpha)B + 2p'(\alpha)C + 6aD &= g_1 \\ p(\alpha)A + p'(\alpha)B + 2aC &= g_0. \end{aligned} \tag{C}$$

Since $e^{\alpha x}$ is not a solution of the complementary equation, $p(\alpha) \neq 0$. Therefore, the triangular system (C) can be solved successively for D , C , B and A .

- (c) Since $e^{\alpha x}$ is a solution of the complementary equation while $xe^{\alpha x}$ is not, $p(\alpha) = 0$ and $p'(\alpha) \neq 0$. Therefore, (B) reduces to (D) $au'' + p'(\alpha)u = G(x)$. Substituting $u_p = Ax + Bx^2 + Cx^3 + Dx^4$ into (D) yields

$$\begin{aligned} a(2B + 6Cx + 12Dx^2) + p'(\alpha)(A + 2Bx + 3Cx^2 + 4Dx^3) \\ = (p'(\alpha)A + 2aB) + (2p'(\alpha)B + 6aC)x + (3p'(\alpha)C + 12aD)x^2 \\ + 4p'(\alpha)Dx^3 = g_0 + g_1x + g_2x^2 + g_3x^3 \end{aligned}$$

if

$$\begin{aligned} 4p'(\alpha)D &= g_3 \\ 3p'(\alpha)C + 12aD &= g_2 \\ 2p'(\alpha)B + 6aC &= g_1 \\ p'(\alpha)A + 2aB &= g_0. \end{aligned}$$

Since $p'(\alpha) \neq 0$ this triangular system can be solved successively for D , C , B and A .

- (d) Since $e^{\alpha x}$ and $xe^{\alpha x}$ are solutions of the complementary equation, $p(\alpha) = 0$ and $p'(\alpha) = 0$. Therefore, (B) reduces to (D) $au'' = G(x)$, so $u'' = \frac{G(x)}{a}$. Integrating this twice and taking the constants of integration yields the particular solution
- $$u_p = x^2 \left(\frac{g_0}{2} + \frac{g_1}{6}x + \frac{g_2}{12}x^2 + \frac{g_3}{20}x^3 \right).$$

5.4.32. If $y_p = Axe^{4x}$, then $y_p'' - 7y_p' + 12y_p = [(8 + 16x) - 7(1 + 4x) + 12x]Ae^{4x} = Ae^{4x} = 5e^{4x}$ if $A = 1$, so $y_p = 5xe^{4x}$.

5.4.34. If $y_p = e^{3x}(A + Bx + Cx^2)$, then

$$\begin{aligned} y_p'' - 3y_p' + 2y_p &= e^{3x}[(9A + 6B + 2C) + (9B + 12C)x + 9Cx^2] \\ &\quad - 3e^{3x}[(3A + B) + (3B + 2C)x + 3Cx^2] \\ &\quad + 2e^{3x}(A + Bx + Cx^2) \\ &= e^{3x}[(2A + 3B + 2C) + (2B + 6C)x + 2Cx^2] \\ &= e^{3x}(-1 + 2x + x^2) \end{aligned}$$

if $2C = 1$, $2B + 6C = 2$, $2A + 3B + 2C = -1$. Therefore, $C = \frac{1}{2}$, $B = -\frac{1}{2}$, $A = -\frac{1}{4}$, and

$$y_p = -\frac{e^{3x}}{4}(1 + 2x - 2x^2).$$

5.4.36. If $y_p = e^{-x/2}(Ax^2 + Bx^3 + Cx^4)$, then

$$\begin{aligned} 4y_p'' + 4y_p' + y_p &= e^{-x/2}[8A - (8A - 24B)x + (A - 12B + 48C)x^2] \\ &\quad + e^{-x/2}[(B - 16C)x^3 + Cx^4] \\ &\quad + e^{-x/2}[8Ax - (2A - 12B)x^2 - (2B - 16C)x^3 - 2Cx^4] \\ &\quad + e^{-x/2}(Ax^2 + Bx^3 + Cx^4) \\ &= e^{-x/2}(8A + 24Bx + 48Cx^2) = e^{-x/2}(-8 + 48x + 144x^2) \end{aligned}$$

if $48C = 144$, $24B = 48$, and $8A = -8$. Therefore, $C = 3$, $B = 2$, $A = -1$, and

$$y_p = x^2 e^{-x/2}(-1 + 2x + 3x^2).$$

5.4.38. If $y = \int e^{\alpha x} P(x) dx$, then $y' = e^{\alpha x} P(x)$. Let $y = ue^{\alpha x}$; then $(u' + \alpha u)e^{\alpha x} = e^{\alpha x} P(x)$, which implies (A). We must show that it is possible to choose $A_0, \dots, A_k$ so that (B) $(A_0 + A_1 x \cdots + A_k x^k)' + \alpha(A_0 + A_1 x \cdots + A_k x^k) = p_0 + p_1 x + \cdots + p_k x^k$. By equating the coefficients of $x^k, x^{k-1}, \dots, 1$ (in that order) on the two sides of (B), we see that (B) holds if and only if $\alpha A_k = p_k$ and $(k - j + 1)A_{k-j+1} + \alpha A_k = p_{k-j}$, $1 \leq j \leq k$.

5.4.40. If $y = \int x^k e^{\alpha x} dx$, then $y' = x^k e^{\alpha x}$. Let $y = ue^{\alpha x}$; then $(u' + \alpha u)e^{\alpha x} = x^k e^{\alpha x}$, so $u' + \alpha u = x^k$. This equation has a particular solution $u_p = A_0 + A_1 x \cdots + A_k x^k$, where (A) $(A_0 + A_1 x \cdots + A_k x^k)' + \alpha(A_0 + A_1 x \cdots + A_k x^k) = x^k$. By equating the coefficients of $x^k, x^{k-1}, \dots, 1$ on the two sides of (A), we see that (A) holds if and only if $\alpha A_k = 1$ and $(k - j + 1)A_{k-j+1} + \alpha A_k - j = 0$, $1 \leq j \leq k$. Therefore, $A_k = \frac{1}{\alpha}$, $A_{k-1} = -\frac{k}{\alpha^2}$,

$A_{k-2} = \frac{k(k-1)}{\alpha^3}$, and, in general,

$$A_{k-j} = (-1)^j \frac{k(k-1) \cdots (k-j+1)}{\alpha^{j+1}} = \frac{(-1)^j k!}{\alpha^{j+1} (k-j)!}, \quad 1 \leq j \leq k. \text{ By introducing the index}$$

$r = k - j$ we can rewrite this as $A_r = \frac{(-1)^{k-r} k!}{\alpha^{k-r+1} r!}$, $0 \leq r \leq k$. Therefore,

$$u_p = \frac{(-1)^k k!}{\alpha^{k+1}} \sum_{r=0}^k \frac{(-\alpha x)^r}{r!} \text{ and } y = \frac{(-1)^k k! e^{\alpha x}}{\alpha^{k+1}} \sum_{r=0}^k \frac{(-\alpha x)^r}{r!} + c.$$

5.5 The Method of Undetermined Coefficients II

5.5.2. Let

$$\begin{aligned} y_p &= (A_0 + A_1 x) \cos x + (B_0 + B_1 x) \sin x; \text{ then} \\ y_p' &= (A_1 + B_0 + B_1 x) \cos x + (B_1 - A_0 - A_1 x) \sin x \\ y_p'' &= (2B_1 - A_0 - A_1 x) \cos x - (2A_1 + B_0 + B_1 x) \sin x, \text{ so} \\ y_p'' + 3y_p' + y_p &= (3A_1 + 3B_0 + 2B_1 + 3B_1 x) \cos x \end{aligned}$$

$$\begin{aligned}
& + (3B_1 - 3A_0 - 2A_1 - 3A_1x) \sin x \\
& = (2 - 6x) \cos x - 9 \sin x
\end{aligned}$$

if $3B_1 = -6$, $-3A_1 = 0$, $3B_0 + 3A_1 + 2B_1 = 2$, $-3A_0 + 3B_1 + 2A_1 = -9$. Therefore, $A_1 = 0$, $B_1 = -2$, $A_0 = 1$, $B_0 = 2$, and $y_p = \cos x + (2 - 2x) \sin x$.

5.5.4. Let $y = ue^{2x}$. Then

$$\begin{aligned}
y'' + 3y' - 2y &= e^{2x} [(u'' + 4u' + 4u) + 3(u' + 2u) - 2u] \\
&= e^{2x} (u'' + 7u' + 8u) = -e^{2x} (5 \cos 2x + 9 \sin 2x)
\end{aligned}$$

if $u'' + 7u' + 8u = -5 \cos 2x - 9 \sin 2x$. Now let $u_p = A \cos 2x + B \sin 2x$. Then

$$\begin{aligned}
u_p'' + 7u_p' + 8u_p &= -4(A \cos 2x + B \sin 2x) + 14(-A \sin 2x + B \cos 2x) \\
&\quad + 8(A \cos 2x + B \sin 2x) \\
&= (4A + 14B) \cos 2x - (14A - 4B) \sin 2x \\
&= -5 \cos 2x - 9 \sin 2x
\end{aligned}$$

if $4A + 14B = -5$, $-14A + 4B = -9$. Therefore, $A = \frac{1}{2}$, $B = -\frac{1}{2}$,

$$u_p = \frac{1}{2}(\cos 2x - \sin 2x), \text{ and } y_p = \frac{e^{2x}}{2}(\cos 2x - \sin 2x).$$

5.5.6. Let $y = ue^{-2x}$. Then

$$\begin{aligned}
y'' + 3y' - 2y &= e^{-2x} [(u'' - 4u' + 4u) + 3(u' - 2u) - 2u] \\
&= e^{-2x} (u'' - u' - 4u) \\
&= e^{-2x} [(4 + 20x) \cos 3x + (26 - 32x) \sin 3x]
\end{aligned}$$

if $u'' - u' - 4u = (4 + 20x) \cos 3x + (26 - 32x) \sin 3x$. Let

$$\begin{aligned}
u_p &= (A_0 + A_1x) \cos 3x + (B_0 + B_1x) \sin 3x; \text{ then} \\
u_p' &= (A_1 + 3B_0 + 3B_1x) \cos 3x + (B_1 - 3A_0 - 3A_1x) \sin 3x \\
u_p'' &= (6B_1 - 9A_0 - 9A_1x) \cos 3x - (2A_1 + 9B_0 + 9B_1x) \sin 3x, \text{ so} \\
u_p'' - u_p' - 4u_p &= -[13A_0 + A_1 + 3B_0 - 6B_1 + (13A_1 + 3B_1)x] \cos 3x \\
&\quad - [13B_0 + B_1 - 3A_0 + 6A_1 + (13B_1 - 3A_1)x] \sin 3x \\
&= (4 + 20x) \cos 3x + (26 - 32x) \sin 3x \quad \text{if}
\end{aligned}$$

$$\begin{cases} -13A_1 - 3B_1 = 20 \\ 3A_1 - 13B_1 = -32 \end{cases} \quad \text{and} \quad \begin{cases} 13A_0 - 3B_0 - A_1 + 6B_1 = 4 \\ 3A_0 - 13B_0 - 6A_1 - B_1 = 26. \end{cases}$$

From the first two equations, $A_1 = -2$, $B_1 = 2$. Substituting these in the last two equations yields $-13A_0 - 3B_0 = -10$, $3A_0 - 13B_0 = 16$. Solving this pair yields $A_0 = 1$, $B_0 = -1$. Therefore,

$$u_p = (1 - 2x)(\cos 3x - \sin 3x) \text{ and } y_p = e^{-2x}(1 - 2x)(\cos 3x - \sin 3x).$$

5.5.8. Let

$$\begin{aligned}
 y_p &= (A_0x + A_1x^2) \cos x + (B_0x + B_1x^2) \sin x; \text{ then} \\
 y'_p &= [A_0 + (2A_1 + B_0)x + B_1x^2] \cos x + [B_0 + (2B_1 - A_0)x - B_1x^2] \sin x \\
 y''_p &= [2A_1 + 2B_0 - (A_0 - 4B_1)x - A_1x^2] \cos x \\
 &\quad + [2B_1 - 2A_0 - (B_0 + 4A_1)x - B_1x^2] \sin x, \text{ so} \\
 y''_p + y_p &= (2A_1 + 2B_0 + 4B_1x) \cos x + (2B_1 - 2A_0 - 4A_1x) \sin x \\
 &= (-4 + 8x) \cos x + (8 - 4x) \sin x
 \end{aligned}$$

if $4B_1 = 8$, $-4A_1 = -4$, $2B_0 + 2A_1 = -4$, $-2A_0 + 2B_1 = 8$. Therefore, $A_1 = 1$, $B_1 = 2$, $A_0 = -2$, $B_0 = -3$, and $y_p = -x[(2 - x) \cos x + (3 - 2x) \sin x]$.

5.5.10. Let $y = ue^{-x}$. Then

$$\begin{aligned}
 y'' + 2y' + 2y &= e^{-x} [(u'' - 2u' + u) + 2(u' - u) + 2u] \\
 &= e^{-x}(u'' + u) = e^{-x}(8 \cos x - 6 \sin x)
 \end{aligned}$$

if $u'' + u = 8 \cos x - 6 \sin x$. Now let

$$\begin{aligned}
 u_p &= Ax \cos x + Bx \sin x; \text{ then} \\
 u'_p &= (A + Bx) \cos x + (B - Ax) \sin x \\
 u''_p &= (2B - Ax) \cos x - (2A + Bx) \sin x, \text{ so} \\
 u''_p + u_p &= 2B \cos x - 2A \sin x = 8 \cos x - 6 \sin x
 \end{aligned}$$

if $2B = 8$, $-2A = -6$. Therefore, $A = 3$, $B = 4$, $u_p = x(3 \cos x + 4 \sin x)$, and $y_p = xe^{-x}(3 \cos x + 4 \sin x)$.

5.5.12. Let

$$\begin{aligned}
 y_p &= (A_0 + A_1x + A_2x^2) \cos x + (B_0 + B_1x + B_2x^2) \sin x; \text{ then} \\
 y'_p &= [A_1 + B_0 + (2A_2 + B_1)x + B_2x^2] \cos x \\
 &\quad + [B_1 - A_0 + (2B_2 - A_1)x - A_2x^2] \sin x, \\
 y''_p &= [-A_0 + 2A_2 + 2B_1 - (A_1 - 4B_2)x - A_2x^2] \cos x \\
 &\quad + [-B_0 + 2B_2 - 2A_1 - (B_1 + 4A_2)x - B_2x^2] \sin x, \text{ so} \\
 y''_p + 2y'_p + y_p &= 2[A_1 + A_2 + B_0 + B_1 + (2A_2 + B_1 + 2B_2)x + B_2x^2] \cos x \\
 &\quad + 2[B_1 + B_2 - A_0 - A_1 + (2B_2 - A_1 - 2A_2)x - A_2x^2] \sin x \\
 &= 8x^2 \cos x - 4x \sin x \quad \text{if} \\
 \text{(i)} \quad \begin{cases} 2B_2 &= 8 \\ -2A_2 &= 0 \end{cases}, \quad \text{(ii)} \quad \begin{cases} 2B_1 + 4A_2 + 4B_2 &= 0 \\ -2A_1 - 4A_2 + 4B_2 &= -4 \end{cases}, \quad \text{and} \\
 \text{(iii)} \quad \begin{cases} 2B_0 + 2A_1 + 2B_1 + 2A_2 &= 0 \\ -2A_0 - 2A_1 + 2B_1 + 2B_2 &= 0 \end{cases}.
 \end{aligned}$$

From (i), $A_2 = 0$, $B_2 = 4$. Substituting these into (ii) and solving for A_1 and B_1 yields $A_1 = 10$, $B_1 = -8$. Substituting the known coefficients into (iii) and solving for A_0 and B_0 yields $A_0 = -14$, $B_0 = -2$. Therefore, $y_p = -(14 - 10x) \cos x - (2 + 8x - 4x^2) \sin x$.

5.5.14. Let

$$\begin{aligned}
 y_p &= (A_0 + A_1x + A_2x^2) \cos 2x + (B_0 + B_1x + B_2x^2) \sin 2x; \text{ then} \\
 y'_p &= [A_1 + 2B_0 + (2A_2 + 2B_1)x + 2B_2x^2] \cos 2x \\
 &\quad + [B_1 - 2A_0 + (2B_2 - 2A_1)x - 2A_2x^2] \sin 2x \\
 y''_p &= [-4A_0 + 2A_2 + 4B_1 - (4A_1 - 8B_2)x - 4A_2x^2] \cos 2x \\
 &\quad + [-4B_0 - 2B_2 - 4A_1 - (4B_1 + 8A_2)x - 4B_2x^2] \sin 2x, \text{ so} \\
 y''_p + 3y'_p + 2y_p &= [-2A_0 + 3A_1 + 4A_2 + 6B_0 + 4B_1 \\
 &\quad - (2A_1 - 6A_2 - 6B_1 - 8B_2)x - (2A_2 - 6B_2)x^2] \cos 2x \\
 &\quad + [-2B_0 + 3B_1 + 4B_2 - 6A_0 - 4A_1 \\
 &\quad - (2B_1 - 6B_2 + 6A_1 + 8A_2)x - (2B_2 + 6A_2)x^2] \sin 2x \\
 &= (1 - x - 4x^2) \cos 2x - (1 + 7x + 2x^2) \sin 2x \quad \text{if} \\
 \text{(i)} \begin{cases} -2A_2 + 6B_2 &= -4 \\ -6A_2 - 2B_2 &= -2 \end{cases}, & \quad \text{(ii)} \begin{cases} -2A_1 + 6B_1 + 6A_2 + 8B_2 &= -1 \\ -6A_1 - 2B_1 - 8A_2 + 6B_2 &= -7 \end{cases}, \quad \text{and} \\
 \text{(iii)} \begin{cases} -2A_0 + 6B_0 + 3A_1 + 4B_1 + 2A_2 &= 1 \\ -6A_0 - 2B_0 - 4A_0 + 3B_1 + 2B_2 &= -1 \end{cases}.
 \end{aligned}$$

From (i), $A_2 = \frac{1}{2}$, $B_2 = -\frac{1}{2}$. Substituting these into (ii) and solving for A_1 and B_1 yields $A_1 = 0$, $B_1 = 0$. Substituting the known coefficients into (iii) and solving for A_0 and B_0 yields $A_0 = 0$, $B_0 = 0$. Therefore, $y_p = \frac{x^2}{2}(\cos 2x - \sin 2x)$.

5.5.16. Let $y = ue^x$. Then

$$\begin{aligned}
 y'' - 2y' + y &= e^x [(u'' + 2u' + u) - 2(u' + u) + u] = e^x u'' \\
 &= -e^x [(3 + 4x - x^2) \cos x + (3 - 4x - x^2) \sin x]
 \end{aligned}$$

if $u'' = -(3 + 4x - x^2) \cos x - (3 - 4x - x^2) \sin x$. Now let

$$\begin{aligned}
 u_p &= (A_0 + A_1x + A_2x^2) \cos x + (B_0 + B_1x + B_2x^2) \sin x; \text{ then} \\
 u'_p &= [A_1 + B_0 + (2A_2 + B_1)x + B_2x^2] \cos x \\
 &\quad + [B_1 - A_0 + (2B_2 - A_1)x - A_2x^2] \sin x, \\
 u''_p &= [-A_0 + 2A_2 + 2B_1 - (A_1 - 4B_2)x - A_2x^2] \cos x \\
 &\quad + [-B_0 + 2B_2 - 2A_1 - (B_1 + 4A_2)x - B_2x^2] \sin x \\
 &= -(3 + 4x - x^2) \cos x - (3 - 4x - x^2) \sin x \quad \text{if} \\
 \text{(i)} \begin{cases} -A_2 &= 1 \\ -B_2 &= 1 \end{cases}, & \quad \text{(ii)} \begin{cases} -A_1 + 4B_2 &= -4 \\ -B_1 - 4A_2 &= 4 \end{cases}, \quad \text{and}
 \end{aligned}$$

$$(iii) \begin{cases} -A_0 + 2B_1 + 2A_2 &= -3 \\ -B_0 - 2A_1 + 2B_2 &= -3 \end{cases}.$$

From (i), $A_2 = -1$, $B_2 = -1$. Substituting these into (ii) and solving for A_1 and B_1 yields $A_1 = 0$, $B_1 = 0$. Substituting the known coefficients into (iii) and solving for A_0 and B_0 yields $A_0 = 1$, $B_0 = 1$. Therefore, $u_p = (1 - x^2)(\cos x + \sin x)$ and $y_p = e^x(1 - x^2)(\cos x + \sin x)$.

5.5.18. Let $y = ue^{-x}$. Then

$$\begin{aligned} y'' + 2y' + y &= e^{-x} [(u'' - 2u' + u) + 2(u' - u) + u] \\ &= e^{-x} u'' = e^{-x} [(5 - 2x) \cos x - (3 + 3x) \sin x] \end{aligned}$$

if $u'' = (5 - 2x) \cos x - (3 + 3x) \sin x$. Let

$$\begin{aligned} u_p &= (A_0 + A_1 x) \cos x + (B_0 + B_1 x) \sin x; \text{ then} \\ u'_p &= (A_1 + B_0 + B_1 x) \cos x + (B_1 - A_0 - A_1 x) \sin x \\ u''_p &= (2B_1 - A_0 - A_1 x) \cos x - (2A_1 + B_0 + B_1 x) \sin x \\ &= (5 - 2x) \cos x - (3 + 3x) \sin x \end{aligned}$$

if $-A_1 = -2$, $-B_1 = -3$, $-A_0 + 2B_1 = 5$, $-B_0 - 2A_1 = -3$. Therefore, $A_1 = 2$, $B_1 = 3$, $A_0 = 1$, $B_0 = -1$, $u_p = e^{-x} [(1 + 2x) \cos x - (1 - 3x) \sin x]$, and $y_p = e^{-x} [(1 + 2x) \cos x - (1 - 3x) \sin x]$.

5.5.20. Let

$$\begin{aligned} y_p &= (A_0 x + A_1 x^2 + A_2 x^3) \cos x + (B_0 x + B_1 x^2 + B_2 x^3) \sin x; \text{ then} \\ y'_p &= [A_0 + (2A_1 + B_0)x + (3A_2 + B_1)x^2 + B_2 x^3] \cos x \\ &\quad + [B_0 + (2B_1 - A_0)x + (3B_2 - A_1)x^2 - A_2 x^3] \sin x \\ y''_p &= [2A_1 + 2B_0 - (A_0 - 6A_2 - 4B_1)x - (A_1 - 6B_2)x^2 - A_2 x^3] \cos x \\ &\quad + [2B_1 - 2A_0 - (B_0 + 6B_2 + 4A_1)x - (B_1 + 6A_2)x^2 - B_2 x^3] \sin x, \text{ so} \\ y''_p + y_p &= [2A_1 + 2B_0 + (6A_2 + 4B_1)x + 6B_2 x^2] \cos x \\ &\quad + [2B_1 - 2A_0 + (6B_2 - 4A_1)x - 6A_2 x^2] \sin x \\ &= (2 + 2x) \cos x + (4 + 6x^2) \sin x \quad \text{if} \\ (i) \begin{cases} 6B_2 &= 0 \\ -6A_2 &= 6 \end{cases}, \quad (ii) \begin{cases} 4B_1 + 6A_2 &= 2 \\ -4A_1 + 6B_2 &= 0 \end{cases}, \quad \text{and} \quad (iii) \begin{cases} 2B_0 + 2A_1 &= 2 \\ -2A_0 + 2B_1 &= 4 \end{cases}. \end{aligned}$$

From (i), $A_2 = -1$, $B_2 = 0$. Substituting these into (ii) and solving for A_1 and B_1 yields $A_1 = 0$, $B_1 = 2$. Substituting the known coefficients into (iii) and solving for A_0 and B_0 yields $A_0 = 0$, $B_0 = 2$. Therefore, $y_p = -x^3 \cos x + (x + 2x^2) \sin x$.

5.5.22. Let $y = ue^x$. Then

$$y'' - 7y' + 6y = e^x [(u'' + 2u' + u) - 7(u' + u) + 6u]$$

$$= e^x(u'' - 5u') = -e^x(17 \cos x - 7 \sin x)$$

if $u'' - 5u' = -17 \cos x + 7 \sin x$. Now let $u_p = A \cos x + B \sin x$. Then

$$\begin{aligned} u_p'' - 5u_p' &= -(A \cos x + B \sin x) - 5(-A \sin x + B \cos x) \\ &= (-A - 5B) \cos x - (B - 5A) \sin x = -17 \cos x + 7 \sin x \end{aligned}$$

if $-A - 5B = -17$, $5A - B = 7$. Therefore, $A = 2$, $B = 3$, $u_p = 2 \cos x + 3 \sin x$, and $y_p = e^x(2 \cos x + 3 \sin x)$. The characteristic polynomial of the complementary equation is $p(r) = r^2 - 7r + 6 = (r - 1)(r - 6)$, so $\{e^x, e^{6x}\}$ is a fundamental set of solutions of the complementary equation. Therefore, (A) $y = e^x(2 \cos x + 3 \sin x) + c_1 e^x + c_2 e^{6x}$ is the general solution of the nonhomogeneous equation. Differentiating (A) yields $y' = e^x(2 \cos x + 3 \sin x) + e^x(-2 \sin x + 3 \cos x) + c_1 e^x + 6c_2 e^{6x}$, so $y(0) = 4$, $y'(0) = 2 \Rightarrow 4 = 2 + c_1 + c_2$, $2 = 2 + 3 + c_1 + 6c_2 \Rightarrow c_1 + c_2 = 2$, $c_1 + 6c_2 = -3$, so $c_1 = 3$, $c_2 = -1$, and $y = e^x(2 \cos x + 3 \sin x) + 3e^x - e^{6x}$.

5.5.24. Let $y = ue^x$. Then

$$\begin{aligned} y'' + 6y' + 10y &= e^x[(u'' + 2u' + u) + 6(u' + u) + 10u] \\ &= e^x(u'' + 8u' + 17u) = -40e^x \sin x \end{aligned}$$

if $u'' + 8u' + 17u = -40 \sin x$. Let $u_p = A \cos x + B \sin x$. Then

$$\begin{aligned} u_p'' + 6u_p' + 17u_p &= -(A \cos x + B \sin x) + 8(-A \sin x + B \cos x) \\ &\quad + 17(A \cos x + B \sin x) \\ &= (16A + 8B) \cos x - (8A - 16B) \sin x = -40 \sin x \end{aligned}$$

if $16A + 8B = 0$, $-8A + 16B = -40$. Therefore, $A = 1$, $B = -2$, and $y_p = e^x(\cos x - 2 \sin x)$. The characteristic polynomial of the complementary equation is $p(r) = r^2 + 6r + 10 = (r + 3)^2 + 1$, so $\{e^{-3x} \cos x, e^{-3x} \sin x\}$ is a fundamental set of solutions of the complementary equation, and (A) $y = e^x(\cos x - 2 \sin x) + e^{-3x}(c_1 \cos x + c_2 \sin x)$ is the general solution of the nonhomogeneous equation. Therefore, $y(0) = 2 \Rightarrow 2 = 1 + c_1$, so $c_1 = 1$. Differentiating (A) yields $y' = e^x(\cos x - 2 \sin x) - e^x(\sin x + 2 \cos x) - 3e^{-3x}(c_1 \cos x + c_2 \sin x) + e^{-3x}(-c_1 \sin x + c_2 \cos x)$. Therefore, $y'(0) = -3 \Rightarrow -3 = 1 - 2 - 3c_1 + c_2$, so $c_2 = 1$, and $y = e^x(\cos x - 2 \sin x) + e^{-3x}(\cos x + \sin x)$.

5.5.26. Let $y = ue^{3x}$. Then

$$\begin{aligned} y'' - 3y' + 2y &= e^{3x}[(u'' + 6u' + 9u) - 3(u' + 3u) + 2u] \\ &= e^{3x}(u'' + 3u' + 2u) = e^{3x}[21 \cos x - (11 + 10x) \sin x] \end{aligned}$$

if $u'' + 3u' + 2u = 21 \cos x - (11 + 10x) \sin x$. Now let

$$\begin{aligned} u_p &= (A_0 + A_1 x) \cos x + (B_0 + B_1 x) \sin x; \text{ then} \\ u_p' &= (A_1 + B_0 + B_1 x) \cos x + (B_1 - A_0 - A_1 x) \sin x \\ u_p'' &= (2B_1 - A_0 - A_1 x) \cos x - (2A_1 + B_0 + B_1 x) \sin x, \text{ so} \end{aligned}$$

$$\begin{aligned}
u'' + 3u' + 2u &= [A_0 + 3A_1 + 3B_0 + 2B_1 + (A_1 + 3B_1)x] \cos x \\
&\quad + [B_0 + 3B_1 - 3A_0 - 2A_1 + (B_1 - 3A_1)x] \sin x \\
&= 21 \cos x - (11 + 10x) \sin x \quad \text{if} \\
\begin{cases} A_1 + 3B_1 = 0 \\ -3A_1 + B_1 = -10 \end{cases} &\quad \text{and} \quad \begin{cases} A_0 + 3B_0 + 3A_1 + 2B_1 = 21 \\ -3A_0 + B_0 - 2A_1 + 3B_1 = -11 \end{cases}.
\end{aligned}$$

From the first two equations $A_1 = 3$, $B_1 = -1$. Substituting these in last two equations and solving for A_0 and B_0 yields $A_0 = 2$, $B_0 = 4$. Therefore, $u_p = (2 + 3x) \cos x + (4 - x) \sin x$ and $y_p = e^{3x} [(2 + 3x) \cos x + (4 - x) \sin x]$. The characteristic polynomial of the complementary equation is $p(r) = r^2 - 3r + 2 = (r - 1)(r - 2)$, so $\{e^x, e^{2x}\}$ is a fundamental set of solutions of the complementary equation, and (A) $y = e^{3x} [(2 + 3x) \cos x + (4 - x) \sin x] + c_1 e^x + c_2 e^{2x}$ is the general solution of the nonhomogeneous equation. Differentiating (A) yields

$$\begin{aligned}
y' &= 3e^{3x} [(2 + 3x) \cos x + (4 - x) \sin x] \\
&\quad + e^{3x} [(7 - x) \cos x - (3 + 3x) \sin x] + c_1 e^x + 2c_2 e^{2x}.
\end{aligned}$$

Therefore, $y(0) = 0$, $y'(0) = 6 \Rightarrow 0 = 2 + c_1 + c_2$, $6 = 6 + 7 + c_1 + 2c_2$, so $c_1 + c_2 = -2$, $c_1 + 2c_2 = -7$. Therefore, $c_1 = 3$, $c_2 = -5$, and $y = e^{3x} [(2 + 3x) \cos x + (4 - x) \sin x] + 3e^x - 5e^{2x}$.

5.5.28. We must find particular solutions y_{p_1} , y_{p_2} , and y_{p_3} of (A) $y'' + y = 4 \cos x - 2 \sin x$ and (B) $y'' + y = xe^x$, and (C) $y'' + y = e^{-x}$, respectively. To find a particular solution of (A) we write

$$\begin{aligned}
y_{p_1} &= Ax \cos x + Bx \sin x; \text{ then} \\
y'_{p_1} &= (A + Bx) \cos x + (B - Ax) \sin x \\
y''_{p_1} &= (2B - Ax) \cos x - (2A + Bx) \sin x, \text{ so}
\end{aligned}$$

$y''_{p_1} + y_{p_1} = 2B \cos x - 2A \sin x = 4 \cos x - 2 \sin x$ if $2B = 4$, $-2A = -2$. Therefore, $A = 1$, $B = 2$, and $y_{p_1} = x(\cos x + 2 \sin x)$. To find a particular solution of (B) we write $y = ue^x$. Then

$$\begin{aligned}
y'' + y &= e^x [(u'' + 2u' + u) + u] \\
&= e^x (u'' + 2u' + 2u) = xe^x
\end{aligned}$$

if $u'' + 2u' + 2u = x$. Now $u_p = A + Bx$, where $2B + 2(A + Bx) = x$. Therefore, $2B = 1$, $2A + 2B = 0$, so $B = \frac{1}{2}$, $A = -\frac{1}{2}$, $u_p = -\frac{1}{2}(1 - x)$, and $y_{p_2} = -\frac{e^x}{2}(1 - x)$. To find a particular solution of (C) we write $y_{p_3} = Ae^{-x}$. Then $y''_{p_3} + y_{p_3} = 2Ae^{-x} = e^{-x}$ if $2A = 1$, so $A = \frac{1}{2}$ and $y_{p_3} = \frac{e^{-x}}{2}$. Now $y_p = y_{p_1} + y_{p_2} + y_{p_3} = x(\cos x + 2 \sin x) - \frac{e^x}{2}(1 - x) + \frac{e^{-x}}{2}$.

5.5.30. We must find particular solutions y_{p_1} , y_{p_2} and y_{p_3} of (A) $y'' - 2y' + 2y = 4xe^x \cos x$, (B) $y'' - 2y' + 2y = xe^{-x}$, and (C) $y'' - 2y' + 2y = 1 + x^2$, respectively. To find a particular solution of (A) we write $y = ue^x$. Then

$y'' - 2y' + 2y = e^x [(u'' + 2u' + u) - 2(u' + u) + 2u] = e^x(u'' + u) = 4xe^x \cos x$ if $u'' + u = 4x \cos x$. Now let

$$\begin{aligned} u_p &= (A_0x + A_1x^2) \cos x + (B_0x + B_1x^2) \sin x; \text{ then} \\ u'_p &= [A_0 + (2A_1 + B_0)x + B_1x^2] \cos x + [B_0 + (2B_1 - A_0)x - B_1x^2] \sin x \\ u''_p &= [2A_1 + 2B_0 - (A_0 - 4B_1)x - A_1x^2] \cos x \\ &\quad + [2B_1 - 2A_0 - (B_0 + 4A_1)x - B_1x^2] \sin x, \text{ so} \\ u''_p + u_p &= (2A_1 + 2B_0 + 4B_1x) \cos x + (2B_1 - 2A_0 - 4A_1x) \sin x \\ &= 4x \cos x \end{aligned}$$

if $4B_1 = 4$, $-4A_1 = 0$, $2B_0 + 2A_1 = 0$, $-2A_0 + 2B_1 = 0$. Therefore, $A_1 = 0$, $B_1 = 1$, $A_0 = 1$, $B_0 = 0$, $u_p = x(\cos x + x \sin x)$, and $y_{p_1} = xe^x(\cos x + x \sin x)$. To find a particular solution of (B) we write $y = ue^{-x}$. Then

$$\begin{aligned} y'' - 2y' + 2y &= e^{-x} [(u'' - 2u' + u) - 2(u' - u) + 2u] \\ &= e^{-x}(u'' - 4u' + 5u) = xe^{-x} \end{aligned}$$

if $u'' - 4u' + 5u = x$. Now $u_p = A + Bx$ where $-4B + 5(A + Bx) = x$. Therefore, $5B = 1$, $5A - 4B = 0$, $B = \frac{1}{5}$, $A = \frac{4}{25}$, $u_p = \frac{1}{25}(4 + 5x)$, and $y_{p_2} = \frac{e^{-x}}{25}(4 + 5x)$. To find a particular solution of (C) we write $y_{p_3} = A + Bx + Cx^2$. Then

$$\begin{aligned} y''_{p_3} - 2y'_{p_3} + 2y_{p_3} &= 2C - 2(B + 2Cx) + 2(A + Bx + Cx^2) \\ &= (2A - 2B + 2C) + (2B - 4C)x + 2Cx^2 = 1 + x^2 \end{aligned}$$

if $2A - 2B + 2C = 1$, $2B - 4C = 0$, $2C = 1$. Therefore, $C = \frac{1}{2}$, $B = 1$, $A = 1$, and

$$y_{p_3} = 1 + x + \frac{x^2}{2}. \text{ Now } y_p = y_{p_1} + y_{p_2} + y_{p_3} = xe^x(\cos x + x \sin x) + \frac{e^{-x}}{25}(4 + 5x) + 1 + x + \frac{x^2}{2}.$$

5.5.32. We must find particular solutions y_{p_1} and y_{p_2} of (A) $y'' - 4y' + 4y = 6e^{2x}$ and (B) $y'' - 4y' + 4y = 25 \sin x$, respectively. To find a particular solution of (A), let $y = ue^{2x}$. Then

$$\begin{aligned} y'' - 4y' + 4y &= e^{2x} [(u'' + 4u' + 4u) - 4(u' + 2u) + 4u] \\ &= e^{2x}u'' = 6e^{2x} \end{aligned}$$

if $u'' = 6$. Integrating twice and taking the constants of integration to be zero yields $u_p = 3x^2$, so $y_{p_1} = 3x^2e^{2x}$. To find a particular solution of (B), let $y_{p_2} = A \cos x + B \sin x$. Then

$$\begin{aligned} y''_{p_2} - 4y'_{p_2} + 4y_{p_2} &= -(A \cos x + B \sin x) - 4(-A \sin x + B \cos x) \\ &\quad + 4(A \cos x + B \sin x) \\ &= (3A - 4B) \cos x + (4A + 3B) \sin x = 25 \sin x \end{aligned}$$

if $3A - 4B = 0$, $4A + 3B = 25$. Therefore, $A = 4$, $B = 3$, and $y_{p_2} = 4 \cos x + 3 \sin x$. Now $y_p = y_{p_1} + y_{p_2} = 3x^2e^{2x} + 4 \cos x + 3 \sin x$. The characteristic polynomial of the

complementary equation is $p(r) = r^2 - 4r + 4 = (r - 2)^2$, so $\{e^{2x}, xe^{2x}\}$ is a fundamental set of solutions of the complementary equation. Therefore, (C)

$y = 3x^2 e^{2x} + 4 \cos x + 3 \sin x + e^{2x}(c_1 + c_2 x)$ is the general solution of the nonhomogeneous equation. Now $y(0) = 5 \Rightarrow 5 = 4 + c_1$, so $c_1 = 1$. Differentiating (C) yields

$y' = 6e^{2x}(x + x^2) - 4 \sin x + 3 \cos x + 2e^{2x}(c_1 + c_2 x) + c_2 e^{2x}$, so $y'(0) = 3 \Rightarrow 3 = 3 + 2 + c_2$. Therefore, $c_2 = -2$, and $y = (1 - 2x + 3x^2)e^{2x} + 4 \cos x + 3 \sin x$.

- 5.5.34. We must find particular solutions y_{p_1} and y_{p_2} of (A) $y'' + 4y' + 4y = 2 \cos 2x + 3 \sin 2x$ and (B) $y'' + 4y' + 4y = e^{-x}$, respectively. To find a particular solution of (A) we write $y_{p_1} = A \cos 2x + B \sin 2x$. Then

$$\begin{aligned} y_{p_1}'' + 4y_{p_1}' + 4y_{p_1} &= -4(A \cos 2x + B \sin 2x) + 8(-A \sin 2x + B \cos 2x) + 4(A \cos 2x + B \sin 2x) \\ &= -8A \sin 2x + 8B \cos 2x = 2 \cos 2x + 3 \sin 2x \end{aligned}$$

if $8B = 2$, $-8A = 3$. Therefore, $A = -\frac{3}{8}$, $B = \frac{1}{4}$, and $y_{p_1} = -\frac{3}{8} \cos 2x + \frac{1}{4} \sin 2x$. To find a particular solution of (B) we write $y_{p_2} = Ae^{-x}$. Then

$y_{p_2}'' + 4y_{p_2}' + 4y_{p_2} = A(1 - 4 + 4)e^{-x} = Ae^{-x} = e^{-x}$ if $A = 1$. Therefore, $y_{p_2} = e^{-x}$. Now

$y_p = y_{p_1} + y_{p_2} = -\frac{3}{8} \cos 2x + \frac{1}{4} \sin 2x + e^{-x}$. The characteristic polynomial of the complementary equation is $p(r) = r^2 + 4r + 4 = (r + 2)^2$, so $\{e^{-2x}, xe^{-2x}\}$ is a fundamental set of solutions of the complementary equation. Therefore, (C)

$y = -\frac{3}{8} \cos 2x + \frac{1}{4} \sin 2x + e^{-x} + e^{-2x}(c_1 + c_2 x)$ is the general solution of the

nonhomogeneous equation. Now $y(0) = -1 \Rightarrow -1 = -\frac{3}{8} + 1 + c_1$, so $c_1 = -\frac{13}{8}$.

Differentiating (C) yields $y' = \frac{3}{4} \sin 2x + \frac{1}{2} \cos 2x - e^{-x} - 2e^{-2x}(c_1 + c_2 x) + c_2 e^{-2x}$, so

$y'(0) = 2 \Rightarrow 2 = \frac{1}{2} - 1 - 2c_1 + c_2$. Therefore, $c_2 = -\frac{3}{4}$, and

$$y = -\frac{3}{8} \cos 2x + \frac{1}{4} \sin 2x + e^{-x} - \frac{13}{8} e^{-2x} - \frac{3}{4} x e^{-2x}.$$

- 5.5.36. (a), (b), and (c) require only routine manipulations. (d) The coefficients of $\sin \omega x$ in y_p' , y_p'' , $ay_p'' + by_p' + cy_p$, and $y_p'' + \omega^2 y_p$ can be obtained by replacing A by B and B by $-A$ in the corresponding coefficients of $\cos \omega x$.

- 5.5.38. Let $y = ue^{\lambda x}$. Then

$$\begin{aligned} ay'' + by' + cy &= e^{\lambda x} [a(u'' + 2\lambda u' + \lambda^2 u) + b(u' + \lambda u) + cu] \\ &= e^{\lambda x} [au'' + (2a\lambda + b)u' + (a\lambda^2 + b\lambda + c)u] \\ &= e^{\lambda x} [au'' + p'(\lambda)u' + p(\lambda)u] \\ &= e^{\lambda x} (P(x) \cos \omega x + Q(x) \sin \omega x) \text{ if} \end{aligned}$$

(A) $au'' + p'(\lambda)u' + p(\lambda)u = P(x) \cos \omega x + Q(x) \sin \omega x$, where $p(r) = ar^2 + br + c$ is that characteristic polynomial of the complementary equation (B) $ay'' + by' + cy = 0$. If $e^{\lambda x} \cos \omega x$ and $e^{\lambda x} \sin \omega x$ are not solutions of (B), then $\cos \omega x$ and $\sin \omega x$ are not solutions of

the complementary equation for (A). Then Theorem 5.5.1 implies that (A) has a particular solution

$$u_p = (A_0 + A_1x + \cdots + A_kx^k) \cos \omega x + (B_0 + B_1x + \cdots + B_kx^k) \sin \omega x,$$

and $y_p = u_p e^{\lambda x}$ is a particular solution of the stated form for the given equation. If $e^{\lambda x} \cos \omega x$ and $e^{\lambda x} \sin \omega x$ are solutions of (B), then $\cos \omega x$ and $\sin \omega x$ are solutions of the complementary equation for (A). Then Theorem 5.5.1 implies that (A) has a particular solution

$$u_p = (A_0x + A_1x^2 + \cdots + A_kx^{k+1}) \cos \omega x + (B_0x + B_1x^2 + \cdots + B_kx^{k+1}) \sin \omega x,$$

and $y_p = u_p e^{\lambda x}$ is a particular solution of the stated form for the given equation.

5.5.40. (a) Let $y = \int x^2 \cos x \, dx$; then $y' = x^2 \cos x$ Now let

$$\begin{aligned} y_p &= (A_0 + A_1x + A_2x^2) \cos x + (B_0 + B_1x + B_2x^2) \sin x; \text{ then} \\ y'_p &= [A_1 + B_0 + (2A_2 + B_1)x + B_2x^2] \cos x \\ &\quad + [B_1 - A_0 + (2B_2 - A_1)x - A_2x^2] \sin x = x^2 \cos x \quad \text{if} \\ \text{(i)} \begin{cases} B_2 &= 1 \\ -A_2 &= 0 \end{cases}, \quad \text{(ii)} \begin{cases} B_1 + 2A_2 &= 0 \\ -A_1 + 2B_2 &= 0 \end{cases}, \quad \text{and} \quad \text{(iii)} \begin{cases} B_0 + A_1 &= 0 \\ -A_0 + B_1 &= 0 \end{cases}. \end{aligned}$$

Solving these equations yields $A_2 = 0$, $B_2 = 1$, $A_1 = 2$, $B_1 = 0$, $A_0 = 0$, $B_0 = -2$. Therefore, $y_p = 2x \cos x - (2 - x^2) \sin x$ and $y = 2x \cos x - (2 - x^2) \sin x + c$.

(b) Let $y = \int x^2 e^x \cos x \, dx = ue^x$; then $y' = (u' + u)e^x = x^2 e^x \cos x$ if $u' + u = x^2 \cos x$. Now let

$$\begin{aligned} u_p &= (A_0 + A_1x + A_2x^2) \cos x + (B_0 + B_1x + B_2x^2) \sin x; \text{ then} \\ u'_p &= [A_1 + B_0 + (2A_2 + B_1)x + B_2x^2] \cos x \\ &\quad + [B_1 - A_0 + (2B_2 - A_1)x - A_2x^2] \sin x, \text{ so} \\ u''_p + u_p &= [A_0 + A_1 + B_0 + (A_1 + 2A_2 + B_1)x + (A_2 + B_2)x^2] \cos x \\ &\quad + [B_0 + B_1 - A_0 + (B_1 + 2B_2 - A_1)x + (B_2 - A_2)x^2] \sin x \\ &= x^2 \cos x \quad \text{if} \\ \text{(i)} \begin{cases} A_2 + B_2 &= 1 \\ -A_2 + B_2 &= 0 \end{cases}, \quad \text{(ii)} \begin{cases} A_1 + B_1 + 2A_2 &= 0 \\ -A_1 + B_1 + 2B_2 &= 0 \end{cases}, \quad \text{and} \\ \text{(iii)} \begin{cases} A_0 + B_0 + A_1 &= 0 \\ -A_0 + B_0 + B_1 &= 0 \end{cases}. \end{aligned}$$

From (i), $A_2 = \frac{1}{2}$, $B_2 = \frac{1}{2}$. Substituting these into (ii) and solving for A_1 and B_1 yields $A_1 = 0$, $B_1 = -1$. Substituting these into (iii) and solving for A_0 and B_0 yields $A_0 = -\frac{1}{2}$, $B_0 = \frac{1}{2}$. Therefore, $u_p = -\frac{1}{2} [(1 - x^2) \cos x - (1 - x)^2 \sin x]$ and $y = -\frac{e^x}{2} [(1 - x^2) \cos x - (1 - x)^2 \sin x]$.

- (c) Let $y = \int x e^{-x} \sin 2x \, dx = u e^{-x}$; then $y' = (u' - u)e^{-x} = x e^{-x} \sin 2x$ if $u' - u = x \sin 2x$. Now let

$$\begin{aligned} u_p &= (A_0 + A_1 x) \cos 2x + (B_0 + B_1 x) \sin 2x; \text{ then} \\ u'_p &= [(A_1 + 2B_0) + 2B_1 x] \cos 2x + [(B_1 - 2A_0) - 2A_1 x] \sin 2x, \text{ so} \\ u''_p - u_p &= [-A_0 + A_1 + 2B_0 - (A_1 - 2B_1)x] \cos 2x \\ &\quad + [-B_0 + B_1 - 2A_0 - (B_1 + 2A_1)x] \sin 2x = x \sin 2x \quad \text{if} \\ \text{(i)} \begin{cases} -A_1 + 2B_1 &= 0 \\ -2A_1 - B_1 &= 1 \end{cases} \quad \text{and} \quad \text{(ii)} \begin{cases} -A_0 + 2B_0 + A_1 &= 0 \\ -2A_0 - B_0 + B_1 &= 0 \end{cases}. \end{aligned}$$

From (i), $A_1 = -\frac{2}{5}$, $B_1 = -\frac{1}{5}$. Substituting these into (ii) and solving for A_0 and B_0 yields $A_0 = -\frac{4}{25}$, $B_0 = \frac{3}{25}$. Therefore,

$$\begin{aligned} u_p &= -\frac{1}{25} [(4 + 10x) \cos 2x - (3 - 5x) \sin 2x] + c \text{ and} \\ y_p &= -\frac{e^{-x}}{25} [(4 + 10x) \cos 2x - (3 - 5x) \sin 2x] + c. \end{aligned}$$

- (d) Let $y = \int x^2 e^{-x} \sin x \, dx = u e^{-x}$; then $y' = (u' - u)e^{-x} = x^2 e^{-x} \sin x$ if $u' - u = x^2 \sin x$. Now let

$$\begin{aligned} u_p &= (A_0 + A_1 x + A_2 x^2) \cos x + (B_0 + B_1 x + B_2 x^2) \sin x; \text{ then} \\ u'_p &= [A_1 + B_0 + (2A_2 + B_1)x + B_2 x^2] \cos x \\ &\quad + [B_1 - A_0 + (2B_2 - A_1)x - A_2 x^2] \sin x, \text{ so} \\ u''_p - u_p &= [-A_0 + A_1 + B_0 - (A_1 - 2A_2 - B_1)x - (A_2 - B_2)x^2] \cos x \\ &\quad + [-B_0 + B_1 - A_0 - (B_1 - 2B_2 + A_1)x - (B_2 + A_2)x^2] \sin x \\ &= x^2 \sin x \quad \text{if} \\ \text{(i)} \begin{cases} -A_2 + B_2 &= 0 \\ -A_2 - B_2 &= 1 \end{cases}, \quad \text{(ii)} \begin{cases} -A_1 + B_1 + 2A_2 &= 0 \\ -A_1 - B_1 + 2B_2 &= 0 \end{cases}, \quad \text{and} \\ \text{(iii)} \begin{cases} -A_0 + B_0 + A_1 &= 0 \\ -A_0 - B_0 + B_1 &= 0 \end{cases}. \end{aligned}$$

From (i), $A_2 = -\frac{1}{2}$, $B_2 = -\frac{1}{2}$. Substituting these into (ii) and solving for A_1 and B_1 yields $A_1 = -1$, $B_1 = 0$. Substituting these into (iii) and solving for A_0 and B_0 yields $A_0 = -\frac{1}{2}$, $B_0 = \frac{1}{2}$. Therefore,

$$\begin{aligned} u_p &= -\frac{e^{-x}}{2} [(1 + x)^2 \cos x - (1 - x^2) \sin x] \text{ and} \\ y &= -\frac{e^{-x}}{2} [(1 + x)^2 \cos x - (1 - x^2) \sin x] + c. \end{aligned}$$

- (e) Let $y = \int x^3 e^x \sin x \, dx = u e^x$; then $y' = (u' + u)e^x = x^3 e^x \sin x$ if $u' + u = x^3 \sin x$.
Now let

$$\begin{aligned}
 u_p &= (A_0 + A_1 x + A_2 x^2 + A_3 x^3) \cos x + (B_0 + B_1 x + B_2 x^2 + B_3 x^3) \sin x; \text{ then} \\
 u'_p &= [A_1 + B_0 + (2A_2 + B_1)x + (3A_3 + B_2)x^2 + B_3 x^3] \cos x \\
 &\quad + [B_1 - A_0 + (2B_2 - A_1)x + (3B_3 - A_2)x^2 - A_3 x^3] \sin x, \text{ so} \\
 u''_p + u_p &= [A_0 + A_1 + B_0 + (A_1 + 2A_2 + B_1)x \\
 &\quad + (A_2 + 3A_3 + B_2)x^2 + (A_3 + B_3)x^3] \cos x \\
 &\quad + [B_0 + B_1 - A_0 + (B_1 + 2B_2 - A_1)x \\
 &\quad + (B_2 + 3B_3 - A_2)x^2 + (B_3 - A_3)x^3] \sin x = x^3 \sin x \quad \text{if} \\
 \text{(i)} \begin{cases} A_3 + B_3 &= 0 \\ -A_3 + B_3 &= 1 \end{cases}, \quad \text{(ii)} \begin{cases} A_2 + B_2 + 3A_3 &= 0 \\ -A_2 + B_2 + 3B_3 &= 0 \end{cases}, \\
 \text{(iii)} \begin{cases} A_1 + B_1 + 2A_2 &= 0 \\ -A_1 + B_1 + 2B_2 &= 0 \end{cases}, \quad \text{and} \quad \text{(iv)} \begin{cases} A_0 + B_0 + A_1 &= 0 \\ -A_0 + B_0 + B_1 &= 0 \end{cases}.
 \end{aligned}$$

From (i), $A_3 = -\frac{1}{2}$, $B_3 = \frac{1}{2}$. Substituting these into (ii) and solving for A_2 and B_2 yields $A_2 = \frac{3}{2}$, $B_2 = 0$. Substituting these into (iii) and solving for A_1 and B_1 yields $A_1 = -\frac{3}{2}$, $B_1 = -\frac{3}{2}$. Substituting these into (iv) and solving for A_0 and B_0 yields $A_0 = 0$, $B_0 = \frac{3}{2}$. Therefore,

$$\begin{aligned}
 u_p &= -\frac{1}{2} [x(3 - 3x + x^2) \cos x - (3 - 3x + x^3) \sin x] \text{ and} \\
 y &= -\frac{e^x}{2} [x(3 - 3x + x^2) \cos x - (3 - 3x + x^3) \sin x] + c.
 \end{aligned}$$

- (f) Let $y = \int e^x [x \cos x - (1 + 3x) \sin x] \, dx = u e^x$; then $y' = (u' + u)e^x = e^x [x \cos x - (1 + 3x) \sin x]$ if $u' + u = x \cos x - (1 + 3x) \sin x$. Now let

$$\begin{aligned}
 u_p &= (A_0 + A_1 x) \cos x + (B_0 + B_1 x) \sin x; \text{ then} \\
 u'_p &= [A_1 + B_0 + B_1 x] \cos x + [B_1 - A_0 - A_1 x] \sin x, \text{ so} \\
 u''_p + u_p &= [A_0 + A_1 + B_0 + (A_1 + B_1)x] \cos x \\
 &\quad + [B_0 + B_1 - A_0 + (B_1 - A_1)x] \sin x \\
 &= x \cos x - (1 + 3x) \sin x \quad \text{if} \\
 \text{(i)} \begin{cases} A_1 + B_1 &= 1 \\ -A_1 + B_1 &= -3 \end{cases}, \quad \text{and} \quad \text{(ii)} \begin{cases} A_0 + B_0 + A_1 &= 0 \\ -A_0 + B_0 + B_1 &= -1 \end{cases}.
 \end{aligned}$$

From (i), $A_1 = 2$, $B_1 = -1$. Substituting these into (ii) and solving for A_0 and B_0 yields $A_0 = -1$, $B_0 = -1$. Therefore, $u_p = -[(1 - 2x) \cos x + (1 + x) \sin x]$ and $y = -e^x [(1 - 2x) \cos x + (1 + x) \sin x] + c$.

(g) Let $y = \int e^{-x} [(1+x^2) \cos x + (1-x^2) \sin x] dx = ue^{-x}$; then

$$y' = (u' - u)e^{-x} = e^{-x} [(1+x^2) \cos x + (1-x^2) \sin x]$$

if $u' - u = (1+x^2) \cos x + (1-x^2) \sin x$. Now let

$$\begin{aligned} u_p &= (A_0 + A_1x + A_2x^2) \cos x + (B_0 + B_1x + B_2x^2) \sin x; \text{ then} \\ u'_p &= [A_1 + B_0 + (2A_2 + B_1)x + B_2x^2] \cos x \\ &\quad + [B_1 - A_0 + (2B_2 - A_1)x - A_2x^2] \sin x, \text{ so} \\ u''_p - u_p &= [-A_0 + A_1 + B_0 - (A_1 - 2A_2 - B_1)x - (A_2 - B_2)x^2] \cos x \\ &\quad + [-B_0 + B_1 - A_0 - (B_1 - 2B_2 + A_1)x - (B_2 + A_2)x^2] \sin x \\ &= (1+x^2) \cos x + (1-x^2) \sin x \quad \text{if} \\ \text{(i)} \quad &\begin{cases} -A_2 + B_2 = 1 \\ -A_2 - B_2 = -1 \end{cases}, \quad \text{(ii)} \quad \begin{cases} -A_1 + B_1 + 2A_2 = 0 \\ -A_1 - B_1 + 2B_2 = 0 \end{cases}, \quad \text{and} \\ \text{(iii)} \quad &\begin{cases} -A_0 + B_0 + A_1 = 1 \\ -A_0 - B_0 + B_1 = 1 \end{cases}. \end{aligned}$$

From (i), $A_2 = 0$, $B_2 = 1$. Substituting these into (ii) and solving for A_1 and B_1 yields $A_1 = 1$, $B_1 = 1$. Substituting these into (iii) and solving for A_0 and B_0 yields $A_0 = 0$, $B_0 = 0$. Therefore, $u_p = x \cos x + x(1+x) \sin x$ and $y = e^{-x} [x \cos x + x(1+x) \sin x] + c$.

5.6 Reduction of Order

(NOTE: The term uy''_1 is indicated by “...” in some of the following solutions, where y''_1 is complicated. Since this term always drops out of the differential equation for u , it is not necessary to include it.)

5.6.2. If $y = ux$, then $y' = u'x + u$ and $y'' = u''x + 2u'$, so $x^2y'' + xy' - y = x^3u'' + 3x^2u' = \frac{4}{x^2}$ if

$u' = z$, where (A) $z' + \frac{3}{x}z = \frac{4}{x^5}$. Since $\int \frac{3}{x} dx = 3 \ln |x|$, $z_1 = \frac{1}{x^3}$ is a solution of the complementary equation for (A). Therefore, the solutions of (A) are of the form (B) $z = \frac{v}{x^3}$,

where $\frac{v'}{x^3} = \frac{4}{x^5}$, so $v' = \frac{4}{x^2}$. Hence, $v = -\frac{4}{x} + C_1$; $u' = z = -\frac{4}{x^4} + \frac{C_1}{x^3}$ (see (B));

$u = \frac{4}{3x^3} - \frac{C_1}{2x^2} + C_2$; $y = ux = \frac{4}{3x^2} - \frac{C_1}{2x} + C_2x$, or $y = \frac{4}{3x^2} + c_1x + \frac{c_2}{x}$. As a byproduct, $\{x, 1/x\}$ is a fundamental set of solutions of the complementary equation.

5.6.4. If $y = ue^{2x}$, then $y' = (u' + 2u)e^{2x}$ and $y'' = (u'' + 4u' + 4u)e^{2x}$, so

$y'' - 3y' + 2y = (u'' + u')e^{2x} = \frac{1}{1+e^{-x}}$ if $u' = z$, where (A) $z' + z = \frac{e^{-2x}}{1+e^{-x}}$. Since $z_1 = e^{-x}$ is a solution of the complementary equation for (A), the solutions of (A) are of the form (B) $z = ve^{-x}$, where $v'e^{-x} = \frac{e^{-2x}}{1+e^{-x}}$, so $v' = \frac{e^{-x}}{1+e^{-x}}$. Hence, $v = -\ln(1+e^{-x}) + C_1$;

$u' = z = -e^{-x} \ln(1 + e^{-x}) + C_1 e^{-x}$ (see (B));
 $u = (1 + e^{-x}) \ln(1 + e^{-x}) - 1 - e^{-x} - C_1 e^{-x} + C_2$;
 $y = u e^{2x} = (e^{2x} + e^x) \ln(1 + e^{-x}) - (C_1 + 1)e^x + (C_2 - 1)e^{2x}$, or
 $y = (e^{2x} + e^x) \ln(1 + e^{-x}) + c_1 e^{2x} + c_2 e^x$. As a byproduct, $\{e^{2x}, e^x\}$ is a fundamental set of solutions of the complementary equation.

5.6.6. If $y = u x^{1/2} e^x$, then $y' = u' x^{1/2} e^x + u \left(x^{1/2} + \frac{x^{-1/2}}{2} \right) e^x$ and

$y'' = u'' x^{1/2} e^x + 2u' \left(x^{1/2} + \frac{x^{-1/2}}{2} \right) e^x + \dots$ so
 $4x^2 y'' + (4x - 8x^2) y' + (4x^2 - 4x - 1) y = e^x (4x^{5/2} u'' + 8x^{3/2} u') = 4x^{1/2} e^x (1 + 4x)$ if $u' = z$,
 where (A) $z' + \frac{2}{x} z = \frac{1 + 4x}{x^2}$. Since $\int \frac{2}{x} dx = 2 \ln |x|$, $z_1 = \frac{2}{x^2}$ is a solution of the
 complementary equation for (A). Therefore, the solutions of (A) are of the form (B) $z = \frac{v}{x^2}$,
 where $\frac{v'}{x^2} = \frac{1 + 4x}{x^2}$, so $v' = 1 + 4x$. Hence, $v = x + 2x^2 + C_1$; $u' = z = \frac{1}{x} + 2 + \frac{C_1}{x^2}$ (see
 (B)); $u = \ln x + 2x - \frac{C_1}{x} + C_2$; $y = u x^{1/2} e^x = e^x (2x^{3/2} + x^{1/2} \ln x - C_1 x^{-1/2} + C_2 x^{1/2})$, or
 $y = e^x (2x^{3/2} + x^{1/2} \ln x + c_1 x^{1/2} + c_2 x^{-1/2})$. As a byproduct, $\{x^{1/2} e^x, x^{-1/2} e^{-x}\}$ is a
 fundamental set of solutions of the complementary equation.

5.6.8. If $y = u e^{-x^2}$, then $y' = u' e^{-x^2} - 2x u e^{-x^2}$ and $y'' = u'' e^{-x^2} - 4x u' e^{-x^2} + \dots$, so
 $y'' + 4x y' + (4x^2 + 2) y = u'' e^{-x^2} = 8e^{-x(x+2)} = 8e^{-x^2} e^{-2x}$ if $u'' = 8e^{-2x}$.
 Therefore, $u' = -4e^{-2x} + C_1$; $u = 2e^{-2x} + C_1 x + C_2$, and
 $y = u e^{-x^2} = e^{-x^2} (2e^{-2x} + C_1 x + C_2)$, or $y = e^{-x^2} (2e^{-2x} + c_1 + c_2 x)$. As a byproduct,
 $\{e^{-x^2}, x e^{-x^2}\}$ is a fundamental set of solutions of the complementary equation.

5.6.10. If $y = u x e^{-x}$, then $y' = u' x e^{-x} - u e^{-x} (x - 1)$ and $y'' = u'' x e^{-x} - 2u' e^{-x} (x - 1) + \dots$, so
 $x^2 y'' + 2x(x - 1) y' + (x^2 - 2x + 2) y = x^3 u'' = x^3 e^{2x}$ if $u'' = e^{2x}$. Therefore, $u' = \frac{e^{3x}}{3} + C_1$;
 $u = \frac{e^{3x}}{9} + C_1 x + C_2$, and $y = u x e^{-x} = \frac{x e^{2x}}{9} + x e^{-x} (C_1 x + C_2)$, or
 $y = \frac{x e^{2x}}{9} + x e^{-x} (c_1 + c_2 x)$. As a byproduct, $\{x e^{-x}, x^2 e^{-x}\}$ is a fundamental set of solutions
 of the complementary equation.

5.6.12. If $y = u e^x$, then $y' = (u' + u) e^x$ and $y'' = (u'' + 2u' + u) e^x$, so
 $(1 - 2x) y'' + 2y' + (2x - 3) y = e^x [(1 - 2x) u'' + (4 - 4x) u'] = (1 - 4x + 4x^2) e^x$ if $u' = z$,
 where (A) $z' + \frac{4 - 4x}{1 - 2x} z = 1 - 2x$. Since
 $\int \frac{4 - 4x}{1 - 2x} dx = \int \left(2 + \frac{2}{1 - 2x} \right) dx = 2x - \ln |1 - 2x|$, $z_1 = (1 - 2x) e^{-2x}$ is a solution of the
 complementary equation for (A). Therefore, the solutions of (A) are of the form (B)
 $z = v(1 - 2x) e^{-2x}$, where $v'(1 - 2x) e^{-2x} = (1 - 2x)$, so $v' = e^{2x}$. Hence, $v = \frac{e^{2x}}{2} + C_1$;
 $u' = z = \left(\frac{1}{2} + C_1 e^{-2x} \right) (1 - 2x)$ (see (B)); $u = -\frac{(2x - 1)^2}{8} + C_1 x e^{-2x} + C_2$;

$y = ue^x = -\frac{(2x-1)^2 e^x}{8} + C_1 x e^{-x} + C_2 e^x$, or $y = -\frac{(2x-1)^2 e^x}{8} + c_1 e^x + c_2 x e^{-x}$. As a byproduct, $\{e^x, x e^{-x}\}$ is a fundamental set of solutions of the complementary equation.

5.6.14. If $y = ue^{-x}$, then $y' = (u' - u)e^{-x}$ and $y'' = (u'' - 2u' + u)e^{-x}$, so $2xy'' + (4x+1)y' + (2x+1)y = e^{-x}(2xu'' + u') = 3x^{1/2}e^{-x}$ if $u' = z$, where (A) $z' + \frac{1}{2x}z = \frac{3}{2}x^{-1/2}$. Since $\int \frac{1}{2x} dx = \frac{1}{2} \ln|x|$, $z_1 = x^{-1/2}$ is a solution of the complementary equation for (A). Therefore, the solutions of (A) are of the form (B) $z = vx^{-1/2}$, where $v'x^{-1/2} = \frac{3}{2}x^{-1/2}$, so $v' = \frac{3}{2}$. Hence, $v = \frac{3x}{2} + C_1$; $u' = z = \frac{3}{2}x^{1/2} + C_1 x^{-1/2}$ (see (B)); $u = x^{3/2} + 2C_1 x^{1/2} + C_2$; $y = ue^{-x} = e^{-x}(x^{3/2} + 2C_1 x^{1/2} + C_2)$, or $y = e^{-x}(x^{3/2} + c_1 + c_2 x^{1/2})$. As a byproduct, is a $\{e^{-x}, x^{1/2}e^{-x}\}$ fundamental set of solutions of the complementary equation.

5.6.16. If $y = ux^{1/2}$, then $y' = u'x^{1/2} + \frac{u}{2x^{1/2}}$ and $y'' = u''x^{1/2} + \frac{u'}{x^{1/2}} + \dots$ so $4x^2y'' - 4x(x+1)y' + (2x+3)y = 4x^{5/2}(u'' - u') = 4x^{5/2}e^{2x}$ if $u' = z$, where (A) $z' - z = e^{2x}$. Since $z_1 = e^x$ is a solution of the complementary equation for (A), the solutions of (A) are of the form (B) $z = ve^x$, where $v'e^x = e^{2x}$, so $v' = e^x$. Hence, $v = e^x + C_1$; $u' = z = e^{2x} + C_1 e^x$ (see (B)); $u = \frac{e^{2x}}{2} + C_1 e^x + C_2$; $y = ux^{1/2} = x^{1/2}\left(\frac{e^{2x}}{2} + C_1 e^x + C_2\right)$, or $y = x^{1/2}\left(\frac{e^{2x}}{2} + c_1 + c_2 e^x\right)$. As a byproduct, $\{x^{1/2}, x^{1/2}e^x\}$ is a fundamental set of solutions of the complementary equation.

5.6.18. If $y = ue^x$, then $y' = (u' + u)e^x$ and $y'' = (u'' + 2u' + u)e^x$, so $xy'' + (2-2x)y' + (x-2)y = e^x(xu'' + 2u') = 0$ if $\frac{u''}{u'} = -\frac{2}{x}$; $\ln|u'| = -2\ln|x| + k$; $u' = \frac{C_1}{x^2}$; $u = -\frac{C_1}{x} + C_2$. Therefore, $y = ue^x = e^x\left(-\frac{C_1}{x} + C_2\right)$ is the general solution, and $\{e^x, e^x/x\}$ is a fundamental set of solutions.

5.6.20. If $y = u \ln|x|$, then $y' = u' \ln|x| + \frac{u}{x}$ and $y'' = u'' \ln|x| + \frac{2u'}{x} \dots$, so $x^2(\ln|x|)^2 y'' - (2x \ln|x|)y' + (2 + \ln|x|)y = x^2(\ln|x|)^3 u'' = 0$ if $u'' = 0$; $u' = C_1$; $u = C_1 x + C_2$. Therefore, $y = u \ln|x| = (C_1 x + C_2) \ln|x|$ is the general solution, and $\{\ln|x|, x \ln|x|\}$ is a fundamental set of solutions.

5.6.22. If $y = ue^x$, then $y' = u'e^x + ue^x$ and $y'' = u''e^x + 2u'e^x + ue^x$, so $xy'' - (2x+2)y' + (x+2)y = e^x(xu'' - 2u') = 0$ if $\frac{u''}{u'} = \frac{2}{x}$; $\ln|u'| = 2\ln|x| + k$; $u' = C_1 x^2$; $u = \frac{C_1 x^3}{3} + C_2$. Therefore, $y = ue^x = \left(\frac{C_1 x^3}{3} + C_2\right) e^x$ is the general solution, and $\{e^x, x^3 e^x\}$ is a fundamental set of solutions.

5.6.24. If $y = ux \sin x$, then $y' = u'x \sin x + u(x \cos x + \sin x)$ and $y'' = u''x \sin x + 2u'(x \cos x + \sin x) + \dots$, so $x^2 y'' - 2xy' + (x^2 + 2)y = (x^3 \sin x)u'' + 2(x^3 \cos x)u' = 0$ if $\frac{u''}{u'} = -\frac{2 \cos x}{\sin x}$;

$\ln |u'| = -2 \ln |\sin x| + k$; $u' = \frac{C_1}{\sin^2 x}$; $u = -C_1 \cot x + C_2$. Therefore,
 $y = ux \sin x = x(-C_1 \cos x + C_2 \sin x)$ is the general solution, and $\{x \sin x, x \cos x\}$ is a
fundamental set of solutions.

5.6.26. If $y = ux^{1/2}$, then $y' = u'x^{1/2} + \frac{u}{2x^{1/2}}$ and $y'' = u''x^{1/2} + \frac{u'}{x^{1/2}} + \dots$ so
 $4x^2(\sin x)y'' - 4x(x \cos x + \sin x)y' + (2x \cos x + 3 \sin x)y = 4x^{5/2}(u'' \sin x - u' \cos x) = 0$ if
 $\frac{u''}{u'} = \frac{\cos x}{\sin x}$; $\ln |u'| = \ln |\sin x| + k$; $u' = C_1 \sin x$; $u = -C_1 \cos x + C_2$.
Therefore, $y = ux^{1/2} = (-C_1 \cos x + C_2)x^{1/2}$ is the general solution, and $\{x^{1/2}, x^{1/2} \cos x\}$ is
a fundamental set of solutions.

5.6.28. If $y = \frac{u}{x}$, then $y' = \frac{u'}{x} - \frac{u}{x^2}$ and $y'' = \frac{u''}{x} - \frac{2u'}{x^2} + \dots$, so
 $(2x+1)xy'' - 2(2x^2-1)y' - 4(x+1)y = (2x+1)u'' - (4x+4)u' = 0$ if
 $\frac{u''}{u'} = \frac{4x+4}{2x+1} = 2 + \frac{2}{2x+1}$; $\ln |u'| = 2x + \ln |2x+1| + k$; $u' = C_1(2x+1)e^{2x}$;
 $u = C_1xe^{2x} + C_2$. Therefore, $y = \frac{u}{x} = C_1e^{2x} + \frac{C_2}{x}$ is the general solution, and $\{1/x, e^{2x}\}$ is a
fundamental set of solutions.

5.6.30. If $y = ue^{2x}$, then $y' = (u' + 2u)e^{2x}$ and $y'' = (u'' + 4u' + 4u)e^{2x}$, so
 $xy'' - (4x+1)y' + (4x+2)y = e^{2x}(xu'' - u') = 0$ if $\frac{u''}{u'} = \frac{1}{x}$; $\ln |u'| = \ln |x| + k$; $u' = C_1x$;
 $u = \frac{C_1x^2}{2} + C_2$. Therefore, $y = ue^{2x} = e^{2x}\left(\frac{C_1x^2}{2} + C_2\right)$ is the general solution, and
 $\{e^{2x}, x^2e^{2x}\}$ is a fundamental set of solutions.

5.6.32. If $y = ue^{2x}$, then $y' = (u' + 2u)e^{2x}$ and $y'' = (u'' + 4u' + 4u)e^{2x}$, so
 $(3x-1)y'' - (3x+2)y' - (6x-8)y = e^{2x}[(3x-1)u'' + (9x-6)u'] = 0$ if
 $\frac{u''}{u'} = -\frac{9x-6}{3x-1} = -3 + \frac{3}{3x-1}$. Therefore, $\ln |u'| = -3x + \ln |3x-1| + k$, so
 $u' = C_1(3x-1)e^{-3x}$, $u = -C_1xe^{-3x} + C_2$. Therefore, the general solution is
 $y = ue^{2x} = -C_1xe^{-x} + C_2e^{2x}$, or (A) $y = c_1e^{2x} + c_2xe^{-x}$. Now $y(0) = 2 \Rightarrow c_1 = 2$.
Differentiating (A) yields $y' = 2c_1e^{2x} + c_2(e^{-x} - xe^{-x})$. Now $y'(0) = 3 \Rightarrow 3 = 2c_1 + c_2$, so
 $c_2 = -1$ and $y = 2e^{2x} - xe^{-x}$.

5.6.34. If $y = ux$, then $y' = u'x + u$ and $y'' = u''x + 2u'$, so $x^2y'' + 2xy' - 2y = x^3u'' + 4x^2u' = x^2$
if $u' = z$, where (A) $z' + \frac{4}{x}z = \frac{1}{x}$. Since $\int \frac{4}{x} dx = 4 \ln |x|$, $z_1 = \frac{1}{x^4}$ is a solution of the
complementary equation for (A). Therefore, the solutions of (A) are of the form (B) $z = \frac{v}{x^4}$,
where $\frac{v'}{x^4} = \frac{1}{x}$, so $v' = x^3$. Hence, $v = \frac{x^4}{4} + C_1$; $u' = z = \frac{1}{4} + \frac{C_1}{x^4}$ (see (B));
 $u = \frac{x}{4} - \frac{C_1}{3x^3} + C_2$. Therefore, the general solution is $y = ux = \frac{x^2}{4} - \frac{C_1}{3x^2} + C_2x$, or (C)
 $y = \frac{x^2}{4} + c_1x + \frac{c_2}{x^2}$. Differentiating (C) yields $y' = \frac{x}{2} + c_1 - 2\frac{c_2}{x^3}$. Now
 $y(1) = \frac{5}{4}$, $y'(1) = \frac{3}{2} \Rightarrow c_1 + c_2 = 1$, $c_1 - 2c_2 = 1$, so $c_1 = 1$, $c_2 = 0$ and $y = \frac{x^2}{4} + x$.

5.6.36. If $y = uy_1$, then $y' = u'y_1 + uy_1'$ and $y'' = u''y_1 + 2u'y_1' + uy_1''$, so

$y'' + p_1(x)y' + p_2(x)y = y_1u'' + (2y_1' + p_1y_1)u' = 0$ if u is any function such that (B)

$\frac{u''}{u'} = -2\frac{y_1'}{y_1} - p_1$. If $\ln |u'(x)| = -2\ln |y_1(x)| - \int_{x_0}^x p_1(t) dt$, then u satisfies (B); therefore, if

(C) $u'(x) = \frac{1}{y_1^2(x)} \exp\left(-\int_{x_0}^x p_1(s) ds\right)$, then u satisfies (B). Since

$u(x) = \int_{x_0}^x \frac{1}{y_1^2(t)} \exp\left(-\int_{x_0}^t p_1(s) ds\right) dt$ satisfies (C), $y_2 = uy_1$ is a solution of (A) on (a, b) .

Since $\frac{y_2}{y_1} = u$ is nonconstant, Theorem 5.1.6 implies that $\{y_1, y_2\}$ is a fundamental set of solutions of (A) on (a, b) .

5.6.38. (a) The associated linear equation is (A) $z'' + k^2z = 0$, with characteristic polynomial $p(r) = r^2 + k^2$. The general solution of (A) is $z = c_1 \cos kx + c_2 \sin kx$. Since

$$z' = -kc_1 \sin kx + kc_2 \cos kx, y = \frac{z'}{z} = \frac{-kc_1 \sin kx + kc_2 \cos kx}{c_1 \cos kx + c_2 \sin kx}.$$

(b) The associated linear equation is (A) $z'' - 3z' + 2z = 0$, with characteristic polynomial $p(r) = r^2 - 3r + 2 = (r-1)(r-2)$. The general solution of (A) is $z = c_1 e^x + c_2 e^{2x}$.

$$\text{Since } z' = c_1 e^x + c_2 e^{2x}, y = \frac{z'}{z} = \frac{c_1 + 2c_2 e^x}{c_1 + c_2 e^x}.$$

(c) The associated linear equation is (A) $z'' + 5z' - 6z = 0$, with characteristic polynomial $p(r) = r^2 + 5r - 6 = (r+6)(r-1)$. The general solution of (A) is $z = c_1 e^{-6x} + c_2 e^x$.

$$\text{Since } z' = -6c_1 e^{-6x} + c_2 e^x, y = \frac{z'}{z} = \frac{-6c_1 + c_2 e^{7x}}{c_1 + c_2 e^{7x}}.$$

(d) The associated linear equation is (A) $z'' + 8z' + 7z = 0$, with characteristic polynomial $p(r) = r^2 + 8r + 7 = (r+7)(r+1)$. The general solution of (A) is $z = c_1 e^{-7x} + c_2 e^{-x}$.

$$\text{Since } z' = -7c_1 e^{-7x} - c_2 e^{-x}, y = \frac{z'}{z} = -\frac{7c_1 + c_2 e^{6x}}{c_1 + c_2 e^{6x}}.$$

(e) The associated linear equation is (A) $z'' + 14z' + 50z = 0$, with characteristic polynomial $p(r) = r^2 + 14r + 50 = (r+7)^2 + 1$. The general solution of (A) is $z = e^{-7x}(c_1 \cos x + c_2 \sin x)$. Since

$$z' = -7e^{-7x}(c_1 \cos x + c_2 \sin x) + e^{-7x}(-c_1 \sin x + c_2 \cos x) = - (7c_1 - c_2) \cos x - (c_1 + 7c_2) \sin x, y = \frac{z'}{z} = -\frac{(7c_1 - c_2) \cos x + (c_1 + 7c_2) \sin x}{c_1 \cos x + c_2 \sin x}.$$

(f) The given equation is equivalent to (A) $y' + y^2 - \frac{1}{6}y - \frac{1}{6} = 0$. The associated linear equation is (B) $z'' - \frac{1}{6}z' - \frac{1}{6}z = 0$, with characteristic polynomial

$$p(r) = r^2 - \frac{1}{6}r - \frac{1}{6} = \left(r + \frac{1}{3}\right)\left(r - \frac{1}{2}\right). \text{ The general solution of (B) is}$$

$$z = c_1 e^{-x/3} + c_2 e^{x/2}. \text{ Since } z' = -\frac{c_1}{3} e^{-x/3} + \frac{c_2}{2} e^{x/2}, y = \frac{z'}{z} = \frac{-2c_1 + 3c_2 e^{5x/6}}{6(c_1 + c_2 e^{5x/6})}.$$

(g) The given equation is equivalent to (A) $y' + y^2 - \frac{1}{3}y + \frac{1}{36} = 0$. The associated linear

equation is (B) $z'' - \frac{1}{3}z' + \frac{1}{36}z = 0$, with characteristic polynomial $p(r) = r^2 - \frac{1}{3}r + \frac{1}{36} = \left(r - \frac{1}{6}\right)^2$. The general solution of (B) is $z = e^{x/6}(c_1 + c_2x)$. Since $z' = \frac{e^{x/6}}{6}(c_1 + c_2x) + c_2e^{x/6} = \frac{e^{x/6}}{6}(c_1 + c_2(x+6))$, $y = \frac{z'}{z} = \frac{c_1 + c_2(x+6)}{6(c_1 + c_2x)}$.

- 5.6.40. (a) Suppose that z is a solution of (B) and let $y = \frac{z'}{rz}$. Then (D) $\frac{z''}{rz} + \left[p(x) - \frac{r'(x)}{r(x)}\right]y + q(x) = 0$ and $y' = \frac{z''}{rz} - \frac{1}{r}\left(\frac{z'}{z}\right)^2 - \frac{r'z'}{r^2z} = \frac{z''}{rz} - ry^2 - \frac{r'}{r}y$, so $\frac{z''}{rz} = y' + ry^2 + \frac{r'}{r}y$. Therefore, (D) implies that y satisfies (A). Now suppose that y is a solution of (A) and let z be any function such that $z' = ryz$. Then $z'' = r'yz + ry'z + ryz' = \frac{r'}{r}z' + (y' + ry^2)rz = \frac{r'}{r}z' - (p(x)y + q(x))rz$, so $z'' - \frac{r'}{r}z' + p(x)ryz + q(x)rz = 0$, which implies that z satisfies (B), since $ryz = z'$.
- (b) If $\{z_1, z_2\}$ is a fundamental set of solutions of (B) on (a, b) , then $z = c_1z_1 + c_2z_2$ is the general solution of (B) on (a, b) . This and (a) imply that (C) is the general solution of (A) on (a, b) .

5.7 Variation of Parameters

5.7.2.

$$y_p = u_1 \cos 2x + u_2 \sin 2x \quad (\text{A})$$

$$u_1' \cos 2x + u_2' \sin 2x = 0 \quad (\text{B})$$

$$-2u_1' \sin 2x + 2u_2' \cos 2x = \sin 2x \sec^2 x. \quad (\text{C})$$

Multiplying (B) by $2 \sin 2x$ and (C) by $\cos 2x$ and adding the resulting equations yields

$2u_2' = \tan 2x$, so $u_2' = \frac{\tan 2x}{2}$. Then (B) implies that

$$u_1' = -u_2' \tan(2x) = -\frac{\tan^2 2x}{2} = \frac{1 - \sec^2 2x}{2}. \text{ Therefore, } u_1 = \frac{x}{2} - \frac{\tan 2x}{4} \text{ and}$$

$$u_2 = -\frac{\ln |\cos 2x|}{4}. \text{ Now (A) yields } y_p = -\frac{\sin 2x \ln |\cos 2x|}{4} + \frac{x \cos 2x}{2} - \frac{\sin 2x}{4}. \text{ Since } \sin 2x \text{ satisfies the complementary equation we redefine } y_p = -\frac{\sin 2x \ln |\cos 2x|}{4} + \frac{x \cos 2x}{2}.$$

5.7.4.

$$y_p = u_1 e^x \cos x + u_2 e^x \sin x \quad (\text{A})$$

$$u_1' e^x \cos x + u_2' e^x \sin x = 0 \quad (\text{B})$$

$$u_1'(e^x \cos x - e^x \sin x) + u_2'(e^x \sin x + e^x \cos x) = 3e^x \sec x. \quad (\text{C})$$

Subtracting (B) from (C) and cancelling e^x from the resulting equations yields

$$u_1' \cos x + u_2' \sin x = 0 \quad (\text{D})$$

$$-u_1' \sin x + u_2' \cos x = 3 \sec x. \quad (\text{E})$$

Multiplying (D) by $\sin x$ and (E) by $\cos x$ and adding the results yields $u_2' = 3$. From (D), $u_1' = -u_2' \tan x = -3 \tan x$. Therefore $u_1 = 3 \ln |\cos x|$, $u_2 = 3x$. Now (A) yields $y_p = 3e^x(\cos x \ln |\cos x| + x \sin x)$.

5.7.6.

$$y_p = u_1 e^x + u_2 e^{-x} \quad (\text{A})$$

$$u_1' e^x + u_2' e^{-x} = 0 \quad (\text{B})$$

$$u_1' e^x - u_2' e^{-x} = \frac{4e^{-x}}{1 + e^{-2x}}. \quad (\text{C})$$

Adding (B) to (C) yields $2u_1' e^x = \frac{4e^{-x}}{1 + e^{-2x}}$, so $u_1' = \frac{2e^{-2x}}{1 - e^{-2x}}$. From (B),

$u_2' = -e^{2x} u_1' = -\frac{2}{1 - e^{-2x}} = \frac{2e^{2x}}{1 - e^{2x}}$. Using the substitution $v = e^{-2x}$ we integrate u_1' to obtain $u_1 = \ln(1 - e^{-2x})$. Using the substitution $v = e^{2x}$ we integrate u_2' to obtain $u_2 = \ln(1 - e^{2x})$. Now (A) yields $y_p = e^x \ln(1 - e^{-2x}) - e^{-x} \ln(e^{2x} - 1)$.

5.7.8.

$$y_p = u_1 e^x + u_2 \frac{e^x}{x} \quad (\text{A})$$

$$u_1' e^x + u_2' \frac{e^x}{x} = 0 \quad (\text{B})$$

$$u_1' e^x + u_2' \left(\frac{e^x}{x} - \frac{e^x}{x^2} \right) = \frac{e^{2x}}{x}. \quad (\text{C})$$

Subtracting (B) from (C) yields $-\frac{u_2' e^x}{x^2} = \frac{e^{2x}}{x}$, so $u_2' = -xe^x$. From (B), $u_1' = \frac{u_2'}{x} = e^x$.

Therefore $u_1 = e^x$, $u_2 = -xe^x + e^x$. Now (A) yields $y_p = \frac{e^{2x}}{x}$.

5.7.10.

$$y_p = u_1 e^{-x^2} + u_2 x e^{-x^2} \quad (\text{A})$$

$$u_1' e^{-x^2} + u_2' x e^{-x^2} = 0 \quad (\text{B})$$

$$-2xu_1' e^{-x^2} + u_2'(e^{-x^2} - 2x^2 e^{-x^2}) = 4e^{-x(x+2)}. \quad (\text{C})$$

Multiplying (B) by $2x$ and adding the result to (C) yields $u_2' e^{-x^2} = 4e^{-x(x+2)}$, so $u_2' = 4e^{-2x}$. From (B), $u_1' = -u_2' x = -4xe^{-2x}$. Therefore $u_1 = (2x + 1)e^{-2x}$, $u_2 = -2e^{-2x}$. Now (A) yields $y_p = e^{-x(x+2)}$.

5.7.12.

$$y_p = u_1 x + u_2 x^3 \quad (\text{A})$$

$$u_1' x + u_2' x^3 = 0 \quad (\text{B})$$

$$u_1' + 3u_2'x^2 = \frac{2x^4 \sin x}{x^2} = 2x^2 \sin x. \quad (C)$$

Multiplying (B) by $\frac{1}{x}$ and subtracting the result from (C) yields $2x^2 u_2' = 2x^2 \sin x$, so $u_2' = \sin x$. From (B), $u_1' = -u_2'x^2 = -x^2 \sin x$. Therefore $u_1 = (x^2 - 2) \cos x - 2x \sin x$, $u_2 = -\cos x$. Now (A) yields $y_p = -2x^2 \sin x - 2x \cos x$.

5.7.14.

$$y_p = u_1 \cos \sqrt{x} + u_2 \sin \sqrt{x} \quad (A)$$

$$u_1' \cos \sqrt{x} + u_2' \sin \sqrt{x} = 0 \quad (B)$$

$$-u_1' \frac{\sin \sqrt{x}}{2\sqrt{x}} + u_2' \frac{\cos \sqrt{x}}{2\sqrt{x}} = \frac{\sin \sqrt{x}}{4x} \quad (C)$$

Multiplying (B) by $\frac{\sin \sqrt{x}}{2\sqrt{x}}$ and (C) by $\cos \sqrt{x}$ and adding the resulting equations yields

$$\frac{u_2'}{2\sqrt{x}} = \frac{\sin \sqrt{x} \cos \sqrt{x}}{4x}, \text{ so } u_2' = \frac{\sin \sqrt{x} \cos \sqrt{x}}{2\sqrt{x}}. \text{ From (B), } u_1' = -u_2' \tan \sqrt{x} = -\frac{\sin^2 \sqrt{x}}{2\sqrt{x}}.$$

Therefore, $u_1 = \frac{\sin \sqrt{x} \cos \sqrt{x}}{2} - \frac{\sqrt{x}}{2}$, $u_2 = \frac{\sin^2 \sqrt{x}}{2}$. Now (A) yields

$$y_p = \frac{\sin \sqrt{x}}{2} - \frac{\sqrt{x} \cos \sqrt{x}}{2}. \text{ Since } \sin \sqrt{x} \text{ satisfies the complementary equation we redefine}$$

$$y_p = -\frac{\sqrt{x} \cos \sqrt{x}}{2}.$$

5.7.16.

$$y_p = u_1 x^a + u_2 x^a \ln x \quad (A)$$

$$u_1' x^a + u_2' x^a \ln x = 0 \quad (B)$$

$$au_1' x^{a-1} + u_2' (ax^{a-1} \ln x + x^{a-1}) = \frac{x^{a+1}}{x^2} = x^{a-1} \quad (C)$$

Multiplying (B) by $\frac{a}{x}$ and subtracting the result from (C) yields $u_2' x^{a-1} = x^{a-1}$, so $u_2' = 1$.

From (B), $u_1' = -u_2' \ln x = -\ln x$. Therefore, $u_1 = x - \ln x$, $u_2 = x$. Now (A) yields

$$y_p = x^{a+1}.$$

5.7.18.

$$y_p = u_1 e^{x^2} + u_2 e^{-x^2} \quad (A)$$

$$u_1' e^{x^2} + u_2' e^{-x^2} = 0 \quad (B)$$

$$2u_1' x e^{x^2} - 2u_2' x e^{-x^2} = \frac{8x^5}{x} = 8x^4. \quad (C)$$

Multiplying (B) by $2x$ and adding the result to (C) yields $4u_1' x e^{x^2} = 8x^4$, so $u_1' = 2x^3 e^{-x^2}$. From (B), $u_2' = -u_1' e^{2x^2} = -2x^3 e^{x^2}$. Therefore $u_1 = -e^{-x^2}(x^2 + 1)$, $u_2 = -e^{x^2}(x^2 - 1)$. Now (A) yields $y_p = -2x^2$.

5.7.20.

$$y_p = u_1 \sqrt{x} e^{2x} + u_2 \sqrt{x} e^{-2x} \quad (\text{A})$$

$$u_1' \sqrt{x} e^{2x} + u_2' \sqrt{x} e^{-2x} = 0 \quad (\text{B})$$

$$u_1' e^{2x} \left(2\sqrt{x} + \frac{1}{2\sqrt{x}} \right) - u_2' e^{-2x} \left(2\sqrt{x} - \frac{1}{2\sqrt{x}} \right) = \frac{8x^{5/2}}{4x^2} = 2\sqrt{x} \quad (\text{C})$$

Multiplying (B) by $\frac{1}{2x}$, subtracting the result from (C), and cancelling common factors from the resulting equations yields

$$u_1' e^{2x} + u_2' e^{-2x} = 0 \quad (\text{D})$$

$$u_1' e^{2x} - u_2' e^{-2x} = 1. \quad (\text{E})$$

Adding (D) to (E) yields $2u_1' e^{2x} = 1$, so $u_1' = \frac{e^{-2x}}{2}$. From (D), $u_2' = -u_1' e^{4x} = -\frac{e^{2x}}{2}$.

Therefore, $u_1 = -\frac{e^{-2x}}{4}$, $u_2 = -\frac{e^{2x}}{4}$. Now (A) yields $y_p = -\frac{\sqrt{x}}{2}$.

5.7.22.

$$y_p = u_1 x e^x + u_2 x e^{-x} \quad (\text{A})$$

$$u_1' x e^x + u_2' x e^{-x} = 0 \quad (\text{B})$$

$$u_1' (x+1) e^x - u_2' (x-1) e^{-x} = \frac{3x^4}{x^2} = 3x^2 \quad (\text{C})$$

Multiplying (B) by $\frac{1}{x}$, subtracting the resulting equation from (C), and cancelling common factors yields

$$u_1' e^x + u_2' e^{-x} = 0 \quad (\text{D})$$

$$u_1' e^x - u_2' e^{-x} = 3x. \quad (\text{E})$$

Adding (D) to (E) yields $2u_1' e^x = 3x$, so $u_1' = \frac{3xe^{-x}}{2}$. From (D), $u_2' = -u_1' e^{2x} = -\frac{3xe^x}{2}$.

Therefore $u_1 = -\frac{3e^x(x+1)}{2}$, $u_2 = -\frac{3e^x(x-1)}{2}$. Now (A) yields $y_p = -3x^2$.

5.7.24.

$$y_p = \frac{u_1}{x} + u_2 x^3 \quad (\text{A})$$

$$\frac{u_1'}{x} + u_2' x^3 = 0 \quad (\text{B})$$

$$-\frac{u_1'}{x^2} + 3u_2' x^2 = \frac{x^{3/2}}{x^2} = x^{-1/2}. \quad (\text{C})$$

Multiplying (B) by $\frac{1}{x}$ and adding the result to (C) yields $4u_2' x^2 = x^{-1/2}$, so $u_2' = \frac{x^{-5/2}}{4}$.

From (B), $u_1' = -u_2' x^4 = -\frac{x^{3/2}}{4}$. Therefore $u_1 = -\frac{x^{5/2}}{10}$, $u_2 = -\frac{x^{-3/2}}{6}$. Now (A) yields

$$y_p = -\frac{4x^{3/2}}{15}.$$

5.7.26.

$$y_p = u_1 x^2 e^x + u_2 x^3 e^x \quad (\text{A})$$

$$u_1' x^2 e^x + u_2' x^3 e^x = 0 \quad (\text{B})$$

$$u_1'(x^2 e^x + 2x e^x) + u_2'(x^3 e^x + 3x^2 e^x) = \frac{2x e^x}{x^2} = \frac{2e^x}{x}. \quad (\text{C})$$

Subtracting (B) from (C) and cancelling common factors in the resulting equations yields

$$u_1' + u_2' x = 0 \quad (\text{D})$$

$$2u_1' x + 3u_2' x^2 = \frac{2}{x}. \quad (\text{E})$$

Multiplying (D) by $2x$ and subtracting the result from (E) yields $x^2 u_2' = \frac{2}{x}$, so $u_2' = \frac{2}{x^3}$.

From (D), $u_1' = -u_2' x = -\frac{2}{x^2}$. Therefore $u_1 = \frac{2}{x}$, $u_2 = -\frac{1}{x^2}$. Now (A) yields $y_p = x e^x$.

5.7.28.

$$y_p = u_1 x + u_2 e^x \quad (\text{A})$$

$$u_1' x + u_2' e^x = 0 \quad (\text{B})$$

$$u_1' + u_2' e^x = \frac{2(x-1)^2 e^x}{x-1} = 2(x-1)e^x. \quad (\text{C})$$

Subtracting (B) from (C) yields $u_1'(1-x) = 2(x-1)e^x$, so $u_1' = -2e^x$. From (B), $u_2' = -u_1' x e^{-x} = 2x$. Therefore, $u_1 = -2e^x$, $u_2 = x^2$. Now (A) yields $y_p = x e^x (x-2)$.

5.7.30.

$$y_p = u_1 e^{2x} + u_2 x e^{-x} \quad (\text{A})$$

$$u_1' e^{2x} + u_2' x e^{-x} = 0 \quad (\text{B})$$

$$2u_1' e^{2x} + u_2'(e^{-x} - x e^{-x}) = \frac{(3x-1)^2 e^{2x}}{3x-1} = (3x-1)e^{2x}. \quad (\text{C})$$

Multiplying (B) by 2 and subtracting the result from (C) yields

$u_2'(1-3x)e^{-x} = (3x-1)e^{2x}$, so $u_2' = -e^{3x}$. From (B), $u_1' = -u_2' x e^{-3x} = x$. Therefore

$u_1 = \frac{x^2}{2}$, $u_2 = -\frac{e^{3x}}{3}$. Now (A) yields $y_p = \frac{x e^{2x}(3x-2)}{3}$. The general solution of the given

equation is $y = \frac{x e^{2x}(3x-2)}{3} + c_1 e^{2x} + c_2 x e^{-x}$. Differentiating this yields

$y' = \frac{e^{2x}(3x^2+x-1)}{3} + 2c_1 e^{2x} + c_2(1-x)e^{-x}$. Now

$y(0) = 1$, $y'(0) = 2 \Rightarrow c_1 = 1$, $2 = -\frac{1}{3} + 2c_1 + c_2$, so $c_2 = \frac{1}{3}$, and

$y = \frac{e^{2x}(3x^2-2x+6)}{6} + \frac{x e^{-x}}{3}$.

5.7.32.

$$y_p = u_1(x-1)e^x + u_2(x-1) \quad (\text{A})$$

$$u_1'(x-1)e^x + u_2'(x-1) = 0 \quad (\text{B})$$

$$u_1'xe^x + u_2' = \frac{(x-1)^3e^x}{(x-1)^2} = (x-1)e^x. \quad (\text{C})$$

From (B), $u_1' = -u_2'e^{-x}$. Substituting this into (C) yields $-u_2'(x-1) = (x-1)e^x$, so $u_2' = -e^x$, $u_1' = 1$. Therefore $u_1 = x$, $u_2 = e^x$. Now (A) yields $y_p = e^x(x-1)^2$. The general solution of the given equation is $y = (x-1)^2e^x + c_1(x-1)e^x + c_2(x-1)$. Differentiating this yields $y' = (x^2-1)e^x + c_1xe^x + c_2$. Now $y(0) = 4$, $y'(0) = -6 \Rightarrow 4 = 1 - c_1 - c_2$, $-6 = -1 + c_2$, so $c_1 = 2$, $c_2 = -5$ and $y = (x^2-1)e^x - 5(x-1)$.

5.7.34.

$$y_p = u_1x + \frac{u_2}{x^2} \quad (\text{A})$$

$$u_1'x + \frac{u_2'}{x^2} = 0 \quad (\text{B})$$

$$u_1' - \frac{2u_2'}{x^3} = -\frac{2x^2}{x^2} = -2. \quad (\text{C})$$

Multiplying (B) by $\frac{2}{x}$ and adding the result to (C) yields $3u_1' = -2$, so $u_1' = -\frac{2}{3}$. From (B), $u_2' = -u_1'x^3 = \frac{2x^3}{3}$. Therefore $u_1 = -\frac{2x}{3}$, $u_2 = \frac{x^4}{6}$. Now (A) yields $y_p = -\frac{x^2}{2}$. The general solution of the given equation is $y = -\frac{x^2}{2} + c_1x + \frac{c_2}{x^2}$. Differentiating this yields $y' = -x + c_1 - \frac{2c_2}{x^3}$. Now $y(1) = 1$, $y'(1) = -1 \Rightarrow 1 = -\frac{1}{2} + c_1 + c_2$, $-1 = -1 + c_1 - 2c_2$, so $c_1 = 1$, $c_2 = \frac{1}{2}$, and $y = -\frac{x^2}{2} + x + \frac{1}{2x^2}$.

5.7.36. Since $\bar{y} = y_p - a_1y_1 - a_2y_2$,

$$\begin{aligned} P_0(x)\bar{y}'' + P_1(x)\bar{y}' + P_2(x)\bar{y} &= P_0(x)(y_p - a_1y_1 - a_2y_2)'' \\ &\quad + P_1(x)(y_p - a_1y_1 - a_2y_2)' \\ &\quad + P_2(x)(y_p - a_1y_1 - a_2y_2) \\ &= (P_0(x)y_p'' + P_1(x)y_p' + P_2(x)y_p) \\ &\quad - a_1[P_0(x)y_1'' + P_1(x)y_1' + P_2(x)y_1] \\ &\quad - a_2[P_0(x)y_2'' + P_1(x)y_2' + P_2(x)y_2] \\ &= F(x) - a_1 \cdot 0 - a_2 \cdot 0 = F(x); \end{aligned}$$

hence $\bar{y}$ is a particular solution of (A).

5.7.38. (a) $y_p = u_1 e^x + u_2 e^{-x}$ is a solution of (A) on (a, ∞) if $u_1' e^x + u_2' e^{-x} = 0$ and $u_1' e^x - u_2' e^{-x} = f(x)$. Solving these two equations yields $u_1' = \frac{e^{-x} f}{2}$, $u_2' = -\frac{e^x f}{2}$. The functions $u_1(x) = \frac{1}{2} \int_0^x e^{-t} f(t) dt$ and $u_2(x) = -\frac{1}{2} \int_0^x e^t f(t) dt$ satisfy these conditions. Therefore,

$$\begin{aligned} y_p(x) &= \frac{e^x}{2} \int_0^x e^{-t} f(t) dt - \frac{e^{-x}}{2} \int_0^x e^t f(t) dt \\ &= \frac{1}{2} \int_0^x f(t) (e^{(x-t)} - e^{-(x-t)}) dt = \int_0^x f(t) \sinh(x-t) dt. \end{aligned}$$

is a particular solution of $y'' - y = f(x)$. Differentiating y_p yields

$$\begin{aligned} y_p'(x) &= \frac{e^x}{2} \int_0^x e^{-t} f(t) dt + \frac{e^x}{2} e^{-x} + \frac{e^{-x}}{2} \int_0^x e^t f(t) dt - \frac{e^{-x}}{2} e^x \\ &= \frac{e^x}{2} \int_0^x e^{-t} f(t) dt + \frac{e^{-x}}{2} \int_0^x e^t f(t) dt \\ &= \frac{1}{2} \int_0^x f(t) (e^{(x-t)} + e^{-(x-t)}) dt = \int_0^x f(t) \cosh(x-t) dt. \end{aligned}$$

Since $y_p(x_0) = y_p'(x_0) = 0$, the solution of the initial value problem is

$$\begin{aligned} y &= y_p + k_0 \cosh x + k_1 \sinh x \\ &= k_0 \cosh x + k_1 \sinh x + \int_0^x \sinh(x-t) f(t) dt. \end{aligned}$$

The derivative of the solution is

$$\begin{aligned} y' &= y_p' + k_0 \sinh x + k_1 \cosh x \\ &= k_0 \sinh x + k_1 \cosh x + \int_0^x \cosh(x-t) f(t) dt. \end{aligned}$$

Chapter 6

Applications of Linear Second Order Equations

6.1 Spring Problems I

6.1.2. Since $\frac{k}{m} = \frac{g}{\Delta l} = \frac{32}{.1} = 320$ the equation of motion is (A) $y'' + 320y = 0$. The general solution of (A) is $y = c_1 \cos 8\sqrt{5}t + c_2 \sin 8\sqrt{5}t$, so $y' = 8\sqrt{5}(-c_1 \sin 8\sqrt{5}t + c_2 \cos 8\sqrt{5}t)$. Now $y(0) = -\frac{1}{4} \Rightarrow c_1 = -\frac{1}{4}$ and $y'(0) = -2 \Rightarrow c_2 = -\frac{1}{4\sqrt{5}}$, so

$$y = -\frac{1}{4} \cos 8\sqrt{5}t - \frac{1}{4\sqrt{5}} \sin 8\sqrt{5}t \text{ ft.}$$

6.1.4. Since $\frac{k}{m} = \frac{g}{\Delta l} = \frac{32}{.5} = 64$ the equation of motion is (A) $y'' + 64y = 0$. The general solution of (A) is $y = c_1 \cos 8t + c_2 \sin 8t$, so $y' = 8(-c_1 \sin 8t + c_2 \cos 8t)$. Now $y(0) = \frac{1}{4} \Rightarrow c_1 = \frac{1}{4}$ and $y'(0) = -\frac{1}{2} \Rightarrow c_2 = -\frac{1}{16}$, so $y = \frac{1}{4} \cos 8t - \frac{1}{16} \sin 8t$ ft; $R = \frac{\sqrt{17}}{16}$ ft; $\omega_0 = 8$ rad/s; $T = \pi/4$ s; $\phi \approx -0.245$ rad $\approx -14.04^\circ$.

6.1.6. Since $k = \frac{mg}{\Delta l} = \frac{(9.8)10}{.7} = 140$, the equation of motion of the 2 kg mass is (A) $y'' + 70y = 0$. The general solution of (A) is $y = c_1 \cos \sqrt{70}t + c_2 \sin \sqrt{70}t$, so $y' = \sqrt{70}(-c_1 \sin \sqrt{70}t + c_2 \cos \sqrt{70}t)$. Now $y(0) = -\frac{1}{4} \Rightarrow c_1 = \frac{1}{4}$ and $y'(0) = 2 \Rightarrow c_2 = \frac{2}{\sqrt{70}}$, so $y = -\frac{1}{4} \cos \sqrt{70}t + \frac{2}{\sqrt{70}} \sin \sqrt{70}t$ m; $R = \frac{1}{4}\sqrt{\frac{67}{35}}$ m; $\omega_0 = \sqrt{70}$ rad/s; $T = 2\pi/\sqrt{70}$ s; $\phi \approx 2.38$ rad $\approx 136.28^\circ$.

6.1.8. Since $\frac{k}{m} = \frac{g}{\Delta l} = \frac{32}{1/2} = 64$ the equation of motion is (A) $y'' + 64y = 0$. The general solution of (A) is $y = c_1 \cos 8t + c_2 \sin 8t$, so $y' = 8(-c_1 \sin 8t + c_2 \cos 8t)$. Now

$$y(0) = \frac{1}{2} \Rightarrow c_1 = \frac{1}{2} \text{ and } y'(0) = -3 \Rightarrow c_2 = -\frac{3}{8}, \text{ so } y = \frac{1}{2} \cos 8t - \frac{3}{8} \sin 8t \text{ ft.}$$

6.1.10. $m = \frac{64}{32} = 2$, so the equation of motion is $2y'' + 8y = 2 \sin t$, or (A) $y'' + 4y = \sin t$. Let $y_p = A \cos t + B \sin t$; then $y_p'' = -A \cos t - B \sin t$, so $y_p'' + 4y_p = 3A \cos t + 3B \sin t = \sin t$ if $3A = 0$, $3B = 1$. Therefore, $A = 0$, $B = \frac{1}{3}$, and $y_p = \frac{1}{3} \sin t$. The general solution of (A) is (B) $y = \frac{1}{3} \sin t + c_1 \cos 2t + c_2 \sin 2t$, so $y(0) = \frac{1}{2} \Rightarrow c_1 = \frac{1}{2}$. Differentiating (B) yields $y' = \frac{1}{3} \cos t - 2c_1 \sin 2t + 2c_2 \cos 2t$, so $y'(0) = 2 \Rightarrow 2 = \frac{1}{3} + 2c_2$, so $c_2 = \frac{5}{6}$. Therefore, $y = \frac{1}{3} \sin t + \frac{1}{2} \cos 2t + \frac{5}{6} \sin 2t$ ft.

6.1.12. $m = \frac{4}{32} = \frac{1}{8}$ and $k = \frac{mg}{\Delta l} = 4$, so the equation of motion is $\frac{1}{8}y'' + 4y = \frac{1}{4} \sin 8t$, or (A) $y'' + 32y = 2 \sin 8t$. Let $y_p = A \cos 8t + B \sin 8t$; then $y_p'' = -64A \cos 8t - 64B \sin 8t$, so $y_p'' + 32y_p = -32A \cos 8t - 32B \sin 8t = 2 \sin 8t$ if $-32A = 0$, $-32B = 2$. Therefore, $A = 0$, $B = -\frac{1}{16}$, and $y_p = -\frac{1}{16} \sin 8t$. The general solution of (A) is (B) $y = -\frac{1}{16} \sin 8t + c_1 \cos 4\sqrt{2}t + c_2 \sin 4\sqrt{2}t$, so $y(0) = \frac{1}{3} \Rightarrow c_1 = \frac{1}{3}$. Differentiating (B) yields $y' = -\frac{1}{2} \cos 8t + 4\sqrt{2}(-c_1 \sin 4\sqrt{2}t + c_2 \cos 4\sqrt{2}t)$, so $y'(0) = -1 \Rightarrow -1 = -\frac{1}{2} + 4\sqrt{2}c_2$, so $c_2 = -\frac{1}{8\sqrt{2}}$. Therefore, $y = -\frac{1}{16} \sin 8t + \frac{1}{3} \cos 4\sqrt{2}t - \frac{1}{8\sqrt{2}} \sin 4\sqrt{2}t$ ft.

6.1.14. Since $T = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{m}{k}}$ the period is proportional to the square root of the mass.

Therefore, doubling the mass multiplies the period by $\sqrt{2}$; hence the period of the system with the 20 g mass is $T = 4\sqrt{2}$ s.

6.1.16. $m = \frac{6}{32} = \frac{3}{16}$ and $k = \frac{mg}{\Delta l} = \frac{6}{1/3} = 18$ so the equation of motion is $\frac{3}{16}y'' + 18y = 4 \sin \omega t - 6 \cos \omega t$, or (A) $y'' + 96y = \frac{64}{3} \sin \omega t - 32 \cos \omega t$. The displacement will be unbounded if $\omega = \sqrt{96} = 4\sqrt{6}$, in which case (A) becomes (B) $y'' + 96y = \frac{64}{3} \sin 4\sqrt{6}t - 32 \cos 4\sqrt{6}t$. Let

$$\begin{aligned} y_p &= At \cos 4\sqrt{6}t + Bt \sin 4\sqrt{6}t; \text{ then} \\ y_p' &= (A + 4\sqrt{6}Bt) \cos 4\sqrt{6}t + (B - 4\sqrt{6}At) \sin 4\sqrt{6}t \\ y_p'' &= (8\sqrt{6}B - 96At) \cos 4\sqrt{6}t - (8\sqrt{6}A + 96Bt) \sin 4\sqrt{6}t, \quad \text{so} \\ y_p'' + 96y_p &= 8\sqrt{6}B \cos 4\sqrt{6}t - 8\sqrt{6}A \sin 4\sqrt{6}t = \frac{64}{3} \sin 4\sqrt{6}t - 32 \cos 4\sqrt{6}t \end{aligned}$$

if $8\sqrt{6}B = -32$, $-8\sqrt{6}A = \frac{64}{3}$. Therefore, $A = -\frac{8}{3\sqrt{6}}$, $B = -\frac{4}{\sqrt{6}}$, and

$y_p = -\frac{t}{\sqrt{6}} \left(\frac{8}{3} \cos 4\sqrt{6}t + 4 \sin 4\sqrt{6}t \right)$. The general solution of (B) is

$$y = -\frac{t}{\sqrt{6}} \left(\frac{8}{3} \cos 4\sqrt{6}t + 4 \sin 4\sqrt{6}t \right) + c_1 \cos 4\sqrt{6}t + c_2 \sin 4\sqrt{6}t, \quad (\text{C})$$

so $y(0) = 0 \Rightarrow c_1 = 0$. Differentiating (C) yields

$$y' = -\left(\frac{8}{3\sqrt{6}} \cos 4\sqrt{6}t + \frac{4}{\sqrt{6}} \sin 4\sqrt{6}t \right) - 4t \left(-\frac{8}{3} \sin 4\sqrt{6}t + 4 \cos 4\sqrt{6}t \right) + 4\sqrt{6}(-c_1 \sin 4\sqrt{6}t + c_2 \cos 4\sqrt{6}t),$$

so $y'(0) = 0 \Rightarrow 0 = -\frac{8}{3\sqrt{6}} + 4\sqrt{6}c_2$, and $c_2 = \frac{1}{9}$. Therefore,

$$y = -\frac{t}{\sqrt{6}} \left(\frac{8}{3} \cos 4\sqrt{6}t + 4 \sin 4\sqrt{6}t \right) + \frac{1}{9} \sin 4\sqrt{6}t.$$

6.1.18. The equation of motion is (A) $y'' + \omega_0^2 y = 0$. The general solution of (A) is

$y = c_1 \cos \omega_0 t + c_2 \sin \omega_0 t$. Now $y(0) = y_0 \Rightarrow c_1 = y_0$. Since $y' = \omega_0(-c_1 \sin \omega_0 t + c_2 \cos \omega_0 t)$, $y'(0) = v_0 \Rightarrow c_2 = \frac{v_0}{\omega_0}$. Therefore, $y = y_0 \cos \omega_0 t + \frac{v_0}{\omega_0} \sin \omega_0 t$;

$$R = \frac{1}{\omega_0} \sqrt{(\omega_0 y_0)^2 + (v_0)^2}; \quad \cos \phi = \frac{y_0 \omega_0}{\sqrt{(\omega_0 y_0)^2 + (v_0)^2}}; \quad \sin \phi = \frac{v_0}{\sqrt{(\omega_0 y_0)^2 + (v_0)^2}}.$$

Discussion 6.1.1 In Exercises 19, 20, and 21 we use the fact that in a spring-mass system with mass m and spring constant k the period of the motion is $T = 2\pi\sqrt{\frac{m}{k}}$. Therefore, if we have two systems with masses m_1 and m_2 and spring constants k_1 and k_2 , then the periods are related by $\frac{T_2}{T_1} = \sqrt{\frac{m_2 k_1}{m_1 k_2}}$. We will use this formula in the solutions of these exercises.

6.1.20. Let $m_2 = 2m_1$. Since $k_1 = k_2$, $\frac{T_2}{T_1} = \sqrt{\frac{2m_1}{m_1}} = \sqrt{2}$, so $T_2 = \sqrt{2}T_1$.

6.1.21. Suppose that $T_2 = 3T_1$. Since $m_1 = m_2$, $\sqrt{\frac{k_1}{k_2}} = 3$, $k_1 = 9k_2$.

6.2 Spring Problems II

6.2.2. Since $k = \frac{mg}{\Delta l} = \frac{16}{3.2} = 5$ the equation of motion is $\frac{1}{2}y'' + y' + 5y = 0$, or (A)

$y'' + 2y' + 10y = 0$. The characteristic polynomial of (A) is

$p(r) = r^2 + 2r + 10 = (r+1)^2 + 9$. Therefore, the general solution of (A) is

$y = e^{-t}(c_1 \cos 3t + c_2 \sin 3t)$, so $y' = -y + 3e^{-t}(-c_1 \sin 3t + c_2 \cos 3t)$. Now $y(0) = -3$ and

$y'(0) = 2 \Rightarrow c_1 = -3$ and $2 = 3 + 3c_2$, or $c_2 = -\frac{1}{3}$. Therefore, $y = -e^{-t} \left(3 \cos 3t + \frac{1}{3} \sin 3t \right)$

ft. The time-varying amplitude is $\frac{\sqrt{82}}{3}e^{-t}$ ft.

6.2.4. Since $k = \frac{mg}{\Delta l} = \frac{96}{3.2} = 30$ the equation of motion is $3y'' + 18y' + 30y = 0$, or (A)

$y'' + 6y' + 10y = 0$. The characteristic polynomial of (A) is $p(r) = r^2 + 6r + 10 = (r + 3)^2 + 1$. Therefore, the general solution of (A) is $y = e^{-3t}(c_1 \cos t + c_2 \sin t)$, so

$y' = -3y + e^{-3t}(-c_1 \sin t + c_2 \cos t)$. Now $y(0) = -\frac{5}{4}$ and $y'(0) = -12 \Rightarrow c_1 = -\frac{5}{4}$ and

$-12 = \frac{15}{4} + c_2$, or $c_2 = -\frac{63}{4}$. Therefore, $y = -\frac{e^{-3t}}{4}(5 \cos t + 63 \sin t)$ ft.

6.2.6. Since $k = \frac{mg}{\Delta l} = \frac{8}{.32} = 25$ the equation of motion is $\frac{1}{4}y'' + \frac{3}{2}y' + 25y = 0$, or (A)

$y'' + 6y' + 100y = 0$. The characteristic polynomial of (A) is

$p(r) = r^2 + 6r + 100 = (r + 3)^2 + 91$. Therefore, the general solution of (A) is

$y = e^{-3t}(c_1 \cos \sqrt{91}t + c_2 \sin \sqrt{91}t)$, so $y' = -3y + \sqrt{91}e^{-3t}(-c_1 \sin \sqrt{91}t + c_2 \cos \sqrt{91}t)$. Now

$y(0) = \frac{1}{2}$ and $y'(0) = 4 \Rightarrow c_1 = \frac{1}{2}$ and $4 = -\frac{3}{2} + \sqrt{91}c_2$, or $c_2 = \frac{11}{2\sqrt{91}}$.

Therefore, $y = \frac{1}{2}e^{-3t} \left(\cos \sqrt{91}t + \frac{11}{\sqrt{91}} \sin \sqrt{91}t \right)$ ft.

6.2.8. Since $k = \frac{mg}{\Delta l} = \frac{20 \cdot 980}{5} = 3920$ the equation of motion is $20y'' + 400y' + 3920y = 0$, or (A)

$y'' + 20y' + 196y = 0$. The characteristic polynomial of (A) is

$p(r) = r^2 + 20r + 196 = (r + 10)^2 + 96$. Therefore, the general solution of (A) is

$y = e^{-10t}(c_1 \cos 4\sqrt{6}t + c_2 \sin 4\sqrt{6}t)$, so $y' = -10y + 4\sqrt{6}e^{-10t}(-c_1 \sin 4\sqrt{6}t + c_2 \cos 4\sqrt{6}t)$.

Now $y(0) = 9$ and $y'(0) = 0 \Rightarrow c_1 = 9$ and $0 = -90 + 4\sqrt{6}c_2$, or $c_2 = \frac{45}{2\sqrt{6}}$. Therefore,

$y = e^{-10t} \left(9 \cos 4\sqrt{6}t + \frac{45}{2\sqrt{6}} \sin 4\sqrt{6}t \right)$ cm.

6.2.10. Since $k = \frac{mg}{\Delta l} = \frac{32}{1} = 32$ the equation of motion is (A) $y'' + 3y' + 32y = 0$. The

characteristic polynomial of (A) is $p(r) = r^2 + 3r + 32 = \left(r + \frac{3}{2}\right)^2 + \frac{119}{4}$. Therefore, the

general solution of (A) is $y = e^{-3t/2} \left(c_1 \cos \frac{\sqrt{119}}{2}t + c_2 \sin \frac{\sqrt{119}}{2}t \right)$, so

$y' = -\frac{3}{2}y + \frac{\sqrt{119}}{2}e^{-3t/2} \left(-c_1 \sin \frac{\sqrt{119}}{2}t + c_2 \cos \frac{\sqrt{119}}{2}t \right)$. Now $y(0) = \frac{1}{2}$ and

$y'(0) = -3 \Rightarrow c_1 = \frac{1}{2}$ and $-3 = -\frac{3}{4} + \frac{\sqrt{119}}{2}c_2$, or $c_2 = -\frac{9}{2\sqrt{119}}$. Therefore,

$y = e^{-\frac{3}{2}t} \left(\frac{1}{2} \cos \frac{\sqrt{119}}{2}t - \frac{9}{2\sqrt{119}} \sin \frac{\sqrt{119}}{2}t \right)$ ft.

- 6.2.12. Since $k = \frac{mg}{\Delta l} = \frac{2}{.32} = \frac{25}{4}$ the equation of motion is $\frac{1}{16}y'' + \frac{1}{8}y' + \frac{25}{4}y = 0$, or (A) $y'' + 2y' + 100y = 0$. The characteristic polynomial of (A) is $p(r) = r^2 + 2r + 100 = (r + 1)^2 + 99$. Therefore, the general solution of (A) is $y = e^{-t}(c_1 \cos 3\sqrt{11}t + c_2 \sin 3\sqrt{11}t)$, so $y' = -y + 3\sqrt{11}e^{-t}(-c_1 \sin 3\sqrt{11}t + c_2 \cos 3\sqrt{11}t)$. Now $y(0) = -\frac{1}{3}$ and $y'(0) = 5 \Rightarrow c_1 = -\frac{1}{3}$ and $5 = \frac{1}{3} + 3\sqrt{11}c_2$, or $c_2 = \frac{14}{9\sqrt{11}}$. Therefore, $y = e^{-t} \left(-\frac{1}{3} \cos 3\sqrt{11}t + \frac{14}{9\sqrt{11}} \sin 3\sqrt{11}t \right)$ ft.
- 6.2.14. Since $k = \frac{mg}{\Delta l} = 32$ the equation of motion is (A) $y'' + 12y' + 32y = 0$. The characteristic polynomial of (A) is $p(r) = r^2 + 12r + 32 = (r + 8)(r + 4)$. Therefore, the general solution of (A) is $y = c_1 e^{-8t} + c_2 e^{-4t}$, so $y' = -8c_1 e^{-8t} - 4c_2 e^{-4t}$. Now $y(0) = -\frac{2}{3}$ and $y'(0) = 0 \Rightarrow c_1 + c_2 = \frac{2}{3}$, $8c_1 - 4c_2 = 0$, so $c_1 = -\frac{2}{3}$, $c_2 = \frac{4}{3}$, and $y = -\frac{2}{3}(e^{-8t} - 2e^{-4t})$.
- 6.2.16. Since $k = \frac{mg}{\Delta l} = \frac{100 \cdot 980}{98} = 100$ the equation of motion is $100y'' + 600y' + 1000y = 0$, or (A) $y'' + 6y' + 10y = 0$. The characteristic polynomial of (A) is $p(r) = r^2 + 6r + 10 = (r + 3)^2 + 1$. Therefore, the general solution of (A) is $y = e^{-3t}(c_1 \cos t + c_2 \sin t)$, so $y' = -3y + e^{-t}(-c_1 \sin t + c_2 \cos t)$. Now $y(0) = 10$ and $y'(0) = -100 \Rightarrow c_1 = 10$ and $-100 = -30 + c_2$, or $c_2 = -70$. Therefore, $y = e^{-3t}(10 \cos t - 70 \sin t)$ cm.
- 6.2.18. The equation of motion is (A) $2y'' + 4y' + 20y = 3 \cos 4t - 5 \sin 4t$. The steady state component of the solution of (A) is of the form $y_p = A \cos 4t + B \sin 4t$; therefore $y'_p = -4A \sin 4t + 4B \cos 4t$ and $y''_p = -16A \cos 4t - 16B \sin 4t$, so $2y''_p + 4y'_p + 20y_p = (-12A + 16B) \cos 4t - (16A + 12B) \sin 4t = 3 \cos 4t - 5 \sin 4t$ if $-12A + 16B = 3$, $-16A - 12B = -5$; therefore $A = \frac{11}{100}$, $B = \frac{27}{100}$, and $y_p = \frac{11}{100} \cos 4t + \frac{27}{100} \sin 4t$ cm.
- 6.2.20. Since $k = \frac{mg}{\Delta l} = \frac{9.8}{.49} = 20$ the equation of motion is (A) $y'' + 4y' + 20y = 8 \sin 2t - 6 \cos 2t$. The steady state component of the solution of (A) is of the form $y_p = A \cos 2t + B \sin 2t$; therefore $y'_p = -2A \sin 2t + 2B \cos 2t$ and $y''_p = -4A \cos 2t - 4B \sin 2t$, so $y''_p + 4y'_p + 20y_p = (16A + 8B) \cos 2t - (8A - 16B) \sin 2t = 8 \sin 2t - 6 \cos 2t$ if $16A + 8B = -6$, $-8A + 16B = 8$; therefore $A = -\frac{1}{2}$, $B = \frac{1}{4}$, and $y = -\frac{1}{2} \cos 2t + \frac{1}{4} \sin 2t$ m.
- 6.2.22. If $e^{r_1 t}(c_1 + c_2 t) = 0$, then (A) $c_1 + c_2 t = 0$. If $c_2 = 0$, then $c_1 \neq 0$ (by assumption), so (A) is impossible. If $c_1 \neq 0$, then the left side of (A) is strictly monotonic and therefore cannot have the same value for two distinct values of t .
- 6.2.24. If $y = e^{-ct/2m}(c_1 \cos \omega_1 t + c_2 \sin \omega_1 t)$, then $y' = -\frac{c}{2m}y + \omega_1 e^{-ct/2m}(-c_1 \sin \omega_1 t + c_2 \cos \omega_1 t)$, so $y(0) = y_0$ and $y'(0) = v_0 \Rightarrow c_1 = y_0$ and

$$v_0 = -\frac{cy_0}{2m} + c_2\omega_1, \text{ so } c_2 = \frac{1}{\omega_1} \left(v_0 + \frac{cy_0}{2m} \right) \text{ and}$$

$$y = e^{-ct/2m} \left(y_0 \cos \omega_1 t + \frac{1}{\omega_1} \left(v_0 + \frac{cy_0}{2m} \right) \sin \omega_1 t \right).$$

6.2.26. If $y = e^{r_1 t}(c_1 + c_2 t)$, then $y' = r_1 y + c_2 e^{r_1 t}$, so $y(0) = y_0$ and $y'(0) = v_0 \Rightarrow c_1 = y_0$ and $v_0 = r_1 y_0 + c_2$, so $c_2 = v_0 - r_1 y_0$. Therefore, $y = e^{r_1 t}(y_0 + (v_0 - r_1 y_0)t)$.

6.3 The RLC Circuit

6.3.2. $\frac{1}{20}Q'' + 2Q' + 100Q = 0$; $Q'' + 40Q' + 2000Q = 0$; $r^2 + 40r + 2000 = (r + 20)^2 + 1600 = 0$;
 $r = -20 \pm 40i$; $Q = e^{-20t}(2 \cos 40t + c_2 \sin 40t)$ (since $Q_0 = 2$);
 $I = Q' = e^{-20t}((40c_2 - 40) \cos 40t - (20c_2 + 80) \sin 40t)$;
 $I_0 = 2 \Rightarrow 40c_2 - 40 = 2 \Rightarrow c_2 = \frac{21}{20}$, so $20c_2 + 80 = 101$; $I = e^{-20t}(2 \cos 40t - 101 \sin 40t)$.

6.3.4. $\frac{1}{10}Q'' + 6Q' + 250Q = 0$; $Q'' + 60Q' + 2500Q = 0$; $r^2 + 60r + 2500 = (r + 30)^2 + 1600 = 0$;
 $r = -30 \pm 40i$; $Q = e^{-30t}(3 \cos 40t + c_2 \sin 40t)$ (since $Q_0 = 3$);
 $I = Q' = e^{-30t}((40c_2 - 90) \cos 40t - (30c_2 + 120) \sin 40t)$;
 $I_0 = -10 \Rightarrow 40c_2 - 90 = -10 \Rightarrow c_2 = 2$, so $-30c_2 - 120 = -180$;
 $I = -10e^{-30t}(\cos 40t + 18 \sin 40t)$.

6.3.6. $Q_p = A \cos 10t + B \sin 10t$; $Q'_p = 10B \cos 10t - 10A \sin 10t$;
 $Q''_p = -100A \cos 10t - 100B \sin 10t$;
 $\frac{1}{10}Q''_p + 3Q'_p + 100Q_p = (90A + 30B) \cos 10t - (30A - 90B) \sin 10t = 5 \cos 10t - 5 \sin 10t$, so
 $90A + 30B = 5$, $-30A + 90B = -5$. Therefore, $A = 1/15$, $B = -1/30$,
 $Q_p = \frac{\cos 10t}{15} - \frac{\sin 10t}{30}$, and $I_p = -\frac{1}{3}(\cos 10t + 2 \sin 10t)$.

6.3.8. $Q_p = A \cos 50t + B \sin 50t$; $Q'_p = 50B \cos 50t - 50A \sin 50t$;
 $Q''_p = -2500A \cos 50t - 2500B \sin 50t$;
 $\frac{1}{10}Q''_p + 2Q'_p + 100Q_p = (-150A + 100B) \cos 50t - (100A + 150B) \sin 50t = 3 \cos 50t - 6 \sin 50t$,
so $-150A + 100B = 3$, $-100A + 150B = -6$. Therefore, $A = 3/650$, $B = 12/325$,
 $Q_p = \frac{3}{650}(\cos 50t + 8 \sin 50t)$, and $I_p = \frac{3}{13}(8 \cos 50t - \sin 50t)$.

6.3.10. $Q_p = A \cos 30t + B \sin 30t$; $Q'_p = 30B \cos 30t - 30A \sin 30t$;
 $Q''_p = -900A \cos 30t - 900B \sin 30t$;
 $\frac{1}{20}Q''_p + 4Q'_p + 125Q_p = (80A + 120B) \cos 30t - (120A - 80B) \sin 30t = 15 \cos 30t - 30 \sin 30t$,
so $80A + 120B = 15$, $-120A + 80B = -30$, $A = 3/13$, $B = -3/104$,
 $Q_p = \frac{3}{104}(8 \cos 30t - \sin 30t)$, and $I_p = -\frac{45}{52}(\cos 30t + 8 \sin 30t)$.

6.3.12. Let $\sigma = \sigma(\omega)$ be the amplitude of I_p . From the solution of Exercise 6.3.11,

$$Q_p = A \cos \omega t + B \sin \omega t, \text{ where } A = \frac{(1/C - L\omega^2)U - R\omega V}{\Delta}, B = \frac{R\omega U + (1/C - L\omega^2)V}{\Delta},$$

and $\Delta = (1/C - L\omega^2)^2 + R^2\omega^2$. Since $I_p = Q'_p = \omega(-A \sin \omega t + B \cos \omega t)$, it follows that $\sigma^2(\omega) = \omega^2(A^2 + B^2) = \frac{U^2 + V^2}{\rho(\omega)}$, with $\rho(\omega) = \frac{\Delta}{\omega^2} = (1/C\omega - L\omega)^2 + R^2$, which attains its minimum value R^2 when $\omega = \omega_0 = \frac{1}{\sqrt{LC}}$. The maximum amplitude of I_p is $\sigma(\omega) = \frac{\sqrt{U^2 + V^2}}{R}$.

6.4 Motion Under a Central Force

6.4.2. Let $h = r_0^2\theta'_0$; then $\rho = \frac{h^2}{k}$. Since $r = \frac{\rho}{1 + e \cos(\theta - \phi)}$, it follows that (A)

$e \cos(\theta - \phi) = \frac{\rho}{r} - 1$. Differentiating this with respect to t yields $-e \sin(\theta - \phi)\theta' = -\frac{\rho r'}{r^2}$, so

(B) $e \sin(\theta - \phi) = \frac{\rho r'}{h}$, since $r^2\theta' \equiv h$. Squaring and adding (A) and (B) and setting $t = 0$ in

the result yields $e = \left[\left(\frac{\rho}{r_0} - 1 \right)^2 + \left(\frac{\rho r'_0}{h} \right)^2 \right]^{1/2}$. If $e = 0$, then θ_0 is undefined, but also

irrelevant; if $e \neq 0$, then set $t = 0$ in (A) and (B) to see that $\phi = \theta_0 - \alpha$, where $-\pi \leq \alpha < \pi$, $\cos \alpha = \frac{1}{e} \left(\frac{\rho}{r_0} - 1 \right)$ and $\sin \alpha = \frac{\rho r'_0}{eh}$

6.4.4. Recall that (A) $\frac{d^2u}{d\theta^2} = -\frac{1}{mh^2u^2}f(1/u)$. Let $u = \frac{1}{r} = \frac{1}{c\theta^2}$; then $\frac{d^2u}{d\theta^2} = \frac{6}{c\theta^4} = 6cu^2$. $6cu^2 + u = -\frac{1}{mh^2u^2}f(1/u)$, so $f(1/u) = -mh^2(6cu^4 + u^3)$ and $f(r) = -mh^2 \left(\frac{6c}{r^4} + \frac{1}{r^3} \right)$.

6.4.6. (a) With $f(r) = -\frac{mk}{r^3}$, Eqn. 6.4.11 becomes (A) $\frac{d^2u}{d\theta^2} + \left(1 - \frac{k}{h^2}\right)u = 0$. The initial conditions imply that $u(\theta_0) = \frac{1}{r_0}$ and $\frac{du}{d\theta}(\theta_0) = -\frac{r'_0}{h}$ (see Eqn. (6.4.)

(b) Let $\gamma = \left|1 - \frac{k}{h^2}\right|^{1/2}$. (i) If $h^2 < k$, then (A) becomes $\frac{d^2u}{d\theta^2} - \gamma^2u = 0$, and the solution of the initial value problem for u is $u = \frac{1}{r_0} \cosh \gamma(\theta - \theta_0) - \frac{r'_0}{\gamma h} \sinh \gamma(\theta - \theta_0)$; therefore

$r = r_0 \left(\cosh \gamma(\theta - \theta_0) - \frac{r'_0}{\gamma h} \sinh \gamma(\theta - \theta_0) \right)^{-1}$. (ii) If $h^2 = k$, then (A) becomes

$\frac{d^2u}{d\theta^2} = 0$, and the solution of the initial value problem for u is $u = \frac{1}{r_0} - \frac{r'_0}{h}(\theta - \theta_0)$;

therefore $r = r_0 \left(1 - \frac{r'_0}{h}(\theta - \theta_0) \right)^{-1}$. (iii) If $h^2 > k$, then (A) becomes

$\frac{d^2u}{d\theta^2} + \gamma^2u = 0$, and the solution of the initial value problem for u is

$u = \frac{1}{r_0} \cos \gamma(\theta - \theta_0) - \frac{r'_0}{\gamma h} \sin \gamma(\theta - \theta_0)$; therefore

$$r = r_0 \left(\cos \gamma(\theta - \theta_0) - \frac{r_0 r'_0}{\gamma h} \sin \gamma(\theta - \theta_0) \right)^{-1}.$$

Chapter 7

Series Solutions of Linear Second Equations

7.1 Review of Power Series

7.1.2. From Theorem 7.1.3, $\sum_{m=0}^{\infty} b_m z^m$ converges if $|z| < 1/L$ and diverges if $|z| > 1/L$.

Therefore, $\sum_{m=0}^{\infty} b_m (x - x_0)^2$ converges if $|x - x_0| < 1/\sqrt{L}$ and diverges if $|x - x_0| > 1/\sqrt{L}$.

7.1.4. From Theorem 7.1.3, $\sum_{m=0}^{\infty} b_m z^m$ converges if $|z| < 1/L$ and diverges if $|z| > 1/L$.

Therefore, $\sum_{m=0}^{\infty} b_m (x - x_0)^{km}$ converges if $|x - x_0| < 1/\sqrt[k]{L}$ and diverges if $|x - x_0| > 1/\sqrt[k]{L}$.

$$\begin{aligned} 7.1.12. \quad (1 + 3x^2)y'' + 3x^2y' - 2y &= \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 3 \sum_{n=2}^{\infty} n(n-1)a_n x^n + 3 \sum_{n=1}^{\infty} n a_n x^{n+1} - \\ 2 \sum_{n=0}^{\infty} a_n x^n &= \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n + 3 \sum_{n=1}^{\infty} n(n-1)a_n x^n + 3 \sum_{n=1}^{\infty} (n-1)a_{n-1} x^n - 2 \sum_{n=0}^{\infty} a_n x^n = \\ 2a_2 - 2a_0 + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + (3n(n-1) - 2)a_n + 3(n-1)a_{n-1}] x^n. \end{aligned}$$

$$\begin{aligned} 7.1.13. \quad (1 + 2x^2)y'' + (2 - 3x)y' + 4y &= \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 2 \sum_{n=2}^{\infty} n(n-1)a_n x^n + 2 \sum_{n=1}^{\infty} n a_n x^{n-1} - \\ 2 \sum_{n=0}^{\infty} a_n x^n + 4 \sum_{n=0}^{\infty} a_n x^n &= \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2} x^n + 2 \sum_{n=0}^{\infty} n(n-1)a_n x^n + 2 \sum_{n=0}^{\infty} (n+1)a_{n+1} x^n - \\ 3 \sum_{n=0}^{\infty} a_n x^n + 4 \sum_{n=0}^{\infty} a_n x^n &= \sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + 2(n+1)a_{n+1} + (2n^2 - 5n + 4)a_n] x^n. \end{aligned}$$

7.1.14.

$$\begin{aligned} (1 + x^2)y'' + (2 - x)y' + 3y &= \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n x^n \\ &\quad + 2 \sum_{n=1}^{\infty} n a_n x^{n-1} - \sum_{n=1}^{\infty} n a_n x^n + 3 \sum_{n=0}^{\infty} a_n x^n \end{aligned}$$

$$\begin{aligned}
&= \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n + \sum_{n=0}^{\infty} n(n-1)a_nx^n \\
&\quad + 2 \sum_{n=0}^{\infty} (n+1)a_{n+1}x^n - \sum_{n=0}^{\infty} na_nx^n + 3 \sum_{n=0}^{\infty} a_nx^n \\
&= \sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + 2(n+1)a_{n+1} + (n^2 - 2n + 3)a_n] x^n.
\end{aligned}$$

7.1.16. Let $t = x + 1$; then

$$\begin{aligned}
xy'' + (4+2x)y' + (2+x)y &= (-1+t)y'' + (2+2t)y' + (1+t)y = - \sum_{n=2}^{\infty} n(n-1)a_nt^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_nt^{n-1} \\
&\quad + 2 \sum_{n=1}^{\infty} na_nt^{n-1} + 2 \sum_{n=1}^{\infty} na_nt^n + \sum_{n=0}^{\infty} a_nt^n + \sum_{n=0}^{\infty} a_nt^{n+1} = - \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}t^n + \\
&\quad \sum_{n=0}^{\infty} (n+1)na_{n+1}t^n + 2 \sum_{n=0}^{\infty} (n+1)a_{n+1}t^n + 2 \sum_{n=0}^{\infty} na_nt^n + \sum_{n=0}^{\infty} a_nt^n + \sum_{n=1}^{\infty} a_{n-1}t^n = (-2a_2 + \\
&\quad 2a_1 + a_0) + \sum_{n=1}^{\infty} [-(n+2)(n+1)a_{n+2} + (n+1)(n+2)a_{n+1} + (2n+1)a_n + a_{n-1}] (x+2)^n.
\end{aligned}$$

$$7.1.20. \quad y'(x) = x^r \sum_{n=0}^{\infty} na_nx^{n-1} + rx^{r-1} \sum_{n=0}^{\infty} a_nx^n = \sum_{n=0}^{\infty} (n+r)x^{n+r-1}$$

$$\begin{aligned}
y'' &= \frac{d}{dx} y'(x) = \frac{d}{dx} \left[x^{r-1} \sum_{n=0}^{\infty} (n+r)a_nx^n \right] = \\
&= x^{r-1} \sum_{n=0}^{\infty} (n+r)na_nx^{n-1} + (r-1)x^{r-2} \sum_{n=0}^{\infty} (n+r)a_nx^n = \sum_{n=0}^{\infty} (n+r)(n+r-1)a_nx^{n+r-2}.
\end{aligned}$$

$$\begin{aligned}
7.1.22. \quad x^2(1+x)y'' + x(1+2x)y' - (4+6x)y &= (x^2y'' + xy' - 4y) + x(x^2y'' + 2xy' - 6y) = \\
&= \sum_{n=0}^{\infty} [(n+r)(n+r-1) + (n+r) - 4]a_nx^{n+r} + \sum_{n=0}^{\infty} [(n+r)(n+r-1) + 2(n+r) - 6]a_nx^{n+r+1} = \\
&= \sum_{n=0}^{\infty} (n+r-2)(n+r+2)a_nx^{n+r} + \sum_{n=0}^{\infty} (n+r+3)(n+r-2)a_nx^{n+r+1} = \\
&= \sum_{n=0}^{\infty} (n+r-2)(n+r+2)a_nx^{n+r} + \sum_{n=1}^{\infty} (n+r+2)(n+r-3)a_{n-1}x^{n+r} = x^r \sum_{n=0}^{\infty} b_nx^n \text{ with} \\
&b_0 = (r-2)(r+2)a_0 \text{ and } b_n = (n+r-2)(n+r+2)a_n + (n+r+2)(n+r-3)a_{n-1}, \quad n \geq 1.
\end{aligned}$$

$$7.1.24. \quad x^2(1+3x)y'' + x(2+12x+x^2)y' + 2x(3+x)y =$$

$$\begin{aligned}
&(x^2y'' + 2xy') + x(3x^2y'' + 12xy' + 6y) + x^2(xy' + 2y) = \sum_{n=0}^{\infty} [(n+r)(n+r-1) + 2(n+r) \\
&\quad + r]a_nx^{n+r} + \sum_{n=0}^{\infty} [3(n+r)(n+r-1) + 12(n+r) + 6]a_nx^{n+r+1} + \sum_{n=0}^{\infty} [(n+r) + 2]a_nx^{n+r+2} = \\
&= \sum_{n=0}^{\infty} (n+r)(n+r+1)a_nx^{n+r} + 3 \sum_{n=0}^{\infty} (n+r+1)(n+r+2)a_nx^{n+r+1} + \sum_{n=0}^{\infty} (n+r+2)a_nx^{n+r+2} = \\
&= \sum_{n=0}^{\infty} (n+r)(n+r+1)a_nx^{n+r} + 3 \sum_{n=1}^{\infty} (n+r)(n+r+1)a_{n-1}x^{n+r} + \sum_{n=2}^{\infty} (n+r)a_{n-2}x^{n+r} = x^r \sum_{n=0}^{\infty} b_nx^n \\
&\text{with } b_0 = r(r+1)a_0, \quad b_1 = (r+1)(r+2)a_1 + 3(r+1)(r+2)a_0, \\
&b_n = (n+r)(n+r+1)a_n + 3(n+r)(n+r+1)a_{n-1} + (n+r)a_{n-2}, \quad n \geq 2.
\end{aligned}$$

$$\begin{aligned}
7.1.26. \quad & x^2(2+x^2)y'' + 2x(5+x^2)y' + 2(3-x^2)y = (2x^2y'' + 10xy' + 6y) + x^2(x^2y'' + 2xy' - 2y) = \\
& \sum_{n=0}^{\infty} [2(n+r)(n+r-1) + 10(n+r) + 6]a_n x^{n+r} + \sum_{n=0}^{\infty} [(n+r)(n+r-1) + 2(n+r) - 2]a_n x^{n+r+2} = \\
& 2 \sum_{n=0}^{\infty} (n+r+1)(n+r+3)a_n x^{n+r} + \sum_{n=0}^{\infty} (n+r-1)(n+r+2)a_n x^{n+r+2} = \\
& 2 \sum_{n=0}^{\infty} (n+r+1)(n+r+3)a_n x^{n+r} + \sum_{n=2}^{\infty} (n+r-3)(n+r)a_{n-2} x^{n+r} = x^r \sum_{n=0}^{\infty} b_n x^n \text{ with} \\
& b_0 = 2(r+1)(r+3)a_0, \quad b_1 = 2(r+2)(r+4)a_1 \\
& b_n = 2(n+r+1)(n+r+3)a_n + (n+r-3)(n+r)a_{n-2}, \quad n \geq 2.
\end{aligned}$$

7.2 Series Solutions Near an Ordinary Point I

$$\begin{aligned}
7.2.2. \quad & p(n) = n(n-1) + 2n - 2 = (n+2)(n-1); \quad a_{n+2} = -\frac{n-1}{n+1}a_n; \quad a_{2m+2} = -\frac{2m-1}{2m+1}a_{2m}, \text{ so} \\
& a_{2m} = \frac{(-1)^m}{2m-1}a_0; \quad a_{2m+3} = -\frac{m}{m+2}a_{2m+1} = 0 \text{ if } m \geq 0; \quad y = a_0 \sum_{m=0}^{\infty} (-1)^{m+1} \frac{x^{2m}}{2m-1} + a_1 x.
\end{aligned}$$

$$\begin{aligned}
7.2.4. \quad & p(n) = -n(n-1) - 8n - 12 = -(n+3)(n+4); \quad a_{n+2} = -\frac{(n+3)(n+4)}{(n+2)(n+1)}a_n; \\
& a_{2m+2} = -\frac{(m+2)(2m+3)}{(m+1)(2m+1)}a_{2m}, \text{ so } a_{2m} = (m+1)(2m+1)a_0; \\
& a_{2m+3} = \frac{(m+2)(2m+5)}{(m+1)(2m+3)}a_{2m+1} \text{ so } a_{2m+1} = \frac{(m+1)(2m+3)}{3}a_1; \\
& y = a_0 \sum_{m=0}^{\infty} (m+1)(2m+1)x^{2m} + \frac{a_1}{3} \sum_{m=0}^{\infty} (m+1)(2m+3)x^{2m+1}.
\end{aligned}$$

$$\begin{aligned}
7.2.6. \quad & p(n) = n(n-1) + 2n + \frac{1}{4} = \frac{(2n+1)^2}{4}; \quad a_{n+2} = -\frac{(2n+1)^2}{4(n+2)(n+1)}a_n; \\
& a_{2m+2} = -\frac{(4m+1)^2}{8(m+1)(2m+1)}a_{2m}, \text{ so } a_{2m} = (-1)^m \left[\prod_{j=0}^{m-1} \frac{(4j+1)^2}{2j+1} \right] \frac{1}{8^m m!} a_0; \\
& a_{2m+3} = -\frac{(4m+3)^2}{8(2m+3)(m+1)}a_{2m+1} \text{ so } a_{2m+1} = (-1)^m \left[\prod_{j=0}^{m-1} \frac{(4j+3)^2}{2j+3} \right] \frac{1}{8^m m!} a_1; \\
& y = a_0 \sum_{m=0}^{\infty} (-1)^m \left[\prod_{j=0}^{m-1} \frac{(4j+1)^2}{2j+1} \right] \frac{x^{2m}}{8^m m!} + a_1 \sum_{m=0}^{\infty} (-1)^m \left[\prod_{j=0}^{m-1} \frac{(4j+3)^2}{2j+3} \right] \frac{x^{2m+1}}{8^m m!}
\end{aligned}$$

$$\begin{aligned}
7.2.8. \quad & p(n) = n(n-1) - 10n + 28 = (n-7)(n-4); \quad a_{n+2} = -\frac{(n-7)(n-4)}{(n+2)(n+1)}a_n; \\
& a_{2m+2} = -\frac{2(2m-7)(m-2)}{2(m+1)(2m+1)}a_{2m}, \text{ so } a_2 = -14a_0, \quad a_4 = -\frac{5}{6}a_2 = \frac{35}{3}a_0, \quad a_{2m} = 0 \text{ if } m \geq 3; \\
& a_{2m+3} = -\frac{(m-3)(2m-3)}{(2m+3)(m+1)}a_{2m+1}, \text{ so } a_3 = -3a_1, \quad a_5 = -\frac{1}{5}a_3 = \frac{3}{5}a_1, \quad a_7 = \frac{1}{21}a_5 = \frac{1}{35}a_1; \\
& y = a_0 \left(1 - 14x^2 + \frac{35}{3}x^4 \right) + a_1 \left(x - 3x^3 + \frac{3}{5}x^5 + \frac{1}{35}x^7 \right).
\end{aligned}$$

$$7.2.10. \quad p(n) = 2n + 3; \quad a_{n+2} = -\frac{2n+3}{(n+2)(n+1)}a_n; \quad a_{2m+2} = -\frac{4m+3}{2(m+1)(2m+1)}a_{2m}, \text{ so}$$

$$a_{2m} = \left[\prod_{j=0}^{m-1} \frac{4j+3}{2j+1} \right] \frac{(-1)^m}{2^m m!} a_0; \quad a_{2m+3} = -\frac{4m+5}{2(2m+3)(m+1)}a_{2m+1} \text{ so}$$

$$a_{2m+1} = \left[\prod_{j=0}^{m-1} \frac{4j+5}{2j+3} \right] \frac{(-1)^m}{2^m m!} a_1;$$

$$y = a_0 \sum_{m=0}^{\infty} (-1)^m \left[\prod_{j=0}^{m-1} \frac{4j+3}{2j+1} \right] \frac{x^{2m}}{2^m m!} + a_1 \sum_{m=0}^{\infty} (-1)^m \left[\prod_{j=0}^{m-1} \frac{4j+5}{2j+3} \right] \frac{x^{2m+1}}{2^m m!}$$

$$7.2.12. \quad p(n) = 2n(n-1) - 9n - 6 = (n-6)(2n+1); \quad a_{n+2} = -\frac{(n-6)(2n+1)}{(n+2)(n+1)}a_n; \quad a_0 = y(0) = 1; \\ a_1 = y'(0) = -1.$$

$$7.2.13. \quad p(n) = 8n(n-1) + 2 = 2(2n-1)^2; \quad a_{n+2} = -\frac{2(2n-1)^2}{(n+2)(n+1)}a_n; \quad a_0 = y(0) = 2; \\ a_1 = y'(0) = -1.$$

$$7.2.16. \quad p(n) = -1; \quad a_{n+2} = \frac{1}{(n+2)(n+1)}a_n; \quad a_{2m+2} = \frac{1}{(2m+2)(2m+1)}a_{2m}, \text{ so } a_{2m} = \frac{1}{(2m)!}a_0; \\ a_{2m+3} = \frac{1}{(2m+3)(2m+1)}a_{2m+1}, \text{ so } a_{2m+1} = \frac{1}{(2m+1)!}a_1; \\ y = a_0 \sum_{m=0}^{\infty} \frac{(x-3)^{2m}}{(2m)!} + a_1 \sum_{m=0}^{\infty} \frac{(x-3)^{2m+1}}{(2m+1)!}.$$

$$7.2.18. \quad \text{Let } t = x - 1; \text{ then } (1 - 2t^2)y'' - 10ty' - 6y = 0;$$

$$p(n) = -2n(n-1) - 10n - 6 = -2(n+1)(n+3); \quad a_{n+2} = \frac{2(n+3)}{n+2}a_n; \quad a_{2m+2} = \frac{2m+3}{m+1}a_{2m},$$

$$\text{so } a_{2m} = \frac{1}{m!} \left[\prod_{j=0}^{m-1} (2j+3) \right] a_0; \quad a_{2m+3} = \frac{4(m+2)}{2m+3}a_{2m+1}, \text{ so } a_{2m+1} = \frac{4^m(m+1)!}{\prod_{j=0}^{m-1} (2j+3)}a_1;$$

$$y = a_0 \sum_{m=0}^{\infty} \left[\prod_{j=0}^{m-1} (2j+3) \right] \frac{(x-1)^{2m}}{m!} + a_1 \sum_{m=0}^{\infty} \frac{4^m(m+1)!}{\prod_{j=0}^{m-1} (2j+3)} (x-1)^{2m+1}.$$

$$7.2.20. \quad \text{Let } t = x + 1; \text{ then } \left(1 + \frac{3t^2}{3}\right)y'' - \frac{9t}{2}y' + \frac{3}{2}y = 0; \quad p(n) = \frac{3}{2}n(n-1) + \frac{9}{2}n + \frac{3}{2} = \frac{3}{2}(n+1)^2;$$

$$a_{n+2} = -\frac{3(n+1)}{2(n+2)}a_n; \quad a_{2m+2} = -\frac{3(2m+1)}{4(m+1)}a_{2m}, \text{ so } a_{2m} = (-1)^m \left[\prod_{j=0}^{m-1} (2j+1) \right] \frac{3^m}{4^m m!}a_0;$$

$$a_{2m+3} = -\frac{3(m+1)}{2m+3}a_{2m+1}, \text{ so } a_{2m+1} = (-1)^m \frac{3^m m!}{\prod_{j=0}^{m-1} (2j+3)}a_1;$$

$$y = a_0 \sum_{m=0}^{\infty} (-1)^m \left[\prod_{j=0}^{m-1} (2j+1) \right] \frac{3^m}{4^m m!} (x+1)^{2m} + a_1 \sum_{m=0}^{\infty} (-1)^m \frac{3^m m!}{\prod_{j=0}^{m-1} (2j+3)} (x+1)^{2m+1}.$$

$$7.2.22. \quad p(n) = n + 3; \quad a_{n+2} = -\frac{n+3}{(n+2)(n+1)}a_n; \quad a_0 = y(3) = -2; \quad a_1 = y'(3) = 3.$$

7.2.24. Let $t = x - 3$; $(1 + 4t^2)y'' + y = 0$; $p(n) = (4n(n-1) + 1 = 2n-1)^2$;

$$a_{n+2} = -\frac{(2n-1)^2}{(n+2)(n+1)}a_n; a_0 = y(3) = 4; a_1 = y'(3) = -6.$$

7.2.26. Let $t = x + 1$; $\left(1 + \frac{2t^2}{3}\right)y'' - \frac{20}{3}ty' + 20y = 0$;

$$p(n) = \frac{2}{3}n(n-1) - \frac{20}{3}n + 20 = \frac{2(n-6)(n-5)}{3}; a_{n+2} = -\frac{2(n-6)(n-5)}{3(n+2)(n+1)}a_n;$$

$$a_0 = y(-1) = 3; a_1 = y'(-1) = -3.$$

7.2.28. From Theorem 7.2.2, $a_{n+2} = -\frac{p(n)}{(n+2)(n+1)}a_n$; $a_{2m+2} = -\frac{p(2m)}{(2m+2)(2m+1)}a_{2m}$, so

$$a_{2m} = \left[\prod_{j=0}^{m-1} p(2j) \right] \frac{(-1)^m}{(2m)!} a_0; a_{2m+3} = -\frac{p(2m+1)}{(2m+3)(2m+2)}a_{2m}, \text{ so}$$

$$a_{2m+1} = \left[\prod_{j=0}^{m-1} p(2j+1) \right] \frac{(-1)^m}{(2m+1)!} a_1.$$

7.2.30. (a) Here $p(n) = -[n(n-1) + 2bn - \alpha(\alpha + 2b - 1)] = -(n-\alpha)(n+\alpha+2b-1)$, so Exercise 7.2.28 implies that y_1 and y_2 have the stated forms. If $\alpha = 2k$, then

$$y_1 = \sum_{m=0}^{\infty} \left[\prod_{j=0}^{m-1} (2j-2k)(2j+2k+2b-1) \right] \frac{x^{2m}}{(2m)!}. \quad (\text{C})$$

If $\alpha = 2k + 1$, then

$$y_2 = \sum_{m=0}^{\infty} \left[\prod_{j=0}^{m-1} (2j-2k)(2j+2k+2b) \right] \frac{x^{2m+1}}{(2m+1)!}. \quad (\text{D})$$

Since $2b$ is not a negative integer and $\prod_{j=0}^{m-1} (2j-2k) = 0$ if $m > k$, y_1 in (C) and y_2 in (D) have the stated properties. This implies the conclusions regarding P_n .

(b) Multiplying (A) through by $(1-x^2)^{b-1}$ yields

$$[(1-x^2)^b P'_n]' = -n(n+2b-1)(1-x^2)^{b-1} P_n. \quad (\text{E})$$

(c) Therefore,

$$[(1-x^2)^b P'_m]' = -m(m+2b-1)(1-x^2)^{b-1} P_m. \quad (\text{F})$$

Subtract P_n times (F) from P_m times (E) to obtain (B).

(d) Integrating the left side of (B) by parts over $[-1, 1]$ yields zero, which implies the conclusion.

7.2.32. (a) Let $Ly = (1 + \alpha x^3)y'' + \beta x^2 y' + \gamma xy$. If $y = \sum_{n=0}^{\infty} a_n x^n$, then $Ly =$

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=0}^{\infty} p(n)a_n x^{n+1} = 2a_2 + \sum_{n=0}^{\infty} [(n+3)(n+2)a_{n+3} + p(n)a_n]x^{n+1} = 0$$

if and only if $a_2 = 0$ and $a_{n+3} = -\frac{p(n)}{(n+3)(n+2)}a_n$ for $n \geq 0$.

7.2.34. $p(r) = -2r(r-1) - 10r - 8 = -2(r+2)^2$; (A)

$$\prod_{j=0}^{m-1} \frac{p(3j)}{3j+2} = \prod_{j=0}^{m-1} \frac{(-2)(3j+2)^2}{3j+2} = (-1)^m 2^m \prod_{j=0}^{m-1} (3j+2); \text{ (B)}$$

$$\prod_{j=0}^{m-1} \frac{p(3j+1)}{3j+4} = \prod_{j=0}^{m-1} \frac{(-2)(3j+3)^2}{3j+4} = \frac{(-1)^m 2^m (m!)^2}{\prod_{j=0}^{m-1} (3j+4)}.$$

Substituting (A) and (B) into the result of Exercise 7.2.32(c) yields

$$y = a_0 \sum_{m=0}^{\infty} \left(\frac{2}{3}\right)^m \left[\prod_{j=0}^{m-1} (3j+2) \right] \frac{x^{3m}}{m!} + a_1 \sum_{m=0}^{\infty} \frac{6^m m!}{\prod_{j=0}^{m-1} (3j+4)} x^{3m+1}.$$

7.2.36. $p(r) = -2r(r-1) + 6r + 24 = -2(r-6)(r+2)$; (A) $\prod_{j=0}^{m-1} \frac{p(3j)}{3j+2} = \prod_{j=0}^{m-1} (-6)(j-2).$

$$\text{(B)} \prod_{j=0}^{m-1} \frac{p(3j+1)}{3j+4} = \prod_{j=0}^{m-1} \frac{(-6)(j+1)(3j-5)}{3j+4} = (-1)^m 6^m m \prod_{j=0}^{m-1} \frac{3j-5}{3j+4}.$$

Substituting (A) and (B) into the result of Exercise 7.2.32(c) yields

$$y = a_0(1 - 4x^3 + 4x^6) + a_1 \sum_{m=0}^{\infty} 2^m \left[\prod_{j=0}^{m-1} \frac{3j-5}{3j+4} \right] x^{3m+1}.$$

7.2.38. (a) Let $Ly = (1 + \alpha x^{k+2})y'' + \beta x^{k+1}y' + \gamma x^k y$. If $y = \sum_{n=0}^{\infty} a_n x^n$, then

$$Ly = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=0}^{\infty} p(n)a_n x^{n+k} = \sum_{n=-k}^{-1} (n+k+2)(n+k-1)a_{n+k+2} x^{n+k} + \sum_{n=0}^{\infty} [(n+k+2)(n+k+1)a_{n+k+2} + p(n)a_n] x^{n+k} = 0$$

if and only if $a_k = 0$ for $2 \leq n \leq k+1$ and (A) $a_{n+k+1} = -\frac{p(n)}{(n+k+2)(n+k+1)} a_n$ for $n \geq 0$.

(b) If $a_n = 0$ the $a_{n+(k+2)m} = 0$ for all $m \geq 0$, from (A).

7.2.40. $k = 2$ and $p(r) = 1$; (A) $\prod_{j=0}^{m-1} \frac{p(4j)}{4j+3} = \frac{1}{\prod_{j=0}^{m-1} (4j+3)}$; (B) $\prod_{j=0}^{m-1} \frac{p(4j+1)}{(4j+5)} = \frac{1}{\prod_{j=0}^{m-1} (4j+5)}.$

Substituting (A) and (B) into the result of Exercise 7.2.38(c) yields

$$y = a_0 \sum_{m=0}^{\infty} (-1)^m \frac{x^{4m}}{4^m m! \prod_{j=0}^{m-1} (4j+3)} + a_1 \sum_{m=0}^{\infty} (-1)^m \frac{x^{4m+1}}{4^m m! \prod_{j=0}^{m-1} (4j+5)}.$$

7.2.42. $k = 6$ and $p(r) = r(r-1) - 16r + 72 = (r-9)(r-8)$; (A)

$$\prod_{j=0}^{m-1} \frac{p(8j)}{8j+7} = \prod_{j=0}^{m-1} \frac{8(j-1)(8j-9)}{8j+7};$$

$$\text{(B)} \prod_{j=0}^{m-1} \frac{p(8j+1)}{(8j+9)} = \prod_{j=0}^{m-1} \frac{8(j-1)(8j-7)}{8j+9};$$

Substituting (A) and (B) into the result of Exercise 7.2.38(c) yields

$$y = a_0 \left(1 - \frac{9}{7}x^8\right) + a_1 \left(x - \frac{7}{9}x^9\right).$$

7.2.44. $k = 4$ and $p(r) = r + 6$; (A) $\prod_{j=0}^{m-1} \frac{p(6j)}{6j+5} = \prod_{j=0}^{m-1} \frac{6(j+1)}{6j+5} = \frac{6^m m!}{\prod_{j=0}^{m-1} (6j+5)}$;

(B) $\prod_{j=0}^{m-1} \frac{p(6j+1)}{(6j+7)} = 1$;

Substituting (A) and (B) into the result of Exercise 7.2.38(c) yields

$$y = a_0 \sum_{m=0}^{\infty} (-1)^m \frac{x^{6m}}{\prod_{j=0}^{m-1} (6j+5)} + a_1 \sum_{m=0}^{\infty} (-1)^m \frac{x^{6m+1}}{6^m m!}.$$

7.3 Series Solutions Near an Ordinary Point II

7.3.2. If $y = \sum_{n=0}^{\infty} a_n x^n$, then

$$(1+x+2x^2)y'' + (2+8x)y' + 4y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} + 2 \sum_{n=2}^{\infty} n(n-1)a_n x^n + 2 \sum_{n=1}^{\infty} na_n x^{n-1} + 8 \sum_{n=1}^{\infty} na_n x^n + 4 \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} (n+2)(n+1)(a_{n+2} + a_{n+1} + 2a_n)x^n = 0$$

if $a_{n+2} = -a_{n+1} - 2a_n$, $a_n \geq 0$. Starting with $a_0 = -1$ and $a_1 = 2$ yields
 $y = -1 + 2x - 4x^3 + 4x^4 + 4x^5 - 12x^6 + 4x^7 + \dots$

7.3.4. If $y = \sum_{n=0}^{\infty} a_n x^n$, then $(1+x+3x^2)y'' + (2+15x)y' + 12y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} + \sum_{n=2}^{\infty} n(n-1)a_n x^n + 2 \sum_{n=1}^{\infty} na_n x^{n-1} + 15 \sum_{n=1}^{\infty} na_n x^n + 12 \sum_{n=0}^{\infty} a_n x^n =$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + (n+1)(n+2)a_{n+1} + 3(n+2)^2 a_n] x^n = 0 \text{ if}$$

$$a_{n+2} = -a_{n+1} - \frac{3(n+2)}{n+1} a_n, \quad a_n \geq 0. \text{ Starting with } a_0 = 0 \text{ and } a_1 = 1 \text{ yields}$$

$$y = x - x^2 - \frac{7}{2}x^3 + \frac{15}{2}x^4 + \frac{45}{8}x^5 - \frac{261}{8}x^6 + \frac{207}{16}x^7 + \dots$$

7.3.6. If $y = \sum_{n=0}^{\infty} a_n x^n$, then

$$(3+3x+x^2)y'' + (6+4x)y' + 2y = 3 \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 3 \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} + \sum_{n=2}^{\infty} n(n-1)a_n x^n + 6 \sum_{n=1}^{\infty} na_n x^{n-1} + 4 \sum_{n=1}^{\infty} na_n x^n + 2 \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} (n+2)(n+1)[3a_{n+2} + 3a_{n+1} + a_n]x^n = 0$$

if $a_{n+2} = -a_{n+1} - a_n/3$, $a_n \geq 0$. Starting with $a_0 = 7$ and $a_1 = 3$ yields
 $y = 7 + 3x - \frac{16}{3}x^2 + \frac{13}{3}x^3 - \frac{23}{9}x^4 + \frac{10}{9}x^5 - \frac{7}{27}x^6 - \frac{1}{9}x^7 + \dots$

7.3.8. The equation is equivalent to $(1 + t + 2t^2)y'' + (2 + 6t)y' + 2y = 0$ with $t = x - 1$. If

$$y = \sum_{n=0}^{\infty} a_n t^n, \text{ then}$$

$$(1 + t + 2t^2)y'' + (2 + 6t)y' + 2y = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + 2 \sum_{n=2}^{\infty} n(n-1)a_n t^n + 2 \sum_{n=1}^{\infty} n a_n t^{n-1} + 6 \sum_{n=1}^{\infty} n a_n t^n + 2 \sum_{n=0}^{\infty} a_n t^n = \sum_{n=0}^{\infty} (n+1)[(n+2)a_{n+2} + (n+2)a_{n+1} + 2(n+1)a_n] t^n = 0$$

if $a_{n+2} = -a_{n+1} - \frac{2(n+1)}{n+2}a_n$, $a_n \geq 0$. Starting with $a_0 = 1$ and $a_1 = -1$ yields

$$y = 1 - (x-1) + \frac{4}{3}(x-1)^3 - \frac{4}{3}(x-1)^4 - \frac{4}{5}(x-1)^5 + \frac{136}{45}(x-1)^6 - \frac{104}{63}(x-1)^7 + \dots$$

7.3.10. The equation is equivalent to $(1 + t + t^2)y'' + (3 + 4t)y' + 2y = 0$ with $t = x - 1$. If

$$y = \sum_{n=0}^{\infty} a_n t^n, \text{ then}$$

$$(1 + t + t^2)y'' + (3 + 4t)y' + 2y = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + \sum_{n=2}^{\infty} n(n-1)a_n t^n + 3 \sum_{n=1}^{\infty} n a_n t^{n-1} + 4 \sum_{n=1}^{\infty} n a_n t^n + 2 \sum_{n=0}^{\infty} a_n t^n = \sum_{n=0}^{\infty} (n+1)[(n+2)a_{n+2} + (n+3)a_{n+1} + (n+2)a_n] t^n = 0$$

if $a_{n+2} = -\frac{n+3}{n+2}a_{n+1} - a_n$, $a_n \geq 0$. Starting with $a_0 = 2$ and $a_1 = -1$ yields $y =$

$$2 - (x-1) - \frac{1}{2}(x-1)^2 + \frac{5}{3}(x-1)^3 - \frac{19}{12}(x-1)^4 + \frac{7}{30}(x-1)^5 + \frac{59}{45}(x-1)^6 - \frac{1091}{630}(x-1)^7 + \dots$$

7.3.12. The equation is equivalent to $(1 + 2t + t^2)y'' + (1 + 7t)y' + 8y = 0$ with $t = x - 1$. If

$$y = \sum_{n=0}^{\infty} a_n t^n, \text{ then } (1 + 2t + t^2)y'' + (1 + 7t)y' + 8y = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + 2 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} +$$

$$1) a_n t^{n-1} + \sum_{n=2}^{\infty} n(n-1)a_n t^n + \sum_{n=1}^{\infty} n a_n t^{n-1} + 7 \sum_{n=1}^{\infty} n a_n t^n + 8 \sum_{n=0}^{\infty} a_n t^n =$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + (n+1)(2n+1)a_{n+1} + (n+2)(n+4)a_n] t^n = 0 \text{ if}$$

$a_{n+2} = -\frac{2n+1}{n+2}a_{n+1} - \frac{n+4}{n+1}a_n$, $a_n \geq 0$. Starting with $a_0 = 1$ and $a_1 = -2$ yields $y =$

$$1 - 2(x-1) - 3(x-1)^2 + 8(x-1)^3 - 4(x-1)^4 - \frac{42}{5}(x-1)^5 + 19(x-1)^6 - \frac{604}{35}(x-1)^7 + \dots$$

7.3.16. If $y = \sum_{n=0}^{\infty} a_n x^n$, then

$$(1-x)y'' - (2-x)y' + y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} - 2 \sum_{n=1}^{\infty} n a_n x^{n-1} +$$

$$\sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} - (n+2)(n+1)a_{n+1} + (n+1)a_n] x^n = 0 \text{ if}$$

$$a_{n+2} = a_{n+1} - \frac{a_n}{n+2}, a_n \geq 0.$$

7.3.18. If $y = \sum_{n=0}^{\infty} a_n x^n$, then

$$(1+x^2)y'' + y' + 2y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n x^n + \sum_{n=1}^{\infty} n a_n x^{n-1} + 2 \sum_{n=0}^{\infty} a_n x^n =$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + (n+1)a_{n+1} + (n^2 - n + 2)a_n] x^n = 0 \text{ if}$$

$$a_{n+2} = -\frac{1}{n+2}a_{n+1} - \frac{n^2 - n + 2}{(n+2)(n+1)}a_n.$$

7.3.20. The equation is equivalent to $(3+2t)y'' + (1+2t)y' - (1-2t)y = 0$ with $t = x - 1$. If

$$y = \sum_{n=0}^{\infty} a_n t^n, \text{ then } (3+2t)y'' + (1+2t)y' - (1-2t)y = 3 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + 2 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + \sum_{n=1}^{\infty} n a_n t^{n-1} + 2 \sum_{n=1}^{\infty} n a_n t^n - \sum_{n=0}^{\infty} a_n t^n + 2 \sum_{n=0}^{\infty} a_n t^{n+1} =$$

$$(6a_2 + a_1 - a_0) + \sum_{n=1}^{\infty} [3(n+2)(n+1)a_{n+2} + (n+1)(2n+1)a_{n+1} + (2n-1)a_n + 2a_{n-1}] t^n = 0$$

if $a_2 = -\frac{a_1 - a_0}{6}$ and $a_{n+2} = -\frac{2n+1}{3(n+2)}a_{n+1} - \frac{2n-1}{3(n+2)(n+1)}a_n - \frac{2}{3(n+2)(n+1)}a_{n-1}$, $n \geq 1$. Starting with $a_0 = 1$ and $a_1 = -2$ yields

$$y = 1 - 2(x-1) + \frac{1}{2}(x-1)^2 - \frac{1}{6}(x-1)^3 + \frac{5}{36}(x-1)^4 - \frac{73}{1080}(x-1)^5 + \dots$$

7.3.22. The equation is equivalent to $(1+t)y'' + (2-2t)y' + (3+t)y = 0$ with $t = x + 3$. If

$$y = \sum_{n=0}^{\infty} a_n t^n, \text{ then } (1+t)y'' + (2-2t)y' + (3+t)y =$$

$$\sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + 2 \sum_{n=1}^{\infty} n a_n t^{n-1} - 2 \sum_{n=1}^{\infty} n a_n t^n + 3 \sum_{n=0}^{\infty} a_n t^n + \sum_{n=0}^{\infty} a_n t^{n+1} =$$

$$(2a_2 + 2a_1 + 3a_0) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + (n+2)(n+1)a_{n+1} - (2n-3)a_n + a_{n-1}] t^n = 0$$

if $a_2 = -\frac{2a_1 + 3a_0}{2}$ and $a_{n+2} = -a_{n+1} + \frac{(2n-3)a_n - a_{n-1}}{(n+2)(n+1)}$, $n \geq 1$. Starting with $a_0 = 2$ and $a_1 = -2$ yields

$$y = 2 - 2(x+3) - (x+3)^2 + (x+3)^3 - \frac{11}{12}(x+3)^4 + \frac{67}{60}(x+3)^5 + \dots$$

7.3.24. The equation is equivalent to $(1+2t)y'' + 3y' + (1-t)y = 0$ with $t = x + 1$. If

$$y = \sum_{n=0}^{\infty} a_n t^n, \text{ then}$$

$$(1+2t)y'' + 3y' + (1-t)y = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + 2 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + 3 \sum_{n=1}^{\infty} n a_n t^{n-1} + \sum_{n=0}^{\infty} a_n t^n -$$

$$\sum_{n=0}^{\infty} a_n t^{n+1} = (2a_2 + 3a_1 + a_0) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + (2n+3)(n+1)a_{n+1} + a_n - a_{n-1}] t^n = 0$$

if $a_2 = -\frac{3a_1 + a_0}{2}$ and $a_{n+2} = -\frac{2n+3}{n+2}a_{n+1} - \frac{a_n - a_{n-1}}{(n+2)(n+1)}$, $n \geq 1$. Starting with $a_0 = 2$ and $a_1 = -3$ yields

$$y = 2 - 3(x+1) + \frac{7}{2}(x+1)^2 - 5(x+1)^3 + \frac{197}{24}(x+1)^4 - \frac{287}{20}(x+1)^5 + \dots$$

7.3.26. The equation is equivalent to $(6-2t)y'' + (3+t)y = 0$ with $t = x-2$. If $y = \sum_{n=0}^{\infty} a_n t^n$, then

$$\begin{aligned} (6-2t)y'' + (3+t)y &= 6 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} - 2 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + 3 \sum_{n=0}^{\infty} a_n t^n + \sum_{n=0}^{\infty} a_n t^{n+1} = \\ &= (12a_2 + 3a_0) + \sum_{n=1}^{\infty} [6(n+2)(n+1)a_{n+2} - 2(n+1)na_{n+1} + 3a_n + a_{n-1}]t^n = 0 \text{ if } a_2 = -\frac{a_0}{4} \text{ and} \\ a_{n+2} &= \frac{n}{3(n+2)}a_{n+1} - \frac{3a_n + a_{n-1}}{6(n+2)(n+1)}, \quad n \geq 1. \text{ Starting with } a_0 = 2 \text{ and } a_1 = -4 \text{ yields} \\ y &= 2 - 4(x-2) - \frac{1}{2}(x-2)^2 + \frac{2}{9}(x-2)^3 + \frac{49}{432}(x-2)^4 + \frac{23}{1080}(x-2)^5 + \dots \end{aligned}$$

7.3.28. The equation is equivalent to $(2+4t)y'' - (1-2t)y = 0$ with $t = x+4$. If $y = \sum_{n=0}^{\infty} a_n t^n$, then

$$\begin{aligned} (2+4t)y'' - (1-2t)y &= 2 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + 4 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} - \sum_{n=0}^{\infty} a_n t^n + 2 \sum_{n=0}^{\infty} a_n t^{n+1} = \\ &= (4a_2 - a_0) + \sum_{n=1}^{\infty} [2(n+2)(n+1)a_{n+2} + 4(n+1)na_{n+1} - a_n + 2a_{n-1}]t^n = 0 \text{ if } a_2 = \frac{a_0}{4} \text{ and} \\ a_{n+2} &= -\frac{2n}{n+2}a_{n+1} + \frac{a_n - 2a_{n-1}}{2(n+2)(n+1)}, \quad n \geq 1. \text{ Starting with } a_0 = -1 \text{ and } a_1 = 2 \text{ yields} \\ y &= -1 + 2(x+1) - \frac{1}{4}(x+1)^2 + \frac{1}{2}(x+1)^3 - \frac{65}{96}(x+1)^4 + \frac{67}{80}(x+1)^5 + \dots \end{aligned}$$

N=5; b=zeros(N,1); b(1)=-1;b(2)=2; b(3)=b(1)/4; for n=1:N-2
b(n+3)=-2*n*b(n+2)/(n+2)+(b(n+1)-2*b(n))/(2*(n+2)*(n+1)); end

7.3.29. Let $Ly = (1 + \alpha x + \beta x^2)y'' + (\gamma + \delta x)y' + \epsilon y$. If $y = \sum_{n=0}^{\infty} a_n x^n$, then

$$\begin{aligned} Ly &= \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \alpha \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} + \beta \sum_{n=2}^{\infty} n(n-1)a_n x^n + \gamma \sum_{n=1}^{\infty} na_n x^{n-1} + \\ &+ \delta \sum_{n=1}^{\infty} na_n x^n + \epsilon \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n + \alpha \sum_{n=0}^{\infty} (n+1)na_{n+1}x^n + \beta \sum_{n=0}^{\infty} n(n-1)a_n x^n + \\ &+ \gamma \sum_{n=0}^{\infty} (n+1)a_{n+1}x^n + \delta \sum_{n=0}^{\infty} na_n x^n + \epsilon \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} b_n x^n, \text{ where} \\ b_n &= (n+1)(n+2)a_{n+2} + (n+1)(\alpha n + \gamma)a_{n+1} + [\beta n(n-1) + \delta n + \epsilon]a_n, \text{ which implies the} \\ &\text{conclusion.} \end{aligned}$$

7.3.30. (a) Let $\gamma = 2\alpha$, $\delta = 4\beta$, and $\epsilon = 2\beta$ in Exercise 7.3.29 to obtain (B).

(b) If $a_n = c_1 r_1^n + c_2 r_2^n$, then
 $a_{n+2} + \alpha a_{n+1} + \beta a_n = c_1 r_1^n (r_1^2 + \alpha r_1 + \beta) + c_2 r_2^n (r_2^2 + \alpha r_2 + \beta) = c_1 r_1^n P_0(r_1) + c_2 r_2^n P_0(r_2) = 0$,
 so $\{a_n\}$ satisfies (B). Since $1/r_1$ and $1/r_2$ are the zeros of P_0 , Theorem 7.2.1 implies
 that $\sum_{n=0}^{\infty} (c_1 r_1^n + c_2 r_2^n)x^n$ is a solution of (A) on $(-\rho, \rho)$.

- (c) If $|x| < \rho$, then $|r_1 x| < \rho$ and $|r_2 x| < 1$, so $\sum_{n=0}^{\infty} r_i^n x^n = \frac{1}{1 - r_i x} = y_i$, $i = 1, 2$. Therefore, (b) implies that $\{y_1, y_2\}$ is a fundamental set of solutions of (A) on $(-\rho, \rho)$.
- (d) (A) can be written as $P_0 y'' + 2P_0' y' + P_0'' y = (P_0 y)'' = 0$. Therefore, $P_0 y = a + bx$ where a and b are arbitrary constants, and a partial fraction expansion shows that the general solution of (A) on any interval not containing $1/r_1$ or $1/r_2$ is
- $$y = \frac{a + bx}{P_0(x)} = \frac{c_1}{1 - r_1 x} + \frac{c_2}{1 - r_2 x} = c_1 y_1 + c_2 y_2.$$
- (e) If $a_n = c_1 r_1^n + c_2 r_2^n$, then
- $$a_{n+2} + \alpha a_{n+1} + \beta a_n = c_1 r_1^n (r_1^2 + \alpha r_1 + \beta) + c_2 r_2^n [(n+2)r_1^2 + \alpha(n+1)r_2 + \beta n] =$$
- $$(c_1 + n c_2) r_1^n P_0'(r_1) + c_2 r_1^n P_0'(r_1) = 0, \text{ so } \{a_n\} \text{ satisfies (B). Since } 1/r_1 \text{ is the only zero of } P_0, \text{ Theorem 7.2.1 implies that } \sum_{n=0}^{\infty} (c_1 + c_2 n) r_1^n x^n \text{ is a solution of (A) on } (-\rho, \rho).$$
- (f) If $|x| < \rho$, then $|r_1 x| < \rho$, so $\sum_{n=0}^{\infty} r_1^n x^n = \frac{1}{1 - r_1 x} = y_1$. Differentiating this and multiplying the result by x shows that $\sum_{n=0}^{\infty} n r_1^n x^n = \frac{r_1 x}{(1 - r_1 x)^2} = r_1 y_2$. Therefore, (e) implies that $\{y_1, y_2\}$ is a fundamental set of solutions of (A) on $(-\rho, \rho)$.
- (g) The argument is the same as in (c), but now the partial fraction expansion can be written as $y = \frac{a + bx}{P_0(x)} = \frac{c_1}{1 - r_1 x} + \frac{c_2 x}{(1 - r_2 x)^2} = c_1 y_1 + c_2 y_2$.

7.3.32. If $y = \sum_{n=0}^{\infty} a_n x^n$, then

$$y'' + 2xy' + (3 + 2x^2)y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 2 \sum_{n=1}^{\infty} n a_n x^n + 3 \sum_{n=0}^{\infty} a_n x^n + 2 \sum_{n=0}^{\infty} a_n x^{n+2} =$$

$$(2a_2 + 3a_0) + (6a_3 + 5a_1)x + \sum_{n=2}^{\infty} [(n+2)(n+1)a_{n+2} + (2n+3)a_n + 2a_{n-2}]x^n = 0 \text{ if}$$

$$a_2 = -3a_0/2, a_3 = -5a_1/6, \text{ and } a_{n+2} = -\frac{(2n+3)a_n + 2a_{n-2}}{(n+2)(n+1)}, n \geq 2. \text{ Starting with } a_0 = 1$$

$$\text{and } a_1 = -2 \text{ yields } y = 1 - 2x - \frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{17}{24}x^4 - \frac{11}{20}x^5 + \dots$$

7.3.34. If $y = \sum_{n=0}^{\infty} a_n x^n$, then

$$y'' + 5xy' - (3 - x^2)y = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 5 \sum_{n=1}^{\infty} n a_n x^n - 3 \sum_{n=0}^{\infty} a_n x^n + \sum_{n=0}^{\infty} a_n x^{n+2} =$$

$$(2a_2 - 3a_0) + (6a_3 + 2a_1)x + \sum_{n=2}^{\infty} [(n+2)(n+1)a_{n+2} + (5n-3)a_n + a_{n-2}]x^n = 0 \text{ if}$$

$$a_2 = 3a_0/2, a_3 = -a_1/3, \text{ and } a_{n+2} = -\frac{(5n-3)a_n + a_{n-2}}{(n+2)(n+1)}, n \geq 2. \text{ Starting with } a_0 = 6 \text{ and}$$

$$a_1 = -2 \text{ yields } y = 6 - 2x + 9x^2 + \frac{2}{3}x^3 - \frac{23}{4}x^4 - \frac{3}{10}x^5 + \dots$$

7.3.36. If $y = \sum_{n=0}^{\infty} a_n x^n$, then

$$\begin{aligned}
y'' - 3xy' + (2 + 4x^2)y &= \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - 3 \sum_{n=1}^{\infty} na_n x^n + 2 \sum_{n=0}^{\infty} a_n x^n + 4 \sum_{n=0}^{\infty} a_n x^{n+2} = \\
(2a_2 + 2a_0) + (6a_3 - a_1)x + \sum_{n=2}^{\infty} [(n+2)(n+1)a_{n+2} - (3n-2)a_n + 4a_{n-2}]x^n &= 0 \text{ if } a_2 = -a_0, \\
a_3 = a_1/6, \text{ and } a_{n+2} = \frac{(3n-2)a_n - 4a_{n-2}}{(n+2)(n+1)}, n \geq 2. \text{ Starting with } a_0 = 3 \text{ and } a_1 = 6 \text{ yields} \\
y = 3 + 6x - 3x^2 + x^3 - 2x^4 - \frac{17}{20}x^5 + \dots
\end{aligned}$$

7.3.38. If $y = \sum_{n=0}^{\infty} a_n x^n$, then

$$\begin{aligned}
3y'' + 2xy' + (4 - x^2)y &= 3 \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + 2 \sum_{n=1}^{\infty} na_n x^n + 4 \sum_{n=0}^{\infty} a_n x^n - \sum_{n=0}^{\infty} a_n x^{n+2} = \\
(6a_2 + 4a_0) + (18a_3 + 6a_1)x + \sum_{n=2}^{\infty} [3(n+2)(n+1)a_{n+2} + (2n+4)a_n - a_{n-2}]x^n &= 0 \text{ if} \\
a_2 = -2a_0/3, a_3 = -a_1/3, \text{ and } a_{n+2} = -\frac{(2n+4)a_n - a_{n-2}}{3(n+2)(n+1)}, n \geq 2. \text{ Starting with } a_0 = -2 \\
\text{and } a_1 = 3 \text{ yields } y = -2 + 3x + \frac{4}{3}x^2 - x^3 - \frac{19}{54}x^4 + \frac{13}{60}x^5 + \dots
\end{aligned}$$

7.3.40. If $y = \sum_{n=0}^{\infty} a_n x^n$, then $(1+x)y'' + x^2y' + (1+2x)y =$

$$\begin{aligned}
\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1} + \sum_{n=1}^{\infty} na_n x^{n+1} + \sum_{n=0}^{\infty} a_n x^n + 2 \sum_{n=0}^{\infty} a_n x^{n+1} &= \\
(2a_2 + a_0) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + (n+1)na_{n+1} + a_n + (n+1)a_{n-1}]x^n &= 0 \text{ if } a_2 = -a_0/2 \\
\text{and } a_{n+2} = -\frac{(n+1)na_{n+1} + a_n + (n+1)a_{n-1}}{(n+2)(n+1)}, n \geq 1. \text{ Starting with } a_0 = -2 \text{ and } a_1 = 3 \\
\text{yields } y = -2 + 3x + x^2 - \frac{1}{6}x^3 - \frac{3}{4}x^4 + \frac{31}{120}x^5 + \dots
\end{aligned}$$

7.3.42. If $y = \sum_{n=0}^{\infty} a_n x^n$, then $(1+x^2)y'' + (2+x^2)y' + xy =$

$$\begin{aligned}
\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n x^n + 2 \sum_{n=1}^{\infty} na_n x^{n-1} + \sum_{n=1}^{\infty} na_n x^{n+1} + \sum_{n=1}^{\infty} a_n x^{n+1} &= \\
(2a_2 + 2a_1) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + 2(n+1)a_{n+1} + n(n-1)a_n + na_{n-1}]x^n &= 0 \text{ if} \\
a_2 = -a_1 \text{ and } a_{n+2} = -\frac{[2(n+1)a_{n+1} + n(n-1)a_n + na_{n-1}]}{(n+2)(n+1)}, n \geq 1. \text{ Starting with } a_0 = -3 \\
\text{and } a_1 = 5 \text{ yields } y = -3 + 5x - 5x^2 + \frac{23}{6}x^3 - \frac{23}{12}x^4 + \frac{11}{30}x^5 + \dots
\end{aligned}$$

7.3.44. The equation is equivalent to $y'' + (1+3t^2)y' + (1+2t)y = 0$ with $t = x - 2$. If

$$y = \sum_{n=0}^{\infty} a_n t^n, \text{ then}$$

$$y'' + (1 + 3t^2)y' + (1 + 2t)y = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + \sum_{n=1}^{\infty} na_n t^{n-1} + 3 \sum_{n=1}^{\infty} na_n t^{n+1} + \sum_{n=0}^{\infty} a_n t^n + 2 \sum_{n=0}^{\infty} a_n t^{n+1} = (2a_2 + a_1 + a_0) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + (n+1)a_{n+1} + a_n + (3n-1)a_{n-1}]t^n = 0$$

if $a_2 = -(a_1 + a_0)/2$ and $a_{n+2} = -\frac{[(n+1)a_{n+1} + a_n + (3n-1)a_{n-1}]}{(n+2)(n+1)}$, $n \geq 1$. Starting with $a_0 = 2$ and $a_1 = -3$ yields

$$y = 2 - 3(x+2) + \frac{1}{2}(x+2)^2 - \frac{1}{3}(x+2)^3 + \frac{31}{24}(x+2)^4 - \frac{53}{120}(x+2)^5 + \dots$$

7.3.46. The equation is equivalent to $(1 - t^2)y'' - (7 - 8t + t^2)y' + ty = 0$ with $t = x + 2$. If

$$y = \sum_{n=0}^{\infty} a_n t^n, \text{ then } (1 - t^2)y'' - (7 - 8t + t^2)y' + ty = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} - \sum_{n=2}^{\infty} n(n-1)a_n t^n - 7 \sum_{n=1}^{\infty} na_n t^{n-1} + 8 \sum_{n=1}^{\infty} na_n t^n - \sum_{n=1}^{\infty} na_n t^{n+1} + \sum_{n=0}^{\infty} a_n t^{n+1} = (2a_2 - 7a_1) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} - 7(n+1)a_{n+1} - n(n-9)a_n - (n-2)a_{n-1}]t^n = 0$$

if $a_2 = 7a_1/2$ and $a_{n+2} = \frac{[7(n+1)a_{n+1} + n(n-9)a_n + (n-2)a_{n-1}]}{(n+2)(n+1)}$, $n \geq 1$. Starting with $a_0 = 2$ and $a_1 = -1$ yields

$$y = 2 - (x+2) - \frac{7}{2}(x+2)^2 - \frac{43}{6}(x+2)^3 - \frac{203}{24}(x+2)^4 - \frac{167}{30}(x+2)^5 + \dots$$

7.3.48. The equation is equivalent to $(1 + 3t + 2t^2)y'' - (3 + t - t^2)y' - (3 + t)y = 0$ with $t = x - 1$.

If $y = \sum_{n=0}^{\infty} a_n t^n$, then

$$(1 + 3t + 2t^2)y'' - (3 + t - t^2)y' - (3 + t)y = \sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + 3 \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + 2 \sum_{n=2}^{\infty} n(n-1)a_n t^n - 3 \sum_{n=1}^{\infty} na_n t^{n-1} - \sum_{n=1}^{\infty} na_n t^n + \sum_{n=1}^{\infty} na_n t^{n+1} - 3 \sum_{n=0}^{\infty} a_n t^n - \sum_{n=0}^{\infty} a_n t^{n+1} = (2a_2 - 3a_1 - 3a_0) + \sum_{n=1}^{\infty} [(n+2)(n+1)a_{n+2} + 3(n^2 - 1)a_{n+1} + (2n^2 - 3n - 3)(n+1)a_n + (n-2)a_{n-1}]t^n = 0$$

if $a_2 = 3(a_1 + a_0)/2$ and $a_{n+2} = -\frac{[3(n^2 - 1)a_{n+1} + (2n^2 - 3n - 3)(n+1)a_n + (n-2)a_{n-1}]}{(n+2)(n+1)}$, $n \geq 1$. Starting with $a_0 = 1$ and $a_1 = 0$ yields

$$y = 1 + \frac{3}{2}(x-1)^2 + \frac{1}{6}(x-1)^3 - \frac{1}{8}(x-1)^5 + \dots$$

7.4 Regular Singular Points; Euler Equations

7.4.2. $p(r) = r(r-1) - 7r + 7 = (r-7)(r-1)$; $y = c_1 x + c_2 x^7$.

7.4.4. $p(r) = r(r-1) + 5r + 4 = (r+2)^2$; $y = x^{-2}(c_1 + c_2 \ln x)$

7.4.6. $p(r) = r(r-1) - 3r + 13 = (r-2)^2 + 9$; $y = x^2[c_1 \cos(3 \ln x) + c_2 \sin(3 \ln x)]$.

$$7.4.8. \quad p(r) = 12r(r-1) - 5r + 6 = (3r-2)(4r-3); \quad y = c_1 x^{2/3} + c_2 x^{3/4}.$$

$$7.4.10. \quad p(r) = 3r(r-1) - r + 1 = (r-1)(3r-1); \quad y = c_1 x + c_2 x^{1/3}.$$

$$7.4.12. \quad p(r) = r(r-1) + 3r + 5 = (r+1)^2 + 4; \quad y = \frac{1}{x} [c_1 \cos(2 \ln x) + c_2 \sin(2 \ln x)]$$

$$7.4.14. \quad p(r) = r(r-1) - r + 10 = (r-1)^2 + 9; \quad y = x [c_1 \cos(3 \ln x) + c_2 \sin(3 \ln x)].$$

$$7.4.16. \quad p(r) = 2r(r-1) + 3r - 1 = (r+1)(2r-1); \quad y = \frac{c_1}{x} + c_2 x^{1/2}.$$

$$7.4.18. \quad p(r) = 2r(r-1) + 10r + 9 = 2(r+2)^2 + 1; \quad y = \frac{1}{x^2} \left[c_1 \cos \left(\frac{1}{\sqrt{2}} \ln x \right) + c_2 \sin \left(\frac{1}{\sqrt{2}} \ln x \right) \right].$$

$$7.4.20. \quad \text{If } p(r) = ar(r-1) + br + c = a(r-r_1)^2, \text{ then (A) } p(r_1) = p'(r_1) = 0. \text{ If } y = ux^{r_1}, \text{ then}$$

$$y' = u'x^{r_1} + r_1 u x^{r_1-1} \text{ and } y'' = u''x^{r_1} + 2r_1 u'x^{r_1-1} + r_1(r_1-1)u x^{r_1-2}, \text{ so}$$

$$\begin{aligned} ax^2 y'' + bxy' + cy &= ax^{r_1+2}u'' + (2ar_1 + b)x^{r_1+1}u' + (ar_1(r_1-1) + br_1 + c)x^{r_1}u \\ &= ax^{r_1+2}u'' + p'(r_1)x^{r_1+1}u' + p(r)x^{r_1}u = ax^{r_1+2}u'', \end{aligned}$$

from (A). Therefore, $u'' = 0$, so $u = c_1 + c_2 x$ and $y = x^{r_1}(c_1 + c_2 x)$.

$$7.4.22. \quad (a) \quad \text{If } t = x - 1 \text{ and } Y(t) = y(t+1) = y(x), \text{ then}$$

$$(1-x^2)y'' - 2xy' + \alpha(\alpha+1)y = -t(2+t)\frac{d^2Y}{dt^2} - 2(1+t)\frac{dY}{dt} + \alpha(\alpha+1)Y = 0, \text{ so } y$$

satisfies Legendre's equation if and only if Y satisfies (A)

$$t(2+t)\frac{d^2Y}{dt^2} + 2(1+t)\frac{dY}{dt} - \alpha(\alpha+1)Y = 0. \text{ Since (A) can be rewritten as}$$

$$t^2(2+t)\frac{d^2Y}{dt^2} + 2t(1+t)\frac{dY}{dt} - \alpha(\alpha+1)tY = 0, \text{ (A) has a regular singular point at } t = 0_0.$$

$$(b) \quad \text{If } t = x + 1 \text{ and } Y(t) = y(t-1) = y(x), \text{ then}$$

$$(1-x^2)y'' - 2xy' + \alpha(\alpha+1)y = t(2-t)\frac{d^2Y}{dt^2} + 2(1-t)\frac{dY}{dt} + \alpha(\alpha+1)Y, \text{ so } y \text{ satisfies}$$

Legendre's equation if and only if Y satisfies (B)

$$t(2-t)\frac{d^2Y}{dt^2} + 2(1-t)\frac{dY}{dt} + \alpha(\alpha+1)Y, \text{ Since (B) can be rewritten as (B)}$$

$$t^2(2-t)\frac{d^2Y}{dt^2} + 2t(1-t)\frac{dY}{dt} + \alpha(\alpha+1)tY, \text{ (B) has a regular singular point at } t = 0_0.$$

7.5 The Method of Frobenius I

$$7.5.2. \quad p_0(r) = r(3r-1); \quad p_1(r) = 2(r+1); \quad p_2(r) = -4(r+2).$$

$$a_1(r) = -\frac{2}{3r+2}; \quad a_n(r) = -\frac{2a_{n-1}(r) - 4a_{n-2}(r)}{3n+3r-1}, \quad n \geq 1.$$

$$r_1 = 1/3; \quad a_1(1/3) = -2/3; \quad a_n(1/3) = -\frac{2a_{n-1}(1/3) - 4a_{n-2}(1/3)}{3n}, \quad n \geq 1;$$

$$y_1 = x^{1/3} \left(1 - \frac{2}{3}x + \frac{8}{9}x^2 - \frac{40}{81}x^3 + \dots \right).$$

$$r_2 = 0; a_1(0) = -1; a_n(0) = -\frac{2a_{n-1}(0) - 4a_{n-2}(0)}{3n-1}, n \geq 1;$$

$$y_2 = 1 - x + \frac{6}{5}x^2 - \frac{4}{5}x^3 + \dots$$

$$7.5.4. p_0(r) = (r+1)(4r-1); p_1(r) = 2(r+2); p_2(r) = 4r+7.$$

$$a_1(r) = -\frac{2}{4r+3}; a_n(r) = -\frac{2}{4n+4r-1}a_{n-1}(r) - \frac{1}{n+r+1}a_{n-2}(r), n \geq 1.$$

$$r_1 = 1/4; a_1(1/4) = -1/2; a_n(1/4) = -\frac{1}{2n}a_{n-1}(1/4) - \frac{4}{4n+5}a_{n-2}(1/4), n \geq 1;$$

$$y_1 = x^{1/4} \left(1 - \frac{1}{2}x - \frac{19}{104}x^2 + \frac{1571}{10608}x^3 + \dots \right).$$

$$r_2 = -1; a_1(-1) = 2; a_n(-1) = -\frac{2}{4n-5}a_{n-1}(-1) - \frac{1}{n}a_{n-2}(-1), n \geq 1;$$

$$y_2 = x^{-1} \left(1 + 2x - \frac{11}{6}x^2 - \frac{1}{7}x^3 + \dots \right).$$

$$7.5.6. p_0(r) = r(5r-1); p_1(r) = (r+1)^2; p_2(r) = 2(r+2)(5r+9).$$

$$a_1(r) = -\frac{r+1}{5r+4}; a_n(r) = -\frac{n+r}{5n+5r-1}a_{n-1}(r) - 2a_{n-2}(r), n \geq 1.$$

$$r_1 = 1/5; a_1(1/5) = -6/25; a_n(1/5) = -\frac{5n+1}{25n}a_{n-1}(1/5) - 2a_{n-2}(1/5), n \geq 1;$$

$$y_1 = x^{1/5} \left(1 - \frac{6}{25}x - \frac{1217}{625}x^2 + \frac{41972}{46875}x^3 + \dots \right).$$

$$r_2 = 0; a_1(0) = -1/4; a_n(0) = -\frac{n}{5n-1}a_{n-1}(0) - 2a_{n-2}(0), n \geq 1;$$

$$y_2 = x - \frac{1}{4}x^2 - \frac{35}{18}x^3 + \frac{11}{12}x^4 + \dots$$

$$7.5.8. p_0(r) = (3r-1)(6r+1); p_1(r) = (3r+2)(6r+1); p_2(r) = 3r+5.$$

$$a_1(r) = -\frac{6r+1}{6r+7}; a_n(r) = -\frac{6n+6r-5}{6n+6r+1}a_{n-1}(r) - \frac{1}{6n+6r+1}a_{n-2}(r), n \geq 1.$$

$$r_1 = 1/3; a_1(1/3) = -1/3; a_n(1/3) = -\frac{2n-1}{2n+1}a_{n-1}(1/3) - \frac{1}{6n+3}a_{n-2}(1/3), n \geq 1;$$

$$y_1 = x^{1/3} \left(1 - \frac{1}{3}x + \frac{2}{15}x^2 - \frac{5}{63}x^3 + \dots \right).$$

$$r_2 = -1/6; a_1(-1/6) = 0; a_n(-1/6) = -\frac{n-1}{n}a_{n-1}(-1/6) - \frac{1}{6n}a_{n-2}(-1/6), n \geq 1;$$

$$y_2 = x^{-1/6} \left(1 - \frac{1}{12}x^2 + \frac{1}{18}x^3 + \dots \right).$$

$$7.5.10. p_0(r) = (2r+1)(5r-1); p_1(r) = (2r-1)(5r+4); p_2(r) = 2(2r+5)(5r-1).$$

$$a_1(r) = -\frac{2r-1}{2r+3}; a_n(r) = -\frac{2n+2r-3}{2n+2r+1}a_{n-1}(r) - \frac{10n+10r-22}{5n+5r-1}a_{n-2}(r), n \geq 1.$$

$$r_1 = 1/5; a_1(1/5) = 3/17; a_n(1/5) = -\frac{10n-13}{10n+7}a_{n-1}(1/5) - \frac{2n-4}{n}a_{n-2}(1/5), n \geq 1;$$

$$y_1 = x^{1/5} \left(1 + \frac{3}{17}x - \frac{7}{153}x^2 - \frac{547}{5661}x^3 + \dots \right).$$

$$r_2 = -1/2; a_1(-1/2) = 1; a_n(-1/2) = -\frac{n-2}{n}a_{n-1}(-1/2) - \frac{20n-54}{10n-7}a_{n-2}(-1/2), n \geq 1;$$

$$y_2 = x^{-1/2} \left(1 + x + \frac{14}{13}x^2 - \frac{556}{897}x^3 + \dots \right).$$

$$7.5.14. p_0(r) = (r+1)(2r-1); p_1(r) = 2r+1; a_n(r) = -\frac{1}{n+r+1}a_{n-1}(r).$$

$$r_1 = 1/2; a_n(1/2) = -\frac{2}{2n+3}a_{n-1}(1/2); y_1 = x^{1/2} \sum_{n=0}^{\infty} \frac{(-2)^n}{\prod_{j=1}^n (2j+3)} x^n.$$

$$r_2 = -1; a_n(-1) = -\frac{1}{n}a_{n-1}(-1); y_2 = x^{-1} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^n.$$

$$7.5.16. p_0(r) = (r+2)(2r-1); p_1(r) = r+3; a_n(r) = -\frac{1}{2n+2r-1}a_{n-1}(r).$$

$$r_1 = 1/2; a_n(1/2) = -\frac{1}{2n}a_{n-1}(1/2); y_1 = x^{1/2} \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n n!} x^n.$$

$$r_2 = -2; a_n(-2) = -\frac{1}{2n-5}a_{n-1}(-2); y_2 = \frac{1}{x^2} \sum_{n=0}^{\infty} \frac{(-1)^n}{\prod_{j=1}^n (2j-5)} x^n.$$

$$7.5.18. p_0(r) = (r-1)(2r-1); p_1(r) = -2; a_n(r) = \frac{2}{(n+r-1)(2n+2r-1)}a_{n-1}(r).$$

$$r_1 = 1; a_n(1) = \frac{2}{n(2n+1)}a_{n-1}(1); y_1 = x \sum_{n=0}^{\infty} \frac{2^n}{n! \prod_{j=1}^n (2j+1)} x^n.$$

$$r_2 = 1/2; a_n(1/2) = \frac{2}{n(2n-1)}a_{n-1}(1/2); y_2 = x^{1/2} \sum_{n=0}^{\infty} \frac{2^n}{n! \prod_{j=1}^n (2j-1)} x^n.$$

$$7.5.20. p_0(r) = (r-1)(3r+1); p_1(r) = r-3; a_n(r) = -\frac{n+r-4}{(n+r-1)(3n+3r+1)}a_{n-1}(r).$$

$$r_1 = ; a_n(1) = -\frac{n-3}{n(3n+4)}a_{n-1}(1); y_1 = x \left(1 + \frac{2}{7}x + \frac{1}{70}x^2 \right).$$

$$r_2 = -1/3; a_n(-1/3) = -\frac{3n-13}{3n(3n-4)}a_{n-1}(-1/3); y_2 = x^{-1/3} \sum_{n=0}^{\infty} \frac{(-1)^n}{3^n n!} \left(\prod_{j=1}^n \frac{3j-13}{3j-4} \right) x^n.$$

$$7.5.22. p_0(r) = (r-1)(4r-1); p_1(r) = r(r+2); a_n(r) = -\frac{n+r+1}{4n+4r-1}a_{n-1}(r).$$

$$r_1 = 1; a_n(1) = -\frac{n+2}{4n+3}a_{n-1}(1); y_1 = x \sum_{n=0}^{\infty} \frac{(-1)^n (n+2)!}{2 \prod_{j=1}^n (4j+3)} x^n.$$

$$r_2 = 1/4; a_n(1/4) = -\frac{4n+5}{16n}a_{n-1}(1/4); y_2 = x^{1/4} \sum_{n=0}^{\infty} \frac{(-1)^n}{16^n n!} \prod_{j=1}^n (4j+5) x^n$$

$$7.5.24. \quad p_0(r) = (r+1)(3r-1); p_1(r) = 2(r+2)(2r+3); a_n(r) = -2 \frac{2n+2r+1}{3n+3r-1} a_{n-1}(r).$$

$$r_1 = 1/3; a_n(1/3) = -2 \frac{6n+5}{9n} a_{n-1}(1/3); y_1 = x^{1/3} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left(\frac{2}{9}\right)^n \left(\prod_{j=1}^n (6j+5)\right) x^n;$$

$$r_2 = -1; a_n(-1) = -2 \frac{2n-1}{3n-4} a_{n-1}(-1); y_2 = x^{-1} \sum_{n=0}^{\infty} (-1)^n 2^n \left(\prod_{j=1}^n \frac{2j-1}{3j-4}\right) x^n$$

$$7.5.28. \quad p_0(r) = (2r-1)(4r-1); p_1(r) = (r+1)^2; a_n(r) = -\frac{(n+r)^2}{(2n+2r-1)(4n+4r-1)} a_{n-1}(r).$$

$$r_1 = 1/2; a_n(1/2) = -\frac{4n^2+4n+1}{8n(4n+1)} a_{n-1}(1/2);$$

$$y_1 = x^{1/2} \left(1 - \frac{9}{40}x + \frac{5}{128}x^2 - \frac{245}{39936}x^3 + \dots\right).$$

$$r_2 = 1/4; a_n(1/4) = -\frac{16n^2+8n+1}{32n(4n-1)} a_{n-1}(1/4);$$

$$y_2 = x^{1/4} \left(1 - \frac{25}{96}x + \frac{675}{14336}x^2 - \frac{38025}{5046272}x^3 + \dots\right).$$

$$7.5.30. \quad p_0(r) = (2r-1)(2r+1); p_1(r) = (2r+1)(3r+1); a_n(r) = -\frac{(3n+3r-2)}{(2n+2r+1)} a_n(r).$$

$$r_1 = 1/2; a_n(1/2) = -\frac{6n-1}{4(n+1)} a_{n-1}(1/2); y_1 = x^{1/2} \left(1 - \frac{5}{8}x + \frac{55}{96}x^2 - \frac{935}{1536}x^3 + \dots\right).$$

$$r_2 = -1/2; a_n(-1/2) = -\frac{6n-7}{4n} a_{n-1}(-1/2); y_2 = x^{-1/2} \left(1 + \frac{1}{4}x - \frac{5}{32}x^2 - \frac{55}{384}x^3 + \dots\right).$$

$$7.5.32. \quad p_0(r) = (2r+1)(3r+1); p_1(r) = (r+1)(r+2); a_n(r) = \frac{(n+r)(n+r+1)}{(2n+2r+1)(3n+3r+1)} a_n(r).$$

$$r_1 = -1/3; a_n(-1/3) = -\frac{(3n-1)(3n+2)}{9n(6n+1)} a_{n-1}(-1/3);$$

$$y_1 = x^{-1/3} \left(1 - \frac{10}{63}x + \frac{200}{7371}x^2 - \frac{17600}{3781323}x^3 + \dots\right).$$

$$r_2 = -1/2; a_n(-1/2) = -\frac{(2n-1)(2n+1)}{4n(6n-1)} a_{n-1}(-1/2);$$

$$y_2 = x^{-1/2} \left(1 - \frac{3}{20}x + \frac{9}{352}x^2 - \frac{105}{23936}x^3 + \dots\right).$$

$$7.5.34. \quad p_0(r) = (2r-1)(4r-1); p_2(r) = -(2r+3)(4r+3); a_{2m}(r) = \frac{8m+4r-5}{8m+4r-1} a_{2m-2}(r).$$

$$r_1 = 1/2; a_{2m}(1/2) = \frac{8m-3}{8m+1} a_{2m-2}(1/2); y_1 = x^{1/2} \sum_{m=0}^{\infty} \left(\prod_{j=1}^m \frac{8j-3}{8j+1}\right) x^{2m}.$$

$$r_2 = 1/4; a_{2m}(1/4) = \frac{2m-1}{2m} a_{2m-2}(1/4); y_2 = x^{1/4} \sum_{m=0}^{\infty} \frac{1}{2^m m!} \left(\prod_{j=1}^m (2j-1)\right) x^{2m}$$

$$7.5.36. \quad p_0(r) = r(3r-1); p_2(r) = (r-4)(r+2); a_{2m}(r) = -\frac{2m+r-6}{6m+3r-1}a_{2m-2}(r).$$

$$r_1 = 1/3; a_{2m}(1/3) = -\frac{6m-17}{18m}a_{2m-2}(1/3); y_1 = x^{1/3} \sum_{m=0}^{\infty} \frac{(-1)^m}{18^m m!} \left(\prod_{j=1}^m (6j-17) \right) x^{2m}.$$

$$r_2 = 0; a_{2m}(0) = -\frac{2m-6}{6m-1}a_{2m-2}(0); y_2 = 1 + \frac{4}{5}x^2 + \frac{8}{55}x^4$$

$$7.5.38. \quad p_0(r) = (2r-1)(3r-1); p_2(r) = -(r+1)(3r+5); a_{2m}(r) = \frac{2m+r-1}{4m+2r-1}a_{2m-2}(r).$$

$$r_1 = 1/2; a_{2m}(1/2) = \frac{4m-1}{8m}a_{2m-2}(1/2); y_1 = x^{1/2} \sum_{m=0}^{\infty} \frac{1}{8^m m!} \left(\prod_{j=1}^m (4j-1) \right) x^{2m}.$$

$$r_2 = 1/3; a_{2m}(1/3) = \frac{6m-2}{12m-1}a_{2m-2}(1/3); y_2 = x^{1/3} \sum_{m=0}^{\infty} 2^m \left(\prod_{j=1}^m \frac{3j-1}{12j-1} \right) x^{2m}.$$

$$7.5.40. \quad p_0(r) = (2r-1)(2r+1); p_1(r) = (r+1)(2r+3); a_{2m}(r) = -\frac{2m+r-1}{4m+2r+1}a_{2m-2}(r).$$

$$r_1 = 1/2; a_{2m}(1/2) = -\frac{4m-1}{4(2m+1)}a_{2m-2}(1/2); y_1 = x^{1/2} \sum_{m=0}^{\infty} \frac{(-1)^m}{4^m} \left(\prod_{j=1}^m \frac{4j-1}{2j+1} \right) x^{2m}.$$

$$r_2 = -1/2; a_{2m}(-1/2) = -\frac{4m-3}{8m}a_{2m-2}(-1/2); y_2 = x^{-1/2} \sum_{m=0}^{\infty} \frac{(-1)^m}{8^m m!} \left(\prod_{j=1}^m (4j-3) \right) x^{2m}$$

$$7.5.42. \quad p_0(r) = (r+1)(3r-1); p_1(r) = (r-1)(3r+5); a_{2m}(r) = -\frac{2m+r-3}{2m+r+1}a_{2m-2}(r).$$

$$r_1 = 1/3; a_{2m}(1/3) = -\frac{3m-4}{3m+2}a_{2m-2}(1/3); y_1 = x^{1/3} \sum_{m=0}^{\infty} (-1)^m \left(\prod_{j=1}^m \frac{3j-4}{3j+2} \right) x^{2m}.$$

$$r_2 = -1; a_{2m}(-1) = -\frac{m-2}{m}a_{2m-2}(-1); y_2 = x^{-1}(1+x^2)$$

$$7.5.44. \quad p_0(r) = (r+1)(2r-1); p_1(r) = r^2; a_{2m}(r) = -\frac{(2m+r-2)^2}{(2m+r+1)(4m+2r-1)}a_{2m-2}(r).$$

$$r_1 = 1/2; a_{2m}(1/2) = -\frac{(4m-3)^2}{8m(4m+3)}a_{2m-2}(1/2); y_1 = x^{1/2} \sum_{m=0}^{\infty} \frac{(-1)^m}{8^m m!} \left(\prod_{j=1}^m \frac{(4j-3)^2}{4j+3} \right) x^{2m}.$$

$$r_2 = -1; a_{2m}(-1) = -\frac{(2m-3)^2}{2m(4m-3)}a_{2m-2}(-1); y_2 = x^{-1} \sum_{m=0}^{\infty} \frac{(-1)^m}{2^m m!} \left(\prod_{j=1}^m \frac{(2j-3)^2}{4j-3} \right) x^{2m}.$$

$$7.5.46. \quad p_0(r) = (3r-1)(3r+1); p_1(r) = 3r+5; a_{2m}(r) = -\frac{1}{6m+3r+1}a_{2m-2}(r).$$

$$r_1 = 1/3; a_{2m}(1/3) = -\frac{1}{2(3m+1)}a_{2m-2}(1/3); y_1 = x^{1/3} \sum_{m=0}^{\infty} \frac{(-1)^m}{2^m \prod_{j=1}^m (3j+1)} x^{2m}.$$

$$r_2 = -1/3; a_{2m}(-1/3) = -\frac{1}{6m}a_{2m-2}(-1/3); y_2 = x^{-1/3} \sum_{m=0}^{\infty} \frac{(-1)^m}{6^m m!} x^{2m}$$

$$7.5.48. \quad p_0(r) = 2(r+1)(4r-1); \quad p_2(r) = (r+3)^2; \quad a_{2m}(r) = -\frac{2m+r+1}{2(8m+4r-1)}a_{2m-2}(r).$$

$$\begin{aligned} r_1 &= 1/4; \quad a_{2m}(1/4) = -\frac{8m+5}{64m}a_{2m-2}(1/4); \\ y_1 &= x^{1/4} \left(1 - \frac{13}{64}x^2 + \frac{273}{8192}x^4 - \frac{2639}{524288}x^6 + \dots \right). \\ r_2 &= -1; \quad a_{2m}(-1) = -\frac{m}{8m-5}a_{2m-2}(-1); \quad y_2 = x^{-1} \left(1 - \frac{1}{3}x^2 + \frac{2}{33}x^4 - \frac{2}{209}x^6 + \dots \right). \end{aligned}$$

$$7.5.50. \quad p_0(r) = (2r-1)(2r+1); \quad p_2(r) = (2r+5)^2; \quad a_{2m}(r) = -\frac{4m+2r+1}{4m+2r-1}a_{2m-2}(r).$$

$$\begin{aligned} r_1 &= 1/2; \quad a_{2m}(1/2) = -\frac{2m+1}{2m}a_{2m-2}(1/2); \quad y_1 = x^{1/2} \left(1 - \frac{3}{2}x^2 + \frac{15}{8}x^4 - \frac{35}{16}x^6 + \dots \right). \\ r_2 &= -1/2; \quad a_{2m}(-1/2) = -\frac{2m}{2m-1}a_{2m-2}(-1/2); \quad y_2 = x^{-1/2} \left(1 - 2x^2 + \frac{8}{3}x^4 - \frac{16}{5}x^6 + \dots \right). \end{aligned}$$

7.5.52. (a) Multiplying (A) $c_1y_1 + c_2y_2 \equiv 0$ by x^{-r_2} yields $c_1x^{r_1-r_2}\sum_{n=0}^{\infty}a_nx^n + c_2\sum_{n=0}^{\infty}b_nx^n = 0$, $0 < x < \rho$. Letting $x \rightarrow 0+$ shows that $c_2 = 0$, since $b_0 = 1$. Now (A) reduces to $c_1y_1 \equiv 0$, so $c_1 = 0$. Therefore, y_1 and y_2 are linearly independent on $(0, \rho)$.

(b) Since $y_1 = \sum_{n=0}^{\infty}a_n(r_1)x^n$ and $y_2 = \sum_{n=0}^{\infty}a_n(r_2)x^n$ are linearly independent solutions of $Ly = 0$ $(0, \rho)$, $\{y_1, y_2\}$ is a fundamental set of solutions of $Ly = 0$ on $(0, \rho)$, by Theorem 5.1.6.

7.5.54. (a) If $x > 0$, then $|x|^rx^n = x^{n+r}$, so the assertions are obvious. If $x < 0$, then

$$\begin{aligned} |x|^r &= (-x)^r, \text{ so } \frac{d}{dx}|x|^r = -r(-x)^{r-1} = \frac{r(-x)^r}{x} = \frac{r|x|^r}{x}. \text{ Therefore, (A)} \\ \frac{d}{dx}(|x|^rx^n) &= \frac{r|x|^r}{x}x^n + |x|^r(nx^{n-1}) = (n+r)|x|^rx^{n-1} \text{ and} \\ \frac{d^2}{dx^2}(|x|^rx^n) &= (n+r)\frac{d}{dx}(|x|^rx^{n-1}) = (n+r)(n+r-1)|x|^rx^{n-2}, \text{ from (A) with } n \\ &\text{replaced by } n-1. \end{aligned}$$

7.5.56. (a) Here $p_1 \equiv 0$, so Eqn. (7.5.12) reduces to $a_0(r) = 1$, $a_1(r) = 0$,

$$\begin{aligned} a_n(r) &= -\frac{p_2(n+r-2)}{p_0(n+r)}a_{n-2}(r), \quad r \geq 0, \text{ which implies that } a_{2m+1}(r) = 0 \text{ for} \\ m &= 1, 2, 3, \dots \text{ Therefore, Eqn. (7.5.12) actually reduces to } a_0(r) = 1, \\ a_{2m}(r) &= -\frac{p_2(2m+r-2)}{p_0(2m+r)}, \text{ which holds because of condition (A).} \end{aligned}$$

(b) Similar to the proof of Exercise 7.5.55(a).

(c) $p_0(2m+r_1) = 2m\alpha_0(2m+r_1-r_2)$, which is nonzero if $m > 0$, since $r_1 - r_2 \geq 0$. Therefore, the assumptions of Theorem 7.5.2 hold with $r = r_1$, and $Ly_1 = p_0(r_1)x^{r_1} = 0$. If $r_1 - r_2$ is not an even integer, then $p_0(2m+r_2) = 2m\alpha_0(2m-r_1+r_2) \neq 0$, $m = 1, 2, \dots$. Hence, the assumptions of Theorem 7.5.2 hold with $r = r_2$ and $Ly_2 = p_0(r_2)x^{r_2} = 0$. From Exercise 7.5.52, $\{y_1, y_2\}$ is a fundamental set of solutions.

(d) Similar to the proof of Exercise 7.5.55(c).

7.5.58. (a) From Exercise 7.5.57, $b_n = 0$ for $n \geq 1$.

7.5.60. (a)

$$(\alpha_0 + \alpha_1 x + \alpha_2 x^2) \sum_{n=0}^{\infty} a_n x^n = \alpha_0 a_0 + (\alpha_0 a_1 + \alpha_1 a_0) x + \sum_{n=2}^{\infty} (\alpha_0 a_n + \alpha_1 a_{n-1} + \alpha_2 a_{n-2}) x^n = 1,$$

$$\text{so } \sum_{n=0}^{\infty} a_n x^n = \frac{\alpha_0 a_0}{\alpha_0 + \alpha_1 x + \alpha_2 x^2}.$$

(b) If $\frac{p_1(r-1)}{p_0(r)} = \frac{\alpha_1}{\alpha_0}$ and $\frac{p_2(r-2)}{p_0(r)} = \frac{\alpha_2}{\alpha_0}$, then Eqn. (7.5.12) is equivalent to $a_0(r) = 1$, $\alpha_0 a_1(r) + \alpha_1 a_0(r) = 0$, $\alpha_0 a_n(r) + \alpha_1 a_{n-1}(r) + \alpha_2 a_{n-2}(r) = 0$, $n \geq 2$. Therefore, Theorem 7.5.2 implies the conclusion.

$$7.5.62. \quad p_0(r) = (2r-1)(3r-1); p_1(r) = 0; p_2(r) = 2(2r+3)(3r+5); \frac{p_1(r-1)}{p_0(r)} = 0 = \frac{\alpha_1}{\alpha_0};$$

$$\frac{p_2(r-2)}{p_0(r)} = 2 = \frac{\alpha_2}{\alpha_0}; y_1 = \frac{x^{1/3}}{1+2x^2}; y_2 = \frac{x^{1/2}}{1+2x^2}.$$

$$7.5.64. \quad p_0(r) = 5(3r-1)(3r+1); p_1(r) = (3r+2)(3r+4); p_2(r) = 0; \frac{p_1(r-1)}{p_0(r)} = \frac{1}{5} = \frac{\alpha_1}{\alpha_0};$$

$$\frac{p_2(r-2)}{p_0(r)} = 0 = \frac{\alpha_2}{\alpha_0}; y_1 = \frac{x^{1/3}}{5+x}; y_2 = \frac{x^{-1/3}}{5+x}.$$

$$7.5.66. \quad p_0(r) = (2r-3)(2r-1); p_1(r) = 3(2r-1)(2r+1); p_2(r) = (2r+1)(2r+3);$$

$$\frac{p_1(r-1)}{p_0(r)} = 3 = \frac{\alpha_1}{\alpha_0}; \frac{p_2(r-2)}{p_0(r)} = 1 = \frac{\alpha_2}{\alpha_0}; y_1 = \frac{x^{1/2}}{1+3x+x^2}; y_2 = \frac{x^{3/2}}{1+3x+x^2}.$$

$$7.5.68. \quad p_0(r) = 3(r-1)(4r-1); p_1(r) = 2r(4r+3); p_2(r) = (r+1)(4r+7); \frac{p_1(r-1)}{p_0(r)} = \frac{2}{3} = \frac{\alpha_1}{\alpha_0};$$

$$\frac{p_2(r-2)}{p_0(r)} = \frac{1}{3} = \frac{\alpha_2}{\alpha_0}; y_1 = \frac{x}{3+2x+x^2}; y_2 = \frac{x^{1/4}}{3+2x+x^2}.$$

7.6 The Method of Frobenius II

$$7.6.2. \quad p_0(r) = (r+1)^2; p_1(r) = (r+2)(r+3); p_2(r) = (r+3)(2r-1);$$

$$a_1(r) = -\frac{r+3}{r+2}; a_n(r) = -\frac{n+r+2}{n+r+1} a_{n-1}(r) - \frac{2n+2r-5}{n+r+1} a_{n-2}(r), \quad n \geq 2.$$

$$a'_1(r) = \frac{1}{(r+2)^2}; a'_n(r) =$$

$$-\frac{n+r+2}{n+r+1} a'_{n-1}(r) - \frac{2n+2r-5}{n+r+1} a'_{n-2}(r) + \frac{1}{(n+r+1)^2} a_{n-1}(r) - \frac{7}{(n+r+1)^2} a_{n-2}(r),$$

$$n \geq 2.$$

$$r_1 = -1; a_1(-1) = -2; a_n(-1) = -\frac{n+1}{n} a_{n-1}(-1) - \frac{2n-7}{n} a_{n-2}(-1), \quad n \geq 2;$$

$$y_1 = x^{-1} \left(1 - 2x + \frac{9}{2}x^2 - \frac{20}{3}x^3 + \dots \right);$$

$$a'_1(-1) = 1; a'_n(-1) = -\frac{n+1}{n}a_{n-1}(-1) - \frac{2n-7}{n}a_{n-2}(-1) + \frac{1}{n^2}a_{n-1}(-1) - \frac{7}{n^2}a_{n-2}(-1),$$

$$n \geq 2;$$

$$y_2 = y_1 \ln x + 1 - \frac{15}{4}x + \frac{133}{18}x^2 + \dots$$

$$7.6.4. \quad p_0(r) = (2r-1)^2; p_1(r) = (2r+1)(2r+3); p_2(r) = (2r+1)(2r+3);$$

$$a_1(r) = -\frac{2r+3}{2r+1}; a_n(r) = -\frac{(2n+2r+1)a_{n-1}(r) - (2n+2r-3)a_{n-2}(r)}{2n+2r-1}, n \geq 2.$$

$$a'_1(r) = \frac{4}{(2r+1)^2};$$

$$a'_n(r) = -\frac{(2n+2r+1)a'_{n-1}(r) - (2n+2r-3)a'_{n-2}(r)}{2n+2r-1} + \frac{4(a_{n-1}(r) - a_{n-2}(r))}{(2n+2r-1)^2}; n \geq 2.$$

$$r_1 = 1/2; a_1(1/2) = -2; a_n(1/2) = -\frac{(n+1)a_{n-1}(1/2) + (n-1)a_{n-2}(1/2)}{n}; n \geq 2;$$

$$y_1 = x^{1/2} \left(1 - 2x + \frac{5}{2}x^2 - 2x^3 + \dots \right);$$

$$a'_1(1/2) = 1; a'_n(1/2) = -\frac{(n+1)a'_{n-1}(1/2) + (n-1)a'_{n-2}(1/2)}{n} + \frac{a_{n-1}(1/2) - a_{n-2}(1/2)}{n^2},$$

$$n \geq 2;$$

$$y_2 = y_1 \ln x + x^{3/2} \left(1 - \frac{9}{4}x + \frac{17}{6}x^2 + \dots \right).$$

$$7.6.6. \quad p_0(r) = (3r+1)^2; p_1(r) = 3(3r+4); p_2(r) = -2(3r+7);$$

$$a_1(r) = -\frac{3}{3r+4}; a_n(r) = \frac{-3a_{n-1}(r) + 2a_{n-2}(r)}{3n+3r+1}; n \geq 2;$$

$$a'_1(r) = \frac{9}{(3r+4)^2}; a'_n(r) = \frac{-3a'_{n-1}(r) + 2a'_{n-2}(r)}{3n+3r+1} + \frac{9a_{n-1}(r) - 6a_{n-2}(r)}{(3n+3r+1)^2}; n \geq 2.$$

$$r_1 = -1/3; a_1(-1/3) = -1; a_n(-1/3) = \frac{-3a_{n-1}(-1/3) + 2a_{n-2}(-1/3)}{3n}, n \geq 2;$$

$$y_1 = x^{-1/3} \left(1 - x + \frac{5}{6}x^2 - \frac{1}{2}x^3 + \dots \right);$$

$$a'_1(-1/3) = 1; a'_n(-1/3) = \frac{-3a'_{n-1}(r) + 2a'_{n-2}(r)}{3n} + \frac{3a_{n-1}(r) - 2a_{n-2}(r)}{3n^2}; n \geq 2;$$

$$y_2 = y_1 \ln x + x^{2/3} \left(1 - \frac{11}{12}x + \frac{25}{36}x^2 + \dots \right).$$

$$7.6.8. \quad p_0(r) = (r+2)^2; p_1(r) = 2(r+3)^2; p_2(r) = 3(r+4);$$

$$a_1(r) = -2; a_n(r) = -2a_{n-1}(r) - \frac{3a_{n-2}(r)}{n+r+2}; n \geq 2;$$

$$a'_1(r) = 0; a'_n(r) = -2a'_{n-1}(r) - \frac{3a'_{n-2}(r)}{n+r+2} + \frac{3a_{n-2}(r)}{(n+r+2)^2}, n \geq 2.$$

$$r_1 = -2; a_1(-2) = -2; a_n(-2) = -2a_{n-1}(-2) - \frac{3a_{n-2}(-2)}{n}, n \geq 2;$$

$$y_1 = x^{-2} \left(1 - 2x + \frac{5}{2}x^2 - 3x^3 + \dots \right);$$

$$a'_1(-2) = 0; a'_n(-2) = -2a_{n-1}(-2) - \frac{3a_{n-2}(-2)}{n} + \frac{3a_{n-2}(-2)}{n^2}; n \geq 2;$$

$$y_2 = y_1 \ln x + \frac{3}{4} - \frac{13}{6}x + \dots.$$

$$7.6.10. p_0(r) = (4r+1)^2; p_1(r) = 4r+5; p_2(r) = 2(4r+9);$$

$$a_1(r) = -\frac{1}{4r+5}; a_n(r) = -\frac{a_{n-1}(r) + 2a_{n-2}(r)}{4n+4r+1}; n \geq 2;$$

$$a'_1(r) = \frac{4}{(4r+5)^2}; a'_n(r) = -\frac{a'_{n-1}(r) + 2a'_{n-2}(r)}{4n+4r+1} + \frac{4a_{n-1}(r) + 8a_{n-2}(r)}{(4n+4r+1)^2}; n \geq 2.$$

$$r_1 = -1/4; a_1(-1/4) = -1/4; a_n(-1/4) = -\frac{a_{n-1}(-1/4) + 2a_{n-2}(-1/4)}{4n}; n \geq 2;$$

$$y_1 = x^{-1/4} \left(1 - \frac{1}{4}x - \frac{7}{32}x^2 + \frac{23}{384}x^3 + \dots \right);$$

$$a'_1(-1/4) = 1/4; a'_n(-1/4) = -\frac{a'_{n-1}(-1/4) + 2a'_{n-2}(-1/4)}{4n} + \frac{a_{n-1}(-1/4) + 2a_{n-2}(-1/4)}{4n^2}; n \geq 2;$$

$$y_2 = y_1 \ln x + x^{3/4} \left(\frac{1}{4} + \frac{5}{64}x - \frac{157}{2304}x^2 + \dots \right).$$

$$7.6.12. p_0(r) = (2r-1)^2; p_1(r) = 4;$$

$$a_n(r) = -\frac{4}{(2n+2r-1)^2} a_{n-1}(r);$$

$$a_n(r) = \frac{(-4)^n}{\prod_{j=1}^n (2j+2r-1)}.$$

$$\text{By logarithmic differentiation, } a'_n(r) = a_n(r) \sum_{j=1}^n \frac{2}{2j+2r-1};$$

$$r_1 = 1/2; a_n(1/2) = \frac{(-1)^n}{(n!)^2};$$

$$a'_n(1/2) = a_n(1/2) \left(-2 \sum_{j=1}^n \frac{1}{j} \right);$$

$$y_1 = x^{1/2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} x^n;$$

$$y_2 = y_1 \ln x - 2x^{1/2} \sum_{n=1}^{\infty} \frac{(-1)^n}{(n!)^2} \left(\sum_{j=1}^n \frac{1}{j} \right) x^n;$$

$$7.6.14. p_0(r) = (r-2)^2; p_1(r) = r^2; a_n(r) = -\frac{(n+r-1)^2}{(n+r-2)^2} a_{n-1}(r); a_n(r) = (-1)^n \frac{(n+r-1)^2}{(r-1)^2};$$

$$a'_n(r) = (-1)^{n+1} \frac{2n(r+n-1)}{(r-1)^3}; r_1 = 2; a_n(2) = (-1)^n (n+1)^2; a'_n(2) = (-1)^{n+1} 2n(n+1);$$

$$y_1 = x^2 \sum_{n=0}^{\infty} (-1)^n (n+1)^2 x^n; \quad y_2 = y_1 \ln x - 2x^2 \sum_{n=1}^{\infty} (-1)^n n(n+1) x^n.$$

$$7.6.16. \quad p_0(r) = (5r-1)^2; \quad p_1(r) = r+1;$$

$$a_n(r) = -\frac{(n+r)}{(5n+5r-1)^2} a_{n-1}(r);$$

$$a_n(r) = (-1)^n \prod_{j=1}^n \frac{(j+r)}{(5j+5r-1)^2};$$

By logarithmic differentiation,

$$a'_n(r) = -a_n(r) \sum_{j=1}^n \frac{(5j+5r+1)}{(j+r)(5j+5r-1)};$$

$$r_1 = 1/5; \quad a_n(1/5) = (-1)^n \prod_{j=1}^n \frac{(5j+1)}{125^n (n!)^2};$$

$$a'_n(1/5) = a_n(1/5) \sum_{j=1}^n \frac{5j+2}{j(5j+1)};$$

$$y_1 = x^{1/5} \sum_{n=0}^{\infty} \frac{(-1)^n \prod_{j=1}^n (5j+1)}{125^n (n!)^2} x^n;$$

$$y_2 = y_1 \ln x - x^{1/5} \sum_{n=1}^{\infty} \frac{(-1)^n \prod_{j=1}^n (5j+1)}{125^n (n!)^2} \left(\sum_{j=1}^n \frac{5j+2}{j(5j+1)} \right) x^n.$$

$$7.6.18. \quad p_0(r) = (3r-1)^2; \quad p_1(r) = (2r-1)^2;$$

$$a_n(r) = -\frac{(2n+2r-3)^2}{(3n+3r-1)^2} a_{n-1}(r);$$

$$a_n(r) = (-1)^n \prod_{j=1}^n \frac{(2j+2r-3)^2}{(3j+3r-1)^2};$$

By logarithmic differentiation,

$$a'_n(r) = 14a_n(r) \sum_{j=1}^n \frac{1}{(2j+2r-3)(3j+3r-1)};$$

$$r_1 = 1/3; \quad a_n(1/3) = \frac{(-1)^n \prod_{j=1}^n (6j-7)^2}{81^n (n!)^2};$$

$$a'_n(1/3) = 14a_n(1/3) \sum_{j=1}^n \frac{1}{j(6j-7)};$$

$$y_1 = x^{1/3} \sum_{n=0}^{\infty} \frac{(-1)^n \prod_{j=1}^n (6j-7)^2}{81^n (n!)^2} x^n;$$

$$y_2 = y_1 \ln x + 14x^{1/3} \sum_{n=1}^{\infty} \frac{(-1)^n \prod_{j=1}^n (6j-7)^2}{81^n (n!)^2} \left(\sum_{j=1}^n \frac{1}{j(6j-7)} \right) x^n.$$

$$7.6.20. \quad p_0(r) = (r+1)^2; \quad p_1(r) = -2(r+2)(2r+3);$$

$$a_n(r) = \frac{2(2n+2r+1)}{n+r+1} a_{n-1}(r), \quad n \geq 1; \quad a_n(r) = 2^n \prod_{j=1}^n \frac{2j+2r+1}{j+r+1};$$

By logarithmic differentiation,

$$a'_n(r) = a_n(r) \sum_{j=1}^n \frac{1}{(j+r+1)(2j+2r+1)};$$

$$r_1 = -1; \quad a_n(-1) = \frac{2^n \prod_{j=1}^n (2j-1)}{n!};$$

$$a'_n(-1) = a_n(-1) \sum_{j=1}^n \frac{1}{j(2j-1)};$$

$$y_1 = \frac{1}{x} \sum_{n=0}^{\infty} \frac{2^n \prod_{j=1}^n (2j-1)}{n!} x^n;$$

$$y_2 = y_1 \ln x + \frac{1}{x} \sum_{n=1}^{\infty} \frac{2^n \prod_{j=1}^n (2j-1)}{n!} \left(\sum_{j=1}^n \frac{1}{j(2j-1)} \right) x^n.$$

$$7.6.22. \quad p_0(r) = 2(r-2)^2; \quad p_1(r) = (r-1)(2r+1);$$

$$a_n(r) = -\frac{2n+2r-1}{2(n+r-2)} a_{n-1}(r);$$

$$a_n(r) = \frac{(-1)^n}{2^n} \prod_{j=1}^n \frac{2j+2r-1}{j+r-2};$$

By logarithmic differentiation,

$$a'_n(r) = -3a_n(r) \sum_{j=1}^n \frac{1}{(j+r-2)(2j+2r-1)};$$

$$r_1 = 2; \quad a_n(2) = \frac{(-1)^n \prod_{j=1}^n (2j+3)}{2^n n!};$$

$$a'_n(2) = -3a_n(2) \sum_{j=1}^n \frac{1}{j(2j+3)};$$

$$y_1 = x^2 \sum_{n=0}^{\infty} \frac{(-1)^n \prod_{j=1}^n (2j+3)}{2^n n!} x^n;$$

$$y_2 = y_1 \ln x - 3x^2 \sum_{n=0}^{\infty} \frac{(-1)^n \prod_{j=1}^n (2j+3)}{2^n n!} \left(\sum_{j=1}^n \frac{1}{j(2j+3)} \right) x^n.$$

$$7.6.24. \quad p_0(r) = (r-3)^2; \quad p_1(r) = -2(r-1)(r+2);$$

$$a_n(r) = \frac{2(n+r-2)(n+r+1)}{(n+r-3)^2} a_{n-1}(r);$$

$$a'_n(r) = \frac{2(n+r-2)(n+r+1)}{(n+r-3)^2} a'_{n-1}(r) - \frac{2(5n+5r-7)}{(n+r-3)^3} a_{n-1}(r);$$

$$\begin{aligned}
r_1 &= 3; a_n(3) = \frac{2(n+1)(n+4)}{n^2} a_{n-1}(3); \\
y_1 &= x^3(1 + 20x + 180x^2 + 1120x^3 + \dots); \\
a'_n(3) &= \frac{2(n+1)(n+4)}{n^2} a'_{n-1}(3) - \frac{2(5n+8)}{n^3} a_{n-1}(3); \\
y_2 &= y_1 \ln x - x^4 \left(26 + 324x + \frac{6968}{3}x^2 + \dots \right)
\end{aligned}$$

$$7.6.26. \quad p_0(r) = r^2; p_1(r) = r^2 + r + 1;$$

$$\begin{aligned}
a_n(r) &= -\frac{(n^2 + n(2r-1) + r^2 - r + 1)}{(n+r)^2} a_{n-1}(r); \\
a'_n(r) &= -\frac{(n^2 + n(2r-1) + r^2 - r + 1)}{(n+r)^2} a'_{n-1}(r) - \frac{(n+r-2)}{(n+r)^3} a_{n-1}(r); \\
r_1 &= 0; a_n(0) = -\frac{(n^2 - n + 1)}{n^2} a_{n-1}(0); \\
y_1 &= 1 - x + \frac{3}{4}x^2 - \frac{7}{12}x^3 + \dots; \\
a'_n(0) &= -\frac{(n^2 - n + 1)}{n^2} a'_{n-1}(0) - \frac{(n-2)}{n^3} a_{n-1}(0); \\
y_2 &= y_1 \ln x + x \left(1 - \frac{3}{4}x + \frac{5}{9}x^2 + \dots \right).
\end{aligned}$$

$$7.6.28. \quad p_0(r) = (r-1)^2; p_2(r) = r+1;$$

$$a_{2m}(r) = -\frac{1}{2m+r-1} a_{2m-2}(r), \quad n \geq 1; \quad a_{2m}(r) = \frac{(-1)^m}{\prod_{j=1}^m (2j+r-1)};$$

By logarithmic differentiation,

$$\begin{aligned}
a'_{2m}(r) &= a_{2m}(r) \sum_{j=1}^m; \\
r_1 &= 1; a_{2m}(1) = \frac{(-1)^m}{2^m m!}; \\
a'_{2m}(1) &= -\frac{1}{2} a_{2m}(1) \sum_{j=1}^m \frac{1}{j}; \\
y_1 &= x \sum_{m=0}^{\infty} \frac{(-1)^m}{2^m m!} x^{2m}; \\
y_2 &= y_1 \ln x - \frac{x}{2} \sum_{m=1}^{\infty} \frac{(-1)^m}{2^m m!} \left(\sum_{j=1}^m \frac{1}{j} \right) x^{2m}.
\end{aligned}$$

$$7.6.30. \quad p_0(r) = (2r-1)^2; p_2(r) = 2r+3;$$

$$a_{2m}(r) = -\frac{1}{4m+2r-1} a_{2m-2}(r);$$

$$a_{2m}(r) = \frac{(-1)^m}{\prod_{j=1}^m (4j + 2r - 1)};$$

By logarithmic differentiation,

$$a'_{2m}(r) = -2a_{2m}(r) \sum_{j=1}^m \frac{1}{4j + 2r - 1};$$

$$r_1 = 1/2; a_{2m}(1/2) = \frac{(-1)^m}{4^m m!};$$

$$a'_{2m}(1/2) = -\frac{1}{2}a_{2m}(1/2) \sum_{j=1}^m \frac{1}{j};$$

$$y_1 = x^{1/2} \sum_{m=0}^{\infty} \frac{(-1)^m}{4^m m!} x^{2m};$$

$$y_2 = y_1 \ln x - \frac{x^{1/2}}{2} \sum_{m=1}^{\infty} \frac{(-1)^m}{4^m m!} \left(\sum_{j=1}^m \frac{1}{j} \right) x^{2m}.$$

$$7.6.32. \quad p_0(r) = (2r - 1)^2; p_2(r) = (r + 1)(2r + 3); a_{2m}(r) = -\frac{2m + r - 1}{4m + 2r - 1} a_{2m-2}(r);$$

$$a_{2m}(r) = (-1)^m \prod_{j=1}^m \frac{2j + r - 1}{4j + 2r - 1};$$

By logarithmic differentiation,

$$a'_{2m}(r) = a_{2m}(r) \sum_{j=1}^m \frac{1}{(2j + r - 1)(4j + 2r - 1)};$$

$$r_1 = 1/2; a_{2m}(1/2) = \frac{(-1)^m \prod_{j=1}^m (4j - 1)}{8^m m!};$$

$$a'_{2m}(1/2) = a_{2m}(1/2) \sum_{j=1}^m \frac{1}{2j(4j - 1)};$$

$$y_1 = x^{1/2} \sum_{m=0}^{\infty} \frac{(-1)^m \prod_{j=1}^m (4j - 1)}{8^m m!} x^{2m};$$

$$y_2 = y_1 \ln x + \frac{x^{1/2}}{2} \sum_{m=1}^{\infty} \frac{(-1)^m \prod_{j=1}^m (4j - 1)}{8^m m!} \left(\sum_{j=1}^m \frac{1}{j(4j - 1)} \right) x^{2m}.$$

$$7.6.34. \quad p_0(r) = (4r + 1)^2; p_2(r) = (r - 1)(4r + 9);$$

$$a_{2m}(r) = -\frac{2m + r - 3}{8m + 4r + 1} a_{2m-2}(r);$$

$$a_{2m}(r) = (-1)^m \prod_{j=1}^m \frac{2j + r - 3}{8j + 4r + 1};$$

By logarithmic differentiation,

$$a'_{2m}(r) = a_{2m}(r) \sum_{j=1}^m \frac{13}{(2j + r - 3)(8j + 4r + 1)};$$

$$r_1 = -1/4; a_{2m}(-1/4) = \frac{(-1)^m \prod_{j=1}^m (8j-13)}{(32)^m m!};$$

$$a'_{2m}(-1/4) = a_{2m}(-1/4) \sum_{j=1}^m \frac{13}{2j(8j-13)};$$

$$y_1 = x^{-1/4} \sum_{m=0}^{\infty} \frac{(-1)^m \prod_{j=1}^m (8j-13)}{(32)^m m!} x^{2m};$$

$$y_2 = y_1 \ln x + \frac{13}{2} x^{-1/4} \sum_{m=1}^{\infty} \frac{(-1)^m \prod_{j=1}^m (8j-13)}{(32)^m m!} \left(\sum_{j=1}^m \frac{1}{j(8j-13)} \right) x^{2m}.$$

$$7.6.36. p_0(r) = (2r-1)^2; p_2(r) = 16r(r+1);$$

$$a_{2m}(r) = -\frac{16(2m+r-2)(2m+r-1)}{(4m+2r-1)^2} a_{2m-2}(r);$$

$$a_{2m}(r) = (-16)^m \prod_{j=1}^m \frac{(2j+r-2)(2j+r-1)}{(4j+2r-1)^2};$$

By logarithmic differentiation,

$$a'_{2m}(r) = a_{2m}(r) \sum_{j=1}^m \frac{8j+4r-5}{(2j+r-2)(2j+r-1)(4j+2r-1)};$$

$$r_1 = 1/2; a_{2m}(1/2) = \frac{(-1)^m \prod_{j=1}^m (4j-3)(4j-1)}{4^m (m!)^2};$$

$$a'_{2m}(1/2) = a_{2m}(1/2) \sum_{j=1}^m \frac{8j-3}{j(4j-3)(4j-1)};$$

$$y_1 = x^{1/2} \sum_{m=0}^{\infty} \frac{(-1)^m \prod_{j=1}^m (4j-3)(4j-1)}{4^m (m!)^2} x^{2m};$$

$$y_2 = y_1 \ln x + x^{1/2} \sum_{m=1}^{\infty} \frac{(-1)^m \prod_{j=1}^m (4j-3)(4j-1)}{4^m (m!)^2} \left(\sum_{j=1}^m \frac{8j-3}{j(4j-3)(4j-1)} \right) x^{2m}.$$

$$7.6.38. p_0(r) = (r+1)^2; p_2(r) = (r+3)(2r-1);$$

$$a_{2m}(r) = -\frac{4m+2r-5}{2m+r+1} a_{2m-2}(r);$$

$$a_{2m}(r) = (-1)^m \prod_{j=1}^m \frac{4j+2r-5}{2j+r+1};$$

By logarithmic differentiation,

$$a'_{2m}(r) = a_{2m}(r) \sum_{j=1}^m \frac{7}{(2j+r+1)(4j+2r-5)};$$

$$r_1 = -1; a_{2m}(-1) = \frac{(-1)^m \prod_{j=1}^m (4j-7)}{2^m m!};$$

$$\begin{aligned}
a'_{2m}(-1) &= a_{2m}(-1) \sum_{j=1}^m \frac{7}{2j(4j-7)}; \\
y_1 &= \frac{1}{x} \sum_{m=0}^{\infty} \frac{(-1)^m \prod_{j=1}^m (4j-7)}{2^m m!} x^{2m}; \\
y_2 &= y_1 \ln x + \frac{7}{2x} \sum_{m=1}^{\infty} \frac{(-1)^m \prod_{j=1}^m (4j-7)}{2^m m!} \left(\sum_{j=1}^m \frac{1}{j(4j-7)} \right) x^{2m}.
\end{aligned}$$

$$7.6.40. \quad p_0(r) = (r-1)^2; \quad p_2(r) = r+1;$$

$$\begin{aligned}
a_{2m}(r) &= -\frac{1}{2m+r-1} a_{2m-2}(r); \\
a'_{2m}(r) &= -\frac{1}{2m+r-1} a'_{2m-2}(r) + \frac{1}{(2m+r-1)^2} a_{2m-2}(r); \\
r_1 &= 1; \quad a_{2m}(1) = -\frac{1}{2m} a_{2m-2}(1); \\
y_1 &= x \left(1 - \frac{1}{2}x^2 + \frac{1}{8}x^4 - \frac{1}{48}x^6 + \dots \right); \\
a'_{2m}(1) &= -\frac{1}{2m} a'_{2m-2}(1) + \frac{1}{4m^2} a_{2m-2}(1), \quad m \geq 1; \\
y_2 &= y_1 \ln x + x^3 \left(\frac{1}{4} - \frac{3}{32}x^2 + \frac{11}{576}x^4 + \dots \right).
\end{aligned}$$

$$7.6.42. \quad p_0(r) = 2(r+3)^2; \quad p_2(r) = r^2 - 2r + 2;$$

$$\begin{aligned}
a_{2m}(r) &= -\frac{4m^2 + 4m(r-3) + r^2 - 6r + 10}{2(2m+r+3)^2} a_{2m-2}(r); \\
a'_{2m}(r) &= -\frac{4m^2 + 4m(r-3) + r^2 - 6r + 10}{2(2m+r+3)^2} a'_{2m-2}(r) - \frac{12m+6r-19}{(2m+r+3)^3} a_{2m-2}(r); \\
r_1 &= -3; \quad a_{2m}(-3) = -\frac{4m^2 - 24m + 37}{8m^2} a_{2m-2}(-3); \\
y_1 &= x^{-3} \left(1 - \frac{17}{8}x^2 + \frac{85}{256}x^4 - \frac{85}{18432}x^6 + \dots \right); \\
a'_{2m}(-3) &= -\frac{4m^2 - 24m + 37}{8m^2} a'_{2m-2}(-3) + \frac{37-12m}{8m^3} a_{2m-2}(-3), \quad m \geq 1; \\
y_2 &= y_1 \ln x + x^{-1} \left(\frac{25}{8} - \frac{471}{512}x^2 + \frac{1583}{110592}x^4 + \dots \right).
\end{aligned}$$

$$7.6.44. \quad p_0(r) = (r+1)^2; \quad p_1(r) = 2(2-r)(r+1); \quad r_1 = -1.$$

$$a_n(r) = \frac{2(n+r)(n+r-3)}{(n+r+1)^2} a_{n-1}(r); \quad a_n(r) = 2^n \prod_{j=1}^n \frac{(j+r)(j+r-3)}{(j+r+1)^2}, \quad n \geq 0.$$

Therefore, $a_n(-1) = 0$ if $n \geq 1$ and $y_1 = 1/x$. If $n \geq 4$, then $a_n(r) = (r+1)^2 b_n(r)$, where $b'_n(-1)$ exists; therefore $a'_n(-1) = 0$ if $n \geq 4$. For $r = 1, 2, 3$, $a_n(r) = (r+1)c_n(r)$, where $c_1(r) = \frac{2(r-2)}{(r+2)^2}$, $c_2(r) = \frac{4(r-2)(r-1)}{(r+2)(r+3)^2}$, $c_3(r) = \frac{8r(r-2)(r-1)}{(r+2)(r+3)(r+4)^2}$. Hence,

$$a'_1(-1) = c_1(-1) = -6, a'_2(-1) = c_2(-1) = 6, a'_3(-1) = c_3(-1) = -8/3, \text{ and} \\ y_2 = y_1 \ln x - 6 + 6x - \frac{8}{3}x^2.$$

$$7.6.46. p_0(r) = (r+1)^2; p_1(r) = -(r-1)(r+2); r_1 = -1.$$

$$a_n(r) = \frac{n+r-2}{n+r+1}a_{n-1}(r); a_n(r) = \prod_{j=1}^m \frac{j+r-2}{j+r+1}, n \geq 0. \text{ Therefore, } a_1(-1) = -2,$$

$$a_2(-1) = 1, \text{ and } a_n(-1) = 0 \text{ if } n \geq 3, \text{ so } y_1 = \frac{(x-1)^2}{x}.$$

$$a_1(r) = \frac{r-1}{r+2}, a'_1(r) = \frac{3}{(r+2)^2}, a'_1(-1) = 3; a_2(r) = \frac{r(r-1)}{(r+2)(r+3)},$$

$$a'_2(r) = \frac{6(r^2+2r-1)}{(r+2)^2(r+3)^2}, a'_2(-1) = -3; \text{ if } n \geq 3 \text{ } a_n(r) = (r+1)c_n(r) \text{ where}$$

$$c_n(r) = \frac{r(r-1)}{(n+r)(n+r-1)(n+r+1)}, \text{ so } a'_n(-1) = c_n(-1) = \frac{2}{n(n-2)(n-1)} \text{ and}$$

$$y_2 = y_1 \ln x + 3 - 3x + 2 \sum_{n=2}^{\infty} \frac{1}{n(n^2-1)} x^n.$$

$$7.6.48. p_0(r) = (r-2)^2; p_1(r) = -(r-5)(r-1); r_1 = 2.$$

$$a_n(r) = \frac{n+r-6}{n+r-2}a_{n-1}(r);$$

$$a_n(r) = \prod_{j=1}^m \frac{j+r-6}{j+r-2}, n \geq 0. \text{ Therefore, } a_1(2) = -3, a_2(2) = 3, a_3(2) = -1, \text{ and } a_n(2) = 0 \text{ if} \\ n \geq 4, \text{ so } y_1 = x^2(1-x)^3.$$

$$a_1(r) = \frac{r-5}{r-1}, a'_1(r) = \frac{4}{(r-1)^2}, a'_1(2) = 4;$$

$$a_2(r) = \frac{(r-5)(r-4)}{r(r-1)}, a'_2(r) = \frac{4(2r^2-10r+5)}{r^2(r-1)^2}, a'_2(2) = -7;$$

$$a_3(r) = \frac{(r-5)(r-4)(r-3)}{r(r-1)(r+1)}, a'_3(r) = \frac{12(r^4-8r^3+16r^2-5)}{r^2(r-1)^2(r+1)^2}, a'_3(2) = 11/3; \text{ if } n \geq 4, \text{ then}$$

$$a_n(r) = (r-2)c_n(r) \text{ where } c_n(r) = \frac{(r-5)(r-4)(r-3)}{(n+r-5)(n+r-4)(n+r-3)(n+r-2)}, \text{ so}$$

$$a'_n(2) = c_n(2) = -\frac{6}{n(n-2)(n^2-1)} \text{ and}$$

$$y_2 = y_1 \ln x + x^3 \left(4 - 7x + \frac{11}{3}x^2 - 6 \sum_{n=3}^{\infty} \frac{1}{n(n-2)(n^2-1)} x^n \right).$$

$$7.6.50. p_0(r) = (3r-1)^2; p_2(r) = 7-3r; r_1 = 1/3.$$

$$a_{2m}(r) = \frac{6m+3r-13}{(6m+3r-1)^2}a_{2m-2}(r);$$

$$a_{2m}(r) = \prod_{j=1}^m \frac{6j+3r-13}{(6j+3r-1)^2}, m \geq 0. \text{ Therefore, } a_2(1/3) = 1/6 \text{ and } a_{2m}(1/3) = 0 \text{ if } m \geq 2, \text{ so}$$

$$y_1 = x^{1/3} \left(1 - \frac{1}{6}x^2 \right).$$

$$\begin{aligned}
a_2(r) &= \frac{3r-7}{(3r+5)^2}; a'_2(r) = \frac{3(19-3r)}{(3r+5)^3}; a'_2(1/3) = 1/4. \text{ If } m \geq 2, \text{ then} \\
a_{2m}(r) &= (r-1/3)c_{2m}(r) \text{ where } c_{2m}(r) = \frac{3(3r-7)}{(6m+3r-7)(6m+3r-1)\prod_{j=1}^m(6j+3r-1)}, \text{ so} \\
a'_{2m}(1/3) &= c_{2m}(1/3) = -\frac{1}{12} \frac{1}{6^{m-1}(m-1)m!}, \text{ and} \\
y_2 &= y_1 \ln x + x^{7/3} \left(\frac{1}{4} - \frac{1}{12} \sum_{m=1}^{\infty} \frac{1}{6^m m(m+1)(m+1)!} x^{2m} \right).
\end{aligned}$$

7.6.52. $p_0(r) = (2r+1)^2$; $p_2(r) = 7-2r$; $r_1 = -1/2$.

$$\begin{aligned}
a_{2m}(r) &= \frac{4m+2r-11}{(4m+2r+1)^2} a_{2m-2}(r); \\
a_{2m}(r) &= \prod_{j=1}^m \frac{4j+2r-11}{(4j+2r+1)^2}, \quad m \geq 0. \text{ Therefore, } a_2(-1/2) = -1/2, \quad a_4(-1/2) = 1/32, \text{ and} \\
a_{2m}(-1/2) &= 0 \text{ if } m \geq 3, \text{ so } y_1 = x^{-1/2} \left(1 - \frac{1}{2}x^2 + \frac{1}{32}x^4 \right). \\
a_2(r) &= \frac{2r-7}{(2r+5)^2}, \quad a'_2(r) = \frac{2(19-2r)}{(2r+5)^3}, \quad a'_2(-1/2) = 5/8, \\
a_4(r) &= \frac{(2r-7)(2r-3)}{(2r+5)^2(2r+9)^2}, \quad a'_4(r) = -\frac{4(8r^3-60r^2-146r+519)}{(2r+5)^3(2r+9)^3}, \quad a'_4(-1/2) = -9/128; \text{ if} \\
m \geq 3, \text{ then } a_{2m}(r) &= (r+1/2)c_{2m}(r) \text{ where} \\
c_{2m}(r) &= \frac{2(2r-7)(2r-3)}{(4m+2r-7)(4m+2r-3)(4m+2r+1)\prod_{j=1}^m(4j+2r+1)}, \text{ so} \\
a'_{2m}(-1/2) &= c_{2m}(-1/2) = \frac{1}{4^m(m-2)(m-1)m!}, \text{ and} \\
y_2 &= y_1 \ln x + x^{3/2} \left(\frac{5}{8} - \frac{9}{128}x^2 + \sum_{m=2}^{\infty} \frac{1}{4^{m+1}(m-1)m(m+1)(m+1)!} x^{2m} \right).
\end{aligned}$$

7.6.54. (a) If $p_0(r) = \alpha_0(r-r_1)^2$, then (A) $a_n(r) = \frac{(-1)^n}{\alpha_0^n} \prod_{j=1}^n \frac{p_1(j+r-1)}{(j+r-r_1)^2}$. Therefore,

$$a_n(r_1) = \frac{(-1)^n}{\alpha_0^n(n!)^2} \prod_{j=1}^n p_1(j+r_1-1). \text{ Theorem 7.6.2 implies } Ly_1 = 0.$$

(b) From (A), $\ln |a_n(r)| = -n \ln |\alpha_0| + \sum_{j=1}^n (\ln |p_1(j+r-1)| - 2 \ln |j+r-r_1|)$, so

$$\begin{aligned}
a'_n(r) &= a_n(r) \sum_{j=1}^n \left(\frac{p'_1(j+r-1)}{p_1(j+r-1)} - \frac{2}{j+r-r_1} \right) \text{ and} \\
a'_n(r_1) &= a_n(r_1) \sum_{j=1}^n \left(\frac{p'_1(j+r_1-1)}{p_1(j+r_1-1)} - \frac{2}{j} \right). \text{ Theorem 7.6.2 implies that } Ly_2 = 0.
\end{aligned}$$

(c) Since $p_1(r) = \gamma_1$, y_1 and y_2 reduce to the stated forms. If $\gamma_1 = 0$, then $y_1 = x^{r_1}$ and $y_2 = x^{r_1} \ln x$, which are solutions of the Euler equation $\alpha_0 x^2 y'' + \beta_0 x y' + \gamma_0 y = 0$.

7.6.56. (a) $Ly_1 = p_0(r_1)x^{r_1} = 0$. Now use the fact that $p_0(j+r_1) = \alpha_0 j^2$, so $\prod_{j=1}^n p_0(j+r_1) = \alpha_0^n (n!)^2$.

(b) From Theorem 7.6.2, $y_2 = y_1 \ln x + x^{r_1} \sum_{n=1}^{\infty} a'_n(r_1)x^n$ is a second solution of $Ly = 0$.

$$\text{Since } a_n(r) = \frac{(-1)^n}{\alpha_0^n} \prod_{j=1}^n \frac{p_1(j+r-1)}{(j+r-r_1)^2}, \quad (\text{A})$$

$$\ln |a_n(r)| = -n \ln |\alpha_0| + \sum_{j=1}^n \ln |p_1(j+r-1)| - 2 \sum_{j=1}^n \ln |j+r-r_1|, \text{ provided that}$$

$p_1(j+r-1)$ and $j+r-r_1$ are nonzero for all positive integers j . Differentiating (A) and then setting $r = r_1$ yields $\frac{a'_n(r_1)}{a_n(r_1)} = \sum_{j=1}^n \frac{p'_1(j+r_1-1)}{p_2(j+r_1-1)} - 2 \sum_{j=1}^n \frac{1}{j}$, which implies the conclusion.

(c) In this case $p_1(r) = \gamma_1$ and $p'_1(r) = 0$, so $a_n(r_1) = \frac{(-1)^n}{(n!)^2} \left(\frac{\gamma_1}{\alpha_0} \right)^n$ and $J_n = -2 \sum_{j=1}^n \frac{1}{j}$. If $\gamma_1 = 0$, then $y_1 = x^{r_1}$ and $y_2 = x^{r_1} \ln x$, while the differential equation is an Euler equation with indicial polynomial $\alpha_0(r-r_1^2)$. See Theorem 7.4.3.

$$7.6.58. \quad p_0(r) = r^2; \quad p_1(r) = 1; \quad r_1 = 0. \quad a_{2m}(r) = -\frac{a_{2m-1}(r)}{(2m+r)^2}, \quad m \geq 1; \quad a_{2m}(r) = \frac{(-1)^m}{\prod_{j=1}^m (2j+r)^2},$$

$$m \geq 0. \text{ Therefore, } a_{2m}(0) = \frac{(-1)^m}{4^m (m!)^2}, \text{ so } y_1 = \sum_{m=0}^{\infty} \frac{(-1)^m}{4^m (m!)^2} x^{2m}.$$

$$\text{By logarithmic differentiation, } a'_{2m}(r) = -2a_{2m}(r) \sum_{j=1}^m \frac{1}{2m+r}, \text{ so } a'_{2m}(0) = -a_{2m}(0) \sum_{j=1}^m \frac{1}{j}$$

$$\text{and } y_2 = y_1 \ln x - \sum_{m=1}^{\infty} \frac{(-1)^m}{4^m (m!)^2} \left(\sum_{j=1}^m \frac{1}{j} \right) x^{2m}.$$

$$7.6.60. \quad p_0(r) = (2r-1)^2; \quad p_1(r) = (2r+1)^2; \quad p_2(r) = 0; \quad \frac{p_1(r-1)}{p_0(r)} = 1 = \frac{\alpha_1}{\alpha_0}; \quad \frac{p_2(r-2)}{p_0(r)} = 0 = \frac{\alpha_2}{\alpha_0};$$

$$y_1 = \frac{x^{1/2}}{1+x}; \quad y_2 = \frac{x^{1/2} \ln x}{1+x}.$$

$$7.6.62. \quad p_0(r) = 2(r-1)^2; \quad p_1(r) = 0; \quad p_2(r) = -(r+1)^2; \quad \frac{p_1(r-1)}{p_0(r)} = 0 = \frac{\alpha_1}{\alpha_0};$$

$$\frac{p_2(r-2)}{p_0(r)} = -1/2 = \frac{\alpha_2}{\alpha_0}; \quad y_1 = \frac{x}{2-x^2}; \quad y_2 = \frac{x \ln x}{2-x^2}.$$

$$7.6.64. \quad p_0(r) = 4(r-1)^2; \quad p_1(r) = 3r^2; \quad p_2(r) = 0; \quad \frac{p_1(r-1)}{p_0(r)} = 3/4 = \frac{\alpha_1}{\alpha_0}; \quad \frac{p_2(r-2)}{p_0(r)} = 0 = \frac{\alpha_2}{\alpha_0};$$

$$y_1 = \frac{x}{4+3x}; \quad y_2 = \frac{x \ln x}{4+3x}.$$

$$7.6.66. \quad p_0(r) = (r-1)^2; \quad p_1(r) = -2r^2; \quad p_2(r) = (r+1)^2; \quad \frac{p_1(r-1)}{p_0(r)} = -2 = \frac{\alpha_1}{\alpha_0};$$

$$\frac{p_2(r-2)}{p_0(r)} = 1 = \frac{\alpha_2}{\alpha_0}; \quad y_1 = \frac{x}{(1-x)^2}; \quad y_2 = \frac{x \ln x}{(1-x)^2}.$$

7.6.68. See the proofs of Theorems 7.6.1 and 7.6.2.

7.7 The Method of Frobenius III

7.7.2. $p_0(r) = r(r-1)$; $p_1(r) = 1$; $r_1 = 1$; $r_2 = 0$; $k = r_1 - r_2 = 1$;

$$a_n(r) = -\frac{1}{(n+r)(n+r-1)}a_{n-1}(r);$$

$$a_n(r) = \frac{(-1)^n}{\prod_{j=1}^n (j+r)(j+r-1)};$$

$$a_n(1) = \frac{(-1)^n}{n!(n+1)!};$$

$$y_1 = x \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+1)!} x^n;$$

$$z = 1; C = -p_1(0)a_0(0) = -1.$$

By logarithmic differentiation,

$$a'_n(r) = -a_n(r) \sum_{j=1}^n \frac{2n+2r-1}{(n+r)(n+r-1)};$$

$$a'_n(1) = -a_n(1) \sum_{j=1}^n \frac{2j+1}{j(j+1)};$$

$$y_2 = 1 - y_1 \ln x + x \sum_{n=1}^{\infty} \frac{(-1)^n}{n!(n+1)!} \left(\sum_{j=1}^n \frac{2j+1}{j(j+1)} \right) x^n.$$

7.7.4. $p_0(r) = r(r-1)$; $p_1(r) = r+1$; $r_1 = 1$; $r_2 = 0$; $k = r_1 - r_2 = 1$; $a_n(r) = -\frac{a_{n-1}(r)}{n+r-1}$;

$$a_n(r) = \frac{(-1)^n}{\prod_{j=1}^n (j+r-1)}; a_n(1) = \frac{(-1)^n}{n!}; y_1 = x \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^n = xe^{-x}; z = 1;$$

$$C = -p_1(0)a_0(0) = -1. \text{ By logarithmic differentiation, } a'_n(r) = -a_n(r) \sum_{j=1}^n \frac{1}{j+r-1};$$

$$a'_n(1) = -a_n(1) \sum_{j=1}^n \frac{1}{j}; y_2 = 1 - y_1 \ln x + x \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \left(\sum_{j=1}^n \frac{1}{j} \right) x^n.$$

7.7.6. $p_0(r) = (r-1)(r+2)$; $p_1(r) = r+3$; $r_1 = 1$; $r_2 = -2$; $k = r_1 - r_2 = 3$.

$$a_n(r) = -\frac{1}{n+r-1}a_{n-1}(r); a_n(r) = \frac{(-1)^n}{\prod_{j=1}^n (j+r-1)}; a_n(1) = \frac{(-1)^n}{n!};$$

$$y_1 = x \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^n = xe^{-x}; z = x^{-2} \left(1 + \frac{1}{2}x + \frac{1}{2}x^2 \right); C = -\frac{p_1(0)}{3}a_2(-2) = -1/2. \text{ By}$$

$$\text{logarithmic differentiation, } a'_n(r) = -a_n(r) \sum_{j=1}^n \frac{1}{j+r-1}; a'_n(1) = a_n(1) \sum_{j=1}^n \frac{1}{j};$$

$$y_2 = x^{-2} \left(1 + \frac{1}{2}x + \frac{1}{2}x^2 \right) - \frac{1}{2} \left(y_1 \ln x - x \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \left(\sum_{j=1}^n \frac{1}{j} \right) x^n \right);$$

7.7.8. $p_0(r) = (r+2)(r+7)$; $p_1(r) = 1$; $r_1 = -2$; $r_2 = -7$; $k = r_1 - r_2 = 5$;

$$a_n(r) = -\frac{a_{n-1}(r)}{(n+r+2)(n+r+7)}; a_n(r) = \frac{(-1)^n}{\prod_{j=1}^n (j+r+2)(j+r+7)};$$

$$a_n(-2) = 120 \prod_{j=1}^n \frac{(-1)^n}{n!(n+5)!}; y_1 = \frac{120}{x^2} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+5)!} x^n;$$

$$z = x^{-7} \left(1 + \frac{1}{4}x + \frac{1}{24}x^2 + \frac{1}{144}x^3 + \frac{1}{576}x^4 \right); C = -\frac{p_1(-3)}{5}a_4(-7) = -1/2880. \text{ By}$$

$$\text{logarithmic differentiation, } a'_n(r) = -a_n(r) \sum_{j=1}^n -\frac{2j+2r+9}{(j+r+2)(j+r+7)};$$

$$a'_n(-2) = -a_n(-2) \sum_{j=1}^n \frac{2j+5}{j(j+5)};$$

$$y_2 = x^{-7} \left(1 + \frac{1}{4}x + \frac{1}{24}x^2 + \frac{1}{144}x^3 + \frac{1}{576}x^4 \right) - \frac{1}{2880} \left(y_1 \ln x - \frac{120}{x^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n!(n+5)!} \left(\sum_{j=1}^n \frac{2j+5}{j(j+5)} \right) x^n \right).$$

7.7.10. $p_0(r) = r(r-4)$; $p_1(r) = (r-6)(r-5)$; $r_1 = 4$; $r_2 = 0$; $k = r_1 - r_2 = 4$;

$$a_n(r) = -\frac{(n+r-7)(n+r-6)}{(n+r)(n+r-4)} a_{n-1}(r); a_n(r) = (-1)^n \prod_{j=1}^n \frac{(j+r-7)(j+r-6)}{(j+r)(j+r-4)}. \text{ Setting}$$

$$r = 4 \text{ yields } y_1 = x^4 \left(1 - \frac{2}{5}x \right). z = 1 + 10x + 50x^2 + 200x^3; C = -\frac{p_1(3)}{4}a_3(0) = -300.$$

$$a_1(r) = -\frac{(r-6)(r-5)}{(r-3)(r+1)}; a'_1(r) = -\frac{3(3r^2-22r+31)}{(r-3)^2(r+1)^2}; a'_1(4) = 27/25. a_2(r) = (r-4)c_2(r),$$

$$\text{with } c_2(r) = \frac{(r-6)(r-5)^2}{(r-3)(r-2)(r+1)(r+2)}, \text{ so } a'_2(4) = c_2(4) = -1/30. \text{ If } n \geq 3, \text{ then}$$

$$a_n(r) = (r-4)^2 b_n(r) \text{ where } b'_n(4) \text{ exists, so } a'_n(4) = 0 \text{ and}$$

$$y_2 = 1 + 10x + 50x^2 + 200x^3 - 300 \left(y_1 \ln x + \frac{27}{25}x^5 - \frac{1}{30}x^6 \right).$$

7.7.12. $p_0(r) = (r-2)(r+2)$; $p_1(r) = -2r-1$; $r_1 = 2$; $r_2 = -2$; $k = r_1 - r_2 = 4$;

$$a_n(r) = \frac{2j+2r-1}{(j+r-2)(j+r+2)} a_{n-1}(r); a_n(r) = \prod_{j=1}^n \frac{2n+2r-1}{(n+r-2)(n+r+2)};$$

$$a_n(2) = \frac{1}{n!} \left(\prod_{j=1}^n \frac{2j+3}{j+4} \right); y_1 = x^2 \sum_{n=0}^{\infty} \frac{1}{n!} \left(\prod_{j=1}^n \frac{2j+3}{j+4} \right) x^n; z = x^{-2} \left(1 + x + \frac{1}{4}x^2 - \frac{1}{12}x^3 \right);$$

$$C = -\frac{p_1(1)}{4}a_3(-2) = -1/16. \text{ By logarithmic differentiation,}$$

$$a'_n(r) = -2a_n(r) \sum_{j=1}^n \frac{j^2 + j(2r-1) + r^2 - r + 4}{(j+r-2)(j+r+2)(2j+2r-1)};$$

$$a'_n(2) = -2a_n(2) \sum_{j=1}^n \frac{(j^2 + 3j + 6)}{j(j+4)(2j+3)}; y_2 =$$

$$x^{-2} \left(1 + x + \frac{1}{4}x^2 - \frac{1}{12}x^3 \right) - \frac{1}{16}y_1 \ln x + \frac{x^2}{8} \sum_{n=1}^{\infty} \frac{1}{n!} \left(\prod_{j=1}^n \frac{2j+3}{j+4} \right) \left(\sum_{j=1}^n \frac{(j^2 + 3j + 6)}{j(j+4)(2j+3)} \right) x^n.$$

$$\begin{aligned} 7.7.14. \quad & p_0(r) = (r+1)(r+7); p_1(r) = (r+5)(2r+1); r_1 = -1; r_2 = -7; k = r_1 - r_2 = 6; \\ & a_n(r) = -\frac{(n+r+4)(2n+2r-1)}{(n+r+1)(n+r+7)} a_{n-1}(r); a_n(r) = (-1)^n \prod_{j=1}^n \frac{(j+r+4)(2j+2r-1)}{(j+r+1)(j+r+7)}; \\ & a_n(-1) = \frac{(-1)^n}{n!} \left(\prod_{j=1}^n \frac{(j+3)(2j-3)}{j+6} \right); y_1 = \frac{1}{x} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left(\prod_{j=1}^n \frac{(j+3)(2j-3)}{j+6} \right) x^n; \\ & z = x^{-7} \left(1 + \frac{26}{5}x + \frac{143}{20}x^2 \right); C = -\frac{p_1(-2)}{6} a_5(-7) = 0; y_2 = x^{-7} \left(1 + \frac{26}{5}x + \frac{143}{20}x^2 \right). \end{aligned}$$

$$\begin{aligned} 7.7.16. \quad & p_0(r) = (3r-10)(3r+2); p_1(r) = r(3r-4); r_1 = 10/3; r_2 = -2/3; k = r_1 - r_2 = 4; \\ & a_n(r) = -\frac{(n+r-1)(3n+3r-7)}{(3n+3r-10)(3n+3r+2)} a_{n-1}(r); \\ & a_n(r) = (-1)^n \prod_{j=1}^n \frac{(j+r-1)(3j+3r-7)}{(3j+3r-10)(3j+3r+2)}; a_n(10/3) = \frac{(-1)^n(n+1)}{9^n} \left(\prod_{j=1}^n \frac{3j+7}{j+4} \right); \\ & y_1 = x^{10/3} \sum_{n=0}^{\infty} \frac{(-1)^n(n+1)}{9^n} \left(\prod_{j=1}^n \frac{3j+7}{j+4} \right) x^n; z = x^{-2/3} \left(1 + \frac{4}{27}x - \frac{1}{243}x^2 \right); \\ & C = -\frac{p_1(7/3)}{36} a_3(-2/3) = 0; y_2 = x^{-2/3} \left(1 + \frac{4}{27}x - \frac{1}{243}x^2 \right). \end{aligned}$$

$$\begin{aligned} 7.7.18. \quad & p_0(r) = (r-3)(r+2); p_1(r) = (r+1)^2; r_1 = 3; r_2 = -2; k = r_1 - r_2 = 5; \\ & a_n(r) = -\frac{(n+r)^2}{(n+r-3)(n+r+2)} a_{n-1}(r); a_n(r) = (-1)^n \prod_{j=1}^n \frac{(j+r)^2}{(j+r-3)(j+r+2)}; \\ & a_n(3) = \frac{(-1)^n}{n!} \left(\prod_{j=1}^n \frac{(j+3)^2}{j+5} \right); y_1 = x^3 \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left(\prod_{j=1}^n \frac{(j+3)^2}{j+5} \right) x^n; z = x^{-2} \left(1 + \frac{1}{4}x \right); \\ & C = -\frac{p_1(2)}{5} a_4(-2) = 0; y_2 = x^{-2} \left(1 + \frac{1}{4}x \right). \end{aligned}$$

$$\begin{aligned} 7.7.20. \quad & p_0(r) = (r-6)(r-1); p_1(r) = (r-8)(r-4); r_1 = 6; r_2 = 1; k = r_1 - r_2 = 5; \\ & a_n(r) = -\frac{(n+r-9)(n+r-5)}{(n+r-6)(n+r-1)} a_{n-1}(r); y_1 = x^6 \left(1 + \frac{2}{3}x + \frac{1}{7}x^2 \right); \\ & z = x \left(1 + \frac{21}{4}x + \frac{21}{2}x^2 + \frac{35}{4}x^3 \right); C = -\frac{p_1(5)}{6} a_5(1) = 0; y_2 = x \left(1 + \frac{21}{4}x + \frac{21}{2}x^2 + \frac{35}{4}x^3 \right). \end{aligned}$$

$$\begin{aligned} 7.7.22. \quad & p_0(r) = r(r-10); p_1(r) = 2(r-6)(r+1); r_1 = 10; r_2 = 0; k = r_1 - r_2 = 10; \\ & a_n(r) = -\frac{2(n+r-7)}{n+r-10} a_{n-1}(r); a_n(r) = (-2)^n \frac{(n+r-9)(n+r-8)(n+r-7)}{(r-9)(r-8)(r-7)}; \\ & a_n(10) = \frac{(-1)^n 2^n (n+1)(n+2)(n+3)}{6}; y_1 = \frac{x^{10}}{6} \sum_{n=0}^{\infty} (-1)^n 2^n (n+1)(n+2)(n+3) x^n; \end{aligned}$$

$$z = \left(1 - \frac{4}{3}x + \frac{5}{3}x^2 - \frac{40}{21}x^3 + \frac{40}{21}x^4 - \frac{32}{21}x^5 + \frac{16}{21}x^6\right); C = -\frac{p_1(9)}{10}a_9(0) = 0;$$

$$y_2 = \left(1 - \frac{4}{3}x + \frac{5}{3}x^2 - \frac{40}{21}x^3 + \frac{40}{21}x^4 - \frac{32}{21}x^5 + \frac{16}{21}x^6\right).$$

Note: in the solutions to Exercises 7.7.23–7.7.40, $z = x^{r_2} \sum_{m=0}^{k-1} a_{2m}(r_2)x^{2m}$.

7.7.24. $p_0(r) = (r-6)(r-2)$; $p_2(r) = r$; $r_1 = 6$; $r_2 = 2$; $k = (r_1 - r_2)/2 = 2$;

$$a_{2m}(r) = -\frac{a_{2m-2}(r)}{2m+r-6}; a_{2m}(r) = \frac{(-1)^m}{\prod_{j=1}^m (2j+r-6)}; a_{2m}(6) = \frac{(-1)^m}{2^m m!};$$

$$y_1 = x^6 \sum_{m=0}^{\infty} \frac{(-1)^m}{2^m m!} x^{2m} = x^6 e^{-x^2/2}; z = x^2 \left(1 + \frac{1}{2}x^2\right); C = -\frac{p_2(4)}{4}a_2(2) = -1/2. \text{ By}$$

$$\text{logarithmic differentiation, } a'_{2m}(r) = -a_{2m}(r) \sum_{j=1}^m \frac{1}{2j+r-6}; a'_{2m}(6) = -a_{2m}(6) \sum_{j=1}^m \frac{1}{2j};$$

$$y_2 = x^2 \left(1 + \frac{1}{2}x^2\right) - \frac{1}{2}y_1 \ln x + \frac{x^6}{4} \sum_{m=1}^{\infty} \frac{(-1)^m}{2^m m!} \left(\sum_{j=1}^m \frac{1}{j}\right) x^{2m}.$$

7.7.26. $p_0(r) = (r-1)(r+1)$; $p_2(r) = 2r+10$; $r_1 = 1$; $r_2 = -1$; $k = (r_1 - r_2)/2 = 1$;

$$a_{2m}(r) = -\frac{2(2m+r+3)}{(2m+r-1)(2m+r+1)} a_{2m-2}(r); a_{2m}(r) = (-2)^m \prod_{j=1}^m \frac{2j+r+3}{(2j+r-1)(2j+r+1)};$$

$$a_{2m}(1) = \frac{(-1)^m(m+2)}{2^m m!}; y_1 = \frac{x}{2} \sum_{m=0}^{\infty} \frac{(-1)^m(m+2)}{m!} x^{2m}; z = x^{-1};$$

$$C = -\frac{p_2(-1)}{2}a_0(-1) = -4. \text{ By logarithmic differentiation,}$$

$$a'_{2m}(r) = -a_{2m}(r) \sum_{j=1}^m \frac{(4j^2 + 4j(r+3) + r^2 + 6r + 1)}{(2j+r-1)(2j+r+1)(2j+r+3)};$$

$$a'_{2m}(1) = -a_{2m}(1) \sum_{j=1}^m \frac{j^2 + 4j + 2}{2j(j+1)(j+2)};$$

$$y_2 = x^{-1} - 4y_1 \ln x + x \sum_{m=1}^{\infty} \frac{(-1)^m(m+2)}{m!} \left(\sum_{j=1}^m \frac{j^2 + 4j + 2}{j(j+1)(j+2)}\right) x^{2m}.$$

7.7.28. $p_0(r) = (2r+1)(2r+5)$; $p_2(r) = 2r+3$; $r_1 = -1/2$; $r_2 = -5/2$; $k = (r_1 - r_2)/2 = 1$;

$$a_{2m}(r) = -\frac{(4m+2r-1)}{(4m+2r+1)(4m+2r+5)} a_{2m-2}(r);$$

$$a_{2m}(r) = (-1)^m \prod_{j=1}^m \frac{(4j+2r-1)}{(4j+2r+1)(4j+2r+5)}; a_{2m}(-1/2) = \frac{(-1)^m \prod_{j=1}^m (2j-1)}{8^m m!(m+1)!};$$

$$y_1 = x^{-1/2} \sum_{m=0}^{\infty} \frac{(-1)^m \prod_{j=1}^m (2j-1)}{8^m m!(m+1)!} x^{2m}; z = x^{-5/2}; C = -\frac{p_2(-5/2)}{8}a_0(-5/2) = 1/4. \text{ By}$$

logarithmic differentiation,

$$a'_{2m}(r) = -2a_{2m}(r) \sum_{j=1}^m \frac{(16j^2 + 8j(2r-1) + 4r^2 - 4r - 11)}{(4j+2r-1)(4j+2r+1)(4j+2r+5)};$$

$$a'_{2m}(-1/2) = -a_{2m}(-1/2) \sum_{j=1}^m \frac{2j^2 - 2j - 1}{2j(j+1)(2j-1)};$$

$$y_2 = x^{-5/2} + \frac{1}{4}y_1 \ln x - x^{-1/2} \sum_{m=1}^{\infty} \frac{(-1)^m \prod_{j=1}^m (2j-1)}{8^{m+1} m! (m+1)!} \left(\sum_{j=1}^m \frac{2j^2 - 2j - 1}{j(j+1)(2j-1)} \right) x^{2m}.$$

7.7.30. $p_0(r) = (r-2)(r+2)$; $p_2(r) = -2(r+4)$; $r_1 = 2$; $r_2 = -2$; $k = (r_1 - r_2)/2 = 2$;
 $a_{2m}(r) = \frac{2}{2m+r-2} a_{2m-2}(r)$; $a_{2m}(r) = \frac{2^m}{\prod_{j=1}^m (2j+r-2)}$; $a_{2m}(2) = \frac{1}{m!}$;
 $y_1 = x^2 \sum_{m=0}^{\infty} \frac{1}{m!} x^{2m} = x^2 e^{x^2}$; $z = x^{-2}(1-x^2)$; $C = -\frac{p_2(0)}{4} a_2(-2) = -2$. By logarithmic
differentiation, $a'_{2m}(r) = -a_{2m}(r) \sum_{j=1}^m \frac{1}{2j+r-2}$; $a'_{2m}(2) = -a_{2m}(2) \sum_{j=1}^m \frac{1}{2j}$;
 $y_2 = x^{-2}(1-x^2) - 2y_1 \ln x + x^2 \sum_{m=1}^{\infty} \frac{1}{m!} \left(\sum_{j=1}^m \frac{1}{j} \right) x^{2m}.$

7.7.32. $p_0(r) = (3r-13)(3r-1)$; $p_2(r) = 2(5-3r)$; $r_1 = 13/3$; $r_2 = 1/3$; $k = (r_1 - r_2)/2 = 2$;
 $a_{2m}(r) = \frac{2(6m+3r-11)}{(6m+3r-13)(6m+3r-1)} a_{2m-2}(r)$;
 $a_{2m}(r) = 2^m \prod_{j=1}^m \frac{(6j+3r-11)}{(6j+3r-13)(6j+3r-1)}$; $a_{2m}(13/3) = \frac{\prod_{j=1}^m (3j+1)}{9^m m! (m+2)!}$;
 $y_1 = 2x^{13/3} \sum_{m=0}^{\infty} \frac{\prod_{j=1}^m (3j+1)}{9^m m! (m+2)!} x^{2m}$; $z = x^{1/3} \left(1 + \frac{2}{9} x^2 \right)$; $C = -\frac{p_2(7/3)}{36} a_2(1/3) = 2/81$. By
logarithmic differentiation,
 $a'_{2m}(r) = -9a_{2m}(r) \sum_{j=1}^m \frac{(12j^2 + 4j(3r-11) + 3r^2 - 22r + 47)}{(6j+3r-13)(6j+3r-11)(6j+3r-1)}$;
 $a'_{2m}(13/3) = -a_{2m}(13/3) \sum_{j=1}^m \frac{3j^2 + 2j + 2}{2j(j+2)(3j+1)}$;
 $y_2 = x^{1/3} \left(1 + \frac{2}{9} x^2 \right) + \frac{2}{81} \left(y_1 \ln x - x^{13/3} \sum_{m=0}^{\infty} \frac{\prod_{j=1}^m (3j+1)}{9^m m! (m+2)!} \left(\sum_{j=1}^m \frac{3j^2 + 2j + 2}{j(j+2)(3j+1)} \right) x^{2m} \right).$

7.7.34. $p_0(r) = (r-2)(r+2)$; $p_2(r) = -3(r-4)$; $r_1 = 2$; $r_2 = -2$; $k = (r_1 - r_2)/2 = 2$;
 $a_{2m}(r) = \frac{3(2m+r-6)}{(2m+r-2)(2m+r+2)} a_{2m-2}(r)$; $y_1 = x^2 \left(1 - \frac{1}{2} x^2 \right)$; $z = x^{-2} \left(1 + \frac{9}{2} x^2 \right)$;
 $C = -\frac{p_2(0)}{4} a_2(-2) = -27/2$; $a_2(r) = \frac{3(r-4)}{r(r+4)}$, $a'_2(r) = -\frac{3(r^2-8r-16)}{r^2(r+4)^2}$, $a'_2(2) = 7/12$. If
 $m \geq 2$, then $a_{2m}(r) = \frac{(r-2)c_{2m}(r)}{3^m(r-4)}$ where
 $c_{2m}(r) = \frac{c_{2m}(2)}{(2m+r-4)(2m+r-2) \prod_{j=1}^m (2j+r+2)}$, so
 $a'_{2m}(2) = c_{2m}(2) = -\frac{\left(\frac{3}{2}\right)^m}{m(m-1)(m+2)!}$;
 $y_2 = x^{-2} \left(1 + \frac{9}{2} x^2 \right) - \frac{27}{2} \left(y_1 \ln x + \frac{7}{12} x^4 - x^2 \sum_{m=2}^{\infty} \frac{\left(\frac{3}{2}\right)^m}{m(m-1)(m+2)!} x^{2m} \right).$

7.7.36. $p_0(r) = (2r-5)(2r+7)$; $p_2(r) = (2r-1)^2$; $r_1 = 5/2$; $r_2 = -7/2$; $k = (r_1 - r_2)/2 = 3$;

$$a_{2m}(r) = -\frac{4m+2r-5}{4m+2r+7}a_{2m-2}(r); a_{2m}(r) = \frac{(2r-1)(2r+3)(2r+7)}{(4m+2r-1)(4m+2r+3)(4m+2r+7)};$$

$$a_{2m}(5/2) = \frac{(-1)^m}{(m+1)(m+2)(m+3)}; y_1 = x^{5/2} \sum_{m=0}^{\infty} \frac{(-1)^m}{(m+1)(m+2)(m+3)} x^{2m};$$

$$z = x^{-7/2}(1+x^2)^2 \quad C = -\frac{p_2(1/2)}{24}a_4(-7/2) = 0; y_2 = x^{-7/2}(1+x^2)^2.$$

7.7.38. $p_0(r) = (r-3)(r+7); p_2(r) = r(r+1); r_1 = 3; r_2 = -7; k = (r_1 - r_2)/2 = 5;$

$$a_{2m}(r) = -\frac{(2m+r-2)(2m+r-1)}{(2m+r-3)(2m+r+7)}a_{2m-2}(r); a_{2m}(r) = (-1)^m \prod_{j=1}^m \frac{(2j+r-2)(2j+r-1)}{(2j+r-3)(2j+r+7)};$$

$$a_{2m}(3) = (-1)^m \frac{m+1}{2^m} \left(\prod_{j=1}^m \frac{2j+1}{j+5} \right); y_1 = x^3 \sum_{m=0}^{\infty} (-1)^m \frac{m+1}{2^m} \left(\prod_{j=1}^m \frac{2j+1}{j+5} \right) x^{2m};$$

$$z = x^{-7} \left(1 + \frac{21}{8}x^2 + \frac{35}{16}x^4 + \frac{35}{64}x^6 \right) \quad C = -\frac{p_2(1)}{10}a_8(-7) = 0;$$

$$y_2 = x^{-7} \left(1 + \frac{21}{8}x^2 + \frac{35}{16}x^4 + \frac{35}{64}x^6 \right).$$

7.7.40. $p_0(r) = (2r-3)(2r+5); p_2(r) = (2r-1)(2r+1); r_1 = 3/2; r_2 = -5/2;$

$$k = (r_1 - r_2)/2 = 2; a_{2m}(r) = -\frac{4m+2r-5}{4m+2r+5}a_{2m-2}(r); a_{2m}(r) = (-1)^m \prod_{j=1}^m \frac{4j+2r-5}{4j+2r+5};$$

$$a_{2m}(3/2) = \frac{(-1)^m \prod_{j=1}^m (2j-1)}{2^{m-1}(m+2)!}; y_1 = x^{3/2} \sum_{m=0}^{\infty} \frac{(-1)^m \prod_{j=1}^m (2j-1)}{2^{m-1}(m+2)!} x^{2m};$$

$$z = x^{-5/2} \left(1 + \frac{3}{2}x^2 \right) \quad C = -\frac{p_2(-1/2)}{16}a_2(-5/2) = 0; y_2 = x^{-5/2} \left(1 + \frac{3}{2}x^2 \right).$$

7.7.42. $p_0(r) = r^2 - \nu^2; p_2(r) = 1; r_1 = \nu; r_2 = -\nu; k = (r_1 - r_2)/2 = \nu;$

$$a_{2m}(r) = -\frac{a_{2m-2}(r)}{(2m+r+\nu)(2m+r-\nu)}; a_{2m}(r) = \frac{(-1)^m}{\prod_{j=1}^m (2j+r+\nu)(2j+r-\nu)};$$

$$a_{2m}(\nu) = \frac{(-1)^m}{4^m m! \prod_{j=1}^m (j+\nu)}; a_{2m}(-\nu) = \frac{(-1)^m}{4^m m! \prod_{j=1}^m (j-\nu)}, j = 0, \dots, \nu-1;$$

$$y_1 = x^\nu \sum_{m=0}^{\infty} \frac{(-1)^m}{4^m m! \prod_{j=1}^m (j+\nu)} x^{2m}; z = x^{-\nu} \sum_{m=0}^{\nu-1} \frac{(-1)^m}{4^m m! \prod_{j=1}^m (j-\nu)} x^{2m};$$

$$C = -\frac{p_2(\nu-2)}{2\nu}a_{2\nu-2}(-\nu) = -\frac{a_{2\nu-2}(-\nu)}{2\nu} = -\frac{2}{4^\nu \nu! (\nu-1)!}. \text{ By logarithmic differentiation,}$$

$$a'_{2m}(r) = -2a_{2m}(r) \sum_{j=1}^m \frac{2j+\nu}{(2j+r+\nu)(2j+r-\nu)}; a'_{2m}(\nu) = -a_{2m}(\nu) \sum_{j=1}^m \frac{2j+\nu}{2j(j+\nu)};$$

$$y_2 = x^{-\nu} \sum_{m=0}^{\nu-1} \frac{(-1)^m}{4^m m! \prod_{j=1}^m (j-\nu)} x^{2m}$$

$$- \frac{2}{4^\nu \nu! (\nu-1)!} \left(y_1 \ln x - \frac{x^\nu}{2} \sum_{m=1}^{\infty} \frac{(-1)^m}{4^m m! \prod_{j=1}^m (j+\nu)} \left(\sum_{j=1}^m \frac{2j+\nu}{j(j+\nu)} \right) x^{2m} \right).$$

7.7.44. Since $a_n(r_2) = -\frac{p_1(n+r_2-1)}{p_0(n+r_2)}a_{n-1}(r_2), 1 \leq n \leq k-1,$

$$a_{k-1}(r_2) = (-1)^{k-1} \prod_{j=1}^{k-1} \frac{p_1(r_2 + j - 1)}{p_0(r_2 + j)}. \text{ But}$$

$$C = -\frac{p_1(r_1 - 1)}{k\alpha_0} a_{k-1}(r_2) = -\frac{p_1(r_2 + k - 1)}{k\alpha_0} a_{k-1}(r_2) = (-1)^k \frac{\prod_{j=1}^k p_1(r_2 + j - 1)}{k\alpha_0 \prod_{j=1}^{k-1} p_0(r_2 + j)} = 0 \text{ if and}$$

$$\text{only if } \prod_{j=1}^k p_1(r_2 + j - 1) = 0.$$

7.7.46. Since $p_1(r) = \gamma_1$, $a_n(r) = -\frac{\gamma_1}{\alpha_0(n+r-r_1)(n+r-r_2)} a_{n-1}(r)$ and (A)

$$a_n(r) = (-1)^n \left(\frac{\gamma_1}{\alpha_0} \right)^n \frac{1}{\prod_{j=1}^n (j+r-r_1)(j+r-r_2)}. \text{ Therefore,}$$

$$a_n(r_1) = \frac{(-1)^n}{n!} \left(\frac{\gamma_1}{\alpha_0} \right)^n \frac{1}{\prod_{j=1}^n (j+k)} \text{ for } n \geq 0 \text{ (so } Ly_1 = 0) \text{ and}$$

$$a_n(r_2) = \frac{(-1)^n}{n!} \left(\frac{\gamma_1}{\alpha_0} \right)^n \frac{1}{\prod_{j=1}^n (j-k)} \text{ for } n = 0, \dots, k-1. \text{ } Ly_2 = 0 \text{ if}$$

$$y_2 = x^{r_2} \sum_{n=0}^{k-1} a_n(r_2) x^n + C \left(y_1 \ln x + x^{r_1} \sum_{n=1}^{\infty} a'_n(r_1) x^n \right) \text{ if}$$

$$C = -\frac{\gamma_1}{k\alpha_0} a_{k-1}(r_2) = -\frac{\gamma_1}{k\alpha_0} \frac{(-1)^{k-1}}{(k-1)!} \left(\frac{\gamma_1}{\alpha_0} \right)^{k-1} \frac{(-1)^{k-1}}{(k-1)!} = -\frac{1}{k!(k-1)!} \left(\frac{\gamma_1}{\alpha_0} \right)^k. \text{ From (A),}$$

$$\ln |a_n(r)| = -n \ln \left| \frac{\gamma_1}{\alpha_0} \right| - \sum_{j=1}^n ((\ln |j+r-r_1| + \ln |j+r-r_2|)), \text{ so}$$

$$a'_n(r) = -a_n(r) \sum_{j=1}^n \left(\frac{1}{j+r-r_1} + \frac{1}{j+r-r_2} \right) \text{ and } a'_n(r_1) = -a_n(r_1) \sum_{j=1}^n \frac{2j+k}{j(j+k)}.$$

7.7.48. (a) From Exercise 7.6.66(a) of Section 7.6, $L \left(\frac{\partial y}{\partial r}(x, r) \right) = p'_0(r)x^r + x^r p_0(r) \ln x$. Setting

$$r = r_1 \text{ yields } L \left(y_1 \ln x + x^{r_1} \sum_{n=1}^{\infty} a'_n(r_1) x^n \right) = p'_0(r_1)x^{r_1}. \text{ Since } p'_0(r) = \alpha_0(2r - r_1 - r_2),$$

$$p'_0(r_1) = k\alpha_0.$$

(b) From Exercise 7.5.57 of Section 7.5, $L \left(x^{r_2} \sum_{n=0}^{\infty} a_n(r_2) x^n \right) = x^{r_2} \sum_{n=0}^{\infty} b_n x^n$, where

$$b_0 = p_0(r_2) = 0 \text{ and } b_n = \sum_{j=0}^n p_j(n+r_2-j) a_{n-j}(r_2) \text{ if } n \geq 1. \text{ From the definition of}$$

$$\{a_n(r_2)\}, b_n = 0 \text{ if } n \neq k, \text{ while } b_k = \sum_{j=0}^k p_k(k+r_2-j) a_{k-j}(r_2) = \sum_{j=1}^k p_j(r_1-j) a_{k-j}(r_2).$$

(d) Let $\{\tilde{a}_n(r_2)\}$ be the coefficients that would be obtained if $\tilde{a}_k(r_2) = 0$. Then

$$a_n(r_2) = \tilde{a}_n(r_2) \text{ if } n = 0, \dots, k-1, \text{ and (A)}$$

$$a_n(r_2) - \tilde{a}_n(r_2) = -\frac{1}{p_0(n+r_2)} \sum_{j=0}^{n-k} p_j(n+r_2-j) (a_{n-j}(r_2) - \tilde{a}_{n-j}(r_2)) \text{ if } n > k. \text{ Now let}$$

$$c_m = a_{k+m}(r_2) - \tilde{a}_{k+m}(r_2). \text{ Setting } n = m+k \text{ in (A) and recalling the } k+r_2 = r_1$$

yields (B) $c_m = -\frac{1}{p_0(m+r_1)} \sum_{j=0}^m p_j(m+r_1-j)c_{m-j}$. Since $c_k = a_k(r_2)$, (B) implies that $c_m = a_k(r_2)a_m(r_1)$ for all $m \geq 0$, which implies the conclusion.

Chapter 8

Laplace Transforms

8.1 Introduction to the Laplace Transform

- 8.1.2. (a) $\cosh t \sin t = \frac{1}{2} (e^t \sin t + e^{-t} \sin t) \leftrightarrow \frac{1}{2} \left[\frac{1}{(s-1)^2 + 1} + \frac{1}{(s+1)^2 + 1} \right] = \frac{s^2 + 2}{[(s-1)^2 + 1][(s+1)^2 + 1]}.$
- (b) $\sin^2 t = \frac{1 - \cos 2t}{2} \leftrightarrow \frac{1}{2} \left[\frac{1}{s} - \frac{s}{(s^2 + 4)} \right] = \frac{2}{s(s^2 + 4)}.$
- (c) $\cos^2 2t = \frac{1}{2} \left[\frac{1}{s} + \frac{s}{s^2 + 16} \right] = \frac{s^2 + 8}{s(s^2 + 16)}.$
- (d) $\cosh^2 t = \frac{(e^t + e^{-t})^2}{4} = \frac{(e^{2t} + 2 + e^{-2t})}{4} \leftrightarrow \frac{1}{4} \left(\frac{1}{s-2} + \frac{2}{s} + \frac{1}{s+2} \right) = \frac{s^2 - 2}{s(s^2 - 4)}.$
- (e) $t \sinh 2t = \frac{te^{2t} - te^{-2t}}{2} \leftrightarrow \frac{1}{2} \left(\frac{1}{(s-2)^2} - \frac{1}{(s+2)^2} \right) = \frac{4s}{(s^2 - 4)^2}.$
- (f) $\sin t \cos t = \frac{\sin 2t}{2} \leftrightarrow \frac{1}{s^2 + 4}.$
- (g) $\sin(t + \pi/4) = \sin t \cos(\pi/4) + \cos t \sin(\pi/4) \leftrightarrow \frac{1}{\sqrt{2}} \frac{s+1}{s^2 + 1}.$
- (h) $\cos 2t - \cos 3t \leftrightarrow \frac{s}{s^2 + 4} - \frac{s}{s^2 + 9} = \frac{5s}{(s^2 + 4)(s^2 + 9)}.$
- (i) $\sin 2t + \cos 4t \leftrightarrow \frac{2}{s^2 + 4} + \frac{s}{s^2 + 16} = \frac{s^3 + 2s^2 + 4s + 32}{(s^2 + 4)(s^2 + 16)}.$

8.1.6. If $F(s) = \int_0^\infty e^{-st} f(t) dt$, then $F'(s) = \int_0^\infty (-te^{-st}) f(t) dt = - \int_0^\infty e^{-st} (tf(t)) dt.$

Applying this argument repeatedly yields the assertion.

8.1.8. Let $f(t) = 1$ and $F(s) = 1/s$. From Exercise 8.1.6, $t^n \leftrightarrow (-1)^n F^{(n)}(s) = n!/s^{n+1}.$

8.1.10. If $|f(t)| \leq Me^{s_0 t}$ for $t \geq t_0$, then $|f(t)e^{-st}| \leq Me^{-(s-s_0)t}$ for $t \geq t_0$. Let $g(t) = e^{-st}f(t)$, $w(t) = Me^{-(s-s_0)t}$, and $\tau = t_0$. Since $\int_{t_0}^{\infty} w(t) dt$ converges if $s > s_0$, $F(s)$ is defined for $s > s_0$.

8.1.12. $\int_0^T e^{-st} \left(\int_0^t f(\tau) d\tau \right) dt = -\frac{e^{-st}}{s} \int_0^t f(\tau) d\tau \Big|_0^T + \frac{1}{s} \int_0^T e^{-st} f(t) dt =$
 $-\frac{e^{-sT}}{s} \int_0^T f(\tau) d\tau + \frac{1}{s} \int_0^T e^{-st} f(t) dt$. Since f is of exponential order s_0 , the second
 integral on the right converges to $\frac{1}{s} \mathcal{L}(f)$ as $T \rightarrow \infty$ (Exercise 8.1.10). Now it suffices to
 show that (A) $\lim_{T \rightarrow \infty} e^{-sT} \int_0^T f(\tau) d\tau = 0$ if $s > s_0$. Suppose that $|f(t)| \leq Me^{s_0 t}$ if $t \geq t_0$ and
 $|f(t)| \leq K$ if $0 \leq t \leq t_0$, and let $T > t_0$. Then
 $\left| \int_0^T f(\tau) d\tau \right| \leq \left| \int_0^{t_0} f(\tau) d\tau \right| + \left| \int_{t_0}^T f(\tau) d\tau \right| < Kt_0 + M \int_{t_0}^T e^{s_0 \tau} d\tau < Kt_0 + \frac{Me^{s_0 T}}{s_0}$, which
 proves (A).

8.1.14. (a) If $T > 0$, then $\int_0^T e^{-st} f(t) dt = \int_0^T e^{-(s-s_0)t} (e^{-s_0 t} f(t)) dt$. Use integration by parts
 with $u = e^{-(s-s_0)t}$, $dv = e^{-s_0 t} f(t) dt$, $du = -(s-s_0)e^{-(s-s_0)t}$, and $v = g$ to obtain
 $\int_0^T e^{-st} f(t) dt = e^{-(s-s_0)t} g(t) \Big|_0^T + (s-s_0) \int_0^T e^{-(s-s_0)t} g(t) dt$. Since $g(0) = 0$ this
 reduces to $\int_0^T e^{-st} f(t) dt = e^{-(s-s_0)T} g(T) + (s-s_0) \int_0^T e^{-(s-s_0)t} g(t) dt$. Since
 $|g(t)| \leq M$ for all $t \geq 0$, we can let $t \rightarrow \infty$ to conclude that
 $\int_0^{\infty} e^{-st} f(t) dt = (s-s_0) \int_0^{\infty} e^{-(s-s_0)t} g(t) dt$ if $s > s_0$.
 (b) If $F(s_0)$ exists, then $g(t)$ is bounded on $[0, \infty)$. Now apply (a).
 (c) Since $f(t) = \frac{1}{2} \frac{d}{dt} \sin(e^{t^2})$, $\left| \int_0^t f(\tau) d\tau \right| = \frac{|\sin(e^{t^2}) - \sin(1)|}{2} \leq 1$ for all $t \geq 0$. Now
 apply (a) with $s_0 = 0$.

8.1.16. (a) $\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx = \frac{x^{\alpha} e^{-x}}{\alpha} \Big|_0^{\infty} + \frac{1}{\alpha} \int_0^{\infty} x^{\alpha} e^{-x} dx = \frac{\Gamma(\alpha+1)}{\alpha}$.

(b) Use induction. $\Gamma(1) = \int_0^{\infty} e^{-x} dx = 1$. If (A) $\Gamma(n+1) = n!$, then
 $\Gamma(n+2) = (n+1)\Gamma(n+1)$ (from (a)) $= (n+1)n!$ (from (A)) $= (n+1)!$.

(c) $\Gamma(\alpha+1) = \int_0^{\infty} x^{\alpha} e^{-x} dx$. Let $x = st$. Then $\Gamma(\alpha+1) = \int_0^{\infty} (st)^{\alpha} e^{-st} s dt$, so
 $\int_0^{\infty} e^{-st} t^{\alpha} dt = \frac{\Gamma(\alpha+1)}{\alpha}$.

8.1.18. (a) $\int_0^2 e^{-st} f(t) dt = \int_0^1 e^{-st} t dt + \int_1^2 e^{-st} (2-t) dt =$
 $\left(\frac{1}{s^2} - \frac{e^{-s}(s+1)}{s^2} \right) + \left(\frac{e^{-s}(s-1)}{s^2} + \frac{e^{-2s}}{s^2} \right) = -\frac{2e^{-s}}{s^2} + \frac{e^{-2s}}{s^2} + \frac{1}{s^2} = \frac{(1-e^{-s})^2}{s^2}.$
 Therefore, $F(s) = \frac{(1-e^{-s})^2}{s^2(1-e^{-2s})} = \frac{1-e^{-s}}{s^2(1+e^{-s})} = \frac{1}{s^2} \tanh \frac{s}{2}.$

(b) $\int_0^1 e^{-st} f(t) dt = \int_0^{1/2} e^{-st} dt - \int_{1/2}^1 e^{-st} dt = \frac{1}{s} - \frac{e^{-s/2}}{s} + \frac{e^{-s}}{s} - \frac{e^{-s/2}}{s} =$
 $-\frac{2e^{-s/2}}{s} + \frac{e^{-s}}{s} + \frac{1}{s} = \frac{(1-e^{-s/2})^2}{s}.$
 Therefore, $F(s) = \frac{(1-e^{-s/2})^2}{s(1-e^{-s})} = \frac{1-e^{-s/2}}{s(1+e^{-s/2})} = \frac{1}{s} \tanh \frac{s}{4}.$

(c) $\int_0^\pi e^{-st} f(t) dt = \int_0^\pi e^{-st} \sin t dt = \frac{1+e^{-\pi s}}{(s^2+1)}.$
 Therefore, $F(s) = \frac{1+e^{-\pi s}}{(s^2+1)(1-e^{-\pi s})} \frac{1}{s^2+1} \coth \frac{\pi s}{2}.$

(d) $\int_0^{2\pi} e^{-st} f(t) dt = \int_0^\pi e^{-st} \sin t dt = \frac{1+e^{-\pi s}}{(s^2+1)}.$
 Therefore, $F(s) = \frac{1+e^{-\pi s}}{(s^2+1)(1+e^{-2\pi s})} = \frac{1}{(s^2+1)(1-e^{-\pi s})}.$

8.2 The Inverse Laplace Transform

8.2.2. (a) $\frac{2s+3}{(s-7)^4} = \frac{2(s-7)+17}{(s-7)^4} = \frac{2}{(s-7)^3} + \frac{17}{(s-7)^4} = \frac{2!}{(s-7)^3} + \frac{17}{6} \frac{3!}{(s-7)^4} \leftrightarrow$
 $e^{7t} \left(t^2 + \frac{17}{6} t^3 \right).$

(b) $\frac{s^2-1}{(s-2)^6} = \frac{[(s-2)+2]^2-1}{(s-2)^6} = \frac{(s-2)^2+4(s-2)+3}{(s-2)^6} = \frac{1}{(s-2)^4} + \frac{4}{(s-2)^5} +$
 $\frac{3}{(s-2)^6} = \frac{1}{6} \frac{3!}{(s-2)^4} + \frac{1}{6} \frac{4!}{(s-2)^5} + \frac{1}{40} \frac{5!}{(s-2)^6} \leftrightarrow \left(\frac{1}{6} t^3 + \frac{1}{6} t^4 + \frac{1}{40} t^5 \right) e^{2t}.$

(c) $\frac{s+5}{s^2+6s+18} = \frac{(s+3)}{(s+3)^2+9} + \frac{2}{3} \frac{3}{(s+3)^2+9} \leftrightarrow e^{-3t} \left(\cos 3t + \frac{2}{3} \sin 3t \right).$

(d) $\frac{2s+1}{s^2+9} = 2 \frac{s}{s^2+9} + \frac{1}{3} \frac{3}{s^2+9} \leftrightarrow 2 \cos 3t + \frac{1}{3} \sin 3t.$

(e) $\frac{s}{s^2+2s+1} = \frac{(s+1)-1}{(s+1)^2} = \frac{1}{s+1} - \frac{1}{(s+1)^2} \leftrightarrow (1-t)e^{-t}.$

(f) $\frac{s+1}{s^2-9} = \frac{s}{s^2-9} + \frac{1}{3} \frac{3}{s^2-9} \leftrightarrow \cosh 3t + \frac{1}{3} \sinh 3t.$

(g) Expand the numerator in powers of $s+1$:
 $s^3+2s^2-s-3 = [(s+1)-1]^3+2[(s+1)-1]^2-[(s+1)-1]-3 = (s+1)^3-(s+1)^2-2(s+1)-1;$

therefore

$$\frac{s^3 + 2s^2 - s - 3}{(s+1)^4} = \frac{1}{s+1} - \frac{1}{(s+1)^2} - \frac{2}{(s+1)^3} - \frac{1}{6} \frac{6}{(s+1)^4} \leftrightarrow \left(1 - t - t^2 - \frac{1}{6}t^3\right)e^{-t}.$$

$$(h) \quad \frac{2s+3}{(s-1)^2+4} = 2 \frac{(s-1)}{(s-1)^2+4} + \frac{5}{2} \frac{2}{(s-1)^2+4} \leftrightarrow e^t \left(2 \cos 2t + \frac{5}{2} \sin 2t\right).$$

$$(i) \quad \frac{1}{s} - \frac{s}{s^2+1} \leftrightarrow 1 - \cos t.$$

$$(j) \quad \frac{3s+4}{s^2-1} = \frac{3s}{s^2-1} + \frac{4}{s^2-1} \leftrightarrow 3 \cosh t + 4 \sinh t. \text{ Alternatively,}$$

$$\frac{3s+4}{s^2-1} = \frac{3s+4}{(s-1)(s+1)} = \frac{1}{2} \left[\frac{7}{s-1} - \frac{1}{s+1} \right] \leftrightarrow \frac{7e^t - e^{-t}}{2}.$$

$$(k) \quad \frac{3}{s-1} + \frac{4s+1}{s^2+9} = 3 \frac{1}{s-1} + 4 \frac{s}{s^2+9} + \frac{1}{3} \frac{3}{s^2+9} \leftrightarrow 3e^t + 4 \cos 3t + \frac{1}{3} \sin 3t.$$

$$(l) \quad \frac{3}{(s+2)^2} - \frac{2s+6}{s^2+4} = 3 \frac{1}{(s+2)^2} - 2 \frac{s}{s^2+4} - 3 \frac{2}{s^2+4} \leftrightarrow 3te^{-2t} - 2 \cos 2t - 3 \sin 2t.$$

8.2.4. (a)

$$\frac{2+3s}{(s^2+1)(s+2)(s+1)} = \frac{A}{s+2} + \frac{B}{s+1} + \frac{Cs+D}{s^2+1},$$

where

$$A(s^2+1)(s+1) + B(s^2+1)(s+2) + (Cs+D)(s+2)(s+1) = 2+3s.$$

$$\begin{aligned} -5A &= -4 & (\text{when } s = -2); \\ 2B &= -1 & (\text{when } s = -1); \\ A+2B+2D &= 2 & (\text{when } s = 0); \\ A+B+C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields $A = \frac{4}{5}$, $B = -\frac{1}{2}$, $C = -\frac{3}{10}$, $D = \frac{11}{10}$. Therefore,

$$\begin{aligned} \frac{2+3s}{(s^2+1)(s+2)(s+1)} &= \frac{4}{5} \frac{1}{s+2} - \frac{1}{2} \frac{1}{s+1} - \frac{1}{10} \frac{3s-11}{s^2+1} \\ &\leftrightarrow \frac{4}{5} e^{-2t} - \frac{1}{2} e^{-t} - \frac{3}{10} \cos t + \frac{11}{10} \sin t. \end{aligned}$$

(b)

$$\frac{3s^2+2s+1}{(s^2+1)(s^2+2s+2)} = \frac{As+B}{s^2+1} + \frac{C(s+1)+D}{(s+1)^2+1},$$

where

$$(As+B)((s+1)^2+1) + (C(s+1)+D)(s^2+1) = 3s^2+2s+1.$$

$$\begin{aligned} 2B+C+D &= 1 & (\text{when } s = 0); \\ -A+B+2D &= 2 & (\text{when } s = -1); \\ 2B+C+D &= 1 & (\text{when } s = 0); \\ A+C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields $A = 6/5$, $B = 2/5$, $C = -6/5$, $D = 7/5$. Therefore,

$$\begin{aligned}\frac{3s^2 + 2s + 1}{(s^2 + 1)(s^2 + 2s + 2)} &= \frac{1}{5} \left[\frac{6s + 2}{s^2 + 1} - \frac{6(s + 1) - 7}{(s + 1)^2 + 1} \right] \\ &\leftrightarrow \frac{6}{5} \cos t + \frac{2}{5} \sin t - \frac{6}{5} e^{-t} \cos t + \frac{7}{5} e^{-t} \sin t.\end{aligned}$$

(c) $s^2 + 2s + 5 = (s + 1)^2 + 4$;

$$\frac{3s + 2}{(s - 2)((s + 1)^2 + 4)} = \frac{A}{s - 2} + \frac{B(s + 1) + C}{(s + 1)^2 + 4},$$

where

$$\begin{aligned}A((s + 1)^2 + 4) + (B(s + 1) + C)(s - 2) &= 3s + 2. \\ 13A &= 8 \quad (\text{when } s = 2); \\ 4A - 3C &= -1 \quad (\text{when } s = -1); \\ A + B &= 0 \quad (\text{equate coefficients of } s^2).\end{aligned}$$

Solving this system yields $A = \frac{8}{13}$, $B = -\frac{8}{13}$, $C = \frac{15}{13}$. Therefore,

$$\begin{aligned}\frac{3s + 2}{(s - 2)((s + 1)^2 + 4)} &= \frac{1}{13} \left[\frac{8}{s - 2} - \frac{8(s - 1) - 15}{(s + 1)^2 + 4} \right] \\ &\leftrightarrow \frac{8}{13} e^{2t} - \frac{8}{13} e^{-t} \cos 2t + \frac{15}{26} e^{-t} \sin 2t.\end{aligned}$$

(d)

$$\frac{3s^2 + 2s + 1}{(s - 1)^2(s + 2)(s + 3)} = \frac{A}{s - 1} + \frac{B}{(s - 1)^2} + \frac{C}{s + 2} + \frac{D}{s + 3},$$

where

$$\begin{aligned}(A(s - 1) + B)(s + 2)(s + 3) + (C(s + 3) + D(s + 2))(s - 1)^2 &= 3s^2 + 2s + 1. \\ 12B &= 6 \quad (\text{when } s = 1); \\ 9C &= 9 \quad (\text{when } s = -2); \\ -16D &= 22 \quad (\text{when } s = -3); \\ A + C + D &= 0 \quad (\text{equate coefficients of } s^3).\end{aligned}$$

Solving this system yields $A = 3/8$, $B = 1/2$, $C = 1$, $D = -11/8$. Therefore,

$$\begin{aligned}\frac{3s^2 + 2s + 1}{(s - 1)^2(s + 2)(s + 3)} &= \frac{3}{8} \frac{1}{s - 1} + \frac{1}{2} \frac{1}{(s - 1)^2} + \frac{1}{s + 2} - \frac{11}{8} \frac{1}{s + 3} \\ &\leftrightarrow \frac{3}{8} e^t + \frac{1}{2} t e^t + e^{-2t} - \frac{11}{8} e^{-3t}.\end{aligned}$$

(e)

$$\frac{2s^2 + s + 3}{(s-1)^2(s+2)^2} = \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{C}{s+2} + \frac{D}{(s+2)^2},$$

where

$$\begin{aligned} (A(s-1) + B)(s+2)^2 + (C(s+2) + D)(s-1)^2 &= 2s^2 + s + 3. \\ 9B &= 6 \quad (\text{when } s = 1); \\ 9D &= 9 \quad (\text{when } s = -2); \\ -4A + 4B + 2C + D &= 3 \quad (\text{when } s = 0); \\ A + C &= 0 \quad (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields $A = 1/9$, $B = 2/3$, $C = -1/9$, $D = 1$. Therefore,

$$\begin{aligned} \frac{2s^2 + s + 3}{(s-1)^2(s+2)^2} &= \frac{1}{9} \frac{1}{s-1} + \frac{2}{3} \frac{1}{(s-1)^2} - \frac{1}{9} \frac{1}{s+2} + \frac{1}{(s+2)^2} \\ &\leftrightarrow \frac{1}{9}e^t + \frac{2}{3}te^t - \frac{1}{9}e^{-2t} + te^{-2t}. \end{aligned}$$

(f)

$$\frac{3s+2}{(s^2+1)(s-1)^2} = \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{Cs+D}{s^2+1},$$

where

$$A(s-1)(s^2+1) + B(s^2+1) + (Cs+D)(s-1)^2 = 3s+2. \quad (\text{A})$$

Setting $s = 1$ yields $2B = 5$, so $B = \frac{5}{2}$. Substituting this into (A) shows that

$$\begin{aligned} A(s-1)(s^2+1) + (Cs+D)(s-1)^2 &= 3s+2 - \frac{5}{2}(s^2+1) \\ &= -\frac{5s^2-6s+1}{2} = -\frac{(s-1)(5s-1)}{2}. \end{aligned}$$

Therefore,

$$\begin{aligned} A(s^2+1) + (Cs+D)(s-1) &= \frac{1-5s}{2}. \\ 2A &= -2 \quad (\text{when } s = 1); \\ A-D &= 1/2 \quad (\text{when } s = 0); \\ A+C &= 0 \quad (\text{equate coefficients of } s^2). \end{aligned}$$

Solving this system yields $A = -1$, $C = 1$, $D = -\frac{3}{2}$. Therefore,

$$\begin{aligned} \frac{3s+2}{(s^2+1)(s-1)^2} &= -\frac{1}{s-1} + \frac{5}{2} \frac{1}{(s-1)^2} + \frac{s-3/2}{s^2+1} \\ &\leftrightarrow -e^t + \frac{5}{2}te^t + \cos t - \frac{3}{2}\sin t. \end{aligned}$$

8.2.6. (a)

$$\frac{17s - 15}{(s^2 - 2s + 5)(s^2 + 2s + 10)} = \frac{A(s - 1) + B}{(s - 1)^2 + 4} + \frac{C(s + 1) + D}{(s + 1)^2 + 9}$$

where

$$(A(s - 1) + B)((s + 1)^2 + 9) + (C(s + 1) + D)((s - 1)^2 + 4) = 17s - 15.$$

$$\begin{aligned} 13B + 8C + 4D &= 2 & (\text{when } s = 1); \\ -18A + 9B + 8D &= -32 & (\text{when } s = -1); \\ -10A + 10B + 5C + 5D &= -15 & (\text{when } s = 0); \\ A + C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields $A = 1$, $B = 2$, $C = -1$, $D = -4$. Therefore,

$$\begin{aligned} \frac{17s - 15}{(s^2 - 2s + 5)(s^2 + 2s + 10)} &= \frac{(s - 1) + 2}{(s - 1)^2 + 4} - \frac{(s + 1) + 4}{(s + 1)^2 + 9} \\ &\leftrightarrow e^t(\cos 2t + \sin 2t) - e^{-t}\left(\cos 3t + \frac{4}{3}\sin 3t\right). \end{aligned}$$

(b)

$$\frac{8s + 56}{(s^2 - 6s + 13)(s^2 + 2s + 5)} = \frac{A(s - 3) + B}{(s - 3)^2 + 4} + \frac{C(s + 1) + D}{(s + 1)^2 + 4}$$

where

$$(A(s - 3) + B)((s + 1)^2 + 4) + (C(s + 1) + D)((s - 3)^2 + 4) = 8s + 56.$$

$$\begin{aligned} 20B + 16C + 4D &= 80 & (\text{when } s = 3); \\ -16A + 4B + 20D &= 48 & (\text{when } s = -1); \\ -15A + 5B + 13C + 13D &= 56 & (\text{when } s = 0); \\ A + C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields $A = -1$, $B = 3$, $C = 1$, $D = 1$. Therefore,

$$\begin{aligned} \frac{8s + 56}{(s^2 - 6s + 13)(s^2 + 2s + 5)} &= \frac{-(s - 3) + 3}{(s - 3)^2 + 4} + \frac{(s + 1) + 1}{(s + 1)^2 + 4} \\ &\leftrightarrow e^{3t}\left(-\cos 2t + \frac{3}{2}\sin 2t\right) + e^{-t}\left(\cos 2t + \frac{1}{2}\sin 2t\right). \end{aligned}$$

(c)

$$\frac{s + 9}{(s^2 + 4s + 5)(s^2 - 4s + 13)} = \frac{A(s + 2) + B}{(s + 2)^2 + 1} + \frac{C(s - 2) + D}{(s - 2)^2 + 9}$$

where

$$(A(s + 2) + B)((s - 2)^2 + 9) + (C(s - 2) + D)((s + 2)^2 + 1) = s + 9.$$

$$\begin{aligned} 25B - 4C + D &= 7 & (\text{when } s = -2); \\ 36A + 9B + 17D &= 11 & (\text{when } s = 2); \\ 26A + 13B - 10C + 5D &= 9 & (\text{when } s = 0); \\ A + C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields $A = 1/8$, $B = 1/4$, $C = -1/8$, $D = 1/4$. Therefore,

$$\begin{aligned} \frac{s+9}{(s^2+4s+5)(s^2-4s+13)} &= \left[\frac{1}{8} \frac{(s+2)+2}{(s+2)^2+1} - \frac{(s-2)-2}{(s-2)^2+3} \right] \\ &\leftrightarrow e^{-2t} \left(\frac{1}{8} \cos t + \frac{1}{4} \sin t \right) - e^{2t} \left(\frac{1}{8} \cos 3t - \frac{1}{12} \sin 3t \right). \end{aligned}$$

(d)

$$\frac{3s-2}{(s^2-4s+5)(s^2-6s+13)} = \frac{A(s-2)+B}{(s-2)^2+1} + \frac{C(s-3)+D}{(s-3)^2+4}$$

where

$$\begin{aligned} (A(s-2)+B)((s-3)^2+4) + (C(s-3)+D)((s-2)^2+1) &= 3s-2. \\ 5B-C+D &= 4 \quad (\text{when } s=2); \\ 4A+4B+2D &= 7 \quad (\text{when } s=3); \\ -26A+13B-15C+5D &= -2 \quad (\text{when } s=0); \\ A+C &= 0 \quad (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields $A = 1$, $B = 1/2$, $C = -1$, $D = 1/2$. Therefore,

$$\begin{aligned} \frac{3s-2}{(s^2-4s+5)(s^2-6s+13)} &= \frac{1}{2} \left[\frac{2(s-2)+1}{(s-2)^2+1} - \frac{2(s-3)-1}{(s-3)^2+4} \right] \\ &\leftrightarrow e^{2t} \left(\cos t + \frac{1}{2} \sin t \right) - e^{3t} \left(\cos 2t - \frac{1}{4} \sin 2t \right). \end{aligned}$$

(e)

$$\frac{3s-1}{(s^2-2s+2)(s^2+2s+5)} = \frac{A(s-1)+B}{(s-1)^2+1} + \frac{C(s+1)+D}{(s+1)^2+4}$$

where

$$\begin{aligned} (A(s-1)+B)((s+1)^2+4) + (C(s+1)+D)((s-1)^2+1) &= 3s-1. \\ 8B+2C+D &= 2 \quad (\text{when } s=1); \\ -8A+4B+5D &= -4 \quad (\text{when } s=-1); \\ -5A+5B+2C+2D &= -1 \quad (\text{when } s=0); \\ A+5B+C &= 0 \quad (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields $A = 1/5$, $B = 2/5$, $C = -1/5$, $D = -4/5$. Therefore,

$$\begin{aligned} \frac{3s-1}{(s^2-2s+2)(s^2+2s+5)} &= \frac{1}{5} \left[\frac{(s-1)+2}{(s-1)^2+1} - \frac{(s+1)+4}{(s+1)^2+4} \right] \\ &\leftrightarrow e^t \left(\frac{1}{5} \cos t + \frac{2}{5} \sin t \right) - e^{-t} \left(\frac{1}{5} \cos 2t + \frac{2}{5} \sin 2t \right). \end{aligned}$$

(f)

$$\frac{20s+40}{(4s^2-4s+5)(4s^2+4s+5)} = \frac{A(s-1/2)+B}{(s-1/2)^2+1} + \frac{C(s+1/2)+D}{(s+1/2)^2+1}$$

where

$$\begin{aligned} (A(s-1/2) + B)((s+1/2)^2 + 1) + (C(s+1/2) + D)((s-1/2)^2 + 1) &= \frac{5s+10}{4}. \\ \begin{aligned} 2B + C + D &= 25/8 & (\text{when } s = 1/2); \\ -A + B + 2D &= 15/8 & (\text{when } s = -1/2); \\ -5A + 10B + 5C + 10D &= 20 & (\text{when } s = 0); \\ A + C &= 0 & (\text{equate coefficients of } s^3). \end{aligned} \end{aligned}$$

Solving this system yields $A = -1$, $B = 9/8$, $C = 1$, $D = -1/8$. Therefore,

$$\begin{aligned} \frac{20s+40}{(4s^2-4s+5)(4s^2+4s+5)} &= \frac{1}{8} \left[\frac{-8(s-1/2)+9}{(s-1/2)^2+1} + \frac{8(s+1/2)-1}{(s+1/2)^2+1} \right] \\ &\leftrightarrow e^{t/2} \left(-\cos t + \frac{9}{8} \sin t \right) + e^{-t/2} \left(\cos t - \frac{1}{8} \sin t \right). \end{aligned}$$

8.2.8. (a)

$$\frac{2s+1}{(s^2+1)(s-1)(s-3)} = \frac{A}{s-1} + \frac{B}{s-3} + \frac{Cs+D}{s^2+1}$$

where

$$\begin{aligned} (A(s-3) + B(s-1))(s^2+1) + (Cs+D)(s-1)(s-3) &= 2s+1. \\ \begin{aligned} -4A &= 3 & (\text{when } s = 1); \\ 20B &= 7 & (\text{when } s = 3); \\ -3A - B + 3D &= 1 & (\text{when } s = 0); \\ A + B + C &= 0 & (\text{equate coefficients of } s^3). \end{aligned} \end{aligned}$$

Solving this system yields $A = -3/4$, $B = 7/20$, $C = 2/5$, $D = -3/10$. Therefore,

$$\begin{aligned} \frac{2s+1}{(s^2+1)(s-1)(s-3)} &= -\frac{3}{4} \frac{1}{s-1} + \frac{7}{20} \frac{1}{s-3} + \frac{2}{5} \frac{s}{s^2+1} - \frac{3}{10} \frac{1}{s^2+1} \\ &\leftrightarrow -\frac{3}{4} e^t + \frac{7}{20} e^{3t} + \frac{2}{5} \cos t - \frac{3}{10} \sin t. \end{aligned}$$

(b)

$$\frac{s+2}{(s^2+2s+2)(s^2-1)} = \frac{A}{s-1} + \frac{B}{s+1} + \frac{C(s+1)+D}{(s+1)^2+1}$$

where

$$\begin{aligned} (A(s+1) + B(s-1))((s+1)^2+1) + (C(s+1)+D)(s^2-1) &= s+2. \\ \begin{aligned} 10A &= 3 & (\text{when } s = 1); \\ -2B &= 1 & (\text{when } s = -1); \\ 2A - 2B - C - D &= 2 & (\text{when } s = 0); \\ A + B + C &= 0 & (\text{equate coefficients of } s^3). \end{aligned} \end{aligned}$$

Solving this system yields $A = 3/10$, $B = -1/2$, $C = 1/5$, $D = -3/5$. Therefore,

$$\begin{aligned}\frac{s+2}{(s^2+2s+2)(s^2-1)} &= \frac{3}{10} \frac{1}{s-1} - \frac{1}{2} \frac{1}{s+1} + \frac{1}{5} \frac{s+1}{(s+1)^2+1} - \frac{3}{5} \frac{1}{(s+1)^2+1} \\ &\leftrightarrow \frac{3}{10}e^t - \frac{1}{2}e^{-t} + \frac{1}{5}e^{-t} \cos t e^{-t} \sin t.\end{aligned}$$

(c)

$$\frac{2s-1}{(s^2-2s+2)(s+1)(s-2)} = \frac{A}{s-2} + \frac{B}{s+1} + \frac{C(s-1)+D}{(s-1)^2+1}$$

where

$$(A(s+1) + B(s-2))((s-1)^2+1) + (C(s-1)+D)(s-2)(s+1) = 2s-1.$$

$$\begin{aligned}6A &= 3 && \text{(when } s=2\text{);}\\ -15B &= -3 && \text{(when } s=-1\text{);}\\ 2A-4B+2C-2D &= -1 && \text{(when } s=0\text{);}\\ A+B+C &= 0 && \text{(equate coefficients of } s^3\text{).}\end{aligned}$$

Solving this system yields $A = 1/2$, $B = 1/5$, $C = -7/10$, $D = -1/10$. Therefore,

$$\begin{aligned}\frac{2s-1}{(s^2-2s+2)(s+1)(s-2)} &= \frac{1}{2} \frac{1}{s-2} + \frac{1}{5} \frac{1}{s+1} - \frac{7}{10} \frac{s-1}{(s-1)^2+1} - \frac{1}{10} \frac{1}{(s-1)^2+1} \\ &\leftrightarrow \frac{1}{2}e^{2t} + \frac{1}{5}e^{-t} - \frac{7}{10}e^t \cos t - \frac{1}{10}e^t \sin t.\end{aligned}$$

(d)

$$\frac{s-6}{(s^2-1)(s^2+4)} = \frac{A}{s-1} + \frac{B}{s+1} + \frac{Cs+D}{s^2+4}$$

where

$$(A(s+1) + B(s-1))(s^2+4) + (Cs+D)(s^2-1) = s-6.$$

$$\begin{aligned}10A &= -5 && \text{(when } s=1\text{);}\\ -10B &= -7 && \text{(when } s=-1\text{);}\\ 4A-4B-D &= -6 && \text{(when } s=0\text{);}\\ A+B+C &= 0 && \text{(equate coefficients of } s^3\text{).}\end{aligned}$$

Solving this system yields $A = -1/2$, $B = 7/10$, $C = -1/5$, $D = 6/5$. Therefore,

$$\begin{aligned}\frac{s-6}{(s^2-1)(s^2+4)} &= -\frac{1}{2} \frac{1}{s-1} + \frac{7}{10} \frac{1}{s+1} - \frac{1}{5} \frac{s}{s^2+4} + \frac{3}{5} + \frac{1}{s^2+4} \\ &\leftrightarrow -\frac{1}{2}e^t + \frac{7}{10}e^{-t} - \frac{1}{5} \cos 2t + \frac{3}{5} \sin 2t.\end{aligned}$$

(e)

$$\frac{2s-3}{s(s-2)(s^2-2s+5)} = \frac{A}{s} + \frac{B}{s-2} + \frac{C(s-1)+D}{(s-1)^2+4}$$

where

$$\begin{aligned} (A(s-2) + Bs)((s-1)^2 + 4) + (C(s-1) + D)s(s-2) &= 2s - 3. \\ -10A &= -3 \quad (\text{when } s = 0); \\ 10B &= 1 \quad (\text{when } s = 2); \\ -4A + 4B - D &= -1 \quad (\text{when } s = 1); \\ A + B + C &= 0 \quad (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields $A = 3/10$, $B = 1/10$, $C = -2/5$, $D = 1/5$. Therefore,

$$\begin{aligned} \frac{2s-3}{s(s-2)(s^2-2s+5)} &= \frac{3}{10s} + \frac{1}{10} \frac{1}{s-2} - \frac{2}{5} \frac{s-1}{(s-1)^2+4} + \frac{1}{5} \frac{1}{(s-1)^2+4} \\ &\leftrightarrow \frac{3}{10} + \frac{1}{10}e^{2t} - \frac{2}{5}e^t \cos 2t + \frac{1}{10}e^t \sin 2t. \end{aligned}$$

(f)

$$\frac{5s-15}{(s^2-4s+13)(s-2)(s-1)} = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C(s-2)+D}{(s-2)^2+9}$$

where

$$\begin{aligned} (A(s-2) + B(s-1))((s-2)^2 + 9) + (C(s-2) + D)(s-1)(s-2) &= 5s - 15. \\ -10A &= -10 \quad (\text{when } s = 1); \\ 9B &= -5 \quad (\text{when } s = 2); \\ -26A - 13B - 4C + 2D &= -15 \quad (\text{when } s = 0); \\ A + B + C &= 0 \quad (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields $A = 1$, $B = -5/9$, $C = -4/9$, $D = 1$. Therefore,

$$\begin{aligned} \frac{5s-15}{(s^2-4s+13)(s-2)(s-1)} &= \frac{1}{s-1} - \frac{5}{9} \frac{1}{s-2} - \frac{4}{9} \frac{s-2}{(s-2)^2+9} + \frac{1}{(s-2)^2+9} \\ &\leftrightarrow e^t - \frac{5}{9}e^{2t} - \frac{4}{9}e^{2t} \cos 3t + \frac{1}{3}e^{2t} \sin 3t. \end{aligned}$$

8.2.10. (a) Let $i = 1$. (The proof for $i = 2, \dots, n$) is similar. Multiplying the given equation through by $s - s_1$ yields

$$\frac{P(s)}{(s-s_2)\cdots(s-s_n)} = A_1 + (s-s_1) \left[\frac{A_2}{s-s_2} + \cdots + \frac{A_n}{s-s_n} \right],$$

and setting $s = s_1$ yields $A_1 = \frac{P(s_1)}{(s_1-s_2)\cdots(s_1-s_n)}$.

(b) From calculus we know that F has a partial fraction expansion of the form

$$\frac{P(s)}{(s-s_1)Q_1(s)} = \frac{A}{s-s_1} + G(s) \text{ where } G \text{ is continuous at } s_1. \text{ Multiplying through by } s-s_1 \text{ shows that } \frac{P(s)}{Q_1(s)} = A + (s-s_1)G(s). \text{ Now set } s = s_1 \text{ to obtain } A = \frac{P(s_1)}{Q(s_1)}.$$

(c) The result in (b) is generalization of the result in (a), since it shows that if s_1 is a simple zero of the denominator of the rational function, then Heaviside's method can be used to determine the coefficient of $1/(s-s_1)$ in the partial fraction expansion even if some of the other zeros of the denominator are repeated or complex.

8.3 Solution of Initial Value Problems

8.3.2.

$$(s^2 - s - 6)Y(s) = \frac{2}{s} + s - 1 = \frac{2 + s(s-1)}{s}.$$

Since $(s^2 - s - 6) = (s-3)(s+2)$,

$$Y(s) = \frac{2 + s(s-1)}{s(s-3)(s+2)} = -\frac{1}{3s} + \frac{8}{15} \frac{1}{s-3} + \frac{4}{5} \frac{1}{s+2}$$

$$\text{and } y = -\frac{1}{3} + \frac{8}{15}e^{3t} + \frac{4}{5}e^{-2t}.$$

8.3.4.

$$(s^2 - 4)Y(s) = \frac{2}{s-3} + (-1 + s) = \frac{2 + (s-1)(s-3)}{s-3}.$$

Since $s^2 - 4 = (s-2)(s+2)$,

$$Y(s) = \frac{2 + (s-1)(s-3)}{(s-2)(s+2)(s-3)} = -\frac{1}{4} \frac{1}{s-2} + \frac{17}{20} \frac{1}{s+2} + \frac{2}{5} \frac{1}{s-3}$$

$$\text{and } y = -\frac{1}{4}e^{2t} + \frac{17}{20}e^{-2t} + \frac{2}{5}e^{3t}.$$

8.3.6.

$$(s^2 + 3s + 2)Y(s) = \frac{6}{s-1} + (-1 + s) + 3 = \frac{6 + (s-1)(s+2)}{s-1}.$$

Since $s^2 + 3s + 2 = (s+2)(s+1)$,

$$Y(s) = \frac{6 + (s-1)(s+2)}{(s-1)(s+2)(s+1)} = \frac{1}{s-1} + \frac{2}{s+2} - \frac{2}{s+1}$$

$$\text{and } y = e^t + 2e^{-2t} - 2e^{-t}.$$

8.3.8.

$$(s^2 - 3s + 2)Y(s) = \frac{2}{s-3} + (-1 + s) - 3 = \frac{2 + (s-3)(s-4)}{s-3}.$$

Since $s^2 - 3s + 2 = (s-1)(s-2)$,

$$Y(s) = \frac{2 + (s-3)(s-4)}{(s-1)(s-2)(s-3)} = \frac{4}{s-1} - \frac{4}{s-2} + \frac{1}{s-3}$$

$$\text{and } y = 4e^t - 4e^{2t} + e^{3t}.$$

8.3.10.

$$(s^2 - 3s + 2)Y(s) = \frac{1}{s-3} + (-4 - s) + 3 = \frac{1 - (s-3)(s+1)}{s-3}.$$

Since $s^2 - 3s + 2 = (s-1)(s-2)$,

$$Y(s) = \frac{1 - (s-3)(s+1)}{(s-1)(s-2)(s-3)} = \frac{5}{2} \frac{1}{s-1} - \frac{4}{s-2} + \frac{1}{2} \frac{1}{s-3}$$

$$\text{and } y = \frac{5}{2}e^t - 4e^{2t} + \frac{1}{2}e^{3t}.$$

8.3.12.

$$(s^2 + s - 2)Y(s) = -\frac{4}{s} + (3 + 2s) + 2 = \frac{-4 + s(5 + 2s)}{s}.$$

Since $(s^2 + s - 2) = (s + 2)(s - 1)$,

$$Y(s) = \frac{-4 + s(5 + 2s)}{s(s + 2)(s - 1)} = \frac{2}{s} - \frac{1}{s + 2} + \frac{1}{s - 1},$$

and $y = 2 - e^{-2t} + e^t$.

8.3.14.

$$(s^2 - s - 6)Y(s) = \frac{2}{s} + s - 1 = \frac{2 + s(s - 1)}{s}.$$

Since $s^2 - s - 6 = (s - 3)(s + 2)$,

$$Y(s) = \frac{2 + s(s - 1)}{s(s - 3)(s + 2)} = -\frac{1}{3s} + \frac{8}{15} \frac{1}{s - 3} + \frac{4}{5} \frac{1}{s + 2}$$

and $y = -\frac{1}{3} + \frac{8}{15}e^{3t} + \frac{4}{5}e^{-2t}$.

8.3.16.

$$(s^2 - 1)Y(s) = \frac{1}{s} + s = \frac{1 + s^2}{s}.$$

Since $s^2 - 1 = (s - 1)(s + 1)$,

$$Y(s) = \frac{1 + s^2}{s(s - 1)(s + 1)} = -\frac{1}{s} + \frac{1}{s - 1} + \frac{1}{s + 1}$$

and $y = -1 + e^t + e^{-t}$.

8.3.18.

$$(s^2 + s)Y(s) = \frac{2}{s - 3} + (4 - s) - 1 = \frac{2 - (s - 3)^2}{s - 3}.$$

Since $s^2 + s = s(s + 1)$,

$$Y(s) = \frac{2 - (s - 3)^2}{s(s + 1)(s - 3)} = \frac{7}{3s} - \frac{7}{2} \frac{1}{s + 1} + \frac{1}{6} \frac{1}{s - 3}$$

and $y = \frac{7}{3} - \frac{7}{2}e^{-t} + \frac{1}{6}e^{3t}$.

8.3.20.

$$(s^2 + 1)Y(s) = \frac{1}{s^2} + 2, \quad \text{so} \quad Y(s) = \frac{1}{(s^2 + 1)s^2} + \frac{2}{s^2 + 1}.$$

Substituting $x = s^2$ into

$$\frac{1}{(x + 1)x} = \frac{1}{x + 1} - \frac{1}{x} \quad \text{yields} \quad \frac{1}{(s^2 + 1)s^2} = \frac{1}{s^2} - \frac{1}{s^2 + 1},$$

so $Y(s) = \frac{1}{s^2} + \frac{1}{s^2 + 1}$ and $y = t + \sin t$.

8.3.22.

$$(s^2 + 5s + 6)Y(s) = \frac{2}{s+1} + (3+s) + 5 = \frac{2 + (s+1)(s+8)}{s+1}.$$

Since $s^2 + 5s + 6 = (s+2)(s+3)$,

$$Y(s) = \frac{2 + (s+1)(s+8)}{(s+1)(s+2)(s+3)} = \frac{1}{s+1} + \frac{4}{s+2} - \frac{4}{s+3}$$

and $y = e^{-t} + 4e^{-2t} - 4e^{-3t}$.

8.3.24.

$$(s^2 - 2s - 3)Y(s) = \frac{10s}{s^2 + 1} + (7 + 2s) - 4 = \frac{10s}{s^2 + 1} + (2s + 3).$$

Since $s^2 - 2s - 3 = (s-3)(s+1)$,

$$Y(s) = \frac{10s}{(s-3)(s+1)(s^2+1)} + \frac{2s+3}{(s-3)(s+1)}. \quad (\text{A})$$

$$\frac{2s+3}{(s-3)(s+1)} = \frac{9}{4} \frac{1}{s-3} - \frac{1}{4} \frac{1}{s+1} \leftrightarrow \frac{9}{4} e^{3t} - \frac{1}{4} e^{-t}. \quad (\text{B})$$

$$\frac{10s}{(s-3)(s+1)(s^2+1)} = \frac{A}{s-3} + \frac{B}{s+1} + \frac{Cs+D}{s^2+1}$$

where

$$(A(s+1) + B(s-3))(s^2+1) + (Cs+D)(s-3)(s+1) = 10s.$$

$$\begin{aligned} 40A &= 30 & (\text{when } s=3); \\ -8B &= -10 & (\text{when } s=-1); \\ A-3B-3D &= 0 & (\text{when } s=0); \\ A+B+C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields $A = 3/4$, $B = 5/4$, $C = -2$, $D = -1$. Therefore,

$$\begin{aligned} \frac{10s}{(s-3)(s+1)(s^2+1)} &= \frac{3}{4} \frac{1}{s-3} + \frac{5}{4} \frac{1}{s+1} - \frac{2s+1}{s^2+1} \\ &\leftrightarrow \frac{3}{4} e^{3t} + \frac{5}{4} e^{-t} - 2 \cos t - \sin t. \end{aligned}$$

From this, (A), and (B), $y = -\sin t - 2 \cos t + 3e^{3t} + e^{-t}$.

8.3.26.

$$\begin{aligned} (s^2 + 4)Y(s) &= \frac{16}{s^2 + 4} + \frac{9s}{s^2 + 1} + s, \quad \text{so} \\ Y(s) &= \frac{16}{(s^2 + 4)^2} + \frac{9s}{(s^2 + 4)(s^2 + 1)} + \frac{s}{s^2 + 4}. \end{aligned}$$

From the table of Laplace transforms,

$$t \cos 2t \leftrightarrow \frac{s^2 - 4}{(s^2 + 4)^2} = \frac{s^2 + 4}{(s^2 + 4)^2} - \frac{8}{(s^2 + 4)^2}$$

$$= \frac{1}{s^2 + 4} - \frac{8}{(s^2 + 4)^2}.$$

Therefore,

$$\frac{8}{(s^2 + 4)^2} = \frac{1}{s^2 + 4} - \mathcal{L}(t \cos 2t), \text{ so } \frac{16}{(s^2 + 4)^2} \leftrightarrow \sin 2t - 2t \cos 2t. \quad (\text{A})$$

Substituting $x = s^2$ into

$$\frac{9}{(x + 4)(x + 1)} = \frac{3}{x + 1} - \frac{3}{x + 4}$$

and multiplying by s yields

$$\frac{9s}{(s^2 + 4)(s^2 + 1)} = \frac{3s}{s^2 + 1} - \frac{3s}{s^2 + 4} \leftrightarrow 3 \cos t - 3 \cos 2t. \quad (\text{B})$$

Finally,

$$\frac{s}{s^2 + 4} \leftrightarrow \cos 2t. \quad (\text{C})$$

Adding (A), (B), and (C) yields $y = -(2t + 2) \cos 2t + \sin 2t + 3 \cos t$.

8.3.28.

$$(s^2 + 2s + 2)Y(s) = \frac{2}{s^2} + (-7 + 2s) + 4.$$

Since $(s^2 + 2s + 2) = (s + 1)^2 + 1$,

$$Y(s) = \frac{2}{s^2((s + 1)^2 + 1)} + \frac{2s - 3}{(s + 1)^2 + 1}. \quad (\text{A})$$

$$\frac{2s - 3}{(s + 1)^2 + 1} = \frac{2(s + 1) - 5}{(s + 1)^2 + 1} \leftrightarrow e^{-t}(2 \cos t - 5 \sin t). \quad (\text{B})$$

$$\frac{2}{s^2((s + 1)^2 + 1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C(s + 1) + D}{(s + 1)^2 + 1},$$

where $(As + B)((s + 1)^2 + 1) + s^2(C(s + 1) + D) = 2$.

$$\begin{aligned} 2B &= 2 & (\text{when } s = 0); \\ -A + B + D &= 2 & (\text{when } s = -1); \\ A + C &= 0 & (\text{equate coefficients of } s^3); \\ 2A + B + C + D &= 0 & (\text{equate coefficients of } s^2). \end{aligned}$$

Solving this system yields $A = -1$, $B = 1$, $C = 1$, $D = 0$. Therefore,

$$\frac{2}{s^2((s + 1)^2 + 1)} = -\frac{1}{s} + \frac{1}{s^2} + \frac{(s + 1)}{(s + 1)^2 + 1} \leftrightarrow -1 + t + e^{-t} \cos t.$$

From this, (A), and (B), $y = -1 + t + e^{-t}(\cos t - 5 \sin t)$.

8.3.30. $(s^2 + 4s + 5)Y(s) = \frac{(s+1)+3}{(s+1)^2+1} + 4$. Since $(s^2 + 4s + 5) = (s+2)^2 + 1$,

$$Y(s) = \frac{s+4}{((s+1)^2+1)((s+2)^2+1)} + \frac{4}{(s+2)^2+1}. \quad (\text{A})$$

$$\frac{4}{(s+2)^2+1} \leftrightarrow 4e^{-2t} \sin t. \quad (\text{B})$$

$$\frac{s+4}{((s+1)^2+1)((s+2)^2+1)} = \frac{A(s+1)+B}{(s+1)^2+1} + \frac{C(s+2)+D}{(s+2)^2+1},$$

where $(A(s+1)+B)((s+2)^2+1) + (C(s+2)+D)((s+1)^2+1) = 4+s$.

$$\begin{aligned} 5A+5B+4C+2D &= 4 & (\text{when } s=0); \\ 2B+C+D &= 3 & (\text{when } s=-1); \\ -A+B+2D &= 2 & (\text{when } s=-2); \\ A+C &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields $A = -1$, $B = 1$, $C = 1$, $D = 0$. Therefore,

$$\frac{s+4}{((s+1)^2+1)((s+2)^2+1)} = \frac{-(s+1)+1}{(s+1)^2+1} + \frac{s+2}{(s+2)^2+1}, \leftrightarrow e^{-t}(-\cos t + \sin t) + e^{-2t} \cos t.$$

From this, (A), and (B), $y = e^t(-\cos t + \sin t) + e^{-2t}(\cos t + 4 \sin t)$.

8.3.32.

$$(2s^2 - 3s - 2)Y(s) = \frac{4}{s-1} + 2(-2+s) - 3 = \frac{4 + (2s-7)(s-1)}{s-1}$$

Since $2s^2 - 3s - 2 = (s-2)(2s+1)$,

$$Y(s) = \frac{4 + (2s-7)(s-1)}{2(s-2)(s-1)(s+1/2)} = \frac{1}{5} \frac{1}{s-2} - \frac{4}{3} \frac{1}{s-1} + \frac{32}{15} \frac{1}{s+1/2}$$

$$\text{and } y = \frac{1}{5}e^{2t} - \frac{4}{3}e^t + \frac{32}{15}e^{-t/2}.$$

8.3.34.

$$(2s^2 + 2s + 1)Y(s) = \frac{2}{s^2} + 2(-1+s) + 2 = \frac{2}{s^2} + 2s.$$

Since $2s^2 + 2s + 1 = 2((s+1/2)^2 + 1/4)$,

$$Y(s) = \frac{1}{s^2((s+1/2)^2 + 1/4)} + \frac{s}{((s+1/2)^2 + 1/4)}. \quad (\text{A})$$

$$\frac{s}{((s+1/2)^2 + 1/4)} \leftrightarrow e^{-t/2}(\cos(t/2) - \sin(t/2)). \quad (\text{B})$$

$$\frac{1}{s^2((s+1/2)^2 + 1/4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C(s+1/2)+D}{((s+1/2)^2 + 1/4)}$$

where

$$(As+B)((s+1/2)^2 + 1/4) + (C(s+1/2)+D)s^2 = 1.$$

$$\begin{aligned}
B &= 2 & (\text{when } s = 0); \\
-A + 2B + 2D &= 8 & (\text{when } s = -1/2); \\
5A + 10B + 2C + 2D &= 8 & (\text{when } s = 1/2); \\
A + C &= 0 & (\text{equate coefficients of } s^3).
\end{aligned}$$

Solving this system yields $A = -4$, $B = 2$, $C = 4$, $D = 0$. Therefore,

$$\begin{aligned}
\frac{1}{s^2((s+1/2)^2+1/4)} &= -\frac{4}{s} + \frac{2}{s^2} + \frac{4(s+1/2)}{(s+1/2)^2+1/4} \\
&\leftrightarrow -4 + 2t + 4e^{-t/2} \cos(t/2).
\end{aligned}$$

This, (A), and (B) imply that $y = e^{-t/2}(5 \cos(t/2) - \sin(t/2)) + 2t - 4$.

8.3.36.

$$(4s^2 + 4s + 1)Y(s) = \frac{3+s}{s^2+1} + 4(-1+2s) + 8 = \frac{3+s}{s^2+1} + 4(-1+2s) + 8s + 4.$$

Since $4s^2 + 4s + 1 = 4(s+1/2)^2$,

$$\begin{aligned}
Y(s) &= \frac{3+s}{4(s+1/2)^2(s^2+1)} + \frac{2}{s+1/2}. \\
\frac{3+s}{4(s+1/2)^2(s^2+1)} &= \frac{A}{s+1/2} + \frac{B}{(s+1/2)^2} + \frac{Cs+D}{s^2+1}
\end{aligned} \tag{A}$$

where

$$\begin{aligned}
(A(s+1/2) + B)(s^2+1) + (Cs+D)(s+1/2)^2 &= \frac{3+s}{4}. \\
10B &= 5 & (\text{when } s = -1/2); \\
2A + 4B + D &= 3 & (\text{when } s = 0); \\
12A + 8B + 9C + 9D &= 4 & (\text{when } s = 1); \\
A + C &= 0 & (\text{equate coefficients of } s^3).
\end{aligned}$$

Solving this system yields $A = 3/5$, $B = 1/2$, $C = -3/5$, $D = -1/5$. Therefore,

$$\begin{aligned}
\frac{3+s}{4(s+1/2)^2(s^2+1)} &= \frac{3}{5} \frac{1}{s+1/2} + \frac{1}{2} \frac{1}{(s+1/2)^2} - \frac{1}{5} \frac{3s+1}{s^2+1}. \\
&\leftrightarrow \frac{3}{5} e^{-t/2} + \frac{1}{2} t e^{-t/2} - \frac{1}{5} (3 \cos t + \sin t).
\end{aligned}$$

Since $\frac{2}{s+1/2} \leftrightarrow 2e^{-t/2}$, this and (A) imply that $y = \frac{e^{-t/2}}{10}(5t+26) - \frac{1}{5}(3 \cos t + \sin t)$.

8.3.38. Transforming the initial value problem

$$ay'' + by' + cy = 0, \quad y(0) = 1, \quad y'(0) = 0$$

yields $(as^2 + bs + c)Y(s) = as + b$, so $Y(s) = \frac{as+b}{as^2+bs+c}$. Therefore,

$y_1 = \mathcal{L}^{-1}\left(\frac{as+b}{as^2+bs+c}\right)$ satisfies the initial conditions $y_1(0) = 1$, $y_1'(0) = 0$.

Transforming the initial value problem

$$ay'' + by' + cy = 0, \quad y(0) = 0, \quad y'(0) = 1$$

yields $(as^2 + bs + c)Y(s) = a$, so $Y(s) = \frac{a}{as^2 + bs + c}$. Therefore, $y_2 = \mathcal{L}^{-1}\left(\frac{a}{as^2 + bs + c}\right)$ satisfies the initial conditions $y_1(0) = 0$, $y_1'(0) = 1$.

8.4 The Unit Step Function

8.4.2.

$$\mathcal{L}(f) = \int_0^\infty e^{-st} f(t) dt = \int_0^1 e^{-st} t dt + \int_1^\infty e^{-st} dt. \quad (\text{A})$$

To relate the first term to a Laplace transform we add and subtract $\int_1^\infty e^{-st} t dt$ in (A) to obtain

$$\mathcal{L}(f) = \int_0^\infty e^{-st} t dt + \int_1^\infty e^{-st} (1 - t) dt = \mathcal{L}(t) - \int_1^\infty e^{-st} (t - 1) dt. \quad (\text{B})$$

Letting $t = x + 1$ in the last integral yields

$$\int_1^\infty e^{-st} (t - 1) dt = - \int_0^\infty e^{-s(x+1)} x dx = e^{-s} \mathcal{L}(t).$$

This and (B) imply that $\mathcal{L}(f) = (1 - e^{-s})\mathcal{L}(t) = \frac{1 - e^{-s}}{s^2}$.

Alternatively, $f(t) = t - u(t-1)(t-1) \leftrightarrow (1 - e^{-s})L(t) = \frac{1 - e^{-s}}{s^2}$.

8.4.4.

$$\mathcal{L}(f) = \int_0^\infty e^{-st} f(t) dt = \int_0^1 e^{-st} dt + \int_1^\infty e^{-st} (t + 2) dt. \quad (\text{A})$$

To relate the first term to a Laplace transform we add and subtract $\int_1^\infty e^{-st} dt$ in (A) to obtain

$$\mathcal{L}(f) = \int_0^\infty e^{-st} dt + \int_1^\infty e^{-st} (t + 1) dt = \mathcal{L}(t) + \int_1^\infty e^{-st} (t + 1) dt. \quad (\text{B})$$

Letting $t = x + 1$ in the last integral yields

$$\int_1^\infty e^{-st} (t + 1) dt = \int_0^\infty e^{-s(x+1)} (x + 2) dx = e^{-s} \mathcal{L}(t + 2).$$

This and (B) imply that $\mathcal{L}(f) = \mathcal{L}(1) + e^{-s} \mathcal{L}(t + 2) = \frac{1}{s} + e^{-s} \left(\frac{1}{s^2} + \frac{2}{s} \right)$.

Alternatively,

$$f(t) = 1 + u(t-1)(t+1) \leftrightarrow \mathcal{L}(1) + e^{-s} L(t+2) = \frac{1}{s} + e^{-s} \left(\frac{1}{s^2} + \frac{2}{s} \right).$$

8.4.6.

$$\mathcal{L}(f) = \int_0^\infty e^{-st} f(t) dt = \int_0^1 e^{-st} t^2 dt = \mathcal{L}(t^2) - \int_1^\infty t^2 dt. \quad (\text{A})$$

Letting $t = x + 1$ in the last integral yields

$$\int_1^\infty e^{-st} t^2 dt = \int_0^\infty e^{-s(x+1)} (t^2 + 2t + 1) dx = e^{-s} \mathcal{L}(t^2 + 2t + 1).$$

This and (A) imply that

$$\mathcal{L}(f) = \mathcal{L}(t^2) + e^{-s} \mathcal{L}(t^2 + 2t + 1) = \frac{2}{s^3} - e^{-s} \left(\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s} \right).$$

Alternatively,

$$f(t) = t^2 (1 - u(t-1)) \leftrightarrow \mathcal{L}(t^2) + e^{-s} \mathcal{L}(t^2 + 2t + 1) = \frac{2}{s^3} - e^{-s} \left(\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s} \right)$$

8.4.8. $f(t) = t^2 + 2 + u(t-1)(t - t^2 - 2)$. Since $t^2 + 2 \leftrightarrow \frac{2}{s^3} + \frac{2}{s}$ and

$$\begin{aligned} \mathcal{L}(u(t-1)(t - t^2 - 2)) &= e^{-s} L((t+1) - (t+1)^2 - 2) \\ &= -e^{-s} L(t^2 + t + 2) = -e^{-s} \left(\frac{2}{s^3} + \frac{1}{s^2} + \frac{2}{s} \right), \end{aligned}$$

it follows that $F(s) = \frac{2}{s^3} + \frac{2}{s} - e^{-s} \left(\frac{2}{s^3} + \frac{1}{s^2} + \frac{2}{s} \right)$.

$$\begin{aligned} 8.4.10. \quad f(t) &= e^{-t} + u(t-1)(e^{-2t} - e^{-t}) \leftrightarrow \mathcal{L}(e^{-t}) + e^{-s} L(e^{-2(t+1)}) - e^{-s} \mathcal{L}(e^{-t-1}) = \\ &= \mathcal{L}(e^{-t}) + e^{-(s+2)} L(e^{-2t}) - e^{-(s+1)} \mathcal{L}(e^{-t}) = \frac{1 - e^{-(s+1)}}{s+1} + \frac{e^{-(s+2)}}{s+2}. \end{aligned}$$

8.4.12.

$$f(t) = [u(t-1) - u(t-2)]t \leftrightarrow e^{-s} \mathcal{L}(t+1) - e^{-2s} \mathcal{L}(t+2) = e^{-s} \left(\frac{1}{s^2} + \frac{1}{s} \right) - e^{-2s} \left(\frac{1}{s^2} + \frac{2}{s} \right).$$

8.4.14.

$$\begin{aligned} f(t) &= t - 2u(t-1)(t-1) + u(t-2)(t+4) \leftrightarrow \frac{1}{s^2} - 2e^{-s} \mathcal{L}(t) + e^{-2s} \mathcal{L}(t+6) \\ &= \frac{1}{s^2} - \frac{2e^{-s}}{s^2} + e^{-2s} \left(\frac{1}{s^2} + \frac{6}{s} \right). \end{aligned}$$

$$\begin{aligned} 8.4.16. \quad f(t) &= 2 - 2u(t-1)t + u(t-3)(5t-2) \leftrightarrow \mathcal{L}(2) - 2e^{-s} \mathcal{L}(t+1) + e^{-3s} L(5t+13) = \\ &= \frac{2}{s} - e^{-s} \left(\frac{2}{s^2} + \frac{2}{s} \right) + e^{-3s} \left(\frac{5}{s^2} + \frac{13}{s} \right). \end{aligned}$$

$$\begin{aligned} 8.4.18. \quad f(t) &= (t+1)^2 + u(t-1)((t+2)^2 - (t+1)^2) = t^2 + 2t + 1 + u(t-1)(2t+3) \leftrightarrow \\ &= \mathcal{L}(t^2 + 2t + 1) + e^{-s} \mathcal{L}(2t+5) = \frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s} + e^{-s} \left(\frac{2}{s^2} + \frac{5}{s} \right). \end{aligned}$$

$$8.4.20. \frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1} \leftrightarrow 1 - e^{-t} \Rightarrow e^{-s} \frac{1}{s(s+1)} \leftrightarrow u(t-1)(1 - e^{-(t-1)}) = \begin{cases} 0, & 0 \leq t < 1, \\ 1 - e^{-(t-1)}, & t \geq 1. \end{cases}$$

8.4.22.

$$\begin{aligned} \frac{3}{s} - \frac{1}{s^2} &\leftrightarrow 3 - t \Rightarrow e^{-s} \left(\frac{3}{s} - \frac{1}{s^2} \right) \leftrightarrow u(t-1)(3 - (t-1)) = u(t-1)(4-t); \\ \frac{1}{s} + \frac{1}{s^2} &\leftrightarrow 1 + t \Rightarrow e^{-3s} \left(\frac{1}{s} + \frac{1}{s^2} \right) \leftrightarrow u(t-3)(1 + (t-3)) = u(t-3)(t-2); \end{aligned}$$

therefore

$$h(t) = 2 + t + u(t-1)(4-t) + u(t-3)(t-2) = \begin{cases} 2+t, & 0 \leq t < 1, \\ 6, & 1 \leq t < 3, \\ t+4, & t \geq 3. \end{cases}$$

8.4.24.

$$\frac{1-2s}{s^2+4s+5} = \frac{5-2(s+2)}{(s+2)^2+1} \leftrightarrow e^{-2t}(5\sin t - 2\cos t);$$

therefore,

$$\begin{aligned} h(t) &= u(t-\pi)e^{-2(t-\pi)}(5\sin(t-\pi) - 2\cos(t-\pi)) \\ &= u(t-\pi)e^{-2(t-\pi)}(2\cos t - 5\sin t) \\ &= \begin{cases} 0, & 0 \leq t < \pi, \\ e^{-2(t-\pi)}(2\cos t - 5\sin t), & t \geq \pi. \end{cases} \end{aligned}$$

$$8.4.26. \text{ Denote } F(s) = \frac{3(s-3)}{(s+1)(s-2)} - \frac{s+1}{(s-1)(s-2)}. \text{ Since } \frac{3(s-3)}{(s+1)(s-2)} = \frac{4}{s+1} - \frac{1}{s-2} \text{ and } \frac{s+1}{(s-1)(s-2)} = \frac{3}{s-2} - \frac{2}{s-1}, F(s) = \frac{4}{s+1} - \frac{4}{s-2} + \frac{2}{s-1} \leftrightarrow 4e^{-t} - 4e^{2t} + 2e^t. \text{ Therefore,}$$

$$e^{-2s}F(s) \leftrightarrow u(t-2)(4e^{-(t-2)} - 4e^{2(t-2)} + 2e^{(t-2)}) = \begin{cases} 0, & 0 \leq t < 2, \\ 4e^{-(t-2)} - 4e^{2(t-2)} + 2e^{(t-2)}, & t \geq 2. \end{cases}$$

8.4.28.

$$\begin{aligned} \frac{3}{s} - \frac{1}{s^3} &\leftrightarrow 3 - \frac{t^2}{2} \Rightarrow e^{-2s} \left(\frac{3}{s} - \frac{1}{s^3} \right) \leftrightarrow u(t-2) \left(3 - \frac{(t-2)^2}{2} \right) = u(t-2) \left(-\frac{t^2}{2} + 2t + 1 \right); \\ \frac{1}{s^2} &\leftrightarrow t \Rightarrow \frac{e^{-4s}}{s^2} \leftrightarrow u(t-4)(t-4); \end{aligned}$$

therefore

$$\begin{aligned} h(t) &= 1 - t^2 + u(t-2) \left(-\frac{t^2}{2} + 2t + 1 \right) + u(t-4)(t-4) \\ &= \begin{cases} 1 - t^2, & 0 \leq t < 2 \\ -\frac{3t^2}{2} + 2t + 2, & 2 \leq t < 4, \\ -\frac{3t^2}{2} + 3t - 2, & t \geq 4. \end{cases} \end{aligned}$$

8.4.30. Let T be an arbitrary positive number. Since $\lim_{m \rightarrow \infty} t_m = \infty$, only finitely many members of $\{t_m\}$ are in $[0, T]$. Since f_m is continuous on $[t_m, \infty)$ for each m , f is piecewise continuous on $[0, T]$. If $t_M \leq t < t_{M+1}$, then $u(t - t_m) = 1$ if $m \leq M$, while $u(t - t_m) = 0$ if $m > M$. Therefore, $f(t) = f_0(t) + \sum_{m=1}^M (f_m(t) - f_{m-1}(t)) = f_M(t)$

8.4.32. Since $\sum_{m=0}^{\infty} e^{-\rho K m}$ converges if $\rho > 0$, $\sum_{m=0}^{\infty} e^{-\rho t_m}$ converges if $\rho > 0$, by the comparison test. Therefore, (C) of Exercise 8.3.31 holds if $s > s_0 + \rho$ if ρ is any positive number. This implies that it holds if $s > s_0$.

8.4.34. Let $t_m = m$ and $f_m(t) = (-1)^m$, $m = 0, 1, 2, \dots$. Then $f_m(t) - f_{m-1}(t) = (-1)^m 2$, so $f(t) = 1 + 2 \sum_{m=1}^{\infty} (-1)^m u(t - m)$ and $F(s) = \frac{1}{s} \left(1 + 2 \sum_{m=1}^{\infty} (-1)^m e^{-ms} \right)$. Substituting $x = e^{-s}$ in the identity $\sum_{m=1}^{\infty} (-1)^m x^m = -\frac{x}{1+x}$ ($|x| < 1$) yields $F(s) = \frac{1}{s} \left(1 - \frac{2e^{-s}}{1+e^{-s}} \right) = \frac{1}{s} \frac{1-e^{-s}}{1+e^{-s}}$.

8.4.36. Let $t_m = m$ and $f_m(t) = (-1)^m m$, $m = 0, 1, 2, \dots$. Then $f_m(t) - f_{m-1}(t) = (-1)^m (2m-1)$, so $f(t) = \sum_{m=1}^{\infty} (-1)^m (2m-1) u(t-m)$ and $F(s) = \frac{1}{s} \sum_{m=1}^{\infty} (-1)^m (2m-1) e^{-ms}$. Substituting $x = e^{-s}$ in the identities $\sum_{m=1}^{\infty} (-1)^m x^m = -\frac{x}{1+x}$ and $\sum_{m=1}^{\infty} (-1)^m m x^m = -\frac{x}{(1+x)^2}$ ($|x| < 1$) yields $F(s) = \frac{1}{s} \left[\frac{e^{-s}}{1+e^{-s}} - \frac{2e^{-s}}{(1+e^{-s})^2} \right] = \frac{1}{s} \frac{(1-e^{-s})}{(1+e^{-s})^2}$.

8.5 Constant Coefficient Equations with Piecewise Continuous Forcing Functions

8.5.2. $y'' + y = 3 + u(t-4)(2t-8)$, $y(0) = 1$, $y'(0) = 0$. Since

$$\begin{aligned} \mathcal{L}(u(t-4)(2t-8)) &= e^{-4s} \mathcal{L}(2(t+4)-8) = e^{-4s} \mathcal{L}(2t) = \frac{2e^{-4s}}{s^2}, \\ (s^2 + 1)Y(s) &= \frac{3}{s} + \frac{2e^{-4s}}{s^2} + s. \end{aligned}$$

$$\begin{aligned}
Y(s) &= \frac{3}{s(s^2+1)} + \frac{2e^{-4s}}{s^2(s^2+1)} + \frac{s}{s^2+1} \\
&= 3\left(\frac{1}{s} - \frac{s}{s^2+1}\right) + 2e^{-4s}\left(\frac{1}{s^2} - \frac{1}{s^2+1}\right) + \frac{s}{s^2+1} \\
&= \frac{3}{s} - \frac{2s}{s^2+1} + 2e^{-4s}\left(\frac{1}{s^2} - \frac{1}{s^2+1}\right).
\end{aligned}$$

Since

$$\frac{1}{s^2} - \frac{1}{s^2+1} \leftrightarrow t - \sin t \Rightarrow e^{-4s}\left(\frac{1}{s^2} - \frac{1}{s^2+1}\right) \leftrightarrow u(t-4)(t-4 - \sin(t-4)),$$

$$y = 3 - 2\cos t + 2u(t-4)(t-4 - \sin(t-4)).$$

8.5.4. $y'' - y = e^{2t} + u(t-2)(1 - e^{2t})$, $y(0) = 3$, $y'(0) = -1$. Since

$$\begin{aligned}
\mathcal{L}(u(t-2)(1 - e^{2t})) &= e^{-2s}\mathcal{L}(1 - e^{2(t+2)}) = e^{-2s}\left(\frac{1}{s} - \frac{e^4}{s-2}\right), \\
(s^2 - 1)Y(s) &= \frac{1}{s-2} + e^{-2s}\left(\frac{1}{s} - \frac{e^4}{s-2}\right) + (-1 + 3s).
\end{aligned}$$

Therefore,

$$\begin{aligned}
Y(s) &= \frac{1}{(s-1)(s+1)(s-2)} + \frac{3s-1}{(s-1)(s+1)} \\
&\quad + e^{-2s}\left(\frac{1}{s(s-1)(s+1)} - \frac{e^4}{(s-1)(s+1)(s-2)}\right). \\
\frac{1}{(s-1)(s+1)(s-2)} &= -\frac{1}{2}\frac{1}{s-1} + \frac{1}{6}\frac{1}{s+1} + \frac{1}{3}\frac{1}{s-2} \\
&\leftrightarrow -\frac{1}{2}e^t + \frac{1}{6}e^{-t} + \frac{1}{3}e^{2t}; \\
\frac{e^{-2s}e^4}{(s-1)(s+1)(s-2)} &\leftrightarrow u(t-2)\left(-\frac{1}{2}e^{t+2} + \frac{1}{6}e^{-(t-6)} + \frac{1}{3}e^{2t}\right); \\
\frac{1}{s(s-1)(s+1)} &= -\frac{1}{s} + \frac{1}{2}\frac{1}{s-1} + \frac{1}{2}\frac{1}{s+1} \leftrightarrow -1 + \frac{1}{2}e^t + \frac{1}{2}e^{-t}; \\
\frac{e^{-2s}}{s(s-1)(s+1)} &\leftrightarrow u(t-2)\left(-1 + \frac{1}{2}e^{t-2} + \frac{1}{2}e^{-(t-2)}\right); \\
\frac{3s-1}{(s-1)(s+1)} &= \frac{1}{s-1} + \frac{2}{s+1} \leftrightarrow e^t + 2e^{-t}.
\end{aligned}$$

Therefore,

$$y = \frac{1}{2}e^t + \frac{13}{6}e^{-t} + \frac{1}{3}e^{2t} + u(t-2)\left(-1 + \frac{1}{2}e^{t-2} + \frac{1}{2}e^{-(t-2)} + \frac{1}{2}e^{t+2} - \frac{1}{6}e^{-(t-6)} - \frac{1}{3}e^{2t}\right).$$

8.5.6. Note that $|\sin t| = \sin t$ if $0 \leq t < \pi$, while $|\sin t| = -\sin t$ if $\pi \leq t < 2\pi$. Rewrite the initial value problem as

$$y'' + 4y = \sin t - 2u(t-\pi)\sin t + u(t-2\pi)\sin t, \quad y(0) = -3, \quad y'(0) = 1.$$

Since

$$\mathcal{L}(u(t - \pi) \sin t) = e^{-\pi s} L(\sin(t + \pi)) = -e^{-\pi s} \mathcal{L}(\sin t)$$

and

$$\begin{aligned} \mathcal{L}(u(t - 2\pi) \sin t) &= e^{-2\pi s} L(\sin(t + 2\pi)) = e^{-2\pi s} \mathcal{L}(\sin t), \\ (s^2 + 4)Y(s) &= \frac{1 + 2e^{-\pi s} + e^{-2\pi s}}{(s^2 + 1)} + 1 - 3s, \text{ so } Y(s) = \frac{1 + 2e^{-\pi s} + e^{-2\pi s}}{(s^2 + 1)(s^2 + 4)} + \frac{1 - 3s}{s^2 + 4}. \\ \frac{1}{(s^2 + 1)(s^2 + 4)} &= \left(\frac{1}{s^2 + 1} - \frac{1}{s^2 + 4} \right) \leftrightarrow \frac{1}{3} \sin t - \frac{1}{6} \sin 2t; \end{aligned}$$

therefore

$$\begin{aligned} \frac{e^{-\pi s}}{(s^2 + 1)(s^2 + 4)} &\leftrightarrow u(t - \pi) \left(\frac{1}{3} \sin(t - \pi) - \frac{1}{6} \sin 2(t - \pi) \right) \\ &= -u(t - \pi) \left(\frac{1}{3} \sin t + \frac{1}{6} \sin 2t \right) \end{aligned}$$

and

$$\begin{aligned} \frac{e^{-2\pi s}}{(s^2 + 1)(s^2 + 4)} &\leftrightarrow u(t - 2\pi) \left(\frac{1}{3} \sin(t - 2\pi) - \frac{1}{6} \sin 2(t - 2\pi) \right) \\ &= u(t - 2\pi) \left(\frac{1}{3} \sin t - \frac{1}{6} \sin 2t \right); \end{aligned}$$

therefore

$$y = \frac{1}{3} \sin 2t - 3 \cos 2t + \frac{1}{3} \sin t - 2u(t - \pi) \left(\frac{1}{3} \sin t + \frac{1}{6} \sin 2t \right) + u(t - 2\pi) \left(\frac{1}{3} \sin t - \frac{1}{6} \sin 2t \right).$$

8.5.8. $y'' + 9y = \cos t + u(t - 3\pi/2)(\sin t - \cos t)$, $y(0) = 0$, $y'(0) = 0$. Since

$$\begin{aligned} \mathcal{L}(u(t - 3\pi/2)(\sin t - \cos t)) &= e^{-3\pi s/2} L(\sin(t + 3\pi/2) - \cos(t + 3\pi/2)) \\ &\quad - e^{-3\pi s/2} \mathcal{L}(\cos t + \sin t), \\ (s^2 + 9)Y(s) &= \frac{1}{s^2 + 1} - e^{-3\pi s/2} \frac{s + 1}{s^2 + 1}, \quad \text{so} \\ Y(s) &= \frac{1}{(s^2 + 1)(s^2 + 9)} - e^{-3\pi s/2} \frac{s + 1}{(s^2 + 1)(s^2 + 9)}. \\ \frac{1}{(s^2 + 1)(s^2 + 9)} &= \frac{1}{8} \left(\frac{1}{s^2 + 1} - \frac{1}{s^2 + 9} \right) \leftrightarrow \frac{1}{8} \left(\sin t - \frac{1}{3} \sin 3t \right) \quad \text{and} \\ \frac{s}{(s^2 + 1)(s^2 + 9)} &= \frac{1}{8} \left(\frac{s}{s^2 + 1} - \frac{s}{s^2 + 9} \right) \leftrightarrow \frac{1}{8} (\cos t - \cos 3t). \\ \frac{s + 1}{(s^2 + 1)(s^2 + 9)} &= \frac{s + 1}{(s^2 + 1)(s^2 + 9)} = \frac{s + 1}{8} \left(\frac{s + 1}{s^2 + 1} - \frac{s + 1}{s^2 + 9} \right) \\ &\leftrightarrow \frac{1}{8} \left(\cos t + \sin t - \cos 3t - \frac{1}{3} \sin 3t \right), \quad \text{so} \\ e^{-3\pi s/2} \frac{s + 1}{(s^2 + 1)(s^2 + 9)} &\leftrightarrow \frac{u(t - 3\pi/2)}{8} (\cos(t - 3\pi/2) + \sin(t - 3\pi/2)) \end{aligned}$$

$$\begin{aligned}
& -\cos 3(t - 3\pi/2) - \frac{1}{3} \sin 3(t - \pi/2) \\
& = \frac{u(t - 3\pi/2)}{8} \left(\sin t - \cos t + \sin 3t - \frac{1}{3} \cos 3t \right).
\end{aligned}$$

Therefore, $y = \frac{1}{8} (\cos t - \cos 3t) - \frac{1}{8} u(t - 3\pi/2) \left(\sin t - \cos t + \sin 3t - \frac{1}{3} \cos 3t \right)$.

8.5.10. $y'' + y = t - 2u(t - \pi)t$, $y(0) = 0$, $y'(0) = 0$. Since

$$\begin{aligned}
\mathcal{L}(u(t - \pi)t) &= e^{-\pi s} \mathcal{L}(t + \pi) = e^{-\pi s} \left(\frac{1}{s^2} + \frac{\pi}{s} \right), \\
(s^2 + 1)Y(s) &= \frac{1}{s^2} - 2e^{-\pi s} \left(\frac{1}{s^2} + \frac{\pi}{s} \right);
\end{aligned}$$

$$\begin{aligned}
Y(s) &= \frac{1}{s^2(s^2 + 1)} - 2e^{-\pi s} \left(\frac{1}{s^2(s^2 + 1)} + \frac{\pi}{s(s^2 + 1)} \right) \\
&= \left(\frac{1}{s^2} - \frac{1}{s^2 + 1} \right) - 2e^{-\pi s} \left(\frac{1}{s^2} - \frac{1}{s^2 + 1} \right) - 2\pi e^{-\pi s} \left(\frac{1}{s} - \frac{s}{s^2 + 1} \right).
\end{aligned}$$

Since

$$\begin{aligned}
\frac{1}{s^2} - \frac{1}{s^2 + 1} &\leftrightarrow t - \sin t \Rightarrow e^{-\pi s} \left(\frac{1}{s^2} - \frac{1}{s^2 + 1} \right) \\
&\leftrightarrow u(t - \pi)(t - \pi - \sin(t - \pi)) = u(t - \pi)(t - \pi + \sin t)
\end{aligned}$$

and

$$\begin{aligned}
\frac{1}{s} - \frac{s}{s^2 + 1} &\leftrightarrow 1 - \cos t \Rightarrow e^{-\pi s} \left(\frac{1}{s} - \frac{s}{s^2 + 1} \right) \\
&\leftrightarrow u(t - \pi)(1 - \cos(t - \pi)) = u(t - \pi)(1 + \cos t),
\end{aligned}$$

$$y = t - \sin t - 2u(t - \pi)(t + \sin t + \pi \cos t).$$

8.5.12. $y'' + y = t - 3u(t - 2\pi)t$, $y(0) = 1$, $y'(0) = 2$;

$$\begin{aligned}
\mathcal{L}(u(t - 2\pi)t) &= e^{-2\pi s} \mathcal{L}(t + 2\pi) = e^{-2\pi s} \left(\frac{1}{s^2} + \frac{2\pi}{s} \right); \\
(s^2 + 1)Y(s) &= \frac{1 - 3e^{-2\pi s}}{s^2} - \frac{6\pi e^{-2\pi s}}{s} + 2 + s; \\
Y(s) &= \frac{1 - 3e^{-2\pi s}}{s^2(s^2 + 1)} - \frac{6\pi e^{-2\pi s}}{s(s^2 + 1)} + \frac{2 + s}{s^2 + 1}; \\
\frac{1}{s^2(s^2 + 1)} &= \frac{1}{s^2} - \frac{1}{s^2 + 1} \leftrightarrow t - \sin t; \\
\frac{e^{-2\pi s}}{s^2(s^2 + 1)} &\leftrightarrow u(t - 2\pi)((t - 2\pi - \sin(t - 2\pi)) = u(t - 2\pi)(t - 2\pi - \sin t); \\
\frac{1}{s(s^2 + 1)} &= \frac{1}{s} - \frac{s}{s^2 + 1} \leftrightarrow 1 - \cos t;
\end{aligned}$$

$$\frac{e^{-2\pi s}}{s(s^2+1)} \leftrightarrow u(t-2\pi)(1-\cos(t-2\pi)) = u(t-2\pi)(1-\cos t);$$

$$\frac{2+s}{s^2+1} \leftrightarrow 2\sin t + \cos t;$$

$$y = t + \sin t + \cos t - u(t-2\pi)(3t-3\sin t-6\pi\cos t).$$

8.5.14. $y'' - 4y' + 3y = -1 + 2u(t-1)$, $y(0) = 0$, $y'(0) = 0$;

$$(s^2 - 4s + 3)Y(s) = \frac{-1 + 2e^{-s}}{s}; \quad Y(s) = \frac{-1 + 2e^{-s}}{s(s-1)(s-3)};$$

$$\frac{1}{s(s-1)(s-3)} = \frac{1}{3s} + \frac{1}{6} \frac{1}{s-3} - \frac{1}{2} \frac{1}{s-1} \leftrightarrow \frac{1}{3} + \frac{1}{6}e^{3t} - \frac{1}{2}e^t;$$

$$\frac{e^{-s}}{s(s-1)(s-3)} \leftrightarrow u(t-1) \left(\frac{1}{3} + \frac{1}{6}e^{3(t-1)} - \frac{1}{2}e^{t-1} \right);$$

$$y = -\frac{1}{3} - \frac{1}{6}e^{3t} + \frac{1}{2}e^t + u(t-1) \left(\frac{2}{3} + \frac{1}{3}e^{3(t-1)} - e^{t-1} \right).$$

8.5.16. $y'' + 2y' + y = 4e^t - 4u(t-1)e^t$, $y(0) = 0$, $y'(0) = 0$. Since

$$\mathcal{L}(4u(t-1)e^t) = e^{-s}L(4e^{(t+1)}) = \frac{4e^{-s+1}}{s-1},$$

$$(s^2 + 2s + 1)Y(s) = \frac{4}{s-1} - \frac{4e^{-s+1}}{s-1}, \quad \text{so}$$

$$Y(s) = \frac{4}{(s-1)(s+1)^2} - \frac{4e^{-s+1}}{(s-1)(s+1)^2}.$$

$$\frac{1}{(s-1)(s+1)^2} = \frac{A}{s-1} + \frac{B}{s+1} + \frac{C}{(s+1)^2},$$

where

$$A(s+1)^2 + B(s-1)(s+1) + C(s-1) = 4.$$

$$A = 1 \quad (\text{when } s = 1);$$

$$C = -2 \quad (\text{when } s = -1);$$

$$A + B = 0 \quad (\text{equate coefficients of } s^2).$$

Solving this system yields $A = 1$, $B = -1$, $C = -2$. Therefore,

$$\frac{1}{(s-1)(s+1)^2} = \frac{1}{s-1} - \frac{1}{s+1} - \frac{2}{(s+1)^2} \quad \text{and}$$

$$\begin{aligned} y &= e^t - e^{-t} - 2te^{-t} - eu(t-1)(e^{t-1} - e^{-(t-1)} - 2(t-1)e^{-(t-1)}) \\ &= e^t - e^{-t} - 2te^{-t} - u(t-1)(e^t - e^{-(t-2)} - 2(t-1)e^{-(t-2)}). \end{aligned}$$

8.5.18. $y'' - 4y' + 4y = e^{2t} - 2u(t-2)e^{2t}$, $y(0) = 0$, $y'(0) = -1$. Since

$$\mathcal{L}(u(t-2)e^{2t}) = e^{-2s}L(e^{2t+4}) = \frac{e^{-2s+4}}{s-2},$$

$$(s^2 - 4s + 4)Y(s) = \frac{1}{s-2} - \frac{2e^{-2s+4}}{s-2} - 1, \quad \text{so}$$

$$Y(s) = \frac{1}{(s-2)^3} - \frac{2e^{-2s+4}}{(s-2)^3} - \frac{1}{(s-2)^2}.$$

$$\frac{1}{(s-2)^3} \leftrightarrow \frac{t^2 e^{2t}}{2} \Rightarrow \frac{e^{-2s+4}}{(s-2)^3} \leftrightarrow \frac{e^4}{2} u(t-2) e^{2(t-2)} (t-2)^2 = u(t-2) \frac{(t-2)^2 e^{2t}}{2};$$

$$\text{therefore } y = \frac{t^2 e^{2t}}{2} - t e^{2t} - u(t-2)(t-2)^2 e^{2t}.$$

$$8.5.20. \quad y'' + 2y' + 2y = 1 + u(t-2\pi)(t-1) - u(t-3\pi)(t+1), \quad y(0) = 2, \quad y'(0) = -1;$$

$$\mathcal{L}(u(t-2\pi)(t-1)) = e^{-2\pi s} \mathcal{L}((t+2\pi-1)) = e^{-2\pi s} \left(\frac{1}{s^2} + \frac{2\pi-1}{s} \right);$$

$$\mathcal{L}(u(t-3\pi)(t+1)) = e^{-3\pi s} \mathcal{L}((t+3\pi+1)) = e^{-3\pi s} \left(\frac{1}{s^2} + \frac{3\pi+1}{s} \right);$$

$$(s^2 + 2s + 2)Y(s) = \frac{1}{s} + e^{-2\pi s} \left(\frac{1}{s^2} + \frac{2\pi-1}{s} \right) - e^{-3\pi s} \left(\frac{1}{s^2} + \frac{3\pi+1}{s} \right) + (-1 + 2s) + 4.$$

$$\text{Let } G(s) = \frac{1}{s(s^2 + 2s + 2)}, \quad H(s) = \frac{1}{s(s^2 + 2s + 2)}; \text{ then}$$

$$Y(s) = Y_1(s) + e^{-2\pi s} Y_2(s) - e^{-3\pi s} Y_3(s), \quad (\text{A})$$

where

$$Y_1(s) = G(s) + \frac{2s+3}{s^2+2s+2}, \quad (\text{B})$$

$$Y_2(s) = H(s) + (2\pi-1)G(s), \quad (\text{C})$$

$$Y_3(s) = H(s) + (3\pi+1)G(s). \quad (\text{D})$$

Let $y_i(t) = \mathcal{L}^{-1}(Y_i(s))$, ($i = 1, 2, 3$). From (A),

$$y(t) = y_1(t) + u(t-2\pi)y_2(t-2\pi) - u(t-3\pi)y_3(t-3\pi). \quad (\text{E})$$

Find $\mathcal{L}^{-1}(G(s))$:

$$G(s) = \frac{A}{s} + \frac{B(s+1)+C}{(s+1)^2+1}$$

where $A((s+1)^2+1) + (B(s+1)+C)s = 1$. Setting $s = 0$ yields $A = 1/2$; setting $s = -1$ yields $A - C = 1$, so $C = -1/2$; since $A + B = 0$ (coefficient of x^2), $B = -1/2$. Therefore,

$$G(s) = \frac{1}{2} \left(\frac{1}{s} - \frac{(s+1)+1}{(s+1)^2+1} \right) \leftrightarrow \frac{1}{2} - \frac{1}{2} e^{-t} (\cos t + \sin t). \quad (\text{F})$$

Find $\mathcal{L}^{-1}(H(s))$:

$$H(s) = \frac{A}{s} + \frac{B}{s+2} + \frac{C(s+1)+D}{(s+1)^2+1}$$

where $(As + B)((s + 1)^2 + 1) + (C(s + 1) + D)s^2 = 1$.

$$\begin{aligned} 2B &= 1 & (\text{when } s = 0); \\ -A + B + D &= 1 & (\text{when } s = -1); \\ 5A + 5B + 2C + D &= 1 & (\text{when } s = 1); \\ A + B &= 0 & (\text{equate coefficients of } s^3). \end{aligned}$$

Solving this system yields $A = -1/2$, $B = 1/2$, $C = 1/2$, $D = 0$; therefore

$$H(s) = -\frac{1}{2} \left(\frac{1}{s} - \frac{1}{s^2} - \frac{s+1}{(s+1)^2 + 1} \right) \leftrightarrow -\frac{1}{2}(1 - t - e^{-t} \cos t). \quad (\text{G})$$

Since

$$\frac{2s+3}{s^2+2s+2} = \frac{2(s+1)+1}{(s+1)^2+1} \leftrightarrow e^{-t}(2 \cos t + \sin t),$$

(B) and (F)) imply that

$$y_1(t) = \frac{1}{2}e^{-t}(3 \cos t + \sin t) + \frac{1}{2}. \quad (\text{H})$$

From (C), (F), and (G),

$$y_2(t) = \pi - 1 + \frac{t}{2} + (\pi - 1)e^{-t} \cos t - \frac{2\pi - 1}{2}e^{-t} \sin t,$$

so

$$y_2(t - 2\pi) = - \left(e^{-(t-2\pi)} \left((\pi - 1) \cos t + \frac{2\pi - 1}{2} \sin t \right) + 1 - \frac{t}{2} \right). \quad (\text{I})$$

From (D), (F)), and (G),

$$y_3(t) = \frac{1}{2}(-e^{-t}(3\pi \cos t + (3\pi + 1) \sin t + t + 3\pi)),$$

so

$$y_3(t - 3\pi) = \frac{1}{2}(e^{-(t-3\pi)}(3\pi \cos t + (3\pi + 1) \sin t + t)). \quad (\text{J})$$

Now (E), (20), (I), and (J)

$$y = \frac{1}{2}e^{-t}(3 \cos t + \sin t) + \frac{1}{2} \text{ imply that}$$

$$\begin{aligned} & -u(t - 2\pi) \left(e^{-(t-2\pi)} \left((\pi - 1) \cos t + \frac{2\pi - 1}{2} \sin t \right) + 1 - \frac{t}{2} \right) \\ & - \frac{1}{2}u(t - 3\pi) (e^{-(t-3\pi)}(3\pi \cos t + (3\pi + 1) \sin t + t)). \end{aligned}$$

$$8.5.22. \quad (\text{a}) \quad f(t) = \sum_{n=0}^{\infty} u(t - n\pi); \quad F(s) = \frac{1}{s} \sum_{n=0}^{\infty} e^{-n\pi s}; \quad Y(s) = \frac{1}{s(s^2 + 1)} \sum_{n=0}^{\infty} e^{-n\pi s};$$

$$\frac{1}{s(s^2 + 1)} = \frac{1}{s} - \frac{s}{s^2 + 1} \leftrightarrow 1 - \cos t;$$

$$\frac{e^{-n\pi s}}{s(s^2 + 1)} \leftrightarrow u(t - n\pi)(1 - \cos(t - n\pi)) = u(t - n\pi)(1 - (-1)^n \cos t);$$

$y(t) = \sum_{n=0}^{\infty} u(t - n\pi)(1 - (-1)^n \cos t)$. If $m\pi \leq t < (m+1)\pi$, $y(t) = \sum_{n=0}^m (1 - (-1)^n \cos t)$.
Therefore,

$$y(t) = \begin{cases} 2m+1 - \cos t, & 2m\pi \leq t < (2m+1)\pi \quad (m = 0, 1, \dots) \\ 2m, & (2m-1)\pi \leq t < 2m\pi \quad (m = 1, 2, \dots). \end{cases}$$

$$(b) \quad f(t) = \sum_{n=0}^{\infty} u(t - 2n\pi)t; \quad F(s) = \sum_{n=0}^{\infty} e^{-2n\pi s} \mathcal{L}(t + 2n\pi s) = \sum_{n=0}^{\infty} e^{-2n\pi s} \left(\frac{1}{s^2} + \frac{2n\pi}{s} \right);$$

$$Y(s) = \sum_{n=0}^{\infty} e^{-2n\pi s} Y_n(s), \text{ where } Y_n(s) = \frac{1}{s^2(s^2 + 1)} + \frac{2n\pi}{s(s^2 + 1)} =$$

$$\frac{1}{s^2} - \frac{1}{s^2 + 1} + \frac{2n\pi}{s} - \frac{2n\pi}{s^2 + 1} \leftrightarrow y_n(t) = t - \sin t + 2n\pi - 2n\pi \cos t. \text{ Since}$$

$$\cos(t - 2n\pi) = \cos t \text{ and } \sin(t - 2n\pi) = \sin t,$$

$$e^{-2n\pi s} Y_n(s) \leftrightarrow u(t - 2n\pi) y_n(t) = u(t - 2n\pi)(t - \sin t - 2n\pi \cos t); \text{ therefore}$$

$$y(t) = \sum_{n=0}^{\infty} u(t - 2n\pi)(t - \sin t - 2n\pi \cos t). \text{ If } 2m\pi \leq t < 2(m+1)\pi, \text{ then}$$

$$y(t) = \sum_{n=0}^m (t - \sin t - 2n\pi \cos t) = (m+1)(t - \sin t - m\pi \cos t).$$

$$(c) \quad f(t) = 1 + 2 \sum_{n=1}^{\infty} (-1)^n u(t - n\pi); \quad F(s) = \frac{1}{s} \left(1 + 2 \sum_{n=1}^{\infty} (-1)^n e^{-n\pi s} \right);$$

$$Y(s) = \frac{1}{s(s^2 + 1)} \left(1 + 2 \sum_{n=1}^{\infty} (-1)^n e^{-n\pi s} \right); \quad \frac{1}{s(s^2 + 1)} = \frac{1}{s} - \frac{s}{s^2 + 1} \leftrightarrow 1 - \cos t;$$

$$\frac{e^{-n\pi s}}{s(s^2 + 1)} \leftrightarrow u(t - n\pi)(1 - \cos(t - n\pi)) = u(t - n\pi)(1 - (-1)^n \cos t);$$

$$y(t) = 1 - \cos t + 2 \sum_{n=1}^{\infty} (-1)^n u(t - n\pi)(1 - (-1)^n \cos t). \text{ If } m\pi \leq t < (m+1)\pi,$$

$$y(t) = 1 - \cos t + 2 \sum_{n=1}^m (-1)^n (1 - (-1)^n \cos t) = (-1)^m - (2m+1) \cos t.$$

$$(d) \quad f(t) = \sum_{n=0}^{\infty} u(t - n); \quad F(s) = \frac{1}{s} \sum_{n=0}^{\infty} e^{-ns}; \quad Y(s) = \frac{1}{s(s^2 - 1)} \sum_{n=0}^{\infty} e^{-ns};$$

$$\frac{1}{s(s^2 - 1)} = \frac{1}{2} \frac{1}{s-1} + \frac{1}{2} \frac{1}{s+1} - \frac{1}{s} \leftrightarrow \frac{1}{2}(e^t + e^{-t} - 2);$$

$$\frac{e^{-ns}}{s(s^2 - 1)} \leftrightarrow \frac{u(t - n)}{2} (e^t + e^{-t} - 2); \quad y(t) = \frac{1}{2} \sum_{n=0}^{\infty} u(t - n) (e^{t-n} + e^{-(t-n)} - 2). \text{ If}$$

$$m \leq t < (m+1),$$

$$\begin{aligned} y(t) &= \frac{1}{2} \sum_{n=0}^m (e^{t-n} + e^{-(t-n)} - 2) = \frac{1}{2} (e^{t-m} + e^{-t}) \sum_{n=0}^m e^n - m - 1 \\ &= \frac{1 - e^{m+1}}{2(1 - e)} (e^{t-m} + e^{-t}) - m - 1. \end{aligned}$$

$$\begin{aligned}
\text{(e)} \quad f(t) &= (\sin t + 2 \cos t) \sum_{n=0}^{\infty} u(t - 2n\pi); \quad F(s) = \frac{1+2s}{s^2+1} \sum_{n=0}^{\infty} e^{-2n\pi s}; \\
Y(s) &= \frac{1+2s}{(s^2+1)(s^2+2s+2)} \sum_{n=0}^{\infty} e^{-2n\pi s}; \\
\frac{1+2s}{(s^2+1)(s^2+2s+2)} &= \frac{As+B}{s^2+1} + \frac{C(s+1)+D}{(s+1)^2+1}
\end{aligned}$$

where

$$\begin{aligned}
(As+B)((s+1)^2+1) + (C(s+1)+D)(s^2+1) &= 1+2s. \\
2B+C+D &= 1 \quad (\text{when } s=0); \\
-A+B+2D &= -1 \quad (\text{when } s=-1); \\
5A+5B+4C+2D &= 3 \quad (\text{when } s=1); \\
A+C &= 0 \quad (\text{equate coefficients of } s^3).
\end{aligned}$$

Solving this system yields $A=0$, $B=1$, $C=0$, $D=-1$. Therefore,

$$\begin{aligned}
\frac{1+2s}{(s^2+1)(s^2+2s+2)} &= \frac{1}{s^2+1} - \frac{1}{(s+1)^2+1} \\
&\leftrightarrow (1 - e^{-t}) \sin t.
\end{aligned}$$

Since $\sin(t - 2n\pi) = \sin t$,

$$e^{-2n\pi s} \frac{1+2s}{(s^2+1)(s^2+2s+2)} \leftrightarrow u(t - 2n\pi) (1 - e^{-(t-2n\pi)}) \sin t,$$

so

$$y(t) = \sin t \sum_{n=0}^{\infty} u(t - 2n\pi) (1 - e^{-(t-2n\pi)}).$$

If $2m\pi \leq t < 2(m+1)\pi$,

$$y(t) = \sin t \sum_{n=0}^m (1 - e^{-(t-2n\pi)}) = \left(m+1 - \left(\frac{1 - e^{2(m+1)\pi}}{1 - e^{2\pi}} \right) e^{-t} \right) \sin t.$$

$$\text{(f)} \quad f(t) = \sum_{n=0}^{\infty} u(t - n); \quad F(s) = \frac{1}{s} \sum_{n=0}^{\infty} e^{-ns}; \quad Y(s) = \frac{1}{s(s-1)(s-2)};$$

$$\begin{aligned}
\frac{1}{s(s-1)(s-2)} &= \frac{1}{2s} - \frac{1}{s-1} + \frac{1}{2(s-2)} \leftrightarrow \frac{1}{2} (1 - 2e^t + e^{2t}); \\
\frac{e^{-ns}}{s(s-1)(s-2)} &\leftrightarrow \frac{1}{2} u(t-n) (1 - 2e^{t-n} + e^{2(t-n)}); \\
y(t) &= \frac{1}{2} \sum_{n=0}^{\infty} u(t-n) (1 - 2e^{t-n} + e^{2(t-n)}).
\end{aligned}$$

If $m \leq t < m+1$,

$$y(t) = \sum_{n=0}^m (1 - 2e^{t-n} + e^{2(t-n)}) = \frac{m+1}{2} - e^{t-m} \sum_{n=0}^m e^n + \frac{1}{2} e^{2(t-m)} \sum_{n=0}^m e^{2n}$$

$$= \frac{m+1}{2} - e^{t-m} \frac{1-e^{m+1}}{1-e} + \frac{1}{2} e^{2(t-m)} \frac{1-e^{2m+2}}{1-e^2}.$$

- 8.5.24. (a) The assumptions imply that $y''(t) = \frac{f(t) - by'(t) - cy(t)}{a}$ on (α, t_0) and (t_0, β) ,
 $y''(t_0+) = \frac{f(t_0+) - by'(t_0) - cy(t_0)}{a}$, and $y''(t_0-) = \frac{f(t_0-) - by'(t_0) - cy(t_0)}{a}$. This implies the conclusion.
- (b) Since y'' has a jump discontinuity at t_0 , applying Exercise 8.4.23(c) to y' shows that y' is not differentiable at t_0 . Therefore, y cannot satisfy (A) on (α, β) if f has a jump discontinuity at some t_0 in (α, β) .

8.5.26. If $0 \leq t < t_0$, then $y(t) = z_0(t)$. Therefore, $y(0) = z_0(0) = k_0$ and $y'(0) = z'_0(0) = k_1$, and

$$ay'' + by' + cy = az''_0 + bz'_0 + cz_0 = f_0(t) = f(t), \quad 0 < t < t_0.$$

Now suppose that $1 \leq m \leq n$. For convenience, define $t_{n+1} = \infty$. If $t_m \leq t < t_{m+1}$, then $y(t) = \sum_{k=0}^m z_m(t)$, so

$$ay'' + by' + cy = \sum_{k=0}^m (az''_k + bz'_k + cz_k) = f_0 + \sum_{k=1}^m (f_k - f_{k-1}) = f_m = f, \quad t_m < t < t_{m+1}.$$

Thus, y satisfies $ay'' + by' + cy = f$ on any open interval that does not contain any of the points $t_1, t_2, \dots, t_n$.

Since $z(t_m) = z'(t_m)$ for $m = 1, 2, \dots$, y and y' are continuous on $[0, \infty)$. Since $y''(t) = -(by'(t) + cy(t))/a$ if $t \neq t_m$ ($m = 1, 2, \dots$), y'' has limits from the left at $t_1, \dots, t_n$.

8.6 Convolution

8.6.2. (a) $\sin at \leftrightarrow \frac{a}{s^2 + a^2}$ and $\cos bt \leftrightarrow \frac{s}{s^2 + b^2}$, so $H(s) = \frac{as}{(s^2 + a^2)(s^2 + b^2)}$.

(b) $e^t \leftrightarrow \frac{1}{s-1}$ and $\sin at \leftrightarrow \frac{a}{s^2 + a^2}$, so $H(s) = \frac{a}{(s-1)(s^2 + a^2)}$.

(c) $\sinh at \leftrightarrow \frac{a}{s^2 - a^2}$ and $\cosh at \leftrightarrow \frac{1}{s^2 - a^2}$, so $H(s) = \frac{as}{(s^2 - a^2)^2}$.

(d) $t \sin \omega t \leftrightarrow \frac{2\omega s}{(s^2 + \omega^2)^2}$ and $t \cos \omega t \leftrightarrow \frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$, so $H(s) = \frac{2\omega s(s^2 - \omega^2)}{(s^2 + \omega^2)^4}$.

(e) $e^t \int_0^t \sin \omega \tau \cos \omega(t-\tau) d\tau = \int_0^t (e^\tau \sin \omega \tau) (e^{(t-\tau)} \cos \omega(t-\tau)) d\tau$;
 $e^t \sin \omega t \leftrightarrow \frac{\omega}{(s-1)^2 + \omega^2}$ and $e^t \cos \omega t \leftrightarrow \frac{s-1}{(s-1)^2 + \omega^2}$, so $H(s) = \frac{(s-1)\omega}{((s-1)^2 + \omega^2)^2}$.

(f) $e^t \int_0^t \tau^2(t-\tau)e^\tau d\tau = \int_0^t \tau^2 e^{2\tau}(t-\tau)e^{(t-\tau)} d\tau$; $t^2 e^{2t} \leftrightarrow \frac{2}{(s-2)^3}$ and $te^t \leftrightarrow \frac{1}{(s-1)^2}$, so
 $H(s) = \frac{2}{(s-2)^3(s-1)^2}$.

- (g) $e^{-t} \int_0^t e^{-\tau} \tau \cos \omega(t-\tau) d\tau = \int_0^t \tau e^{-2\tau} e^{-(t-\tau)} \cos \omega(t-\tau) d\tau$; $te^{-2t} \leftrightarrow \frac{1}{(s+2)^2}$ and $e^{-t} \cos \omega t \leftrightarrow \frac{s+1}{(s+1)^2 + \omega^2}$, so $H(s) = \frac{s+1}{(s+2)^2 [(s+1)^2 + \omega^2]}$.
- (h) $e^t \int_0^t e^{2\tau} \sinh(t-\tau) d\tau = \int_0^t e^{3\tau} (e^{(t-\tau)} \sinh(t-\tau)) d\tau$; $e^{3t} \leftrightarrow \frac{1}{s-3}$ and $e^t \sinh t \leftrightarrow \frac{1}{(s-1)^2 - 1}$, so $H(s) = \frac{1}{(s-3)((s-1)^2 - 1)}$.
- (i) $te^{2t} \leftrightarrow \frac{1}{(s-2)^2}$ and $\sin 2t \leftrightarrow \frac{2}{s^2 + 4}$, so $H(s) = \frac{2}{(s-2)^2(s^2 + 4)}$.
- (j) $t^3 \leftrightarrow \frac{6}{s^4}$ and $e^t \leftrightarrow \frac{1}{s-1}$, so $H(s) = \frac{6}{s^4(s-1)}$.
- (k) $t^6 \leftrightarrow \frac{6!}{s^7}$ and $e^{-t} \sin 3t \leftrightarrow \frac{3}{(s+1)^2 + 9}$, so $H(s) = \frac{3 \cdot 6!}{s^7 [(s+1)^2 + 9]}$.
- (l) $t^2 \leftrightarrow \frac{2}{s^3}$ and $t^3 \leftrightarrow \frac{6}{s^4}$, so $H(s) = \frac{12}{s^7}$.
- (m) $t^7 \leftrightarrow \frac{7!}{s^8}$ and $e^{-t} \sin 2t \leftrightarrow \frac{2}{(s+1)^2 + 4}$, so $H(s) = \frac{2 \cdot 7!}{s^8 [(s+1)^2 + 4]}$.
- (n) $t^4 \leftrightarrow \frac{24}{s^5}$ and $\sin 2t \leftrightarrow \frac{2}{s^2 + 4}$, so $H(s) = \frac{48}{s^5(s^2 + 4)}$.

8.6.4. (a) $Y(s) = \frac{1}{s^2} - \frac{Y(s)}{s^2}$; $Y(s) \left(1 + \frac{1}{s^2}\right) = \frac{1}{s^2}$; $Y(s) \frac{s^2 + 1}{s^2} = \frac{1}{s^2}$; $Y(s) = \frac{1}{s^2 + 1}$, so $y = \sin t$.

(b) $Y(s) = \frac{1}{s^2 + 1} - \frac{2sY(s)}{s^2 + 1}$; $Y(s) \left(1 + \frac{2s}{s^2 + 1}\right) = \frac{1}{s^2 + 1}$; $Y(s) \frac{(s+1)^2}{s^2 + 1} = \frac{1}{s^2 + 1}$;
 $Y(s) = \frac{1}{(s+1)^2}$, so $y = te^{-t}$.

(c) $Y(s) = \frac{1}{s} + \frac{2sY(s)}{s^2 + 1}$; $Y(s) \left(1 - \frac{2s}{s^2 + 1}\right) = \frac{1}{s}$; $Y(s) \frac{(s-1)^2}{s^2 + 1} = \frac{1}{s}$;
 $Y(s) = \frac{(s^2 + 1)}{s(s-1)^2} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{(s-1)^2}$, where $A(s-1)^2 + Bs(s-1) + Cs = s^2 + 1$.
 Setting $s = 0$ and $s = 1$ shows that $A = 1$ and $C = 2$; equating coefficients of s^2 yields $A + B = 1$, so $B = 0$. Therefore, $Y(s) = \frac{1}{s} + \frac{1}{(s-1)^2}$, so $y = 1 + 2te^t$.

(d) $Y(s) = \frac{1}{s^2} + \frac{Y(s)}{s+1}$; $Y(s) \left(1 - \frac{1}{s+1}\right) = \frac{1}{s^2}$; $Y(s) \left(\frac{s}{s+1}\right) = \frac{1}{s^2}$;
 $Y(s) = \frac{s+1}{s^3} = \frac{1}{s^2} + \frac{1}{s^3}$, so $y = t + \frac{t^2}{2}$.

(e) $sY(s) - 4 = \frac{1}{s^2} + \frac{sY(s)}{s^2 + 1}$; $Y(s) \left(s - \frac{s}{s^2 + 1}\right) = 4 + \frac{1}{s^2}$; $Y(s) \frac{s^3}{s^2 + 1} = \frac{4s^2 + 1}{s^2}$;
 $Y(s) = \frac{(4s^2 + 1)(s^2 + 1)}{s^5} = \frac{4s^4 + 5s^2 + 1}{s^5} = \frac{4}{s} + \frac{5}{s^3} + \frac{1}{s^5}$, so $y = 4 + \frac{5}{2}t^2 + \frac{1}{24}t^4$.

$$(f) \quad Y(s) = \frac{s-1}{s^2+1} + \frac{Y(s)}{s^2+1}; \quad Y(s) \left(1 - \frac{1}{s^2+1}\right) = \frac{s-1}{s^2+1}; \quad Y(s) \frac{s^2}{s^2+1} = \frac{s-1}{s^2+1};$$

$$Y(s) = \frac{s-1}{s^2} = \frac{1}{s} - \frac{1}{s^2}, \text{ so } y = 1 - t.$$

8.6.6. Substituting $x = t - \tau$ yields

$$\int_0^t f(t-\tau)g(\tau) d\tau = - \int_t^0 f(x)g(t-x)(-dx) = \int_0^t f(x)g(t-x) dx = \int_0^t f(\tau)g(t-\tau) d\tau.$$

8.6.8. $p(s)Y(s) = F(s) + a(k_1 + k_0s) + bk_0$, so (A) $Y(s) = \frac{F(s)}{p(s)} + \frac{k_0(as+b) + k_1a}{p(s)}$. Since $p(s) = a(s-r_1)(s-r_2)$ and therefore $b = -a(r_1+r_2)$, (A) can be rewritten as

$$Y(s) = \frac{F(s)}{a(s-r_1)(s-r_2)} + \frac{k_0(s-r_1-r_2)}{(s-r_1)(s-r_2)} + \frac{k_1}{(s-r_1)(s-r_2)}.$$

$$\frac{1}{(s-r_1)(s-r_2)} = \frac{1}{r_2-r_1} \left(\frac{1}{s-r_2} - \frac{1}{s-r_1} \right) \leftrightarrow \frac{e^{r_2t} - e^{r_1t}}{r_2-r_1},$$

so the convolution theorem implies that

$$\frac{F(s)}{a(s-r_1)(s-r_2)} \leftrightarrow \frac{1}{a} \int_0^t \frac{e^{r_2\tau} - e^{r_1\tau}}{r_2-r_1} f(t-\tau) d\tau.$$

$$\frac{s-r_1-r_2}{(s-r_1)(s-r_2)} = \frac{r_2}{r_2-r_1} \frac{1}{s-r_1} - \frac{r_1}{r_2-r_1} \frac{1}{s-r_2} \leftrightarrow \frac{r_2e^{r_1t} - r_1e^{r_2t}}{r_2-r_1}.$$

Therefore,

$$y(t) = k_0 \frac{r_2e^{r_1t} - r_1e^{r_2t}}{r_2-r_1} + k_1 \frac{e^{r_2t} - e^{r_1t}}{r_2-r_1} + \frac{1}{a} \int_0^t \frac{e^{r_2\tau} - e^{r_1\tau}}{r_2-r_1} f(t-\tau) d\tau.$$

8.6.10. $p(s)Y(s) = F(s) + a(k_1 + k_0s) + bk_0$, so (A) $Y(s) = \frac{F(s)}{p(s)} + \frac{k_0(as+b) + k_1a}{p(s)}$. Since $p(s) = a(s-\lambda)^2 + \omega^2$ and therefore $b = -2a\lambda$, (A) can be rewritten as

$$Y(s) = \frac{F(s)}{a[(s-\lambda)^2 + \omega^2]} + \frac{k_0(s-2\lambda)}{(s-\lambda)^2 + \omega^2} + \frac{k_1}{(s-\lambda)^2 + \omega^2}.$$

$\frac{1}{(s-\lambda)^2 + \omega^2} \leftrightarrow \frac{1}{\omega} e^{\lambda t} \sin \omega t$, so the convolution theorem implies that

$$\frac{F(s)}{a[(s-\lambda)^2 + \omega^2]} \leftrightarrow \frac{1}{a\omega} \int_0^t e^{\lambda\tau} f(t-\tau) \sin \omega\tau d\tau.$$

$$\frac{s-2\lambda}{(s-\lambda)^2 + \omega^2} = \frac{(s-\lambda) - \lambda}{(s-\lambda)^2 + \omega^2} \leftrightarrow e^{\lambda t} \left(\cos \omega t - \frac{\lambda}{\omega} \sin \omega t \right).$$

Therefore,

$$y(t) = e^{\lambda t} \left[k_0 \left(\cos \omega t - \frac{\lambda}{\omega} \sin \omega t \right) + \frac{k_1}{\omega} \sin \omega t \right] + \frac{1}{a\omega} \int_0^t e^{\lambda\tau} f(t-\tau) \sin \omega\tau d\tau.$$

8.6.12. (a)

$$\begin{aligned}
ay'' + by' + cy &= f_0(t) + u(t - t_1)(f_1(t) - f_0(t)), \quad y(0) = 0, \quad y'(0) = 0; \\
p(s)Y(s) &= F_0(s) + \mathcal{L}(u(t - t_1)(f_1(t) - f_0(t))) = F_0(s) + e^{-st_1}\mathcal{L}(g); \\
Y(s) &= \frac{F_0(s) + e^{-st_1}G(s)}{p(s)}. \tag{B}
\end{aligned}$$

(b) Since $F_0(s) \leftrightarrow f_0(t)$, $G(s) \leftrightarrow g(t)$, and $\frac{1}{p(s)} \leftrightarrow w(t)$, the convolution theorem implies that

$$\frac{F_0(s)}{p(s)} \leftrightarrow \int_0^t w(t - \tau)f_0(\tau) d\tau \quad \text{and} \quad \frac{G(s)}{p(s)} \leftrightarrow \int_0^t w(t - \tau)g(\tau) d\tau.$$

Now Theorem 8.4.2 implies that $\frac{e^{-st_1}G(s)}{p(s)} \leftrightarrow u(t - t_1) \int_0^t w(t - t_1 - \tau)g(\tau) d\tau$, and (B) implies that

$$y(t) = \int_0^t w(t - \tau)f_0(\tau) d\tau + u(t - t_1) \int_0^{t-t_1} w(t - t_1 - \tau)g(\tau) d\tau.$$

(c) Let $z_0(t) = \int_0^t w(t - \tau)f_0(\tau) d\tau$ and $z_1(t) = \int_0^t w(t - \tau)g(\tau) d\tau$. Then $y(t) = z_0(t) + u(t - t_1)z_1(t - t_1)$. Using Leibniz's rule as in the solution of Exercise 8.6.11(b) shows that

$$\begin{aligned}
z'_0(t) &= \int_0^t w'(t - \tau)f_0(\tau) d\tau, \quad z'_1(t) = \int_0^t w'(t - \tau)g(\tau) d\tau, \quad t > 0, \\
z''_0(t) &= \frac{f_0(t)}{a} + \int_0^t w''(t - \tau)f_0(\tau) d\tau, \quad z''_1(t) = \frac{g(t)}{a} + \int_0^t w''(t - \tau)g(\tau) d\tau, \quad t > 0,
\end{aligned}$$

if $t > 0$, and that

$$az''_0 + bz'_0 + cz_0 = f_0(t) \quad \text{and} \quad az''_1 + bz'_1 + cz_1 = f_1(t + t_1) - f_0(t + t_1), \quad t > 0.$$

This implies the stated conclusion for y' and y'' on $(0, t)$ and (t, ∞) , and that $ay'' + by' + cy = f(t)$ on these intervals.

(d) Since the functions $z_0(t)$ and $h(t) = u(t - t_1)z_1(t - t_1)$ are both continuous on $[0, \infty)$ and $h(t) = 0$ if $0 \leq t \leq t_1$, y is continuous on $[0, \infty)$. From (c), y' is continuous on $[0, t_1)$ and (t_1, ∞) , so we need only show that y' is continuous at t_1 . For this it suffices to show that $h'(t_1) = 0$. Since $h(t_1) = 0$ if $t \leq t_1$, (B) $\lim_{t \rightarrow t_1^-} \frac{h(t) - h(t_1)}{t - t_1} = 0$. If $t > t_1$, then $h(t) = \int_0^{t-t_1} w(t - t_1 - \tau)g(\tau) d\tau$. Since $h(t_1) = 0$,

$$\left| \frac{h(t) - h(t_1)}{t - t_1} \right| \leq \int_0^{t-t_1} |w(t - t_1 - \tau)g(\tau)| d\tau. \tag{B}$$

Since g is continuous from the right at 0, we can choose constants $T > 0$ and $M > 0$ so that $|g(\tau)| < M$ if $0 \leq \tau \leq T$. Then (B) implies that

$$\left| \frac{h(t) - h(t_1)}{t - t_1} \right| \leq M \int_0^{t-t_1} |w(t - t_1 - \tau)| d\tau, \quad t_1 < t < t_1 + T. \quad (C)$$

Now suppose $\epsilon > 0$. Since $w(0) = 0$, we can choose T_1 such that $0 < T_1 < T$ and $|w(x)| < \epsilon/M$ if $0 \leq x < T_1$. If $t_1 < t < t_1 + T_1$ and $0 \leq \tau \leq t - t_1$, then $0 \leq t - t_1 - \tau < T_1$, so (C) implies that

$$\left| \frac{h(t) - h(t_1)}{t - t_1} \right| < \epsilon, \quad t_1 < t < t_1 + T.$$

Therefore, $\lim_{t \rightarrow t_1+} \frac{h(t) - h(t_1)}{t - t_1} = 0$. This and (B) imply that $h'(t_1) = 0$.

8.7 Constant Coefficient Equations with Impulses

$$\begin{aligned} 8.7.2. \quad (s^2 + s - 2)\hat{Y}(s) &= -\frac{10}{s+1} + (-9 + 7s) + 7; \quad \hat{Y}(s) = \frac{-10 + (s+1)(7s-2)}{(s-1)(s+2)(s+1)} = \frac{2}{s+2} + \frac{5}{s+1}; \\ \hat{y} &= 2e^{-2t} + 5e^{-t}; \quad \frac{1}{p(s)} = \frac{1}{(s+2)(s-1)} = \frac{1}{3} \left(\frac{1}{s-1} - \frac{1}{s+2} \right); \\ w &= \mathcal{L}^{-1} \left(\frac{1}{p(s)} \right) = \frac{e^t - e^{-2t}}{3}; \quad y = 2e^{-2t} + 5e^{-t} + \frac{5}{3}u(t-1)(e^{(t-1)} - e^{-2(t-1)}). \end{aligned}$$

$$\begin{aligned} 8.7.4. \quad (s^2 + 1)\hat{Y}(s) &= \frac{3}{s^2 + 9} - 1 + s; \\ \hat{Y}(s) &= \frac{3}{(s^2 + 1)(s^2 + 9)} + \frac{s-1}{s^2 + 1} = \frac{3}{8} \left(\frac{1}{s^2 + 1} - \frac{1}{s^2 + 9} \right) + \frac{s-1}{s^2 + 1} = \frac{1}{8} \left(\frac{8s-5}{s^2 + 1} - \frac{3}{s^2 + 9} \right); \\ \hat{y} &= \frac{1}{8}(8 \cos t - 5 \sin t - \sin 3t); \quad \frac{1}{p(s)} = \frac{1}{s^2 + 1}; \quad w = \mathcal{L}^{-1} \left(\frac{1}{p(s)} \right) = \sin t; \\ y &= \frac{1}{8}(8 \cos t - 5 \sin t - \sin 3t) - 2u(t - \pi/2) \cos t. \end{aligned}$$

$$\begin{aligned} 8.7.6. \quad (s^2 - 1)\hat{Y}(s) &= \frac{8}{s} + 1 - s; \quad \hat{Y}(s) = \frac{8 + s(1-s)}{s(s-1)(s+1)} = \frac{4}{s-1} + \frac{3}{s+1} - \frac{8}{s}; \quad \hat{y} = 4e^t + 3e^{-t} - 8; \\ \frac{1}{p(s)} &= \frac{1}{(s-1)(s+1)} = \frac{1}{2} \frac{1}{s-1} - \frac{1}{s+1}; \quad w = \mathcal{L}^{-1} \left(\frac{1}{p(s)} \right) = \frac{e^t + e^{-t}}{2} = \sinh t; \\ y &= 4e^t + 3e^{-t} - 8 + 2u(t-2) \sinh(t-2); \end{aligned}$$

$$\begin{aligned} 8.7.8. \quad (s^2 + 4)\hat{Y}(s) &= \frac{8}{s-2} + 8s; \quad (A) \quad \hat{Y}(s) = \frac{8}{(s-2)(s^2 + 4)} + \frac{8s}{s^2 + 4}; \\ \frac{8}{(s-2)(s^2 + 4)} &= \frac{A}{s-2} + \frac{Bs+C}{s^2 + 4} \text{ where } A(s^2 + 4) + (Bs+C)(s-2) = 8. \text{ Setting } s = 2 \\ \text{yields } A &= 1; \text{ setting } s = 0 \text{ yields } 4A - 2C = 8, \text{ so } C = -2; A + B = 0 \text{ (coefficient of } x^2), \text{ so} \\ B &= -A = -1; \text{ therefore } \frac{8}{(s-2)(s^2 + 4)} = \frac{1}{s-2} - \frac{s+2}{s^2 + 4}, \text{ so (A) implies that} \end{aligned}$$

$$\hat{y} = e^{2t} + 7 \cos 2t - \sin 2t; \frac{1}{p(s)} = \frac{1}{s^2 + 4}; w = \mathcal{L}^{-1} \left(\frac{1}{p(s)} \right) = \frac{1}{2} \sin 2t. \text{ Since}$$

$$\sin(2t - \pi) = -\sin 2t, y = e^{2t} + 7 \cos 2t - \sin 2t - \frac{1}{2} u(t - \pi/2) \sin 2t.$$

$$8.7.10. (s^2 + 2s + 1)\hat{Y}(s) = \frac{1}{s-1} + (2-s) - 2; \hat{Y}(s) = \frac{1-s(s-1)}{(s-1)(s+1)^2} = \frac{A}{s-1} + \frac{B}{s+1} + \frac{C}{(s+1)^2}$$

where $A(s+1)^2 + (B(s+1) + C)(s-1) = 1 - s(s-1)$. Setting $s = 1$ yields $A = 1/4$; setting $s = -1$ yields $C = 1/2$; since $A + B = -1$ (coefficient of s^2), $B = -1 - A = -5/4$.

Therefore, $\hat{Y}(s) = \frac{1}{4} \frac{1}{s-1} - \frac{5}{4} \frac{1}{s+1} + \frac{1}{2} \frac{1}{(s+1)^2}$; $\hat{y} = \frac{1}{4} e^t + \frac{1}{4} e^{-t}(2t - 5)$; $\frac{1}{p(s)} = \frac{1}{(s+1)^2}$;

$$w = \mathcal{L}^{-1} \left(\frac{1}{p(s)} \right) = te^{-t}; y = \frac{1}{4} e^t + \frac{1}{4} e^{-t}(2t - 5) + 2u(t-2)(t-2)e^{-(t-2)}.$$

$$8.7.12. (s^2 + 2s + 2)\hat{Y}(s) = (2-s) - 2; Y(s) = \frac{-(s+1)+1}{(s+1)^2+1}; \hat{y} = e^{-t}(\sin t - \cos t);$$

$$\frac{1}{p(s)} = \frac{1}{(s+1)^2+1}; w = \mathcal{L}^{-1} \left(\frac{1}{p(s)} \right) = e^{-t} \sin t. \text{ Since } \sin(t-\pi) = -\sin t \text{ and}$$

$$\sin(t-2\pi) = \sin t, y = e^{-t}(\sin t - \cos t) - e^{-(t-\pi)}u(t-\pi)\sin t - 3u(t-2\pi)e^{-(t-2\pi)}\sin t.$$

$$8.7.14. (2s^2 - 3s - 2)\hat{Y}(s) = \frac{1}{s} + 2(2-s) + 3;$$

$$\hat{Y}(s) = \frac{1+s(7-2s)}{2s(s+1/2)(s-2)} = \frac{7}{10} \frac{1}{s-2} - \frac{6}{5} \frac{1}{s+1/2} - \frac{1}{2s}; \hat{y} = \frac{7}{10} e^{2t} - \frac{6}{5} e^{-t/2} - \frac{1}{2};$$

$$\frac{1}{p(s)} = \frac{1}{2(s+1/2)(s-2)} = \frac{1}{5} \left(\frac{1}{s-2} - \frac{1}{s+1/2} \right); w = \mathcal{L}^{-1} \left(\frac{1}{p(s)} \right) = \frac{1}{5} (e^{2t} - e^{-t/2});$$

$$y = \frac{7}{10} e^{2t} - \frac{6}{5} e^{-t/2} - \frac{1}{2} + \frac{1}{5} u(t-2) (e^{2(t-2)} - e^{-(t-2)/2});$$

$$8.7.16. (s^2 + 1)\hat{Y}(s) = \frac{s}{s^2+4} - 1;$$

$$\hat{Y}(s) = \frac{s}{(s^2+1)(s^2+4)} - \frac{1}{s^2+1} = \frac{1}{3} \left(\frac{s}{s^2+1} - \frac{s}{s^2+4} \right) - \frac{1}{s^2+1};$$

$$\hat{y} = \frac{1}{3} (\cos t - \cos 2t - 3 \sin t); \frac{1}{p(s)} = \frac{1}{s^2+1}; w = \mathcal{L}^{-1} \left(\frac{1}{p(s)} \right) = \sin t. \text{ Since}$$

$$\sin(t - \pi/2) = -\cos t \text{ and } \sin(t - \pi) = -\sin t,$$

$$y = \frac{1}{3} (\cos t - \cos 2t - 3 \sin t) - 2u(t - \pi/2) \cos t + 3u(t - \pi) \sin t.$$

$$8.7.18. (s^2 + 2s + 1)\hat{Y}(s) = \frac{1}{s-1} - 1; (A) \hat{Y}(s) = \frac{1}{(s-1)(s+1)^2} - \frac{1}{(s+1)^2};$$

$$\frac{1}{(s-1)(s+1)^2} = \frac{A}{s-1} + \frac{B}{s+1} + \frac{C}{(s+1)^2}$$

where $A(s+1)^2 + (B(s+1) + C)(s-1) = 1$. Setting $s = 1$ yields $A = 1/4$; setting $s = -1$ yields $C = -1/2$; since $A + B = 0$ (coefficient of s^2), $B = -A = -1/4$. Therefore,

$$\frac{1}{(s-1)(s+1)^2} = \frac{1}{4} \frac{1}{s-1} - \frac{1}{4} \frac{1}{s+1} - \frac{1}{2} \frac{1}{(s+1)^2}.$$

This and (A) imply that

$$\hat{Y}(s) = \frac{1}{4} \frac{1}{s-1} - \frac{1}{4} \frac{1}{s+1} - \frac{3}{2} \frac{1}{(s+1)^2};$$

$$\hat{y} = \frac{1}{4} (e^t - e^{-t}(1+6t)); \frac{1}{p(s)} = \frac{1}{(s+1)^2}; w = \mathcal{L}^{-1} \left(\frac{1}{p(s)} \right) = te^{-t};$$

$$y = \frac{1}{4} (e^t - e^{-t}(1+6t)) - u(t-1)(t-1)e^{-(t-1)} + 2u(t-2)(t-2)e^{-(t-2)}.$$

$$8.7.20. \quad y'' + 4y = 1 - 2u(t - \pi/2) + \delta(t - \pi) - 3\delta(t - 3\pi/2), \quad y(0) = 1, \quad y'(0) = -1.$$

$$(s^2 + 4)\hat{Y}(s) = \frac{1 - 2e^{-\pi s/2}}{s} + s - 1; \hat{Y}(s) = \frac{1 - 2e^{-\pi s/2}}{s(s^2 + 4)} + \frac{s-1}{s^2 + 4}. \text{ Since}$$

$$\frac{1}{s(s^2 + 4)} = \frac{1}{4} \left(\frac{1}{s} - \frac{s}{s^2 + 4} \right), \quad \hat{Y}(s) = \frac{1}{4s} + \frac{3}{4} \frac{s}{s^2 + 4} - \frac{1}{s^2 + 4} - 2e^{-\pi s/2} \left(\frac{1}{s} - \frac{s}{s^2 + 4} \right).$$

$$\hat{y} = \frac{3}{4} \cos 2t - \frac{1}{2} \sin 2t + \frac{1}{4} + \frac{1}{4} u(t - \pi/2)(1 + \cos 2t). \quad w = \mathcal{L}^{-1} \left(\frac{1}{p(s)} \right) = \frac{1}{2} \sin 2t. \text{ Since}$$

$$\sin 2(t - \pi) = \sin 2t \text{ and } \sin 2(t - 3\pi/2) = -\sin 2t,$$

$$y = \frac{3}{4} \cos 2t - \frac{1}{2} \sin 2t + \frac{1}{4} + \frac{1}{4} u(t - \pi/2)(1 + \cos 2t) + \frac{1}{2} u(t - \pi) \sin 2t + \frac{3}{2} u(t - 3\pi/2) \sin 2t.$$

$$8.7.26. \quad w(t) = e^{-t} \sin t; f_h(t) = \frac{u(t - t_0) - u(t - t_0 - h)}{h}; (s^2 + 2s + 2)Y_h(s) = \frac{1}{h} \frac{e^{-st_0} - e^{-s(t_0+h)}}{s};$$

$$Y_h(s) = \frac{1}{h} \frac{e^{-st_0} - e^{-s(t_0+h)}}{s(s^2 + 2s + 2)};$$

$$\frac{1}{s(s^2 + 2s + 2)} = -\frac{(s+1)+1}{2((s+1)^2 + 1)} + \frac{1}{2s} \leftrightarrow \frac{1}{2} (1 - e^{-t}(\cos t + \sin t));$$

$$y_h(t) = \begin{cases} 0, & 0 \leq t < t_0, \\ \frac{1}{2h} [1 - e^{-(t-t_0)}(\cos(t-t_0) + \sin(t-t_0))], & t_0 \leq t < t_0 + h, \\ \frac{e^{-(t-t_0)}}{2h} [e^h(\cos(t-t_0-h) + \sin(t-t_0-h)) - \cos(t-t_0) - \sin(t-t_0)], & t \geq t_0 + h. \end{cases}$$

$$8.7.28. \quad w(t) = e^{-t} - e^{-2t}; f_h(t) = \frac{u(t - t_0) - u(t - t_0 - h)}{h};$$

$$(s^2 + 3s + 2)Y_h(s) = \frac{1}{h} \frac{e^{-st_0} - e^{-s(t_0+h)}}{s}; Y_h(s) = \frac{1}{h} \frac{e^{-st_0} - e^{-s(t_0+h)}}{s(s+1)(s+2)};$$

$$\frac{1}{s(s+1)(s+2)} = \frac{1}{2(s+2)} - \frac{1}{s+1} + \frac{1}{2s} \leftrightarrow \frac{e^{-2t}}{2} - e^{-t} + \frac{1}{2} = \frac{(e^{-t} - 1)^2}{2};$$

$$y_h(t) = \begin{cases} 0, & 0 \leq t < t_0, \\ \frac{(e^{-(t-t_0)} - 1)^2}{2h}, & t_0 \leq t < t_0 + h, \\ \frac{(e^{-(t-t_0)} - 1)^2 - (e^{-(t-t_0-h)} - 1)^2}{2h}, & t \geq t_0 + h. \end{cases}$$

8.7.30. (a) $(s^2 - 1)\hat{Y}(s) = 1$, so $\hat{y} = w = \mathcal{L}^{-1}\left(\frac{1}{s^2 - 1}\right) = \frac{1}{2}(e^t - e^{-t})$;

$$y = \hat{y} + \sum_{k=0}^{\infty} u(t-k)w(t-k) = \frac{1}{2} \sum_{k=0}^{\infty} u(t-k)(e^{t-k} - e^{-t-k}). \text{ If } m \leq t < m+1, \text{ then}$$

$$y = \frac{1}{2} \sum_{k=0}^m (e^{t-k} - e^{-t-k}) = \frac{1}{2}(e^{t-m} - e^{-t}) \sum_{k=0}^m e^k \frac{e^{m+1} - 1}{2(e-1)}(e^{t-m} - e^{-t}).$$

(b) $(s^2 + 1)\hat{Y}(s) = 1$, so $\hat{y} = w = \mathcal{L}^{-1}\left(\frac{1}{s^2 + 1}\right) = \sin t$;

$$y = \hat{y} + \sum_{k=0}^{\infty} u(t-2k\pi)w(t-2k\pi) = \sin t \sum_{k=0}^{\infty} u(t-2k\pi). \text{ If } 2m\pi \leq t < 2(m+1)\pi, \text{ then}$$

$$y = (m+1) \sin t.$$

(c) $(s^2 - 3s + 2)\hat{Y}(s) = 1$, so $\hat{y} = w = \mathcal{L}^{-1}\left(\frac{1}{(s-1)(s-2)}\right) = (e^{2t} - e^t)$;

$$y = \hat{y} + \sum_{k=0}^{\infty} u(t-k)w(t-k) = \sum_{k=0}^{\infty} u(t-k)(e^{2(t-k)} - e^{t-k}). \text{ If } m \leq t < m+1, \text{ then}$$

$$y = \sum_{k=0}^m (e^{2(t-k)} - e^{t-k}) = e^{2(t-m)} \sum_{k=0}^m e^{2k} - e^{t-m} \sum_{k=0}^m e^k = e^{2(t-m)} \frac{e^{2m+2} - 1}{e^2 - 1} - e^{(t-m)} \frac{e^{m+1} - 1}{e - 1}.$$

(d) $w = \mathcal{L}^{-1}\left(\frac{1}{s^2 + 1}\right) = \sin t$; $y = \sum_{k=1}^{\infty} u(t-k\pi)w(t-k\pi) = \sin t \sum_{k=1}^{\infty} (-1)^k u(t-k\pi)$, so

$$y = \begin{cases} 0, & 2m\pi \leq t < (2m+1)\pi, \\ -\sin t, & (2m+1)\pi \leq t < (2m+2)\pi, \end{cases} \quad (m = 0, 1, \dots).$$

Chapter 9

Linear Higher Order Equations

9.1 Introduction to Linear Higher Order Equations

9.1.2. From Example 9.1.1, $y = c_1x^2 + c_2x^3 + \frac{c_3}{x}$ $y' = 2c_1x + 3c_2x^2 - \frac{c_3}{x^2}$, and

$$y'' = 2c_1 + 6c_2x + \frac{2c_3}{x^3}, \text{ where}$$

$$\begin{aligned}c_1 - c_2 - c_3 &= 4 \\ -2c_1 + 3c_2 - c_3 &= -14 \\ 2c_1 - 6c_2 - 2c_3 &= 20,\end{aligned}$$

$$\text{so } c_1 = 2, c_2 = -3, c_3 = 1, \text{ and } y = 2x^2 - 3x^3 + \frac{1}{x}.$$

9.1.4. The general solution of $y^{(n)} = 0$ can be written as $y(x) = \sum_{m=0}^{n-1} c_m(x-x_0)^m$. Since

$$\begin{aligned}y^{(j)}(x) &= \sum_{m=j}^{n-1} m(m-1)\cdots(m-j+1)c_m(x-x_0)^{m-j}, \quad y^{(j)}(x_0) = j!c_j. \text{ Therefore,} \\ y_i &= \frac{(x-x_0)^{i-1}}{(i-1)!}, \quad 1 \leq i \leq n.\end{aligned}$$

9.1.6. We omit the verification that the given functions are solutions of the given equations.

(a) The equation is normal on $(-\infty, \infty)$. $W(x) = \begin{vmatrix} e^x & e^{-x} & xe^{-x} \\ e^x & -e^{-x} & e^{-x}(1-x) \\ e^x & e^{-x} & e^{-x}(x-2) \end{vmatrix};$

$$W(0) = \begin{vmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & 1 & -2 \end{vmatrix} = 4. \text{ Apply Theorem 9.1.4.}$$

(b) The equation is normal on $(-\infty, \infty)$.

$$W(x) = \begin{vmatrix} e^x & e^x \cos 2x & e^x \sin 2x \\ e^x & e^x(\cos 2x - 2 \sin 2x) & e^x(2 \cos 2x + \sin 2x) \\ e^x & -e^x(3 \cos 2x + 4 \sin 2x) & e^x(4 \cos 2x - 3 \sin 2x) \end{vmatrix};$$

$$W(0) = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & -3 & 4 \end{vmatrix} = 8. \text{ Apply Theorem 9.1.4.}$$

(c) The equation is normal on $(-\infty, 0)$ and $(0, \infty)$.

$$W(x) = \begin{vmatrix} e^x & e^{-x} & x \\ e^x & -e^{-x} & 1 \\ e^x & e^{-x} & 0 \end{vmatrix} = 2x. \text{ Apply Theorem 9.1.4.}$$

(d) The equation is normal on $(-\infty, 0)$ and $(0, \infty)$.

$$W(x) = \begin{vmatrix} e^x/x & e^{-x}/x & 1 \\ e^x(1/x - 1/x^2) & -e^{-x}(x+1)/x^2 & 0 \\ e^x(1/x - 2/x^2 + 2/x^3) & e^{-x}(x^2 + 2x + 2)/x^3 & 0 \end{vmatrix} = 2/x^2. \text{ Apply Theorem 9.1.4.}$$

(e) The equation is normal on $(-\infty, \infty)$. $W(x) = \begin{vmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 2 & e^x \end{vmatrix}$; $W(0) = \begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix} = 2$;

Apply Theorem 9.1.4.

(f) The equation is normal on $(-\infty, 1/2)$ and $(1/2, \infty)$.

$$W(x) = \begin{vmatrix} x & e^x & e^{-x} & e^{2x} \\ 1 & e^x & -e^{-x} & 2e^{2x} \\ 0 & e^x & e^{-x} & 4e^{2x} \\ 0 & e^x & -e^{-x} & 8e^{2x} \end{vmatrix} = e^{2x}(12x - 6). \text{ Apply Theorem 9.1.4.}$$

(g) The equation is normal on $(-\infty, 0)$ and $(0, \infty)$.

$$W(x) = \begin{vmatrix} 1 & x^2 & e^{2x} & e^{-2x} \\ 0 & 2x & 2e^{2x} & -2e^{-2x} \\ 0 & 2 & 4e^{2x} & 4e^{-2x} \\ 0 & 0 & 8e^{2x} & -8e^{-2x} \end{vmatrix} = -128x. \text{ Apply Theorem 9.1.4.}$$

9.1.8. From Abel's formula, (A) $W(x) = W(\pi/2) \exp\left(-\int_{\pi/4}^x \tan t \, dt\right)$;

$$\int_{\pi/4}^x \tan t \, dt = -\ln \cos t \Big|_{\pi/4}^x = -\ln(\sqrt{2} \cos x); \text{ therefore (A) implies that } W(x) = \sqrt{2}K \cos x.$$

$$9.1.10. \text{ (a) } W(x) = \begin{vmatrix} 1 & e^x & e^{-x} \\ 0 & e^x & -e^{-x} \\ 0 & e^x & e^{-x} \end{vmatrix} = (e^x)(e^{-x}) \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{vmatrix} = 2.$$

(b)

$$\begin{aligned} W(x) &= \begin{vmatrix} e^x & e^x \sin x & e^x \cos x \\ e^x & e^x(\cos x + \sin x) & e^x(\cos x - \sin x) \\ e^x & 2e^x \cos x & -2e^x \sin x \end{vmatrix} = \\ &= e^{3x} \begin{vmatrix} 1 & \sin x & \cos x \\ 1 & \cos x + \sin x & \cos x - \sin x \\ 1 & 2 \cos x & -2 \sin x \end{vmatrix} \end{aligned}$$

$$\begin{aligned}
&= e^{3x} \begin{vmatrix} 1 & \sin x & \cos x \\ 0 & \cos x & -\sin x \\ 1 & 2\cos x - \sin x & -2\sin x - \cos x \end{vmatrix} \\
&= e^{3x} \begin{vmatrix} 1 & \sin x & \cos x \\ 0 & \cos x & -\sin x \\ 0 & -\sin x & -\cos x \end{vmatrix} = -e^{3x}
\end{aligned}$$

$$(c) \quad W(x) = \begin{vmatrix} 2 & x+1 & x^2+2 \\ 0 & 1 & 2x \\ 0 & 0 & 2 \end{vmatrix} = 4.$$

(d)

$$\begin{aligned}
W(x) &= \begin{vmatrix} x & x \ln |x| & 1/x \\ 1 & \ln |x| + 1 & -1/x^2 \\ 0 & 1/x & 2/x^3 \end{vmatrix} = \begin{vmatrix} 1 & \ln |x| & 1/x^2 \\ 1 & \ln |x| + 1 & -1/x^2 \\ 0 & 1 & 2/x^2 \end{vmatrix} \\
&= \frac{1}{x^2} \begin{vmatrix} 1 & \ln |x| & 1 \\ 1 & \ln |x| + 1 & -1 \\ 0 & 1 & 2 \end{vmatrix} = \frac{1}{x^2} \begin{vmatrix} 1 & \ln |x| & 1 \\ 0 & 1 & -2 \\ 0 & 1 & 2 \end{vmatrix} \\
&= \frac{1}{x^2} \begin{vmatrix} 1 & \ln |x| & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 4 \end{vmatrix} = 4/x^2.
\end{aligned}$$

$$(e) \quad W(x) = \begin{vmatrix} 1 & x & x^2/2 & x^3/3 & \cdots & x^n/n! \\ 0 & 1 & x & x^2/2 & \cdots & x^{n-1}/(n-1)! \\ 0 & 0 & 1 & x & \cdots & x^{n-2}/(n-2)! \\ 0 & 0 & 0 & 1 & \cdots & x^{n-3}/(n-3)! \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \end{vmatrix} = 1.$$

$$(f) \quad W(x) = \begin{vmatrix} e^x & e^{-x} & x \\ e^x & -e^{-x} & 1 \\ e^x & e^{-x} & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & x \\ 1 & -1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & x \\ 0 & -2 & 1-x \\ 0 & 0 & -x \end{vmatrix} = 2x.$$

(g)

$$\begin{aligned}
W(x) &= \begin{vmatrix} e^x/x & e^{-x}/x & 1 \\ e^x/x - e^x/x^2 & -e^{-x}/x - e^{-x}/x^2 & 0 \\ e^x/x - 2e^x/x^2 + 2e^x/x^3 & e^{-x}/x + 2e^{-x}/x^2 + 2e^{-x}/x^3 & 0 \end{vmatrix} \\
&= \begin{vmatrix} 1/x & 1/x & 1 \\ 1/x - 1/x^2 & -1/x - 1/x^2 & 0 \\ 1/x - 2/x^2 + 2/x^3 & 1/x + 2/x^2 + 2/x^3 & 0 \end{vmatrix} \\
&= \begin{vmatrix} 1/x - 1/x^2 & -1/x - 1/x^2 \\ 1/x - 2/x^2 + 2/x^3 & 1/x + 2/x^2 + 2/x^3 \end{vmatrix} \\
&= \begin{vmatrix} 1/x - 1/x^2 & -2/x \\ 1/x - 2/x^2 + 2/x^3 & 4/x^2 \end{vmatrix} = 2/x^2.
\end{aligned}$$

$$(h) \quad W(x) = \begin{vmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 2 & e^x \end{vmatrix} = e^x \begin{vmatrix} x & x^2 & 1 \\ 1 & 2x & 1 \\ 0 & 2 & 1 \end{vmatrix} = e^x \left(x \begin{vmatrix} 2x & 1 \\ 2 & 1 \end{vmatrix} - \begin{vmatrix} x^2 & 1 \\ 2 & 1 \end{vmatrix} \right) = e^x(x^2 - 2x + 2).$$

(i)

$$\begin{aligned} W(x) &= \begin{vmatrix} x & x^3 & 1/x & 1/x^2 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \end{vmatrix} = \begin{vmatrix} 0 & -2x^3 & 2/x & 3/x^2 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \end{vmatrix} \\ &= - \begin{vmatrix} -2x^3 & 2/x & 3/x^2 \\ 6x & 2/x^3 & 6/x^4 \\ 6 & -6/x^4 & -24/x^5 \end{vmatrix} = -x^4 \begin{vmatrix} -2 & 2/x^4 & 3/x^5 \\ 6 & 2/x^4 & 6/x^5 \\ 6 & -6/x^4 & -24/x^5 \end{vmatrix} \\ &= -x^4 \begin{vmatrix} -2 & 2/x^4 & 3/x^5 \\ 0 & 8/x^4 & 15/x^5 \\ 0 & 0 & -15/x^5 \end{vmatrix} = -240/x^5. \end{aligned}$$

(j)

$$\begin{aligned} W(x) &= \begin{vmatrix} e^x & e^{-x} & x & e^{2x} \\ e^x & -e^{-x} & 1 & 2e^{2x} \\ e^x & e^{-x} & 0 & 4e^{2x} \\ e^x & -e^{-x} & 0 & 8e^{2x} \end{vmatrix} = e^{2x} \begin{vmatrix} 1 & 1 & x & 1 \\ 1 & -1 & 1 & 2 \\ 1 & 1 & 0 & 4 \\ 1 & -1 & 0 & 8 \end{vmatrix} \\ &= e^{2x} \begin{vmatrix} 1-x & 1+x & 0 & 1-2x \\ 1 & -1 & 1 & 2 \\ 1 & 1 & 0 & 4 \\ 1 & -1 & 0 & 8 \end{vmatrix} = -e^{2x} \begin{vmatrix} 1-x & 1+x & 1-2x \\ 1 & 1 & 4 \\ 1 & -1 & 8 \end{vmatrix} \\ &= -e^{2x} \begin{vmatrix} 2 & 1+x & 1-2x \\ 2 & 1 & 4 \\ 0 & -1 & 8 \end{vmatrix} = -e^{2x} \begin{vmatrix} 2 & 1+x & 1-2x \\ 0 & -x & 3+2x \\ 0 & -1 & 8 \end{vmatrix} = 6e^{2x}(2x-1). \end{aligned}$$

(k)

$$\begin{aligned} W(x) &= \begin{vmatrix} e^{2x} & e^{-2x} & 1 & x^2 \\ 2e^{2x} & -2e^{-2x} & 0 & 2x \\ 4e^{2x} & 4e^{-2x} & 0 & 2 \\ 8e^{2x} & -8e^{-2x} & 0 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & x^2 \\ 2 & -2 & 0 & 2x \\ 4 & 4 & 0 & 2 \\ 8 & -8 & 0 & 0 \end{vmatrix} \\ &= \begin{vmatrix} 2 & -2 & 2x \\ 4 & 4 & 2 \\ 8 & -8 & 0 \end{vmatrix} = \begin{vmatrix} 0 & -2 & 2x \\ 8 & 4 & 2 \\ 0 & -8 & 0 \end{vmatrix} = -128x. \end{aligned}$$

9.1.12. Let y be an arbitrary solution of $Ly = 0$ on (a, b) . Since $\{z_1, \dots, z_n\}$ is a fundamental set

of solutions of $Ly = 0$ on (a, b) , there are constants $c_1, c_2, \dots, c_n$ such that $y = \sum_{i=1}^n c_i y_i$.

Therefore, $y = \sum_{i=1}^n c_i \sum_{j=1}^n a_{ij} y_j = \sum_{j=1}^n C_j y_j$, with $C_j = \sum_{i=1}^n a_{ij} c_i$. Hence $\{y_1, \dots, y_n\}$ is a fundamental set of solutions of $Ly = 0$ on (a, b) .

9.1.14. Let y be a given solution of $Ly = 0$ and $z = \sum_{j=1}^n y^{(j-1)}(x_0)y_j$. Then $z^{(r)}(x_0) = y^{(r)}(x_0)$,

$r = 0, \dots, n-1$. Since the solution of every initial value problem is unique (Theorem 9.1.1), $z = y$.

9.1.16. If $\{y_1, y_2, \dots, y_n\}$ is linearly dependent on (a, b) there are constants $c_1, \dots, c_n$, not all zeros, such that $c_1 y_1 + c_2 y_2 + \dots + c_n y_n = 0$. Let k be the smallest integer such that $c_k \neq 0$. If $k = 1$, then $y_1 = -\frac{1}{c_1}(c_2 y_2 + \dots + c_n y_n)$; if $1 < k < n$, then

$$y_k = 0 \cdot y_1 + \dots + 0 \cdot y_{k-1} + \frac{1}{c_k}(c_{k+1} y_{k+1} + \dots + c_n y_n); \text{ if } k = n, \text{ then } y_n = 0, \text{ so}$$

$$y_n = 0 \cdot y_1 + 0 \cdot y_2 + \dots + 0 \cdot y_n.$$

9.1.18. Since $F = \sum \pm f_{1i_1} f_{2i_2}, \dots, f_{ni_n}$,

$$F' = \sum \pm f'_{1i_1} f_{2i_2}, \dots, f_{ni_n} + \sum \pm f_{1i_1} f'_{2i_2}, \dots, f_{ni_n} + \dots + \sum \pm f_{1i_1} f_{2i_2}, \dots, f'_{ni_n}$$

$$= F_1 + F_2 + \dots + F_n.$$

9.1.20. Since $y_j^{(n)} = -\sum_{k=1}^n (P_k/P_0) y_j^{(n-k)}$, Exercise 9.1.19 implies that

$$W' = - \begin{vmatrix} y_1 & y_2 & \dots & y_n \\ y'_1 & y'_2 & \dots & y'_n \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-2)} & y_2^{(n-2)} & \dots & y_n^{(n-2)} \\ \sum_{k=1}^n (P_k/P_0) y_1^{(n-k)} & \sum_{k=1}^n (P_k/P_0) y_2^{(n-k)} & \dots & \sum_{k=1}^n (P_k/P_0) y_n^{(n-k)} \end{vmatrix},$$

so Exercise 9.1.17 implies that

$$W' = - \sum_{k=1}^n \frac{P_k}{P_0} \begin{vmatrix} y_1 & y_2 & \dots & y_n \\ y'_1 & y'_2 & \dots & y'_n \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-2)} & y_2^{(n-2)} & \dots & y_n^{(n-2)} \\ y_1^{(n-k)} & y_2^{(n-k)} & \dots & y_n^{(n-k)} \end{vmatrix}.$$

However, the determinants on the right each have two identical rows if $k = 2, \dots, n$.

Therefore, $W' = -\frac{P_1 W}{P_0}$. Separating variables yields $\frac{W'}{W} = -\frac{P_1}{P_0}$; hence

$$\ln \frac{W(x)}{W(x_0)} = - \int_{x_0}^x \frac{P_1(t)}{P_0(t)} dt, \text{ which implies Abel's formula.}$$

9.1.22. See the proof of Theorem 5.3.3.

9.1.24. (a)

$$P_0(x) = - \begin{vmatrix} x & x^2 - 1 & x^2 + 1 \\ 1 & 2x & 2x \\ 0 & 2 & 2 \end{vmatrix} = - \begin{vmatrix} x & x^2 - 1 & x^2 + 1 \\ 1 & 0 & 0 \\ 0 & 2 & 2 \end{vmatrix} = \begin{vmatrix} x^2 - 1 & x^2 + 1 \\ 2 & 2 \end{vmatrix} = -4;$$

$$P_1(x) = \begin{vmatrix} x & x^2 - 1 & x^2 + 1 \\ 1 & 2x & 2x \\ 0 & 0 & 0 \end{vmatrix} = 0; \quad P_2(x) = -\begin{vmatrix} x & x^2 - 1 & x^2 + 1 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{vmatrix} = 0;$$

$$P_3(x) = \begin{vmatrix} 1 & 2x & 2x \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{vmatrix} = 0. \text{ Therefore, } -4y''' = 0, \text{ which is equivalent to } y''' = 0.$$

(b)

$$P_0 = -\begin{vmatrix} e^x & e^{-x} & x \\ e^x & -e^{-x} & 1 \\ e^x & e^{-x} & 0 \end{vmatrix} = -\begin{vmatrix} 1 & 1 & x \\ 1 & -1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = -\begin{vmatrix} 0 & 0 & x \\ 0 & -2 & 1 \\ 1 & 1 & 0 \end{vmatrix} = -2x;$$

$$P_1 = \begin{vmatrix} e^x & e^{-x} & x \\ e^x & -e^{-x} & 1 \\ e^x & -e^{-x} & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & x \\ 1 & -1 & 1 \\ 1 & -1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 2 & x \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{vmatrix} = 2;$$

$$P_2 = -\begin{vmatrix} e^x & e^{-x} & x \\ e^x & e^{-x} & 0 \\ e^x & -e^{-x} & 0 \end{vmatrix} = -\begin{vmatrix} 1 & 1 & x \\ 1 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix} = -\begin{vmatrix} 0 & 0 & x \\ 1 & 1 & 0 \\ 0 & -2 & 0 \end{vmatrix} = 2x;$$

$$P_3 = \begin{vmatrix} e^x & -e^{-x} & 1 \\ e^x & e^{-x} & 0 \\ e^x & -e^{-x} & 0 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 1 \\ 1 & 2 & 0 \\ 1 & -1 & 0 \end{vmatrix} = -2.$$

Therefore, $-2xy''' + 2y'' + 2xy' - 2y = 0$, which is equivalent to $xy''' - y'' - xy' + y = 0$.

(c)

$$P_0(x) = -\begin{vmatrix} e^x & xe^{-x} & 1 \\ e^x & e^{-x}(1-x) & 0 \\ e^x & e^{-x}(x-2) & 0 \end{vmatrix} = -\begin{vmatrix} 1 & x & 1 \\ 1 & 1-x & 0 \\ 1 & x-2 & 0 \end{vmatrix} = -\begin{vmatrix} 1 & 1-x \\ 1 & x-2 \end{vmatrix} = 3 - 2x;$$

$$P_1(x) = \begin{vmatrix} e^x & xe^{-x} & 1 \\ e^x & e^{-x}(1-x) & 0 \\ e^x & e^{-x}(3-x) & 0 \end{vmatrix} = \begin{vmatrix} 1 & x & 1 \\ 1 & 1-x & 0 \\ 1 & 3-x & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1-x \\ 1 & 3-x \end{vmatrix} = 2;$$

$$P_2(x) = -\begin{vmatrix} e^x & xe^{-x} & 1 \\ e^x & e^{-x}(x-2) & 0 \\ e^x & e^{-x}(3-x) & 0 \end{vmatrix} = -\begin{vmatrix} 1 & x & 1 \\ 1 & x-2 & 0 \\ 1 & 3-x & 0 \end{vmatrix} = -\begin{vmatrix} 1 & x-2 \\ 1 & 3-x \end{vmatrix} = 2x - 5;$$

$$P_3(x) = \begin{vmatrix} e^x & e^{-x}(1-x) & 0 \\ e^x & e^{-x}(x-2) & 0 \\ e^x & e^{-x}(3-x) & 0 \end{vmatrix} = 0.$$

Therefore, $(3 - 2x)y''' + 2y'' + (2x - 5)y' = 0$.

(d)

$$P_0(x) = -\begin{vmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 2 & e^x \end{vmatrix} = -e^x \begin{vmatrix} x & x^2 & 1 \\ 1 & 2x & 1 \\ 0 & 2 & 1 \end{vmatrix} = -e^x \begin{vmatrix} x & x^2 - 2 & 0 \\ 1 & 2x - 2 & 0 \\ 0 & 2 & 1 \end{vmatrix}$$

$$= -e^x \begin{vmatrix} x & x^2 - 2 \\ 1 & 2x - 2 \end{vmatrix} = -e^x(x^2 - 2x + 2);$$

$$\begin{aligned}
P_1(x) &= \begin{vmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 0 & e^x \end{vmatrix} = e^x \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = x^2 e^x; \\
P_2(x) &= - \begin{vmatrix} x & x^2 & e^x \\ 0 & 2 & e^x \\ 0 & 0 & e^x \end{vmatrix} = -e^x \begin{vmatrix} x & x^2 \\ 0 & 2 \end{vmatrix} = -2x e^x; \\
P_3(x) &= \begin{vmatrix} 1 & 2x & e^x \\ 0 & 2 & e^x \\ 0 & 0 & e^x \end{vmatrix} = e^x \begin{vmatrix} 1 & 2x \\ 0 & 2 \end{vmatrix} = 2e^x.
\end{aligned}$$

Therefore, $-e^x(x^2 - 2x + 2)y''' + x^2 e^x y'' - 2x e^x y' + 2e^x y = 0$; which is equivalent to $(x^2 - 2x + 2)y''' - x^2 y'' + 2xy' - 2y = 0$.

(e)

$$\begin{aligned}
P_0(x) &= - \begin{vmatrix} x & x^2 & 1/x \\ 1 & 2x & -1/x^2 \\ 0 & 2 & 2/x^3 \end{vmatrix} = -x \begin{vmatrix} 1 & x & 1/x^2 \\ 1 & 2x & -1/x^2 \\ 0 & 2 & 2/x^3 \end{vmatrix} = -x \begin{vmatrix} 1 & x & 1/x^2 \\ 0 & x & -2/x^2 \\ 0 & 2 & 2/x^3 \end{vmatrix} \\
&= -x \begin{vmatrix} x & -2/x^2 \\ 2 & 2/x^3 \end{vmatrix} = -\frac{6}{x}; \\
P_1(x) &= \begin{vmatrix} x & x^2 & 1/x \\ 1 & 2x & -1/x^2 \\ 0 & 0 & -6/x^4 \end{vmatrix} = -\frac{6}{x^4} \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = -\frac{6}{x^2}; \\
P_2(x) &= - \begin{vmatrix} x & x^2 & 1/x \\ 0 & 2 & 2/x^3 \\ 0 & 0 & -6/x^4 \end{vmatrix} = \frac{6}{x^4} \begin{vmatrix} x & x^2 \\ 0 & 2 \end{vmatrix} = \frac{12}{x^3}; \\
P_3(x) &= \begin{vmatrix} 1 & 2x & -1/x^2 \\ 0 & 2 & 2/x^3 \\ 0 & 0 & -6/x^4 \end{vmatrix} = -\frac{6}{x^4} \begin{vmatrix} 1 & 2x \\ 0 & 2 \end{vmatrix} = -\frac{12}{x^4}.
\end{aligned}$$

Therefore, $-\frac{6}{x}y''' - \frac{6}{x^2}y'' + \frac{12}{x^3}y' - \frac{12}{x^4}y = 0$, which is equivalent to $x^3 y''' + x^2 y'' - 2xy' + 2y = 0$.

(f)

$$\begin{aligned}
P_0(x) &= - \begin{vmatrix} x+1 & e^x & e^{3x} \\ 1 & e^x & 3e^{3x} \\ 0 & e^x & 9e^{3x} \end{vmatrix} = -e^{4x} \begin{vmatrix} x+1 & 1 & 1 \\ 1 & 1 & 3 \\ 0 & 1 & 9 \end{vmatrix} = -e^{4x} \begin{vmatrix} x+1 & 0 & -8 \\ 1 & 0 & -6 \\ 0 & 1 & 9 \end{vmatrix} \\
&= e^{4x} \begin{vmatrix} x+1 & -8 \\ 1 & -6 \end{vmatrix} = 2e^{4x}(1-3x); \\
P_1(x) &= \begin{vmatrix} x+1 & e^x & e^{3x} \\ 1 & e^x & 3e^{3x} \\ 0 & e^x & 27e^{3x} \end{vmatrix} = e^{4x} \begin{vmatrix} x+1 & 1 & 1 \\ 1 & 1 & 3 \\ 0 & 1 & 27 \end{vmatrix} = e^{4x} \begin{vmatrix} x+1 & 0 & -26 \\ 1 & 0 & -24 \\ 0 & 1 & 27 \end{vmatrix} \\
&= -e^{4x} \begin{vmatrix} x+1 & -26 \\ 1 & -24 \end{vmatrix} = 2e^{4x}(12x-1);
\end{aligned}$$

$$\begin{aligned}
P_2(x) &= - \begin{vmatrix} x+1 & e^x & e^{3x} \\ 0 & e^x & 9e^{3x} \\ 0 & e^x & 27e^{3x} \end{vmatrix} = -e^{4x} \begin{vmatrix} x+1 & 1 & 1 \\ 0 & 1 & 9 \\ 0 & 1 & 27 \end{vmatrix} \\
&= -e^{4x} \begin{vmatrix} x+1 & 1 & 1 \\ 0 & 1 & 9 \\ 0 & 0 & 18 \end{vmatrix} = -18e^{4x}(x+1); \\
P_3(x) &= \begin{vmatrix} 1 & e^x & 3e^{3x} \\ 0 & e^x & 9e^{3x} \\ 0 & e^x & 27e^{3x} \end{vmatrix} = e^{4x} \begin{vmatrix} 1 & 1 & 3 \\ 0 & 1 & 9 \\ 0 & 1 & 27 \end{vmatrix} = e^{4x} \begin{vmatrix} 1 & 1 & 3 \\ 0 & 1 & 9 \\ 0 & 0 & 18 \end{vmatrix} = 18e^{4x}.
\end{aligned}$$

Therefore,

$$2e^{4x}(1-3x)y''' + 2e^{4x}(12x-1)y'' - 18e^{4x}(x+1)y' + 18e^{4x}y = 0,$$

which is equivalent to

$$(3x-1)y''' - (12x-1)y'' + 9(x+1)y' - 9y = 0.$$

(g)

$$\begin{aligned}
P_0(x) &= \begin{vmatrix} x & x^3 & 1/x & 1/x^2 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \end{vmatrix} = x \begin{vmatrix} 1 & x^2 & 1/x^2 & 1/x^3 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \end{vmatrix} \\
&= x \begin{vmatrix} 1 & x^2 & 1/x^2 & 1/x^3 \\ 0 & 2x^2 & -2/x^2 & -3/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \end{vmatrix} = x \begin{vmatrix} 2x^2 & -2/x^2 & -3/x^3 \\ 6x & 2/x^3 & 6/x^4 \\ 6 & -6/x^4 & -24/x^5 \end{vmatrix} \\
&= x^2 \begin{vmatrix} 2x & -2/x^3 & -3/x^4 \\ 6x & 2/x^3 & 6/x^4 \\ 6 & -6/x^4 & -24/x^5 \end{vmatrix} = x^2 \begin{vmatrix} 2x & -2/x^3 & -3/x^4 \\ 0 & 8/x^3 & 15/x^4 \\ 0 & 0 & -15/x^5 \end{vmatrix} = -\frac{240}{x^5} \\
P_1(x) &= - \begin{vmatrix} x & x^3 & 1/x & 1/x^2 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = -x \begin{vmatrix} 1 & x^2 & 1/x^2 & 1/x^3 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} \\
&= -x \begin{vmatrix} 1 & x^2 & 1/x^2 & 1/x^3 \\ 0 & 2x^2 & -2/x^2 & -3/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = -x^2 \begin{vmatrix} 2x & -2/x^3 & -3/x^4 \\ 6x & 2/x^3 & 6/x^4 \\ 0 & 24/x^5 & 120/x^6 \end{vmatrix} \\
&= -x^2 \begin{vmatrix} 2x & -2/x^3 & -3/x^4 \\ 0 & 8/x^3 & 15/x^4 \\ 0 & 24/x^5 & 120/x^6 \end{vmatrix} = -2x^3 \begin{vmatrix} 8/x^3 & 15/x^4 \\ 24/x^5 & 120/x^6 \end{vmatrix} = -\frac{1200}{x^6}; \\
P_2(x) &= \begin{vmatrix} x & x^3 & 1/x & 1/x^2 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6 & -6/x^4 & -24/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = x \begin{vmatrix} 1 & x^2 & 1/x^2 & 1/x^3 \\ 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6 & -6/x^4 & -24/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix}
\end{aligned}$$

$$\begin{aligned}
&= x \begin{vmatrix} 1 & x^2 & 1/x^2 & 1/x^3 \\ 0 & 2x^2 & -2/x^2 & -3/x^3 \\ 0 & 6 & -6/x^4 & -24/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = x^3 \begin{vmatrix} 2 & -2/x^4 & -3/x^5 \\ 6 & -6/x^4 & -24/x^5 \\ 0 & 24/x^5 & 120/x^6 \end{vmatrix} \\
&= x^3 \begin{vmatrix} 2 & -2/x^4 & -3/x^5 \\ 0 & 0 & -15/x^5 \\ 0 & 24/x^5 & 120/x^6 \end{vmatrix} = 2x^3 \begin{vmatrix} 0 & -15/x^5 \\ 24/x^5 & 120/x^6 \end{vmatrix} = \frac{720}{x^7}; \\
P_3(x) &= - \begin{vmatrix} x & x^3 & 1/x & 1/x^2 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = -x^2 \begin{vmatrix} 6 & 2/x^4 & 6/x^5 \\ 6 & -6/x^4 & -24/x^5 \\ 0 & 24/x^5 & 120/x^6 \end{vmatrix} \\
&= -x^2 \begin{vmatrix} 6 & 2/x^4 & 6/x^5 \\ 0 & -8/x^4 & -30/x^5 \\ 0 & 24/x^5 & 120/x^6 \end{vmatrix} = -6x^2 \begin{vmatrix} -8/x^4 & -30/x^5 \\ 24/x^5 & 120/x^6 \end{vmatrix} = \frac{1440}{x^6}; \\
P_4(x) &= \begin{vmatrix} 1 & 3x^2 & -1/x^2 & -2/x^3 \\ 0 & 6x & 2/x^3 & 6/x^4 \\ 0 & 6 & -6/x^4 & -24/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = x \begin{vmatrix} 6 & 2/x^4 & 6/x^5 \\ 6 & -6/x^4 & -24/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} \\
&= x \begin{vmatrix} 6 & 2/x^4 & 6/x^5 \\ 0 & -8/x^4 & -30/x^5 \\ 0 & 0 & 24/x^5 & 120/x^6 \end{vmatrix} = 6x \begin{vmatrix} -8/x^4 & -30/x^5 \\ 24/x^5 & 120/x^6 \end{vmatrix} = -\frac{1440}{x^9}.
\end{aligned}$$

Therefore,

$$-\frac{240}{x^5}y^{(4)} - \frac{1200}{x^6}y''' + \frac{720}{x^7}y'' + \frac{1440}{x^8}y' - \frac{1440}{x^9}y = 0,$$

which is equivalent to $x^4y^{(4)} + 5x^3y''' - 3x^2y'' - 6xy' + 6y = 0$.

(h)

$$\begin{aligned}
P_0(x) &= \begin{vmatrix} x & x \ln |x| & 1/x & x^2 \\ 1 & \ln |x| + 1 & -1/x^2 & 2x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & -1/x^2 & -6/x^4 & 0 \end{vmatrix} = x \begin{vmatrix} 1 & \ln |x| & 1/x^2 & x \\ 1 & \ln |x| + 1 & -1/x^2 & 2x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & -1/x^2 & -6/x^4 & 0 \end{vmatrix} \\
&= x \begin{vmatrix} 1 & \ln |x| & 1/x^2 & x \\ 0 & 1 & -2/x^2 & x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & -1/x^2 & -6/x^4 & 0 \end{vmatrix} = x \begin{vmatrix} 1 & -2/x^2 & x \\ 1/x & 2/x^3 & 2 \\ -1/x^2 & -6/x^4 & 0 \end{vmatrix} \\
&= x \begin{vmatrix} 1 & -2/x^2 & x \\ 1/x & 2/x^3 & 2 \\ -1/x^2 & -6/x^4 & 0 \end{vmatrix} = x \begin{vmatrix} 1 & -2/x^2 & x \\ 0 & 4/x^3 & 1 \\ 0 & -8/x^4 & 1/x \end{vmatrix} = x \begin{vmatrix} 4/x^3 & 1 \\ -8/x^4 & 1/x \end{vmatrix} = \frac{12}{x^3}; \\
P_1(x) &= - \begin{vmatrix} x & x \ln |x| & 1/x & x^2 \\ 1 & \ln |x| + 1 & -1/x^2 & 2x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} = -x \begin{vmatrix} 1 & \ln |x| & 1/x^2 & x \\ 1 & \ln |x| + 1 & -1/x^2 & 2x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix}
\end{aligned}$$

$$\begin{aligned}
&= -x \begin{vmatrix} 1 & \ln|x| & 1/x^2 & x \\ 0 & 1 & -2/x^2 & x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} = -x \begin{vmatrix} 1 & -2/x^2 & x \\ 1/x & 2/x^3 & 2 \\ 2/x^3 & 24/x^5 & 0 \end{vmatrix} \\
&= -x \begin{vmatrix} 1 & -2/x^2 & x \\ 0 & 4/x^3 & 1 \\ 0 & 28/x^5 & -2/x^2 \end{vmatrix} = -x \begin{vmatrix} 4/x^3 & 1 \\ 28/x^5 & -2/x^2 \end{vmatrix} = \frac{36}{x^4}; \\
P_2(x) &= \begin{vmatrix} x & x \ln|x| & 1/x & x^2 \\ 1 & \ln|x| + 1 & -1/x^2 & 2x \\ 0 & -1/x^2 & -6/x^4 & 0 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} = x \begin{vmatrix} 1 & \ln|x| & 1/x^2 & x \\ 1 & \ln|x| + 1 & -1/x^2 & 2x \\ 0 & -1/x^2 & -6/x^4 & 0 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} \\
&= x \begin{vmatrix} 1 & \ln|x| & 1/x^2 & x \\ 0 & 1 & -2/x^2 & x \\ 0 & -1/x^2 & -6/x^4 & 0 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} = x \begin{vmatrix} 1 & -2/x^2 & x \\ -1/x^2 & -6/x^4 & 0 \\ 2/x^3 & 24/x^5 & 0 \end{vmatrix} \\
&= x \begin{vmatrix} 1 & -2/x^2 & x \\ 0 & -8/x^4 & 1/x \\ 0 & 28/x^5 & -2/x^2 \end{vmatrix} = x \begin{vmatrix} -8/x^4 & 1/x \\ 28/x^5 & -2/x^2 \end{vmatrix} = -\frac{12}{x^5}; \\
P_3(x) &= - \begin{vmatrix} x & x \ln|x| & 1/x & x^2 \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & -1/x^2 & -6/x^4 & 0 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} = -x \begin{vmatrix} 1/x & 2/x^3 & 2 \\ -1/x^2 & -6/x^4 & 0 \\ 2/x^3 & 24/x^5 & 0 \end{vmatrix} \\
&= -x \begin{vmatrix} 1/x & 2/x^3 & 2 \\ 0 & -4/x^4 & 2/x \\ 0 & 20/x^5 & -4/x^2 \end{vmatrix} = - \begin{vmatrix} -4/x^4 & 2/x \\ 20/x^5 & -4/x^2 \end{vmatrix} = \frac{24}{x^6}; \\
P_4(x) &= \begin{vmatrix} 1 & \ln|x| + 1 & -1/x^2 & 2x \\ 0 & 1/x & 2/x^3 & 2 \\ 0 & -1/x^2 & -6/x^4 & 0 \\ 0 & 2/x^3 & 24/x^5 & 0 \end{vmatrix} = \begin{vmatrix} 1/x & 2/x^3 & 2 \\ -1/x^2 & -6/x^4 & 0 \\ 2/x^3 & 24/x^5 & 0 \end{vmatrix} \\
&= 2 \begin{vmatrix} -1/x^2 & -6/x^4 \\ 2/x^3 & 24/x^5 \end{vmatrix} = -\frac{24}{x^7}.
\end{aligned}$$

Therefore,

$$\frac{12}{x^3}y^{(4)} + \frac{36}{x^4}y''' - \frac{12}{x^5}y'' + \frac{24}{x^6}y' - \frac{24}{x^7}y = 0,$$

which is equivalent to $x^4y^{(4)} + 3x^2y''' - x^2y'' + 2xy' - 2y = 0$.

(i)

$$P_0(x) = \begin{vmatrix} e^x & e^{-x} & x & e^{2x} \\ e^x & -e^{-x} & 1 & 2e^{2x} \\ e^x & e^{-x} & 0 & 4e^{2x} \\ e^x & -e^{-x} & 0 & 8e^{2x} \end{vmatrix} = e^{2x} \begin{vmatrix} 1 & 1 & x & 1 \\ 1 & -1 & 1 & 2 \\ 1 & 1 & 0 & 4 \\ 1 & -1 & 0 & 8 \end{vmatrix}$$

$$\begin{aligned}
&= e^{2x} \begin{vmatrix} 2 & 1 & x & 1 \\ 0 & -1 & 1 & 2 \\ 2 & 1 & 0 & 4 \\ 0 & -1 & 0 & 8 \end{vmatrix} = e^{2x} \begin{vmatrix} 2 & 1 & x & 1 \\ 0 & -1 & 1 & 2 \\ 0 & 0 & -x & 3 \\ 0 & -1 & 0 & 8 \end{vmatrix} \\
&= 2e^{2x} \begin{vmatrix} -1 & 1 & 2 \\ 0 & -x & 3 \\ -1 & 0 & 8 \end{vmatrix} = 2e^{2x} \begin{vmatrix} -1 & 1 & 2 \\ 0 & -x & 3 \\ 0 & -1 & 6 \end{vmatrix} \\
&= -2e^{2x} \begin{vmatrix} -x & 3 \\ -1 & 6 \end{vmatrix} = 6e^{2x}(2x-1); \\
P_1(x) &= - \begin{vmatrix} e^x & e^{-x} & x & e^{2x} \\ e^x & -e^{-x} & 1 & 2e^{2x} \\ e^x & e^{-x} & 0 & 4e^{2x} \\ e^x & e^{-x} & 0 & 16e^{2x} \end{vmatrix} = -e^{2x} \begin{vmatrix} 1 & 1 & x & 1 \\ 1 & -1 & 1 & 2 \\ 1 & 1 & 0 & 4 \\ 1 & 1 & 0 & 16 \end{vmatrix} \\
&= -e^{2x} \begin{vmatrix} 0 & 1 & x & 1 \\ 2 & -1 & 1 & 2 \\ 0 & 1 & 0 & 4 \\ 0 & 1 & 0 & 16 \end{vmatrix} = 2e^{2x} \begin{vmatrix} 1 & x & 1 \\ 1 & 0 & 4 \\ 1 & 0 & 16 \end{vmatrix} \\
&= 2e^{2x} \begin{vmatrix} 1 & x & 1 \\ 0 & -x & 3 \\ 0 & -x & 15 \end{vmatrix} = 2e^{2x} \begin{vmatrix} -x & 3 \\ -x & 15 \end{vmatrix} = -24xe^{2x}; \\
P_2(x) &= \begin{vmatrix} e^x & e^{-x} & x & e^{2x} \\ e^x & -e^{-x} & 1 & 2e^{2x} \\ e^x & -e^{-x} & 0 & 8e^{2x} \\ e^x & e^{-x} & 0 & 16e^{2x} \end{vmatrix} = e^{2x} \begin{vmatrix} 1 & 1 & x & 1 \\ 1 & -1 & 1 & 2 \\ 1 & -1 & 0 & 8 \\ 1 & 1 & 0 & 16 \end{vmatrix} \\
&= e^{2x} \begin{vmatrix} 0 & 1 & x & 1 \\ 2 & -1 & 1 & 2 \\ 2 & -1 & 0 & 8 \\ 0 & 1 & 0 & 16 \end{vmatrix} = e^{2x} \begin{vmatrix} 0 & 1 & x & 1 \\ 2 & -1 & 1 & 2 \\ 0 & 0 & -1 & 6 \\ 0 & 1 & 0 & 16 \end{vmatrix} = -2e^{2x} \begin{vmatrix} 1 & x & 1 \\ 0 & -1 & 6 \\ 1 & 0 & 16 \end{vmatrix} \\
&= -2e^{2x} \begin{vmatrix} 1 & x & 1 \\ 0 & -1 & 6 \\ 0 & -x & 15 \end{vmatrix} = -2e^{2x} \begin{vmatrix} -1 & 6 \\ -x & 15 \end{vmatrix} = 6e^{2x}(5-2x); \\
P_3(x) &= - \begin{vmatrix} e^x & e^{-x} & x & e^{2x} \\ e^x & e^{-x} & 0 & 4e^{2x} \\ e^x & -e^{-x} & 0 & 8e^{2x} \\ e^x & e^{-x} & 0 & 16e^{2x} \end{vmatrix} = -e^{2x} \begin{vmatrix} 1 & 1 & x & 1 \\ 1 & 1 & 0 & 4 \\ 1 & -1 & 0 & 8 \\ 1 & 1 & 0 & 16 \end{vmatrix} \\
&= -e^{2x} \begin{vmatrix} 0 & 1 & x & 1 \\ 0 & 1 & 0 & 4 \\ 2 & -1 & 0 & 8 \\ 0 & 1 & 0 & 16 \end{vmatrix} = -2e^{2x} \begin{vmatrix} 1 & x & 1 \\ 0 & -x & 3 \\ 0 & -x & 15 \end{vmatrix} = -2e^{2x} \begin{vmatrix} -x & 3 \\ -x & 15 \end{vmatrix} = 24xe^{2x}; \\
P_4(x) &= \begin{vmatrix} e^x & -e^{-x} & 1 & 2e^{2x} \\ e^x & e^{-x} & 0 & 4e^{2x} \\ e^x & -e^{-x} & 0 & 8e^{2x} \\ e^x & e^{-x} & 0 & 16e^{2x} \end{vmatrix} = e^{2x} \begin{vmatrix} 1 & -1 & 1 & 2 \\ 1 & 1 & 0 & 4 \\ 1 & -1 & 0 & 8 \\ 1 & 1 & 0 & 16 \end{vmatrix}
\end{aligned}$$

$$\begin{aligned}
&= e^{2x} \begin{vmatrix} 0 & -1 & 1 & 2 \\ 2 & 1 & 0 & 4 \\ 0 & -1 & 0 & 8 \\ 2 & 1 & 0 & 16 \end{vmatrix} = e^{2x} \begin{vmatrix} 0 & -1 & 1 & 2 \\ 2 & 1 & 0 & 4 \\ 0 & -1 & 0 & 8 \\ 0 & 0 & 0 & 12 \end{vmatrix} \\
&= -2e^{2x} \begin{vmatrix} -1 & 1 & 2 \\ -1 & 0 & 8 \\ 0 & 0 & 12 \end{vmatrix} = -2e^{2x} \begin{vmatrix} -1 & 1 & 2 \\ 0 & -1 & 6 \\ 0 & 0 & 12 \end{vmatrix} = -24e^{2x}.
\end{aligned}$$

Therefore,

$$6e^{2x}(2x-1)y^{(4)} - 24xe^{2x}y''' + 6e^{2x}(5-2x)y'' + 24xe^{2x}y' - 24e^{2x}y = 0,$$

which is equivalent to $(2x-1)y^{(4)} - 4xy''' + (5-2x)y'' + 4xy' - 4y = 0$.

(j)

$$\begin{aligned}
P_0(x) &= \begin{vmatrix} e^{2x} & e^{-2x} & 1 & x^2 \\ 2e^{2x} & -2e^{-2x} & 0 & 2x \\ 4e^{2x} & 4e^{-2x} & 0 & 2 \\ 8e^{2x} & -8e^{-2x} & 0 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & x^2 \\ 2 & -2 & 0 & 2x \\ 4 & 4 & 0 & 2 \\ 8 & -8 & 0 & 0 \end{vmatrix} \\
&= \begin{vmatrix} 2 & -2 & 2x \\ 4 & 4 & 2 \\ 8 & -8 & 0 \end{vmatrix} = \begin{vmatrix} 2 & -2 & 2x \\ 0 & 8 & 2-4x \\ 0 & 0 & -8x \end{vmatrix} = -128x \\
P_1(x) &= - \begin{vmatrix} e^{2x} & e^{-2x} & 1 & x^2 \\ 2e^{2x} & -2e^{-2x} & 0 & 2x \\ 4e^{2x} & 4e^{-2x} & 0 & 2 \\ 16e^{2x} & 16e^{-2x} & 0 & 0 \end{vmatrix} = - \begin{vmatrix} 1 & 1 & 1 & x^2 \\ 2 & -2 & 0 & 2x \\ 4 & 4 & 0 & 2 \\ 16 & 16 & 0 & 0 \end{vmatrix} \\
&= - \begin{vmatrix} 2 & -2 & 2x \\ 4 & 4 & 2 \\ 16 & 16 & 0 \end{vmatrix} = - \begin{vmatrix} 2 & -2 & 2x \\ 0 & 8 & 2-4x \\ 0 & 0 & -8 \end{vmatrix} = 128; \\
P_2(x) &= \begin{vmatrix} e^{2x} & e^{-2x} & 1 & x^2 \\ 2e^{2x} & -2e^{-2x} & 0 & 2x \\ 8e^{2x} & -8e^{-2x} & 0 & 0 \\ 16e^{2x} & 16e^{-2x} & 0 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & x^2 \\ 2 & -2 & 0 & 2x \\ 8 & -8 & 0 & 0 \\ 16 & 16 & 0 & 0 \end{vmatrix} \\
&= \begin{vmatrix} 2 & -2 & 2x \\ 8 & -8 & 0 \\ 16 & 16 & 0 \end{vmatrix} = 2x \begin{vmatrix} 8 & -8 \\ 16 & 16 \end{vmatrix} = 512x; \\
P_3(x) &= - \begin{vmatrix} e^{2x} & e^{-2x} & 1 & x^2 \\ 4e^{2x} & 4e^{-2x} & 0 & 2 \\ 8e^{2x} & -8e^{-2x} & 0 & 0 \\ 16e^{2x} & 16e^{-2x} & 0 & 0 \end{vmatrix} = - \begin{vmatrix} 1 & 1 & 1 & x^2 \\ 4 & 4 & 0 & 2 \\ 8 & -8 & 0 & 0 \\ 16 & 16 & 0 & 0 \end{vmatrix} \\
&= - \begin{vmatrix} 4 & 4 & 2 \\ 8 & -8 & 0 \\ 16 & 16 & 0 \end{vmatrix} = -2 \begin{vmatrix} 8 & -8 \\ 16 & 16 \end{vmatrix} = -512;
\end{aligned}$$

$$P_4(x) = \begin{vmatrix} 2e^{2x} & -2e^{-2x} & 0 & 2x \\ 4e^{2x} & 4e^{-2x} & 0 & 2 \\ 8e^{2x} & -8e^{-2x} & 0 & 0 \\ 16e^{2x} & 16e^{-2x} & 0 & 0 \end{vmatrix} = 0.$$

Therefore, $-128xy^{(4)} + 128y''' + 512xy'' - 512y = 0$, which is equivalent to $xy^{(4)} - y''' - 4xy'' + 4y' = 0$.

9.2 Higher Order Constant Coefficient Homogeneous Equations

$$9.2.2. \quad p(r) = r^4 + 8r^2 - 9 = (r-1)(r+1)(r^2+9); \quad y = c_1e^x + c_2e^{-x} + c_3\cos 3x + c_4\sin 3x.$$

$$9.2.4. \quad p(r) = 2r^3 + 3r^2 - 2r - 3 = (r-1)(r+1)(2r+3); \quad y = c_1e^x + c_2e^{-x} + c_3e^{-3x/2}.$$

$$9.2.6. \quad p(r) = 4r^3 - 8r^2 + 5r - 1 = (r-1)(2r-1)^2; \quad y = c_1e^x + e^{x/2}(c_2 + c_3x).$$

$$9.2.8. \quad p(r) = r^4 + r^2 = r^2(r^2+1); \quad y = c_1 + c_2x + c_3\cos x + c_4\sin x.$$

$$9.2.10. \quad p(r) = r^4 + 12r^2 + 36 = (r^2+6)^2; \quad y = (c_1 + c_2x)\cos\sqrt{6}x + (c_3 + c_4x)\sin\sqrt{6}x.$$

$$9.2.12. \quad p(r) = 6r^4 + 5r^3 + 7r^2 + 5r + 1 = (2r+1)(3r+1)(r^2+1);$$

$$y = c_1e^{-x/2} + c_2e^{-x/3} + c_3\cos x + c_4\sin x.$$

$$9.2.14. \quad p(r) = r^4 - 4r^3 + 7r^2 - 6r + 2 = (r-1)^2(r^2-2r+2); \quad y = e^x(c_1 + c_2x + c_3\cos x + c_4\sin x).$$

$$9.2.16. \quad p(r) = r^3 + 3r^2 - r - 3 = (r-1)(r+1)(r+3);$$

$$\begin{aligned} y &= c_1e^x + c_2e^{-x} + c_3e^{-3x} & c_1 + c_2 + c_3 &= 0 \\ y' &= c_1e^x - c_2e^{-x} - 3c_3e^{-3x}; & c_1 - c_2 - 3c_3 &= 14 \quad ; \\ y'' &= c_1e^x + c_2e^{-x} + 9c_3e^{-3x} & c_1 + c_2 + 9c_3 &= -40 \end{aligned}$$

$$c_1 = 2, \quad c_2 = 3, \quad c_3 = -5; \quad y = 2e^x + 3e^{-x} - 5e^{-3x}.$$

$$9.2.18. \quad p(r) = r^3 - 2r - 4 = (r-2)(r^2+2r+2);$$

$$\begin{aligned} y &= e^{-x}(c_1\cos x + c_2\sin x) + c_3e^{2x} & c_1 + c_3 &= 6 \\ y' &= -e^{-x}((c_1 - c_2)\cos x + (c_1 + c_2)\sin x) + 2c_3e^{2x}; & -c_1 + c_2 + 2c_3 &= 3 \quad ; \\ y'' &= e^{-x}(2c_1\sin x - 2c_2\cos x) + 4c_3e^{2x} & -2c_2 + 4c_3 &= 22 \end{aligned}$$

$$c_1 = 2, \quad c_2 = -3, \quad c_3 = 4; \quad y = 2e^{-x}\cos x - 3e^{-x}\sin x + 4e^{2x}.$$

$$9.2.20. \quad p(r) = r^3 - 6r^2 + 12r - 8 = (r-2)^3;$$

$$\begin{aligned} y &= e^{2x}(c_1 + c_2x + c_3x^2) & c_1 &= 1 \\ y' &= e^{2x}(2c_1 + c_2 + (2c_2 + 2c_3)x + 2c_3x^2) & ; & \quad 2c_1 + c_2 = -1 \\ y'' &= 2e^{2x}(2c_1 + 2c_2 + c_3 + 2(c_2 + 2c_3)x + 2c_3x^2) & 4c_1 + 4c_2 + 2c_3 &= -4 \end{aligned}$$

$$c_1 = 1, \quad c_2 = -3, \quad c_3 = 2; \quad y = e^{2x}(1 - 3x + 2x^2).$$

$$9.2.22. \quad p(r) = 8r^3 - 4r^2 - 2r + 1 = (2r + 1)(2r - 1)^2;$$

$$\begin{aligned} y &= e^{x/2}(c_1 + c_2x) + c_3e^{-x/2} & c_1 + c_3 &= 4 \\ y' &= \frac{1}{2}e^{x/2}(c_1 + 2c_2 + c_2x) - \frac{1}{2}c_3e^{-x/2}; & \frac{1}{2}c_1 + c_2 - \frac{1}{2}c_3 &= -3, \\ y'' &= \frac{1}{4}e^{x/2}(c_1 + 4c_2 + c_2x) + \frac{1}{4}c_3e^{-x/2} & \frac{1}{4}c_1 + c_2 + \frac{1}{4}c_3 &= -1 \end{aligned}$$

$$c_1 = 1, c_2 = -2, c_3 = 3; y = e^{x/2}(1 - 2x) + 3e^{-x/2}.$$

$$9.2.24. \quad p(r) = r^4 - 6r^3 + 7r^2 + 6r - 8 = (r - 1)(r - 2)(r - 4)(r + 1);$$

$$\begin{aligned} y &= c_1e^x + c_2e^{2x} + c_3e^{4x} + c_4e^{-x} & c_1 + c_2 + c_3 + c_4 &= -2 \\ y' &= c_1e^x + 2c_2e^{2x} + 4c_3e^{4x} - c_4e^{-x} & c_1 + 2c_2 + 4c_3 - c_4 &= -8 \\ y'' &= c_1e^x + 4c_2e^{2x} + 16c_3e^{4x} + c_4e^{-x}; & c_1 + 4c_2 + 16c_3 + c_4 &= -14, \\ y''' &= c_1e^x + 8c_2e^{2x} + 64c_3e^{4x} - c_4e^{-x} & c_1 + 8c_2 + 64c_3 - c_4 &= -62 \end{aligned}$$

$$c_1 = -4, c_2 = 1, c_3 = -1, c_4 = 2; y = -4e^x + e^{2x} - e^{4x} + 2e^{-x}.$$

$$9.2.26. \quad p(r) = r^4 + 2r^3 - 2r^2 - 8r - 8 = (r - 2)(r + 2)(r^2 + 2r + 2);$$

$$\begin{aligned} y &= c_1e^{2x} + c_2e^{-2x} + e^{-x}(c_3 \cos x + c_4 \sin x) \\ y' &= 2c_1e^{2x} - 2c_2e^{-2x} - e^{-x}((c_3 - c_4) \cos x + (c_3 + c_4) \sin x) \\ y'' &= 4c_1e^{2x} + 4c_2e^{-2x} + e^{-x}(2c_3 \sin x - 2c_4 \cos x) \\ y''' &= 8c_1e^{2x} - 8c_2e^{-2x} + e^{-x}((2c_3 + 2c_4) \cos x + 2(c_4 - c_3) \sin x); \end{aligned}$$

$$c_1 + c_2 + c_3 = 5$$

$$2c_1 - 2c_2 - c_3 + c_4 = -2$$

$$4c_1 + 4c_2 - 2c_4 = 6$$

$$8c_1 - 8c_2 + 2c_3 + 2c_4 = 8;$$

$$c_1 = 1, c_2 = 1, c_3 = 3, c_4 = 1; y = e^{2x} + e^{-2x} + e^{-x}(3 \cos x + \sin x).$$

$$9.2.28. \quad (a) \quad W(x) = \begin{vmatrix} e^x & xe^x & e^{2x} \\ e^x & e^x(x+1) & 2e^{2x} \\ e^x & e^x(x+2) & 4e^{2x} \end{vmatrix}; \quad W(0) = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} = 1.$$

$$(b) \quad W(x) = \begin{vmatrix} \cos 2x & \sin 2x & e^{3x} \\ -2 \sin 2x & 2 \cos 2x & 3e^{3x} \\ -4 \cos 2x & -4 \sin 2x & 9e^{3x} \end{vmatrix};$$

$$W(0) = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 2 & 3 \\ -4 & 0 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 2 & 3 \\ -4 & 0 & 13 \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 0 & 13 \end{vmatrix} = 26.$$

$$(c) \quad W(x) = \begin{vmatrix} e^{-x} \cos x & e^{-x} \sin x & e^x \\ -e^{-x}(\cos x + \sin x) & e^{-x}(\cos x - \sin x) & e^x \\ 2e^{-x} \sin x & -2e^{-x} \cos x & e^x \end{vmatrix};$$

$$W(0) = \begin{vmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & -2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ -2 & 1 \end{vmatrix} = 5.$$

$$(d) \quad W(x) = \begin{vmatrix} 1 & x & x^2 & e^x \\ 0 & 1 & 2x & e^x \\ 0 & 0 & 2 & e^x \\ 0 & 0 & 0 & e^x \end{vmatrix}; \quad W(0) = \begin{vmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 1.$$

$$(e) \quad W(x) = \begin{vmatrix} e^x & e^{-x} & \cos x & \sin x \\ e^x & -e^{-x} & -\sin x & \cos x \\ e^x & e^{-x} & -\cos x & -\sin x \\ e^x & -e^{-x} & \sin x & -\cos x \end{vmatrix};$$

$$\begin{aligned} W(0) &= \begin{vmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \\ 1 & 1 & -1 & 0 \\ 1 & -1 & 0 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \\ 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & -2 \end{vmatrix} = -2 \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & -1 \end{vmatrix} \\ &= -2 \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 0 & -2 \end{vmatrix} = 4 \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -8. \end{aligned}$$

$$(f) \quad W(x) = \begin{vmatrix} \cos x & \sin x & e^x \cos x & e^x \sin x \\ -\sin x & \cos x & e^x(\cos x - \sin x) & e^x(\cos x + \sin x) \\ -\cos x & -\sin x & -2e^x \sin x & 2e^x \cos x \\ \sin x & -\cos x & -e^x(2 \cos x + 2 \sin x) & e^x(2 \cos x - 2 \sin x) \end{vmatrix};$$

$$\begin{aligned} W(0) &= \begin{vmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ -1 & 0 & 0 & 2 \\ 0 & -1 & -2 & 2 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & -1 & -2 & 2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ -1 & -2 & 2 \end{vmatrix} \\ &= \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & -1 & 3 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 5 \end{vmatrix} = 5. \end{aligned}$$

9.2.30. (a) Since $y = Q_1(D)P_1(D)y + Q_2(D)P_2(D)y$ and $P_1(D)y = P_2(D)y = 0$, it follows that $y = 0$.

(b) Suppose that (A) $a_1u_1 + \cdots + a_ru_r + b_1v_1 + \cdots + b_sv_s = 0$, where $a_1, \dots, a_r$ and $b_1, \dots, b_s$ are constants. Denote $u = a_1u_1 + \cdots + a_ru_r$ and $v = b_1v_1 + \cdots + b_sv_s$. Then (B) $P_1(D)u = 0$ and (C) $P_2(D)v = 0$. Since $u + v = 0$, $P_2(D)(u + v) = 0$. Therefore, $0 = P_2(D)(u + v) = P_2(D)u + P_2(D)v$. Now (C) implies that $P_2(D)u = 0$. This, (B), and (a) imply that $u = a_1u_1 + \cdots + a_ru_r = 0$, so $a_1 = \cdots = a_r = 0$, since $u_1, \dots, u_r$ are linearly independent. Now (A) reduces to $b_1v_1 + \cdots + b_sv_s = 0$, so $b_1 = \cdots = b_s = 0$, since $v_1, \dots, v_s$ are linearly independent. Therefore, $u_1, \dots, u_r, v_1, \dots, v_s$ are linearly independent.

(c) It suffices to show that $\{y_1, y_2, \dots, y_n\}$ is linearly independent. Suppose that $c_1y_1 + \cdots + c_ny_n = 0$. We may assume that $y_1, \dots, y_r$ are linearly independent solutions of $p_1(D)y = 0$ and $y_{r+1}, \dots, y_n$ are solutions of $P_2(D)y = p_2(D) \cdots p_k(D)y = 0$. Since $p_1(r)$ and $P_2(r)$ have no common factors, (b) implies that (A) $c_1y_1 + \cdots + c_ry_r = 0$ and

(B) $c_{r+1}y_{r+1} + \dots + c_n y_n = 0$. Now (A) implies that $c_1 = \dots = c_r = 0$, since $y_1, \dots, y_r$ are linearly independent. If $k = 2$, then $y_{r+1}, \dots, y_n$ are linearly independent, so $c_{r+1} = \dots = c_n = 0$, and the proof is complete. If $k > 2$ repeat this argument, starting from (B), with p_1 replaced by p_2 , and P_2 replaced by $P_3 = p_3 \dots p_n$.

9.2.32. (a)

$$\begin{aligned} (\cos A + i \sin A)(\cos B + i \sin B) &= (\cos A \cos B - \sin A \sin B) \\ &\quad + (\cos A \sin B + \sin A \cos B) \\ &= \cos(A + B) + i \sin(A + B). \end{aligned}$$

(b) Obvious for $n = 0$. If $n = -1$ write

$$\begin{aligned} \frac{1}{\cos \theta + i \sin \theta} &= \frac{1}{\cos \theta + i \sin \theta} \frac{\cos \theta - i \sin \theta}{\cos \theta - i \sin \theta} \\ &= \frac{\cos \theta - i \sin \theta}{\cos^2 \theta + \sin^2 \theta} = \cos \theta - i \sin \theta = \cos(-\theta) + i \sin(-\theta). \end{aligned}$$

(d) If n is a negative integer, then (B) $(\cos \theta + i \sin \theta)^n = \frac{1}{(\cos \theta + i \sin \theta)^{|n|}}$. From the hint,

(C) $\frac{1}{(\cos \theta + i \sin \theta)^{|n|}} = (\cos \theta - i \sin \theta)^{|n|} = (\cos(-\theta) + i \sin(-\theta))^{|n|}$. Replacing θ by $-\theta$ and n by $|n|$ in (A) shows that (D) $(\cos(-\theta) + i \sin(-\theta))^{|n|} = \cos(-|n|\theta) + i \sin(-|n|\theta)$. Since $|n| = -n$, (E) $\cos(-|n|\theta) + i \sin(-|n|\theta) = \cos n\theta + i \sin n\theta$. Now (B), (C), (D), and (E) imply (A).

(e) From (A), $z_k^n = \cos 2k\pi + i \sin 2k\pi = 1$ and

$$\zeta_k^n = \cos(2k+1)\pi + i \sin(2k+1)\pi = \cos(2k+1)\pi = \cos \pi = -1.$$

(f) From (e), $\rho^{1/n}z_0, \dots, \rho^{1/n}z_{n-1}$ are all zeros of $z^n - \rho$. Since they are distinct numbers, $z^n - \rho$ has the stated factorization.

From (e), $\rho^{1/n}\zeta_0, \dots, \rho^{1/n}\zeta_{n-1}$ are all zeros of $z^n + \rho$. Since they are distinct numbers, $z^n + \rho$ has the stated factorization.

9.2.33. (a) $p(r) = r^3 - 1 = (r - z_0)(r - z_1)(r - z_2)$ where $z_k = \cos \frac{2k\pi}{3} + i \sin \frac{2k\pi}{3}$, $k = 0, 1, 2$.

Hence, $z_0 = 1$, $z_1 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, and $z_2 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$. Therefore,

$p(r) = (r - 1) \left(\left(r + \frac{1}{2} \right)^2 + \frac{3}{4} \right)$, so $\left\{ e^x, e^{-x/2} \cos \left(\frac{\sqrt{3}}{2}x \right), e^{-x/2} \sin \left(\frac{\sqrt{3}}{2}x \right) \right\}$ is a fundamental set of solutions.

(b) $p(r) = r^3 + 1 = (r - \zeta_0)(r - \zeta_1)(r - \zeta_2)$ where $\zeta_k = \cos \frac{(2k+1)\pi}{3} + i \sin \frac{(2k+1)\pi}{3}$,

$k = 0, 1, 2$. Hence, $\zeta_0 = \frac{1}{2} + i\frac{\sqrt{3}}{2}$, $\zeta_1 = -1$, $\zeta_2 = \frac{1}{2} - i\frac{\sqrt{3}}{2}$. Therefore,

$p(r) = (r + 1) \left(\left(r - \frac{1}{2} \right)^2 + \frac{3}{4} \right)$, so $\left\{ e^{-x}, e^{x/2} \cos \left(\frac{\sqrt{3}}{2}x \right), e^{x/2} \sin \left(\frac{\sqrt{3}}{2}x \right) \right\}$ is a fundamental set of solutions.

- (c) $p(r) = r^4 + 64 = (r - 2\sqrt{2}\zeta_0)(r - 2\sqrt{2}\zeta_1)(r - 2\sqrt{2}\zeta_2)(r - 2\sqrt{2}\zeta_3)$, where $\zeta_k = \cos \frac{(2k+1)\pi}{4} + i \sin \frac{(2k+1)\pi}{4}$, $k = 0, 1, 2, 3$. Therefore, $\zeta_0 = \frac{1+i}{\sqrt{2}}$, $\zeta_1 = \frac{-1+i}{\sqrt{2}}$, $\zeta_2 = \frac{-1-i}{\sqrt{2}}$, and $\zeta_3 = \frac{1-i}{\sqrt{2}}$, so $p(r) = ((r-2)^2 + 4)((r+2)^2 + 4)$ and $\{e^{2x} \cos 2x, e^{2x} \sin 2x, e^{-2x} \cos 2x, e^{-2x} \sin 2x\}$ is a fundamental set of solutions.
- (d) $p(r) = r^6 - 1 = (r - z_0)(r - z_1)(r - z_2)(r - z_3)(r - z_4)(r - z_5)$ where $z_k = \cos \frac{2k\pi}{6} + i \sin \frac{2k\pi}{6}$, $k = 0, 1, 2, 3, 4, 5$. Therefore, $z_0 = 1$, $z_1 = \frac{1}{2} + i\frac{\sqrt{3}}{2}$, $z_2 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, $z_3 = -1$, $z_4 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$, and $z_5 = \frac{1}{2} - i\frac{\sqrt{3}}{2}$, so $p(r) = (r-1)(r+1) \left(\left(r - \frac{1}{2} \right)^2 + \frac{3}{4} \right) \left(\left(r + \frac{1}{2} \right)^2 + \frac{3}{4} \right)$ and $\left\{ e^x, e^{-x}, e^{x/2} \cos \left(\frac{\sqrt{3}}{2}x \right), e^{x/2} \sin \left(\frac{\sqrt{3}}{2}x \right), e^{-x/2} \cos \left(\frac{\sqrt{3}}{2}x \right), e^{-x/2} \sin \left(\frac{\sqrt{3}}{2}x \right) \right\}$ is a fundamental set of solutions.
- (e) $p(r) = r^6 + 64 = (r - 2\zeta_0)(r - 2\zeta_1)(r - 2\zeta_2)(r - 2\zeta_3)(r - 2\zeta_4)(r - 2\zeta_5)$ where $\zeta_k = \cos \frac{(2k+1)\pi}{6} + i \sin \frac{(2k+1)\pi}{6}$, $k = 0, 1, 2, 3, 4, 5$. Therefore, $\zeta_0 = \frac{\sqrt{3}}{2} + \frac{i}{2}$, $\zeta_1 = i$, $\zeta_2 = -\frac{\sqrt{3}}{2} + \frac{i}{2}$, $\zeta_3 = -\frac{\sqrt{3}}{2} - \frac{i}{2}$, $\zeta_4 = -i$, and $\zeta_5 = \frac{\sqrt{3}}{2} - \frac{i}{2}$, so $p(r) = (r^2 + 4)((r - \sqrt{3})^2 + 1)((r + \sqrt{3})^2 + 1)$ and $\{\cos 2x, \sin 2x, e^{-\sqrt{3}x} \cos x, e^{-\sqrt{3}x} \sin x, e^{\sqrt{3}x} \cos x, e^{\sqrt{3}x} \sin x\}$ is a fundamental set of solutions.
- (f) $p(r) = (r-1)^6 - 1 = (r-1-z_0)(r-1-z_1)(r-1-z_2)(r-1-z_3)(r-1-z_4)(r-1-z_5)$ where $z_k = \cos \frac{2k\pi}{6} + i \sin \frac{2k\pi}{6}$, $k = 0, 1, 2, 3, 4, 5$. Therefore, $z_0 = 1$, $z_1 = \frac{1}{2} + i\frac{\sqrt{3}}{2}$, $z_2 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, $z_3 = -1$, $z_4 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$, and $z_5 = \frac{1}{2} - i\frac{\sqrt{3}}{2}$, so $p(r) = r(r-2) \left(\left(r - \frac{3}{2} \right)^2 + \frac{3}{4} \right) \left(\left(r - \frac{1}{2} \right)^2 + \frac{3}{4} \right)$ and $\left\{ 1, e^{2x}, e^{3x/2} \cos \left(\frac{\sqrt{3}}{2}x \right), e^{3x/2} \sin \left(\frac{\sqrt{3}}{2}x \right), e^{x/2} \cos \left(\frac{\sqrt{3}}{2}x \right), e^{x/2} \sin \left(\frac{\sqrt{3}}{2}x \right) \right\}$ is a fundamental set of solutions.
- (g) $p(r) = r^5 + r^4 + r^3 + r^2 + r + 1 = \frac{r^6 - 1}{r - 1}$. Therefore, from the solution of (d) $p(r) = (r+1) \left(\left(r - \frac{1}{2} \right)^2 + \frac{3}{4} \right) \left(\left(r + \frac{1}{2} \right)^2 + \frac{3}{4} \right)$ and $\left\{ e^{-x}, e^{x/2} \cos \left(\frac{\sqrt{3}}{2}x \right), e^{x/2} \sin \left(\frac{\sqrt{3}}{2}x \right), e^{-x/2} \cos \left(\frac{\sqrt{3}}{2}x \right), e^{-x/2} \sin \left(\frac{\sqrt{3}}{2}x \right) \right\}$ is a fundamental set of solutions.

9.3 Undetermined Coefficients for Higher Order Equations

9.3.2. If $y = u^{-3x}$, then

$$y''' - 2y'' - 5y' + 6y = e^{-3x}[(u''' - 11u'' + 34u' - 24u) - 2(u'' - 6u' + 9u) - 5(u' - 3u) + 6u] = e^{-3x}(u''' - 11u'' + 34u' - 24u). \text{ Let } u_p = A + Bx + Cx^2, \text{ where } (-24A + 34B - 22C) + (-24B + 68C)x - 24Cx^2 = 32 - 23x + 6x^2. \text{ Then } C = -1/4, B = 1/4, A = -3/4 \text{ and } y_p = -\frac{e^{-3x}}{4}(3 - x + x^2).$$

9.3.4. If $y = ue^{-2x}$, then $y''' + 3y'' - y' - 3y =$

$$e^{-2x}[(u''' - 6u'' + 12u' - 8u) + 3(u'' - 4u' + 4u) - (u' - 2u) - 3u] = e^{-2x}(u''' - 3u'' - u' + 3u). \text{ Let } u_p = A + Bx + Cx^2, \text{ where } (3A - B - 6C) + (3B - 2C)x + 3Cx^2 = 2 - 17x + 3x^2. \text{ Then } C = 1, B = -5, A = 1, \text{ and } y_p = e^{-2x}(1 - 5x + x^2).$$

9.3.6. If $y = ue^x$, then

$$y''' + y'' - 2y = e^x[(u''' + 3u'' + 3u' + u) + (u'' + 2u' + u) - 2u] = e^x(u''' + 4u'' + 5u'). \text{ Let } u_p = x(A + Bx + Cx^2), \text{ where } (5A + 8B + 6C) + (10B + 24C)x + 15Cx^2 = 14 + 34x + 15x^2. \text{ Then } C = 1, B = 1, A = 0, \text{ and } y_p = x^2e^x(1 + x).$$

9.3.8. If $y = ue^x$, then

$$y''' - y'' - y' + y = e^x[(u''' + 3u'' + 3u' + u) - (u'' + 2u' + u) - (u' + u) + u] = e^x(u''' + 2u''). \text{ Let } u_p = x^2(A + Bx) \text{ where } (4A + 6B) + 12Bx = 7 + 6x. \text{ Then } B = 1/2, A = 1, \text{ and } y_p = \frac{x^2e^x}{2}(2 + x).$$

9.3.10. If $y = ue^{3x}$, then $y''' - 5y'' + 3y' + 9y =$

$$e^{3x}[(u''' + 9u'' + 27u' + 27u) - 5(u'' + 6u' + 9u) + 3(u' + 3u) + 9u] = e^{3x}(u''' + 4u''). \text{ Let } u_p = x^2(A + Bx + Cx^2), \text{ where } (8A + 6B) + (24B + 24C)x + 48Cx^2 = 22 - 48x^2. \text{ Then } C = -1, B = 1, A = 2, \text{ and } y_p = x^2e^{3x}(2 + x - x^2).$$

9.3.12. If $y = ue^{x/2}$, then $8y''' - 12y'' + 6y' - y =$

$$e^{x/2}[8(u''' + 3u''/2 + 3u'/4 + u/8) - 12(u'' + u' + u/4) + 6(u' + u/2) - u] = 8e^{x/2}u''', \text{ so } u''' = \frac{1 + 4x}{8}. \text{ Integrating three times and taking the constants of integration to be zero yields } u_p = \frac{x^3}{48}(1 + x). \text{ Therefore, } y_p = \frac{x^3e^{x/2}}{48}(1 + x).$$

9.3.14. If $y = ue^{2x}$, then

$$y^{(4)} + 3y''' + y'' - 3y' - 2y = e^{2x}[(u^{(4)} + 8u''' + 24u'' + 32u' + 16u) + 3(u''' + 6u'' + 12u' + 8u) + (u'' + 4u' + 4u) - 3(u' + 2u) - 2u] = e^{2x}(u^{(4)} + 11u''' + 43u'' + 69u' + 36u). \text{ Let } u_p = A + Bx \text{ where } (36A + 69B) + 36Bx = -33 - 36x. \text{ Then } B = -1, A = 1, \text{ and } y_p = e^{2x}(1 - x).$$

9.3.16. If $y = ue^x$, then

$$4y^{(4)} - 11y''' - 9y'' - 2y = e^x[4(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 11(u'' + 2u' + u) - 9(u' + u) - 2u] = e^x(4u^{(4)} + 16u''' + 13u'' - 15u' - 18u). \text{ Let } u_p = A + Bx \text{ where } -(18A + 15B) - 18Bx = -1 + 6x. \text{ Then } B = -1/3, A = 1/3, \text{ and } y_p = \frac{e^x}{3}(1 - x).$$

9.3.18. If $y = ue^x$, then $y^{(4)} - 4y''' + 6y'' - 4y' + 2y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 4(u''' + 3u'' + 3u' + u) + 6(u'' + 2u' + u) - 4(u' + u) + 2u] = e^x(u^{(4)} + u)$. Let $u_p = A + Bx + Cx^2 + Dx^3 + Ex^4$ where $(A + 24E) + Bx + Cx^2 + Dx^3 + Ex^4 = 24 + x + x^4$. Then $E = 1$, $D = 0$, $C = 0$, $B = 1$, $A = 0$, and $y_p = xe^x(1 + x^3)$.

9.3.20. If $y = ue^{2x}$, then $y^{(4)} + y''' - 2y'' - 6y' - 4y = e^{2x}[(u^{(4)} + 8u''' + 24u'' + 32u' + 16u) + (u''' + 6u'' + 12u' + 8u) - 2(u'' + 4u' + 4u) - 6(u' + 2u) - 4u] = e^{2x}(u^{(4)} + 9u''' + 28u'' + 30u')$. Let $u_p = x(A + Bx + Cx^2)$ where $(30A + 56B + 54C) + (60B + 168C)x + 90Cx^2 = -(4 + 28x + 15x^2)$. Then $C = -1/6$, $B = 0$, $A = 1/6$, and $y_p = \frac{xe^{2x}}{6}(1 - x^2)$.

9.3.22. If $y = ue^x$, then $y^{(4)} - 5y'' + 4y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 5(u'' + 2u' + u) + 4u] = e^x(u^{(4)} + 4u''' + u'' - 6u')$. Let $u_p = x(A + Bx + Cx^2)$ where $(-6A + 2B + 24C) + (-12B + 6C)x - 18Cx^2 = 3 + x - 3x^2$, so $C = 1/6$, $B = 0$, $A = 1/6$. Then $y_p = \frac{xe^x}{6}(1 + x^2)$.

9.3.24. If $y = ue^{2x}$, then $y^{(4)} - 3y''' + 4y' = e^{2x}[(u^{(4)} + 8u''' + 24u'' + 32u' + 16u) - 3(u''' + 6u'' + 12u' + 8u) + 4(u' + 2u)] = e^{2x}(u^{(4)} + 5u''' + 6u'')$. Let $u_p = x^2(A + Bx + Cx^2)$ where $(12A + 30B + 24C) + (36B + 120C)x + 72Cx^3 = 15 + 26x + 12x^2$. Then $C = 1/6$, $B = 1/6$, $A = 1/2$, and $y_p = \frac{x^2e^{2x}}{6}(3 + x + x^2)$.

9.3.26. If $y = ue^x$, then $2y^{(4)} - 5y''' + 3y'' + y' - y = e^x[2(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 5(u''' + 3u'' + 3u' + u) + 3(u'' + 2u' + u) + (u' + u) - u] = e^x(2u^{(4)} + 3u''')$. Let $u_p = x^3(A + Bx)$, where $(18A + 48B) + 72Bx = 11 + 12x$. Then $B = 1/6$, $A = 1/6$, and $y_p = \frac{x^3e^x}{6}(1 + x)$.

9.3.28. If $y = ue^{2x}$, then $y^{(4)} - 7y''' + 18y'' - 20y' + 8y = e^{2x}[(u^{(4)} + 8u''' + 24u'' + 32u' + 16u) - 7(u''' + 6u'' + 12u' + 8u) + 18(u'' + 4u' + 4u) - 20(u' + 2u) + 8u] = e^{2x}(u^{(4)} + u''')$. Let $u_p = x^3(A + Bx + Cx^2)$ where $(6A + 24B) + (24B + 120C)x + 60Cx^2 = 3 - 8x - 5x^2$. Then $so C = -1/12$, $B = 1/12$, $A = 1/6$, and $y_p = \frac{x^3e^{2x}}{12}(2 + x - x^2)$.

9.3.30. If $y = ue^{-x}$, then $y''' + y'' - 4y' - 4y = e^{-x}[(u''' - 3u'' + 3u' - u) + (u'' - 2u' + u) - 4(u' - u) - 4u] = e^{-x}(u''' - 2u'' - 3u')$. Let $u_p = (A_0 + A_1x) \cos 2x + (B_0 + B_1x) \sin 2x$, where

$$\begin{aligned} 8A_1 - 14B_1 &= -22 \\ 14A_1 + 8B_1 &= -6 \\ 8A_0 - 14B_0 - 15A_1 - 8B_1 &= 1 \\ 14A_0 + 8B_0 + 8A_1 - 15B_1 &= -1. \end{aligned}$$

Then $A_1 = -1$, $B_1 = 1$, $A_0 = 1$, $B_0 = 1$, and $y_p = e^{-x}[(1 - x) \cos 2x + (1 + x) \sin 2x]$.

9.3.32. If $y = ue^x$, then

$$y''' - 2y'' + y' - 2y = e^x[(u''' + 3u'' + 3u' + u) - 2(u'' + 2u' + u) + (u' + u) - 2u] = e^x(u''' + u'' - 2u).$$

Let $u_p = (A_0 + A_1x + A_2x^2) \cos 2x + (B_0 + B_1x + B_2x^2) \sin 2x$ where

$$\begin{aligned} -6A_2 - 8B_2 &= -4 \\ 8A_2 - 6B_2 &= -3 \\ -6A_1 - 8B_1 - 24A_2 + 8B_2 &= 5 \\ 8A_1 - 6B_1 - 8A_2 - 24B_2 &= -5 \\ -6A_0 - 8B_0 - 12A_1 + 4B_1 + 2A_2 + 12B_2 &= -9 \\ 8A_0 - 6B_0 - 4A_1 - 12B_1 - 12A_2 + 2B_2 &= 6. \end{aligned}$$

Then $A_2 = 0$, $B_2 = 1/2$; $A_1 = 1/2$, $B_1 = -1/2$; $A_0 = 1/1$, $B_0 = 1/2$; and

$$y_p = \frac{e^x}{2}[(1+x) \cos 2x + (1-x+x^2) \sin 2x].$$

9.3.34. If $y = ue^x$, then

$$y''' - y'' + 2y = e^x[(u''' + 3u'' + 3u' + u) - (u'' + 2u' + u) + 2u] = e^x(u''' + 2u'' + u' + 2u).$$

Since $\cos x$ and $\sin x$ satisfy $u''' + 2u'' + u' + 2u = 0$, let

$u_p = x[(A_0 + A_1x) \cos x + (B_0 + B_1x) \sin x]$ where

$$\begin{aligned} -4A_1 + 8B_1 &= 4 \\ -8A_1 - 4B_1 &= -12 \\ -2A_0 + 4B_0 + 4A_1 + 6B_1 &= 20 \\ -4A_0 - 2B_0 - 6A_1 + 4B_1 &= -12. \end{aligned}$$

Then $A_1 = 1$, $B_1 = 1$; $A_0 = 1$, $B_0 = 3$; and $y_p = xe^x[(1+x) \cos x + (3+x) \sin x]$.

9.3.36. If $y = ue^{3x}$, then

$$= e^{3x}[(u''' + 9u'' + 27u' + 27u) - 6(u'' + 6u' + 9u) + 18(u' + 3u)] = e^{3x}(u''' + 3u'' + 9u' + 27u).$$

Since $\cos 3x$ and $\sin 3x$ satisfy $u''' + 3u'' + 9u' + 27u = 0$, let

$u_p = x[(A_0 + A_1x) \cos 3x + (B_0 + B_1x) \sin 3x]$ where

$$\begin{aligned} -36A_1 + 36B_1 &= 3 \\ -36A_1 - 36B_1 &= 3 \\ -18A_0 + 18B_0 + 6A_1 + 18B_1 &= -2 \\ -18A_0 - 18B_0 - 18A_1 + 6B_1 &= 3. \end{aligned}$$

Then $A_1 = -1/12$, $B_1 = 0$; $A_0 =$, $B_0 = -1/12$; and $y_p = -\frac{xe^{3x}}{12}(x \cos 3x + \sin 3x)$.

9.3.38. If $y = ue^x$, then $y^{(4)} - 3y''' + 2y'' + 2y' - 4y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 3(u''' + 3u'' + 3u' + u) + 2(u'' + 2u' + u) + 4(u' + u) + u] = e^x(u^{(4)} + u''' - u'' + u' - 2u)$. Let

$u_p = A \cos 2x + B \sin 2x$ where $18A - 6B = 2$ and $6A + 18B = -1$. Then $A = 1/12$,

$B = -1/12$, and $y_p = \frac{e^x}{12}(\cos 2x - \sin 2x)$.

9.3.40. If $y = ue^{-x}$, then $y^{(4)} + 6y''' + 13y'' + 12y' + 4y = e^{-x}[(u^{(4)} - 4u''' + 6u'' - 4u' + u) + 6(u''' - 3u'' + 3u' - u) + 13(u'' - 2u' + u) + 12u' - u] + 4u = e^{-x}(u^{(4)} + 2u''' + u'')$. Let $u_p = (A_0 + A_1x) \cos x + (B_0 + B_1x) \sin x$ where

$$\begin{aligned} -2B_1 &= -1 \\ 2A_1 &= -1 \\ -2B_0 - 6A_1 - 2B_1 &= 4 \\ 2A_0 + 2A_1 - 6B_1 &= -5. \end{aligned}$$

Then $A_1 = -1/2$, $B_1 = 1/2$, $A_0 = -1/2$, $B_0 = -1$, and

$$y_p = -\frac{e^{-x}}{2}[(1+x) \cos x + (2-x) \sin x].$$

9.3.42. If $y = ue^x$, then $y^{(4)} - 5y''' + 13y'' - 19y' + 10y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 5(u''' + 3u'' + 3u' + u) + 13(u'' + 2u' + u) - 19(u' + u) + 10u] = e^x(u^{(4)} - u''' + 4u'' - 4u')$. Since $\cos 2x$ and $\sin 2x$ satisfy $u^{(4)} - u''' + 4u'' - 4u' = 0$, let $u_p = x(A \cos 2x + B \sin 2x)$ where $8A - 16B = 1$ and $16A + 8B = 1$. Then $A = 3/40$, $B_0 = -1/40$, and

$$y_p = \frac{xe^x}{40}(3 \cos 2x - \sin 2x).$$

9.3.44. If $y = ue^x$, then $y^{(4)} - 5y''' + 13y'' - 19y' + 10y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 5(u''' + 3u'' + 3u' + u) + 13(u'' + 2u' + u) - 19(u' + u) + 10u] = e^x(u^{(4)} - u''' + 4u'' - 4u')$. Since $\cos 2x$ and $\sin 2x$ satisfy $u^{(4)} - u''' + 4u'' - 4u' = 0$, let $u_p = x[(A_0 + A_1x) \cos 2x + (B_0 + B_1x) \sin 2x]$ where

$$\begin{aligned} 16A_1 - 32B_1 &= 8 \\ 32A_1 + 16B_1 &= -4 \\ 8A_0 - 16B_0 - 40A_1 - 12B_1 &= 7 \\ 16A_0 + 8B_0 + 12A_1 - 40B_1 &= 8. \end{aligned}$$

Then $A_1 = 0$, $B_1 = -1/4$; $A_0 = 0$, $B_0 = -1/4$, and $y_p = -\frac{xe^x}{4}(1+x) \sin 2x$.

9.3.46. If $y = ue^{2x}$, then

$y^{(4)} - 8y''' + 32y'' - 64y' + 64y = e^{2x}[(u^{(4)} + 8u''' + 24u'' + 32u' + 16u) - 8(u''' + 6u'' + 12u' + 8u) + 32(u'' + 4u' + 4u) - 64(u' + 2u) + 64u] = e^{2x}(u^{(4)} + 8u'' + 16u)$. Since $\cos 2x$, $\sin 2x$, $x \cos 2x$, and $x \sin 2x$ satisfy $u^{(4)} + 8u'' + 16u = 0$, let $u_p = x^2(A \cos 2x + B \sin 2x)$ where $-32A = 1$ and $-32B = -1$. Then $A = -1/32$, $B = 1/32$, and

$$y_p = -\frac{x^2 e^{2x}}{32}(\cos 2x - \sin 2x).$$

9.3.48. Find particular solutions of (a) $y''' - 4y'' + 5y' - 2y = -4e^x$, (b)

$y''' - 4y'' + 5y' - 2y = e^{2x}$, and (c) $y''' - 4y'' + 5y' - 2y = -2 \cos x + 4 \sin x$.

(a) If $y = ue^x$, then

$$y''' - 4y'' + 5y' - 2y = e^x[(u''' + 3u'' + 3u' + u) - 4(u'' + 2u' + u) + 5(u' + u) - 2u] = e^x(u''' - u'').$$

Let $u_{1p} = Ax^2$ where $-2A = -4$. Then $A = 2$, and $y_{1p} = 2x^2 e^x$.

- (b) If $y = ue^{2x}$, then $y''' - 4y'' + 5y' - 2y = e^{2x}[(u''' + 6u'' + 12u' + 8u) - 4(u'' + 4u' + 4u) + 5(u' + 2u) - 2u] = e^{2x}(u''' + 2u'' + u')$. Let $u_{2p} = x$. Then $y_{2p} = xe^{2x}$.
- (c) If $y_{3p} = A \cos x + B \sin x$, then $y_{3p}''' - 4y_{3p}'' + 5y_{3p}' - 2y_{3p} = (2A + 4B) \cos x + (-4A + 2B) \sin x = -2 \cos x + 4 \sin x$ if $A = -1$ and $B = 0$, so $y_{3p} = -\cos x$.

From the principle of superposition, $y_p = 2x^2e^x + xe^{2x} - \cos x$.

- 9.3.50. Find particular solutions of (a) $y''' - y' = -2(1 + x)$, (b) $y''' - y' = 4e^x$, (c) $y''' - y' = -6e^{-x}$, and (d) $y''' - y' = 96e^{3x}$
- (a) Let $y_{1p} = x(A + Bx)$. Then $y_{1p}''' - y_{1p}' = -A - 2Bx = -2(1 + x)$ if $A = 2$ and $B = 1$; therefore $y_{1p} = 2x + 2x^2$.
- (b) If $y = ue^x$, then $y''' - y' = e^x[(u''' + 3u'' + 3u' + u) - (u' + u)] = e^x(u''' + 3u'' + 2u')$. Let $u_{2p} = 4x$. Then $y_{2p} = 4xe^x$.
- (c) If $y = ue^{-x}$, then $y''' - y' = e^{-x}[(u''' - 3u'' + 3u' - u) - (u' - u)] = e^{-x}(u''' - 3u'' + 2u')$. Let $u_{2p} = -3x$. Then $y_{2p} = -6xe^{-x}$.
- (d) Since e^{3x} does not satisfy the complementary equation, let $y_{4p} = Ae^{3x}$. Then $y_{4p}''' - y_{4p}' = 24Ae^{3x}$. Let $A = 4$; then $y_{4p} = 4e^{4x}$.

From the principle of superposition, $y_p = 2x + x^2 + 2xe^x - 3xe^{-x} + 4e^{3x}$

- 9.3.52. Find particular solutions of (a) $y''' + 3y'' + 3y' + y = 12e^{-x}$ and (b) $y''' + 3y'' + 3y' + y = 9 \cos 2x - 13 \sin 2x$.
- (a) If $y = ue^{-2x}$, then $y''' + 3y'' + 3y' + y = e^{-2x}[(u''' - 3u'' + 3u' - u) + 3(u'' - 2u' + u) + 3(u' - u) + u] = e^{-2x}u''$. Let $u_{1p} = 12$. Integrating three times and taking the constants of integration to be zero yields $u_{1p} = 2x^3$. Therefore, $y_{1p} = 2x^3$.
- (b) Let $y_{2p} = A \cos 2x + B \sin 2x$ where $-11A - 2B = 9$ and $2A - 11B = -13$. Then $A = -1$, $B = 1$, and $y_{2p} = -\cos 2x + \sin 2x$.

From the principle of superposition, $y_p = 2x^3e^{-2x} - \cos 2x + \sin 2x$.

- 9.3.54. Find particular solutions of (a) $y^{(4)} - 5y'' + 4y = -12e^x$, (b) $y^{(4)} - 5y'' + 4y = 6e^{-x}$, and (c) $y^{(4)} - 5y'' + 4y = 10 \cos x$.
- (a) If $y = ue^x$, then $y^{(4)} - 5y'' + 4y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 5(u'' + 2u' + u) + 4u] = e^x(u^{(4)} + 4u''' + u'' - 6u')$. Let $u_{1p} = 2x$. Then $y_{1p} = 2xe^x$.
- (b) If $y = ue^{-x}$, then $y^{(4)} - 5y'' + 4y = e^{-x}[(u^{(4)} - 4u''' + 6u'' - 4u' + u) - 5(u'' - 2u' + u) + 4u] = e^{-x}(u^{(4)} - 4u''' + u'' + 6u')$. Let $u_{2p} = x$. Then $y_{2p} = xe^{-x}$.
- (c) Let $y_{3p} = A \cos x + B \sin x$ where $10A = 10$ and $10B = 0$. Then $A = 1$, $B = 0$, and $y_{3p} = \cos x$.

From the principle of superposition, $y_p = 2xe^x + xe^{-x} + \cos x$.

9.3.56. Find particular solutions of (a) $y^{(4)} + 2y''' - 3y'' - 4y' + 4y = 2e^x(1+x)$ and (b)

$$y^{(4)} + 2y''' - 3y'' - 4y' + 4y = e^{-2x}.$$

(a) If $y = ue^x$, then $y^{(4)} + 2y''' - 3y'' - 4y' + 4y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) + 2(u''' + 3u'' + 3u' + u) - 3(u'' + 2u' + u) - 4(u' + u) + 4u] = e^x(u^{(4)} + 6u''' + 9u'')$. Let $u_{1p} = x^2(A + Bx)$ where $(18A + 36B) + 54Bx = 2 + 2x$. Then $B = 1/27$, $A = 1/27$, and $y_{1p} = \frac{x^2}{27}(1+x)e^x$.

(b) If $y = ue^{-2x}$, then $y^{(4)} + 2y''' - 3y'' - 4y' + 4y = e^{-2x}[(u^{(4)} - 8u''' + 24u'' - 32u' + 16u) + 2(u''' - 6u'' + 12u' - 8u) - 3(u'' - 4u' + 4u) - 4(u' - 2u) + 4u] = e^{-2x}(u^{(4)} - 6u''' + 9u'')$.

Let $u_{2p} = Ax^2$ where $18A = 1$. Then $A = 1/18$ and $y_p = \frac{x^2}{18}e^{-2x}$.

From the principle of superposition, $y_p = \frac{x^2}{54}[(2+2x)e^x + 3e^{-2x}]$.

9.3.58. Find particular solutions of (a) $y^{(4)} + 5y''' + 9y'' + 7y' + 2y = e^{-x}(30 + 24x)$ and (b)

$$y^{(4)} + 5y''' + 9y'' + 7y' + 2y = -e^{-2x}.$$

(a) If $y = ue^{-x}$, then $y^{(4)} + 5y''' + 9y'' + 7y' + 2y = e^{-x}[(u^{(4)} - 4u''' + 6u'' - 4u' + u) + 5(u''' - 3u'' + 3u' - u) + 9(u'' - 2u' + u) + 7(u' - u) + 2u] = e^{-x}(u^{(4)} + u''')$. Let $u_{1p} = x^3(A + Bx)$ where $(6A + 24B) + 24Bx = 30 + 24x$. The $B = 1$, $A = 1$, and $y_{1p} = x^3(1+x)e^{-x}$.

(b) If $y = ue^{-2x}$, then $y^{(4)} + 5y''' + 9y'' + 7y' + 2y = e^{-2x}[(u^{(4)} - 8u''' + 24u'' - 32u' + 16u) + 5(u''' - 6u'' + 12u' - 8u) + 9(u'' - 4u' + 4u) + 7(u' - 2u) + 2u] = e^{-2x}(u^{(4)} - 3u''' + 3u'' - u')$. Let $u_{2p} = x$. Then $y_{2p} = xe^{-2x}$.

From the principle of superposition, $y_p = x^3(1+x)e^{-x} + xe^{-2x}$.

9.3.60. If $y = ue^{2x}$, then $y''' - y'' - y' + y =$

$$e^{2x}[(u''' + 6u'' + 12u' + 8u) - (u'' + 4u' + 4u) - (u' + 2u) + u] = e^{2x}(u''' + 5u'' + 7u' + 3u).$$

Let $u_p = A + Bx$, where $(3A + 7B) + 3x = 10 + 3x$. Then $B = 1$, $A = 1$ and $y_p = e^{2x}(1+x)$.

Since $p(r) = (r+1)(r-1)^2$, $y = e^{2x}(1+x) + c_1e^{-x} + e^x(c_2 + c_3x)$

9.3.62. If $y = ue^{2x}$, then $y''' - 6y'' + 11y' - 6y =$

$$e^{2x}[(u''' + 6u'' + 12u' + 8u) - 6(u'' + 4u' + 4u) + 11(u' + 2u) - 6u] = e^{2x}(u''' - u').$$

Let $u_p = x(A + Bx + Cx^2)$ where $(-A + 6C) - 2Bx - 3Cx^2 = 5 - 4x - 3x^2$. Then $C = 1$, $B = 2$,

$A = 1$, and $y_p = xe^{2x}(1+x)^2$. Since $p(r) = (r-1)(r-2)(r-3)$,

$$y = xe^{2x}(1+x)^2 + c_1e^x + c_2e^{2x} + c_3e^{3x}.$$

9.3.64. If $y = ue^x$, then

$$y''' - 3y'' + 3y' - y = e^x[(u''' + 3u'' + 3u' + u) - 3(u'' + 2u' + u) + 3(u' + u) - u] = e^xu''.$$

Let $u''' = 1 + x$. Integrating three times and taking the constants of integration to be zero yields

$$u = \frac{x^3}{24}(4+x). \text{ Therefore, } y_p = \frac{x^3e^x}{24}(4+x). \text{ Since } p(r) = (r-1)^3,$$

$$y = \frac{x^3e^x}{24}(4+x) + e^x(c_1 + c_2x + c_3x^2).$$

9.3.66. If $y = ue^{-2x}$, then $y''' + 2y'' - y' - 2y = e^{-2x}[(u''' - 6u'' + 12u' - 8u) + 2(u'' - 4u' + 4u) - (u' - 2u) - 2u] = e^{-2x}(u''' - 4u'' + 3u')$.
Let $u_p = (A_0 + A_1x) \cos x + (B_0 + B_1x) \sin x$ where

$$\begin{aligned} 4A_1 + 2B_1 &= -2 \\ -2A_1 + 4B_1 &= -9 \\ 4A_0 + 2B_0 - 8B_1 &= 23 \\ -2A_0 + 4B_0 + 8A_1 &= 8. \end{aligned}$$

Then $A_1 = 1/2$, $B_1 = -2$; $A_0 = 1$, $B_0 = 3/2$, and

$$\begin{aligned} y_p &= e^{-2x} \left[\left(1 + \frac{x}{2}\right) \cos x + \left(\frac{3}{2} - 2x\right) \sin x \right]. \text{ Since } p(r) = (r-1)(r+1)(r+2), \\ y &= e^{-2x} \left[\left(1 + \frac{x}{2}\right) \cos x + \left(\frac{3}{2} - 2x\right) \sin x \right] + c_1 e^x + c_2 e^{-x} + c_3 e^{-2x}. \end{aligned}$$

9.3.68. If $y = ue^x$, then $y^{(4)} - 4y''' + 14y'' - 20y' + 25y = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 4(u''' + 3u'' + 3u' + u) + 14(u'' + 2u' + u) - 20(u' + u) + 25u] = e^x(u^{(4)} + 8u'' + 16u)$. Since $\cos 2x$, $\sin 2x$, $x \cos 2x$, and $x \sin 2x$ satisfy $u^{(4)} + 8u'' + 16u = 0$, let $u_p = x^2[(A_0 + A_1x) \cos 2x + (B_0 + B_1x) \sin 2x]$ where

$$\begin{aligned} -96A_1 &= 6 \\ -96B_1 &= 0 \\ -32A_0 + 48B_1 &= 2 \\ -48A_1 - 32B_0 &= 3. \end{aligned}$$

Then $A_1 = -1/16$, $B_1 = 0$; $A_0 = -1/16$, $B_0 = 0$, and $y_l = -\frac{x^2 e^x}{16}(1+x) \cos 2x$. Since

$$p(r) = [(r-1)^2 + 1]^2, \quad y = -\frac{x^2 e^x}{16}(1+x) \cos 2x + e^x [(c_1 + c_2 x) \cos 2x + (c_3 + c_4 x) \sin 2x].$$

9.3.70. If $y = ue^{-x}$, then

$$y''' - y'' - y' + y = e^{-x}[(u''' - 3u'' + 3u' - u) - (u'' - 2u' + u) - (u' - u) + u] = e^{-x}(u''' - 4u'' + 4u').$$

Let $u_p = x(A + Bx)$, where $(4A - 8B) + 8Bx = -4 + 8x$. Then $B = 1$, $A = 1$, and

$y_p = x(1+x)e^{-x}$. Since $p(r) = (r+1)(r-1)^2$ the general solution is

$y = x(1+x)e^{-x} + c_1 e^{-x} + c_2 e^x + c_3 x e^x$. Therefore,

$$\begin{bmatrix} y \\ y' \\ y'' \end{bmatrix} = \begin{bmatrix} x(1+x)e^{-x} \\ -e^{-x}(x^2 - x - 1) \\ e^{-x}(x^2 - 3x) \end{bmatrix} + \begin{bmatrix} e^{-x} & e^x & x e^x \\ -e^{-x} & e^x & e^x(x+1) \\ e^{-x} & e^x & e^x(x+2) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Setting $x = 0$ and imposing the initial conditions yields

$$\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix},$$

so $c_1 = 1$, $c_2 = 1$, $c_3 = -1$, and $y = e^{-x}(1+x+x^2) + (1-x)e^x$.

9.3.72. If $y = ue^{-x}$, then $y''' - 2y'' - 5y' + 6y = e^{-x}[(u^{(4)} - 4u''' + 6u'' - 4u' + u) + 2(u''' - 3u'' + 3u' - u) + 2(u'' - 2u' + u) + 2(u' - u) + u] = e^{-x}(u^{(4)} - 2u''' + 2u'')$. Let $u_p = x^2(A + Bx)$, where $(4A - 12B) + 12Bx = 20 - 12x$. Then $B = -1$, $A = 2$, and $y_p = x^2(2 - x)e^{-x}$. Since $p(r) = (r + 1)^2(r^2 + 1)$, the general solution is $y = x^2(2 - x)e^{-x} + e^{-x}(c_1 + c_2x) + c_3 \cos x + c_4 \sin x$. Therefore,

$$\begin{bmatrix} y \\ y' \\ y'' \\ y''' \end{bmatrix} = \begin{bmatrix} x^2(2 - x)e^{-x} \\ x(x^2 - 5x + 4)e^{-x} \\ -(x^3 - 8x^2 + 14x - 4)e^{-x} \\ (x^3 - 11x^2 + 30x - 18)e^{-x} \end{bmatrix} + \begin{bmatrix} e^{-x} & xe^{-x} & \cos x & \sin x \\ -e^{-x} & (1 - x)e^{-x} & -\sin x & \cos x \\ e^{-x} & (x - 2)e^{-x} & -\cos x & -\sin x \\ -e^{-x} & (3 - x)e^{-x} & \sin x & -\cos x \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

Setting $x = 0$ and imposing the initial conditions yields

$$\begin{bmatrix} 3 \\ -4 \\ 7 \\ -22 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 4 \\ -18 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 1 \\ 1 & -2 & -1 & 0 \\ -1 & 3 & 0 & -1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix},$$

so $c_1 = 2$, $c_2 = -1$, $c_3 = 1$, $c_4 = -1$, and $y = (2 - x)(x^2 + 1)e^{-x} + \cos x - \sin x$.

9.3.74. If $y = ue^x$, then $y^{(4)} - 3y''' + 5y'' - 2y' = e^x[(u^{(4)} + 4u''' + 6u'' + 4u' + u) - 3(u''' + 3u'' + 3u' + u) + 4(u'' + 2u' + u) - 2(u' + u)] = e^x(u^{(4)} + u''' + u'' + u')$. Since $\cos x$ and $\sin x$ satisfy $u^{(4)} + u''' + u'' + u' = 0$, let $u_p = x(A \cos x + B \sin x)$ where $-2A - 2B = -2$ and $2A - 2B = 2$. Then $A = 1$, $B = 0$, and $y_p = xe^x \cos x$. Since $p(r) = r(r - 1)[(r - 1)^2 + 1]$ the general solution is $y = xe^x \cos x + c_1 + e^x(c_2 + c_3 \cos x + c_4 \sin x)$. Therefore,

$$\begin{bmatrix} y \\ y' \\ y'' \\ y''' \end{bmatrix} = \begin{bmatrix} xe^x \cos x \\ e^x((x + 1) \cos x - x \sin x) \\ e^x(2 \cos x - 2(x + 1) \sin x) \\ -e^x(2x \cos x + 2(x + 3) \sin x) \end{bmatrix} + \begin{bmatrix} 1 & e^x & e^x \cos x & e^x \sin x \\ 0 & e^x & e^x(\cos x - \sin x) & e^x(\cos x + \sin x) \\ 0 & e^x & -2e^x \sin x & 2e^x \cos x \\ 0 & e^x & -e^x(2 \cos x + 2 \sin x) & e^x(2 \cos x - 2 \sin x) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

Setting $x = 0$ and imposing the initial conditions yields

$$\begin{bmatrix} 2 \\ 0 \\ -1 \\ -5 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & -2 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix},$$

so $c_1 = 2$, $c_2 = -1$, $c_3 = 1$, $c_4 = -1$, and $y = 2 + e^x[(1 + x) \cos x - \sin x - 1]$.

9.4 Variation of Parameters for Higher Order Equations

$$\begin{aligned} 9.4.2. \quad W &= \begin{vmatrix} e^{-x^2} & xe^{-x^2} & x^2e^{-x^2} \\ -2xe^{-x^2} & e^{-x^2}(1 - 2x^2) & 2xe^{-x^2}(1 - x^2) \\ e^{-x^2}(4x^2 - 2) & 2xe^{-x^2}(2x^2 - 3) & 2e^{-x^2}(2x^4 - 5x^2 + 1) \end{vmatrix} = 2e^{-3x^2}; \\ W_1 &= \begin{vmatrix} xe^{-x^2} & x^2e^{-x^2} \\ e^{-x^2}(1 - 2x^2) & 2xe^{-x^2}(1 - x^2) \end{vmatrix} = x^2e^{-2x^2}; \\ W_2 &= \begin{vmatrix} e^{-x^2} & x^2e^{-x^2} \\ -2xe^{-x^2} & 2xe^{-x^2}(1 - x^2) \end{vmatrix} = 2xe^{-2x^2}; \quad W_3 = \begin{vmatrix} e^{-x^2} & xe^{-x^2} \\ -2xe^{-x^2} & e^{-x^2}(1 - 2x^2) \end{vmatrix} = e^{-2x^2}; \end{aligned}$$

$$u'_1 = \frac{FW_1}{P_0W} = \frac{1}{2}x^{5/2}; u'_2 = -\frac{FW_2}{P_0W} = -x^{3/2}; u'_3 = \frac{FW_3}{P_0W} = \sqrt{x}/2; u_1 = x^{7/2}/7; u_2 = -\frac{2}{5}x^{5/2};$$

$$u_3 = x^{3/2}/3; y_p = u_1y_1 + u_2y_2 + u_3y_3 = \frac{8}{105}e^{-x^2}x^{7/2}.$$

$$9.4.4. \quad W = \begin{vmatrix} 1 & \frac{e^x}{x} & \frac{e^{-x}}{x} \\ 0 & \frac{e^x(x-1)}{x^2} & -\frac{e^{-x}(x+1)}{x^2} \\ 0 & \frac{e^x(x^2-2x+2)}{x^3} & \frac{e^{-x}(x^2+2x+2)}{x^3} \end{vmatrix} = 2/x^2;$$

$$W_1 = \begin{vmatrix} \frac{e^x}{x} & \frac{e^{-x}}{x} \\ e^x \frac{x-1}{x^2} & -e^{-x} \frac{x+1}{x^2} \end{vmatrix} = -\frac{2}{x^2}; \quad W_2 = \begin{vmatrix} 1 & \frac{e^{-x}}{x} \\ 0 & -\frac{e^{-x}(x+1)}{x^2} \end{vmatrix} = -\frac{e^{-x}(x+1)}{x^2};$$

$$W_3 = \begin{vmatrix} 1 & \frac{e^x}{x} \\ 0 & \frac{e^x(x-1)}{x^2} \end{vmatrix} = \frac{e^x(x-1)}{x^2}; \quad u'_1 = \frac{FW_1}{P_0W} = -2; \quad u'_2 = -\frac{FW_2}{P_0W} = e^{-x}(x+1);$$

$$u'_3 = \frac{FW_3}{P_0W} = e^x(x-1); \quad u_1 = -2x; \quad u_2 = -e^{-x}(x+2); \quad u_3 = e^x(x-2);$$

$$y_p = u_1y_1 + u_2y_2 + u_3y_3 = -2(x^2+2)/x.$$

$$9.4.6. \quad W = \begin{vmatrix} e^x & e^{-x} & \frac{1}{x} \\ e^x & -e^{-x} & -\frac{1}{x^2} \\ e^x & e^{-x} & \frac{2}{x^3} \end{vmatrix} = \frac{2(x^2-2)}{x^3}; \quad W_1 = \begin{vmatrix} e^{-x} & \frac{1}{x} \\ -e^{-x} & -\frac{1}{x^2} \end{vmatrix} = \frac{e^{-x}(x-1)}{x^2};$$

$$W_2 = \begin{vmatrix} e^x & \frac{1}{x} \\ e^x & -\frac{1}{x^2} \end{vmatrix} = -\frac{e^x(x+1)}{x^2}; \quad W_3 = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -2; \quad u'_1 = \frac{FW_1}{P_0W} = e^{-x}(x-1);$$

$$u'_2 = -\frac{FW_2}{P_0W} = e^x(x+1); \quad u'_3 = \frac{FW_3}{P_0W} = -2x^2; \quad y_p = u_1y_1 + u_2y_2 + u_3y_3 = -2\frac{x^2}{3}.$$

$$9.4.8. \quad W = \begin{vmatrix} \sqrt{x} & \frac{1}{\sqrt{x}} & x^2 \\ \frac{1}{2\sqrt{x}} & -\frac{1}{2x^{3/2}} & 2x \\ -\frac{1}{4x^{3/2}} & \frac{1}{4x^{5/2}} & 2 \end{vmatrix} = -\frac{15}{4x}; \quad W_1 = \begin{vmatrix} \frac{1}{\sqrt{x}} & x^2 \\ -\frac{1}{2x^{3/2}} & 2x \end{vmatrix} = \frac{5\sqrt{x}}{2};$$

$$W_2 = \begin{vmatrix} \sqrt{x} & x^2 \\ \frac{1}{2\sqrt{x}} & 2x \end{vmatrix} = \frac{3x^{3/2}}{2}; \quad W_3 = \begin{vmatrix} \sqrt{x} & \frac{1}{\sqrt{x}} \\ \frac{1}{2\sqrt{x}} & -\frac{1}{2x^{3/2}} \end{vmatrix} = -\frac{1}{x}; \quad u'_1 = \frac{FW_1}{P_0W} = -5\sqrt{x};$$

$$u'_2 = -\frac{FW_2}{P_0W} = 3x^{3/2}; \quad u'_3 = \frac{FW_3}{P_0W} = \frac{2}{x}; \quad u_1 = -\frac{10}{3}x^{3/2}; \quad u_2 = \frac{6}{5}x^{5/2}; \quad u_3 = 2\ln|x|;$$

$$y_p = u_1y_1 + u_2y_2 + u_3y_3 = 2x^2\ln|x| - \frac{32}{15}x^2. \quad \text{Since } -\frac{32}{15}x^2 \text{ satisfies the complementary}$$

equation we take $y_p = \ln|x|$.

$$\begin{aligned}
9.4.10. \quad W &= \begin{vmatrix} x & 1/x & \frac{e^x}{x} \\ 1 & -\frac{1}{x^2} & \frac{e^x(x-1)}{x^2} \\ 0 & \frac{2}{x^3} & \frac{e^x(x^2-2x+2)}{x^3} \end{vmatrix} = \frac{2e^x(1-x)}{x^3}; \quad W_1 = \begin{vmatrix} \frac{1}{x} & \frac{e^x}{x} \\ \frac{1}{x^2} & \frac{e^x(x-1)}{x^2} \end{vmatrix} = \frac{e^x}{x^2}; \\
W_2 &= \begin{vmatrix} x & \frac{e^x}{x} \\ 1 & \frac{e^x(x-1)}{x^2} \end{vmatrix} = \frac{e^x(x-2)}{x}; \quad W_3 = \begin{vmatrix} x & \frac{1}{x} \\ 1 & -\frac{1}{x^2} \end{vmatrix} = -\frac{2}{x}; \quad u'_1 = \frac{FW_1}{P_0W} = 1; \\
u'_2 &= -\frac{FW_2}{P_0W} = x(2-x); \quad u'_3 = \frac{FW_3}{P_0W} = -2xe^{-x}; \quad u_1 = x; \quad u_2 = \frac{x^2(3-x)}{3}; \quad u_3 = 2e^{-x}(x+1); \\
y_p &= u_1y_1 + u_2y_2 + u_3y_3 = \frac{2x^3 + 3x^2 + 6x + 6}{3x}. \quad \text{Since } x + \frac{2}{x} \text{ satisfies the complementary} \\
&\text{equation we take } y_p = \frac{2x^2 + 6}{3}.
\end{aligned}$$

$$\begin{aligned}
9.4.12. \quad W &= \begin{vmatrix} x & e^x & e^{-x} \\ 1 & e^x & -e^{-x} \\ 0 & e^x & e^{-x} \end{vmatrix} = 2x; \quad W_1 = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -2; \quad W_2 = \begin{vmatrix} x & e^{-x} \\ 1 & -e^{-x} \end{vmatrix} = -e^{-x}(x+1); \\
W_3 &= \begin{vmatrix} x & e^x \\ 1 & e^x \end{vmatrix} = e^x(x-1); \quad u'_1 = \frac{FW_1}{P_0W} = -1; \quad u'_2 = -\frac{FW_2}{P_0W} = e^{-x}(x+1)/2; \\
u'_3 &= \frac{FW_3}{P_0W} = e^x(x-1)/2; \quad u_1 = -x; \quad u_2 = -e^{-x}(x+2)/2; \quad u_3 = e^x(x-2)/2; \\
y_p &= u_1y_1 + u_2y_2 + u_3y_3 = -x^2 - 2.
\end{aligned}$$

$$\begin{aligned}
9.4.14. \quad W &= \begin{vmatrix} \sqrt{x} & 1/\sqrt{x} & x^{3/2} & \frac{1}{x^{3/2}} \\ \frac{1}{2\sqrt{x}} & -\frac{1}{2x^{3/2}} & \frac{3\sqrt{x}}{2} & -\frac{2x^{5/2}}{15} \\ -\frac{4x^{3/2}}{3} & \frac{4x^{5/2}}{15} & \frac{4\sqrt{x}}{3} & \frac{4x^{7/2}}{105} \\ \frac{8x^{5/2}}{15} & -\frac{8x^{7/2}}{105} & -\frac{8x^{3/2}}{105} & -\frac{8x^{9/2}}{105} \end{vmatrix} = \frac{12}{x^6}; \\
W_1 &= \begin{vmatrix} \frac{1}{\sqrt{x}} & x^{3/2} & \frac{1}{x^{3/2}} \\ \frac{1}{2\sqrt{x}} & \frac{3\sqrt{x}}{2} & -\frac{2x^{5/2}}{15} \end{vmatrix} = \frac{6}{x^{7/2}}; \quad W_2 = \begin{vmatrix} \sqrt{x} & x^{3/2} & \frac{1}{x^{3/2}} \\ \frac{1}{2\sqrt{x}} & \frac{3\sqrt{x}}{2} & -\frac{2x^{5/2}}{15} \end{vmatrix} = \frac{6}{x^{5/2}}; \\
W_3 &= \begin{vmatrix} \sqrt{x} & \frac{1}{\sqrt{x}} & \frac{1}{x^{3/2}} \\ \frac{1}{2\sqrt{x}} & -\frac{1}{2x^{3/2}} & -\frac{2x^{5/2}}{15} \end{vmatrix} = -\frac{2}{x^{9/2}}; \quad W_4 = \begin{vmatrix} \sqrt{x} & \frac{1}{\sqrt{x}} & x^{3/2} \\ \frac{1}{2\sqrt{x}} & -\frac{1}{2x^{3/2}} & \frac{3\sqrt{x}}{2} \end{vmatrix} = -\frac{2}{x^{3/2}}; \\
u'_1 &= -\frac{FW_1}{P_0W} = -3x; \quad u'_2 = \frac{FW_2}{P_0W} = 3x^2; \quad u'_3 = -\frac{FW_3}{P_0W} = 1; \quad u'_4 = \frac{FW_4}{P_0W} = -x^3; \quad u_1 = -\frac{3x^2}{2};
\end{aligned}$$

$$u_2 = x^3; u_3 = x; u_4 = -\frac{x^4}{4}; y_p = u_1 y_1 + u_2 y_2 + u_3 y_3 = \frac{x^{5/2}}{4}.$$

$$\begin{aligned} 9.4.16. \quad W &= \begin{vmatrix} x & x^2 & x^3 & x^4 \\ 1 & 2x & 3x^2 & 4x^3 \\ 0 & 2 & 6x & 12x^2 \\ 0 & 0 & 6 & 24x \end{vmatrix} = 12x^4; W_1 = \begin{vmatrix} x^2 & x^3 & x^4 \\ 2x & 3x^2 & 4x^3 \\ 2 & 6x & 12x^2 \end{vmatrix} = 2x^6; \\ W_2 &= \begin{vmatrix} x & x^3 & x^4 \\ 1 & 3x^2 & 4x^3 \\ 0 & 6x & 12x^2 \end{vmatrix} = 6x^5; W_3 = \begin{vmatrix} x & x^2 & x^4 \\ 1 & 2x & 4x^3 \\ 0 & 2 & 12x^2 \end{vmatrix} = 6x^4; W_4 = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix} = 2x^3; \\ u'_1 &= -\frac{FW_1}{P_0 W} = -\frac{x^2}{6}; u'_2 = \frac{FW_2}{P_0 W} = \frac{x}{2}; u'_3 = -\frac{FW_3}{P_0 W} = -\frac{1}{2}; u'_4 = \frac{FW_4}{P_0 W} = \frac{1}{6x}; u_1 = -\frac{x^3}{18}; \\ u_2 &= \frac{x^2}{4}; u_3 = -\frac{x}{2}; u_4 = \frac{\ln|x|}{6}; y_p = u_1 y_1 + u_2 y_2 + u_3 y_3 = \frac{x^4 \ln|x|}{6} - \frac{11x^4}{36}. \text{ Since } -\frac{11x^4}{36} \\ &\text{satisfies the complementary equation we take } y_p = \frac{x^4 \ln|x|}{6}. \end{aligned}$$

$$\begin{aligned} 9.4.18. \quad W &= \begin{vmatrix} x & x^2 & 1/x & 1/x^2 \\ 1 & 2x & -1/x^2 & -2/x^3 \\ 0 & 2 & 2/x^3 & 6/x^4 \\ 0 & 0 & -6/x^4 & -24/x^5 \end{vmatrix} = -72/x^6; W_1 = \begin{vmatrix} x^2 & 1/x & 1/x^2 \\ 2x & -1/x^2 & -2/x^3 \\ 2 & 2/x^3 & 6/x^4 \end{vmatrix} = -12/x^4; \\ W_2 &= \begin{vmatrix} x & 1/x & 1/x^2 \\ 1 & -1/x^2 & -2/x^3 \\ 0 & 2/x^3 & 6/x^4 \end{vmatrix} = -6/x^5; W_3 = \begin{vmatrix} x & x^2 & 1/x^2 \\ 1 & 2x & -2/x^3 \\ 0 & 2 & 6/x^4 \end{vmatrix} = 12/x^2; \\ W_4 &= \begin{vmatrix} x & x^2 & 1/x \\ 1 & 2x & -1/x^2 \\ 0 & 2 & 2/x^3 \end{vmatrix} = 6/x; u'_1 = -\frac{FW_1}{P_0 W} = -2; u'_2 = \frac{FW_2}{P_0 W} = 1/x; u'_3 = -\frac{FW_3}{P_0 W} = 2x^2; \\ u'_4 &= \frac{FW_4}{P_0 W} = -x^3; u_1 = -2x; u_2 = \ln|x|; u_3 = 2x^3/3; u_4 = -x^4/4; \\ y_p &= u_1 y_1 + u_2 y_2 + u_3 y_3 = x^2 \ln|x| - 19x^2/12. \text{ Since } -19x^2/12 \text{ satisfies the complementary} \\ &\text{equation we take } y_p = x^2 \ln|x|. \end{aligned}$$

$$\begin{aligned} 9.4.20. \quad W &= \begin{vmatrix} e^x & e^{2x} & \frac{e^x}{x} & \frac{e^{2x}}{x} \\ e^x & 2e^{2x} & \frac{e^x(x-1)}{x} & \frac{e^{2x}(2x-1)}{x} \\ e^x & 4e^{2x} & \frac{e^x(x^2-2x+2)}{x^2} & \frac{2e^{2x}(2x^2-2x+1)}{x^2} \\ e^x & 8e^{2x} & \frac{e^x(x^3-3x^2+6x-6)}{x^3} & \frac{2e^{2x}(4x^3-6x^2+6x-3)}{x^3} \end{vmatrix} = -\frac{e^{6x}}{x^4}; \\ W_1 &= \begin{vmatrix} e^{2x} & \frac{e^x}{x} & \frac{e^{2x}}{x} \\ 2e^{2x} & \frac{e^x(x-1)}{x} & \frac{e^{2x}(2x-1)}{x} \\ 4e^{2x} & \frac{e^x(x^2-2x+2)}{x^2} & \frac{2e^{2x}(2x^2-2x+1)}{x^2} \end{vmatrix} = \frac{e^{5x}}{x^3}; \end{aligned}$$

$$W_2 = \begin{vmatrix} e^x & \frac{e^x}{x} & \frac{e^{2x}}{x} \\ e^x & \frac{e^x(x-1)}{x^2} & \frac{e^{2x}(2x-1)}{x^2} \\ e^x & \frac{e^x(x^2-2x+2)}{x^3} & \frac{2e^{2x}(2x^2-2x+1)}{x^3} \end{vmatrix} = -\frac{e^{4x}}{x^3};$$

$$W_3 = \begin{vmatrix} e^x & e^{2x} & \frac{e^{2x}}{x} \\ e^x & 2e^{2x} & \frac{e^{2x}(2x-1)}{x^2} \\ e^x & 4e^{2x} & \frac{2e^{2x}(2x^2-2x+1)}{x^3} \end{vmatrix} = \frac{e^{5x}(2-x)}{x^3};$$

$$W_4 = \begin{vmatrix} e^x & e^{2x} & \frac{e^x}{x} \\ e^x & 2e^{2x} & \frac{e^x(x-1)}{x^2} \\ e^x & 4e^{2x} & \frac{e^x(x^2-2x+2)}{x^3} \end{vmatrix} = \frac{e^{4x}(x+2)}{x^3};$$

$$u'_1 = -\frac{FW_1}{P_0W} = 3; u'_2 = \frac{FW_2}{P_0W} = 3e^{-x}; u'_3 = -\frac{FW_2}{P_0W} = 3(2-x); u'_4 = \frac{FW_4}{P_0W} = -3e^{-x}(x+2);$$

$$u_1 = 3x; u_2 = -3e^{-x}; u_3 = 3x(4-x)/2; u_4 = 3e^{-x}(x+3);$$

$$y_p = u_1y_1 + u_2y_2 + u_3y_3 = \frac{3e^x(x^2+4x+6)}{2x}. \text{ Since } \frac{3e^x(2x+3)}{x} \text{ is a solution of the}$$

$$\text{complementary equation we take } y_p = \frac{3xe^x}{2}.$$

$$9.4.22. W = \begin{vmatrix} x & x^3 & x \ln x \\ 1 & 3x^2 & 1 + \ln x \\ 0 & 6x & 1/x \end{vmatrix} = -4x^2; W_1 = \begin{vmatrix} x^3 & x \ln x \\ 3x^2 & 1 + \ln x \end{vmatrix} = x^3 - 2x^3 \ln x;$$

$$W_2 = \begin{vmatrix} x & x \ln x \\ 1 & 1 + \ln x \end{vmatrix} = x; W_3 = \begin{vmatrix} x & x^3 \\ 1 & 3x^2 \end{vmatrix} = 2x^3; u'_1 = \frac{FW_1}{P_0W} = 2 \ln x/x - \frac{1}{x};$$

$$u'_2 = -\frac{FW_2}{P_0W} = \frac{1}{x^3}; u'_3 = \frac{FW_2}{P_0W} = -\frac{2}{x}; u_1 = (\ln x)^2 - \ln x; u_2 = -\frac{1}{2x^2}; u_3 = -2 \ln x;$$

$$y_p = u_1y_1 + u_2y_2 + u_3y_3 = -x(\ln x)^2 - x \ln x - \frac{x}{2}. \text{ Since } -x \ln x - \frac{x}{2} \text{ satisfies the}$$

$$\text{complementary equation we take } y_p = -x(\ln x)^2$$

The general solution is $y = -x(\ln x)^2 + c_1x + c_2x^3 + c_3x \ln x$, so

$$\begin{bmatrix} y \\ y' \\ y'' \end{bmatrix} = \begin{bmatrix} -x(\ln x)^2 \\ -(\ln x)^2 - 2 \ln x \\ -\frac{2 \ln x}{x} - \frac{2}{x} \end{bmatrix} + \begin{bmatrix} x & x^3 & x \ln x \\ 1 & 3x^2 & 1 + \ln x \\ 0 & 6x & \frac{1}{x} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

Setting $x = 1$ and imposing the initial conditions yields

$$\begin{bmatrix} 4 \\ 4 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 6 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Solving this system yields $c_1 = 3$, $c_2 = 1$, $c_3 = -2$.

Therefore, $y = -x(\ln x)^2 + 3x + x^3 - 2x \ln x$.

$$\begin{aligned}
 9.4.24. \quad W &= \begin{vmatrix} e^x & e^{2x} & xe^{-x} \\ e^x & 2e^{2x} & e^{-x}(1-x) \\ e^x & 4e^{2x} & e^{-x}(x-2) \end{vmatrix} = e^{2x}(6x-5); \quad W_1 = \begin{vmatrix} e^{2x} & xe^{-x} \\ 2e^{2x} & e^{-x}(1-x) \end{vmatrix} = e^x(1-3x); \\
 W_2 &= \begin{vmatrix} e^x & xe^{-x} \\ e^x & e^{-x}(1-x) \end{vmatrix} = 1-2x; \quad W_3 = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = e^{3x}; \quad u'_1 = \frac{FW_1}{P_0W} = 1-3x; \\
 u'_2 &= -\frac{FW_2}{P_0W} = e^{-x}(2x-1); \quad u'_3 = \frac{FW_3}{P_0W} = e^{2x}; \quad u_1 = \frac{x(2-3x)}{2}; \quad u_2 = -e^{-x}(2x+1); \\
 u_3 &= \frac{e^{2x}}{2}; \quad y_p = u_1y_1 + u_2y_2 + u_3y_3 = -e^x(3x^2+x+2)/2. \quad \text{Since } -\frac{e^x}{2} \text{ is a solution of the} \\
 &\text{complementary equation we take } y_p = -\frac{e^x(3x+1)x}{2}.
 \end{aligned}$$

The general solution is $y = -\frac{e^x(3x+1)x}{2} + c_1e^x + c_2e^{2x} + c_3xe^{-x}$, so

$$\begin{bmatrix} y \\ y' \\ y'' \end{bmatrix} = \begin{bmatrix} -\frac{e^x(3x+1)x}{2} \\ -\frac{e^x(3x^2+7x+1)}{2} \\ -\frac{e^x(3x^2+13x+8)}{2} \end{bmatrix} + \begin{bmatrix} e^x & e^{2x} & xe^{-x} \\ e^x & 2e^{2x} & e^{-x}(1-x) \\ e^x & 4e^{2x} & e^{-x}(x-2) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Setting $x = 0$ and imposing the initial conditions yields

$$\begin{bmatrix} -4 \\ -\frac{3}{2} \\ -19 \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{1}{2} \\ -4 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 4 & -2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Solving this system yields $c_1 = -3$, $c_2 = -1$, $c_3 = 4$. Therefore,

$$y = -\frac{e^x(3x+1)x}{2} - 3e^x - e^{2x} + 4xe^{-x}.$$

$$\begin{aligned}
 9.4.26. \quad W &= \begin{vmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 2 & e^x \end{vmatrix} = e^x(x^2-2x+2); \quad W_1 = \begin{vmatrix} x^2 & e^x \\ 2x & e^x \end{vmatrix} = e^x(x^2-2x); \\
 W_2 &= \begin{vmatrix} x & e^x \\ 1 & e^x \end{vmatrix} = e^x(x-1); \quad W_3 = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = x^2; \quad u'_1 = \frac{FW_1}{P_0W} = x(x-2); \quad u'_2 = -\frac{FW_2}{P_0W} = 1-x; \\
 u'_3 &= \frac{FW_3}{P_0W} = x^2e^{-x}; \quad y_p = u_1y_1 + u_2y_2 + u_3y_3 = -\frac{x^4+6x^2+12x+12}{6}. \quad \text{Since } -\frac{6x^2+12x}{6} \\
 &\text{is a solution of the complementary equations we take } y_p = -\frac{x^4+12}{6}.
 \end{aligned}$$

The general solution is $y = -\frac{x^4+12}{6} + c_1x + c_2x^2 + c_3e^x$, so

$$\begin{bmatrix} y \\ y' \\ y'' \end{bmatrix} = -\begin{bmatrix} (x^4+12)/6 \\ 2x^3/3 \\ 2x^2 \end{bmatrix} + \begin{bmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 2 & e^x \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Setting $x = 0$ and imposing the initial conditions yields

$$\begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} = - \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Solving this system yields $c_1 = 3$, $c_2 = -1$, $c_3 = 2$. Therefore, $y = -\frac{x^4 + 12}{6} + 3x - x^2 + 2e^x$.

$$\begin{aligned} 9.4.28. \quad W &= \begin{vmatrix} x+1 & e^x & e^{3x} \\ 1 & e^x & 3e^{3x} \\ 0 & e^x & 9e^{3x} \end{vmatrix} = 2e^{4x}(3x-1); \quad W_1 = \begin{vmatrix} e^x & e^{3x} \\ e^x & 3e^{3x} \end{vmatrix} = 2e^{4x}; \\ W_2 &= \begin{vmatrix} x+1 & e^{3x} \\ 1 & 3e^{3x} \end{vmatrix} = e^{3x}(3x+2); \quad W_3 = \begin{vmatrix} x+1 & e^x \\ 1 & e^x \end{vmatrix} = xe^x; \quad u'_1 = \frac{FW_1}{P_0W} = 2e^x; \\ u'_2 &= -\frac{FW_2}{P_0W} = -3x-2; \quad u'_3 = \frac{FW_3}{P_0W} = xe^{-2x}; \quad u_1 = 2e^x; \quad u_2 = -\frac{x(3x+4)}{2}; \\ u_3 &= -\frac{e^{-2x}(2x+1)}{4}; \quad y_p = u_1y_1 + u_2y_2 + u_3y_3 = -\frac{e^x(6x^2+2x-7)}{4}. \quad \text{Since } \frac{7e^x}{4} \text{ is a solution} \\ &\text{of the complementary equation we take } y_p = -\frac{xe^x(3x+1)}{2} \end{aligned}$$

The general solution is $y = -\frac{xe^x(3x+1)}{2} + c_1(x+1) + c_2e^x + c_3e^{2x}$, so

$$\begin{bmatrix} y \\ y' \\ y'' \end{bmatrix} = \begin{bmatrix} -xe^x(3x+1)/2 \\ -e^x(3x^2+7x+1)/2 \\ -e^x(3x^2+13x+8)/2 \end{bmatrix} + \begin{bmatrix} x+1 & e^x & e^{3x} \\ 1 & e^x & 3e^{3x} \\ 0 & e^x & 9e^{3x} \end{bmatrix}.$$

Setting $x = 0$ and imposing the initial conditions yields

$$\begin{bmatrix} \frac{3}{4} \\ \frac{5}{4} \\ \frac{1}{4} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{1}{2} \\ -4 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 3 \\ 0 & 1 & 9 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}.$$

Solving this system yields $c_1 = \frac{1}{2}$, $c_2 = -\frac{1}{4}$, $c_3 = \frac{1}{2}$. Therefore,

$$y = -\frac{xe^x(3x+1)}{2} + \frac{x+1}{2} - \frac{e^x}{4} + \frac{e^{2x}}{2}.$$

$$\begin{aligned} 9.4.30. \quad W &= \begin{vmatrix} x & x^2 & \frac{1}{x} & x \ln x \\ 1 & 2x & -\frac{1}{x^2} & \ln x + 1 \\ 0 & 2 & \frac{2}{x^3} & \frac{1}{x} \\ 0 & 0 & -\frac{6}{x^2} & -\frac{1}{x^2} \end{vmatrix} = -\frac{12}{x^3}; \quad W_1 = \begin{vmatrix} x^2 & \frac{1}{x} & x \ln x \\ 2x & -\frac{1}{x^2} & \ln x + 1 \\ 2 & \frac{2}{x^3} & \frac{1}{x} \end{vmatrix} = 6\frac{\ln x}{x} - \frac{3}{x}; \\ W_2 &= \begin{vmatrix} x & \frac{1}{x} & x \ln x \\ 1 & -\frac{1}{x^2} & \ln x + 1 \\ 0 & \frac{2}{x^3} & \frac{1}{x} \end{vmatrix} = -\frac{4}{x^2}; \quad W_3 = \begin{vmatrix} x & x^2 & x \ln x \\ 1 & 2x & \ln x + 1 \\ 0 & 2 & \frac{1}{x} \end{vmatrix} = -x; \quad W_4 = \begin{vmatrix} x & x^2 & \frac{1}{x} \\ 1 & 2x & -\frac{1}{x^2} \\ 0 & 2 & \frac{2}{x^3} \end{vmatrix} = \frac{6}{x}; \end{aligned}$$

$$\begin{aligned}
u'_1 &= -\frac{FW_1}{P_0W} = 9 \ln x / 2 - \frac{9}{4}; \quad u'_2 = \frac{FW_2}{P_0W} = \frac{3}{x}; \quad u'_3 = -\frac{FW_2}{P_0W} = -\frac{3x^2}{4}; \quad u'_4 = \frac{FW_4}{P_0W} = -\frac{9}{2}; \\
u_1 &= \frac{9x \ln x}{2} - \frac{27x}{4}; \quad u_2 = 3 \ln x; \quad u_3 = -\frac{x^3}{4}; \quad u_4 = -\frac{9x}{2}; \\
y_p &= u_1 y_1 + u_2 y_2 + u_3 y_3 = 3x^2 \ln x - 7x^2. \text{ Since } -7x^2 \text{ satisfies the complementary equation} \\
&\text{we take } y_p = 3x^2 \ln x.
\end{aligned}$$

The general solution is $y = 3x^2 \ln x + c_1 x + c_2 x^2 + \frac{c_3}{x} + c_4 x \ln x$, so

$$\begin{bmatrix} y \\ y' \\ y'' \\ y''' \end{bmatrix} = \begin{bmatrix} 3x^2 \ln x \\ 6x \ln x + 3x \\ 6 \ln x + 9 \\ \frac{6}{x} \end{bmatrix} + \begin{vmatrix} x & x^2 & \frac{1}{x} & x \ln x \\ 1 & 2x & -\frac{1}{x^2} & \ln x + 1 \\ 0 & 2 & \frac{2}{x^3} & \frac{1}{x} \\ 0 & 0 & -\frac{6}{x^2} & -\frac{1}{x^2} \end{vmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}.$$

Setting $x = 1$ and imposing the initial conditions yields

$$\begin{bmatrix} -7 \\ -11 \\ -5 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 9 \\ 6 \end{bmatrix} + \begin{vmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & -1 & 1 \\ 0 & 2 & 2 & 1 \\ 0 & 0 & -6 & -1 \end{vmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}.$$

Solving this system yields $c_1 = 0$, $c_2 = -7$, $c_3 = 0$, $c_4 = 0$. Therefore, $y = 3x^2 \ln x - 7x^2$.

$$9.4.32. \quad W = \begin{vmatrix} x & \sqrt{x} & 1/x & 1/\sqrt{x} \\ 1 & 1/2\sqrt{x} & -1/x^2 & -1/2x^{3/2} \\ 0 & -1/4x^{3/2} & 2/x^3 & 3/4x^{5/2} \\ 0 & 3/8x^{5/2} & -6/x^4 & -15/8x^{7/2} \end{vmatrix} = -9/8x^6;$$

$$W_1 = \begin{vmatrix} \sqrt{x} & 1/x & 1/\sqrt{x} \\ 1/2\sqrt{x} & -1/x^2 & -1/2x^{3/2} \\ -1/4x^{3/2} & 2/x^3 & 3/4x^{5/2} \end{vmatrix} = 3/4x^4; \quad W_2 = \begin{vmatrix} x & 1/x & 1/\sqrt{x} \\ 1 & -1/x^2 & -1/2x^{3/2} \\ 0 & 2/x^3 & 3/4x^{5/2} \end{vmatrix} = 3/2x^{7/2};$$

$$W_3 = \begin{vmatrix} x & \sqrt{x} & 1/\sqrt{x} \\ 1 & 1/2\sqrt{x} & -1/2x^{3/2} \\ 0 & -1/4x^{3/2} & 3/4x^{5/2} \end{vmatrix} = -3/4x^2; \quad W_4 = \begin{vmatrix} x & \sqrt{x} & 1/x \\ 1 & 1/2\sqrt{x} & -1/x^2 \\ 0 & -1/4x^{3/2} & 2/x^3 \end{vmatrix} = -3/2x^{5/2};$$

$$u'_1 = -\frac{FW_1}{P_0W} = 1/x; \quad u'_2 = \frac{FW_2}{P_0W} = -2/\sqrt{x}; \quad u'_3 = -\frac{FW_2}{P_0W} = -x; \quad u'_4 = \frac{FW_4}{P_0W} = 2\sqrt{x};$$

$u_1 = \ln x$; $u_2 = -4\sqrt{x}$; $u_3 = -x^2/2$; $u_4 = 4x^{3/2}/3$; $y_p = u_1 y_1 + u_2 y_2 + u_3 y_3 = x \ln x - 19x/6$.
since $-19x/6$ satisfies the complementary equation we take $y_p = x \ln x$.

The general solution is $y = x \ln x + c_1 x + c_2 \sqrt{x} + c_3/x + c_4/\sqrt{x}$, so

$$\begin{bmatrix} y \\ y' \\ y'' \\ y''' \end{bmatrix} = \begin{bmatrix} x \ln x \\ \ln x + 1 \\ 1/x \\ -1/x^2 \end{bmatrix} + \begin{vmatrix} x & \sqrt{x} & 1/x & 1/\sqrt{x} \\ 1 & 1/2\sqrt{x} & -1/x^2 & -1/2x^{3/2} \\ 0 & -1/4x^{3/2} & 2/x^3 & 3/4x^{5/2} \\ 0 & 3/8x^{5/2} & -6/x^4 & -15/8x^{7/2} \end{vmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}.$$

Setting $x = 1$ and imposing the initial conditions yields

$$\begin{bmatrix} 2 \\ 0 \\ 4 \\ \frac{37}{4} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1/2 & -1 & -1/2 \\ 0 & -1/4 & 2 & 3/4 \\ 0 & 3/8 & -6 & -15/8 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}.$$

Solving this system yields $c_1 = 1$, $c_2 = -1$, $c_3 = 1$, $c_4 = 1$. Therefore,
 $y = x \ln x + x - \sqrt{x} + \frac{1}{x} + \frac{1}{\sqrt{x}}.$

- 9.4.34. (a) Since $u'_j = (-1)^{n-j} \frac{FW_j}{P_0 W}$ ($1 \leq j \leq n$), the argument used in the derivation of the method of variation of parameters implies that y_p is a solution of (A).
 (b) Follows immediately from (a), since $u_j(x_0) = 0$, $j = 1, 2, \dots, n$.
 (c) Expand the determinant in cofactors of its n th row.
 (d) Just differentiate the determinant $n - 1$ times.
 (e) If $0 \leq j \leq n - 2$, then $\left. \frac{\partial^j G(x, t)}{\partial x^j} \right|_{x=t}$ has two identical rows, and is therefore zero, while
 $\left. \frac{\partial^{n-1} G(x, t)}{\partial x^j} \right|_{x=t} = W(t)$
 (f) Since $y_p(x) = \int_{x_0}^x G(x, t) F(t) dt$, $y'_p(x) = G(x, x) F(x) + \int_{x_0}^x \frac{\partial G(x, t)}{\partial x} F(t) dt$. But
 $G(x, x) = 0$ from (e), so $y'_p(x) = \int_{x_0}^x \frac{\partial G(x, t)}{\partial x} F(t) dt$. Repeating this argument for
 $j = 1, \dots, n$ and invoking (e) each time yields the conclusion.

9.4.36.

$$\begin{aligned} \begin{vmatrix} y_1(t) & y_1(t) & y_2(t) \\ y'_1(t) & y'_1(t) & y'_2(t) \\ y_1(x) & y_1(x) & y_2(x) \end{vmatrix} &= \begin{vmatrix} t & t^2 & 1/t \\ 1 & 2t & -1/t^2 \\ x & x^2 & 1/x \end{vmatrix} \\ &= x \begin{vmatrix} t^2 & 1/t \\ 2t & -1/t^2 \end{vmatrix} - x^2 \begin{vmatrix} t & 1/t \\ 1 & -1/t^2 \end{vmatrix} + \frac{1}{x} \begin{vmatrix} t & t^2 \\ 1 & 2t \end{vmatrix} \\ &= -3x + 2\frac{x^2}{t} + \frac{t^2}{x} = \frac{(x-t)^2(2x+t)}{xt}. \end{aligned}$$

Since $P_0(t) = t^3$ and $W(t) = \begin{vmatrix} t & t^2 & 1/t \\ 1 & 2t & -1/t^2 \\ x & x^2 & 1/x \end{vmatrix} = \frac{6}{t}$, $G(x, t) = \frac{(x-t)^2(2x+t)}{6xt^3}$, so

$$y_p = \int_{x_0}^x \frac{(x-t)^2(2x+t)}{6xt^3} F(t) dt.$$

9.4.38.

$$\begin{aligned}
\begin{vmatrix} y_1(t) & y_1(t) & y_2(t) \\ y_1'(t) & y_1'(t) & y_2'(t) \\ y_1(x) & y_1(x) & y_2(x) \end{vmatrix} &= \begin{vmatrix} t & 1/t & e^t/t \\ 1 & -1/t^2 & e^t(1/t - 1/t^2) \\ x & 1/x & e^x/x \end{vmatrix} \\
&= x \begin{vmatrix} 1/t & e^t/t \\ -1/t^2 & e^t(1/t - 1/t^2) \end{vmatrix} - \frac{1}{x} \begin{vmatrix} t & e^t/t \\ 1 & e^t(1/t - 1/t^2) \end{vmatrix} + \frac{e^x}{x} \begin{vmatrix} t & 1/t \\ 1 & -1/t^2 \end{vmatrix} \\
&= \frac{xe^t}{t^2} - \frac{e^t(t-2)}{xt} - \frac{2e^x}{xt} = \frac{x^2e^t - e^tt(t-2) - 2te^x}{xt^2}
\end{aligned}$$

$$\begin{aligned}
\text{Since } P_0(t) = t(1-t) \text{ and } W(t) &= \begin{vmatrix} t & 1/t & e^t/t \\ 1 & -1/t^2 & e^t(1/t - 1/t^2) \\ 0 & 2/t^3 & e^t(1/t - 2/t^2 + 2/t^3) \end{vmatrix} = \frac{2e^t(1-t)}{t^3}, \\
G(x, t) &= \frac{x^2 - t(t-2) - 2te^{(x-t)}}{2x(t-1)^2}, \text{ so } y_p = \int_{x_0}^x \frac{x^2 - t(t-2) - 2te^{(x-t)}}{2x(t-1)^2} F(t) dt.
\end{aligned}$$

9.4.40.

$$\begin{aligned}
\begin{vmatrix} y_1(t) & y_1(t) & y_2(t) & y_3(t) \\ y_1'(t) & y_1'(t) & y_2'(t) & y_3'(t) \\ y_1''(t) & y_1''(t) & y_2''(t) & y_3''(t) \\ y_1(x) & y_1(x) & y_2(x) & y_3(x) \end{vmatrix} &= \begin{vmatrix} 1 & t & t^2 & 1/t \\ 0 & 1 & 2t & -1/t^2 \\ 0 & 0 & 2 & 2/t^3 \\ 1 & x & x^2 & 1/x \end{vmatrix} \\
&= - \begin{vmatrix} t & t^2 & 1/t \\ 1 & 2t & -1/t^2 \\ 0 & 2 & 2/t^3 \end{vmatrix} + x \begin{vmatrix} 1 & t^2 & 1/t \\ 0 & 2t & -1/t^2 \\ 0 & 2 & 2/t^3 \end{vmatrix} \\
&\quad - x^2 \begin{vmatrix} 1 & t & 1/t \\ 0 & 1 & -1/t^2 \\ 0 & 0 & 2/t^3 \end{vmatrix} + \frac{1}{x} \begin{vmatrix} 1 & t & t^2 \\ 0 & 1 & 2t \\ 0 & 0 & 2 \end{vmatrix} \\
&= -\frac{6}{t} + \frac{6x}{t^2} - \frac{2x^2}{t^3} + \frac{2}{x} = \frac{2(t-x)^3}{xt^3}.
\end{aligned}$$

$$\text{Since } P_0(t) = t \text{ and } W(t) = \begin{vmatrix} 1 & t & t^2 & 1/t \\ 0 & 1 & 2t & -1/t^2 \\ 0 & 0 & 2 & 2/t^3 \\ 0 & 0 & 0 & -6/t^4 \end{vmatrix} = -\frac{12}{t^4}, \quad G(x, t) = \frac{(x-t)^3}{6x}, \text{ so}$$

$$y_p = \int_{x_0}^x \frac{(x-t)^3}{6x} F(t) dt.$$

9.4.42.

$$\begin{aligned}
\begin{vmatrix} y_1(t) & y_1(t) & y_2(t) & y_3(t) \\ y_1'(t) & y_1'(t) & y_2'(t) & y_3'(t) \\ y_1''(t) & y_1''(t) & y_2''(t) & y_3''(t) \\ y_1(x) & y_1(x) & y_2(x) & y_3(x) \end{vmatrix} &= \begin{vmatrix} 1 & t^2 & e^{2t} & e^{-2t} \\ 0 & 2t & 2e^{2t} & -2e^{-2t} \\ 0 & 2 & 4e^{2t} & 4e^{-2t} \\ 1 & x^2 & e^{2x} & e^{-2x} \end{vmatrix} \\
&= - \begin{vmatrix} t^2 & e^{2t} & e^{-2t} \\ 2t & 2e^{2t} & -2e^{-2t} \\ 2 & 4e^{2t} & 4e^{-2t} \end{vmatrix} + x^2 \begin{vmatrix} 1 & e^{2t} & e^{-2t} \\ 0 & 2e^{2t} & -2e^{-2t} \\ 0 & 4e^{2t} & 4e^{-2t} \end{vmatrix}
\end{aligned}$$

$$\begin{aligned}
& -e^{2x} \begin{vmatrix} 1 & t^2 & e^{-2t} \\ 0 & 2t & -2e^{-2t} \\ 0 & 2 & 4e^{-2t} \end{vmatrix} + e^{-2x} \begin{vmatrix} 1 & t^2 & e^{2t} \\ 0 & 2t & 2e^{2t} \\ 0 & 2 & 4e^{2t} \end{vmatrix} \\
& = -(16t^2 - 8) + 16x^2 - e^{2(x-t)t}(8t + 4) + e^{-2(x-t)}(8t - 4).
\end{aligned}$$

$$\text{Since } P_0(t) = t \text{ and } W(t) = \begin{vmatrix} 1 & t^2 & e^{2t} & e^{-2t} \\ 0 & 2t & 2e^{2t} & -2e^{-2t} \\ 0 & 2 & 4e^{2t} & 4e^{-2t} \\ 0 & 0 & 8e^{2t} & -8e^{-2t} \end{vmatrix} = -128t,$$

$$G(x, t) = \frac{e^{2(x-t)}(1 + 2t) + e^{-2(x-t)}(1 - 2t) - 4x^2 + 4t^2 - 2}{32t^2}, \text{ so}$$

$$y_p = \int_{x_0}^x \frac{e^{2(x-t)}(1 + 2t) + e^{-2(x-t)}(1 - 2t) - 4x^2 + 4t^2 - 2}{32t^2} F(t) dt.$$

Chapter 10

Linear Systems of Differential Equations

10.1 Introduction to Systems of Differential Equations

10.1.2. $Q'_1 = (\text{ratein})_1 - (\text{rateout})_1$ and $Q'_2 = (\text{ratein})_2 - (\text{rateout})_2$.

The volumes of the solutions in T_1 and T_2 are $V_1(t) = 100 + 2t$ and $V_2(t) = 100 + 3t$, respectively. T_1 receives salt from the external source at the rate of $2 \text{ lb/gal} \times 6 \text{ gal/min} = 12 \text{ lb/min}$, and from T_2 at the rate of

$T_2 \text{ lb/gal} \times 1 \text{ gal/min} = \frac{1}{100 + 3t} Q_2 \text{ lb/min}$. Therefore, (A) $(\text{ratein})_1 = 12 + \frac{1}{100 + 3t} Q_2$.

Solution leaves T_1 at 5 gal/min , since 3 gal/min are drained and 2 gal/min are pumped to T_2 ;

hence (B) $(\text{rateout})_1 = T_1 \text{ lb/gal} \times 5 \text{ gal/min} = \frac{1}{100 + 2t} Q_1 \times 5 = \frac{5}{100 + 2t} Q_1$. Now (A) and

(B) imply that (C) $Q'_1 = 12 - \frac{5}{100 + 2t} Q_1 + \frac{1}{100 + 3t} Q_2$.

T_2 receives salt from the external source at the rate of $1 \text{ lb/gal} \times 5 \text{ gal/min} = 5 \text{ lb/min}$, and from T_1 at the rate of $T_1 \text{ lb/gal} \times 2 \text{ gal/min} = \frac{1}{100 + 2t} Q_1 \times 2 = \frac{1}{50 + t} Q_1 \text{ lb/min}$. Therefore,

(D) $(\text{ratein})_2 = 5 + \frac{1}{50 + t} Q_1$. Solution leaves T_2 at 4 gal/min , since 3 gal/min are drained and 1 gal/min is pumped to T_1 ; hence (E)

$(\text{rateout})_2 = T_2 \text{ lb/gal} \times 4 \text{ gal/min} = \frac{1}{100 + 3t} Q_2 \times 4 = \frac{4}{100 + 3t} Q_2$. Now (D) and (E) imply

that (F) $Q'_2 = 5 + \frac{1}{50 + t} Q_1 - \frac{4}{100 + 3t} Q_2$. Now (C) and (F) form the desired system.

10.1.4. $m\mathbf{X}'' = -\alpha\mathbf{X}' - mgR^2 \frac{\mathbf{X}}{\|\mathbf{X}\|^3}$; see Example 10.1.3.

10.1.8.

$$I_{1i} = g_1(t_i, y_{1i}, y_{2i}),$$

$$\begin{aligned}
J_{1i} &= g_2(t_i, y_{1i}, y_{2i}), \\
I_{2i} &= g_1(t_i + h, y_{1i} + hI_{1i}, y_{2i} + hJ_{1i}), \\
J_{2i} &= g_2(t_i + h, y_{1i} + hI_{1i}, y_{2i} + hJ_{1i}), \\
y_{1,i+1} &= y_{1i} + \frac{h}{2}(I_{1i} + I_{2i}), \\
y_{2,i+1} &= y_{2i} + \frac{h}{2}(J_{1i} + J_{2i}).
\end{aligned}$$

10.2 Linear Systems of Differential Equations

10.2.6. Let $y_i = y^{(i-1)}$, $i = 1, 2, \dots, n$; then $y'_i = y_{i+1}$, $i = 1, 2, \dots, n-1$ and $P_0(t)y'_n + P_1(t)y_n + \dots + P_n(t)y_1 = F(t)$, so

$$A = -\frac{1}{P_0} \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ P_n & P_{n-1} & P_{n-2} & \cdots & P_1 \end{bmatrix} \quad \text{and} \quad \mathbf{f} = \frac{1}{P_0} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ F \end{bmatrix}.$$

If $P_0, P_1, \dots, P_n$ and F are continuous and P_0 has no zeros on (a, b) , then $P_1/P_0, \dots, P_n/P_0$ and F/P_0 are continuous on (a, b) .

10.2.7. (a) $(c_1P + c_2Q)'_{ij} = (c_1p_{ij} + c_2q_{ij})' = c_1p'_{ij} + c_2q'_{ij} = (c_1P' + c_2Q')_{ij}$; hence $(c_1P + c_2Q)' = c_1P' + c_2Q'$.

(b) Let P be $k \times r$ and Q be $r \times s$; then PQ is $k \times s$ and $(PQ)_{ij} = \sum_{\ell=1}^r p_{i\ell}q_{\ell j}$. Therefore,

$$(PQ)'_{ij} = \sum_{\ell=1}^r p'_{i\ell}q_{\ell j} + \sum_{\ell=1}^r p_{i\ell}q'_{\ell j} = (P'Q)_{ij} + (PQ')_{ij}. \quad \text{Therefore, } (PQ)' = P'Q + PQ'.$$

10.2.10. (a) From Exercise 10.2.7(b) with $P = Q = X$, $(X^2)' = (XX)' = X'X + XX'$.

(b) By starting from Exercise 10.2.7(b) and using induction it can be shown if $P_1, P_2, \dots, P_n$ are square matrices of the same order, then $(P_1P_2 \cdots P_n)' = P'_1P_2 \cdots P_n + P_1P'_2 \cdots P_n + \dots + P_1P_2 \cdots P'_n$. Taking $P_1 = P_2 = \dots = P_n = X$ yields (A)

$$(Y^n)' = Y'Y^{n-1} + YY'Y^{n-2} + Y^2Y'Y^{n-3} + \dots + Y^{n-1}Y' = \sum_{r=0}^{n-1} Y^r Y' Y^{n-r-1}.$$

(c) If Y is a scalar function, then (A) reduces to the familiar result $(Y^n)' = nY^{n-1}Y'$.

10.2.12. From Exercise 10.2.6, the initial value problem (A)

$$P_0(x)y^{(n)} + P_1(x)y^{(n-1)} + \dots + P_n(x)y = F(x), \quad y(x_0) = k_0, \quad y'(x_0) = k_1, \dots, \quad y^{(n-1)}(x_0) = k_{n-1}$$

is equivalent to the initial value problem (B) $\mathbf{y}' = A(t)\mathbf{y} + \mathbf{f}(t)$, with

$$A = -\frac{1}{P_0} \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ P_n & P_{n-1} & P_{n-2} & \cdots & P_1 \end{bmatrix}, \mathbf{f} = \frac{1}{P_0} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ F \end{bmatrix}, \quad \text{and} \quad \mathbf{k} = \begin{bmatrix} k_0 \\ k_1 \\ \vdots \\ k_{n-1} \end{bmatrix}.$$

Since Theorem 10.2.1 implies that (B) has a unique solution on (a, b) , it follows that (A) does also.

10.3 Basic Theory of Homogeneous Linear System

10.3.2. (a) The system equivalent of (A) is (B) $\mathbf{y}' = -\frac{1}{P_0(x)} \begin{bmatrix} 0 & 1 \\ P_2(x) & P_1(x) \end{bmatrix} \mathbf{y}$, where $\mathbf{y} = \begin{bmatrix} y \\ y' \end{bmatrix}$.

Let $\mathbf{y}_1 = \begin{bmatrix} y_1 \\ y'_1 \end{bmatrix}$ and $\mathbf{y}_2 = \begin{bmatrix} y_2 \\ y'_2 \end{bmatrix}$. Then the Wronskian of $\{\mathbf{y}_1, \mathbf{y}_2\}$ as defined in this section is $\begin{bmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{bmatrix} = W$.

(b) The trace of the matrix in (B) is $-P_1(x)/P_0(x)$, so Eqn. 10.3.6 implies that

$$W(x) = W(x_0) \exp \left\{ - \int_{x_0}^x \frac{P_1(s)}{P_0(s)} ds \right\}.$$

10.3.4. (a) See the solution of Exercise 9.1.18.

$$(b) \begin{vmatrix} y'_{11} & y'_{12} \\ y_{21} & y_{22} \end{vmatrix} = \begin{vmatrix} a_{11}y_{11} + a_{12}y_{21} & a_{11}y_{12} + a_{12}y_{22} \\ y_{21} & y_{22} \end{vmatrix} = a_{11} \begin{vmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{vmatrix} + a_{12} \begin{vmatrix} y_{21} & y_{22} \\ y_{21} & y_{22} \end{vmatrix} = a_{11}W + a_{12}0 = a_{11}W. \text{ Similarly, } \begin{vmatrix} y_{11} & y_{12} \\ y'_{21} & y'_{22} \end{vmatrix} = a_{22}W.$$

10.3.6. (a) From the equivalence of Theorem 10.3.3(b) and (e), $Y(t_0)$ is invertible.

(b) From the equivalence of Theorem 10.3.3(a) and (b), the solution of the initial value problem is $\mathbf{y} = Y(t)\mathbf{c}$, where $\mathbf{c}$ is a constant vector. To satisfy $\mathbf{y}(t_0) = \mathbf{k}$, we must have $Y(t_0)\mathbf{c} = \mathbf{k}$, so $\mathbf{c} = Y^{-1}(t_0)\mathbf{k}$ and $\mathbf{y} = Y^{-1}(t_0)Y(t)\mathbf{k}$.

$$10.3.8. (b) \mathbf{y} = \begin{bmatrix} e^{-4t} \\ e^{-4t} \end{bmatrix} + c_2 \begin{bmatrix} -2e^{3t} \\ 5e^{3t} \end{bmatrix} \text{ where } \begin{cases} c_1 - 2c_2 = 10 \\ c_1 + 5c_2 = -4 \end{cases}, \text{ so } c_1 = 6, c_2 = -2, \text{ and}$$

$$\mathbf{y} = \begin{bmatrix} 6e^{-4t} + 4e^{3t} \\ 6e^{-4t} - 10e^{3t} \end{bmatrix}.$$

$$(c) Y(t) = \begin{bmatrix} e^{-4t} & -2e^{3t} \\ e^{-4t} & 5e^{3t} \end{bmatrix}; Y(0) = \begin{bmatrix} 1 & -2 \\ 1 & 5 \end{bmatrix}; Y^{-1}(0) = \frac{1}{7} \begin{bmatrix} 5 & 2 \\ -1 & 1 \end{bmatrix};$$

$$\mathbf{y} = Y(t)Y^{-1}(0)\mathbf{k} = \frac{1}{7} \begin{bmatrix} 5e^{-4t} + 2e^{3t} & 2e^{-4t} - 2e^{3t} \\ 5e^{-4t} - 5e^{3t} & 2e^{-4t} + 5e^{3t} \end{bmatrix} \mathbf{k}.$$

10.3.10. (b) $\mathbf{y}_1 = c_1 \begin{bmatrix} e^{3t} \\ e^{3t} \end{bmatrix} + c_2 \begin{bmatrix} e^t \\ -e^t \end{bmatrix}$, where $\begin{cases} c_1 + c_2 = 2 \\ c_1 - c_2 = 8 \end{cases}$, so $c_1 = 5$, $c_2 = -3$, and

$$\mathbf{y} = \begin{bmatrix} 5e^{3t} - 3e^t \\ 5e^{3t} + 3e^t \end{bmatrix}.$$

(c) $Y(t) = \begin{bmatrix} e^{3t} & e^t \\ 3e^{3t} & -e^t \end{bmatrix}$; $Y(0) = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$;

$$Y^{-1}(0) = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}; \mathbf{y} = Y(t)Y^{-1}(0)\mathbf{k} = \frac{1}{2} \begin{bmatrix} e^{3t} + e^t & e^{3t} - e^t \\ e^{3t} - e^t & e^{3t} + e^t \end{bmatrix} \mathbf{k}.$$

10.3.12. (b) $\mathbf{y} = c_1 \begin{bmatrix} -e^{-2t} \\ 0 \\ e^{-2t} \end{bmatrix} + c_2 \begin{bmatrix} -e^{-2t} \\ e^{-2t} \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} e^{4t} \\ e^{4t} \\ e^{4t} \end{bmatrix}$, where $\begin{cases} -c_1 - c_2 + c_3 = 0 \\ c_2 + c_3 = -9 \\ c_1 + c_3 = 12 \end{cases}$, so $c_1 = 11$,

$$c_2 = -10, c_3 = 1, \text{ and } \mathbf{y} = \frac{1}{3} \begin{bmatrix} -e^{-2t} + e^{4t} \\ -10e^{-2t} + e^{4t} \\ 11e^{-2t} + e^{4t} \end{bmatrix}.$$

(c) $Y(t) = \begin{bmatrix} -e^{-2t} & -e^{-2t} & e^{4t} \\ 0 & e^{-2t} & e^{4t} \\ e^{-2t} & 0 & e^{4t} \end{bmatrix}$; $Y(0) = \begin{bmatrix} -1 & -1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$; $Y^{-1}(0) = \frac{1}{3} \begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{bmatrix}$;

$$\mathbf{y} = Y(t)Y^{-1}(0)\mathbf{k} = \frac{1}{3} \begin{bmatrix} 2e^{-2t} + e^{4t} & -e^{-2t} + e^{4t} & -e^{-2t} + e^{4t} \\ -e^{-2t} + e^{4t} & 2e^{-2t} + e^{4t} & -e^{-2t} + e^{4t} \\ -e^{-2t} + e^{4t} & -e^{-2t} + e^{4t} & 2e^{-2t} + e^{4t} \end{bmatrix} \mathbf{k}.$$

10.3.14. If Y and Z are both fundamental matrices for $\mathbf{y}' = A(t)\mathbf{y}$, then $Z = CY$, where C is a constant invertible matrix. Therefore, $ZY^{-1} = C$ and $YZ^{-1} = C^{-1}$.

10.3.16. (a) The Wronskian of $\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n\}$ equals one when $t = t_0$. Apply Theorem 10.3.3.

(b) Let Y be the matrix with columns $\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n\}$. From (a), Y is a fundamental matrix for $\mathbf{y}' = A(t)\mathbf{y}$ on (a, b) . From Exercise 10.3.15(b), so is $Z = YC$ if C is any invertible constant matrix.

10.3.18. (a) $\Gamma_1'(t) = Z'(t)Z(s) = AZ(t)Z(s) = A\Gamma(t)$ and $\Gamma_1(0) = Z(s)$, since $Z(0) = I$. $\Gamma_2'(t) = Z'(t+s) = AZ(t+s) = A\Gamma_2(t)$ (since A is constant) and $\Gamma_2(0) = Z(s)$. Applying Theorem 10.2.1 to the columns of Γ_1 and Γ_2 shows that $\Gamma_1 = \Gamma_2$.

(b) With $s = -t$, (a) implies that $Z(t)Z(-t) = Z(0) = I$; therefore $(Z(t))^{-1} = Z(-t)$.

(c) $e^{0 \cdot A} = I$ is analogous to $e^{0 \cdot a} = e^0 = 1$ when a is a scalar, while $e^{(t+s)A} = e^{tA}e^{sA}$ is analogous to $e^{(t+s)a} = e^{ta}e^{sa}$ when a is a scalar.

10.4 Constant Coefficient Homogeneous Systems I

10.4.2. $\frac{1}{4} \begin{vmatrix} -5-4\lambda & 3 \\ 3 & -5-4\lambda \end{vmatrix} = (\lambda + 1/2)(\lambda + 2)$. Eigenvectors associated with $\lambda_1 = -1/2$ satisfy

$$\begin{bmatrix} -3 & 3 \\ -4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \text{ so } x_1 = x_2. \text{ Taking } x_2 = 1 \text{ yields } \mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-t/2}. \text{ Eigenvectors}$$

associated with $\lambda_2 = -2$ satisfy $\begin{bmatrix} \frac{3}{4} & \frac{3}{4} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, so $x_1 = -x_2$. Taking $x_2 = 1$ yields $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-2t}$. Hence $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-t/2} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-2t}$.

10.4.4. $\begin{vmatrix} -1-\lambda & -4 \\ -1 & -1-\lambda \end{vmatrix} = (\lambda-1)(\lambda+3)$. Eigenvectors associated with $\lambda_1 = -3$ satisfy $\begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, so $x_1 = 2x_2$. Taking $x_2 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-3t}$. Eigenvectors associated with $\lambda_2 = 1$ satisfy $\begin{bmatrix} -2 & -4 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, so $x_1 = -2x_2$. Taking $x_2 = 1$ yields $\mathbf{y}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix} e^t$. Hence $\mathbf{y} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-3t} + c_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix} e^t$.

10.4.6. $\begin{vmatrix} 4-\lambda & -3 \\ 2 & -1-\lambda \end{vmatrix} = (\lambda-2)(\lambda-1)$. Eigenvectors associated with $\lambda_1 = 2$ satisfy $\begin{bmatrix} 2 & -3 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, so $x_1 = \frac{3}{2}x_2$. Taking $x_2 = 2$ yields $\mathbf{y}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{2t}$. Eigenvectors associated with $\lambda_2 = 1$ satisfy $\begin{bmatrix} 3 & -3 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, so $x_1 = x_2$. Taking $x_2 = 1$ yields $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t$. Hence $\mathbf{y} = c_1 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t$.

10.4.8. $\begin{vmatrix} 1-\lambda & -1 & -2 \\ 1 & -2-\lambda & -3 \\ -4 & 1 & -1-\lambda \end{vmatrix} = -(\lambda+3)(\lambda+1)(\lambda-2)$. The eigenvectors associated with

$\lambda_1 = -3$ satisfy the system with augmented matrix $\begin{bmatrix} 4 & -1 & -2 & : & 0 \\ 1 & 1 & -3 & : & 0 \\ -4 & 1 & 2 & : & 0 \end{bmatrix}$, which is row

equivalent to $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & -2 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = x_3$ and $x_2 = 2x_3$. Taking $x_3 = 1$ yields

$\mathbf{y}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} e^{-3t}$. The eigenvectors associated with $\lambda_2 = -1$ satisfy the system with

augmented matrix $\begin{bmatrix} 2 & -1 & -2 & : & 0 \\ 1 & -1 & -3 & : & 0 \\ -4 & 1 & 0 & : & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 1 & 4 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$.

Hence $x_1 = -x_3$ and $x_2 = -4x_3$. Taking $x_3 = 1$ yields $\mathbf{y}_2 = \begin{bmatrix} -1 \\ -4 \\ 1 \end{bmatrix} e^{-t}$. The eigenvectors

associated with $\lambda_3 = 2$ satisfy the system with augmented matrix

$\begin{bmatrix} -1 & -1 & -2 & : & 0 \\ 1 & -4 & -3 & : & 0 \\ -4 & 1 & -3 & : & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = -x_3$ and

$x_2 = -x_3$. Taking $x_3 = 1$ yields $\mathbf{y}_3 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{2t}$. Hence

$$\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} e^{-3t} + c_2 \begin{bmatrix} -1 \\ -4 \\ 1 \end{bmatrix} e^{-t} + c_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{2t}.$$

10.4.10. $\begin{vmatrix} 3-\lambda & 5 & 8 \\ 1 & -1-\lambda & -2 \\ -1 & -1 & -1-\lambda \end{vmatrix} = -(\lambda-1)(\lambda+2)(\lambda-2)$. The eigenvectors associated with

with $\lambda_1 = 1$ satisfy the system with augmented matrix $\begin{bmatrix} 2 & 5 & 8 & : & 0 \\ 1 & -2 & -2 & : & 0 \\ -1 & -1 & -2 & : & 0 \end{bmatrix}$, which is row

equivalent to $\begin{bmatrix} 1 & 0 & \frac{2}{3} & : & 0 \\ 0 & 1 & \frac{4}{3} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = -\frac{2}{3}x_3$ and $x_2 = -\frac{4}{3}x_3$. Taking $x_3 = 3$ yields

$\mathbf{y}_1 = \begin{bmatrix} -2 \\ -4 \\ 3 \end{bmatrix} e^t$. The eigenvectors associated with with $\lambda_2 = -2$ satisfy the system with

augmented matrix $\begin{bmatrix} 5 & 5 & 8 & : & 0 \\ 1 & 1 & -2 & : & 0 \\ -1 & -1 & 1 & : & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & 1 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$.

Hence $x_1 = -x_2$ and $x_3 = 0$. Taking $x_2 = 1$ yields $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t}$. The eigenvectors

associated with with $\lambda_3 = 2$ satisfy the system with augmented matrix

$\begin{bmatrix} 1 & 5 & 8 & : & 0 \\ 1 & -3 & -2 & : & 0 \\ -1 & -1 & -3 & : & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & 0 & \frac{7}{4} & : & 0 \\ 0 & 1 & \frac{5}{4} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = -\frac{7}{4}x_3$ and

$x_2 = -\frac{5}{4}x_3$. Taking $x_3 = 4$ yields $\mathbf{y}_3 = \begin{bmatrix} -7 \\ -5 \\ 4 \end{bmatrix} e^{2t}$. Hence

$$\mathbf{y} = c_1 \begin{bmatrix} -2 \\ -4 \\ 3 \end{bmatrix} e^t + c_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t} + c_3 \begin{bmatrix} -7 \\ -5 \\ 4 \end{bmatrix} e^{2t}.$$

10.4.12. $\begin{vmatrix} 4-\lambda & -1 & -4 \\ 4 & -3-\lambda & -2 \\ 1 & -1 & -1-\lambda \end{vmatrix} = -(\lambda-3)(\lambda+2)(\lambda+1)$. The eigenvectors associated with

$\lambda_1 = 3$ satisfy the system with augmented matrix $\begin{bmatrix} 1 & -1 & -4 & : & 0 \\ 4 & -6 & -2 & : & 0 \\ 1 & -1 & -4 & : & 0 \end{bmatrix}$, which is row

equivalent to $\begin{bmatrix} 1 & 0 & -11 & : & 0 \\ 0 & 1 & -7 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = 11x_3$ and $x_2 = 7x_3$. Taking $x_3 = 1$ yields

$\mathbf{y}_1 = \begin{bmatrix} 11 \\ 7 \\ 1 \end{bmatrix} e^{3t}$. The eigenvectors associated with $\lambda_2 = -2$ satisfy the system with augmented

matrix $\begin{bmatrix} 6 & -1 & -4 & : & 0 \\ 4 & -1 & -2 & : & 0 \\ 1 & -1 & 1 & : & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & -2 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = x_3$

and $x_2 = 2x_3$. Taking $x_3 = 1$ yields $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} e^{-2t}$. The eigenvectors associated with

$\lambda_3 = -1$ satisfy the system with augmented matrix $\begin{bmatrix} 5 & -1 & -4 & : & 0 \\ 4 & -2 & -2 & : & 0 \\ 1 & -1 & 0 & : & 0 \end{bmatrix}$, which is row

equivalent to $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = x_3$ and $x_2 = x_3$. Taking $x_3 = 1$ yields

$\mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{-t}$. Hence $\mathbf{y} = c_1 \begin{bmatrix} 11 \\ 7 \\ 1 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} e^{-2t} + c_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{-t}$.

10.4.14. $\begin{vmatrix} 3-\lambda & 2 & -2 \\ -2 & 7-\lambda & -2 \\ -10 & 10 & -5-\lambda \end{vmatrix} = -(\lambda+5)(\lambda-5)^2$. The eigenvectors associated with $\lambda_1 = -5$

satisfy the system with augmented matrix $\begin{bmatrix} 8 & 2 & -2 & : & 0 \\ -2 & 12 & -2 & : & 0 \\ -10 & 10 & 0 & : & 0 \end{bmatrix}$, which is row equivalent

to $\begin{bmatrix} 1 & 0 & -\frac{1}{5} & : & 0 \\ 0 & 1 & -\frac{1}{5} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = \frac{1}{5}x_3$ and $x_2 = \frac{1}{5}x_3$. Taking $x_3 = 5$ yields $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} e^{-5t}$.

The eigenvectors associated with $\lambda_2 = 5$ satisfy the system with augmented matrix

$\begin{bmatrix} -2 & 2 & -2 & : & 0 \\ -2 & 2 & -2 & : & 0 \\ -10 & 10 & -10 & : & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & -1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence

$x_1 = x_2 - x_3$. Taking $x_2 = 0$ and $x_3 = 1$ yields $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{5t}$. Taking $x_2 = 1$ and $x_3 = 0$

yields $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{5t}$. Hence $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} e^{-5t} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{5t} + c_3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{5t}$.

10.4.16. $\begin{vmatrix} -7-\lambda & 4 \\ -6 & 7-\lambda \end{vmatrix} = (\lambda-5)(\lambda+5)$. Eigenvectors associated with $\lambda_1 = 5$ satisfy

$\begin{bmatrix} -12 & 4 \\ -6 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, so $x_1 = \frac{x_2}{3}$. Taking $x_2 = 3$ yields $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{5t}$. Eigenvectors

associated with $\lambda_2 = -5$ satisfy $\begin{bmatrix} -2 & 4 \\ -6 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, so $x_1 = 2x_2$. Taking $x_2 = 1$ yields

$\mathbf{y}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-5t}$. The general solution is $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{-5t}$. Now

$\mathbf{y}(0) = \begin{bmatrix} 2 \\ -4 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$, so $c_1 = -2$ and $c_2 = 2$. Therefore,

$\mathbf{y} = -\begin{bmatrix} 2 \\ 6 \end{bmatrix} e^{5t} + \begin{bmatrix} 4 \\ 2 \end{bmatrix} e^{-5t}$.

10.4.18. $\begin{vmatrix} 21-\lambda & -12 \\ 24 & -15-\lambda \end{vmatrix} = (\lambda-9)(\lambda+3)$. Eigenvectors associated with $\lambda_1 = 9$ satisfy

$\begin{bmatrix} 12 & -12 \\ 24 & -24 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, so $x_1 = x_2$. Taking $x_2 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{9t}$. Eigenvectors associated with $\lambda_2 = -3$ $\begin{bmatrix} 24 & -12 \\ 24 & -12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, so $x_1 = \frac{1}{2}x_2$. Taking $x_2 = 2$ yields

$\mathbf{y}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-3t}$. The general solution is $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{9t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-3t}$. Now

$\mathbf{y}(0) = \begin{bmatrix} 5 \\ 3 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$, so $c_1 = 7$ and $c_2 = -2$. Therefore,

$\mathbf{y} = \begin{bmatrix} 7 \\ 7 \end{bmatrix} e^{9t} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} e^{-3t}$.

10.4.20. $\begin{vmatrix} \frac{1}{6}-\lambda & \frac{1}{3} & 0 \\ \frac{2}{3} & -\frac{1}{6}-\lambda & 0 \\ 0 & 0 & \frac{1}{2}-\lambda \end{vmatrix} = -(\lambda+1/2)(\lambda-1/2)^2$. The eigenvectors associated with

$\lambda_1 = -1/2$ satisfy the system with augmented matrix $\begin{bmatrix} \frac{2}{3} & \frac{1}{3} & 0 & : & 0 \\ \frac{2}{3} & \frac{1}{3} & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \end{bmatrix}$, which is row

equivalent to $\begin{bmatrix} 1 & \frac{1}{2} & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = -\frac{x_2}{2}$ and $x_3 = 0$. Taking $x_2 = 2$ yields

$\mathbf{y}_1 = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} e^{-t/2}$. The eigenvectors associated with $\lambda_2 = \lambda_3 = 1/2$ satisfy the system

with augmented matrix $\begin{bmatrix} -\frac{1}{3} & \frac{1}{3} & 0 & : & 0 \\ \frac{2}{3} & -\frac{2}{3} & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$, which is row equivalent to

$\begin{bmatrix} 1 & -1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = x_2$ and x_3 is arbitrary. Taking $x_2 = 1$ and $x_3 = 0$ yields

$\mathbf{y}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{t/2}$. Taking $x_2 = 0$ and $x_3 = 1$ yields $\mathbf{y}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{t/2}$. The general solution is

$\mathbf{y} = c_1 \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} e^{-t/2} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{t/2} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{t/2}$. Now

$$\mathbf{y}(0) = \begin{bmatrix} 4 \\ 7 \\ 1 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{t/2} = \begin{bmatrix} 4 \\ 7 \\ 1 \end{bmatrix}, \text{ so } c_1 = 1, c_2 = 5, \text{ and } c_3 = 1. \text{ Hence}$$

$$\mathbf{y} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} e^{-t/2} + \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix} e^{t/2} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{t/2}.$$

10.4.22. $\begin{vmatrix} 6-\lambda & -3 & -8 \\ 2 & 1-\lambda & -2 \\ 3 & -3 & -5-\lambda \end{vmatrix} = -(\lambda-1)(\lambda+2)(\lambda-3)$. The eigenvectors associated with

$\lambda_1 = 1$ satisfy the system with augmented matrix $\begin{bmatrix} 5 & -3 & -8 & : & 0 \\ 2 & 0 & -2 & : & 0 \\ 3 & -3 & -6 & : & 0 \end{bmatrix}$, which is row

equivalent to $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = x_3$ and $x_2 = -x_3$. Taking $x_3 = 1$ yields

$\mathbf{y}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} e^t$. The eigenvectors associated with $\lambda_2 = -2$ satisfy the system with augmented

matrix $\begin{bmatrix} 8 & -3 & -8 & : & 0 \\ 2 & 3 & -2 & : & 0 \\ 3 & -3 & -3 & : & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = x_3$

and $x_2 = 0$. Taking $x_3 = 1$ yields $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{-2t}$. The eigenvectors associated with $\lambda_3 = 3$

satisfy the system with augmented matrix $\begin{bmatrix} 3 & -3 & -8 & : & 0 \\ 2 & -2 & -2 & : & 0 \\ 3 & -3 & -8 & : & 0 \end{bmatrix}$, which is row equivalent to

$\begin{bmatrix} 1 & -1 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = x_2$ and $x_3 = 0$. Taking $x_2 = 1$ yields $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{3t}$. The

general solution is $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} e^t + c_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{-2t} + c_3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{3t}$. Now

$\mathbf{y}(0) = \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix}$, so $c_1 = 2$, $c_2 = -3$, and $c_3 = 1$.

Therefore, $\mathbf{y} = \begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix} e^t - \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix} e^{-2t} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{3t}$.

10.4.24. $\begin{vmatrix} 3-\lambda & 0 & 1 \\ 11 & -2-\lambda & 7 \\ 1 & 0 & 3-\lambda \end{vmatrix} = -(\lambda-2)(\lambda+2)(\lambda-4)$. The eigenvectors associated with

$\lambda_1 = 2$ satisfy the system with augmented matrix $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 11 & -4 & 7 & : & 0 \\ 1 & 0 & 1 & : & 0 \end{bmatrix}$, which is row

equivalent to $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = -x_3$ and $x_2 = -x_3$. Taking $x_3 = 1$ yields

$\mathbf{y}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{2t}$. The eigenvectors associated with $\lambda_2 = -2$ satisfy the system with

augmented matrix $\begin{bmatrix} 5 & 0 & 1 & : & 0 \\ 11 & 0 & 7 & : & 0 \\ 1 & 0 & 5 & : & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence

$x_1 = x_3 = 0$ and x_2 is arbitrary. Taking $x_3 = 1$ yields $\mathbf{y}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-2t}$. The eigenvectors

associated with $\lambda_3 = 4$ satisfy the system with augmented matrix

$\begin{bmatrix} -1 & 0 & 1 & : & 0 \\ 11 & -6 & 7 & : & 0 \\ 1 & 0 & -1 & : & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & -3 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = x_3$ and

$x_2 = 3x_3$. Taking $x_3 = 1$ yields $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} e^{4t}$. The general solution is

$\mathbf{y} = c_1 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-2t} + c_3 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} e^{4t}$. Now

$\mathbf{y}(0) = \begin{bmatrix} 2 \\ 7 \\ 6 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ 6 \end{bmatrix}$, so $c_1 = 2$, $c_2 = -3$, and $c_3 = 4$. Hence

$\mathbf{y} = \begin{bmatrix} -2 \\ -2 \\ 2 \end{bmatrix} e^{2t} - \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} e^{-2t} + \begin{bmatrix} 4 \\ 12 \\ 4 \end{bmatrix} e^{4t}$

10.4.26. $\begin{vmatrix} 3-\lambda & -1 & 0 \\ 4 & -2-\lambda & 0 \\ 4 & -4 & 2-\lambda \end{vmatrix} = -(\lambda+1)(\lambda-2)^2$. The eigenvectors associated with $\lambda_1 = -1$

satisfy the system with augmented matrix $\begin{bmatrix} 4 & -1 & 0 & : & 0 \\ 4 & -1 & 0 & : & 0 \\ 4 & -4 & 3 & : & 0 \end{bmatrix}$, which is row equivalent to

$\begin{bmatrix} 1 & 0 & -\frac{1}{4} & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$ Hence $x_1 = x_2/4$ and $x_2 = x_3$. Taking $x_3 = 4$ yields $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix} e^{-t}$. The

eigenvectors associated with $\lambda_2 = \lambda_3 = 2$ satisfy the system with augmented matrix

$\begin{bmatrix} 1 & -1 & 0 & : & 0 \\ 4 & -4 & 0 & : & 0 \\ 4 & -4 & 0 & : & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & -1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = x_2$ and x_3 is

arbitrary. Taking $x_2 = 1$ and $x_3 = 0$ yields $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t}$. Taking $x_2 = 0$ and $x_3 = 1$ yields

$\mathbf{y}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t}$. The general solution is $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{2t} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{2t}$. Now

$\mathbf{y}(0) = \begin{bmatrix} 7 \\ 10 \\ 2 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \\ 2 \end{bmatrix}$, so $c_1 = 1$, $c_2 = 6$, and $c_3 = -2$. Hence

$$\mathbf{y} = \begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix} e^{-t} + \begin{bmatrix} 6 \\ 6 \\ -2 \end{bmatrix} e^{2t}.$$

10.4.28. (a) If $\mathbf{y}(t_0) = 0$, then $\mathbf{y}$ is the solution of the initial value problem $\mathbf{y}' = A\mathbf{y}$, $\mathbf{y}(t_0) = 0$. Since $\mathbf{y} \equiv 0$ is a solution of this problem, Theorem 10.2.1 implies the conclusion.

(b) It is given that $\mathbf{y}'_1(t) = A\mathbf{y}_1(t)$ for all t . Replacing t by $t - \tau$ shows that $\mathbf{y}'_1(t - \tau) = A\mathbf{y}_1(t - \tau) = A\mathbf{y}_2(t)$ for all t . Since $\mathbf{y}'_2(t) = \mathbf{y}'_1(t - \tau)$ by the chain rule, this implies that $\mathbf{y}'_2(t) = A\mathbf{y}_2(t)$ for all t .

(c) If $\mathbf{z}(t) = \mathbf{y}_1(t - \tau)$, then $\mathbf{z}(t_2) = \mathbf{y}_1(t_1) = \mathbf{y}_2(t_2)$; therefore $\mathbf{z}$ and $\mathbf{y}_2$ are both solutions of the initial value problem $\mathbf{y}' = A\mathbf{y}$, $\mathbf{y}(t_2) = \mathbf{k}$, where $\mathbf{k} = \mathbf{y}_2(t_2)$.

10.4.42. The characteristic polynomial of A is $p(\lambda) = \lambda^2 - (a + b)\lambda + ab - \alpha\beta$, so the eigenvalues of

A are $\lambda_1 = \frac{a + b - \gamma}{2}$ and $\lambda_2 = \frac{a + b + \gamma}{2}$, where $\gamma = \sqrt{(a - b)^2 + 4\alpha\beta}$; $\mathbf{x}_1 = \begin{bmatrix} b - a + \gamma \\ 2\beta \end{bmatrix}$

and $\mathbf{x}_2 = \begin{bmatrix} b - a - \gamma \\ 2\beta \end{bmatrix}$ are associated eigenvectors. Since $\gamma > |b - a|$, if L_1 and L_2 are lines

through the origin parallel to $\mathbf{x}_1$ and $\mathbf{x}_2$, then L_1 is in the first and third quadrants and L_2 is in the second and fourth quadrants. The slope of L_1 is $\rho = \frac{2\beta}{b - a + \gamma} > 0$. If $Q_0 = \rho P_0$

there are three possibilities: (i) if $\alpha\beta = ab$, then $\lambda_1 = 0$ and $P(t) = P_0$, $Q(t) = Q_0$ for all $t > 0$; (ii) if $\alpha\beta < ab$, then $\lambda_1 > 0$ and $\lim_{t \rightarrow \infty} P(t) = \lim_{t \rightarrow \infty} Q(t) = \infty$ (monotonically); (iii) if $\alpha\beta > ab$, then $\lambda_1 < 0$ and $\lim_{t \rightarrow \infty} P(t) = \lim_{t \rightarrow \infty} Q(t) = 0$ (monotonically). Now suppose $Q_0 \neq \rho P_0$, so that the trajectory cannot intersect L_1 , and assume for the moment that (A) makes sense for all $t > 0$; that is, even if one or the other of P and Q is negative. Since $\lambda_2 > 0$ it follows that either $\lim_{t \rightarrow \infty} P(t) = \infty$ or $\lim_{t \rightarrow \infty} Q(t) = \infty$ (or both), and the trajectory is asymptotically parallel to L_2 . Therefore, the trajectory must cross into the third quadrant (so $P(T) = 0$ and $Q(T) > 0$ for some finite T) if $Q_0 > \rho P_0$, or into the fourth quadrant (so $Q(T) = 0$ and $P(T) > 0$ for some finite T) if $Q_0 < \rho P_0$.

10.5 Constant Coefficient Homogeneous Systems II

10.5.2. $\begin{vmatrix} -\lambda & -1 \\ 1 & -2 - \lambda \end{vmatrix} = (\lambda + 1)^2$. Hence $\lambda_1 = -1$. Eigenvectors satisfy $\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, so

$x_1 = x_2$. Taking $x_2 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-t}$. For a second solution we need a vector $\mathbf{u}$ such

that $\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Let $u_1 = 1$ and $u_2 = 0$. Then $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{-t} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} te^{-t}$. The general solution is $\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-t} + c_2 \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} e^{-t} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} te^{-t} \right)$.

10.5.4. $\mathbf{y}' = \begin{bmatrix} 3-\lambda & 1 \\ -1 & 1-\lambda \end{bmatrix} = (\lambda-2)^2$. Hence $\lambda_1 = 2$. Eigenvectors satisfy $\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$,

so $x_1 = -x_2$. Taking $x_2 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{2t}$. For a second solution we need a vector $\mathbf{u}$

such that $\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Let $u_1 = -1$ and $u_2 = 0$. Then

$\mathbf{y}_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix} e^{2t} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} te^{2t}$. The general solution is

$\mathbf{y} = c_1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{2t} + c_2 \left(\begin{bmatrix} -1 \\ 0 \end{bmatrix} e^{2t} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} te^{2t} \right)$.

10.5.6. $\begin{bmatrix} -10-\lambda & 9 \\ -4 & 2-\lambda \end{bmatrix} = (\lambda+4)^2$. Hence $\lambda_1 = -4$. Eigenvectors satisfy $\begin{bmatrix} -6 & 9 \\ -4 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$,

so $x_1 = \frac{3}{2}x_2$. Taking $x_2 = 2$ yields $\mathbf{y}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{-4t}$. For a second solution we need a vector $\mathbf{u}$

such that $\begin{bmatrix} -6 & 9 \\ -4 & 6 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$. Let $u_1 = -\frac{1}{2}$ and $u_2 = 0$. Then $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \frac{e^{-4t}}{2} + \begin{bmatrix} 3 \\ 2 \end{bmatrix} te^{-4t}$.

The general solution is $\mathbf{y} = c_1 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{-4t} + c_2 \left(\begin{bmatrix} -1 \\ 0 \end{bmatrix} \frac{e^{-4t}}{2} + \begin{bmatrix} 3 \\ 2 \end{bmatrix} te^{-4t} \right)$.

10.5.8. $\begin{bmatrix} -\lambda & 2 & 1 \\ -4 & 6-\lambda & 1 \\ 0 & 4 & 2-\lambda \end{bmatrix} = -\lambda(\lambda-4)^2$. Hence $\lambda_1 = 0$ and $\lambda_2 = \lambda_3 = 4$. The eigenvectors

associated with $\lambda_1 = 0$ satisfy the system with augmented matrix $\begin{bmatrix} 0 & 2 & 1 & : & 0 \\ -4 & 6 & 1 & : & 0 \\ 0 & 4 & 2 & : & 0 \end{bmatrix}$, which

is row equivalent to $\begin{bmatrix} 1 & 0 & \frac{1}{2} & : & 0 \\ 0 & 1 & \frac{1}{2} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = -\frac{1}{2}x_3$ and $x_2 = -\frac{1}{2}x_3$. Taking $x_3 = 2$

yields $\mathbf{y}_1 = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$. The eigenvectors associated with $\lambda_2 = 4$ satisfy the system with

augmented matrix $\begin{bmatrix} -4 & 2 & 1 & : & 0 \\ -4 & 2 & 1 & : & 0 \\ 0 & 4 & -2 & : & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & 0 & -\frac{1}{2} & : & 0 \\ 0 & 1 & -\frac{1}{2} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$.

Hence $x_1 = \frac{1}{2}x_3$ and $x_2 = \frac{1}{2}x_3$. Taking $x_3 = 2$ yields $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} e^{4t}$. For a third solution we

need a vector $\mathbf{u}$ such that $\begin{bmatrix} -4 & 2 & 1 \\ -4 & 2 & 1 \\ 0 & 4 & -2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$. The augmented matrix of this

system is row equivalent to $\begin{bmatrix} 1 & 0 & -\frac{1}{2} & : & 0 \\ 0 & 1 & -\frac{1}{2} & : & \frac{1}{2} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Let $u_3 = 0$, $u_1 = 0$, and $u_2 = \frac{1}{2}$. Then

$$\mathbf{y}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \frac{e^{4t}}{2} + \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} te^{4t}. \text{ The general solution is}$$

$$\mathbf{y} = c_1 \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} e^{4t} + c_3 \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \frac{e^{4t}}{2} + \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} te^{4t} \right).$$

10.5.10. $\begin{bmatrix} -1-\lambda & 1 & -1 \\ -2 & -\lambda & 2 \\ -1 & 3 & -1-\lambda \end{bmatrix} = -(\lambda-2)(\lambda+2)^2$. Hence $\lambda_1 = 2$ and $\lambda_2 = \lambda_3 = -2$. The

eigenvectors associated with $\lambda_1 = 2$ satisfy the system with augmented matrix

$$\begin{bmatrix} -3 & 1 & -1 & : & 0 \\ -2 & -2 & 2 & : & 0 \\ -1 & 3 & -3 & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}. \text{ Hence } x_1 = 0 \text{ and}$$

$x_2 = x_3$. Taking $x_3 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^{2t}$. The eigenvectors associated with $\lambda_2 = -2$

satisfy the system with augmented matrix $\begin{bmatrix} 1 & 1 & -1 & : & 0 \\ -2 & 2 & 2 & : & 0 \\ -1 & 3 & 1 & : & 0 \end{bmatrix}$, which is row equivalent to

$$\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}. \text{ Hence } x_1 = x_3 \text{ and } x_2 = 0. \text{ Taking } x_3 = 1 \text{ yields } \mathbf{y}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{-2t}. \text{ For a}$$

third solution we need a vector $\mathbf{u}$ such that $\begin{bmatrix} 1 & 1 & -1 \\ -2 & 2 & 2 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. The augmented

matrix of this system is row equivalent to $\begin{bmatrix} 1 & 0 & -1 & : & \frac{1}{2} \\ 0 & 1 & 0 & : & \frac{1}{2} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Let $u_3 = 0$, $u_1 = \frac{1}{2}$, and

$u_2 = \frac{1}{2}$. Then $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \frac{e^{-2t}}{2} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} te^{-2t}$. The general solution is

$$\mathbf{y} = c_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{-2t} + c_3 \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \frac{e^{-2t}}{2} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} te^{-2t} \right).$$

10.5.12. $\begin{bmatrix} 6-\lambda & -5 & 3 \\ 2 & -1-\lambda & 3 \\ 2 & 1 & 1-\lambda \end{bmatrix} = -(\lambda+2)(\lambda-4)^2$. Hence $\lambda_1 = -2$ and $\lambda_2 = \lambda_3 = 4$. The

eigenvectors associated with $\lambda_1 = -2$ satisfy the system with augmented matrix

$\begin{bmatrix} 8 & -5 & 3 & : & 0 \\ 2 & 1 & 3 & : & 0 \\ 2 & 1 & 3 & : & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = -x_3$ and

$x_2 = -x_3$. Taking $x_3 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{-2t}$. The eigenvectors associated with $\lambda_2 = 4$

satisfy the system with augmented matrix $\begin{bmatrix} 2 & -5 & 3 & : & 0 \\ 2 & -5 & 3 & : & 0 \\ 2 & 1 & -3 & : & 0 \end{bmatrix}$, which is row equivalent to

$\begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = x_3$ and $x_2 = x_3$. Taking $x_3 = 1$ yields $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{4t}$. For a

third solution we need a vector $\mathbf{u}$ such that $\begin{bmatrix} 2 & -5 & 3 \\ 2 & -5 & 3 \\ 2 & 1 & -3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. The augmented

matrix of this system is row equivalent to $\begin{bmatrix} 1 & 0 & -1 & : & \frac{1}{2} \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Let $u_3 = 0$, $u_1 = \frac{1}{2}$, and

$u_2 = 0$. Then $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{e^{4t}}{2} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} te^{4t}$. The general solution is

$$\mathbf{y} = c_1 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{-2t} + c_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^{4t} + c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \frac{e^{4t}}{2} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} te^{4t} \right).$$

10.5.14. $\begin{vmatrix} 15 - \lambda & -9 \\ 16 & -9 - \lambda \end{vmatrix} = (\lambda - 3)^2$. Hence $\lambda_1 = 3$. Eigenvectors satisfy $\begin{bmatrix} 12 & -9 \\ 16 & -12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$,

so $x_1 = \frac{3}{4}x_2$. Taking $x_2 = 4$ yields $\mathbf{y}_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix} e^{3t}$. For a second solution we need a vector $\mathbf{u}$

such that $\begin{bmatrix} 12 & -9 \\ 16 & -12 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Let $u_1 = \frac{1}{4}$ and $u_2 = 0$. Then $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \frac{e^{3t}}{4} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} te^{3t}$.

The general solution is $\mathbf{y} = c_1 \begin{bmatrix} 3 \\ 4 \end{bmatrix} e^{3t} + c_2 \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \frac{e^{3t}}{4} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} te^{3t} \right)$. Now

$\mathbf{y}(0) = \begin{bmatrix} 5 \\ 8 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 3 \\ 4 \end{bmatrix} + c_2 \begin{bmatrix} \frac{1}{4} \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$, so $c_1 = 2$ and $c_2 = -4$. Therefore,

$$\mathbf{y} = \begin{bmatrix} 5 \\ 8 \end{bmatrix} e^{3t} - \begin{bmatrix} 12 \\ 16 \end{bmatrix} te^{3t}.$$

10.5.16. $\begin{vmatrix} -7 - \lambda & 24 \\ -6 & 17 - \lambda \end{vmatrix} = (\lambda - 5)^2$. Hence $\lambda_1 = 5$. Eigenvectors satisfy $\begin{bmatrix} -12 & 24 \\ -6 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$,

so $x_1 = 2x_2$. Taking $x_2 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{5t}$. For a second solution we need a vector $\mathbf{u}$

such that $\begin{bmatrix} -12 & 24 \\ -6 & 12 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Let $u_1 = \frac{1}{6}$ and $u_2 = 0$. Then $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \frac{e^{5t}}{6} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} te^{5t}$.

The general solution is $\mathbf{y} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} e^{5t} + c_2 \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \frac{e^{5t}}{6} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} t e^{5t} \right)$. Now

$$\mathbf{y}(0) = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} \frac{1}{6} \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \text{ so } c_1 = 1 \text{ and } c_2 = 6. \text{ Therefore, } \mathbf{y} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} e^{5t} - \begin{bmatrix} 12 \\ 6 \end{bmatrix} t e^{5t}.$$

10.5.18. $\begin{bmatrix} -1-\lambda & 1 & 0 \\ 1 & -1-\lambda & -2 \\ -1 & -1 & -1-\lambda \end{bmatrix} = -(\lambda-1)(\lambda+2)^2$. Hence $\lambda_1 = 1$ and $\lambda_2 = \lambda_3 = -2$. The

eigenvectors associated with $\lambda_1 = 1$ satisfy the system with augmented matrix

$$\begin{bmatrix} -2 & 1 & 0 & : & 0 \\ 1 & -2 & -2 & : & 0 \\ -1 & -1 & -2 & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & 0 & \frac{2}{3} & : & 0 \\ 0 & 1 & \frac{3}{2} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}. \text{ Hence } x_1 = -\frac{2}{3}x_3 \text{ and}$$

$x_2 = -\frac{4}{3}x_3$. Taking $x_3 = 3$ yields The eigenvectors associated with $\lambda_2 = -2$ satisfy the

system with augmented matrix $\begin{bmatrix} 1 & 1 & 0 & : & 0 \\ 1 & 1 & -2 & : & 0 \\ -1 & -1 & 1 & : & 0 \end{bmatrix}$, which is row equivalent to

$$\begin{bmatrix} 1 & 1 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}. \text{ Hence } x_1 = -x_2 \text{ and } x_3 = 0. \text{ Taking } x_2 = 1 \text{ yields } \mathbf{y}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t}. \text{ For a}$$

third solution we need a vector $\mathbf{u}$ such that $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & -2 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$. The augmented

matrix of this system is row equivalent to $\begin{bmatrix} 1 & 1 & 0 & : & -1 \\ 0 & 0 & 1 & : & -1 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Let $u_2 = 0$, $u_1 = -1$, and

$u_3 = -1$. Then $\mathbf{y}_3 = \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix} e^{-2t} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} t e^{-2t}$. The general solution is

$$\mathbf{y} = c_1 \begin{bmatrix} -2 \\ -4 \\ 3 \end{bmatrix} e^t + c_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t} + c_3 \left(\begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix} e^{-2t} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} t e^{-2t} \right). \text{ Now}$$

$$\mathbf{y}(0) = \begin{bmatrix} 6 \\ 5 \\ -7 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} -2 \\ -4 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \\ -7 \end{bmatrix}, \text{ so } c_1 = -2, c_2 = -3, \text{ and } c_3 = 1.$$

$$\text{Therefore, } \mathbf{y} = \begin{bmatrix} 4 \\ 8 \\ -6 \end{bmatrix} e^t + \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix} e^{-2t} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} t e^{-2t}.$$

10.5.20. $\begin{bmatrix} -7-\lambda & -4 & 4 \\ -1 & 0-\lambda & 1 \\ -9 & -5 & 6-\lambda \end{bmatrix} = -(\lambda+3)(\lambda-1)^2$. Hence $\lambda_1 = -3$ and $\lambda_2 = \lambda_3 = 1$. The

eigenvectors associated with $\lambda_1 = -3$ satisfy the system with augmented matrix

$$\begin{bmatrix} -4 & -4 & 4 & : & 0 \\ -1 & 3 & 1 & : & 0 \\ -9 & -5 & 9 & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}. \text{ Hence } x_1 = x_3 \text{ and}$$

$x_2 = 0$. Taking $x_3 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{-3t}$. The eigenvectors associated with $\lambda_2 = 1$ satisfy

the system with augmented matrix $\begin{bmatrix} -8 & -4 & 4 & : & 0 \\ -1 & -1 & 1 & : & 0 \\ -9 & -5 & -5 & : & 0 \end{bmatrix}$, which is row equivalent to

$\begin{bmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = 0$ and $x_2 = x_3$. Taking $x_3 = 1$ yields $\mathbf{y}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^t$. For a

third solution we need a vector $\mathbf{u}$ such that $\begin{bmatrix} -8 & -4 & 4 \\ -1 & -1 & 1 \\ -9 & -5 & 5 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. The augmented

matrix of this system is row equivalent to $\begin{bmatrix} 1 & 0 & 0 & : & 1 \\ 0 & 1 & -1 & : & -2 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Let $u_3 = 0$, $u_1 = 1$, and

$u_2 = -2$. Then $\mathbf{y}_3 = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} e^t + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} te^t$. The general solution is

$\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} e^{-3t} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^t + c_3 \left(\begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} e^t + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} te^t \right)$. Now

$\mathbf{y}(0) = \begin{bmatrix} -6 \\ 9 \\ -1 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} -6 \\ 9 \\ -1 \end{bmatrix}$, so $c_1 = -2$, $c_2 = 1$, and $c_3 = -4$.

Therefore, $\mathbf{y} = -\begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} e^{-3t} + \begin{bmatrix} -4 \\ 9 \\ 1 \end{bmatrix} e^t - \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix} te^t$.

10.5.22. $\begin{bmatrix} 4-\lambda & -8 & -4 \\ -3 & -1-\lambda & -3 \\ 1 & -1 & 9-\lambda \end{bmatrix} = -(\lambda+4)(\lambda-8)^2$. Hence $\lambda_1 = -4$ and $\lambda_2 = \lambda_3 = 8$. The

eigenvectors associated with $\lambda_1 = -4$ satisfy the system with augmented matrix

$\begin{bmatrix} 8 & -8 & -4 & : & 0 \\ -3 & 3 & -3 & : & 0 \\ 1 & -1 & 13 & : & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & -1 & 0 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = x_2$ and

$x_3 = 0$. Taking $x_2 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^t$. The eigenvectors associated with $\lambda_2 = 8$ satisfy

the system with augmented matrix $\begin{bmatrix} -4 & -8 & -4 & : & 0 \\ -3 & -9 & -3 & : & 0 \\ 1 & -1 & 1 & : & 0 \end{bmatrix}$, which is row equivalent to

$\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = -x_3$ and $x_2 = 0$. Taking $x_3 = 1$ yields $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{8t}$. For a

third solution we need a vector $\mathbf{u}$ such that $\begin{bmatrix} -4 & -8 & -4 \\ -3 & -9 & -3 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$. The augmented

matrix of this system is row equivalent to $\begin{bmatrix} 1 & 0 & 1 & : & \frac{3}{4} \\ 0 & 1 & 0 & : & -\frac{1}{4} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Let $u_3 = 0$, $u_1 = \frac{3}{4}$, and

$u_2 = -\frac{1}{4}$. Then $\mathbf{y}_3 = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix} \frac{e^{8t}}{4} + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} te^{8t}$. The general solution is

$c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^t + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{8t} + c_3 \left(\begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix} \frac{e^{8t}}{4} + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} te^{8t} \right)$. Now

$\mathbf{y}(0) = \begin{bmatrix} -4 \\ 1 \\ -3 \end{bmatrix} \Rightarrow c_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} \frac{3}{4} \\ -\frac{1}{4} \\ 0 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \\ -3 \end{bmatrix}$, so $c_1 = -1$, $c_2 = -3$, and $c_3 = -8$.

Therefore, $\mathbf{y} = -\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} e^{-4t} + \begin{bmatrix} -3 \\ 2 \\ -3 \end{bmatrix} e^{8t} + \begin{bmatrix} 8 \\ 0 \\ -8 \end{bmatrix} te^{8t}$.

10.5.24. $\begin{bmatrix} 5-\lambda & -1 & 1 \\ -1 & 9-\lambda & -3 \\ -2 & 2 & 4-\lambda \end{bmatrix} = -(\lambda-6)^3$. Hence $\lambda_1 = 6$. The eigenvectors satisfy the system

with augmented matrix $\begin{bmatrix} 1 & -1 & 1 & : & 0 \\ -1 & 3 & -3 & : & 0 \\ -2 & 2 & -2 & : & 0 \end{bmatrix}$, which is row equivalent to

$\begin{bmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = 0$ and $x_2 = x_3$. Taking $x_3 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^{6t}$. For a

second solution we need a vector $\mathbf{u}$ such that $\begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -3 \\ -2 & 2 & -2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. The augmented

matrix of this system is row equivalent to $\begin{bmatrix} 1 & 0 & 0 & : & -\frac{1}{4} \\ 0 & 1 & -1 & : & \frac{1}{4} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Let $u_3 = 0$, $u_1 = -\frac{1}{4}$, and

$u_2 = \frac{1}{4}$. Then $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \frac{e^{6t}}{4} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} te^{6t}$. For a third solution we need a vector $\mathbf{v}$ such that

$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -3 \\ -2 & 2 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} \\ \frac{1}{4} \\ 0 \end{bmatrix}$. The augmented matrix of this system is row equivalent to

$\begin{bmatrix} 1 & 0 & 0 & : & \frac{1}{8} \\ 0 & 1 & -1 & : & \frac{1}{8} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Let $v_3 = 0$, $v_1 = \frac{1}{8}$, and $v_2 = \frac{1}{8}$. Then

$$\mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \frac{e^{6t}}{8} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \frac{te^{6t}}{4} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \frac{t^2e^{6t}}{2}. \text{ The general solution is}$$

$$\mathbf{y} = c_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} e^{6t} + c_2 \left(\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \frac{e^{6t}}{4} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} te^{6t} \right) \\ + c_3 \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \frac{e^{6t}}{8} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \frac{te^{6t}}{4} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \frac{t^2e^{6t}}{2} \right).$$

10.5.26. $\begin{bmatrix} -6-\lambda & -4 & -4 \\ 2 & -1-\lambda & 1 \\ 2 & 3 & 1-\lambda \end{bmatrix} = -(\lambda+2)^3$. Hence $\lambda_1 = -2$. The eigenvectors satisfy the

system with augmented matrix $\begin{bmatrix} -4 & -4 & -4 & : & 0 \\ 2 & 1 & 1 & : & 0 \\ 2 & 3 & 3 & : & 0 \end{bmatrix}$, which is row equivalent to

$$\begin{bmatrix} 1 & 0 & 0 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}. \text{ Hence } x_1 = 0 \text{ and } x_2 = -x_3. \text{ Taking } x_3 = 1 \text{ yields } \mathbf{y}_1 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} e^{-2t}. \text{ For a}$$

second solution we need a vector $\mathbf{u}$ such that $\begin{bmatrix} -4 & -4 & -4 \\ 2 & 1 & 1 \\ 2 & 3 & 3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$. The

augmented matrix of this system is row equivalent to $\begin{bmatrix} 1 & 0 & 0 & : & -1 \\ 0 & 1 & 1 & : & 1 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Let $u_3 = 0$,

$u_1 = -1$, and $u_2 = 1$. Then $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t} + \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} te^{-2t}$. For a third solution we need a

vector $\mathbf{v}$ such that $\begin{bmatrix} -4 & -4 & -4 \\ 2 & 1 & 1 \\ 2 & 3 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$. The augmented matrix of this system is

row equivalent to $\begin{bmatrix} 1 & 0 & 0 & : & \frac{3}{4} \\ 0 & 1 & 1 & : & -\frac{1}{2} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Let $v_3 = 0$, $v_1 = \frac{3}{4}$, and $v_2 = -\frac{1}{2}$. Then

$$\mathbf{y}_3 = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix} \frac{e^{-2t}}{4} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} te^{-2t} + \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \frac{t^2e^{-2t}}{2}. \text{ The general solution is}$$

$$\mathbf{y} = c_1 \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} e^{-2t} + c_2 \left(\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t} + \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} te^{-2t} \right) \\ + c_3 \left(\begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix} \frac{e^{-2t}}{4} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} te^{-2t} + \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \frac{t^2e^{-2t}}{2} \right)$$

10.5.28. $\begin{bmatrix} -2-\lambda & -12 & 10 \\ 2 & -24-\lambda & 11 \\ 2 & -24 & 8-\lambda \end{bmatrix} = -(\lambda+6)^3$. Hence $\lambda_1 = -6$. The eigenvectors satisfy the

system with augmented matrix $\begin{bmatrix} 4 & -12 & 10 & : & 0 \\ 2 & -18 & 11 & : & 0 \\ 2 & -24 & 14 & : & 0 \end{bmatrix}$, which is row equivalent to

$$\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 1 & -\frac{1}{2} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}. \text{ Hence } x_1 = -x_3 \text{ and } x_2 = \frac{x_3}{2}. \text{ Taking } x_3 = 2 \text{ yields } \mathbf{y}_1 = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} e^{-6t}.$$

For a second solution we need a vector $\mathbf{u}$ such that $\begin{bmatrix} 4 & -12 & 10 \\ 2 & -18 & 11 \\ 2 & -24 & 14 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$. The

augmented matrix of this system is row equivalent to $\begin{bmatrix} 1 & 0 & 1 & : & -1 \\ 0 & 1 & -\frac{1}{2} & : & -\frac{1}{6} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Let $u_3 = 0$,

$u_1 = -1$, and $u_2 = -\frac{1}{6}$. Then $\mathbf{y}_2 = -\begin{bmatrix} 6 \\ 1 \\ 0 \end{bmatrix} \frac{e^{-6t}}{6} + \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} te^{-6t}$. For a third solution we need

a vector $\mathbf{v}$ such that $\begin{bmatrix} 4 & -12 & 10 \\ 2 & -18 & 11 \\ 2 & -24 & 14 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -1 \\ -\frac{1}{6} \\ 0 \end{bmatrix}$. The augmented matrix of this system is

row equivalent to $\begin{bmatrix} 1 & 0 & 1 & : & -\frac{1}{3} \\ 0 & 1 & -\frac{1}{2} & : & -\frac{1}{36} \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Let $v_3 = 0$, $v_1 = -\frac{1}{3}$, and $v_2 = -\frac{1}{36}$. Then

$\mathbf{y}_3 = -\begin{bmatrix} 12 \\ 1 \\ 0 \end{bmatrix} \frac{e^{-6t}}{36} - \begin{bmatrix} 6 \\ 1 \\ 0 \end{bmatrix} \frac{te^{-6t}}{6} + \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} \frac{t^2 e^{-6t}}{2}$. The general solution is

$$\begin{aligned} \mathbf{y} = & c_1 \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} e^{-6t} + c_2 \left(-\begin{bmatrix} 6 \\ 1 \\ 0 \end{bmatrix} \frac{e^{-6t}}{6} + \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} te^{-6t} \right) \\ & + c_3 \left(-\begin{bmatrix} 12 \\ 1 \\ 0 \end{bmatrix} \frac{e^{-6t}}{36} - \begin{bmatrix} 6 \\ 1 \\ 0 \end{bmatrix} \frac{te^{-6t}}{6} + \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} \frac{t^2 e^{-6t}}{2} \right). \end{aligned}$$

10.5.30. $\begin{bmatrix} -4-\lambda & 0 & -1 \\ -1 & -3-\lambda & -1 \\ 1 & 0 & -2-\lambda \end{bmatrix} = -(\lambda+3)^3$. Hence $\lambda_1 = 3$. The eigenvectors satisfy the

system with augmented matrix $\begin{bmatrix} -1 & 0 & -1 & : & 0 \\ -1 & 0 & -1 & : & 0 \\ 1 & 0 & 1 & : & 0 \end{bmatrix}$, which is row equivalent to

$$\begin{bmatrix} 1 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}. \text{ Hence } x_1 = -x_3 \text{ and } x_2 \text{ is arbitrary. Taking } x_2 = 0 \text{ and } x_3 = 1 \text{ yields}$$

$\mathbf{y}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{-3t}$. Taking $x_2 = 1$ and $x_3 = 0$ yields $\mathbf{y}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-3t}$. For a third solution we

need constants α and β and a vector $\mathbf{u}$ such that $\begin{bmatrix} -1 & 0 & -1 \\ -1 & 0 & -1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \alpha \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.

The augmented matrix of this system is row equivalent to $\begin{bmatrix} 1 & 0 & 1 & : & \alpha \\ 0 & 0 & 0 & : & \alpha + \beta \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$; hence the

system has a solution if $\alpha = -\beta = 1$, which yields the eigenvector $\mathbf{x}_3 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$. Taking

$u_1 = 1$ and $u_2 = u_3 = 0$ yields the solution $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-3t} + \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} t e^{-3t}$. The general

solution is $\mathbf{y} = c_1 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{-3t} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{-3t} + c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-3t} + \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} t e^{-3t} \right)$

10.5.32. $\begin{bmatrix} -3-\lambda & -1 & 0 \\ 1 & -1-\lambda & 0 \\ -1 & -1 & -2-\lambda \end{bmatrix} = -(\lambda+2)^3$. Hence $\lambda_1 = -2$. The eigenvectors satisfy the

system with augmented matrix $\begin{bmatrix} -1 & -1 & 0 & : & 0 \\ 1 & 1 & 0 & : & 0 \\ -1 & -1 & 0 & : & 0 \end{bmatrix}$, which is row equivalent to

$\begin{bmatrix} 1 & 1 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$. Hence $x_1 = -x_2$ and x_3 is arbitrary. Taking $x_2 = 1$ and $x_3 = 0$ yields

$\mathbf{y}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t}$. Taking $x_2 = 0$ and $x_3 = 1$ yields $\mathbf{y}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{-2t}$. For a third solution we

need constants α and β and a vector $\mathbf{u}$ such that $\begin{bmatrix} -1 & -1 & 0 \\ 1 & 1 & 0 \\ -1 & -1 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \alpha \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

The augmented matrix of this system is row equivalent to $\begin{bmatrix} 1 & 1 & 0 & : & \alpha \\ 0 & 0 & 0 & : & \alpha + \beta \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$; hence the

system has a solution if $\alpha = -\beta = 1$, which yields the eigenvector $\mathbf{x}_3 = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$. Taking

$u_1 = 1$ and $u_2 = u_3 = 0$ yields the solution $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-2t} + \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} t e^{-2t}$. The general

solution is $\mathbf{y} = c_1 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} e^{-2t} + c_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{-2t} + c_3 \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{-2t} + \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} t e^{-2t} \right)$.

10.5.34.

$$\begin{aligned}
\mathbf{y}'_3 - A\mathbf{y}_3 &= (\lambda_1 I - A)\mathbf{v}e^{\lambda_1 t} + (\lambda_1 I - A)\mathbf{u}te^{\lambda_1 t} + \mathbf{u}e^{\lambda_1 t} \\
&\quad + (\lambda_1 I - A)\mathbf{x}\frac{t^2 e^{\lambda_1 t}}{2} + \mathbf{x}te^{\lambda_1 t} \\
&= -\mathbf{u}e^{\lambda_1 t} - \mathbf{x}te^{\lambda_1 t} + \mathbf{u}e^{\lambda_1 t} + 0 + \mathbf{x}te^{\lambda_1 t} = 0.
\end{aligned}$$

Now suppose that $c_1\mathbf{y}_1 + c_2\mathbf{y}_2 + c_3\mathbf{y}_3 = 0$. Then

$$c_1\mathbf{x} + c_2(\mathbf{u} + t\mathbf{x}) + c_3\left(\mathbf{v} + t\mathbf{u} + \frac{t^2}{2}\mathbf{x}\right) = 0. \quad (\text{A})$$

Differentiating this twice yields $c_3\mathbf{x} = 0$, so $c_3 = 0$ since $\mathbf{x} \neq 0$. Therefore, (A) reduces to (B) $c_1\mathbf{x} + c_2(\mathbf{u} + t\mathbf{x}) = 0$. Differentiating this yields $c_2\mathbf{x} = 0$, so $c_2 = 0$ since $\mathbf{x} \neq 0$. Therefore, (B) reduces to $c_1\mathbf{x} = 0$, so $c_1 = 0$ since $\mathbf{x} \neq 0$. Therefore, $\mathbf{y}_1$, $\mathbf{y}_2$, and $\mathbf{y}_3$ are linearly independent.

10.6 Constant Coefficient Homogeneous Systems III

10.6.2. $\begin{vmatrix} -11-\lambda & 4 \\ -26 & 9-\lambda \end{vmatrix} = (\lambda+1)^2 + 4$. The augmented matrix of $(A - (-1+2i)I)\mathbf{x} = 0$ is $\begin{bmatrix} -10-2i & 4 & \vdots & 0 \\ -26 & 10-2i & \vdots & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & \frac{-5+i}{13} & \vdots & 0 \\ 0 & 0 & \vdots & 0 \end{bmatrix}$.

Therefore, $x_1 = (5-i)x_2/13$. Taking $x_2 = 13$ yields the eigenvector $\mathbf{x} = \begin{bmatrix} 5-i \\ 13 \end{bmatrix}$. Taking real

and imaginary parts of $e^{-t}(\cos 2t + i \sin 2t)$ $\begin{bmatrix} 5-i \\ 13 \end{bmatrix}$ yields

$$\mathbf{y} = c_1 e^{-t} \begin{bmatrix} 5 \cos 2t + \sin 2t \\ 13 \cos 2t \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 5 \sin 2t - \cos 2t \\ 13 \sin 2t \end{bmatrix}.$$

10.6.4. $\begin{vmatrix} 5-\lambda & -6 \\ 3 & -1-\lambda \end{vmatrix} = (\lambda-2)^2 + 9$. Hence, $\lambda = 2+3i$ is an eigenvalue of A . The associated eigenvectors satisfy $(A - (2+3i)I)\mathbf{x} = 0$. The augmented matrix of this system is $\begin{bmatrix} 3-3i & -6 & \vdots & 0 \\ 3 & -3-3i & \vdots & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & -1-i & \vdots & 0 \\ 0 & 0 & \vdots & 0 \end{bmatrix}$.

Therefore, $x_1 = (1+i)x_2$. Taking $x_2 = 1$ yields $x_1 = 1+i$, so $\mathbf{x} = \begin{bmatrix} 1+i \\ 1 \end{bmatrix}$ is an eigenvector.

Taking real and imaginary parts of $e^{2t}(\cos 3t + i \sin 3t)$ $\begin{bmatrix} 1+i \\ 1 \end{bmatrix}$ yields

$$\mathbf{y} = c_1 e^{2t} \begin{bmatrix} \cos 3t - \sin 3t \\ \cos 3t \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} \sin 3t + \cos 3t \\ \sin 3t \end{bmatrix}.$$

10.6.6. $\begin{vmatrix} -3-\lambda & 3 & 1 \\ 1 & -5-\lambda & -3 \\ -3 & 7 & 3-\lambda \end{vmatrix} = -(\lambda+1)((\lambda+2)^2 + 4)$. The augmented matrix of

$(A+I)\mathbf{x} = 0$ is $\begin{bmatrix} -2 & 3 & 1 & \vdots & 0 \\ 1 & -4 & -3 & \vdots & 0 \\ -3 & 7 & 4 & \vdots & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & 0 & 1 & \vdots & 0 \\ 0 & 1 & 1 & \vdots & 0 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}$.

Therefore, $x_1 = x_2 = -x_3$. Taking $x_3 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{-t}$. The augmented matrix of

$(A - (-2 + 2i)I)\mathbf{x} = 0$ is $\begin{bmatrix} -1-2i & 3 & 1 & \vdots & 0 \\ 1 & -3-2i & -3 & \vdots & 0 \\ -3 & 7 & 5-2i & \vdots & 0 \end{bmatrix}$, which is row equivalent to

$\begin{bmatrix} 1 & 0 & -\frac{1+i}{2} & \vdots & 0 \\ 0 & 1 & \frac{1-i}{2} & \vdots & 0 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}$. Therefore, $x_1 = \frac{(1+i)}{2}x_3$ and $x_2 = -\frac{(1-i)}{2}x_3$. Taking $x_3 = 2$ yields

the eigenvector $\mathbf{x}_2 = \begin{bmatrix} 1+i \\ -1+i \\ 2 \end{bmatrix}$. The real and imaginary parts of

$e^{-2t}(\cos 2t + i \sin 2t) \begin{bmatrix} 1+i \\ -1+i \\ 2 \end{bmatrix}$ are $\mathbf{y}_2 = e^{-2t} \begin{bmatrix} \cos 2t - \sin 2t \\ -\cos 2t - \sin 2t \\ 2 \cos 2t \end{bmatrix}$ and

$\mathbf{y}_3 = e^{-2t} \begin{bmatrix} \sin 2t + \cos 2t \\ -\sin 2t + \cos 2t \\ 2 \sin 2t \end{bmatrix}$. Therefore,

$$\mathbf{y} = c_1 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^{-t} + c_2 e^{-2t} \begin{bmatrix} \cos 2t - \sin 2t \\ -\cos 2t - \sin 2t \\ 2 \cos 2t \end{bmatrix} + c_3 e^{-2t} \begin{bmatrix} \sin 2t + \cos 2t \\ -\sin 2t + \cos 2t \\ 2 \sin 2t \end{bmatrix}.$$

10.6.8. $\begin{vmatrix} -3-\lambda & 1 & -3 \\ 4 & -1-\lambda & 2 \\ 4 & -2 & 3-\lambda \end{vmatrix} = -(\lambda-1)((\lambda+1)^2+4)$. The augmented matrix of

$(A - I)\mathbf{x} = 0$ is $\begin{bmatrix} -4 & 1 & -3 & \vdots & 0 \\ 4 & -2 & 2 & \vdots & 0 \\ 4 & -2 & 2 & \vdots & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & 0 & 1 & \vdots & 0 \\ 0 & 1 & 1 & \vdots & 0 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}$.

Therefore, $x_1 = x_2 = -x_3$. Taking $x_3 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} e^t$. The augmented matrix of

$(A - (-1 + 2i)I)\mathbf{x} = 0$ is $\begin{bmatrix} -2-2i & 1 & -3 & \vdots & 0 \\ 4 & -2i & 2 & \vdots & 0 \\ 4 & -2 & 4-2i & \vdots & 0 \end{bmatrix}$, which is row equivalent to

$\begin{bmatrix} 1 & 0 & \frac{1-i}{2} & \vdots & 0 \\ 0 & 1 & -1 & \vdots & 0 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}$. Therefore, $x_1 = -\frac{1-i}{2}x_3$ and $x_2 = x_3$. Taking $x_3 = 2$ yields the

eigenvector $\mathbf{x}_2 = \begin{bmatrix} -1+i \\ 2 \\ 2 \end{bmatrix}$. The real and imaginary parts of $e^{-t}(\cos 2t + i \sin 2t) \begin{bmatrix} -1+i \\ 2 \\ 2 \end{bmatrix}$

are $\mathbf{y}_2 = e^{-t} \begin{bmatrix} -\sin 2t - \cos 2t \\ 2 \cos 2t \\ 2 \cos 2t \end{bmatrix}$ and $\mathbf{y}_3 = e^{-t} \begin{bmatrix} \cos 2t - \sin 2t \\ 2 \sin 2t \\ 2 \sin 2t \end{bmatrix}$. Therefore,

$$\mathbf{y} = c_1 \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} e^{-t} + c_2 e^{-t} \begin{bmatrix} -\sin 2t - \cos 2t \\ 2 \cos 2t \\ 2 \cos 2t \end{bmatrix} + c_3 e^{-t} \begin{bmatrix} \cos 2t - \sin 2t \\ 2 \sin 2t \\ 2 \sin 2t \end{bmatrix}.$$

10.6.10. $\frac{1}{3} \begin{vmatrix} 7-3\lambda & -5 \\ 2 & 5-3\lambda \end{vmatrix} = (\lambda-2)^2 + 1$. The augmented matrix of $(A - (2+i)I)\mathbf{x} = 0$ is

$$\frac{1}{3} \begin{bmatrix} 1-3i & -5 & : & 0 \\ 2 & -1-3i & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & -\frac{1+3i}{2} & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}.$$

Therefore, $x_1 = \frac{1+3i}{2}x_2$. Taking $x_2 = 2$ yields the eigenvector $\mathbf{x} = \begin{bmatrix} 1+3i \\ 2 \end{bmatrix}$. Taking real

and imaginary parts of $e^{2t}(\cos t + i \sin t) \begin{bmatrix} 1+3i \\ 2 \end{bmatrix}$ yields

$$\mathbf{y} = c_1 e^{2t} \begin{bmatrix} \cos t - 3 \sin t \\ 2 \cos t \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} \sin t + 3 \cos t \\ 2 \sin t \end{bmatrix}.$$

10.6.12. $\begin{vmatrix} 34-\lambda & 52 \\ -20 & -30-\lambda \end{vmatrix} = (\lambda-2)^2 + 16$. The augmented matrix of $(A - (2+4i)I)\mathbf{x} = 0$ is

$$\begin{bmatrix} 32-4i & 52 & : & 0 \\ -20 & -32-4i & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & \frac{8+i}{5} & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}.$$

Therefore, $x_1 = -\frac{(8+i)}{5}x_2$. Taking $x_2 = 5$ yields the eigenvector $\mathbf{x} = \begin{bmatrix} -8-i \\ 5 \end{bmatrix}$. Taking real

and imaginary parts of $e^{2t}(\cos 4t + i \sin 4t) \begin{bmatrix} -8-i \\ 5 \end{bmatrix}$ yields

$$\mathbf{y} = c_1 e^{2t} \begin{bmatrix} \sin 4t - 8 \cos 4t \\ 5 \cos 4t \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} -\cos 4t - 8 \sin 4t \\ 5 \sin 4t \end{bmatrix}.$$

10.6.14. $\begin{vmatrix} 3-\lambda & -4 & -2 \\ -5 & 7-\lambda & -8 \\ -10 & 13 & -8-\lambda \end{vmatrix} = -(\lambda+2)((\lambda-2)^2 + 9)$. The augmented matrix of

$$(A + 2I)\mathbf{x} = 0 \text{ is } \begin{bmatrix} 5 & -4 & -2 & : & 0 \\ -5 & 9 & -8 & : & 0 \\ -10 & 13 & -6 & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & 0 & -2 & : & 0 \\ 0 & 1 & -2 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}.$$

Therefore, $x_1 = x_2 = 2x_3$. Taking $x_3 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} e^{-2t}$. The augmented matrix of

$$(A - (2+3i)I)\mathbf{x} = 0 \text{ is } \begin{bmatrix} 1-3i & -4 & -2 & : & 0 \\ -5 & 5-3i & -8 & : & 0 \\ -10 & 13 & -10-3i & : & 0 \end{bmatrix}, \text{ which is row equivalent to}$$

$$\begin{bmatrix} 1 & 0 & 1-i & : & 0 \\ 0 & 1 & -i & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}. \text{ Therefore, } x_1 = -(1-i)x_3 \text{ and } x_2 = ix_3. \text{ Taking } x_3 = 1 \text{ yields the}$$

eigenvector $\mathbf{x}_2 = \begin{bmatrix} -1+i \\ i \\ 1 \end{bmatrix}$. The real and imaginary parts of $e^{2t}(\cos 3t + i \sin 3t) \begin{bmatrix} -1+i \\ i \\ 1 \end{bmatrix}$

are $\mathbf{y}_2 = e^{2t} \begin{bmatrix} -\cos 3t - \sin 3t \\ -\sin 3t \\ \cos 3t \end{bmatrix}$ and $\mathbf{y}_3 = c_3 e^{2t} \begin{bmatrix} -\sin 3t + \cos 3t \\ \cos 3t \\ \sin 3t \end{bmatrix}$. Therefore,

$$\mathbf{y} = c_1 \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} e^{-2t} + c_2 e^{2t} \begin{bmatrix} -\cos 3t - \sin 3t \\ -\sin 3t \\ \cos 3t \end{bmatrix} + c_3 e^{2t} \begin{bmatrix} -\sin 3t + \cos 3t \\ \cos 3t \\ \sin 3t \end{bmatrix}.$$

10.6.16. $\begin{vmatrix} 1-\lambda & 2 & -2 \\ 0 & 2-\lambda & -1 \\ 1 & 0 & -\lambda \end{vmatrix} = -(\lambda-2)((\lambda-1)^2+1)$. The augmented matrix of $(A-I)\mathbf{x} = 0$ is

$$\begin{bmatrix} 0 & 2 & -2 & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 1 & 0 & -1 & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & 0 & -1 & : & 0 \\ 0 & 1 & -1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}. \text{ Therefore, } x_1 = x_2 = 1.$$

Taking $x_3 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^t$. The augmented matrix of $(A - (1+i)I)\mathbf{x} = 0$ is

$$\begin{bmatrix} -i & 2 & -2 & : & 0 \\ 0 & 1-i & -1 & : & 0 \\ 1 & 0 & -1-i & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & 0 & -1-i & : & 0 \\ 0 & 1 & -\frac{1+i}{2} & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}.$$

Therefore, $x_1 = (1+i)x_3$ and $x_2 = \frac{(1+i)}{2}x_3$. Taking $x_3 = 2$ yields the eigenvector

$\mathbf{x}_2 = \begin{bmatrix} 2+2i \\ 1+i \\ 2 \end{bmatrix}$. The real and imaginary parts of $e^{4t}(\cos t + i \sin t) \begin{bmatrix} 2+2i \\ 1+i \\ 2 \end{bmatrix}$ are

$\mathbf{y}_2 = e^t \begin{bmatrix} 2 \cos t - 2 \sin t \\ \cos t - \sin t \\ 2 \cos t \end{bmatrix}$ and $\mathbf{y}_3 = c_3 e^t \begin{bmatrix} 2 \sin t + 2 \cos t \\ \cos t + \sin t \\ 2 \sin t \end{bmatrix}$. Therefore,

$$\mathbf{y} = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} e^t + c_2 e^t \begin{bmatrix} 2 \cos t - 2 \sin t \\ \cos t - \sin t \\ 2 \cos t \end{bmatrix} + c_3 e^t \begin{bmatrix} 2 \sin t + 2 \cos t \\ \cos t + \sin t \\ 2 \sin t \end{bmatrix}.$$

10.6.18. $\begin{vmatrix} 7-\lambda & 15 \\ -3 & 1-\lambda \end{vmatrix} = (\lambda-4)^2 + 36$. The augmented matrix of $(A - (4+6i)I)\mathbf{x} = 0$ is

$$\begin{bmatrix} 3-6i & 15 & : & 0 \\ -3 & -3-6i & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & 1+2i & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}.$$

Therefore, $x_1 = -(1+2i)x_2$. Taking $x_2 = 1$ yields the eigenvector $\mathbf{x} = \begin{bmatrix} -1-2i \\ 1 \end{bmatrix}$. Taking

real and imaginary parts of $e^{4t}(\cos 6t + i \sin 6t) \begin{bmatrix} -1-2i \\ 1 \end{bmatrix}$ yields

$\mathbf{y} = c_1 e^{4t} \begin{bmatrix} 2 \sin 6t - \cos 6t \\ \cos 6t \end{bmatrix} + c_2 e^{4t} \begin{bmatrix} -2 \cos 6t - \sin 6t \\ \sin 6t \end{bmatrix}$. Now

$$\mathbf{y}(0) = \begin{bmatrix} 5 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}, \text{ so } c_1 = 1, c_2 = -3, \text{ and } \mathbf{y} = e^{4t} \begin{bmatrix} 5 \cos 6t + 5 \sin 6t \\ \cos 6t - 3 \sin 6t \end{bmatrix}.$$

10.6.20. $\frac{1}{6} \begin{vmatrix} 4-6\lambda & -2 \\ 5 & 2-6\lambda \end{vmatrix} = \left(\lambda - \frac{1}{2}\right)^2 + \frac{1}{4}$. The augmented matrix of $\left(A - \frac{1+i}{2}I\right)\mathbf{x} = 0$ is $\frac{1}{6} \begin{bmatrix} 1-3i & -2 & \vdots & 0 \\ 5 & -1-3i & \vdots & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & -\frac{1+3i}{5} & \vdots & 0 \\ 0 & 0 & \vdots & 0 \end{bmatrix}$.

Therefore, $x_1 = \frac{1+3i}{5}x_2$. Taking $x_2 = 5$ yields the eigenvector $\mathbf{x} = \begin{bmatrix} 1+3i \\ 5 \end{bmatrix}$. Taking real

and imaginary parts of $e^{t/2}(\cos t/2 + i \sin t/2) \begin{bmatrix} 1+3i \\ 5 \end{bmatrix}$ yields

$$\mathbf{y} = c_1 e^{t/2} \begin{bmatrix} \cos t/2 - 3 \sin t/2 \\ 5 \cos t/2 \end{bmatrix} + c_2 e^{t/2} \begin{bmatrix} \sin t/2 + 3 \cos t/2 \\ 5 \sin t/2 \end{bmatrix}. \text{ Now}$$

$$\mathbf{y}(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 3 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \text{ so } c_1 = -\frac{1}{5}, c_2 = \frac{2}{5}, \text{ and}$$

$$\mathbf{y} = e^{t/2} \begin{bmatrix} \cos(t/2) + \sin(t/2) \\ -\cos(t/2) + 2 \sin(t/2) \end{bmatrix}.$$

10.6.22. $\begin{vmatrix} 4-\lambda & 4 & 0 \\ 8 & 10-\lambda & -20 \\ 2 & 3 & -2-\lambda \end{vmatrix} = -(\lambda-8)((\lambda-2)^2+4)$. The augmented matrix of

$$(A-8I)\mathbf{x} = 0 \text{ is } \begin{bmatrix} 0 & 4 & 0 & \vdots & 0 \\ 8 & 6 & -20 & \vdots & 0 \\ 2 & 3 & -6 & \vdots & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & 0 & -2 & \vdots & 0 \\ 0 & 1 & -2 & \vdots & 0 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}.$$

Therefore, $x_1 = x_2 = 2x_3$. Taking $x_3 = 2$ yields $\mathbf{y}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} e^{8t}$. The augmented matrix of

$$(A - (2+2i)I)\mathbf{x} = 0 \text{ is } \begin{bmatrix} 2-2i & 4 & 0 & \vdots & 0 \\ 8 & 8-2i & -20 & \vdots & 0 \\ 2 & 3 & -4-2i & \vdots & 0 \end{bmatrix}, \text{ which is row equivalent to}$$

$$\begin{bmatrix} 1 & 0 & -2+2i & \vdots & 0 \\ 0 & 1 & -2i & \vdots & 0 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}. \text{ Therefore, } x_1 = (2-2i)x_3 \text{ and } x_2 = 2ix_3. \text{ Taking } x_3 = 1 \text{ yields the}$$

eigenvector $\mathbf{x}_2 = \begin{bmatrix} 2-2i \\ 2i \\ 1 \end{bmatrix}$. The real and imaginary parts of $e^{2t}(\cos 2t + i \sin 2t) \begin{bmatrix} 2-2i \\ 2i \\ 1 \end{bmatrix}$ are

$$\mathbf{y}_2 = e^{2t} \begin{bmatrix} 2 \cos 2t + 2 \sin 2t \\ -2 \sin 2t \\ 2 \cos 2t \end{bmatrix} \text{ and } \mathbf{y}_3 = e^{2t} \begin{bmatrix} 2 \sin 2t + 2 \cos 2t \\ 2 \cos 2t \\ \sin 2t \end{bmatrix}, \text{ so the general solution is}$$

$$\mathbf{y} = c_1 \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} e^{8t} + c_2 e^{2t} \begin{bmatrix} 2 \cos 2t + 2 \sin 2t \\ -2 \sin 2t \\ 2 \cos 2t \end{bmatrix} + c_3 e^{2t} \begin{bmatrix} 2 \sin 2t + 2 \cos 2t \\ 2 \cos 2t \\ \sin 2t \end{bmatrix}. \text{ Now}$$

$$\mathbf{y}(0) = \begin{bmatrix} 8 \\ 6 \\ 5 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 2 & -2 \\ 2 & 0 & 2 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \\ 5 \end{bmatrix}, \text{ so } c_1 = 2, c_2 = 3, c_3 = 1, \text{ and}$$

$$\mathbf{y} = \begin{bmatrix} 4 \\ 4 \\ 2 \end{bmatrix} e^{8t} + e^{2t} \begin{bmatrix} 4 \cos 2t + 8 \sin 2t \\ -6 \sin 2t + 2 \cos 2t \\ 3 \cos 2t + \sin 2t \end{bmatrix}.$$

10.6.24. $\begin{vmatrix} 4-\lambda & -4 & 4 \\ -10 & 3-\lambda & 15 \\ 2 & -3 & 1-\lambda \end{vmatrix} = -(\lambda-8)(\lambda^2+16)$. The augmented matrix of $(A-8I)\mathbf{x} = 0$ is

$$\begin{bmatrix} -4 & -4 & 4 & : & 0 \\ -10 & -5 & 15 & : & 0 \\ 2 & -3 & -7 & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & 0 & -2 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}.$$
 Therefore, $x_1 = 2x_3$

and $x_2 = -x_3$. Taking $x_3 = 1$ yields $\mathbf{y}_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} e^{8t}$. The augmented matrix of

$(A-4iI)\mathbf{x} = 0$ is $\begin{bmatrix} 4-4i & -4 & 4 & : & 0 \\ -10 & 3-4i & 15 & : & 0 \\ 2 & -3 & 1-4i & : & 0 \end{bmatrix}$, which is row equivalent to

$$\begin{bmatrix} 1 & 0 & -1+i & : & 0 \\ 0 & 1 & -1+2i & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}.$$
 Therefore, $x_1 = (1-i)x_3$ and $x_2 = (1-2i)x_3$. Taking $x_3 = 1$ yields

the eigenvector $\mathbf{x}_2 = \begin{bmatrix} 1-i \\ 1-2i \\ 1 \end{bmatrix}$. The real and imaginary parts of $(\cos 4t + i \sin 4t) \begin{bmatrix} 1-i \\ 1-2i \\ 1 \end{bmatrix}$

are $\mathbf{y}_2 = \begin{bmatrix} \cos 4t + \sin 4t \\ \cos 4t + 2 \sin 4t \\ \cos 4t \end{bmatrix}$ and $\mathbf{y}_3 = \begin{bmatrix} \sin 4t - \cos 4t \\ \sin 4t - 2 \cos 4t \\ \sin 4t \end{bmatrix}$, so the general solution is

$$\mathbf{y} = c_1 \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} e^{8t} + c_2 \begin{bmatrix} \cos 4t + \sin 4t \\ \cos 4t + 2 \sin 4t \\ \cos 4t \end{bmatrix} + c_3 \begin{bmatrix} \sin 4t - \cos 4t \\ \sin 4t - 2 \cos 4t \\ \sin 4t \end{bmatrix}.$$
 Now

$$\mathbf{y}(0) = \begin{bmatrix} 16 \\ 14 \\ 6 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 1 & -1 \\ -1 & 1 & -2 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 16 \\ 14 \\ 6 \end{bmatrix}, \text{ so } c_1 = 3, c_2 = 3, c_3 = -7, \text{ and}$$

$$\mathbf{y} = \begin{bmatrix} 6 \\ -3 \\ 3 \end{bmatrix} e^{8t} + \begin{bmatrix} 10 \cos 4t - 4 \sin 4t \\ 17 \cos 4t - \sin 4t \\ 3 \cos 4t - 7 \sin 4t \end{bmatrix}.$$

10.6.28. (a) From the quadratic formula the roots are

$$k_1 = \frac{\|\mathbf{u}\|^2 - \|\mathbf{v}\|^2 + \sqrt{(\|\mathbf{u}\|^2 - \|\mathbf{v}\|^2)^2 + 4(\mathbf{u}, \mathbf{v})^2}}{2(\mathbf{u}, \mathbf{v})}$$

$$k_2 = \frac{\|\mathbf{u}\|^2 - \|\mathbf{v}\|^2 - \sqrt{(\|\mathbf{u}\|^2 - \|\mathbf{v}\|^2)^2 + 4(\mathbf{u}, \mathbf{v})^2}}{2(\mathbf{u}, \mathbf{v})}.$$

Clearly $k_1 > 0$ and $k_2 < 0$. Moreover,

$$k_1 k_2 = \frac{(\|\mathbf{u}\|^2 - \|\mathbf{v}\|^2)^2 - [(\|\mathbf{u}\|^2 - \|\mathbf{v}\|^2)^2 + 4(\mathbf{u}, \mathbf{v})^2]}{4(\mathbf{u}, \mathbf{v})^2} = -1.$$

(b) Since $k_2 = -1/k_1$,

$$\begin{aligned}\mathbf{u}_1^{(2)} &= \mathbf{u} - k_2 \mathbf{v} = \mathbf{u} + \frac{1}{k_1} \mathbf{v} = \frac{1}{k_1} (\mathbf{v} + k_1 \mathbf{u}) = \frac{1}{k_1} \mathbf{v}_1^{(1)} \\ \mathbf{v}_1^{(2)} &= \mathbf{v} + k_2 \mathbf{u} = \mathbf{v} - \frac{1}{k_1} \mathbf{u} = -\frac{1}{k_1} (\mathbf{u} - k_1 \mathbf{v}) = -\frac{1}{k_1} \mathbf{u}_1^{(1)}.\end{aligned}$$

10.6.30. $\begin{vmatrix} -15 - \lambda & 10 \\ -25 & 15 - \lambda \end{vmatrix} = \lambda^2 + 25$. The augmented matrix of $(A - 5iI)\mathbf{x} = 0$ is

$$\begin{bmatrix} -15 - 5i & 10 & : & 0 \\ -25 & 15 - 5i & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & \frac{-3+i}{5} & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}.$$

Therefore, $x_1 = \frac{(3-i)}{5}x_2$. Taking $x_2 = 5$ yields the eigenvector $\mathbf{x} = \begin{bmatrix} 3-i \\ 5 \end{bmatrix}$, so $\mathbf{u} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ and

$\mathbf{v} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$. The quadratic equation is $-3k^2 - 33k + 3 = 0$, with positive root $k \approx .0902$.

Routine calculations yield $\mathbf{U} \approx \begin{bmatrix} .5257 \\ .8507 \end{bmatrix}$, $\mathbf{V} \approx \begin{bmatrix} -.8507 \\ .5257 \end{bmatrix}$.

10.6.32. $\begin{vmatrix} -3 - \lambda & -15 \\ 3 & 3 - \lambda \end{vmatrix} = \lambda^2 + 36$. The augmented matrix of $(A - 6iI)\mathbf{x} = 0$ is

$$\begin{bmatrix} -3 - 6i & -15 & : & 0 \\ 3 & 3 - 6i & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & 1 - 2i & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}.$$

Therefore, $x_1 = -(1 - 2i)x_2$. Taking $x_2 = 1$ yields the eigenvector $\mathbf{x} = \begin{bmatrix} -1 + 2i \\ 1 \end{bmatrix}$, so

$\mathbf{u} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$. The quadratic equation is $-2k^2 + 2k + 2 = 0$, with positive root

$k \approx 1.6180$. Routine calculations yield $\mathbf{U} \approx \begin{bmatrix} -.9732 \\ .2298 \end{bmatrix}$, $\mathbf{V} \approx \begin{bmatrix} .2298 \\ .9732 \end{bmatrix}$.

10.6.34. $\begin{vmatrix} 5 - \lambda & -12 \\ 6 & -7 - \lambda \end{vmatrix} = (\lambda + 1)^2 + 36$. The augmented matrix of $(A - (-1 + 6i)I)\mathbf{x} = 0$ is

$$\begin{bmatrix} 6 - 6i & -12 & : & 0 \\ 6 & -6 - 6i & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & -(1+i) & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}.$$

Therefore, $x_1 = (1 + i)x_2$. Taking $x_2 = 1$ yields the eigenvector $\mathbf{x} = \begin{bmatrix} 1+i \\ 1 \end{bmatrix}$, so $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and

$\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. The quadratic equation is $k^2 - k - 1 = 0$, with positive root $k \approx 1.6180$. Routine

calculations yield $\mathbf{U} \approx \begin{bmatrix} -.5257 \\ .8507 \end{bmatrix}$, $\mathbf{V} \approx \begin{bmatrix} .8507 \\ .5257 \end{bmatrix}$.

10.6.36. $\begin{vmatrix} -4 - \lambda & 9 \\ -5 & 2 - \lambda \end{vmatrix} = (\lambda + 1)^2 + 36$. The augmented matrix of $(A - (-1 + 6i)I)\mathbf{x} = 0$ is

$$\begin{bmatrix} -3 - 6i & 9 & : & 0 \\ -5 & 3 - 6i & : & 0 \end{bmatrix}, \text{ which is row equivalent to } \begin{bmatrix} 1 & -\frac{3-6i}{5} & : & 0 \\ 0 & 0 & : & 0 \end{bmatrix}.$$

Therefore, $x_1 = \frac{3-6i}{5}x_2$. Taking $x_2 = 5$ yields the eigenvector $\mathbf{x} = \begin{bmatrix} 3-6i \\ 5 \end{bmatrix}$, so $\mathbf{u} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ and

$\mathbf{v} = \begin{bmatrix} -6 \\ 0 \end{bmatrix}$. The quadratic equation is $-18k^2 + 2k + 18 = 0$, with positive root $k \approx 1.0571$.

Routine calculations yield $\mathbf{U} \approx \begin{bmatrix} .8817 \\ .4719 \end{bmatrix}$, $\mathbf{V} \approx \begin{bmatrix} -.4719 \\ .8817 \end{bmatrix}$.

10.6.38. $\begin{vmatrix} -1-\lambda & -5 \\ 20 & -1-\lambda \end{vmatrix} = (\lambda+1)^2 + 100$. The augmented matrix of $(A - (-1+10i)I)\mathbf{x} = 0$ is $\begin{bmatrix} -10i & -5 & \vdots & 0 \\ 20 & -10i & \vdots & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & -\frac{i}{2} & \vdots & 0 \\ 0 & 0 & \vdots & 0 \end{bmatrix}$. Therefore, $x_1 = \frac{i}{2}x_2$.

Taking $x_2 = 2$ yields the eigenvector $\mathbf{x} = \begin{bmatrix} i \\ 2 \end{bmatrix}$, so $\mathbf{u} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Since $(\mathbf{u}, \mathbf{v}) = 0$ we just normalize $\mathbf{u}$ and $\mathbf{v}$ to obtain $\mathbf{U} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\mathbf{V} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

10.6.40. $\begin{vmatrix} -7-\lambda & 6 \\ -12 & 5-\lambda \end{vmatrix} = (\lambda+1)^2 + 36$. The augmented matrix of $(A - (-1+6i)I)\mathbf{x} = 0$ is $\begin{bmatrix} -6-6i & 6 & \vdots & 0 \\ -12 & 6-6i & \vdots & 0 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & -\frac{1-i}{2} & \vdots & 0 \\ 0 & 0 & \vdots & 0 \end{bmatrix}$.

Therefore, $x_1 = \frac{1-i}{2}x_2$. Taking $x_2 = 2$ yields the eigenvector $\mathbf{x} = \begin{bmatrix} 1-i \\ 2 \end{bmatrix}$, so $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$. The quadratic equation is $-k^2 - 4k + 1 = 0$, with positive root $k \approx .2361$.

Routine calculations yield $\mathbf{U} \approx \begin{bmatrix} .5257 \\ .8507 \end{bmatrix}$, $\mathbf{V} \approx \begin{bmatrix} -.8507 \\ .5257 \end{bmatrix}$.

10.7 Variation of Parameters for Nonhomogeneous Linear Systems

$$10.7.2. Y = \begin{bmatrix} -3e^{-t} & -e^{-2t} \\ e^{-t} & 2e^{-2t} \end{bmatrix}; \mathbf{u}' = Y^{-1}\mathbf{f} = \frac{1}{5} \begin{bmatrix} -2e^t & -e^t \\ e^{2t} & 3e^{2t} \end{bmatrix} \begin{bmatrix} 50e^{3t} \\ 10e^{-3t} \end{bmatrix} = \begin{bmatrix} -20e^{4t} - 2e^{-2t} \\ 10e^{5t} + 6e^{-t} \end{bmatrix};$$

$$\mathbf{u} = \begin{bmatrix} e^{-2t} - 5e^{4t} \\ 2e^{5t} - 6e^{-t} \end{bmatrix}; \mathbf{y}_p = Y\mathbf{u} = \begin{bmatrix} 13e^{3t} + 3e^{-3t} \\ -e^{3t} - 11e^{-3t} \end{bmatrix}.$$

$$10.7.4. Y = \begin{bmatrix} -e^{2t} & -e^{-t} \\ 2e^{2t} & e^{-t} \end{bmatrix}; \mathbf{u}' = Y^{-1}\mathbf{f} = \begin{bmatrix} e^{-2t} & e^{-2t} \\ -2e^t & -e^t \end{bmatrix} \begin{bmatrix} 2 \\ -2e^t \end{bmatrix} = \begin{bmatrix} 2e^{-2t} - 2e^{-t} \\ 2e^{2t} - 4e^t \end{bmatrix};$$

$$\mathbf{u} = \begin{bmatrix} 2e^{-t} - e^{-2t} \\ e^{2t} - 4e^t \end{bmatrix}; \mathbf{y}_p = Y\mathbf{u} = \begin{bmatrix} 5 - 3e^t \\ 5e^t - 6 \end{bmatrix}.$$

$$10.7.6. Y = \begin{bmatrix} \sin t & -\cos t \\ \cos t & \sin t \end{bmatrix}; \mathbf{u}' = Y^{-1}\mathbf{f} = \begin{bmatrix} \sin t & \cos t \\ -\cos t & \sin t \end{bmatrix} \begin{bmatrix} 1 \\ t \end{bmatrix} = \begin{bmatrix} t \cos t + \sin t \\ t \sin t - \cos t \end{bmatrix}; \mathbf{u} = \begin{bmatrix} t \sin t \\ -t \cos t \end{bmatrix};$$

$$\mathbf{y}_p = Y\mathbf{u} = \begin{bmatrix} t \\ 0 \end{bmatrix}.$$

$$10.7.8. Y = \begin{bmatrix} e^{3t} & e^{2t} & e^{-t} \\ -e^{3t} & 0 & -3e^{-t} \\ e^{3t} & e^{2t} & 7e^{-t} \end{bmatrix};$$

$$\mathbf{u}' = Y^{-1}\mathbf{f} = \frac{1}{6} \begin{bmatrix} 3e^{-3t} & -6e^{-3t} & -3e^{-3t} \\ 4e^{-2t} & 6e^{-2t} & 2e^{-2t} \\ -e^t & 0 & e^t \end{bmatrix} \begin{bmatrix} 1 \\ e^t \\ e^t \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 3e^{-3t} - 9e^{-2t} \\ 8e^{-t} + 4e^{-2t} \\ e^{2t} - e^t \end{bmatrix};$$

$$\mathbf{u} = \frac{1}{12} \begin{bmatrix} 9e^{-2t} - 2e^{-3t} \\ -16e^{-t} - 4e^{-2t} \\ e^{2t} - 2e^t \end{bmatrix}; \mathbf{y}_p = Y\mathbf{u} = -\frac{1}{6} \begin{bmatrix} 3e^t + 4 \\ 6e^t - 4 \\ 10 \end{bmatrix}.$$

$$10.7.10. Y = \begin{bmatrix} -e^t & e^{-t} & te^{-t} \\ e^t & -e^{-t} & 3e^{-t} - te^{-t} \\ e^t & e^{-t} & te^{-t} \end{bmatrix};$$

$$\mathbf{u}' = Y^{-1}\mathbf{f} = \frac{1}{18} \begin{bmatrix} -9e^{-t} & 0 & 9e^{-t} \\ 3e^t(3-2t) & -6te^t & 9e^t \\ 6e^t & 6e^t & 0 \end{bmatrix} \begin{bmatrix} e^t \\ e^t \\ e^t \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 0 \\ e^{2t}(3-2t) \\ 2e^{2t} \end{bmatrix}; \mathbf{u} = \frac{1}{3} \begin{bmatrix} 0 \\ e^{2t}(2-t) \\ e^{2t} \end{bmatrix};$$

$$\mathbf{y}_p = Y\mathbf{u} = \frac{1}{3} \begin{bmatrix} 2e^t \\ e^t \\ 2e^t \end{bmatrix}.$$

$$10.7.12. \mathbf{u}' = Y^{-1}\mathbf{f} = \frac{1}{2t} \begin{bmatrix} e^{-t} & e^{-t} \\ e^t & -e^t \end{bmatrix} \begin{bmatrix} t \\ t^2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} e^{-t}(t+1) \\ e^t(1-t) \end{bmatrix}; \mathbf{u} = \frac{1}{2} \begin{bmatrix} -e^{-t}(t+2) \\ e^t(2-t) \end{bmatrix};$$

$$\mathbf{y}_p = Y\mathbf{u} = \frac{t}{2} \begin{bmatrix} e^t & e^{-t} \\ e^t & -e^{-t} \end{bmatrix} \begin{bmatrix} -e^{-t}(t+2) \\ e^t(2-t) \end{bmatrix} = - \begin{bmatrix} t^2 \\ 2t \end{bmatrix}.$$

$$10.7.14. \mathbf{u}' = Y^{-1}\mathbf{f} = \frac{1}{3} \begin{bmatrix} 2 & -e^{-t} \\ -e^t & 2 \end{bmatrix} \begin{bmatrix} e^{2t} \\ e^{-2t} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2e^{2t} - e^{-3t} \\ 2e^{-2t} - e^{3t} \end{bmatrix}; \mathbf{u} = \frac{1}{9} \begin{bmatrix} 3e^{2t} + e^{-3t} \\ -e^{3t} - 3e^{-2t} \end{bmatrix};$$

$$\mathbf{y}_p = Y\mathbf{u} = \frac{1}{9} \begin{bmatrix} 2 & e^{-t} \\ e^t & 2 \end{bmatrix} \begin{bmatrix} 3e^{2t} + e^{-3t} \\ -e^{3t} - 3e^{-2t} \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 5e^{2t} - e^{-3t} \\ e^{3t} - 5e^{-2t} \end{bmatrix}.$$

$$10.7.16. \mathbf{u}' = Y^{-1}\mathbf{f} = \frac{1}{t^2-1} \begin{bmatrix} t & -e^{-t} \\ -e^t & t \end{bmatrix} \begin{bmatrix} t^2-1 \\ t^2-1 \end{bmatrix} = \begin{bmatrix} t-e^{-t} \\ t-e^t \end{bmatrix}; \mathbf{u} = \frac{1}{2} \begin{bmatrix} 2e^{-t} + t^2 \\ t^2 - 2e^t \end{bmatrix};$$

$$\mathbf{y}_p = Y\mathbf{u} = \frac{1}{2} \begin{bmatrix} t & e^{-t} \\ e^t & t \end{bmatrix} \begin{bmatrix} 2e^{-t} + t^2 \\ t^2 - 2e^t \end{bmatrix} = \frac{1}{2} \begin{bmatrix} te^{-t}(t+2) + t^3 - 2 \\ te^t(t-2) + t^3 + 2 \end{bmatrix}.$$

$$10.7.18. \mathbf{u}' = Y^{-1}\mathbf{f} = \frac{1}{3} \begin{bmatrix} e^{-5t} & e^{-4t} & e^{-3t} \\ 2e^{-2t} & -e^{-t} & -1 \\ -e^{-2t} & 2e^{-t} & -1 \end{bmatrix} \begin{bmatrix} e^t \\ 0 \\ 0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} e^{-4t} \\ 2e^{-t} \\ -e^{-t} \end{bmatrix}; \mathbf{u} = \frac{1}{12} \begin{bmatrix} -e^{-4t} \\ -8e^{-t} \\ 4e^{-t} \end{bmatrix};$$

$$\mathbf{y}_p = Y\mathbf{u} = \frac{1}{12} \begin{bmatrix} e^{5t} & e^{2t} & 0 \\ e^{4t} & 0 & e^t \\ e^{3t} & -1 & -1 \end{bmatrix} \begin{bmatrix} -e^{-4t} \\ -8e^{-t} \\ 4e^{-t} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -3e^t \\ 1 \\ e^{-t} \end{bmatrix}.$$

$$10.7.20. \mathbf{u}' = Y^{-1}\mathbf{f} = \frac{1}{2t} \begin{bmatrix} -e^{-2t} & te^{-t} & te^{-t} + e^{-2t} \\ 1 & -te^t & te^t - 1 \\ 2 & 0 & -2 \end{bmatrix} \begin{bmatrix} e^t \\ 0 \\ e^t \end{bmatrix} = \begin{bmatrix} 1 \\ e^{2t} \\ 0 \end{bmatrix}; \mathbf{u} = \frac{1}{4} \begin{bmatrix} 2t \\ e^{2t} \\ 0 \end{bmatrix};$$

$$\mathbf{y}_p = Y\mathbf{u} = Y = \frac{1}{4t} \begin{bmatrix} e^t & e^{-t} & t \\ e^t & -e^{-t} & e^{-t} \\ e^t & e^{-t} & 0 \end{bmatrix} \begin{bmatrix} 2t \\ e^{2t} \\ 0 \end{bmatrix} = \frac{e^t}{4t} \begin{bmatrix} 2t+1 \\ 2t-1 \\ 2t+1 \end{bmatrix}.$$

10.7.22. (c) If $\mathbf{y}_p = Y\mathbf{u}$, then $\mathbf{y}_p' = Y'\mathbf{u} + Y\mathbf{u}' = AY\mathbf{u} + Y\mathbf{u}'$, so (E) $\mathbf{y}_p' = A\mathbf{y}_p + Y\mathbf{u}'$. However, from the derivation of the method of variation of parameters in Section 9.4, $Y\mathbf{u}' = \mathbf{f}$ as

defined in the solution of (a). This and (E) imply the conclusion.

- (d) Since $Y\mathbf{u}' = \mathbf{f}$ with $\mathbf{f}$ as defined in the solution of (a), $u_1, u_2, \dots, u_n$ satisfy the conditions required in the derivation of the method of variation of parameters in Section 9.4; hence, $y_p = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$ is a particular solution of (A).